Ganong's Review of Medical Physiology

26th Edition

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Ganong's Review of Medical Physiology

TWENTY-SIXTH EDITION

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Dedication to

William Francis Ganong

William Francis ("Fran") Ganong was an outstanding scientist, educator, and writer. He was completely dedicated to the field of physiology and medical education in general. Chairman of the Department of Physiology at the University of California, San Francisco, for many years, he received numerous teaching awards and loved working with medical students.

Over the course of 40 years and some 22 editions, he was the sole author of the best selling *Review of Medical Physiology*, and a co-author of 5 editions of *Pathophysiology of Disease: An Introduction to Clinical Medicine*. He was one of the "deans" of the Lange group of authors who produced concise medical text and review books that to this day remain extraordinarily popular in print and now in digital formats. Dr. Ganong made a gigantic impact on the education of countless medical students and clinicians.

A general physiologist par excellence and a neuroendocrine physiologist by subspecialty, Fran developed and maintained a rare understanding of the entire field of physiology. This allowed him to write each new edition (every 2 years!) of the *Review of Medical Physiology* as a sole author, a feat remarked on and admired whenever the book came up for discussion among physiologists. He was an excellent writer and far ahead of his time with his objective of distilling a complex subject into a concise presentation. Like his good friend, Dr. Jack Lange, founder of the Lange series of books, Fran took great pride in the many different translations of the *Review of Medical Physiology* and was always delighted to receive a copy of the new edition in any language.

He was a model author, organized, dedicated, and enthusiastic. His book was his pride and joy and like other best-selling authors, he would work on the next edition seemingly every day, updating references, rewriting as needed, and always ready and on time when the next edition was due to the publisher. He did the same with his other book, *Pathophysiology of Disease: An Introduction to Clinical Medicine*, a book that he worked on meticulously in the years following his formal retirement and appointment as an emeritus professor at UCSF.

Fran Ganong will always have a seat at the head table of the greats of the art of medical science education and communication. He died on December 23, 2007. All of us who knew him and worked with him miss him greatly.

About the Authors

KIM E. BARRETT

Kim Barrett received her PhD in biological chemistry from University College London in 1982. Following postdoctoral training at the National Institutes of Health, she joined the faculty at the University of California, San Diego, School of Medicine in 1985, rising to the rank of Professor of Medicine in 1996, and was named Distinguished Professor of Medicine in 2015. From 2006 to 2016, she also served the University as Dean of the Graduate Division. Her research interests focus on the physiology and pathophysiology of the intestinal epithelium, and how its function is altered by commensal, probiotic, and pathogenic bacteria as well as in specific disease states, such as inflammatory bowel diseases. She has published more than 250 articles, chapters, and reviews, and has received several honors for her research accomplishments including the Bowditch and Davenport Lectureships from the American Physiological Society (APS), the Bayliss-Starling Lectureship from The Physiological Society of the UK and Ireland, and the degree of Doctor of Medical Sciences, honoris causa, from Queens University, Belfast. She has been very active in scholarly editing, serving currently as the Editor-in-Chief of the *Journal of Physiology*. She is also a dedicated and award-winning instructor of medical, pharmacy, and graduate students, and has taught various topics in medical and systems physiology to

these groups for more than 30 years. Her efforts as a teacher and mentor were recognized with the Bodil M. Schmidt- Nielson Distinguished Mentor and Scientist Award from the APS in 2012, and she also served as the 86th APS President from 2013 to 2014. Her teaching experiences led her to author a prior volume (*Gastrointestinal Physiology*, McGraw-Hill, 2005; second edition published in 2014) and she was honored to have been invited to take over the helm of Ganong in 2007 for the 23rd and subsequent editions, including this one.

SUSAN M. BARMAN

Susan Barman received her PhD in physiology from Loyola University School of Medicine in Maywood, Illinois. Afterward she went to Michigan State University (MSU) where she is currently a Professor in the Department of Pharmacology/Toxicology and the Neuroscience Program. She is also Chair of the Institutional Animal Care and Use Committee and serves on the College of Human Medicine (CHM) Curriculum Development Group for medical school education. She has had a career-long interest in neural control of cardiorespiratory function with an emphasis on the characterization and origin of the naturally occurring discharges of sympathetic and phrenic nerves. She has published about 150 research articles, invited review articles, and book chapters. She was a recipient of a prestigious National Institutes of Health MERIT (Method to Extend Research in Time) Award. She is also a recipient of an MSU Outstanding University Woman Faculty Award, a CHM Distinguished Faculty Award, a Distinguished Service Award from the Association of Chairs of Departments of Physiology, and the Carl Ludwig Distinguished Lecture Award from the Neural Control of Autonomic Regulation section of the American Physiological Society (APS). She is also a Fellow of the APS and served as its 85th President. She has also served as a Councilor of APS and Chair of the Women in Physiology and Section Advisory Committees of the APS. She is also active in the Michigan Physiological Society, a chapter of the APS.

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Heddwen Brooks received her PhD from Imperial College, University of London and is a Professor in the Departments of Physiology and Pharmacology at the University of Arizona (UA). Dr. Brooks is a renal physiologist and is best known for her development of microarray technology to address in vivo signaling pathways involved in the hormonal regulation of renal function. Dr. Brooks' many awards include the American Physiological Society (APS) Lazaro J. Mandel Young Investigator Award, which is for an individual demonstrating outstanding promise in epithelial or renal physiology. In 2009, Dr. Brooks received the APS Renal Young Investigator Award at the annual meeting of the Federation of American Societies for Experimental Biology. Dr. Brooks served as Chair of the APS Renal Section (2011–2014) and currently serves as Associate Editor for the *American Journal of Physiology-Regulatory, Integrative and Comparative Physiology*, and on the Editorial Board for the *American Journal of Physiology-Renal Physiology* (since 2001). Dr. Brooks has served on study sections of the National Institutes of Health, the American Heart Association, and served as a member of the Nephrology Merit Review Board for the Department of Veterans' Affairs.

JASON X.-J. YUAN

Jason Yuan received his medical degree from Suzhou Medical College (Suzhou, China) in 1983, his doctoral degree in cardiovascular physiology from Peking Union Medical College (Beijing, China), and his postdoctoral training at the University of Maryland at Baltimore. He joined the faculty at the University of Maryland School of Medicine in 1993 and then moved to the University of California, San Diego in 1999, rising to the rank of Professor in 2013. His research interests center on pathogenic roles of membrane receptors and ion channels in pulmonary vascular disease. He has published more than 300 articles, reviews, editorials and chapters, and has edited or co-edited nine books. He has received several honors for his research accomplishments including the Cournand and Comroe Young Investigator Award, the Established Investigator Award and the Kenneth D. Bloch Memorial Lectureship from the American Heart Association; the Guggenheim Fellowship Award from the John Simon Guggenheim Memorial Foundation; the Estelle Grover Lectureship from the American Thoracic Society; and the Robert M. Berne Memorial Lectureship from The American Physiological Society. He is an elected Fellow of the American Association for the Advancement of Science and an elected Member of the American Society for Clinical Investigation and the Association of American Physicians. He has served on many advisory committees including Chair of the Respiratory Integrative Biology and Translational Research study section of the National Institutes of Health and Chair of the Pulmonary Circulation Assembly of the American Thoracic Society. He has also been very active in scholarly editing serving currently as the Editor-in-Chief of the journal *Pulmonary Circulation* and Associate Editor of the *American Journal of Physiology-Cell Physiology*. He is a leading editor of the *Textbook of Pulmonary Vascular Disease* (Springer, 2011).

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Preface

It is difficult to believe that this preface signifies the fourth edition *of Ganong's Review of Medical Physiology* that our author group has overseen, and the 26th edition overall of this important reference work aimed at medical and other health professional students. As always, we have tried to maintain the highest standards of excellence that were promulgated by the original author, Fran Ganong, over the 46 years where he served, remarkably, as the sole author of the textbook.

In this new edition, we have cast a fresh eye on the pedagogical approach taken in each chapter and section, and have focused particularly on including only material that is of the highest yield. We have thoroughly revised the learning objectives for every chapter, reorganized and updated the text to ensure that all objectives are clearly addressed in a logical order, aligned chapter summaries so that the take-home messages quickly address each learning objective in turn, and expanded the number of review questions so that readers also have the ability to check their understanding and retention of every objective covered. As a discipline evolves and new information emerges, there is a tendency simply to concatenate these concepts such that chapter structure degrades inevitably over time. With in-depth discussions amongst the author team and significant "spring-cleaning," we believe we have freshened and simplified the volume while also making sure that important new developments are incorporated. We are immensely thankful to Erica Wehrwein, PhD, Assistant Professor of Physiology and an award-winning instructor at Michigan State University, who took on the task of reviewing the book as a whole and providing specific and detailed feedback to us on each chapter.

This new edition also welcomes a new member to the author team. We are delighted to have been able to recruit Jason X.-J. Yuan, MD, PhD, Professor of Medicine and Physiology as well as Chief of the Division of Translational and Regenerative Medicine and Associate Vice President for Translational Health Sciences at the University of Arizona, who has assumed responsibility for some

cell physiology and cardiovascular topics, as well as the respiratory physiology section. We are particularly excited to have a physician-scientist on the team, who can guide us overall to focus on material that is of most benefit to those preparing for a career incorporating patient care. We are most grateful for the past contributions of Scott Boitano, PhD, whose other obligations meant that he could no longer serve as an author.

We continue to be gratified by the many colleagues and students who contact us from all over the world to request clarification of material covered in the text, or to point out errors or omissions. We are especially grateful to Rajan Pandit, Lecturer in Physiology at Nepal Medical College, who has painstakingly offered dozens of suggestions for revision over the years. His efforts, and those of the many others whom we have not named, allow us to engage in a process of continual improvement. While, as always, any errors that remain in the book (inevitable in a complex project such as this) are the sole responsibility of the authorship team, we greatly value critical input, and urge readers once again to contact us with any suggestions or critiques. We thank you in advance both for such feedback, and also for your support of this new edition.

This edition is a revision of the original works of Dr. Francis Ganong.

SECTION I

Cellular & Molecular Basis for Medical Physiology

Study of physiological system structure and function, as well as pathophysiological alterations, has its foundations in physical and chemical laws and the molecular and cellular makeup of each tissue and organ system. *Ganong's Review of Medical Physiology* is structured into seven sections. This first section provides an overview of the basic building blocks or bases that provide the important framework for human physiology. It is important to note here that the seven chapters in this initial section are not meant to provide an exhaustive understanding of biophysics, biochemistry, or cellular and molecular physiology; rather, they are to serve as a reminder of how the basic principles from these disciplines contribute to medical physiology discussed in later sections associated with physiological functions of organs and systems.

In the first two chapters of this section, the following basic building blocks are introduced and discussed: electrolytes; carbohydrates, lipids, and fatty acids; amino acids and proteins; and nucleic acids. Students are reminded of some of the basic principles and building blocks of biophysics and biochemistry and how they fit into the physiologic environment. Examples of direct clinical applications are provided in the clinical boxes to help bridge the gap between basic principles and human cell, tissue, and organ functions. These basic principles are followed up with a discussion of the generic cell and its components. It is important to realize the cell is the basic functional unit within the body, and it is the collection and fine-tuned interactions among and between these fundamental units that allow for proper tissue, organ, and organism function.

In the third to seventh chapters of this introductory section, we take a cellular

approach to lay a groundwork of understanding groups of cells that interact with many of the systems discussed in future chapters. The first group of cells presented contribute to inflammatory reactions in the body. These individual players, their coordinated behavior, and the net effects of the “open system” of inflammation in the body are discussed in detail. The second group of cells discussed are responsible for the excitatory responses in human physiological function and include both neuronal and muscle cells. A fundamental understanding of the inner workings of these cells, and how they are controlled by their neighboring cells, helps the student to understand their eventual integration into individual systems discussed in later sections.

This first section serves as an introduction, refresher, and quick source of material to best understand organ functions and systems physiology presented in the later sections. For detailed understanding of any of the chapters within this section, several excellent and current textbooks that provide more in-depth reviews of principles of biochemistry, biophysics, cell physiology, and muscle and neuronal physiology are provided as resources at the end of each individual chapter. Students are encouraged to visit these texts for a more thorough understanding of these basic principles.

CHAPTER 1

General Principles & Energy Production in Medical Physiology

OBJECTIVES

After studying this chapter, you should be able to:

- Define functional units used in measuring physiological properties.
- Define pH and buffering.
- Understand electrolytes and define diffusion, osmosis, and tonicity.
- Define and explain the significance of resting membrane potential.
- Understand in general terms the basic building blocks of the cell (eg, nucleotides, amino acids, carbohydrates, and fatty acids) to cell metabolism, proliferation, and function.
- Understand higher-order structures of the basic building blocks of the cell (eg, DNA, RNA, proteins, and lipids) to cell replication, proliferation, and signal transduction.
- Understand the basic contributions of the basic building blocks of the cell to its structure, function, and energy balance.

INTRODUCTION

In unicellular organisms, all vital processes occur in a single cell. As the evolution of multicellular organisms progressed, various cell groups organized into tissues, and organs have taken over particular functions. In humans and other vertebrate animals, there are a number of specialized collections of cells that consist in organ systems serving for different functions. For example, a gastrointestinal system to digest and absorb food; a respiratory system to take up O_2 and eliminate CO_2; a urinary system to remove wastes; a cardiovascular system to distribute nutrients, O_2, and the products of metabolism; a reproductive system to perpetuate the species; and nervous and endocrine systems to coordinate and integrate the functions of the other systems. This book is concerned with the way these systems function and the way each contributes to the functions of the body as a whole. This first chapter lays a foundation for the discussion of these organ systems with a review of basic biophysical and biochemical principles at the cellular level and the introduction of the molecular building blocks that contribute to cell physiological function within these organ systems.

GENERAL PRINCIPLES

THE BODY AS ORGANIZED "SOLUTIONS"

In the average young adult male, 18% of the body weight is protein and related substances, 7% is mineral, and 15% is fat. The remaining 60% is water. The distribution of the body water is shown in **Figure 1–1A**.

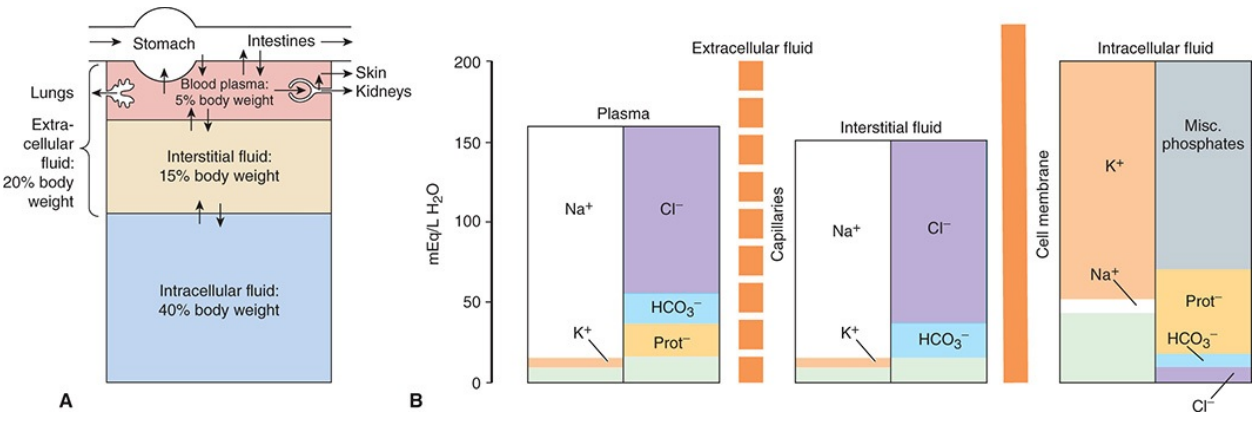


FIGURE 1–1 Organization of body fluids and electrolytes into compartments. A) Body fluids can be divided into intracellular and extracellular fluid compartments (ICF and ECF, respectively). Their contribution

to percentage body weight (based on a healthy young adult male; slight variations exist with age and gender) emphasizes the dominance of fluid makeup of the body. Transcellular fluids, which constitute a very small percentage of total body fluids, are not shown. Arrows represent fluid movement between compartments. **B)** Electrolytes and proteins are unequally distributed among the body fluids. This uneven distribution is crucial to physiology. $Prot^-$, protein, which tends to have a negative charge at physiologic pH.

The cells that make up the bodies of all but the simplest multicellular animals, both aquatic and terrestrial, exist in an "internal sea" of **extracellular fluid (ECF)** enclosed within the integument of the animal. From this fluid, the cells take up O_2 and nutrients; into it, they discharge metabolic waste products. The ECF is more dilute than present-day seawater, but its composition closely resembles that of the primordial oceans in which, presumably, all life originated.

In animals with a closed vascular system, the ECF is divided into the **interstitial fluid,** the circulating **blood plasma,** and **the lymph fluid that bridges these two domains.** The interstitial fluid is the part of the ECF that is outside the vascular and lymph systems, bathing the cells. The plasma and the cellular elements of the blood, principally red blood cells, fill the vascular system, and together they constitute the **total blood volume.** About one-third of the **total body water** is extracellular; the remaining two-thirds is intracellular **(intracellular fluid).** Inappropriate compartmentalization of the body fluids can result in edema **(Clinical Box 1–1)**.

CLINICAL BOX 1–1

Edema

Edema is the buildup of body fluids extracellularly or interstitially in tissues. The increased fluid is related to an increased leak from the blood and/or reduced removal by the lymph system. Edema is often observed in the feet, ankles, and legs, but can happen in many areas of the body in response to disease, including those of the heart, lung, liver, kidney, or thyroid.

THERAPEUTIC HIGHLIGHTS

The best treatment for edema includes reversing the underlying disorder. Thus, proper diagnosis of the cause of edema is the primary first step in therapy. More

general treatments include restricting dietary sodium to minimize fluid retention and using appropriate diuretic therapy.

The intracellular component of the body water accounts for about 40% of body weight and the extracellular component for about 20%. Approximately 25% of the extracellular component is in the vascular system (plasma = 5% of body weight) and 75% outside the blood vessels (interstitial fluid = 15% of body weight). The total blood volume is about 8% of body weight. Flow between these compartments is tightly regulated.

UNITS FOR MEASURING CONCENTRATION OF SOLUTES

In considering the effects of various physiologically important substances and the interactions among them, the number of molecules, electrical charges, or particles of a substance per unit volume of a particular body fluid are often more meaningful than simply the weight of the substance per unit volume. For this reason, physiological concentrations are frequently expressed in moles, equivalents, or osmoles.

Moles

A mole is the gram-molecular weight of a substance, that is, the molecular weight (MW) of the substance in grams. Each mole (mol) consists of 6×10^{23} molecules. The millimole (mmol) is 1/1000 of a mole, and the micromole (µmol) is 1/1,000,000 of a mole. Thus, 1 mol of NaCl = 23 g + 35.5 g = 58.5 g and 1 mmol = 58.5 mg. The mole is the standard unit for expressing the amount of substances in the SI unit system.

The molecular weight of a substance is the ratio of the mass of one molecule of the substance to the mass of one-twelfth the mass of an atom of carbon-12. Because molecular weight is a ratio, it is dimensionless. The dalton (Da) is a unit of mass equal to one-twelfth the mass of an atom of carbon-12. The kilodalton (1 kDa = 1000 Da) is a useful unit for expressing the molecular mass of proteins. Thus, for example, one can speak of a 64-kDa protein or state that the **molecular mass** of the protein is 64,000 Da. However, because molecular weight is a dimensionless ratio, it is incorrect to say that the molecular weight of the protein

is 64 kDa.

Equivalents

The concept of electrical equivalence is important in physiology because many of the solutes in the body are in the form of charged particles. One equivalent (Eq) is 1 mol of an ionized substance divided by its valence. One mole of NaCl dissociates into 1 Eq of Na^+ and 1 Eq of Cl^-. One equivalent of Na^+ = 23 g, but 1 Eq of Ca^{2+} = 40 g ÷ 2 = 20 g. The milliequivalent (mEq) is 1/1000 of 1 Eq.

Electrical equivalence is not necessarily the same as chemical equivalence. A gram equivalent is the weight of a substance that is chemically equivalent to 8.0 g of oxygen. The normality (N) of a solution is the number of gram equivalents in 1 liter (L). A 1 N solution of hydrochloric acid (HCl) contains both H^+ (1.0 g) and Cl^- (35.5 g) equivalents, = (1.0 g + 35.5 g)/L = 36.5 g/L.

WATER, ELECTROLYTES, & ACID/BASE

The water molecule (H_2O) is an ideal solvent for physiological reactions. H_2O has a **dipole moment** where oxygen slightly pulls away electrons from the hydrogen atoms and creates a charge separation that makes the molecule **polar.** This allows water to dissolve a variety of charged atoms and molecules. It also allows the H_2O molecule to interact with other H_2O molecules via hydrogen bonding. The resulting hydrogen bond network in water allows for several key properties relevant to physiology: (1) water has a high surface tension, (2) water has a high heat of vaporization and heat capacity, and (3) water has a high dielectric constant. In layman's terms, H_2O is an excellent biological fluid that serves as a solute; it provides optimal heat transfer and conduction of current.

Electrolytes (eg, NaCl) are molecules that dissociate in water to their cation (Na^+) and anion (Cl^-) equivalents. Because of the net charge on water molecules, these electrolytes tend not to reassociate in water. There are many important electrolytes in physiology, notably Na^+, K^+, Ca^{2+}, Mg^{2+}, Cl^-, and HCO_3^-. It is important to note that electrolytes and other charged compounds (eg, proteins) are unevenly distributed in the body fluids **(Figure 1–1B)**. These separations play an important role in physiology, for example, in the establishment of membrane potential and generation of action potential.

pH & BUFFERING

The maintenance of a stable hydrogen ion concentration ($[H^+]$) in body fluids is essential to life. The **pH** of a solution is defined as the logarithm to the base 10 of the reciprocal of the H^+, that is, the negative logarithm of the $[H^+]$. The pH of water at 25°C, in which H^+ and OH^- ions are present in equal numbers, is 7.0 **(Figure 1–2)**. For each pH unit less than 7.0, the $[H^+]$ is increased 10-fold; for each pH unit above 7.0, it is decreased 10-fold. In the plasma of healthy individuals, pH is slightly alkaline, maintained in the narrow range of 7.35–7.45 **(Clinical Box 1–2)**. Conversely, gastric fluid pH can be quite acidic (on the order of 3.0) and pancreatic secretions can be quite alkaline (on the order of 8.0). Enzymatic activity and protein structure are frequently sensitive to pH; in any given body or cellular compartment, pH is maintained to allow for maximal enzyme/protein efficiency.

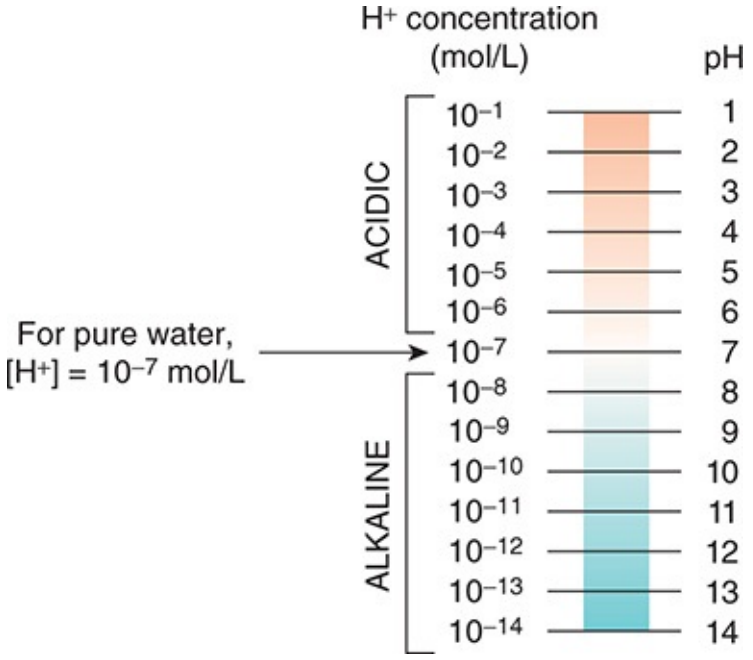


FIGURE 1–2 Proton concentration and pH. Relative proton (H^+) concentrations for solutions on a pH scale are shown.

Molecules that act as H^+ donors in solution are considered acids, while those that tend to remove H^+ from solutions are considered bases. Strong acids (eg, HCl) or bases (eg, NaOH) dissociate completely in water and thus can most change the $[H^+]$ in solution. In physiological compounds, most acids or bases are considered “weak,” that is, they contribute or remove relatively few H^+ from

solution. Body pH is stabilized by the **buffering capacity** of the body fluids. A **buffer** is a substance that has the ability to bind or release H^+ in solution, thus keeping the pH of the solution relatively constant despite the addition of considerable quantities of acid or base. Of course there are a number of buffers at work in biological fluids at any given time. All buffer pairs in a homogenous solution are in equilibrium with the same $[H^+]$; this is known as the **isohydric principle.** One outcome of this principle is that by assaying a single buffer system, we can understand a great deal about all of the biological buffers in that system.

When acids are placed into solution, there is dissociation of some of the component acid (HA) into its proton (H^+) and free acid (A^-). This is frequently written as an equation:

$$HA \rightleftharpoons H^+ + A^-$$

According to the laws of mass action, a relationship for the dissociation can be defined mathematically as:

$$K_a = [H^+][A^-]/[HA]$$

CLINICAL BOX 1–2

Acid–Base Balance and Disorders

Excesses of acid (acidosis) or base (alkalosis) exist when the blood or blood plasma is outside the normal pH range (7.35–7.45). Such changes impair the delivery of O_2 to and removal of CO_2 from tissues. There are a variety of conditions and diseases that can interfere with pH control in the body and cause blood pH to fall outside of healthy limits. Acid–base disorders that result from respiration to alter CO_2 concentration are called respiratory acidosis and respiratory alkalosis. Respiratory acidosis is often caused by respiratory failure or ventilator failure, while respiratory alkalosis is caused by alveolar hyperventilation and often found in patients with chronic liver disease. Nonrespiratory disorders that affect HCO_3^- concentration are referred to as metabolic acidosis and metabolic alkalosis. Metabolic acidosis or alkalosis can be caused by electrolyte disturbances, severe vomiting or diarrhea, ingestion of certain drugs and toxins, kidney disease, and diseases that affect normal metabolism (eg, diabetes).

THERAPEUTIC HIGHLIGHTS

Proper treatments for acid–base disorders are dependent on correctly identifying the underlying causal process(es). This is especially true when mixed disorders are encountered. Treatment of respiratory acidosis should be initially targeted at restoring ventilation, whereas treatment for respiratory alkalosis is focused on the reversal of the primary causes (eg, alveolar hyperventilation associated with head injury and anxiety, hypoxemia due to peripheral chemoreceptor stimulation, pulmonary embolism, and edema). Bicarbonate (via intravenous injection) is typically used as a treatment for acute metabolic acidosis. An adequate amount of a chloride salt can restore acid–base balance to normal over a matter of days for patients with a chloride-responsive metabolic alkalosis whereas chloride-resistant metabolic alkalosis requires treatment of the underlying disease.

where K_a is a constant, and the brackets represent concentrations of the individual species (elements). In layman's terms, the product of the proton concentration ($[H^+]$) and the free acid concentration ($[A^-]$) divided by the bound acid concentration ([HA]) is a defined constant (K). This can be rearranged to read:

$$[H^+] = K_a\,[HA]/[A^-]$$

If the logarithm of each side is taken:

$$\log[H^+] = \log K_a + \log[HA]/[A^-]$$

Both sides can be multiplied by –1 to yield:

$$-\log[H^+] = -\log K_a + \log[A^-]/[HA]$$

This can be written in a more conventional form known as the **Henderson-Hasselbalch equation:**

$$pH = pK_a + \log[A^-]/[HA]$$

This relatively simple equation is quite powerful. One thing that can be discerned right away is that the buffering capacity of a particular weak acid is

best when the pK_a of that acid is equal to the pH of the solution, or when:

$$[A^-] = [HA], pH = pK_a$$

Similar equations can be set up for weak bases. An important buffer in the body is carbonic acid (H_2CO_3). Carbonic acid is a weak acid, and thus is only partly dissociated into H^+ and HCO_3^-:

$$H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

If H^+ is added to a solution of carbonic acid, the equilibrium shifts to the left and most of the added H^+ is removed from solution. If OH^- is added, H^+ and OH^- combine, taking H^+ out of solution. However, the decrease is countered by more dissociation of H_2CO_3, and the decline in H^+ concentration is minimized. A unique feature of HCO_3^- is the linkage between its buffering ability and the ability for the lungs to remove CO_2 from the body. Other important biological buffers include phosphates and proteins.

DIFFUSION

The particles (molecules or atoms) of a substance dissolved in a solvent are in continuous random movement. Diffusion is the process by which a gas or a substance in a solution expands or moves from a region to another, because of the motion of its particles, to fill all the available volume. A given particle is equally likely to move into or out of an area in which it is present in high concentration. However, because there are more particles in the area of high concentration, the total number of particles moving to areas of lower concentration is greater; that is, there is a **net flux** of solute particles from areas of high concentration to areas of low concentration. The time required for equilibrium by diffusion is proportional to the square of the diffusion distance. The magnitude of the diffusing tendency from one region to another separated by a boundary (eg, cell membrane, blood-gas barrier) is directly proportional to the cross-sectional area across which diffusion is taking place and the **concentration gradient,** or **chemical gradient,** which is the difference in concentration of the diffusing substance divided by the thickness of the boundary **(Fick's law of diffusion).** Thus,

$$J = -DA\frac{\Delta c}{\Delta x}$$

where J is the net rate of diffusion, D is the diffusion coefficient, A is the area, and $\Delta c/\Delta x$ is the concentration gradient. The minus sign indicates the direction of diffusion. When considering movement of molecules from a higher to a lower concentration, $\Delta c/\Delta x$ is negative, so multiplying by –DA gives a positive value. The permeabilities of the boundaries across which diffusion occurs in the body vary, but diffusion is still a major force affecting the distribution of water and solutes.

OSMOSIS

When a substance is dissolved in water, the concentration of water molecules in the solution is less than that in pure water, because the addition of solute to water results in a solution that occupies a greater volume than does the water alone. If the solution is placed on one side of a membrane that is permeable to water but not to the solute, and an equal volume of water is placed on the other, water molecules diffuse down their concentration (chemical) gradient into the solution **(Figure 1–3)**. This process—the diffusion of **solvent** molecules into a region in which there is a higher concentration of a **solute** to which the membrane is impermeable—is called **osmosis.** It is an important factor in physiological processes. The tendency for movement of solvent molecules to a region of greater solute concentration can be prevented by applying pressure to the more concentrated solution. The pressure necessary to prevent solvent migration is the **osmotic pressure** of the solution.

Osmotic pressure—like vapor pressure lowering, freezing-point depression, and boiling-point elevation—depends on the number rather than the type of particles in a solution; that is, it is a fundamental colligative property of solutions. In an **ideal solution,** osmotic pressure (P) is related to temperature and volume in the same way as the pressure of a gas:

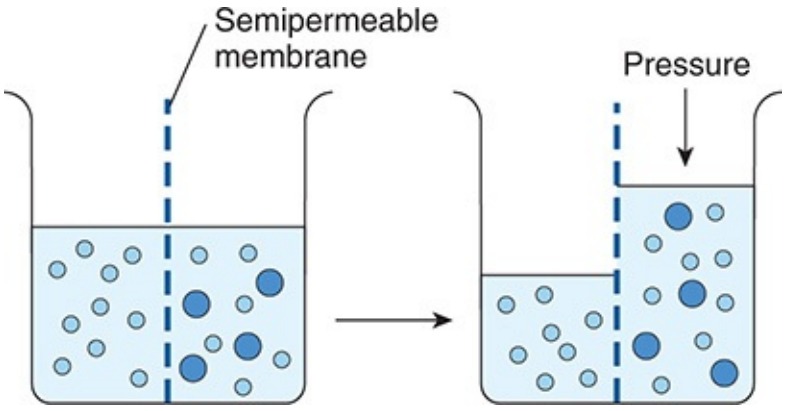


FIGURE 1–3 Diagrammatic representation of osmosis. Water molecules are represented by small open circles, and solute molecules by large solid circles. In the diagram on the left, water is placed on one side of a membrane permeable to water but not to solute, and an equal volume of a solution of the solute is placed on the other. Water molecules move down their concentration (chemical) gradient into the solution, and, as shown in the diagram on the right, the volume of the solution increases. As indicated by the arrow on the right, the osmotic pressure is the pressure that would have to be applied to prevent the movement of the water molecules.

$$P = \frac{nRT}{V}$$

where n is the number of particles, R is the gas constant, T is the absolute temperature, and V is the volume. If T is held constant, it is clear that the osmotic pressure is proportional to the number of particles in solution per unit volume of solution. For this reason, the concentration of osmotically active particles is usually expressed in **osmoles.** One osmole (Osm) equals the gram-molecular weight of a substance divided by the number of freely moving particles that each molecule liberates in solution. For biological solutions, the milliosmole (mOsm; 1/1000 of 1 Osm) is more commonly used.

If a solute is a nonionizing compound such as glucose, the osmotic pressure is a function of the number of glucose molecules present. If the solute ionizes and forms an ideal solution, each ion is an osmotically active particle. For example, NaCl would dissociate into Na^+ and Cl^- ions, so that each mole in solution would supply 2 Osm. One mole of Na_2SO_4 would dissociate into Na^+, Na^+, and SO_4^{2-} supplying 3 Osm. However, the body fluids are not ideal solutions, and although the dissociation of strong electrolytes is complete, the number of

particles free to exert an osmotic effect is reduced owing to interactions between the ions. Thus, it is actually the effective concentration **(activity)** in the body fluids rather than the number of equivalents of an electrolyte in solution that determines its osmotic capacity. This is why, for example, 1 mmol of NaCl per liter in the body fluids contributes somewhat less than 2 mOsm of osmotically active particles per liter. The more concentrated the solution, the greater the deviation from an ideal solution.

The osmolal concentration of a substance in a fluid is measured by the degree to which it depresses the freezing point, with 1 mol of an ideal solution depressing the freezing point by 1.86°C. The number of milliosmoles per liter in a solution equals the freezing point depression divided by 0.00186. The **osmolarity** is the number of osmoles per liter of solution (eg, plasma), whereas the **osmolality** is the number of osmoles per kilogram of solvent. Therefore, osmolarity is affected by the volume of the various solutes in the solution and the temperature, while the osmolality is not. Osmotically active substances in the body are dissolved in water, and the density of water is 1, so osmolal concentrations can be expressed as osmoles per liter (Osm/L) of water. In this book, osmolal (rather than osmolar) concentrations are considered, and osmolality is expressed in milliosmoles per liter (of water).

Note that although a homogeneous solution contains osmotically active particles and can be said to have an osmotic pressure, it can exert an osmotic pressure only when it is in contact with another solution across a membrane permeable to the solvent but not to the solute.

OSMOLAL CONCENTRATION OF PLASMA: TONICITY

The freezing point of normal human plasma averages −0.54°C, which corresponds to an osmolal concentration in plasma of 290 mOsm/L. This is equivalent to an osmotic pressure against pure water of 7.3 atmospheres (atm). The osmolality might be expected to be higher than 290 mOsm/L, because the sum of all the cation and anion equivalents in plasma is over 300 mOsm/L. It is not this high because plasma is not an ideal solution and ionic interactions reduce the number of particles free to exert an osmotic effect. Except when there has been insufficient time after a sudden change in composition for equilibrium to occur, all fluid compartments of the body are in (or nearly in) osmotic equilibrium. The term **tonicity** is used to describe the osmolality of a solution

relative to plasma. Solutions that have the same osmolality as plasma are said to be **isotonic;** those with greater osmolality are **hypertonic;** and those with lesser osmolality are **hypotonic.** All solutions that are initially isosmotic with plasma (ie, that have the same actual osmotic pressure or freezing-point depression as plasma) would remain isotonic if it were not for the fact that some solutes diffuse into cells and others are metabolized. Thus, a 0.9% saline solution remains isotonic because there is no net movement of the osmotically active particles in the solution into cells and the particles are not metabolized. On the other hand, a 5% glucose solution is isotonic when initially infused intravenously, but glucose is metabolized, so the net effect is that of infusing a hypotonic solution.

It is important to note the relative contributions of the various plasma components to the total osmolal concentration of plasma. All but about 20 of the 290 mOsm in each liter of normal plasma are contributed by Na^+ and its accompanying anions, principally Cl^- and HCO_3^-. Other cations and anions make a relatively small contribution. Although the concentration of the plasma proteins is large when expressed in grams per liter, they normally contribute less than 2 mOsm/L because of their very high molecular weights. The major nonelectrolytes of plasma are glucose and urea, which in the steady state are in equilibrium with cells. Their contributions to osmolality are normally about 5 mOsm/L each but can become quite large in hyperglycemia or uremia. The total plasma osmolality is important in assessing dehydration, overhydration, and other fluid and electrolyte abnormalities **(Clinical Box 1–3)**.

NONIONIC DIFFUSION

Some weak acids and bases are quite soluble in cell membranes in the undissociated form, whereas they cannot cross membranes in the charged (ie, dissociated) form. Consequently, if molecules of the undissociated substance diffuse from one side of the membrane to the other and then dissociate, there is appreciable net movement of the undissociated substance from one side of the membrane to the other. This phenomenon is called **nonionic diffusion.**

DONNAN EFFECT

When an ion on one side of a membrane cannot diffuse through the membrane, the distribution of other ions to which the membrane is permeable is affected in a

predictable way. For example, the negative charge of a nondiffusible anion hinders diffusion of the diffusible cations and favors diffusion of the diffusible anions. Consider the situation shown in **Figure 1–4**, in which the membrane (M) between compartment X and compartment Y is impermeable to charged proteins ($Prot^-$) but freely permeable to K^+ and Cl^-.

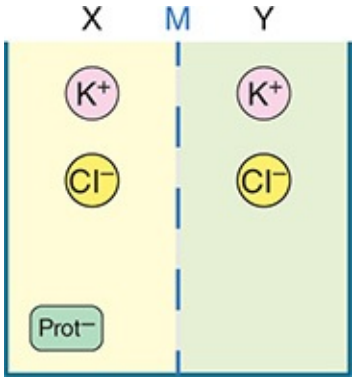


FIGURE 1–4 Equilibrium across a cell membrane. Diagram showing two compartments (X and Y) separated by the membrane (M). Charged K^+ and Cl^- are distributed in both compartments, while charged protein ($prot^-$) is only in X compartment.

CLINICAL BOX 1–3

Plasma Osmolarity & Disease

Unlike plant cells, which have rigid walls, animal cell membranes are flexible. Therefore, animal cells swell when exposed to extracellular hypotonicity and shrink when exposed to extracellular hypertonicity. Cells contain ion channels and pumps that can be activated to offset moderate changes in osmolarity; however, these can be overwhelmed under certain pathological conditions. Hyperosmolality can cause coma (hyperosmolar coma), a prolonged state of deep unconsciousness. Because of the predominant role of the major solutes and the deviation of plasma from an ideal solution, one can ordinarily approximate the plasma osmolarity within a few mOsm/L by using the following formula, in which the constants convert the clinical units to millimoles of solute per liter:

$$\text{Osmolarity (mOsm/L)} = 2[Na^+]\ \text{(mEq/L)} + 0.055[\text{Glucose}]\ \text{(mg/dL)} + 0.36[\text{BUN}]\ \text{(mg/dL)}$$

BUN is the blood urea nitrogen. The formula is also useful in calling attention to abnormally high concentrations of other solutes. An observed plasma osmolarity (measured by freezing-point depression) that greatly exceeds the value predicted by this formula indicates the presence of a foreign substance such as ethanol, mannitol (sometimes injected to shrink swollen cells osmotically), or poisons such as ethylene glycol (component of antifreeze) or methanol (alternative automotive fuel).

Assume that the concentrations of the anions (eg, Cl^-) and of the cations (eg, K^+) on the two sides are initially equal. Cl^- diffuses down its concentration gradient from Y to X, and some K^+ moves with the negatively charged Cl^- because of its opposite charge. Therefore,

$$[K^+_X] > [K^+_Y]$$

Furthermore,

$$[K^+_X] + [Cl^-_X] + [Prot^-_X] > [K^+_Y] + [Cl^-_Y]$$

that is, more osmotically active particles are on side X (or compartment X) than on side Y (or compartment Y).

Donnan and Gibbs showed that in the presence of a nondiffusible ion, the diffusible ions distribute themselves so that at equilibrium their concentration ratios are equal:

$$\frac{[K^+_X]}{[K^+_Y]} = \frac{[Cl^-_Y]}{[Cl^-_X]}$$

Cross-multiplying,

$$[K^+_X][Cl^-_X] = [K^+_Y][Cl^-_Y]$$

This is the **Gibbs–Donnan equation.** It holds for any pair of cations and anions of the same valence.

The Donnan effect on the distribution of ions has three effects in the body introduced here and discussed below. First, because of charged proteins ($Prot^-$) in cells, there are more osmotically active particles in cells than in interstitial or intercellular fluid, and because animal cells have flexible walls, osmosis would

make them swell and eventually rupture if it were not for the sodium-potassium adenosine triphosphatase (**Na, K ATPase)** pumping ions back out of cells. Thus, normal cell volume and pressure largely depend on Na, K ATPase, also known as the Na^+/K^+ pump. Second, because at equilibrium the distribution of permeant ions across the membrane (m, in the example shown in Figure 1–4) is asymmetric, an electrical difference exists across the membrane whose magnitude can be determined by the **Nernst equation** (see below). In the example used here (Figure 1–4), side X will be negative relative to side Y. The charges line up along the membrane, with the concentration gradient for Cl^- exactly balanced by the oppositely directed electrical gradient, and the same holds true for K^+. Third, because there are more proteins in plasma than in interstitial fluid, there is a Donnan effect on ion movement across the capillary wall.

FORCES ACTING ON IONS

The forces acting across the cell membrane on each ion can be analyzed mathematically. Chloride ions (Cl^-) are present in higher concentration in the ECF than in the cell interior, and they tend to diffuse along this **concentration gradient** into the cell. The interior of the cell is negative relative to the exterior, and chloride ions are pushed out of the cell along this **electrical gradient.** An equilibrium is reached between Cl^- influx and Cl^- efflux. The membrane potential at which this equilibrium exists is the **equilibrium potential.** Its magnitude can be calculated from the Nernst equation, as follows:

$$E_{Cl} = \frac{RT}{FZ_{Cl}} \ln \frac{[Cl_o^-]}{[Cl_i^-]}$$

where

E_{Cl} = equilibrium potential for Cl^-

R = gas constant

T = absolute temperature

F = the Faraday number (number of coulombs per mole of charge)

Z_{Cl} = valence of Cl^- (–1)

$[Cl_o^-]$ = Cl^- concentration outside the cell

$[Cl_i^-]$ = Cl^- concentration inside the cell

Converting from the natural log to the base 10 log and replacing some of the constants with numeric values holding temperature at 37°C, the equation becomes:

$$E_{Cl} = 61.5 \log \frac{[Cl_i^-]}{[Cl_o^-]} \text{ at } 37^\circ C$$

Note that in converting to the simplified expression the concentration ratio is reversed because the –1 valence of Cl^- has been removed from the expression.

The equilibrium potential for Cl^- (E_{Cl}) in the mammalian spinal neuron, calculated from the standard values listed in **Table 1–1**, is −70 mV, a value identical to the typical measured resting membrane potential of −70 mV. Therefore, no forces other than those represented by the chemical and electrical gradients need to be invoked to explain the distribution of Cl^- across the membrane.

A similar equilibrium potential can be calculated for K^+ (E_K; again, at 37°C):

$$E_K = \frac{RT}{FZ_K} \ln \frac{[K_o^+]}{[K_i^+]} = 61.5 \log \frac{[K_o^+]}{[K_i^+]} \qquad (\text{at } 37^\circ C)$$

where

E_K = equilibrium potential for K^+

Z_K = valence of K^+ (+1)

$[K_o^+]$ = K^+ concentration outside the cell

$[K_i^+]$ = K^+ concentration inside the cells R, T, and F as above

In this case, the concentration gradient is outward and the electrical gradient inward. In mammalian spinal motor neurons, E_K is −90 mV (Table 1–1). Because the resting membrane potential is −70 mV, there is somewhat more K^+ in the neurons that can be accounted for by the electrical and chemical gradients.

TABLE 1–1 Concentration of some ions inside and outside mammalian spinal motor neurons.

Ion	Concentration (mmol/L of H_2O)		Equilibrium Potential (mV)
	Inside Cell	Outside Cell	
Na^+	15.0	150.0	+60
K^+	150.0	5.5	−90
Cl^-	9.0	125.0	−70

Resting membrane potential = −70 mV.

The situation for Na^+ in the mammalian spinal motor neuron is quite different from that for K^+ or Cl^-. The direction of the chemical gradient for Na^+ is inward, to the area where it is in lesser concentration, and the electrical gradient is in the same direction. E_{Na} is +60 mV (Table 1–1). Because neither E_K nor E_{Na} is equal to the membrane potential, one would expect the cell to gradually gain Na^+ and lose K^+ if only passive electrical and chemical forces were acting across the membrane. However, the intracellular concentrations of Na^+ and K^+ remain constant because of selective permeability of the membrane to different ions (Na^+, K^+) and the action of the Na, K ATPase that actively transports Na^+ out of the cell and K^+ into the cell (against their respective electrochemical gradients).

ESTABLISHMENT OF THE MEMBRANE POTENTIAL

The distribution of ions across the cell membrane and the nature of this membrane provide the explanation for the membrane potential. The concentration gradient for K^+ facilitates its movement out of the cell via K^+ channels, but its electrical gradient is in the opposite (inward) direction. Consequently, an equilibrium is reached in which the tendency of K^+ to move out of the cell is balanced by its tendency to move into the cell, and at that equilibrium there is a slight excess of cations on the outside and anions on the inside. This condition is maintained by Na, K ATPase, which uses the energy of ATP to pump K^+ back into the cell and keeps the intracellular concentration of Na^+ low. Because the Na, K ATPase moves three Na^+ out of the cell for every two K^+ moved in, it also contributes to the membrane potential, and thus is termed an **electrogenic pump.** It should be emphasized that the number of ions

responsible for the membrane potential is a minute fraction of the total number present and that the total concentrations of positive and negative ions are equal everywhere except along the membrane.

ENERGY PRODUCTION

ENERGY TRANSFER

Energy used in cellular processes and cell function is primarily stored in bonds between phosphoric acid residues and certain organic compounds. Because the energy of bond formation in some of these phosphates is particularly high, relatively large amounts of energy (10–12 kcal/mol) are released when the bond is hydrolyzed. Compounds containing such bonds are called **high-energy phosphate compounds.** Not all organic phosphates are of the high-energy type. Many, like glucose 6-phosphate, are low-energy phosphates that on hydrolysis liberate 2–3 kcal/mol. Some of the intermediates formed in carbohydrate metabolism are high-energy phosphates, but the most important energy-rich phosphate compound is **adenosine triphosphate (ATP).** This ubiquitous molecule, ATP **(Figure 1–5)**, is the energy storehouse of the body. On hydrolysis to adenosine diphosphate (ADP), it liberates energy directly to such processes as muscle contraction, active transport, and the synthesis of many chemical compounds. Loss of another phosphate to form adenosine monophosphate (AMP) releases more energy.

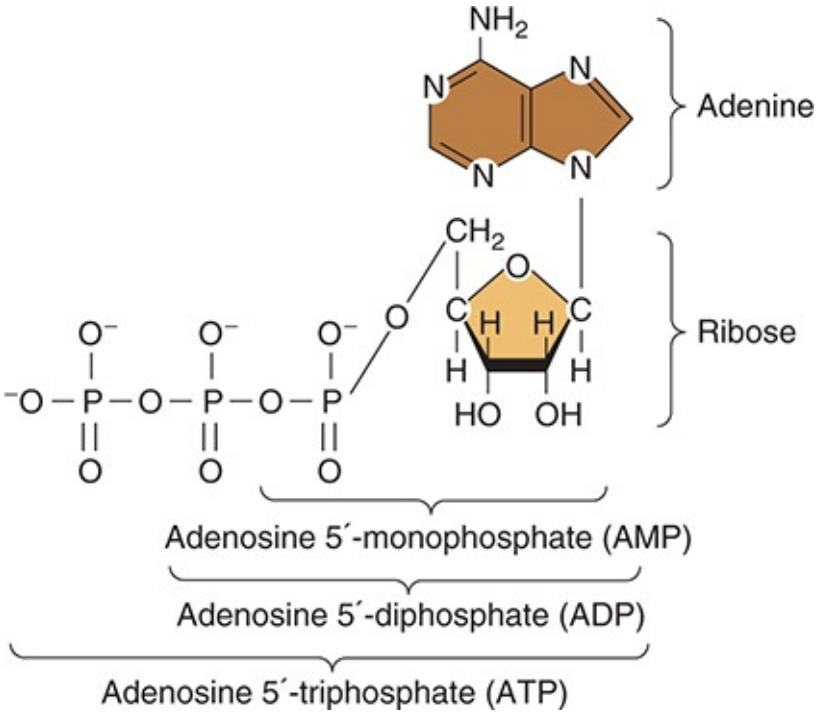


FIGURE 1–5 Energy-rich adenosine derivatives. Adenosine triphosphate is broken down into its backbone purine base and sugar (at right) as well as its high-energy phosphate derivatives (across bottom). (Reproduced with permission from Murray RK, et al: *Harper's Biochemistry*, 28th ed. New York, NY: McGraw-Hill; 2009.)

Another group of energy-rich, or high-energy, compounds are the thioesters, the acyl derivatives of mercaptans. **Coenzyme A (CoA)** is a widely distributed mercaptan-containing adenine, ribose, pantothenic acid, and thioethanolamine **(Figure 1–6)**. Reduced CoA (usually abbreviated HS-CoA) reacts with acyl groups (R–CO–) to form R–CO–S–CoA derivatives. A prime example is the reaction of HS-CoA with acetic acid to form acetylcoenzyme A (acetyl-CoA), a compound of pivotal importance in intermediary metabolism. Because acetyl-CoA has a much higher energy content than acetic acid, it combines readily with substances in reactions that would otherwise require outside energy. Acetyl-CoA is therefore often called "active acetate." From the point of view of energetics, formation of 1 mol of any acyl-CoA compound is equivalent to the formation of 1 mol of ATP.

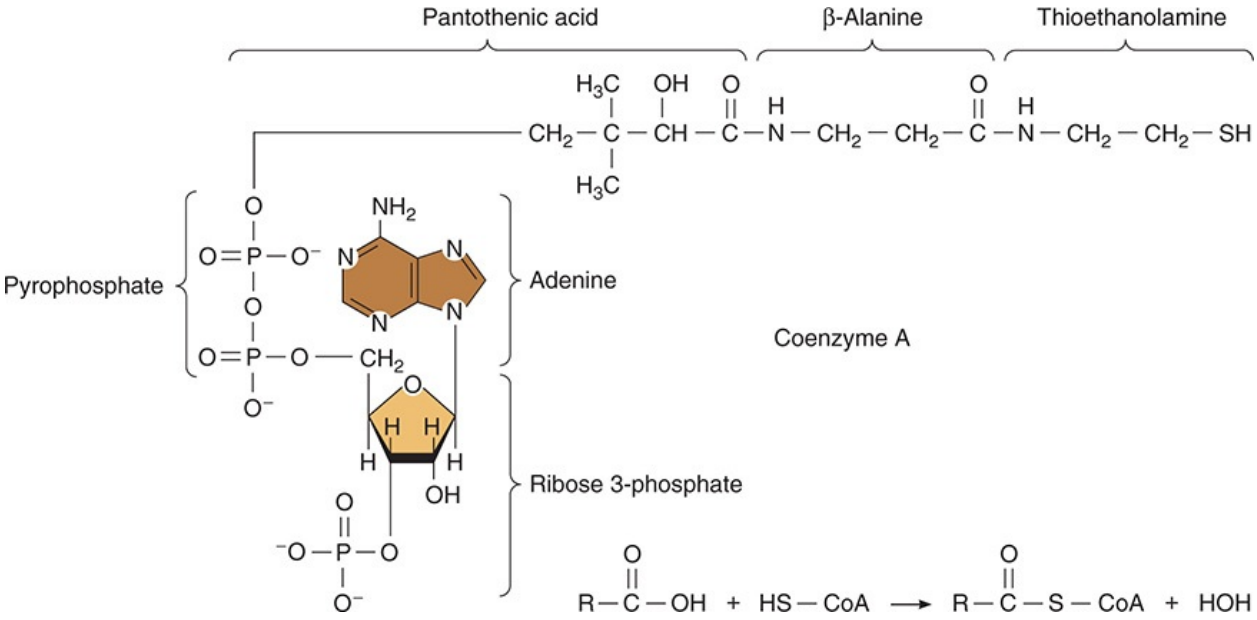


FIGURE 1–6 Coenzyme A (CoA) and its derivatives. Left: Formula of reduced coenzyme A (HS-CoA) with its components highlighted. **Right:** Formula for reaction of CoA with biologically important compounds to form thioesters. R, remainder of molecule.

BIOLOGICAL OXIDATIONS

Oxidation is the combination of a substance with O_2, or loss of hydrogen, or loss of electrons. The corresponding reverse processes are called **reduction.** Biological oxidations are catalyzed by specific enzymes. Cofactors (simple ions) or coenzymes (organic, nonprotein substances) are accessory substances that usually act as carriers for products of the reaction. Unlike the enzymes, the coenzymes may catalyze a variety of reactions.

A number of coenzymes serve as hydrogen acceptors. One common form of biological oxidation is removal of hydrogen from an R–OH group, forming R=O. In such dehydrogenation reactions, nicotinamide adenine dinucleotide (NAD^+) and dihydronicotinamide adenine dinucleotide phosphate ($NADP^+$) pick up hydrogen, forming dihydronicotinamide adenine dinucleotide (NADH) and dihydronicotinamide adenine dinucleotide phosphate (NADPH) **(Figure 1–7)**. The hydrogen is then transferred to the flavoprotein–cytochrome system, reoxidizing the NAD^+ and $NADP^+$. Flavin adenine dinucleotide (FAD) is formed when riboflavin is phosphorylated, forming flavin mononucleotide (FMN). FMN then combines with AMP, forming the dinucleotide. FAD can accept hydrogens

in a similar fashion, forming its hydro (FADH) and dihydro ($FADH_2$) derivatives.

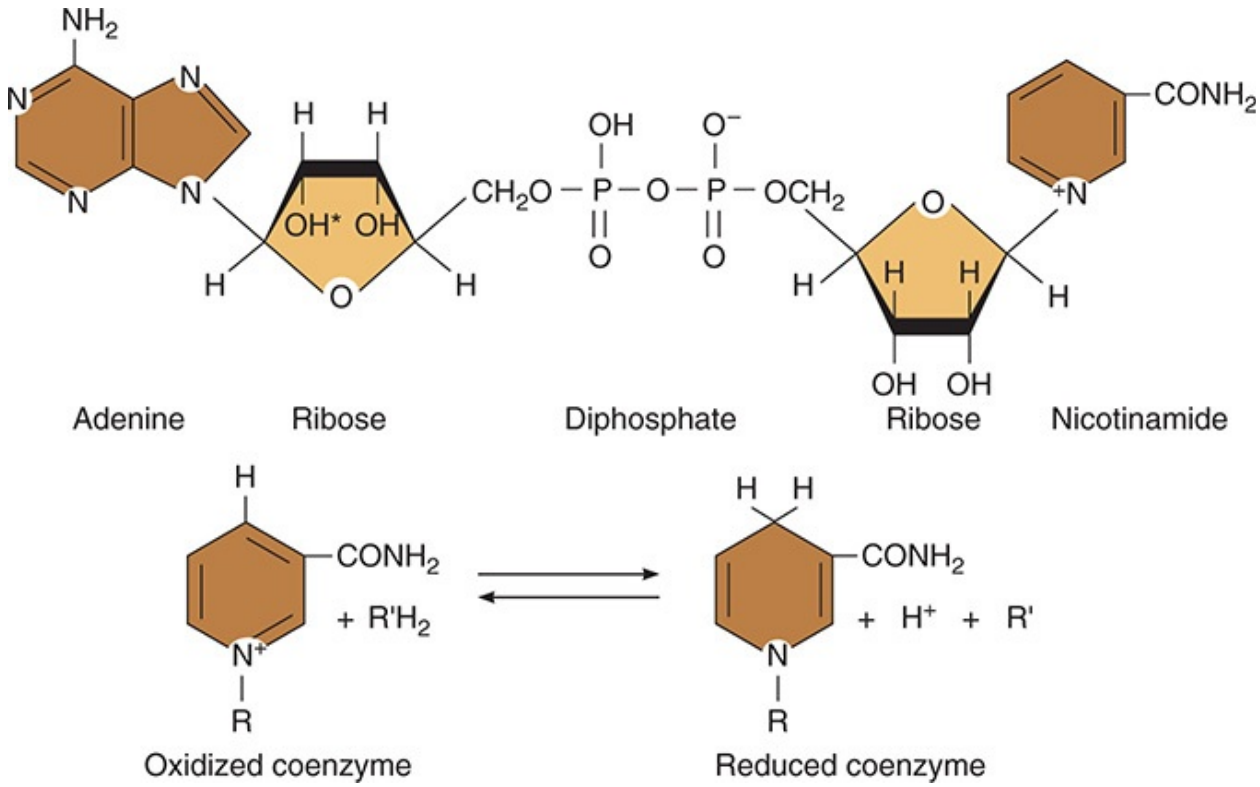


FIGURE 1–7 Structures of molecules important in oxidation–reduction reactions to produce energy. Top: Formula of the oxidized form of nicotinamide adenine dinucleotide (NAD^+). Nicotinamide adenine dinucleotide phosphate ($NADP^+$) has an additional phosphate group at the location marked by the asterisk. **Bottom:** Reaction by which NAD^+ and $NADP^+$ become reduced to form NADH and NADPH. R, remainder of molecule; R′, hydrogen donor.

The flavoprotein–cytochrome system is a chain of enzymes that transfers hydrogen to oxygen, forming water. This process occurs in the mitochondria. Each enzyme in the chain is reduced and then reoxidized as the hydrogen is passed down the line. Each of the enzymes is a protein with an attached nonprotein prosthetic group. The final enzyme in the chain is cytochrome c oxidase, which transfers hydrogens to O_2, forming H_2O. It contains two atoms of Fe and three of Cu and has 13 subunits.

The principal process by which ATP is formed in the body is **oxidative phosphorylation.** This process harnesses the energy from a proton gradient across the mitochondrial membrane to produce the high-energy bond of ATP

(see Figure 2–4 for more detail). Ninety percent of the O_2 consumption, the amount of oxygen used by the body per minute, in the basal state is in mitochondria, and 80% of the mitochondrial O_2 consumption is coupled to ATP synthesis.

ATP is utilized throughout the cell, with the bulk used in a handful of processes: approximately 27% is used for protein synthesis, 24% by Na, K ATPase to help set membrane potential, 9% by gluconeogenesis, 6% by Ca^{2+} ATPase to maintain a low cytosolic Ca^{2+} concentration, 5% by myosin ATPase, and 3% by ureagenesis.

MOLECULAR BUILDING BLOCKS

NUCLEOSIDES, NUCLEOTIDES, & NUCLEIC ACIDS

Nucleosides contain a sugar linked to a nitrogen-containing base. The physiologically important bases, **purines** and **pyrimidines,** have ring structures **(Figure 1–8)**. These structures are bound to a sugar, either ribose or 2-deoxyribose, to complete the nucleoside. When inorganic phosphate is added to the nucleoside, a **nucleotide** is formed **(Figure 1–9)**. Nucleosides and nucleotides form the backbone for RNA and DNA, as well as a variety of coenzymes and regulatory molecules of physiological importance (eg, NAD^+, $NADP^+$, and ATP) **(Table 1–2)**. Nucleic acids in the diet are digested and their constituent purines and pyrimidines absorbed, but most of the purines and pyrimidines are synthesized from amino acids, principally in the liver. The nucleotides and RNA and DNA are then synthesized. RNA is in dynamic equilibrium with the amino acid pool, but DNA, once formed, is metabolically stable throughout life. The purines and pyrimidines released by the breakdown of nucleotides may be reused or catabolized. Minor amounts are excreted unchanged in the urine.

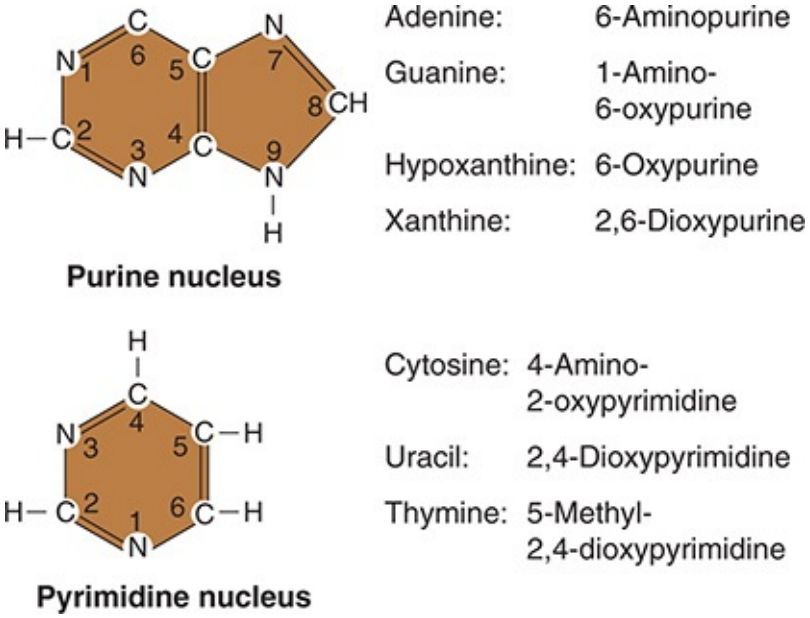


FIGURE 1–8 Principal physiologically important purines and pyrimidines. Purine and pyrimidine structures are shown next to representative molecules from each group. Oxypurines and oxypyrimidines may form enol derivatives (hydroxypurines and hydroxypyrimidines) by migration of hydrogen to the oxygen substituents.

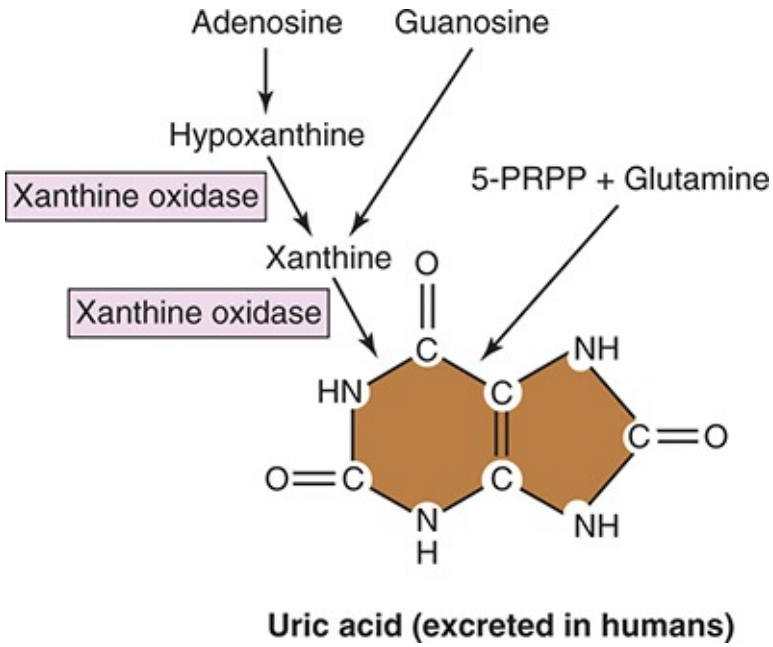


FIGURE 1–9 Synthesis and breakdown of uric acid. Adenosine is converted to hypoxanthine, which is then converted to xanthine, and xanthine is converted to uric acid. The latter two reactions are both catalyzed by xanthine oxidase.

Guanosine is converted directly to xanthine, while 5-PRPP and glutamine can be converted to uric acid.

TABLE 1–2 Purine- and pyrimidine-containing compounds.

Type of Compound	Components
Nucleoside	Purine or pyrimidine plus ribose or 2-deoxyribose
Nucleotide (mononucleotide)	Nucleoside plus phosphoric acid residue
Nucleic acid	Many nucleotides forming double-helical structures of two polynucleotide chains
Nucleoprotein	Nucleic acid plus one or more simple basic proteins
Contain ribose	RNA
Contain 2-deoxyribose	DNA

The pyrimidines are catabolized to the **β-amino acids,** β-alanine, and β-aminoisobutyrate. These amino acids have their amino group on β-carbon, rather than the α-carbon typical to physiologically active amino acids. Because β-aminoisobutyrate is a product of thymine degradation, it can serve as a measure of DNA turnover. The β-amino acids are further degraded to CO_2 and NH_3.

Uric acid is formed by the breakdown of purines and by direct synthesis from 5-phosphoribosyl pyrophosphate (5-PRPP) and glutamine (Figure 1–9). In humans, uric acid is excreted in the urine, but in other mammals, uric acid is further oxidized to allantoin before excretion. The normal blood uric acid level in humans is approximately 4 mg/dL (0.24 mmol/L). In the kidney, uric acid is filtered, reabsorbed, and secreted. Normally, 98% of the filtered uric acid is reabsorbed and the remaining 2% makes up approximately 20% of the amount excreted. The remaining 80% comes from the tubular secretion. The uric acid excretion on a purine-free diet is about 0.5 g/24 h and on a regular diet about 1 g/24 h. Excess uric acid in the blood or urine is a characteristic of gout **(Clinical Box 1–4)**.

CLINICAL BOX 1–4

Gout

Gout is a disease characterized by recurrent attacks of arthritis; urate (a salt derived from uric acid) deposits in the joints, kidneys, and other tissues; and elevated blood and urine uric acid levels. The joint most commonly affected initially is the metatarsophalangeal joint of the great toe. There are two forms of "primary" gout. In one, uric acid production is increased because of various enzyme abnormalities. In the other, there is a selective deficit in renal tubular transport of uric acid. In "secondary" gout, the uric acid levels in the body fluids are elevated as a result of decreased excretion or increased production secondary to some other disease process. For example, excretion is decreased in patients treated with thiazide diuretics and those with renal disease. Production is increased in leukemia and pneumonia because of increased breakdown of uric acid-rich white blood cells.

THERAPEUTIC HIGHLIGHTS

The treatment of gout is aimed at relieving the acute arthritis with drugs such as colchicine or nonsteroidal anti-inflammatory drugs (NSAIDs) and decreasing the uric acid level in the blood. Colchicine does not affect uric acid metabolism, and it apparently relieves gouty attacks by inhibiting the phagocytosis of uric acid crystals by leukocytes, a process that in some way produces the joint symptoms. Phenylbutazone and probenecid inhibit uric acid reabsorption in the renal tubules. Allopurinol, which directly inhibits xanthine oxidase in the purine degradation pathway, is used to decrease uric acid production.

DNA

DNA is found in bacteria, in the nuclei of eukaryotic cells, and in mitochondria. It is made up of two extremely long nucleotide chains containing the bases adenine (A), guanine (G), thymine (T), and cytosine (C) **(Figure 1–10)**. The chains are bound together by hydrogen bonding between the bases, with A bonding to T and G to C. This stable association forms a double-helical structure **(Figure 1–11)**. The double helical structure of DNA is compacted in the cell by association with **histones,** and further compacted into **chromosomes.** A diploid

human cell contains 46 chromosomes.

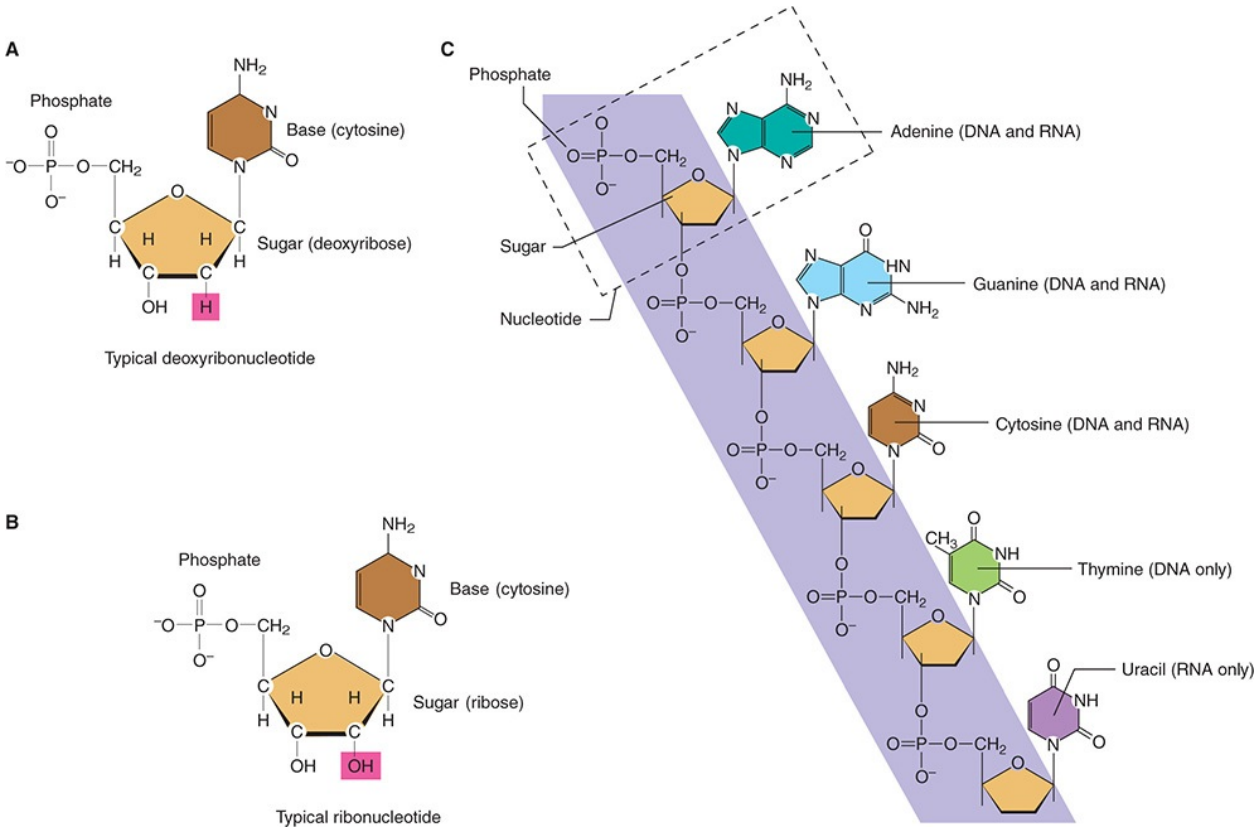


FIGURE 1–10 Basic structure of nucleotides and nucleic acids. A and B) The nucleotide cytosine is shown with deoxyribose and with ribose as the principal sugar. **C)** Purine bases adenine and guanine are bound to each other or to pyrimidine bases, cytosine, thymine, or uracil via a phosphodiester backbone between 2′-deoxyribosyl moieties attached to the nucleobases by an N-glycosidic bond. Note that the backbone has a polarity (ie, a 5′ and a 3′ direction). Thymine is only found in DNA, while uracil is only found in RNA.

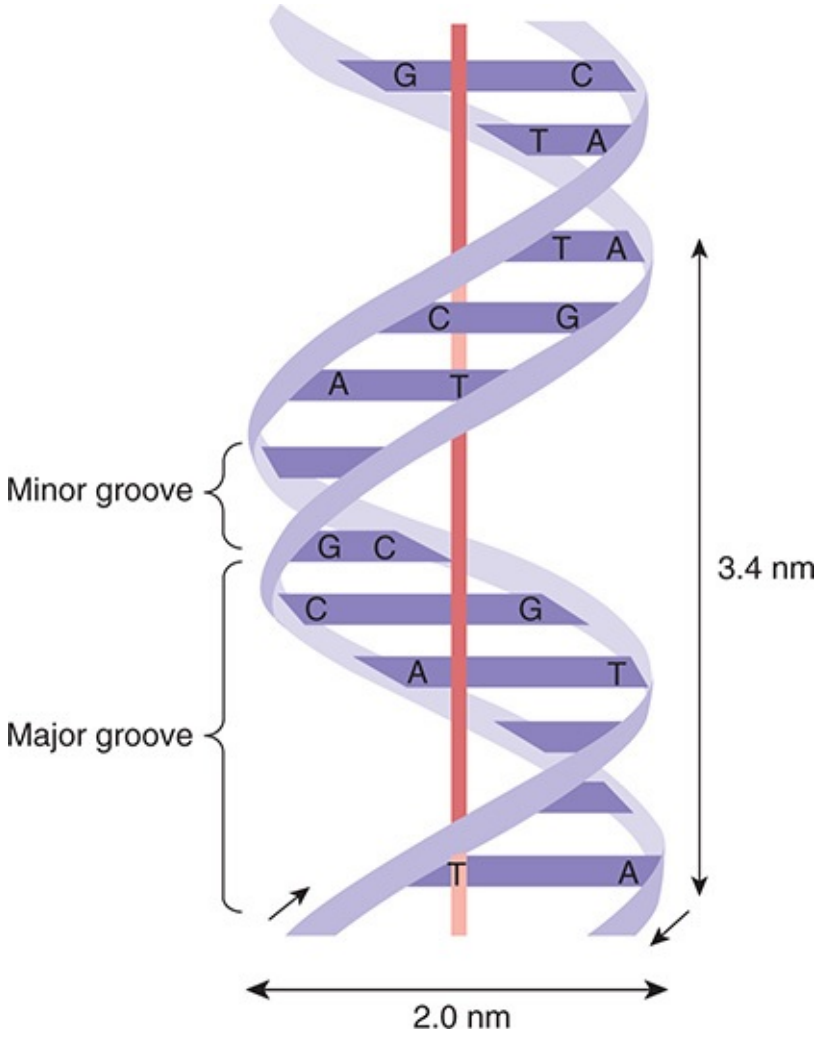


FIGURE 1–11 Double-helical structure of DNA. The compact structure has an approximately 2.0 nm thickness and 3.4 nm between full turns of the helix that contains both major and minor grooves. The structure is maintained in the double helix by hydrogen bonding between purines and pyrimidines across individual strands of DNA. Adenine (A) is bound to thymine (T) and cytosine (C) to guanine (G). (Reproduced with permission from Murray RK et al: *Harper's Biochemistry*, 28th ed. New York, NY: McGraw-Hill; 2009.)

A fundamental unit of DNA, or a **gene,** can be defined as the sequence of DNA nucleotides that contain the information for the production of an ordered amino acid sequence for a single polypeptide chain. Interestingly, the protein encoded by a single gene may be subsequently divided into several different physiologically active proteins. Information is accumulating at an accelerating rate about the structure of genes and their regulation. The basic structure of a typical eukaryotic gene is shown in diagrammatic form in **Figure 1–12**. It is

made up of a strand of DNA that includes coding and noncoding regions. In eukaryotes, unlike prokaryotes, the portions of the genes that dictate the formation of proteins are usually broken into several segments **(exons)** separated by segments that are not translated **(introns).** Near the transcription start site of the gene is a **promoter,** which is the site at which RNA polymerase and its cofactors bind. It often includes a thymidine–adenine–thymidine–adenine (TATA) sequence **(TATA box),** which ensures that transcription starts at the proper point. Farther out in the 5′ region are **regulatory elements,** which include enhancer and silencer sequences. It has been estimated that each gene has an average of five regulatory sites. Regulatory sequences are sometimes found in the 3′-flanking region as well. In a diploid cell, each gene will have two **alleles,** or versions of that gene. Each allele occupies the same position on the homologous chromosome. Individual alleles can confer slightly different properties of the gene when fully transcribed. It is interesting to note that changes in single nucleotides within or outside coding regions of a gene **(single nucleotide polymorphisms; SNPs)** can have great consequences for gene function. The study of SNPs in human disease is a growing and exciting area of genetic research.

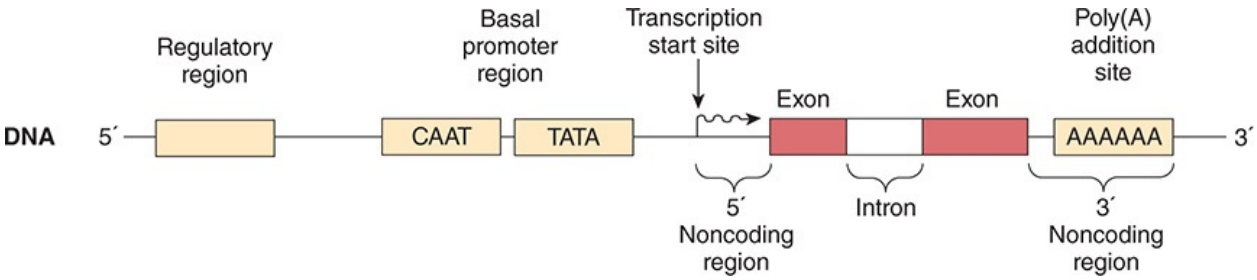


FIGURE 1–12 Diagram of the components of a typical eukaryotic gene. The region that produces introns and exons is flanked by noncoding regions. The 5′-flanking region contains stretches of DNA that interact with proteins to facilitate or inhibit transcription. The 3′-flanking region contains the poly(A) addition site. (Modified with permission from Murray RK et al: *Harper's Biochemistry*, 28th ed. New York, NY: McGraw-Hill; 2009.)

Gene mutations occur when the base sequence in the DNA is altered from its original sequence. Alterations can be through insertions, deletions, or duplications. Such alterations can affect protein structure and be passed on to daughter cells after cell division. **Point mutations** are single base substitutions. A variety of chemical modifications (eg, alkylating or intercalating agents, or ionizing radiation) can lead to changes in DNA sequences and mutations. The

collection of genes within the full expression of DNA from an organism is termed its **genome.** An indication of the complexity of DNA in the human haploid genome (the total genetic message) is its size; it is made up of 3×10^9 base pairs that can code for approximately 30,000 genes. This genetic message is the blueprint for the heritable characteristics of the cell and its descendants. The proteins formed from the DNA blueprint include all the enzymes, and these in turn control the metabolism of the cell.

Each nucleated somatic cell in the body contains the full genetic message, yet there is great differentiation and specialization in the functions of the various types of adult cells. Only small parts of the message are normally transcribed. Thus, the genetic message is normally maintained in a repressed state. However, genes are controlled both spatially and temporally. The double helix requires highly regulated interaction by proteins to unravel for **replication, transcription,** or both.

REPLICATION: MITOSIS & MEIOSIS

At the time of each somatic cell division **(mitosis),** the two DNA chains separate, each serving as a template for the synthesis of a new complementary chain. DNA polymerase catalyzes this reaction. One of the double helices thus formed goes to one daughter cell and one goes to the other, so the amount of DNA in each daughter cell is the same as that in the parent cell. The life cycle of the cell that begins after mitosis is highly regulated and is termed the **cell cycle** **(Figure 1–13)**. The G_1 (or Gap 1) phase represents a period of cell growth and divides the end of mitosis from the DNA synthesis (or S) phase. Following DNA synthesis, the cell enters another period of cell growth, the G_2 (Gap 2) phase. The ending of this stage is marked by chromosome condensation and the beginning of mitosis (M stage).

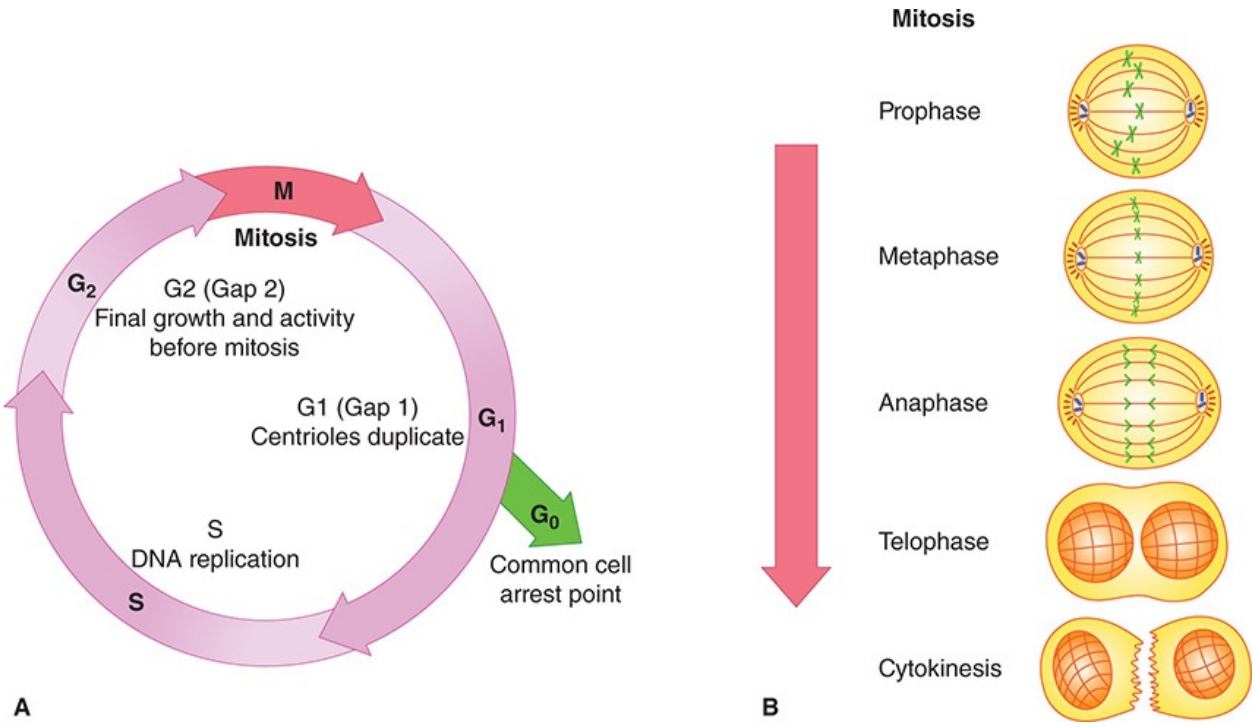


FIGURE 1–13 Sequence of events during the cell cycle. A) Immediately following mitosis (M) the cell enters a gap phase (G_1). At this point many cells will undergo cell arrest (G_0 phase). G_1 is followed by a DNA synthesis phase (S) a second gap phase (G_2) and back to mitosis. **B)** Stages of mitosis are highlighted.

In germ cells, reductive division **(meiosis)** takes place during maturation. The net result is that one of each pair of chromosomes ends up in each mature germ cell; consequently, each mature germ cell contains half the amount of chromosomal material found in somatic cells. Therefore, when a sperm unites with an ovum, the resulting zygote has the full complement of DNA, half of which came from the father and half from the mother. The term “ploidy” is sometimes used to refer to the number of chromosomes in cells. Normal resting diploid cells are **euploid** and become **tetraploid** just before division. **Aneuploidy** is the condition in which a cell contains other than the haploid number of chromosomes or an exact multiple of it, and this condition is common in cancerous cells.

RNA

The strands of the DNA double helix not only replicate themselves but also serve

as templates by lining up complementary bases for the formation in the nucleus of **RNA.** RNA differs from DNA in that it is single-stranded, has **uracil** (U) in place of thymine (T), and its sugar moiety is ribose rather than 2′-deoxyribose (Figure 1–10). The production of RNA from DNA is called **transcription.** Transcription can lead to several types of RNA including: **messenger RNA (mRNA), transfer RNA (tRNA), ribosomal RNA (rRNA),** and other RNAs. Transcription is catalyzed by various forms of **RNA polymerase.**

Typical transcription of an mRNA is shown in **Figure 1–14**. When suitably activated, transcription of the gene into a pre-mRNA starts at the **cap site** and ends about 20 bases beyond the AATAAA sequence. The RNA transcript is capped in the nucleus by addition of 7-methylguanosine triphosphate to the 5′ end; this cap is necessary for proper binding to the ribosome. A **poly(A) tail** of about 100 bases is added to the untranslated segment at the 3′ end to help maintain the stability of the mRNA. The pre-mRNA formed by capping and addition of the poly(A) tail is then processed by elimination of the introns, and once this posttranscriptional modification is complete, the mature mRNA moves to the cytoplasm. Posttranscriptional modification of the pre-mRNA is a regulated process where differential splicing can occur to form more than one mRNA from a single pre-mRNA. The introns of some genes are eliminated by **spliceosomes,** complex units that are made up of small RNAs and proteins. Other introns are eliminated by **self-splicing** by the RNA they contain. Because of introns and splicing, more than one mRNA can be formed from the same gene.

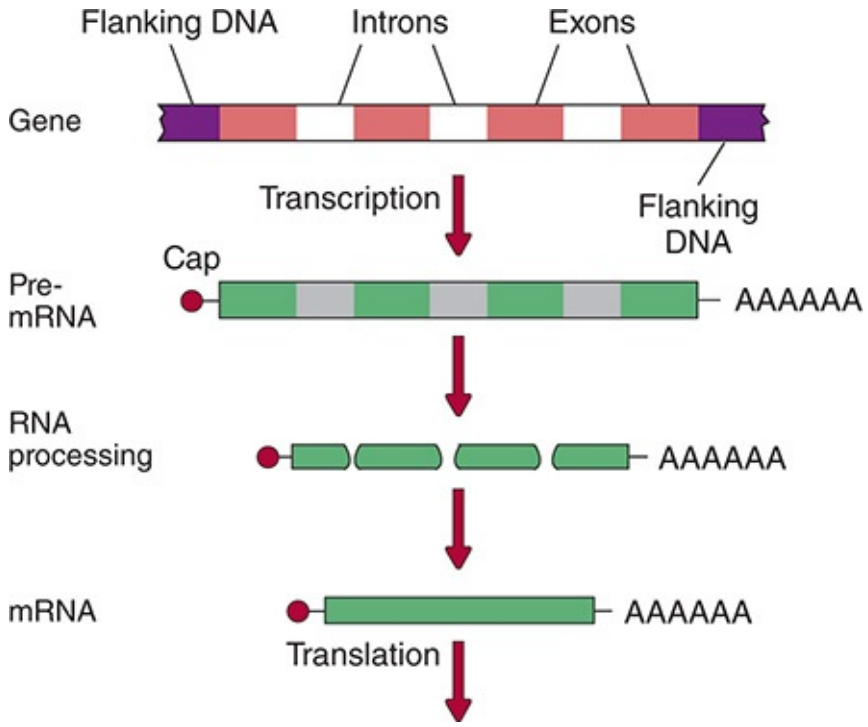


FIGURE 1–14 Transcription of a typical mRNA. Steps in transcription from a typical gene to a processed mRNA are shown. Cap, cap site; AAAAAA, poly(A) site.

Most forms of RNA in the cell are involved in **translation,** or protein synthesis. A brief outline of the transition from transcription to translation is shown in **Figure 1–15**. In the cytoplasm, ribosomes provide a template for tRNA to deliver specific amino acids to a growing polypeptide chain based on specific sequences in mRNA. The mRNA molecules are smaller than the DNA molecules, and each represents a transcript of a small segment of the DNA chain. For comparison, the molecules of tRNA contain only 70–80 nitrogenous bases, compared with hundreds in mRNA and 3 billion in DNA. A newer class of RNA, **microRNAs,** have recently been reported. MicroRNAs measure approximately 21–25-nucleotides in length and have been shown to negatively regulate gene expression at the posttranscriptional level. It is expected that roles for these small RNAs will continue to expand as research into their function continues.

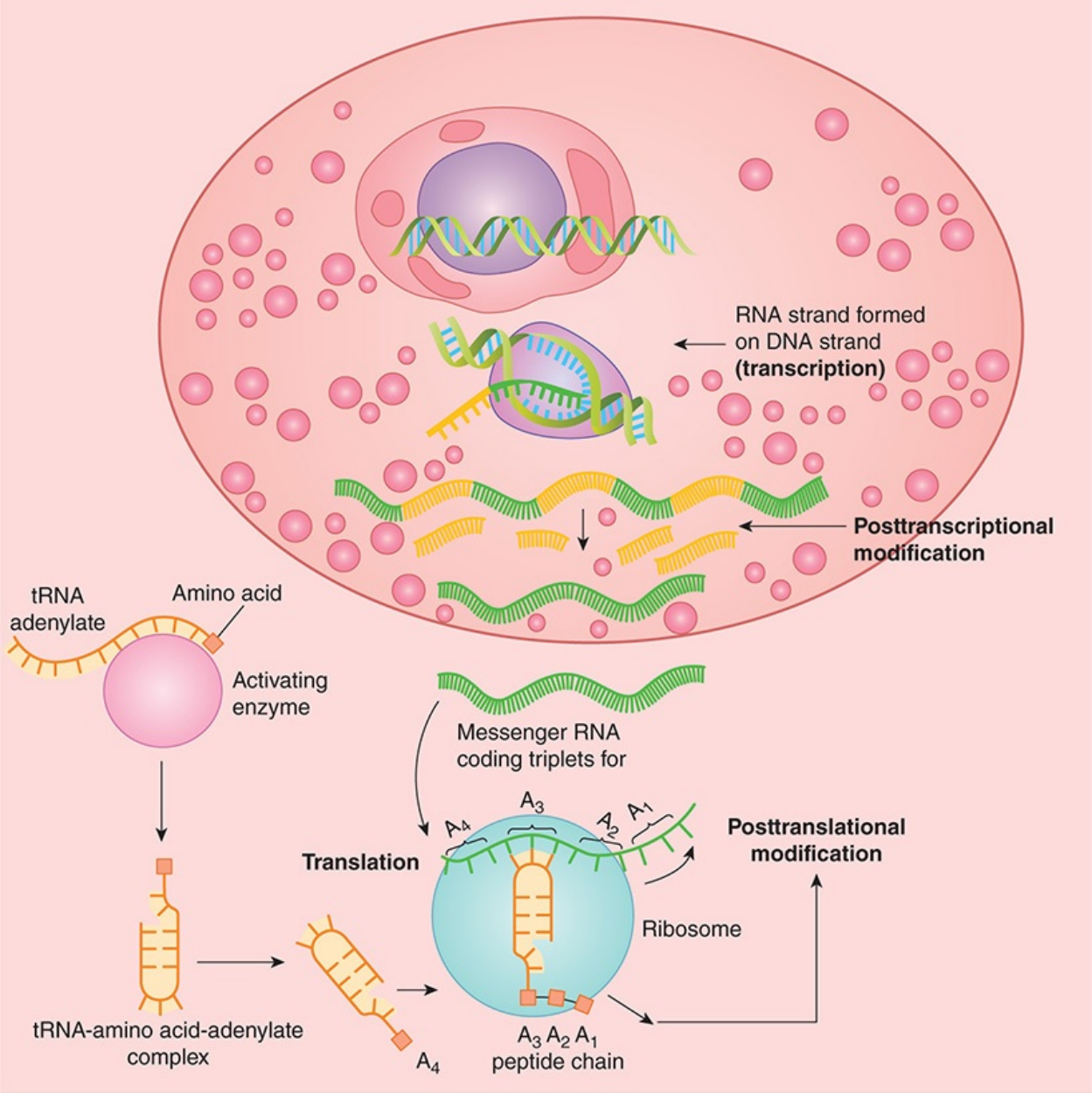


FIGURE 1–15 Diagrammatic outline of transcription to translation. In the nucleus, a messenger RNA is produced from the DNA molecule. This messenger RNA is processed and moved to the cytosol where it is presented to the ribosome. It is at the ribosome where charged tRNA match up with their complementary codons of mRNA to position the amino acid for growth of the polypeptide chain. The lines with multiple short projections in DNA and RNA represent individual bases. Small boxes labeled A represent individual amino acids.

AMINO ACIDS & PROTEINS

AMINO ACIDS

Amino acids that form the basic building blocks for proteins are identified in Table 1–3. These amino acids are often referred to by their corresponding three-letter, or single-letter abbreviations (eg, Ala or A for alanine). Various other important amino acids such as ornithine, 5-hydroxytryptophan, L-dopa, taurine, and thyroxine (T_4) are present in the body but are not found in proteins. In higher animals, the L isomers of the amino acids are the only naturally occurring forms in proteins. The L isomers of hormones such as thyroxine are much more active than the D isomers. The amino acids are acidic, neutral, or basic, depending on the relative proportions of free acidic (–COOH) or basic ($–NH_2$) groups in the molecule. Some of the amino acids are **nutritionally essential amino acids,** that is, they must be obtained in the diet, because they cannot be made in the body. Arginine and histidine must be provided through diet during times of rapid growth or recovery from illness and are termed **conditionally essential.** All others are **nonessential amino acids** in the sense that they can be synthesized in vivo in amounts sufficient to meet metabolic needs.

TABLE 1–3 Amino acids found in proteins.

Amino acids with aliphatic side chains
- Alanine (Ala, A)
- **Valine** (Val, V)
- **Leucine** (Leu, L)
- **Isoleucine** (Ile, I)

Hydroxyl-substituted amino acids
- Serine (Ser, S)
- **Threonine** (Thr, T)

Sulfur-containing amino acids
- Cysteine (Cys, C)
- **Methionine** (Met, M)
- Selenocysteine[a]

Amino acids with aromatic ring side chains
- **Phenylalanine** (Phe, F)
- Tyrosine (Tyr, Y)
- **Tryptophan** (Trp, W)

Amino acids with acidic side chains, or their amides
- Aspartic acid (Asp, D)
- Asparagine (Asn, N)
- Glutamine (Gln, Q)
- Glutamic acid (Glu, E)
- γ-Carboxyglutamic acid[b] (Gla)

Amino acids with side chains containing basic groups
- ***Arginine***[c] (Arg, R)
- **Lysine** (Lys, K)
- Hydroxylysine[b] (Hyl)
- ***Histidine***[c] (His, H)

Imino acids (contain imino group but no amino group)
- Proline (Pro, P)
- 4-Hydroxyproline[b] (Hyp)
- 3-Hydroxyproline[b]

Those in bold type are the nutritionally essential amino acids or conditionally essential amino acids. The generally accepted three-letter and one-letter abbreviations for the amino acids are shown in parentheses.

[a]Selenocysteine is a rare amino acid in which the sulfur of cysteine is replaced by selenium. The codon UGA is usually a stop codon, but in certain situations it codes for selenocysteine.

[b]There are no tRNAs for these four amino acids; they are formed by posttranslational modification of the corresponding unmodified amino acid in peptide linkage. There are tRNAs for selenocysteine and the remaining 20 amino acids, and they are incorporated into peptides and proteins under direct genetic control.

[c]Arginine and histidine are sometimes called "conditionally essential"—they are not necessary for maintenance of nitrogen balance, but are needed for normal growth (especially for children).

THE AMINO ACID POOL

Although small amounts of proteins are absorbed from the gastrointestinal tract and some peptides are also absorbed, most ingested proteins are digested into their constituent amino acids before absorption. Traditionally, peptides are

defined as molecules that consist of 2–100 amino acids, while proteins are made up of 100 or more amino acids. The body's proteins are being continuously hydrolyzed to amino acids and resynthesized. The turnover rate of endogenous proteins averages 80–100 g/day, being highest in the intestinal mucosa and practically nil in the extracellular structural protein, collagen. The amino acids formed by endogenous protein breakdown are identical to those derived from ingested protein. Together they form a common **amino acid pool** that supplies the needs of the body **(Figure 1–16)**.

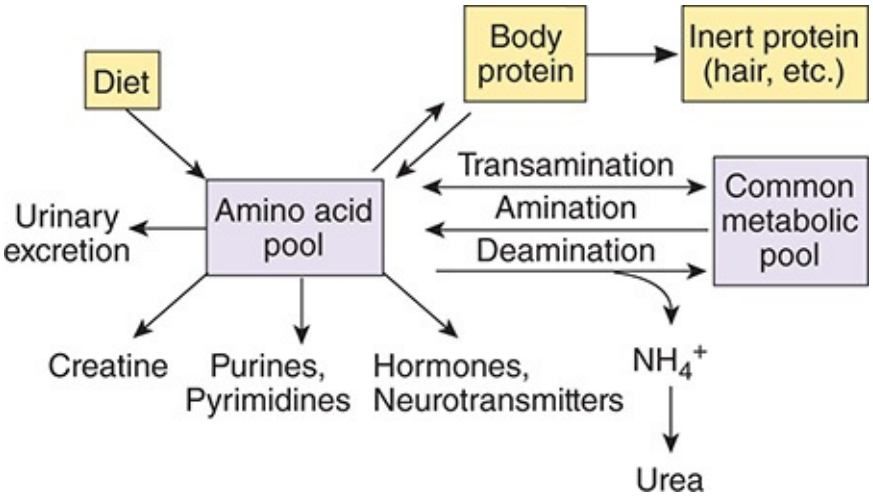


FIGURE 1–16 Amino acids in the body. There is an extensive network of amino acid turnover in the body. Boxes represent large pools of amino acids and some of the common interchanges are represented by arrows. Note that most amino acids come from the diet and end up in protein; however, a large portion of amino acids are interconverted and can feed into and out of a common metabolic pool through amination reactions.

PROTEINS

Proteins are made up of large numbers of amino acids linked into chains by **peptide bonds** joining the amino group of one amino acid to the carboxyl group of the next **(Figure 1–17)**. In addition, some proteins contain carbohydrates (glycoproteins) and lipids (lipoproteins). Smaller chains of amino acids are called **peptides** or **polypeptides.** The boundaries between peptides, polypeptides, and proteins are not well defined. For this text, amino acid chains containing 2–10 amino acid residues are called peptides, chains containing more than 10 but fewer than 100 amino acid residues are called polypeptides, and

chains containing 100 or more amino acid residues are called proteins.

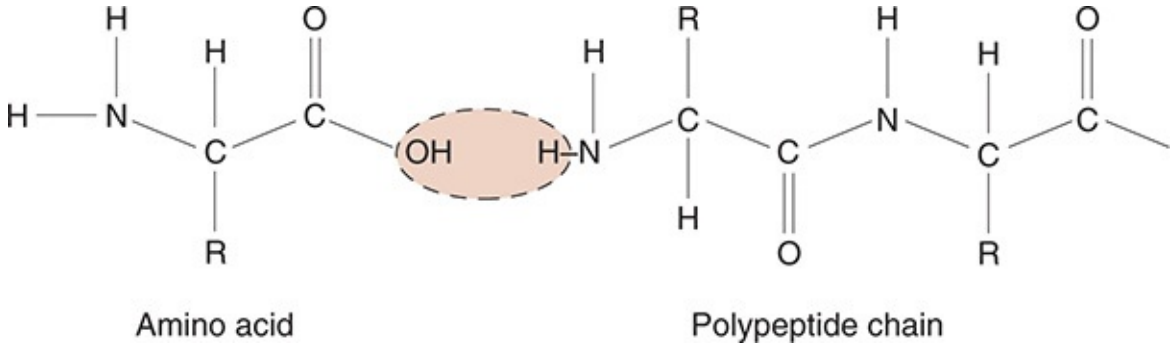


FIGURE 1–17 Amino acid structure and formation of peptide bonds. The dashed line shows where peptide bonds are formed between two amino acids. The highlighted area is released as H_2O. R, remainder of the amino acid. For example, in glycine, R = H; in glutamate, R = —$(CH_2)_2$—COO^-.

The order of the amino acids in the peptide chains is called the **primary structure** of a protein. The chains are twisted and folded in complex ways, and the term **secondary structure** of a protein refers to the spatial arrangement produced by the twisting and folding. A common secondary structure is a regular coil with 3.7 amino acid residues per turn (α-helix). Another common secondary structure is a β-sheet. An antiparallel β-sheet is formed when extended polypeptide chains fold back and forth on one another and hydrogen bonding occurs between the peptide bonds on neighboring chains. Parallel β-sheets between polypeptide chains also occur. The **tertiary structure** of a protein is the arrangement of the twisted chains into layers, crystals, or fibers. Many protein molecules are made of several proteins, or subunits (eg, hemoglobin), and the term **quaternary structure** is used to refer to the arrangement of the subunits into a functional structure.

PROTEIN SYNTHESIS

The process of protein synthesis, **translation,** is the conversion of information encoded in mRNA to a protein (Figure 1–15). As described previously, when a definitive mRNA reaches a ribosome in the cytoplasm, it dictates the formation of a polypeptide chain. Amino acids in the cytoplasm are activated by combination with an enzyme and AMP (adenylate), and each **activated amino acid** then combines with a specific molecule of tRNA. There is at least one

tRNA for each of the 20 unmodified amino acids found in large quantities in the body proteins of animals, but some amino acids have more than one tRNA. The tRNA–amino acid–adenylate complex is next attached to the mRNA template, a process that occurs in the ribosomes. The tRNA "recognizes" the proper spot to attach on the mRNA template because it has on its active end a set of three bases that are complementary to a set of three bases in a particular spot on the mRNA chain. The genetic code is made up of such triplets **(codons),** sequences of three purine, pyrimidine, or purine and pyrimidine bases; each codon stands for a particular amino acid.

Translation typically starts in the ribosomes with an AUG (transcribed from ATG in the gene), which codes for methionine. The amino terminal (or N-terminal) amino acid is then added, and the chain is lengthened one amino acid at a time. The mRNA attaches to the 40S subunit of the ribosome during protein synthesis, the polypeptide chain being formed attaches to the 60S subunit, and the tRNA attaches to both. As the amino acids are added in the order dictated by the codon, the ribosome moves along the mRNA molecule like a bead on a string. Translation stops at one of three stop, or nonsense, codons (UGA, UAA, or UAG), and the polypeptide chain is released. The tRNA molecules are used again. The mRNA molecules are typically reused approximately 10 times before being replaced. It is common to have more than one ribosome on a given mRNA chain at a time. The mRNA chain plus its collection of ribosomes is visible under the electron microscope as an aggregation of ribosomes called a **polyribosome.**

POSTTRANSLATIONAL MODIFICATION

After the polypeptide chain is formed, it "folds" into its biological form and can be further modified to the final protein by one or more of a combination of reactions that include hydroxylation, carboxylation, glycosylation, or phosphorylation of amino acid residues; cleavage of peptide bonds that converts a larger polypeptide to a smaller form; and the further folding, packaging, or folding and packaging of the protein into its ultimate, often complex configuration. Protein folding is a complex process that is dictated primarily by the sequence of the amino acids in the polypeptide chain. In some instances, however, nascent proteins associate with other proteins called **chaperones,** which prevent inappropriate contacts with other proteins and ensure that the final "proper" conformation of the nascent protein is reached.

Proteins also contain information that helps direct them to individual cell

compartments. Many proteins that are destined to be secreted or stored in organelles and most transmembrane proteins have at their amino terminal a **signal peptide (leader sequence)** that guides them into the endoplasmic reticulum. The sequence is made up of 15–30 predominantly hydrophobic amino acid residues. The signal peptide, once synthesized, binds to a **signal recognition particle (SRP),** a complex molecule made up of six polypeptides and 7S RNA, one of the small RNAs. The SRP stops translation until it binds to a **translocon,** a pore in the endoplasmic reticulum that is a heterotrimeric structure made up of Sec 61 proteins. The ribosome also binds, and the signal peptide leads the growing peptide chain into the cavity of the endoplasmic reticulum **(Figure 1–18)**. The signal peptide is next cleaved from the rest of the peptide by a signal peptidase while the rest of the peptide chain is still being synthesized. SRPs are not the only signals that help direct proteins to their proper place in or out of the cell; other signal sequences, posttranslational modifications, or both (eg, glycosylation) can serve this function.

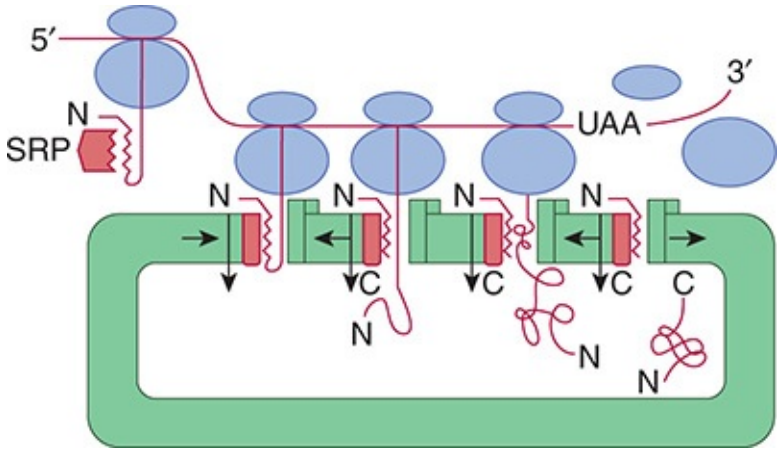


FIGURE 1–18 Translation of protein into the endoplasmic reticulum according to the signal hypothesis. The ribosomes synthesizing a protein move along the mRNA from the 5′ to the 3′ end. When the signal peptide of a protein destined for secretion, the cell membrane, or lysosomes emerges from the large unit of the ribosome, it binds to a signal recognition particle (SRP), and this arrests further translation until it binds to the translocon on the endoplasmic reticulum. C, carboxyl end of protein; N, amino end of protein. (Reproduced with permission from Perara E, Lingappa VR: Transport of proteins into and across the endoplasmic reticulum membrane. In: *Protein Transfer and Organelle Biogenesis*. Das RC, Robbins PW (editors). Academic Press, 1988.)

UBIQUITINATION & PROTEIN DEGRADATION

Like protein synthesis, protein degradation is a carefully regulated, complex process. It has been estimated that overall, up to 30% of newly produced proteins are abnormal, such as can occur during improper folding. Aged normal proteins also need to be removed as they are replaced. Conjugation of proteins to the 74-amino-acid polypeptide **ubiquitin** marks them for degradation. This polypeptide is highly conserved and is present in species ranging from bacteria to humans. The process of binding ubiquitin is called **ubiquitination,** and in some instances, multiple ubiquitin molecules bind **(polyubiquitination).** Ubiquitination of cytoplasmic proteins, including integral proteins of the endoplasmic reticulum, can mark the proteins for degradation in multisubunit proteolytic particles, or **proteasomes.** Ubiquitination of membrane proteins, such as the growth hormone receptors, also marks them for degradation; however, these can be degraded in lysosomes as well as via the proteasomes. Alteration of proteins by ubiquitin or the small ubiquitin-related modifier **(SUMO),** however, does not necessarily lead to degradation. More recently it has been shown that these posttranslational modifications can play important roles in protein–protein interactions and various cellular signaling pathways.

There is an obvious balance between the rate of production of a protein and its destruction, so ubiquitin conjugation is of major importance in cellular physiology. The rates at which individual proteins are metabolized vary, and the body has mechanisms by which abnormal proteins are recognized and degraded more rapidly than normal body constituents. For example, abnormal hemoglobins are metabolized rapidly in individuals with congenital hemoglobinopathies (see Chapter 31).

CATABOLISM OF AMINO ACIDS

The short-chain fragments produced by amino acid, carbohydrate, and fat catabolism are very similar (see below). From this **common metabolic pool** (Figure 1–16) of intermediates, carbohydrates, proteins, and fats can be synthesized. These fragments can enter the citric acid cycle (or tricarboxylic acid cycle or Krebs cycle), a final common pathway of catabolism, in which they are broken down to hydrogen atoms and CO_2. Interconversion of amino acids involves transfer, removal, or formation of amino groups. **Transamination** reactions, conversion of one amino acid to the corresponding keto acid with simultaneous conversion of another keto acid to an amino acid, occur in many

tissues:

$$\text{Alanine} + \alpha\text{–Ketoglutarate} \rightleftharpoons \text{Pyruvate} + \text{Glutamate}$$

Oxidative deamination of amino acids occurs in the liver. An imino acid is formed by dehydrogenation, and this compound is hydrolyzed to the corresponding keto acid, with production of NH_4^+:

$$\text{Amino acid} + NAD^+ \rightarrow \text{Imino acid} + NADH + H^+$$

$$\text{Imino acid} + H_2O \rightarrow \text{Keto acid} + NH_4^+$$

Interconversions between the amino acid pool and the common metabolic pool are summarized in **Figure 1–19**. Leucine, isoleucine, phenylalanine, and tyrosine are said to be **ketogenic** because they are converted to the ketone body acetoacetate (see below). Alanine and many other amino acids are **glucogenic** or **gluconeogenic;** that is, they give rise to compounds that can readily be converted to glucose.

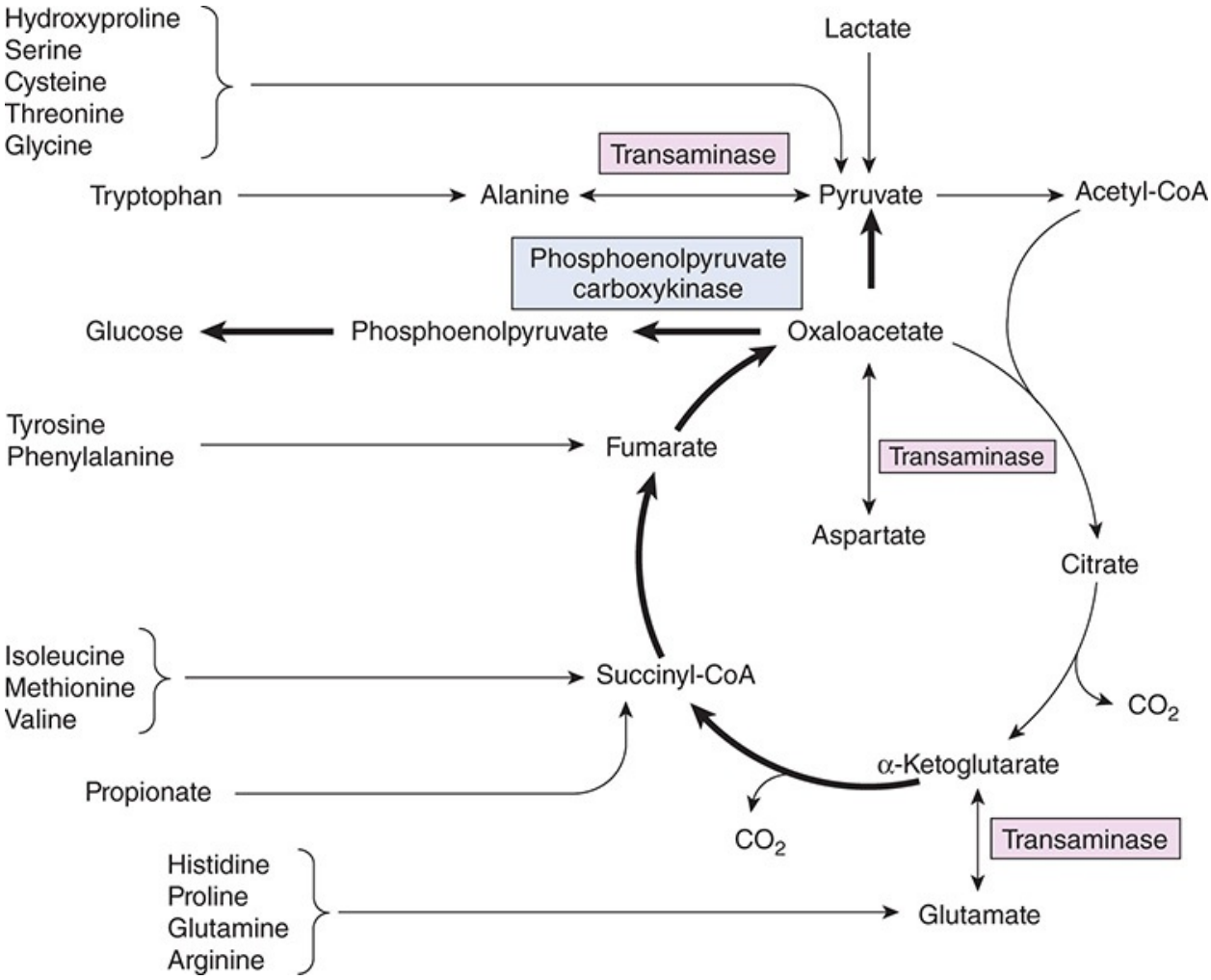


FIGURE 1–19 Involvement of the citric acid cycle in transamination and gluconeogenesis. The bold arrows indicate the main pathway of gluconeogenesis. Note the many entry positions for groups of amino acids into the citric acid cycle. (Reproduced with permission from Murray RK et al: *Harper's Biochemistry*, 28th ed. New York, NY: McGraw-Hill; 2009.)

UREA FORMATION

Most of the NH_4^+ formed by deamination of amino acids in the liver is converted to urea, and the urea is excreted in the urine. The NH_4^+ forms carbamoyl phosphate, and in the mitochondria it is transferred to ornithine, forming citrulline. The enzyme involved is ornithine carbamoyltransferase. Citrulline is converted to arginine, after which urea is split off and ornithine is regenerated (urea cycle; **Figure 1–20**). The overall reaction in the urea cycle

consumes 3 ATP (not shown) and thus requires significant energy. Most of the urea is formed in the liver, and in severe liver disease, the blood urea nitrogen (BUN) falls and blood NH_3 rises (see Chapter 28). Congenital deficiency of ornithine carbamoyltransferase can also lead to NH_3 intoxication.

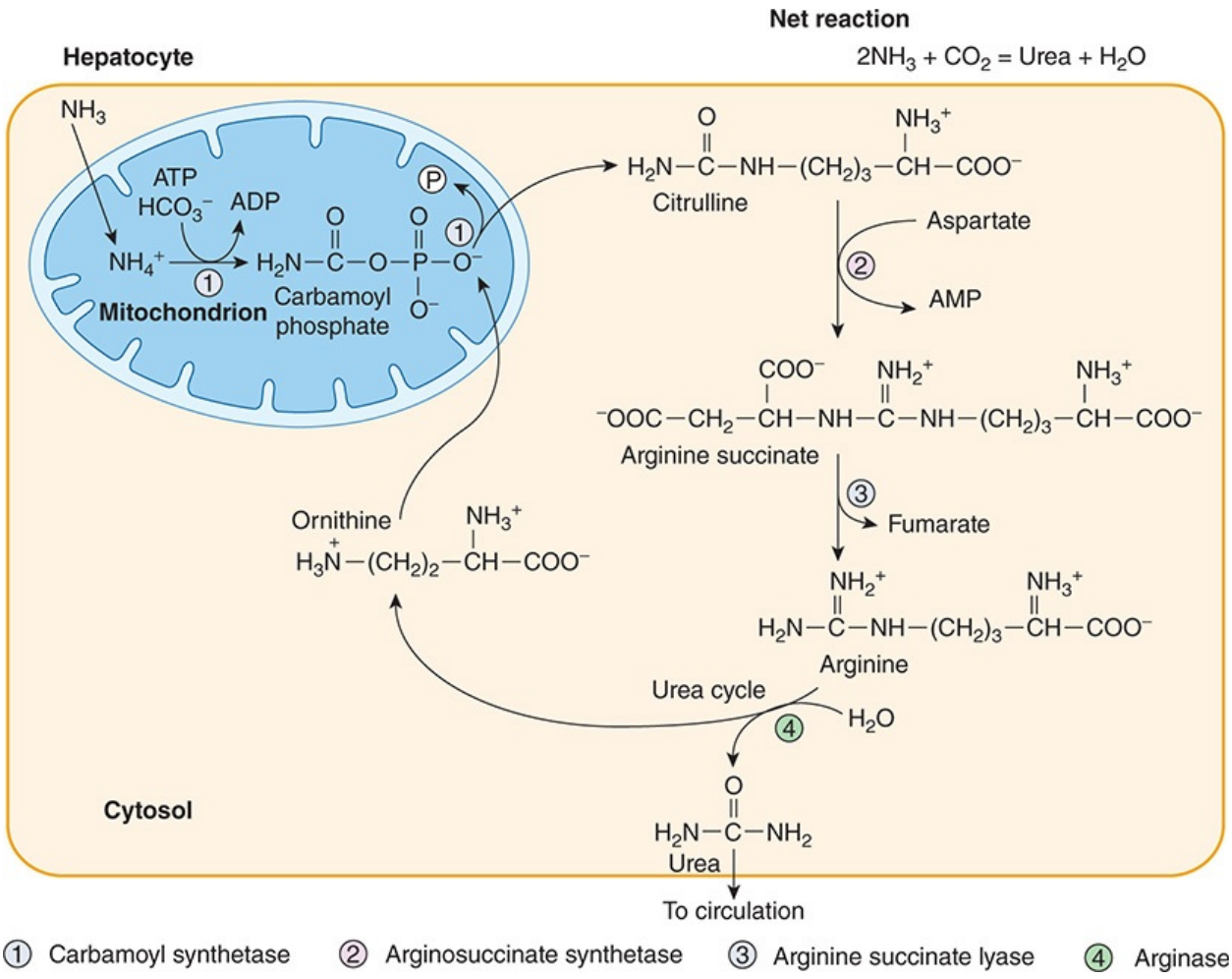


FIGURE 1–20 Urea cycle. The processing of NH_3 to urea for excretion contains coordinative steps in both the cytosol and the mitochondrion of a hepatocyte. Note that the production of carbamoyl phosphate and its conversion to citrulline occurs in the mitochondria, whereas other processes are in the cytoplasm.

METABOLIC FUNCTIONS OF AMINO ACIDS

In addition to providing the basic building blocks for proteins, amino acids also have metabolic functions. Thyroid hormones, catecholamines, histamine,

serotonin, melatonin, and intermediates in the urea cycle are formed from specific amino acids. Methionine and cysteine provide the sulfur contained in proteins, CoA, taurine, and other biologically important compounds. Methionine is converted into S-adenosylmethionine, which is the active methylating agent in the synthesis of compounds such as epinephrine.

CARBOHYDRATES

Carbohydrates are organic molecules made of equal amounts of carbon and H_2O. The simple sugars, or **monosaccharides,** including **pentoses** (five carbons; eg, ribose) and **hexoses** (six carbons; eg, glucose) perform both structural (eg, as part of nucleotides discussed previously) and functional (eg, inositol 1,4,5 trisphosphate acts as a cellular signaling molecule) roles in the body. Monosaccharides can be linked together to form disaccharides (eg, sucrose), or polysaccharides (eg, glycogen). The placement of sugar moieties onto proteins (glycoproteins) aids in cellular targeting, and in the case of some receptors, recognition of signaling molecules. In this section, the major role of carbohydrates in the production and storage of energy will be discussed.

Dietary carbohydrates are for the most part polymers of hexoses, of which the most important are glucose, galactose, and fructose. Most of the monosaccharides occurring in the body are the D isomers. The principal product of carbohydrate digestion and the principal circulating sugar is glucose. The normal fasting level of plasma glucose in peripheral venous blood is 70–110 mg/dL (3.9–6.1 mmol/L). In arterial blood, the plasma glucose level is 15–30 mg/dL higher than in venous blood.

Once it enters cells, glucose is normally phosphorylated to form glucose-6-phosphate. The enzyme that catalyzes this reaction is **hexokinase.** In the liver, there is an additional enzyme, **glucokinase,** which has greater specificity for glucose and which, unlike hexokinase, is increased by insulin and decreased in starvation and diabetes. The glucose-6- phosphate is either polymerized into glycogen or catabolized. The process of glycogen formation is called **glycogenesis,** and glycogen breakdown is called **glycogenolysis.** Glycogen, the storage form of glucose, is present in most body tissues, but the major supplies are in the liver and skeletal muscle. The breakdown of glucose to pyruvate or lactate (or both) is called **glycolysis.** Glucose catabolism proceeds via cleavage through fructose to trioses or via oxidation and decarboxylation to pentoses. The pathway to pyruvate through the trioses is the **Embden–Meyerhof pathway,** and that through 6-phosphogluconate and the pentoses is the **direct oxidative**

pathway (hexose monophosphate shunt). Pyruvate is converted to acetyl-CoA. Interconversions between carbohydrate, fat, and protein include conversion of the glycerol from fats to dihydroxyacetone phosphate and conversion of a number of amino acids with carbon skeletons resembling intermediates in the Embden–Meyerhof pathway and citric acid cycle to these intermediates by deamination. In this way, and by conversion of lactate to glucose, nonglucose molecules can be converted to glucose **(gluconeogenesis).** Glucose can be converted to fats through acetyl-CoA, but because the conversion of pyruvate to acetyl-CoA, unlike most reactions in glycolysis, is irreversible, fats are not converted to glucose via this pathway. There is therefore very little net conversion of fats to carbohydrates in the body because, except for the quantitatively unimportant production from glycerol, there is no pathway for conversion.

CITRIC ACID CYCLE

The **citric acid cycle** (Krebs cycle, tricarboxylic acid cycle) produces ATP through a sequence of reactions in which acetyl-CoA is metabolized to CO_2 and H atoms. Acetyl-CoA is first condensed with the anion of a four-carbon acid, oxaloacetate, to form citrate and HS-CoA. In a series of seven subsequent reactions, 2 CO_2 molecules are split off, regenerating oxaloacetate **(Figure 1–21)**. Four pairs of H atoms are transferred to the flavoprotein–cytochrome chain, producing 12 ATP and 4 H_2O, of which 2 H_2O is used in the cycle. The citric acid cycle is the common pathway for oxidation to CO_2 and H_2O of carbohydrate, fat, and some amino acids. The major entry into it is through acetyl CoA, but a number of amino acids can be converted to citric acid cycle intermediates by deamination. The citric acid cycle requires O_2 and does not function under anaerobic conditions.

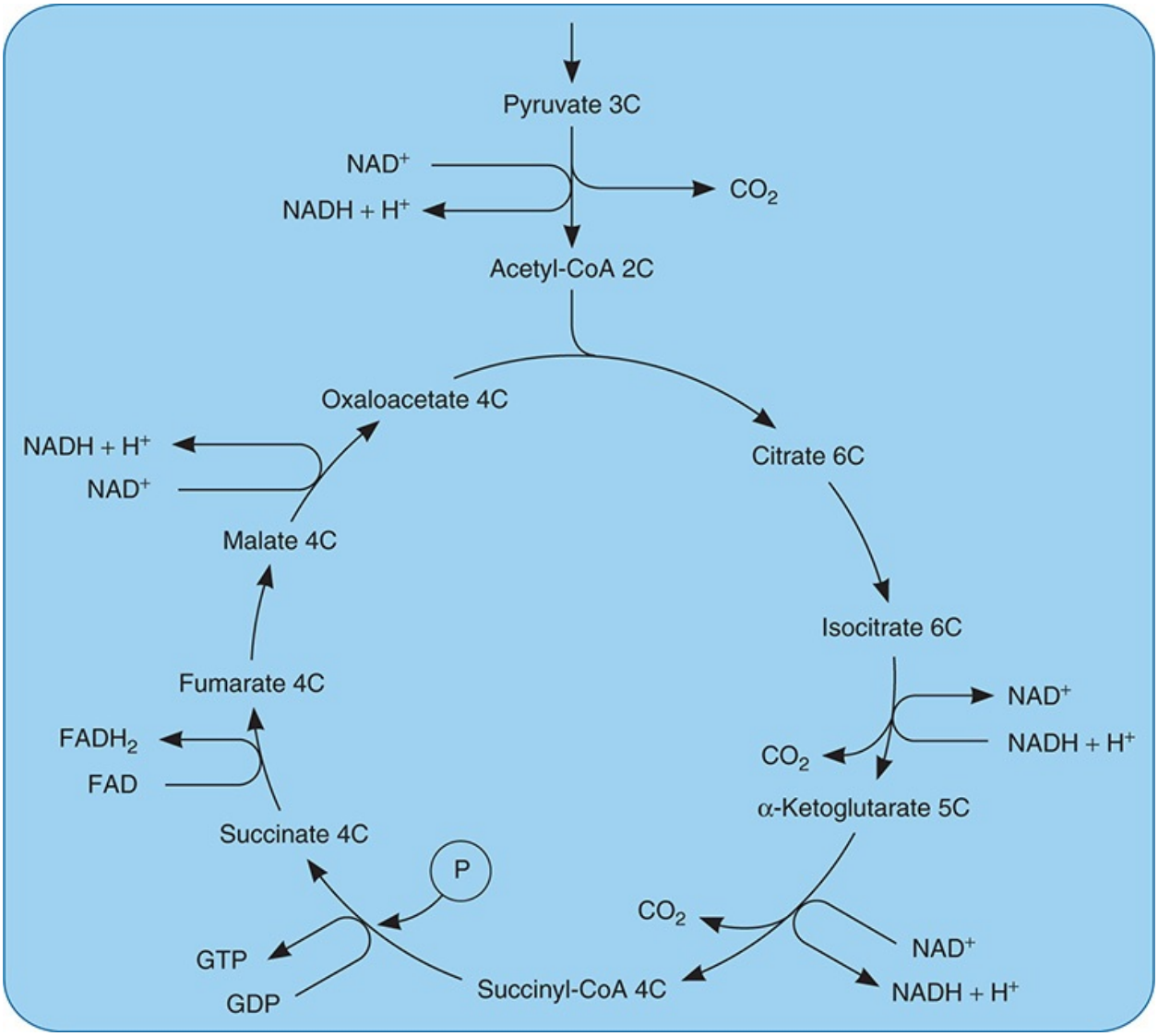


FIGURE 1–21 Citric acid cycle. The numbers (6C, 5C, etc.) indicate the number of carbon atoms in each of the intermediates. The conversion of pyruvate to acetyl-CoA and each turn of the cycle provide four NADH and one $FADH_2$ for oxidation via the flavoprotein-cytochrome chain plus formation of one GTP that is readily converted to ATP.

ENERGY PRODUCTION

The net production of energy-rich phosphate compounds during the metabolism of glucose and glycogen to pyruvate depends on whether metabolism occurs via the Embden–Meyerhof pathway or the hexose monophosphate shunt. By oxidation at the substrate level, the conversion of 1 mol of

phosphoglyceraldehyde to phosphoglycerate generates 1 mol of ATP, and the conversion of 1 mol of phosphoenolpyruvate to pyruvate generates another. Because 1 mol of glucose-6-phosphate produces, via the Embden–Meyerhof pathway, 2 mol of phosphoglyceraldehyde, 4 mol of ATP is generated per mole of glucose metabolized to pyruvate. All these reactions occur in the absence of O_2 and consequently represent anaerobic production of energy. However, 1 mol of ATP is used in forming fructose 1,6-diphosphate from fructose 6-phosphate and 1 mol in phosphorylating glucose when it enters the cell. Consequently, when pyruvate is formed anaerobically from glycogen, there is a *net* production of 3 mol of ATP per mole of glucose-6-phosphate; however, when pyruvate is formed from 1 mol of blood glucose, the net gain is only 2 mol of ATP.

A supply of NAD^+ is necessary for the conversion of phosphoglyceraldehyde to phosphoglycerate. Under anaerobic conditions (anaerobic glycolysis), a block of glycolysis at the phosphoglyceraldehyde conversion step might be expected to develop as soon as the available NAD^+ is converted to NADH. However, pyruvate can accept hydrogen from NADH, forming NAD^+ and lactate:

$$\text{Pyruvate} + \text{NADH} \rightleftharpoons \text{Lactate} + \text{NAD}^+$$

In this way, glucose metabolism and energy production can continue for a while without O_2. The lactate that accumulates is converted back to pyruvate when the O_2 supply is restored, with NADH transferring its hydrogen to the flavoprotein–cytochrome chain.

During aerobic glycolysis, the net production of ATP is 19 times greater than the 2 ATPs formed under anaerobic conditions. Six ATPs are formed by oxidation, via the flavoprotein–cytochrome chain, of the 2 NADHs produced when 2 molecules of phosphoglyceraldehyde are converted to phosphoglycerate (Figure 1–21), 6 ATPs are formed from the 2 NADHs produced when 2 molecules of pyruvate are converted to acetyl-CoA, and 24 ATPs are formed during the subsequent two turns of the citric acid cycle. Of these, 18 are formed by oxidation of 6 NADHs, 4 by oxidation of 2 $FADH_2$s, and 2 by oxidation at the substrate level, when succinyl-CoA is converted to succinate (this reaction actually produces guanosine triphosphate [GTP], but the GTP is converted to ATP). Thus, the net production of ATP per mol of blood glucose metabolized aerobically via the Embden–Meyerhof pathway and citric acid cycle is $38 = 2 + [2 \times 3] + [2 \times 3] + [2 \times 12]$.

Glucose oxidation via the hexose monophosphate shunt generates large amounts of NADPH. A supply of this reduced coenzyme is essential for many

metabolic processes. The pentoses formed in the process are building blocks for nucleotides (see below). The amount of ATP generated depends on the amount of NADPH converted to NADH and then oxidized.

“DIRECTIONAL-FLOW VALVES” IN METABOLISM

Metabolism is regulated by a variety of hormones and other factors. To bring about any net change in a particular metabolic process, regulatory factors obviously must drive a chemical reaction in one direction. Most of the reactions in intermediary metabolism are freely reversible, but there are a number of “directional-flow valves,” that is, reactions that proceed in one direction under the influence of one enzyme or transport mechanism and in the opposite direction under the influence of another. Five examples in the intermediary metabolism of carbohydrate are shown in **Figure 1–22**. The different pathways for fatty acid synthesis and catabolism (see below) are another example. Regulatory factors exert their influence on metabolism by acting directly or indirectly at these directional-flow valves.

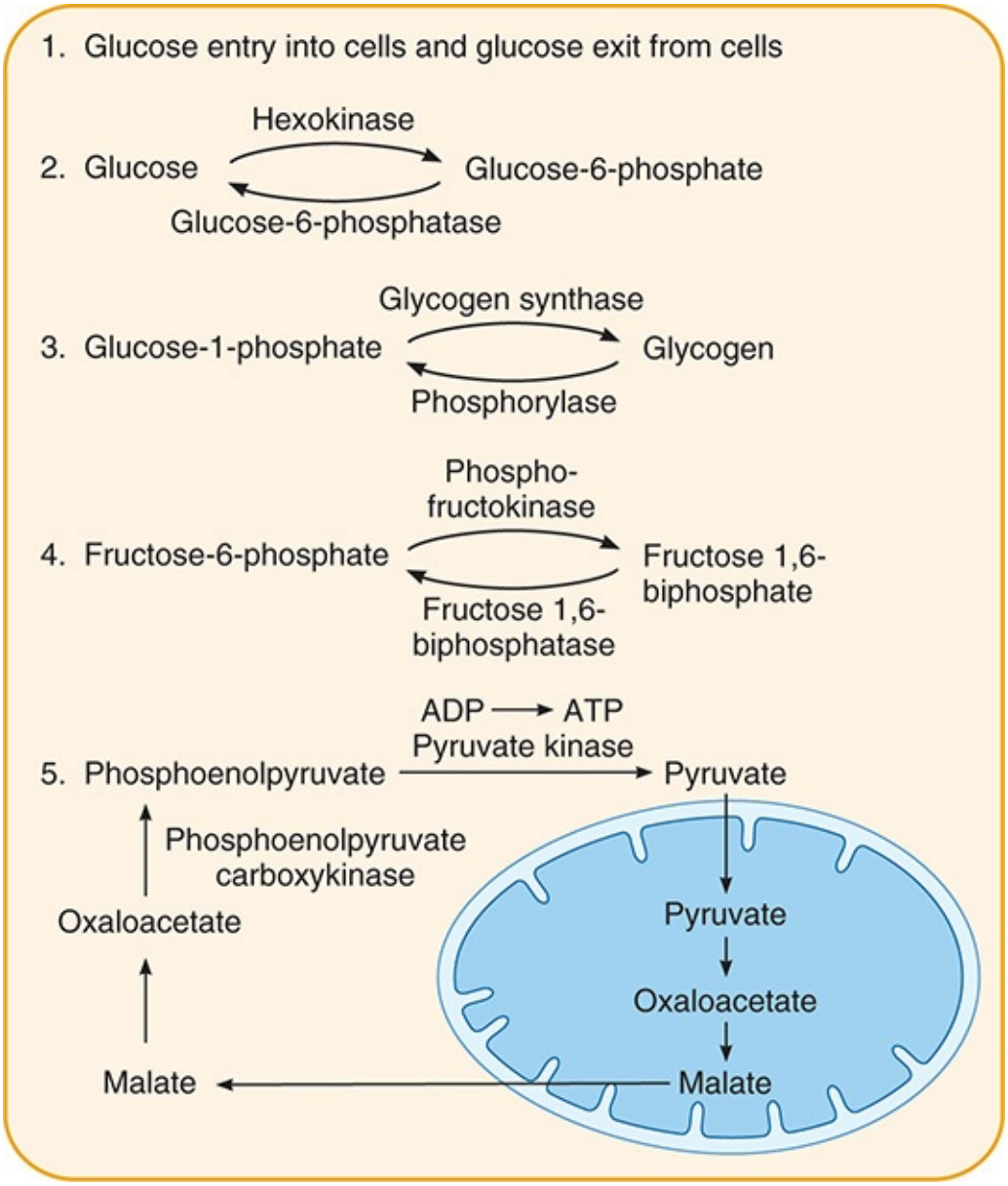


FIGURE 1–22 Directional-flow valves in energy production reactions. In carbohydrate metabolism there are several reactions that proceed in one direction by one mechanism and in the other direction by a different mechanism, termed "directional-flow valves." Five examples of these reactions are illustrated (numbered at left). The double line in example 5 represents the mitochondrial membrane. Pyruvate is converted to malate in mitochondria, and the malate diffuses out of the mitochondria to the cytosol, where it is converted to phosphoenolpyruvate.

GLYCOGEN SYNTHESIS & BREAKDOWN

Glycogen is a branched glucose polymer with two types of glycoside linkages: 1:4α and 1:6α **(Figure 1–23)**. It is synthesized on **glycogenin,** a protein primer, from glucose-1-phosphate via uridine diphosphoglucose (UDPG). The enzyme **glycogen synthase** catalyses the final synthetic step. The availability of glycogenin is one of the factors determining the amount of glycogen synthesized. The breakdown of glycogen in 1:4α linkage is catalyzed by phosphorylase, whereas another enzyme catalyzes the breakdown of glycogen in 1:6α linkage.

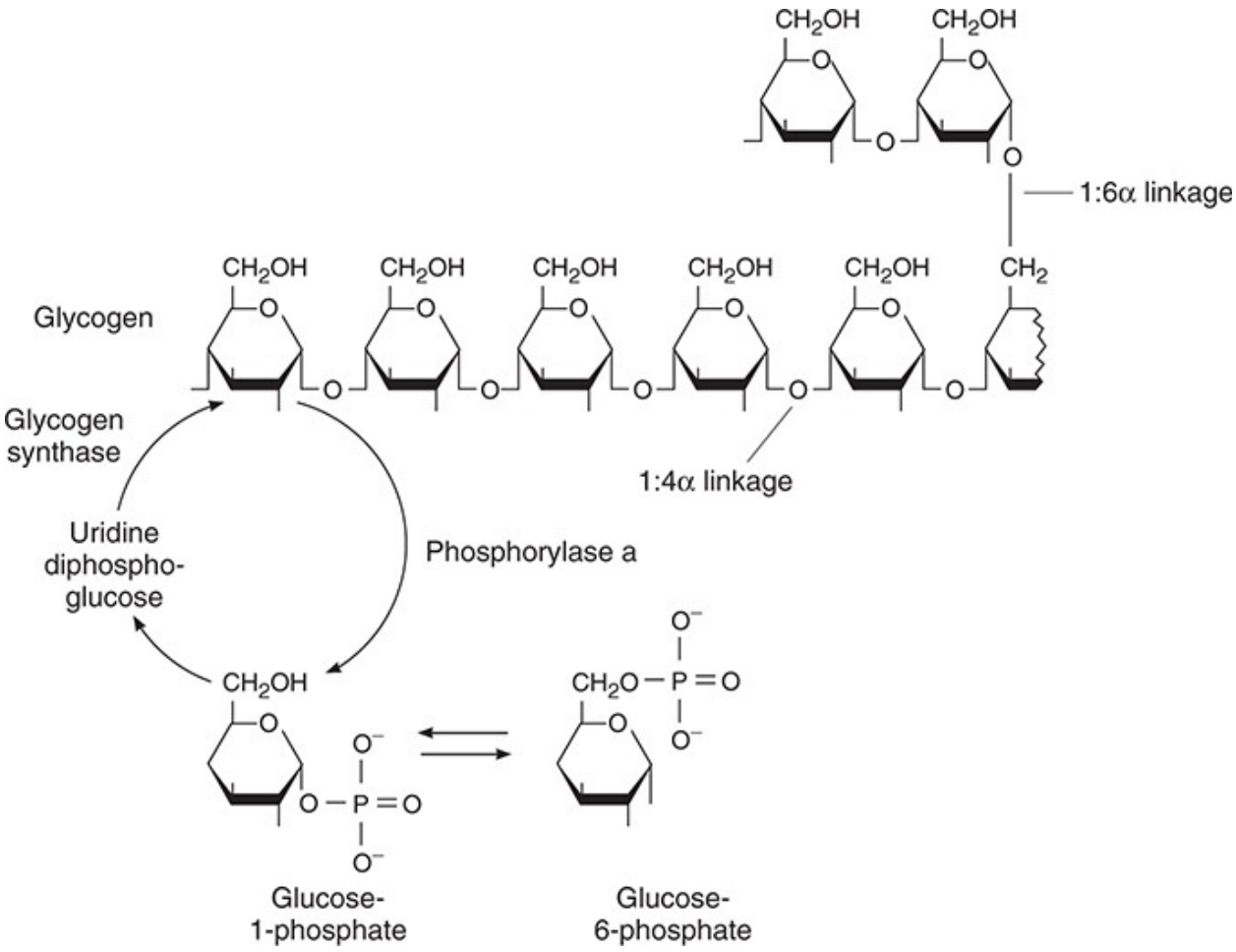


FIGURE 1–23 Glycogen synthesis and breakdown. Glycogen is the main storage for glucose in the cell. It is cycled: built up from glucose-6-phosphate when energy is stored and broken down to glucose-6-phosphate when energy is required. Note the intermediate glucose-1-phosphate and enzymatic control by phosphorylase a and glycogen kinase.

FACTORS DETERMINING THE PLASMA GLUCOSE LEVEL

The blood plasma glucose level at any given time is determined by the balance between the amount of glucose entering the bloodstream and the amount of glucose leaving the bloodstream. The principal determinants are therefore the dietary intake; the rate of entry into the cells of muscle, adipose tissue, and other organs; and the glucostatic activity of the liver **(Figure 1–24)**. Five percent of ingested glucose is promptly converted into glycogen in the liver, and 30–40% is converted into fat. The remainder is metabolized in muscle and other tissues. During fasting, liver glycogen is broken down and the liver adds glucose to the bloodstream. With more prolonged fasting, glycogen is depleted and there is increased gluconeogenesis from amino acids and glycerol in the liver. Plasma glucose declines modestly to about 60 mg/dL during prolonged starvation in normal individuals, but symptoms of hypoglycemia do not occur because gluconeogenesis prevents any further fall.

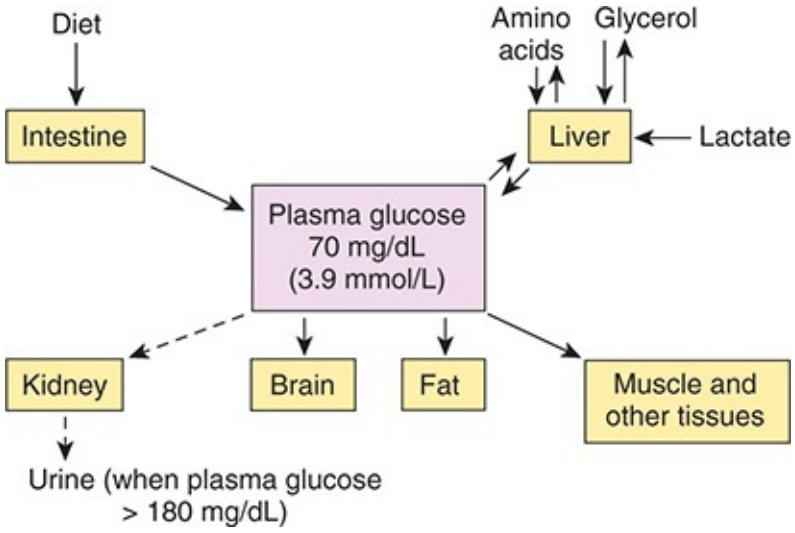


FIGURE 1–24 Plasma glucose homeostasis. Note the glucostatic function of the liver, as well as the loss of glucose in the urine when the renal threshold is exceeded (dashed arrows).

METABOLISM OF HEXOSES OTHER THAN GLUCOSE

Other hexoses that are absorbed from the intestine include galactose, which is

liberated by the digestion of lactose and converted to glucose in the body; and fructose, part of which is ingested and part produced by hydrolysis of sucrose. After phosphorylation, galactose reacts with UDPG to form uridine diphosphogalactose. The uridine diphosphogalactose is converted back to UDPG, and the UDPG functions in glycogen synthesis. This reaction is reversible, and conversion of UDPG to uridine diphosphogalactose provides the galactose necessary for formation of glycolipids and mucoproteins when dietary galactose intake is inadequate. The utilization of galactose, like that of glucose, depends on insulin. The inability to make UDPG can have serious health consequences **(Clinical Box 1–5)**.

CLINICAL BOX 1–5

Galactosemia

In the inborn error of metabolism known as **galactosemia**, there is a congenital deficiency of galactose-1-phosphate uridyl transferase, the enzyme responsible for the reaction between galactose-1-phosphate and UDPG, so that ingested galactose accumulates in the circulation; serious disturbances of growth and development result.

THERAPEUTIC HIGHLIGHTS

Treatment with galactose-free diets improves galactosemia without leading to galactose deficiency. This occurs because the enzyme necessary for the formation of uridine diphosphogalactose from UDPG is present.

Fructose is converted in part to fructose 6-phosphate and then metabolized via fructose 1,6-diphosphate. The enzyme catalyzing the formation of fructose 6-phosphate is hexokinase, the same enzyme that catalyzes the conversion of glucose to glucose-6-phosphate. However, much more fructose is converted to fructose 1-phosphate in a reaction catalyzed by fructokinase. Most of the fructose 1-phosphate is then split into dihydroxyacetone phosphate and glyceraldehyde. The glyceraldehyde is phosphorylated, and it and the dihydroxyacetone phosphate enter the pathways for glucose metabolism. Because the reactions proceeding through phosphorylation of fructose in the 1 position can occur at a normal rate in the absence of insulin, it had been

recommended that fructose be given to diabetics to replenish their carbohydrate stores. However, most of the fructose is metabolized in the intestines and liver, so its value in replenishing carbohydrate elsewhere in the body is limited.

Fructose 6-phosphate can also be phosphorylated in the 2 position, forming fructose 2,6-diphosphate. This compound is an important regulator of hepatic gluconeogenesis. When the fructose 2,6-diphosphate level is high, conversion of fructose 6-phosphate to fructose 1,6-diphosphate is facilitated, and thus breakdown of glucose to pyruvate is increased. A decreased level of fructose 2,6-diphosphate facilitates the reverse reaction and consequently aids gluconeogenesis.

FATTY ACIDS & LIPIDS

The biologically important lipids are the fatty acids and their derivatives, the neutral fats (triglycerides), the phospholipids and related compounds, and the sterols. The triglycerides are made up of three fatty acids bound to glycerol **(Table 1–4)**. Naturally occurring fatty acids contain an even number of carbon atoms. They may be saturated (no double bonds) or unsaturated (dehydrogenated, with various numbers of double bonds). The phospholipids are constituents of cell membranes and provide structural components of the cell membrane, as well as an important source of intracellular and intercellular signaling molecules. Fatty acids also are an important source of energy in the body.

TABLE 1–4 Lipids.

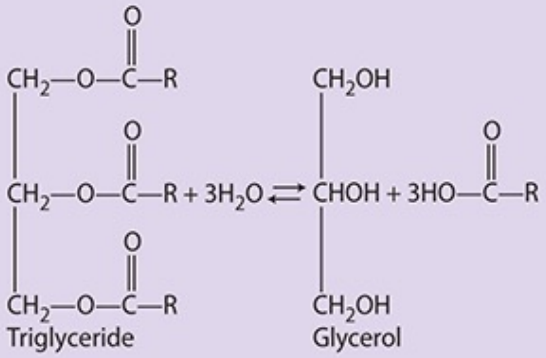
Typical fatty acids:

Palmitic acid: $CH_5(CH_2)_{14}—C(=O)—OH$

Stearic acid: $CH_5(CH_2)_{16}—C(=O)—OH$

Oleic acid: $CH_5(CH_2)_7CH{=}CH(CH_2)_7—C(=O)—OH$ (Unsaturated)

Triglycerides (triacylglycerols): Esters of glycerol and three fatty acids.



R = Aliphatic chain of various lengths and degrees of saturation.

Phospholipids:

A. Esters of glycerol, two fatty acids, and
   1. Phosphate = phosphatidic acid
   2. Phosphate plus inositol = phosphatidylinositol
   3. Phosphate plus choline = phosphatidylcholine (lecithin)
   4. Phosphate plus ethanolamine = phosphatidyl-ethanolamine (cephalin)
   5. Phosphate plus serine = phosphatidylserine

B. Other phosphate-containing derivatives of glycerol.

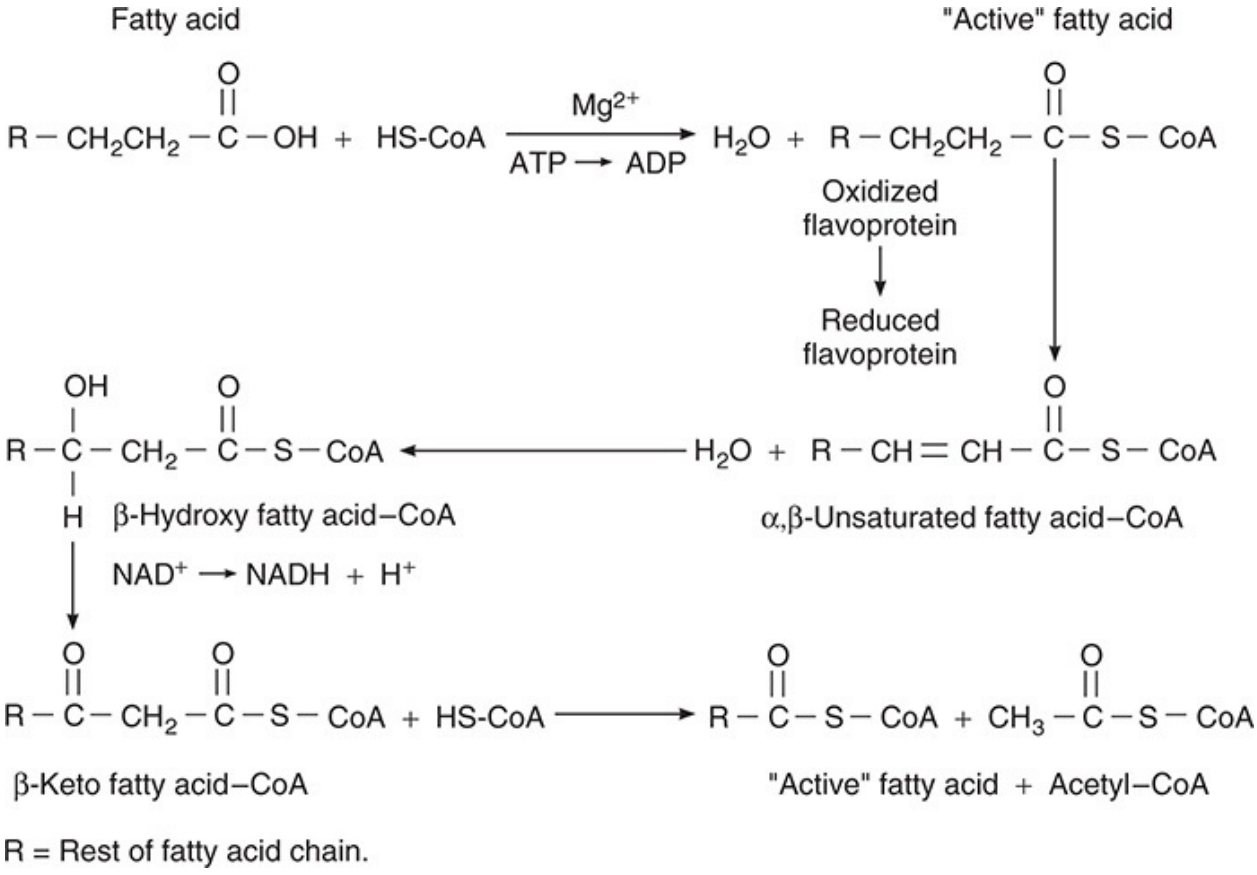
C. Sphingomyelins: Esters of fatty acid, phosphate, choline, and the amino alcohol sphingosine.

Cerebrosides: Compounds containing galactose, fatty acid, and sphingosine.

Sterols: Cholesterol and its derivatives, including steroid hormones, bile acids, and various vitamins.

FATTY ACID OXIDATION & SYNTHESIS

In the body, fatty acids are broken down to acetyl-CoA, which enters the citric acid cycle. The main breakdown occurs in the mitochondria by β-oxidation. Fatty acid oxidation begins with activation (formation of the CoA derivative) of the fatty acid, a reaction that occurs both inside and outside the mitochondria. Medium- and short-chain fatty acids can enter the mitochondria without difficulty, but long-chain fatty acids must be bound to **carnitine** in ester linkage before they can cross the inner mitochondrial membrane. Carnitine is β-hydroxy-γ-trimethylammonium butyrate, and it is synthesized in the body from lysine and methionine. A translocase moves the fatty acid–carnitine ester into the matrix space. The ester is hydrolyzed, and the carnitine recycles. β-Oxidation proceeds by serial removal of two carbon fragments from the fatty acid **(Figure 1–25)**. The energy yield of this process is large. For example, catabolism of 1 mol of a six-carbon fatty acid through the citric acid cycle to CO_2 and H_2O generates 44 mol of ATP, compared with the 38 mol generated by catabolism of 1 mol of the six-carbon carbohydrate glucose.



R = Rest of fatty acid chain.

FIGURE 1–25 Fatty acid oxidation. This process, splitting off two carbon fragments at a time, is repeated to the end of the chain.

KETONE BODIES

In many tissues, acetyl-CoA units condense to form acetoacetyl- CoA **(Figure 1–26)**. In the liver, which (unlike other tissues) contains a deacylase, free acetoacetate is formed. This β-keto acid is converted to β-hydroxybutyrate and acetone, and because these compounds are metabolized with difficulty in the liver, they diffuse into the circulation. Acetoacetate is also formed in the liver via the formation of 3-hydroxy-3-methylglutaryl-CoA, and this pathway is quantitatively more important than deacylation. Acetoacetate, β-hydroxybutyrate, and acetone are called **ketone bodies.** Tissues other than liver transfer CoA from succinyl-CoA to acetoacetate and metabolize the "active" acetoacetate to CO_2 and H_2O via the citric acid cycle. Ketone bodies are also metabolized via other pathways. Acetone is discharged in the urine and expired air. An imbalance of ketone bodies can lead to serious health problems **(Clinical Box 1–6)**.

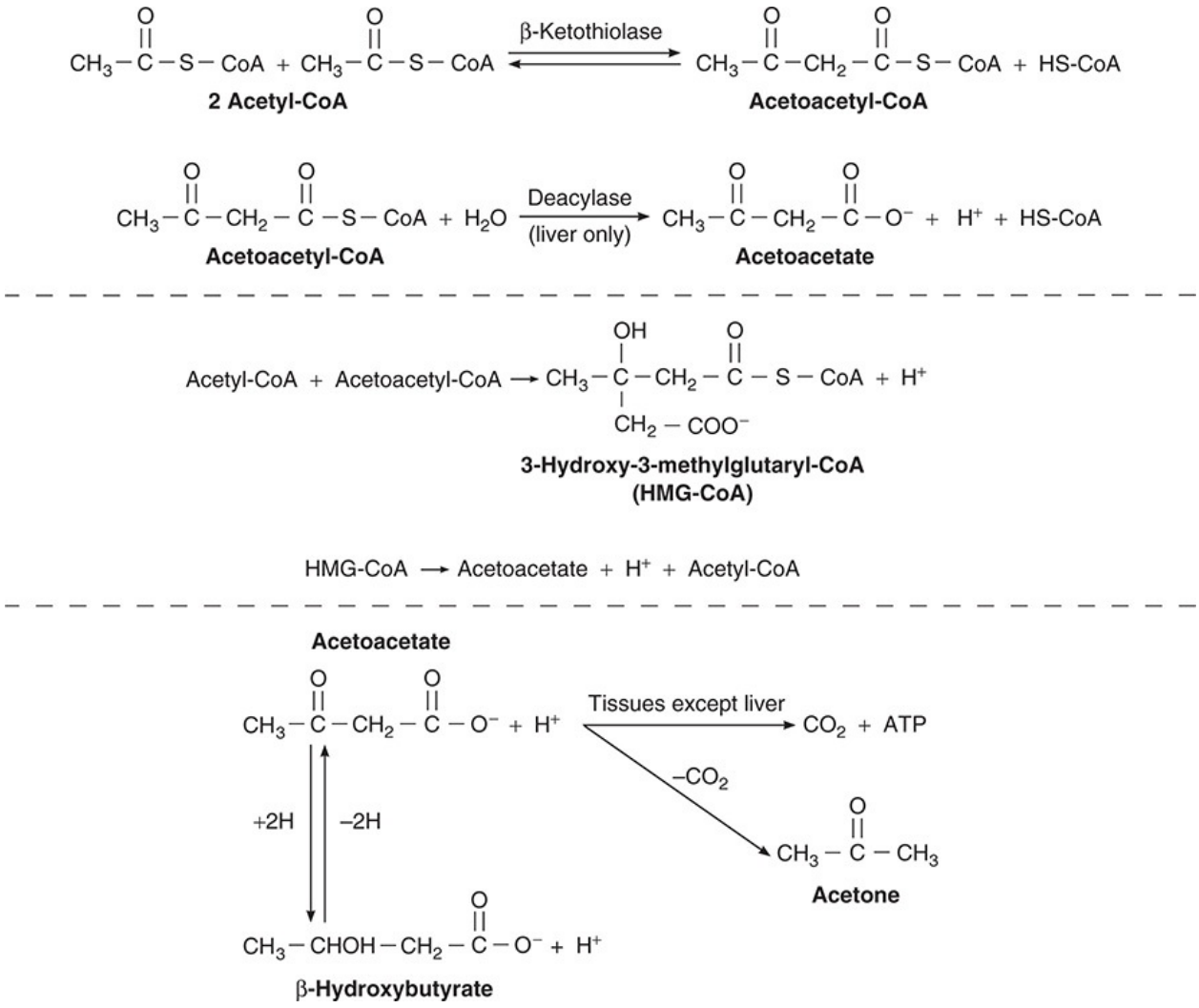


FIGURE 1–26 Formation and metabolism of ketone bodies. Note the two pathways for the formation of acetoacetate.

CLINICAL BOX 1–6

Diseases Associated with Imbalance of β-oxidation of Fatty Acids

Ketoacidosis

The normal blood ketone level in humans is low (about 1 mg/dL) and less than 1 mg is excreted per 24 h, because the ketones are normally metabolized as rapidly as they are formed. However, if the entry of acetyl-CoA into the citric acid cycle is depressed because of a decreased supply of the products of glucose metabolism, or if the entry does not increase when the supply of

acetyl-CoA increases, acetyl-CoA accumulates, the rate of condensation to acetoacetyl-CoA increases, and more acetoacetate is formed in the liver. The ability of the tissues to oxidize the ketones is soon exceeded, and they accumulate in the bloodstream (ketosis). Two of the three ketone bodies, acetoacetate and β-hydroxybutyrate, are anions of the moderately strong acids acetoacetic acid and β-hydroxybutyric acid. Many of their protons are buffered, reducing the decline in pH that would otherwise occur. However, the buffering capacity can be exceeded, and the metabolic acidosis that develops in conditions such as diabetic ketosis can be severe and even fatal. Three conditions lead to deficient intracellular glucose supplies, and hence to ketoacidosis: starvation; diabetes mellitus; and a high-fat, low-carbohydrate diet. The acetone odor on the breath of children who have been vomiting is due to the ketosis of starvation. Parenteral administration of relatively small amounts of glucose abolishes the ketosis, and it is for this reason that carbohydrate is said to be antiketogenic.

Carnitine Deficiency

Deficient β-oxidation of fatty acids can be produced by carnitine deficiency or genetic defects in the translocase or other enzymes involved in the transfer of long-chain fatty acids into the mitochondria. This causes cardiomyopathy. In addition, it causes **hypoketonemic hypoglycemia** with coma, a serious and often fatal condition triggered by fasting, in which glucose stores are used up because of the lack of fatty acid oxidation to provide energy. Ketone bodies are not formed in normal amounts because of the lack of adequate CoA in the liver.

THERAPEUTIC HIGHLIGHTS

The primary treatment for diabetic ketoacidosis is to replace the lost fluids and electrolytes, while insulin is generally given to reverse the processes that cause diabetic ketoacidosis. The main treatment for primary carnitine deficiency is lifelong use of L-carnitine, a natural substance that helps body cells make energy and get rid of harmful wastes. L-carnitine can also reverse the heart problems and muscle weakness caused by carnitine deficiency. Pediatric patients with carnitine deficiency also benefit from a low-fat, high carbohydrate diet and drinking more fluids.

CELLULAR LIPIDS

The lipids in cells are of two main types: **structural lipids,** which are an inherent part of the membranes and can serve as progenitors for cellular signaling molecules; and **neutral fat,** stored in the adipose cells of the fat depots. Neutral fat is mobilized during starvation, but structural lipid is preserved. The fat depots obviously vary in size, but in nonobese individuals they make up about 15% of body weight in men and 21% in women. They are not the inert structures they were once thought to be but, rather, active dynamic tissues undergoing continuous breakdown and resynthesis. In the depots, glucose is metabolized to fatty acids, and neutral fats are synthesized. Neutral fat is also broken down, and free fatty acids (FFAs) are released into the circulation.

A third, special type of lipid is **brown fat,** which makes up a small percentage of total body fat. Brown fat, which is somewhat more abundant in infants but is present in adults as well, is located between the scapulas, at the nape of the neck, along the great vessels in the thorax and abdomen, and in other scattered locations in the body. In brown fat depots, the fat cells as well as the blood vessels have an extensive sympathetic innervation. This is in contrast to white fat depots, in which some fat cells may be innervated but the principal sympathetic innervation is solely on blood vessels. In addition, ordinary lipocytes have only a single large droplet of white fat, whereas brown fat cells contain several small droplets of fat. Brown fat cells also contain many mitochondria. In these mitochondria, an inward proton conductance that generates ATP takes places as usual, but in addition there is a second proton conductance that does not generate ATP. This “short-circuit” conductance depends on a 32-kDa uncoupling protein (UCP1). It causes uncoupling of metabolism and generation of ATP, so that more heat is produced.

PLASMA LIPIDS & LIPID TRANSPORT

The major lipids are relatively insoluble in aqueous solutions and do not circulate in the free form. **FFAs** are bound to albumin, whereas cholesterol, triglycerides, and phospholipids are transported in the form of **lipoprotein** complexes. The complexes greatly increase the solubility of the lipids. The six families of lipoproteins **(Table 1–5)** are graded in size and lipid content. The density of these lipoproteins is inversely proportionate to their lipid content. In general, the lipoproteins consist of a hydrophobic core of triglycerides and cholesteryl esters surrounded by phospholipids and protein. These lipoproteins

can be transported from the intestine to the liver via an **exogenous pathway,** and between other tissues via an **endogenous pathway.**

TABLE 1–5 The principal lipoproteins.[a]

Lipoprotein	Size (nm)	Composition (%)					Origin
		Protein	Free Cholesteryl	Cholesterol Esters	Triglyceride	Phospholipid	
Chylomicrons	75–1000	2	2	3	90	3	Intestine
Chylomicron remnants	30–80	…	…	…	…	…	Capillaries
Very low-density lipoproteins (VLDL)	30–80	8	4	16	55	17	Liver and intestine
Intermediate-density lipoproteins (IDL)	25–40	10	5	25	40	20	VLDL
Low-density lipoproteins (LDL)	20	20	7	46	6	21	IDL
High-density lipoproteins (HDL)	7.5–10	50	4	16	5	25	Liver and intestine

[a]The plasma lipids include these components plus free fatty acids from adipose tissue, which circulate bound to albumin.

Dietary lipids are processed by several pancreatic lipases in the intestine to form mixed micelles of predominantly FFA, **2-monoacylglycerols,** and cholesterol derivatives (see Chapter 26). These micelles additionally can contain important water-insoluble molecules such as **vitamins A, D, E,** and **K**. These mixed micelles are taken up into cells of the intestinal mucosa where large lipoprotein complexes, **chylomicrons,** are formed. The chylomicrons and their remnants constitute a transport system for ingested exogenous lipids (exogenous pathway). Chylomicrons can enter the circulation via the lymphatic ducts. The chylomicrons are cleared from the circulation by the action of **lipoprotein lipase,** which is located on the surface of the endothelium of the capillaries. The enzyme catalyzes the breakdown of the triglyceride in the chylomicrons to FFA and glycerol, which then enter adipose cells and are reesterified. Alternatively, the FFA can remain in the circulation bound to albumin. Lipoprotein lipase, which requires heparin as a cofactor, also removes triglycerides from circulating **very low-density lipoproteins (VLDL).** Chylomicrons depleted of their triglyceride remain in the circulation as cholesterol-rich lipoproteins called **chylomicron remnants,** which are 30–80 nm in diameter. The remnants are carried to the liver, where they are internalized and degraded.

The endogenous system, made up of VLDL, **intermediate-density lipoproteins (IDL), low-density lipoproteins (LDL),** and **high-density lipoproteins (HDL),** also transports triglycerides and cholesterol throughout the

body. VLDL are formed in the liver and transport triglycerides formed from fatty acids and carbohydrates in the liver to extrahepatic tissues. After their triglyceride is largely removed by the action of lipoprotein lipase, they become IDL. The IDL give up phospholipids and, through the action of the plasma enzyme **lecithin-cholesterol acyltransferase (LCAT),** pick up cholesteryl esters formed from cholesterol in the HDL. Some IDL are taken up by the liver. The remaining IDL then lose more triglyceride and protein, probably in the sinusoids of the liver, and become LDL. LDL provide cholesterol to the tissues. The cholesterol is an essential constituent in cell membranes and is used by gland cells to make steroid hormones.

FREE FATTY ACID METABOLISM

In addition to the exogenous and endogenous pathways described above, FFA are also synthesized in the fat depots in which they are stored. They can circulate as lipoproteins bound to albumin and are a major source of energy for many organs. They are used extensively in the heart, but probably all tissues can oxidize FFA to CO_2 and H_2O.

The supply of FFA to the tissues is regulated by two lipases. As noted above, lipoprotein lipase on the surface of the endothelium of the capillaries hydrolyzes the triglycerides in chylomicrons and VLDL, providing FFA and glycerol, which are reassembled into new triglycerides in the fat cells. The intracellular **hormone-sensitive lipase** of adipose tissue catalyzes the breakdown of stored triglycerides into glycerol and fatty acids, with the latter entering the circulation. Hormone-sensitive lipase is increased by fasting and stress and decreased by feeding and insulin. Conversely, feeding increases and fasting and stress decrease the activity of lipoprotein lipase.

CHOLESTEROL METABOLISM

Cholesterol is the precursor of the steroid hormones and bile acids and is an essential constituent of cell membranes. It is found only in animals. Related sterols occur in plants, but plant sterols are poorly absorbed from the gastrointestinal tract. Most of the dietary cholesterol is contained in egg yolks and animal fat.

Cholesterol is absorbed from the intestine and incorporated into the chylomicrons formed in the intestinal mucosa. After the chylomicrons discharge

their triglyceride in adipose tissue, the chylomicron remnants bring cholesterol to the liver. The liver and other tissues also synthesize cholesterol. Some of the cholesterol in the liver is excreted in the bile, both in the free form and as bile acids. Some of the biliary cholesterol is reabsorbed from the intestine. Most of the cholesterol in the liver is incorporated into VLDL and circulates in lipoprotein complexes.

The biosynthesis of cholesterol from acetate is summarized in **Figure 1–27**. Cholesterol feeds back to inhibit its own synthesis by inhibiting **HMG-CoA reductase,** the enzyme that converts 3-hydroxy-3-methylglutaryl-coenzyme A (HMG-CoA) to mevalonic acid. Thus, when dietary cholesterol intake is high, hepatic cholesterol synthesis is decreased, and vice versa. However, the feedback compensation is incomplete, because a diet that is low in cholesterol and saturated fat leads to only a modest decline in circulating plasma cholesterol. The most effective and most commonly used cholesterol-lowering drugs are lovastatin and other **statins,** which reduce cholesterol synthesis by inhibiting HMG-CoA. The relationship between cholesterol and vascular disease is discussed in **Clinical Box 1–7**.

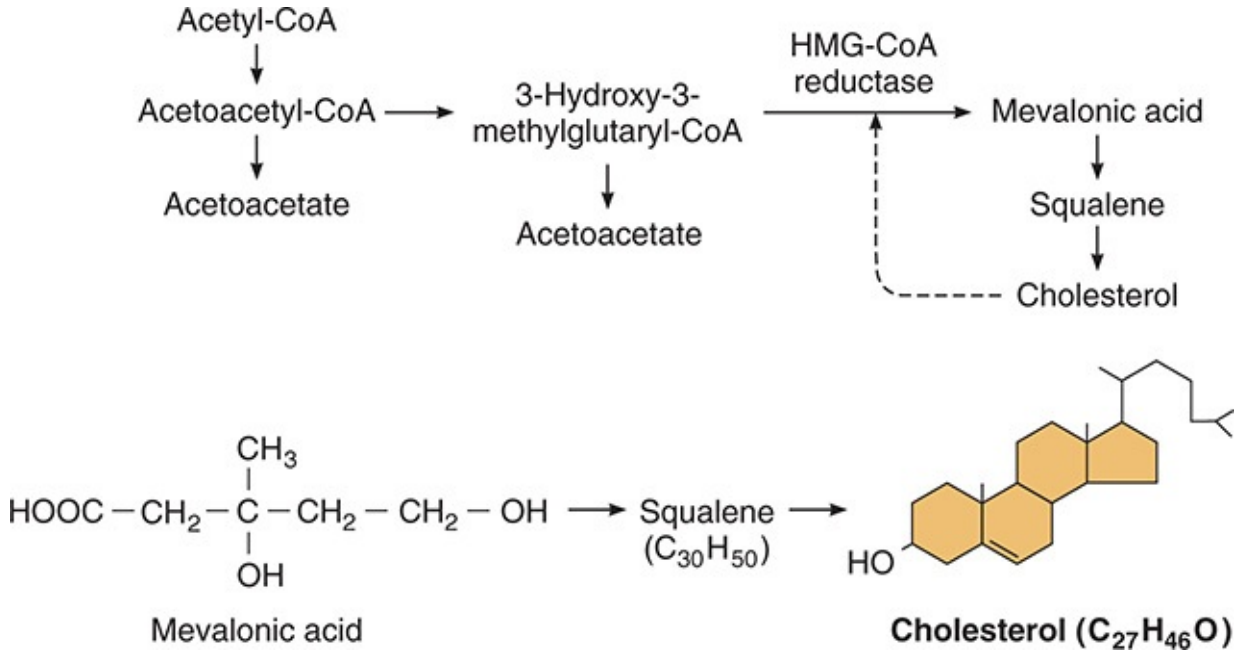


FIGURE 1–27 Biosynthesis of cholesterol. Six mevalonic acid molecules condense to form squalene, which is then hydroxylated to cholesterol. The dashed arrow indicates feedback inhibition by cholesterol of HMG-CoA reductase, the enzyme that catalyzes mevalonic acid formation.

ESSENTIAL FATTY ACIDS

Animals fed a fat-free diet fail to grow, develop skin and kidney lesions, and become infertile. Adding linolenic, linoleic, and arachidonic acids to the diet cures all the deficiency symptoms. These three acids are polyunsaturated fatty acids and because of their action are called **essential fatty acids.** Similar deficiency symptoms have not been unequivocally demonstrated in humans, but there is reason to believe that some unsaturated fats are essential dietary constituents, especially for children. Dehydrogenation of fats is known to occur in the body, but there does not appear to be any synthesis of carbon chains with the arrangement of double bonds found in the essential fatty acids.

EICOSANOIDS

One of the reasons that essential fatty acids are necessary for health is that they are the precursors of prostaglandins (including prostacyclin, thromboxanes), lipoxins, leukotrienes, and related compounds. These substances are called **eicosanoids,** reflecting their origin from the 20-carbon (eicosa-) polyunsaturated fatty acid **arachidonic acid (arachidonate)** and the 20-carbon derivatives of linoleic and linolenic acids.

The **prostaglandins** are a series of 20-carbon unsaturated fatty acids containing a cyclopentane ring. They were first isolated from semen but are synthesized in most and possibly in all organs in the body. Prostaglandin H_2 (PGH_2) is the precursor for various other prostaglandins, thromboxanes, and prostacyclin. Arachidonic acid is formed from tissue phospholipids by **phospholipase A_2**. It is converted to prostaglandin H_2 (PGH_2) by **prostaglandin G/H synthases** 1 and 2. These are bifunctional enzymes that have both cyclooxygenase and peroxidase activity, but they are more commonly known by the names cyclooxygenase 1 **(COX1)** and cyclooxygenase 2 **(COX2).** Their structures are very similar, but COX1 is constitutive whereas COX2 is induced by growth factors, cytokines, and tumor promoters. PGH_2 is converted to prostacyclin, thromboxanes, and other prostaglandins (eg, PGE_2, $PGF_{2\alpha}$, PGD_2) by various tissue isomerases. The effects of prostaglandins are multitudinous and varied. They are particularly important in the female reproductive cycle, in parturition, in the cardiovascular system, in inflammatory responses, and in the causation of pain. Drugs that target production of prostaglandins are among the most common over-the-counter drugs available **(Clinical Box 1–8)**.

CLINICAL BOX 1–7

Cholesterol & Atherosclerosis

The interest in cholesterol-lowering drugs stems from the role of cholesterol in the development and progression of **atherosclerosis**. This extremely widespread disease predisposes to myocardial infarction, cerebral thrombosis, ischemic gangrene of the extremities, and other serious illnesses. It is characterized by infiltration of cholesterol and oxidized cholesterol into macrophages, converting them into foam cells in lesions of the arterial walls. This is followed by a complex sequence of changes involving platelets, macrophages, and smooth muscle cells, as well as growth factors and inflammatory mediators that produces proliferative lesions that eventually ulcerate and may calcify. The lesions distort the vessels and make them rigid. In individuals with elevated plasma cholesterol levels, the incidence of atherosclerosis and its complications is increased. The normal range for plasma cholesterol is said to be 120–200 mg/dL, but in men, there is a clear, tight, positive correlation between the death rate from ischemic heart disease and plasma cholesterol levels above 180 mg/dL. Furthermore, it is now clear that lowering plasma cholesterol by diet and drugs slows and may even reverse the progression of atherosclerotic lesions and the complications they cause.

In evaluating plasma cholesterol levels in relation to atherosclerosis, it is important to analyze the LDL and HDL levels as well. LDL delivers cholesterol to peripheral tissues, including atheromatous lesions, and the LDL plasma concentration correlates positively with myocardial infarctions and ischemic strokes. On the other hand, HDL picks up cholesterol from peripheral tissues and transports it to the liver, thus lowering plasma cholesterol. It is interesting that women, who have a lower incidence of myocardial infarction than men, have higher HDL levels. In addition, HDL levels are increased in individuals who exercise and those who drink one or two alcoholic drinks per day, whereas they are decreased in individuals who smoke, are obese, or live sedentary lives. Moderate drinking decreases the incidence of myocardial infarction, and obesity and smoking are risk factors that increase it. Plasma cholesterol and the incidence of cardiovascular diseases are increased in **familial hypercholesterolemia**, due to various loss-of-function mutations in the genes for LDL receptors.

THERAPEUTIC HIGHLIGHTS

Although atherosclerosis is a progressive disease, it is also preventable in many cases by limiting risk factors, including lowering “bad” cholesterol through a healthy diet and exercise. Drug treatments for high cholesterol, including the statins among others, provide additional relief that can complement a healthy diet and exercise. If atherosclerosis is advanced, invasive techniques, such as angioplasty and stenting, can be used to unblock arteries.

CLINICAL BOX 1–8

Pharmacology of Prostaglandins

Because prostaglandins play a prominent role in the genesis of pain, inflammation, and fever, pharmacologists have long sought drugs to inhibit their synthesis. Glucocorticoids inhibit phospholipase A_2 and thus inhibit the formation of all eicosanoids. A variety of NSAIDs inhibit both cyclooxygenases, inhibiting the production of PGH_2 and its derivatives. Aspirin is the best-known of these, but ibuprofen, indomethacin, and others are also used. However, there is evidence that prostaglandins synthesized by COX2 are more involved in the production of pain and inflammation, and prostaglandins synthesized by COX1 are more involved in protecting the gastrointestinal mucosa from ulceration. Several novel NSAIDs have been introduced in an attempt to specifically target COX enzymes. However, in many cases, significant side effects, including increased incidence of stroke and heart attack, have led to drug withdrawals from the market. More research is underway to better understand all the effects of the COX enzymes, their products, and their inhibitors. Some prosglandins can however be used as drugs for cardiovascular disease based on the vasodilative and antiproliferative effects. For example, intravenous perfusion of prostacyclin is an effective treatment for idiopathic pulmonary arterial hypertension, a progressive and fatal disease that predominantly affects young women.

Arachidonic acid also serves as a substrate for the production of several physiologically important **leukotrienes** and **lipoxins.** The leukotrienes, thromboxanes, lipoxins, and prostaglandins have been called local hormones.

They have short half-lives and are inactivated in many different tissues. They undoubtedly act mainly in the tissues at sites in which they are produced. The leukotrienes are mediators of allergic responses and inflammation. Their release is provoked when specific allergens combine with IgE antibodies on the surfaces of mast cells (see Chapter 3). They produce bronchoconstriction, constrict arterioles, increase vascular permeability, and attract neutrophils and eosinophils to inflammatory sites. Diseases in which they may be involved include asthma, psoriasis, acute respiratory distress syndrome, allergic rhinitis, rheumatoid arthritis, Crohn disease, and ulcerative colitis.

CHAPTER SUMMARY

- Cells contain approximately two-thirds of the body fluids, while the remaining extracellular fluid is found between cells (interstitial fluid) or in the circulating lymph and blood plasma.
- The number of molecules, electrical charges, and particles of substances in solution are important in physiology.
- Biological buffers including bicarbonate, proteins, and phosphates can bind or release protons in solution to help maintain pH. Biological buffering capacity of a weak acid or base is greatest when pK_a = pH.
- Although the osmolality of solutions can be similar across a plasma membrane, the distribution of individual molecules and distribution of charge across the plasma membrane can be quite different. The separation of concentrations of charged species sets up an electrical gradient at the plasma membrane (inside negative). The electrochemical gradient is in large part maintained by the Na, K ATPase. These are affected by the Gibbs–Donnan equilibrium and can be calculated using the Nernst equation.
- Cellular energy can be stored in high-energy or energy-rich phosphate compounds, including adenosine triphosphate (ATP). Coordinated oxidation–reduction reactions allow for the production of a proton gradient at the inner mitochondrial membrane that ultimately yields to the production of ATP in the cell.
- Nucleotides made from purine or pyrimidine bases linked to ribose or 2-deoxyribose sugars with inorganic phosphates are the basic building blocks for nucleic acids, DNA, and RNA. The fundamental unit of DNA is the gene, which encodes information to make proteins in the cell. Genes are transcribed into messenger RNA, and with the help of ribosomal RNA and

transfer RNAs, translated into proteins.

- Amino acids are the basic building blocks for proteins in the cell and can also serve as sources for several biologically active molecules. Translation is the process of protein synthesis. After synthesis, proteins can undergo a variety of posttranslational modifications prior to obtaining their full function in the cell.
- Carbohydrates are organic molecules that contain equal amounts of carbon (C) and H_2O. Carbohydrates can be attached to proteins (glycoproteins) or fatty acids (glycolipids) and are critically important for the production and storage of cellular and body energy. The breakdown of glucose to generate energy, or glycolysis, can occur in the presence or absence of O_2 (aerobically or anaerobically). The net production of ATP during aerobic glycolysis is 19 times higher than anaerobic glycolysis.
- Fatty acids are carboxylic acids with extended hydrocarbon chains. They are an important energy source for cells and fatty acid derivatives—including triglycerides, phospholipids, and sterols—have additional important cellular applications.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. The membrane potential of a particular cell is at the K^+ equilibrium. The intracellular concentration for K^+ is at 150 mmol/L and the extracellular concentration for K^+ is at 5.5 mmol/L. What is the resting potential?

   A. −70 mV

   B. −90 mV

   C. +70 mV

   D. +90 mV

2. The difference in concentration of H^+ in a solution of pH 2.0 compared with one of pH 7.0 is

   A. 5-fold

   B. 1/5 as much

   C. 10^5-fold

   D. 10^{-5} as much

3. Transcription refers to
   A. the process where an mRNA is used as a template for protein production.
   B. the process where a DNA sequence is copied into RNA for the purpose of gene expression.
   C. the process where DNA wraps around histones to form a nucleosome.
   D. the process of replication of DNA prior to cell division.
4. The primary structure of a protein refers to
   A. the twist, folds, or twist and folds of the amino acid sequence into stabilized structures within the protein (ie, α-helices and β-sheets).
   B. the arrangement of subunits to form a functional structure.
   C. the amino acid sequence of the protein.
   D. the arrangement of twisted chains and folds within a protein into a stable structure.
5. Fill in the blanks: Glycogen is a storage form of glucose. _____ refers to the process of making glycogen and _____ refers to the process of breakdown of glycogen.
   A. Glycogenolysis, glycogenesis
   B. Glycolysis, glycogenolysis
   C. Glycogenesis, glycogenolysis
   D. Glycogenolysis, glycolysis
6. The major lipoprotein source of the cholesterol used in cells is
   A. chylomicrons.
   B. intermediate-density lipoproteins (IDL).
   C. albumin-bound free fatty acids.
   D. low-density lipoproteins (LDL).
   E. high-density lipoproteins (HDL).
7. Which of the following produces the most high-energy phosphate compounds?
   A. Aerobic metabolism of 1 mol of glucose
   B. Anaerobic metabolism of 1 mol of glucose
   C. Metabolism of 1 mol of galactose
   D. Metabolism of 1 mol of amino acid
   E. Metabolism of 1 mol of long-chain fatty acid
8. When LDL enters cells by receptor-mediated endocytosis, which of the

following does not occur?

A. Decrease in the formation of cholesterol from mevalonic acid
B. Increase in the intracellular concentration of cholesteryl esters
C. Increase in the transfer of cholesterol from the cell to HDL
D. Decrease in the rate of synthesis of LDL receptors
E. Decrease in the cholesterol in endosomes

CHAPTER 2

Overview of Cellular Physiology

OBJECTIVES

After studying this chapter, you should be able to:

- Name the prominent cellular organelles and state their functions in cells.
- Name the building blocks of the cellular cytoskeleton and state their contributions to cell structure and function.
- Name the intercellular connections and intracellular to extracellular connections.
- Define the processes of exocytosis and endocytosis, and describe the contribution of each to normal cell function.
- Define proteins that contribute to membrane permeability and transport.
- Recognize various forms of intercellular communication and describe ways in which chemical messengers (including second messengers) affect cellular functions.

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INTRODUCTION

The cell is the fundamental working unit of all organisms. In humans, cells can be highly specialized in both structure and function; alternatively, cells from

different organs can share features and function. In the previous chapter, some basic principles of biophysics and the catabolism and metabolism of building blocks found in the cell were examined. In some of those discussions, how the building blocks could contribute to basic cell function (eg, DNA replication, transcription, and translation) were discussed. In this chapter, more of the fundamental aspects of cellular and molecular physiology will be reviewed. Additional aspects that concern specialization of cellular and molecular function are considered in the next chapters concerning immune function and excitable cells and within the sections that highlight each physiological system.

FUNCTIONAL MORPHOLOGY OF THE CELL AND HOMEOSTASIS

The actual environment of the cells of the body is the interstitial component of the extracellular fluid (ECF). Because normal cell function depends on the constancy of this fluid, it is not surprising that in multicellular animals, an immense number of regulatory mechanisms have evolved to maintain it. To describe “the various physiologic arrangements which serve to restore the normal state, once it has been disturbed,” W.B. Cannon coined the term **homeostasis.** The buffering properties of the body fluids and the renal and respiratory adjustments to the presence of excess acid or alkali are examples of homeostatic mechanisms. There are countless other examples, and a large part of physiology is concerned with regulatory mechanisms that act to maintain the constancy of the internal environment. Many of these regulatory mechanisms operate on the principle of negative feedback; deviations from a given normal set point are detected by a sensor, and signals from the sensor trigger compensatory changes that continue until the set point is again reached.

A basic knowledge of cell function and structure is essential to an understanding of the homeostasis, the organ systems and the way they function in the body. A key tool for examining cellular constituents is the microscope. A light microscope can resolve structures as close as 0.2 μm, while an electron microscope can resolve structures as close as 0.002 μm. Although cell dimensions are quite variable, this resolution can provide a good look at the inner workings of the cell. The advent of common access to phase contrast, fluorescent, confocal, and many other microscopy techniques along with specialized probes for both static and dynamic cellular structures further expanded the examination of cell structure and function. Equally revolutionary advances in modern biophysical, biochemical, and molecular biological

techniques have also greatly contributed to our knowledge of the cell.

The specialization of the cells in the various organs is considerable, and no cell can be called “typical” of all cells in the body. However, a number of cell structures **(organelles)** are common to most cells. These structures are shown in **Figure 2–1**. Many of them can be isolated by ultracentrifugation combined with other techniques. When cells are homogenized and the resulting suspension is centrifuged, the nuclei sediment first, followed by the mitochondria. High-speed centrifugation that generates forces of 100,000 times gravity or more causes a fraction made up of granules called the **microsomes** to sediment. This fraction includes organelles such as the **ribosomes** and **peroxisomes.**

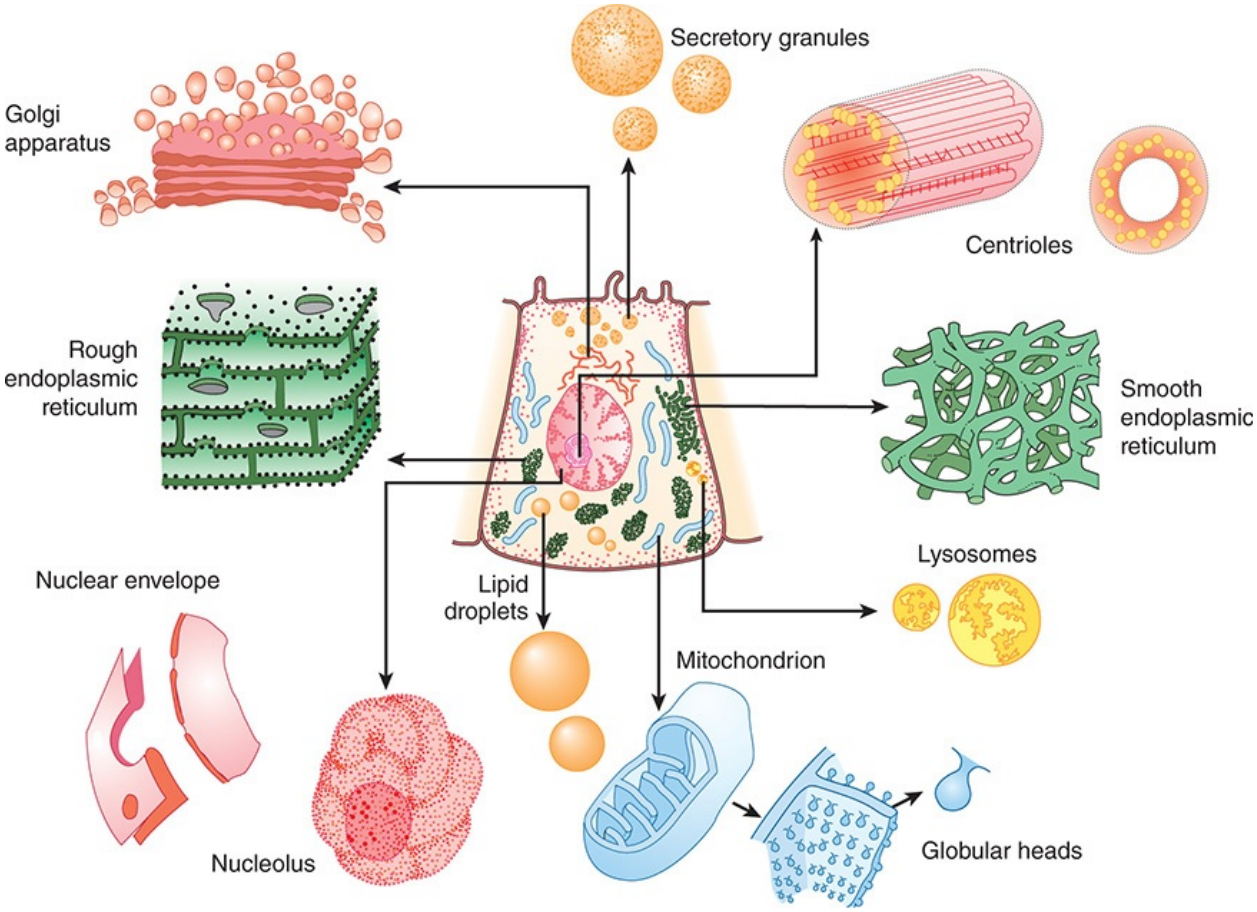


FIGURE 2–1 Cross-sectional diagram of a hypothetical cell as seen with the light microscope. Individual organelles are expanded for closer examination. (Adapted with permission from Bloom and Fawcett. Reproduced with permission from Junqueira LC, Carneiro J, Kelley RO: *Basic Histology*, 9th ed. New York, NY: McGraw-Hill; 1998.)

CELL MEMBRANES

The membrane that surrounds the cell is a remarkable structure. It is made up of lipids and proteins and is semipermeable, allowing some substances to pass through it and excluding others. However, its permeability can also be varied because it contains numerous regulated ion channels and other transport proteins that can change the amounts of substances moving across it. It is generally referred to as the **plasma membrane.** The nucleus and other organelles in the cell are bound by similar membranous structures.

Although the chemical structures of membranes and their properties vary considerably from one location to another, they have certain common features. They are generally about 7.5 nm (75 angstroms [Å]) thick. The major lipids are phospholipids such as phosphatidylcholine, phosphatidylserine, and phosphatidylethanolamine. The shape of the phospholipid molecule reflects its solubility properties: the "head" end of the molecule contains the phosphate portion and is relatively soluble in water (polar, **hydrophilic**) and the "tail" ends are relatively insoluble (nonpolar, **hydrophobic**). The possession of both hydrophilic and hydrophobic properties makes the lipid an **amphipathic** molecule. In the membrane, the hydrophilic ends of the molecules are exposed to the aqueous environment that bathes the exterior of the cells and the aqueous cytoplasm; the hydrophobic ends meet in the water-poor interior of the membrane **(Figure 2–2)**. In **prokaryotes** (ie, bacteria in which there is no nucleus), the membranes are relatively simple, but in **eukaryotes** (ie, cells containing nuclei), cell membranes contain various glycosphingolipids, sphingomyelin, and cholesterol in addition to phospholipids and phosphatidylcholine.

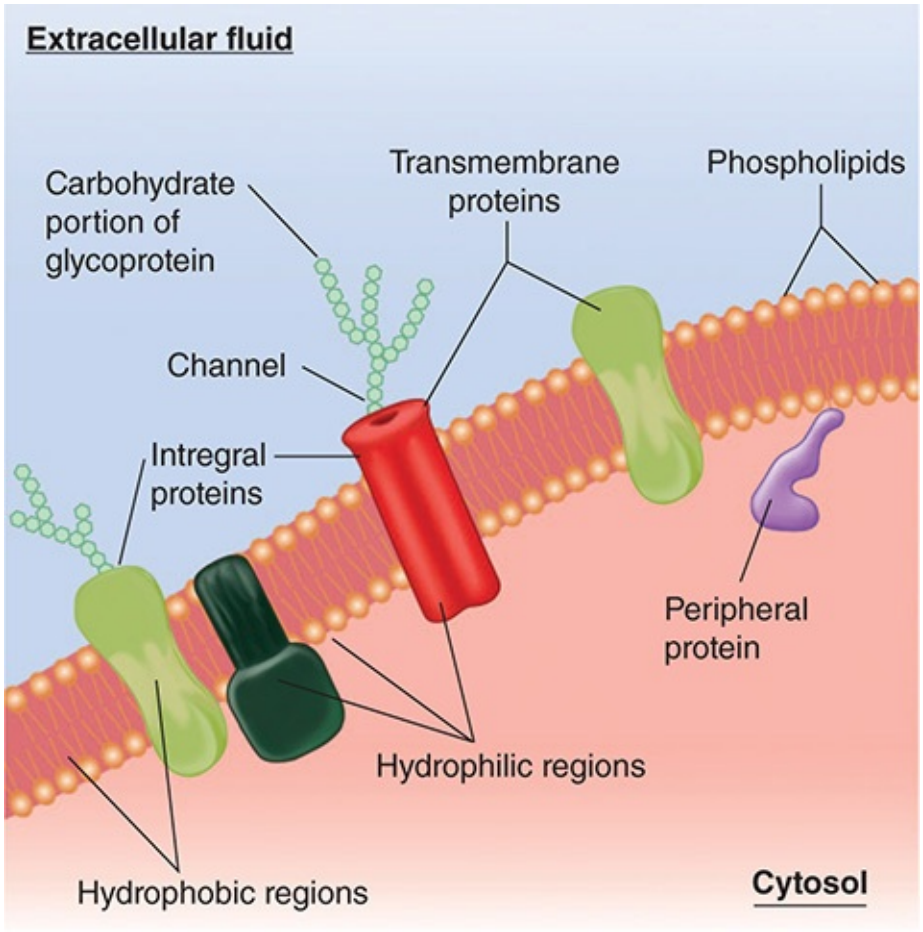


FIGURE 2–2 Organization of the phospholipid bilayer and associated proteins in a biologic membrane. The phospholipid molecules that make up the membrane each have two hydrophobic fatty acid chains attached to a hydrophilic phosphate head. Individual proteins take on different shapes and positions in the cell. Many are integral proteins, extending into the membrane or peripheral proteins that are attached to the inside or outside (not shown) of the membrane. Proteins can be modified (eg, with carbohydrate chains). Many specific protein attachments and cholesterol that are commonly found in the bilayer are omitted for clarity.

Many different proteins are embedded in the membrane. The membrane proteins are classified into two categories: **integral proteins** or intrinsic membrane proteins and **peripheral proteins** or extrinsic membrane proteins. They exist as separate globular units and many pass through or are embedded in one leaflet of the membrane (eg, integral proteins), whereas peripheral proteins are associated with the inside or outside of the membrane (Figure 2–2). The amount of protein varies significantly with the function of the cell but makes up

on average 50% of the mass of the membrane; that is, there is about one protein molecule per 50 of the much smaller phospholipid molecules. The proteins in the membrane carry out many functions. Some are **cell adhesion molecules** (CAMs) that anchor cells to their neighbors or to basal laminas. Some proteins function as **pumps,** actively transporting ions across the membrane. Other proteins function as **carriers,** transporting substances down electrochemical gradients by facilitated diffusion. Still others are **ion channels,** which, when activated, permit the passage of ions into or out of the cell. The role of the pumps, carriers, and ion channels in transport across the cell membrane is discussed below. Proteins in another group function as **receptors** that bind **ligands** or messenger molecules, initiating physiological changes inside the cell. Proteins also function as **enzymes,** catalyzing reactions at the surfaces of the membrane. Examples from each of these groups are discussed later in this chapter.

The hydrophobic portions of the proteins are usually located in the interior of the membrane, whereas the charged, hydrophilic portions are located on the surfaces. Peripheral proteins are attached to the surfaces of the membrane in various ways. One common way is attachment to glycosylated forms of phosphatidylinositol. Proteins held by these **glycosylphosphatidylinositol (GPI) anchors** **(Figure 2–3)** include enzymes such as alkaline phosphatase, various antigens, a number of CAMs, and proteins that combat cell lysis by complement. Over 250 GPI-linked cell surface proteins have now been described in humans. Other proteins are **lipidated,** that is, they have specific lipids attached to them (Figure 2–3). Proteins may be **myristoylated, palmitoylated,** or **prenylated** (ie, attached to geranylgeranyl or farnesyl groups).

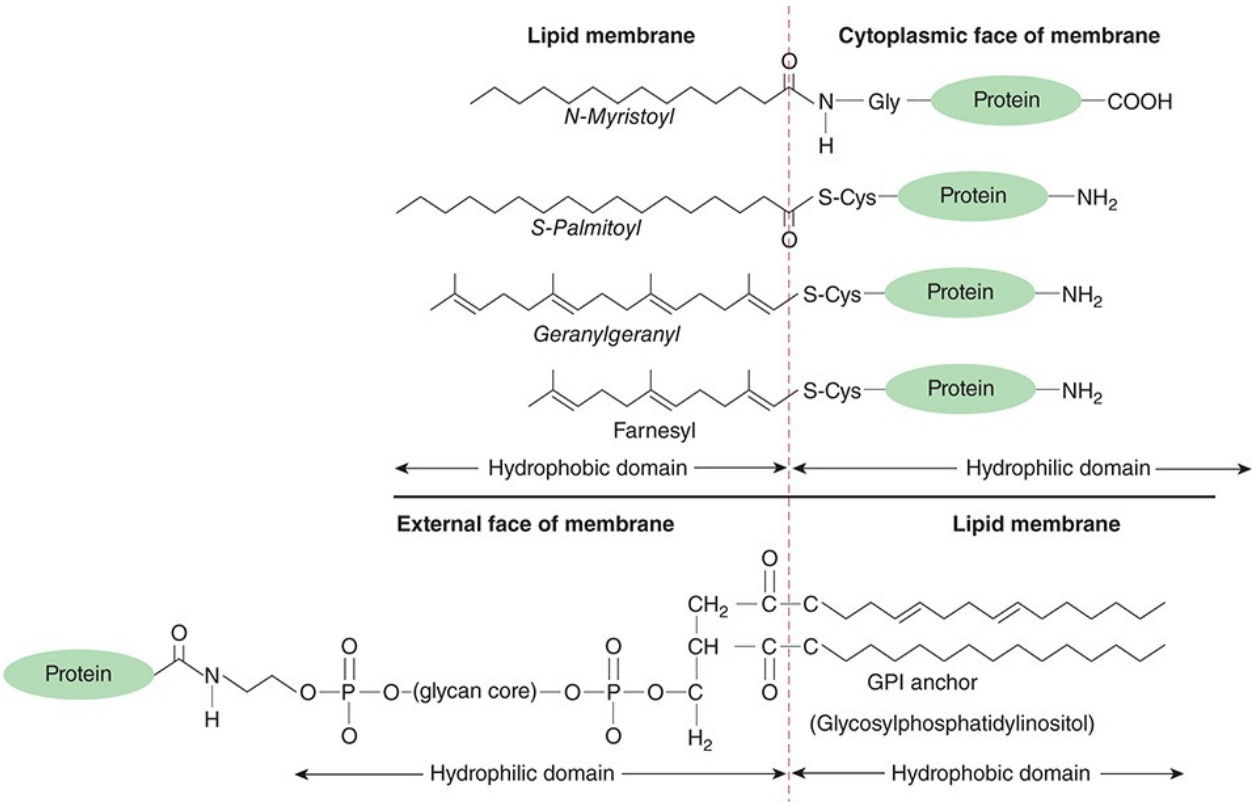


FIGURE 2–3 Protein linkages to membrane lipids. A variety of lipid modifications can occur at amino or carboxy terminals of proteins attached to cytosolic side of the plasma membrane. Many proteins associated with the external side of the plasma membrane can be attached via glycosylated forms of phosphatidylinositol (eg, GPI anchors).

The protein structure—and particularly the enzyme content—of biologic membranes varies not only from cell to cell, but also within the same cell. For example, some of the enzymes embedded in cell membranes are different from those in mitochondrial membranes. In epithelial cells, the enzymes in the cell membrane on the mucosal surface differ from those in the cell membrane on the basal and lateral margins of the cells; that is, the cells are **polarized.** Such polarization makes directional transport across epithelia possible. The membranes are dynamic structures, and their constituents are being constantly renewed at different rates. Some proteins are anchored to the cytoskeleton, but others move laterally in the membrane.

Underlying most cells is a thin, “fuzzy” layer plus some fibrils that collectively make up the **basement membrane** or, more properly, the **basal lamina.** The basal lamina and, more generally, the extracellular matrix are made up of many proteins that hold cells together, regulate their development, and

determine their growth. These include collagens, laminins, fibronectin, tenascin, and various proteoglycans.

MITOCHONDRIA

Over a billion years ago, aerobic bacteria were engulfed by eukaryotic cells and evolved into **mitochondria,** providing the eukaryotic cells with the ability to form the energy-rich compound ATP by **oxidative phosphorylation.** Mitochondria perform other functions, including a role in the regulation of **apoptosis** (programmed cell death), but oxidative phosphorylation is the most crucial. Each eukaryotic cell can have hundreds to thousands of mitochondria. In mammals, they are generally depicted as sausage-shaped organelles (Figure 2–1), but their shape can be quite dynamic. Each has an outer membrane, an intermembrane space, an inner membrane, which is folded to form shelves **(cristae),** and a central matrix space. The enzyme complexes responsible for oxidative phosphorylation are lined up on the cristae **(Figure 2–4)**.

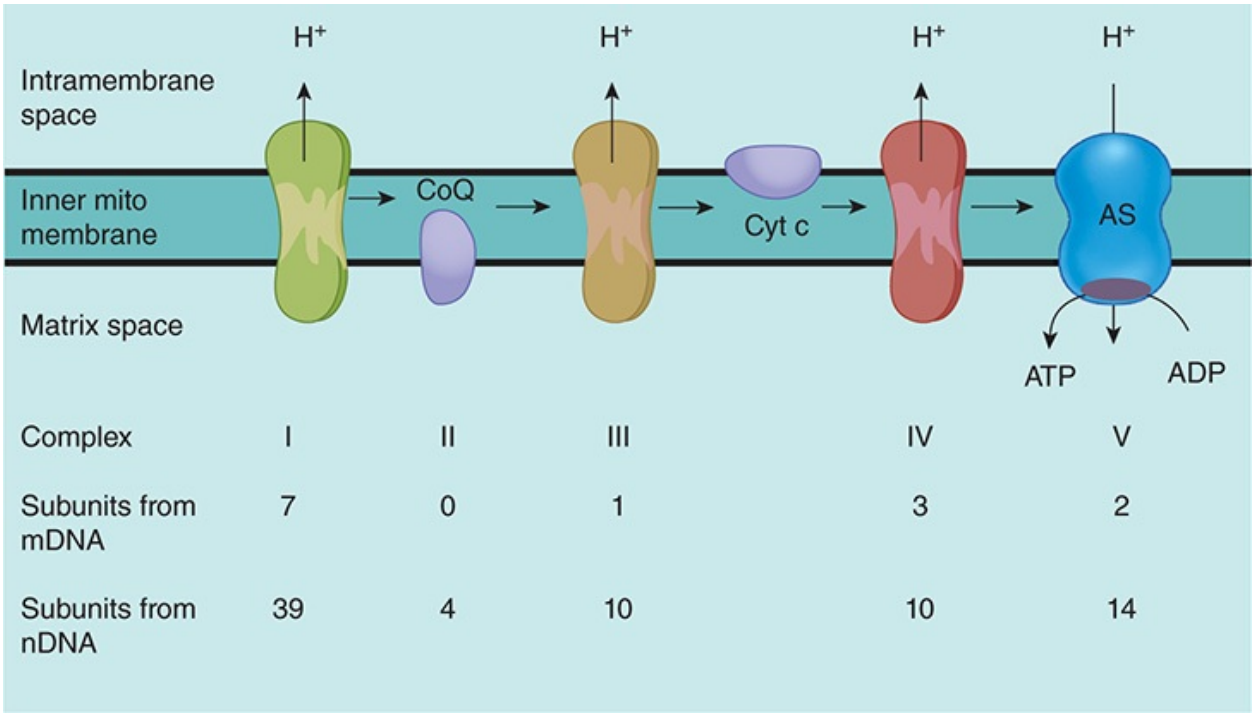


FIGURE 2–4 Components involved in oxidative phosphorylation in mitochondria and their origins. As enzyme complexes I through IV convert 2-carbon metabolic fragments to CO_2 and H_2O, protons (H^+) are pumped into the intermembrane space. The protons diffuse back to the matrix space via complex

V, ATP synthase (AS), in which ADP is converted to ATP. The enzyme complexes are made up of subunits coded by mitochondrial DNA (mDNA) and nuclear DNA (nDNA), and the figures document the contribution of each DNA to the complexes.

Consistent with their origin from aerobic bacteria, the mitochondria have their own genome. There is much less DNA in the mitochondrial genome than in the nuclear genome, and 99% of the proteins in the mitochondria are the products of nuclear genes, but mitochondrial DNA is responsible for certain key components of the pathway for oxidative phosphorylation. Specifically, human mitochondrial DNA is a double-stranded circular molecule containing approximately 16,500 base pairs (compared with over a billion in nuclear DNA). It codes for 13 protein subunits that are associated with proteins encoded by nuclear genes to form four enzyme complexes plus two ribosomal and 22 transfer RNAs that are needed for protein production by the intramitochondrial ribosomes.

CLINICAL BOX 2–1

Mitochondrial Diseases

Mitochondrial diseases encompass at least 40 diverse disorders that are grouped because of their links to mitochondrial failure. These diseases can occur following inheritance or spontaneous mutations in mitochondrial or nuclear DNA that lead to altered functions of the mitochondrial proteins (or RNA). Depending on the target cell and/or tissues affected, symptoms resulting from mitochondrial diseases may include altered motor control; altered muscle output; gastrointestinal dysfunction; altered growth; diabetes; seizures; visual/hearing problems; lactic acidosis; developmental delays; and susceptibility to infection or cardiac, liver, and respiratory disease. Although there is evidence for tissue-specific isoforms of mitochondrial proteins, mutations in these proteins do not fully explain the highly variable patterns or targeted organ systems observed with mitochondrial diseases.

THERAPEUTIC HIGHLIGHTS

With the diversity of disease types and the overall importance of mitochondria in energy production, it is not surprising that there is no single cure for mitochondrial diseases and focus remains on treating the symptoms when possible. For example, in some mitochondrial myopathies (ie, mitochondrial

diseases associated with neuromuscular function), physical therapy may help extend the range of movement of muscles and improve dexterity.

The enzyme complexes responsible for oxidative phosphorylation illustrate the interactions between the products of the mitochondrial genome and the nuclear genome. For example, complex I, reduced nicotinamide adenine dinucleotide dehydrogenase (NADH), is made up of seven protein subunits coded by mitochondrial DNA and 39 subunits coded by nuclear DNA. The origin of the subunits in the other complexes is shown in Figure 2–4. Complex II, succinate dehydrogenase- ubiquinone oxidoreductase; complex III, ubiquinonecytochrome c oxidoreductase; and complex IV, cytochrome c oxidase, act with complex I, coenzyme Q, and cytochrome c to convert metabolites to CO_2 and H_2O. Complexes I, III, and IV pump protons (H^+) into the intermembrane space during this electron transfer. The protons then flow down their electrochemical gradient through complex V, ATP synthase, which harnesses this energy to generate ATP.

As zygote mitochondria are derived from the ovum, their inheritance is maternal. This maternal inheritance has been used as a tool to track evolutionary descent. Mitochondria have an ineffective DNA repair system, and the mutation rate for mitochondrial DNA is over 10 times the rate for nuclear DNA. A large number of relatively rare diseases have been traced to mutations in mitochondrial DNA. These include disorders of tissues with high metabolic rates in which energy production is defective as a result of abnormalities in the production of ATP, as well as other disorders **(Clinical Box 2–1)**.

LYSOSOMES

In the cytoplasm of the cell there are large, somewhat irregular structures surrounded by membranes (Figure 2–1). The interior of these structures, which are called **lysosomes,** is more acidic than the rest of the cytoplasm, and external material such as endocytosed bacteria, as well as worn-out cell components, are digested in them. The interior is kept acidic by the action of a **proton pump,** or **H^+ ATPase.** This integral membrane protein uses the energy of ATP to move protons from the cytosol up their electrochemical gradient and keep the lysosome relatively acidic, near pH 5.0. Lysosomes can contain over 40 types of hydrolytic enzymes, some of which are listed in **Table 2–1**. Not surprisingly,

these enzymes are all acid hydrolases, in that they function best at the acidic pH of the lysosomal compartment. This can be a safety feature for the cell; if the lysosomes were to break open and release their contents, the enzymes would not be efficient at the near neutral cytosolic pH 7.2, and thus would be unable to digest cytosolic targets they may encounter. Diseases associated with lysosomal dysfunction are discussed in **Clinical Box 2–2**.

TABLE 2–1 Some of the enzymes found in lysosomes and the cell components that are their substrates.

Enzyme	Substrate
Ribonuclease	RNA
Deoxyribonuclease	DNA
Phosphatase	Phosphate esters
Glycosidases	Complex carbohydrates: glycosides and polysaccharides
Arylsulfatases	Sulfate esters
Collagenase	Collagens
Cathepsins	Proteins

CLINICAL BOX 2–2

Lysosomal Diseases

When a lysosomal enzyme is congenitally absent, the lysosomes become engorged with the material the enzyme normally degrades. This eventually leads to one of the **lysosomal diseases** (also called lysosomal storage diseases). There are over 50 such diseases currently recognized. For example, Fabry disease is caused by a deficiency in α-galactosidase; Gaucher disease is caused by a deficiency in β-galactocerebrosidase; and Tay-Sachs disease, which causes mental retardation and blindness, is caused by the loss of hexosaminidase A, a lysosomal enzyme that catalyzes the biodegradation of gangliosides (fatty acid derivatives). Such individual lysosomal diseases are rare, but they are serious and can be fatal.

THERAPEUTIC HIGHLIGHTS

Since there are many different lysosomal disorders, treatments vary considerably and “cures” remain elusive for most of these diseases. Much of the care is focused on managing symptoms of each specific disorder. Enzyme replacement therapy has shown to be effective for certain lysosomal diseases, including Gaucher disease and Fabry disease. However, the long-term effectiveness and the tissue-specific effects of many of the enzyme replacement treatments have not yet been established. Recent alternative approaches include bone marrow or stem cell transplantation. Again, medical advances are still necessary to fully combat this group of diseases.

PEROXISOMES

Peroxisomes are 0.5 µm in diameter (Figure 2–1), are surrounded by a membrane, and contain enzymes that can either produce H_2O_2 **(oxidases)** or break it down **(catalases).** Proteins are directed to the peroxisome by a unique signal sequence with the help of protein chaperones, **peroxins.** The peroxisome membrane contains a number of peroxisome-specific proteins that are concerned with transport of substances into and out of the matrix of the peroxisome. The matrix contains more than 40 enzymes, which operate in concert with enzymes outside the peroxisome to catalyze a variety of anabolic and catabolic reactions (eg, breakdown of lipids). Peroxisomes can form by budding of the endoplasmic reticulum, or by division. A number of synthetic compounds were found to cause proliferation of peroxisomes by acting on receptors in the nuclei of cells. These **peroxisome proliferator-activated receptors (PPARs)** are members of the nuclear receptor superfamily. When activated, they bind to DNA, producing changes in the production of mRNAs. The known effects for PPARs are extensive and can affect most tissues and organs.

CYTOSKELETON

All cells have a **cytoskeleton,** a system of fibers that not only maintains the structure of the cell but also permits it to change shape and move. The cytoskeleton is made up primarily of **microtubules, intermediate filaments,** and **microfilaments (Figure 2–5)**, along with proteins that anchor them and tie

them together. In addition, proteins and organelles move along microtubules and microfilaments from one part of the cell to another, propelled by molecular motors.

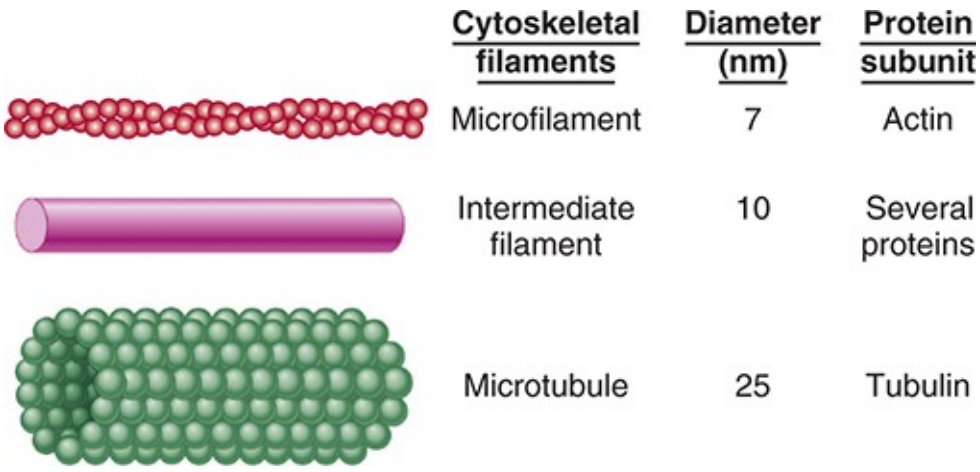


FIGURE 2–5 Cytoskeletal elements of the cell. Artistic impressions that depict the major cytoskeletal elements are shown on the left, with approximate diameters and protein subunits of these elements are listed for comparison. (Reproduced with permission from Widmaier EP, Raff H, Strang KT: *Vander's Human Physiology: The Mechanisms of Body Function*, 11th ed. New York, NY: McGraw-Hill; 2008.)

Microtubules (Figure 2–5 and **Figure 2–6**) are long, hollow structures with 5 nm walls surrounding a cavity 15 nm in diameter. They are made up of two globular protein subunits: α- and β-tubulin. A third subunit, γ-tubulin, is associated with the production of microtubules by the centrosomes. The α and β subunits form heterodimers, which aggregate to form long tubes made up of stacked rings, with each ring usually containing 13 subunits. The tubules interact with guanosine triphosphate (GTP) to facilitate their formation. Although microtubule subunits can be added to either end, microtubules are polar with assembly predominating at the plus ("+") end and disassembly predominating at the minus ("–") end. Both processes occur simultaneously in vitro. The growth of microtubules is temperature sensitive (disassembly is favored under cold conditions) as well as under the control of a variety of cellular factors that can directly interact with microtubules in the cell.

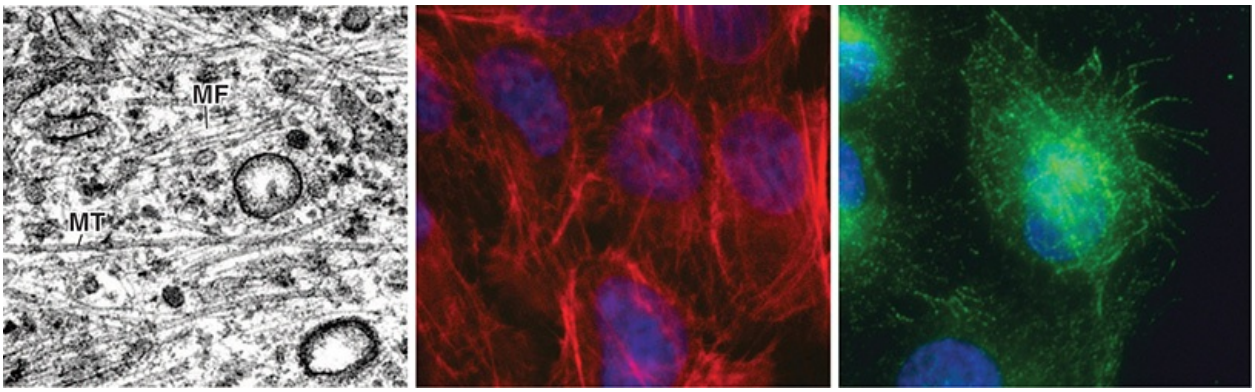


FIGURE 2–6 Microfilaments and microtubules. Electron micrograph **(Left)** of the cytoplasm of a fibroblast, displaying actin microfilaments (MF) and microtubules (MT). Fluorescent micrographs of airway epithelial cells displaying actin microfilaments stained with phalloidin **(Middle)** and microtubules visualized with an antibody to β-tubulin **(Right)**. Both fluorescent micrographs are counterstained with Hoechst dye (blue) to visualize nuclei. Note the distinct differences in cytoskeletal structure. (Electron 70 micrograph used with permission of E Katchburian; fluorescent micrographs used with permission of Stephanie Manberg.)

Because of their constant assembly and disassembly, microtubules are a dynamic portion of the cytoskeleton. They provide the tracks along which several different molecular motors move transport vesicles, organelles such as secretory granules, and mitochondria from one part of the cell to another. They also form the spindle, which moves the chromosomes in mitosis. Cargo can be transported in either direction on microtubules.

There are several drugs available that disrupt cellular function through interaction with microtubules. Microtubule assembly is prevented by colchicine and vinblastine. The anticancer drug **paclitaxel (Taxol)** binds to microtubules and makes them so stable that organelles cannot move. Mitotic spindles cannot form, and the cells die.

Intermediate filaments (Figure 2–5) are 8–14 nm in diameter and are made up of various subunits. Some of these filaments connect the nuclear membrane to the cell membrane. They form a flexible scaffolding for the cell and help it resist external pressure. In their absence, cells rupture more easily, and when they are abnormal in humans, blistering of the skin is common. The proteins that make up intermediate filaments are cell-type specific, and are thus frequently used as cellular markers. For example, vimentin is a major intermediate filament in fibroblasts, whereas cytokeratin is expressed in epithelial cells.

Microfilaments (Figures 2–5 and 2–6) are long solid fibers with a 5–9 nm diameter that are made up of **actin.** Although actin is most often associated with muscle contraction, it is present in all types of cells. It is the most abundant protein in mammalian cells, sometimes accounting for as much as 15% of the total protein in the cell. Its structure is highly conserved; for example, 88% of the amino acid sequences in yeast and rabbit actin are identical. Actin filaments polymerize and depolymerize in vivo, and it is not uncommon to find polymerization occurring at one end of the filament while depolymerization is occurring at the other end. **Filamentous (F) actin** refers to intact microfilaments and **globular (G) actin** refers to the unpolymerized protein actin subunits. F-actin fibers attach to various parts of the cytoskeleton and can interact directly or indirectly with membrane-bound proteins. They reach to the tips of the microvilli on the epithelial cells of the intestinal mucosa. They are also abundant in the lamellipodia that cells put out when they crawl along surfaces. The actin filaments interact with integrin receptors and form **focal adhesion complexes,** which serve as points of traction with the surface over which the cell pulls itself. In addition, some molecular motors use microfilaments as tracks.

MOLECULAR MOTORS

The molecular motors that move proteins, organelles, and other cell parts (collectively referred to as "cargo") to all parts of the cell are 100–500 kDa ATPases. They attach to their cargo at one end of the molecule and to microtubules or actin polymers with the other end, sometimes referred to as the "head." They convert the energy of ATP into movement along the cytoskeleton, taking their cargo with them. There are three super families of molecular motors: **kinesin, dynein,** and **myosin.** Examples of individual proteins from each superfamily are shown in **Figure 2–7**. It is important to note that there is extensive variation among superfamily members, allowing for the specialization of function (eg, choice of cargo, cytoskeletal filament type, and/or direction of movement).

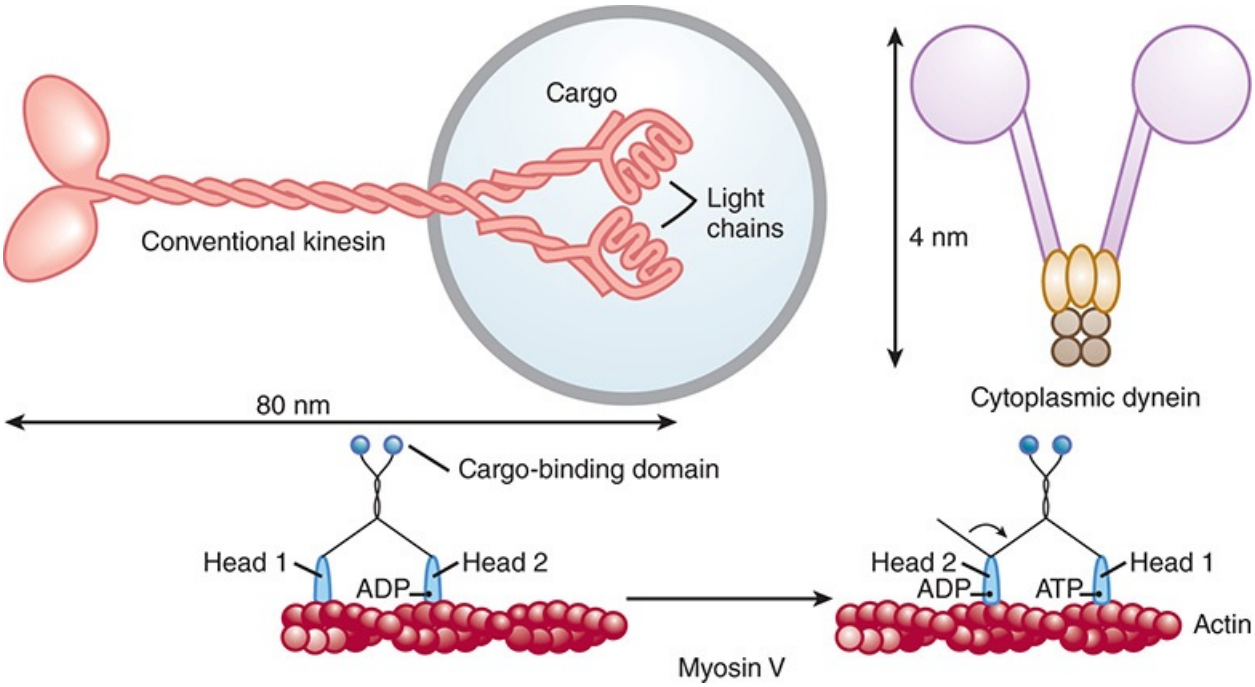


FIGURE 2–7 Examples of molecular motors. Conventional kinesin is shown attached to cargo, in this case a membrane-bound organelle (light blue). Cytoplasmic dynein is shown in isolation. Myosin V and its ability to "walk" along a microfilament are displayed in a two-part sequence. Note that the "heads" of each of the motors hydrolyze ATP and use the energy to produce motion.

The conventional form of **kinesin** is a double-headed molecule that tends to move its cargo toward the plus ("+") ends of microtubules. One head binds to the microtubule and then bends its neck while the other head swings forward and binds, producing almost continuous movement. Some kinesins are associated with mitosis and meiosis. Other kinesins perform different functions, including, in some instances, moving cargo to the minus ("–") end of microtubules. **Dyneins** have two heads, with their neck pieces embedded in a complex of proteins. **Cytoplasmic dyneins** have a function like that of conventional kinesin, except they tend to move particles and membranes to the minus (–) end of the microtubules. The multiple forms of **myosin** in the body are divided into 18 classes. The heads of myosin molecules bind to actin and produce motion by bending their neck regions (myosin II) or walking along microfilaments, one head after the other (myosin V). In these ways, they perform functions as diverse as contraction of muscle and cell migration.

CENTROSOMES

Near the nucleus in the cytoplasm of eukaryotic animal cells is a **centrosome.** The centrosome is made up of two **centrioles** and surrounding amorphous **pericentriolar material.** The centrioles are short cylinders arranged so that they are at right angles to each other. Microtubules in groups of three run longitudinally in the walls of each centriole (Figure 2–1). Nine of these triplets are spaced at regular intervals around the circumference.

The centrosomes are **microtubule-organizing centers (MTOCs)** that contain γ-tubulin. The microtubules grow out of this γ-tubulin in the pericentriolar material. When a cell divides, the centrosomes duplicate themselves, and the pairs move apart to the poles of the mitotic spindle, where they monitor the steps in cell division. In multinucleate cells, a centrosome is near each nucleus.

CILIA

Cilia are specialized cellular projections that are used by unicellular organisms to propel themselves through liquid and by multicellular organisms to propel mucus and other substances over the surface of various epithelia. Additionally, virtually all cells in the human body contain a primary cilium that emanates from the surface. The primary cilium serves as a sensory organelle that receives both mechanical and chemical signals from other cells and the environment. Cilia are functionally indistinct from the eukaryotic flagella of sperm cells. Within the cilium there is an **axoneme** that comprises a unique arrangement of nine outer microtubule doublets and two inner microtubules (“9+2” arrangement). Along this cytoskeleton is **axonemal dynein.** Coordinated dynein– microtubule interactions within the axoneme are the basis of ciliary and sperm movement. At the base of the axoneme and just inside lies the **basal body.** It has nine circumferential triplet microtubules, like a centriole, and there is evidence that basal bodies and centrioles are interconvertible. A wide variety of diseases and disorders arise from dysfunctional cilia **(Clinical Box 2–3)**.

CELL ADHESION MOLECULES

Cells are attached to the basal lamina and to each other by **CAMs** that are prominent parts of the intercellular connections described below. The unique structural and signaling functions of these adhesion proteins have been found to

be important in embryonic development and formation of the nervous system and other tissues, in holding tissues together in adults, in inflammation and wound healing, and in the metastasis of tumors. Many CAMs pass through the cell membrane and are anchored to the cytoskeleton inside the cell. Some bind to like molecules on other cells (homophilic binding), whereas others bind to nonself molecules (heterophilic binding). Many bind to **laminins,** a family of large cross-shaped molecules with multiple receptor domains in the extracellular matrix.

Nomenclature in the CAM field is somewhat chaotic, partly because the field is growing so rapidly and partly because of the extensive use of acronyms, as in other areas of modern biology. However, the CAMs can be divided into four broad families: (1) **integrins,** heterodimers that bind to various receptors; (2) adhesion molecules of the **IgG superfamily** of immunoglobulins; (3) **cadherins,** Ca^{2+}-dependent molecules that mediate cell-to-cell adhesion by homophilic reactions; and (4) **selectins,** which have lectin-like domains that bind carbohydrates.

CLINICAL BOX 2–3

Ciliary Diseases

Primary ciliary dyskinesia refers to a set of inherited disorders that limit ciliary structure and/or function. Disorders associated with ciliary dysfunction have long been recognized in the conducting airway. Altered ciliary function in the conducting airway can slow the mucociliary escalator and result in airway obstruction and increased infection. Dysregulation of ciliary function in sperm cells has also been well characterized to result in loss of motility and infertility. Ciliary defects in the function or structure of primary cilia have been shown to have effects on a variety of tissues/organs. As would be expected, such diseases are quite varied in their presentation, largely due to the affected tissue, and include mental retardation, retinal blindness, obesity, polycystic kidney disease, liver fibrosis, ataxia, and some forms of cancer.

THERAPEUTIC HIGHLIGHTS

The severity in ciliary disorders can vary widely, and treatments targeted to individual organs also vary. Treatment of ciliary dyskinesia in the conducting airway is focused on keeping the airways clear and free of infection. Strategies

include routine washing and suctioning of the sinus cavities and ear canals and liberal use of antibiotics. Other treatments that keep the airway from being obstructed (eg, bronchodilators, mucolytics, and corticosteroids) are also commonly used.

The CAMs not only fasten cells to their neighbors, but they also transmit signals into and out of the cell. For example, cells that lose their contact with the extracellular matrix via integrins have a higher rate of apoptosis than anchored cells, and interactions between integrins and the cytoskeleton are involved in cell movement.

INTERCELLULAR CONNECTIONS

Intercellular junctions that form between the cells in tissues can be broadly split into two groups: junctions that fasten the cells to one another and to surrounding tissues, and junctions that permit transfer of ions and other molecules from one cell to another. The types of junctions that tie cells together and endow tissues with strength and stability include **tight junctions,** which are also known as the **zonula occludens** **(Figure 2–8)**. The **desmosome** and **zonula adherens** also help hold cells together, and the **hemidesmosome** and **focal adhesions** attach cells to their basal laminas. The **gap junction** forms a cytoplasmic “tunnel” for diffusion of small molecules (< 1000 Da) between two neighboring cells.

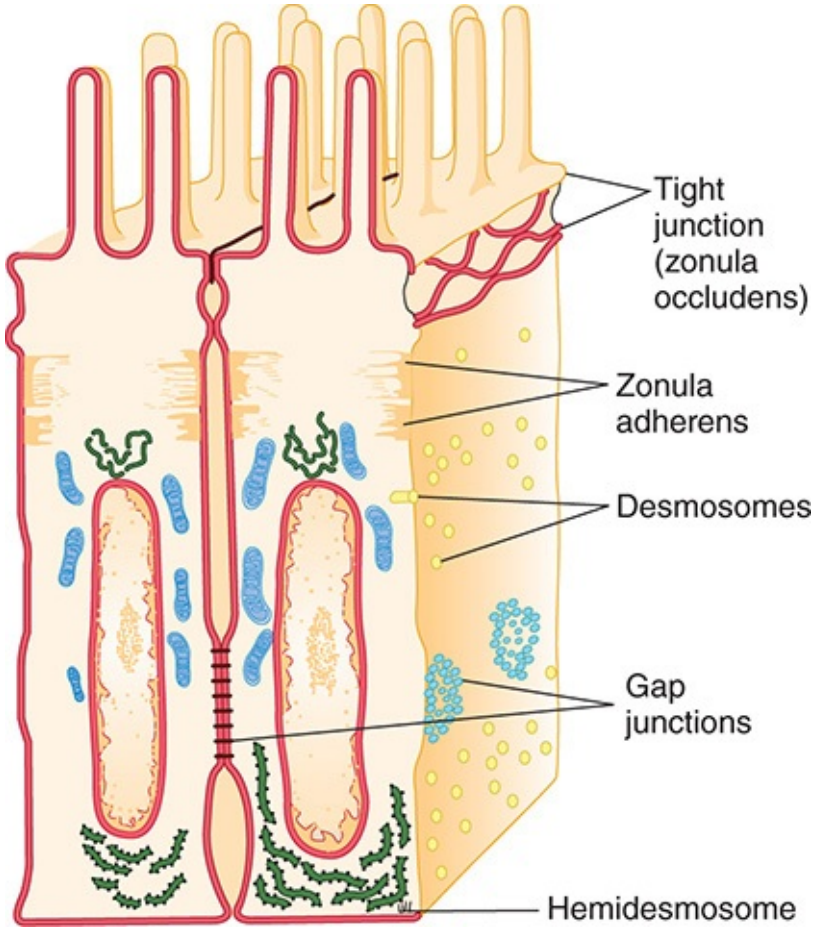


FIGURE 2–8 Intercellular junctions in the mucosa of the small intestine. Tight junctions (zonula occludens), adherens junctions (zonula adherens), desmosomes, gap junctions, and hemidesmosomes are all shown in relative positions in a polarized epithelial cell.

Tight junctions characteristically surround the apical margins of the cells in epithelia such as the intestinal mucosa, the walls of the renal tubules, and the choroid plexus. They are also important to endothelial barrier function and endothelium-dependent vasodilation. They are made up of ridges—half from one cell and half from the other—which adhere so strongly at cell junctions that they almost obliterate the space between the cells. There are three main families of transmembrane proteins that contribute to tight junctions: **occludin, junctional adhesion molecules,** and **claudins;** there are several more proteins that interact from the cytosolic side. Tight junctions permit the passage of ions, solute and intracellular signaling molecules in between adjacent cells **(paracellular pathway)** and the degree of this “leakiness” varies, depending in part on the protein makeup of the tight junction. Extracellular fluxes of ions and solute

across epithelia at these junctions are a significant part of overall ion and solute flux. In addition, tight junctions prevent the movement of proteins in the plane of the membrane, helping maintain the different distribution of transporters and channels in the apical and basolateral cell membranes that make transport across epithelia possible.

In epithelial cells, each zonula adherens is usually a continuous structure on the basal side of the zonula occludens, and it is a major site of attachment for intracellular microfilaments. It contains cadherins.

Desmosomes are patches characterized by apposed thickenings of the membranes of two adjacent cells. Attached to the thickened area in each cell are intermediate filaments, some running parallel to the membrane and others radiating away from it. Between the two membrane thickenings, the intercellular space contains filamentous material that includes cadherins and the extracellular portions of several other transmembrane proteins.

Hemidesmosomes look like half-desmosomes that attach cells to the underlying basal lamina and are connected intracellularly to intermediate filaments. However, they contain integrins rather than cadherins. Focal adhesions also attach cells to their basal laminas. As noted previously, they are labile structures associated with actin filaments inside the cell, and they play an important role in cell movement.

GAP JUNCTIONS

The intercellular space is up to 4 nm at gap junctions. Here, units called **connexons** in the membrane of each cell are lined up with one another to form the dodecameric gap junction **(Figure 2–9)**. Each connexon is made up of six protein subunits called **connexins.** They surround a channel that, when lined up with the channel in the corresponding connexon in the adjacent cell, permits substances to pass between the cells without entering the ECF. The pore diameter in the channel is estimated between 0.8 and 1.4 nm, which permits the passage of ions, sugars, amino acids, and other solutes with molecular weights up to about 1000 Da. Gap junctions thus permit the rapid propagation of electrical activity from cell to cell, as well as the exchange of various chemical messengers and intracellular signaling molecules. However, the gap junction channels are not simply passive, nonspecific conduits. At least 20 different genes code for connexins in humans, and mutations in these genes can lead to diseases that are highly selective in terms of the tissues involved and the type of communication between cells produced **(Clinical Box 2–4**). Experiments in

which particular connexins are deleted by gene manipulation or replaced with different connexins confirm that the particular connexin subunits that make up connexons determine their permeability and selectivity. It should be noted that connexons can also provide a conduit (ie, connexin semichannels) for regulated passage of small molecules between the cytoplasm and the ECF. Such movement can allow additional signaling pathways between and among cells in a tissue.

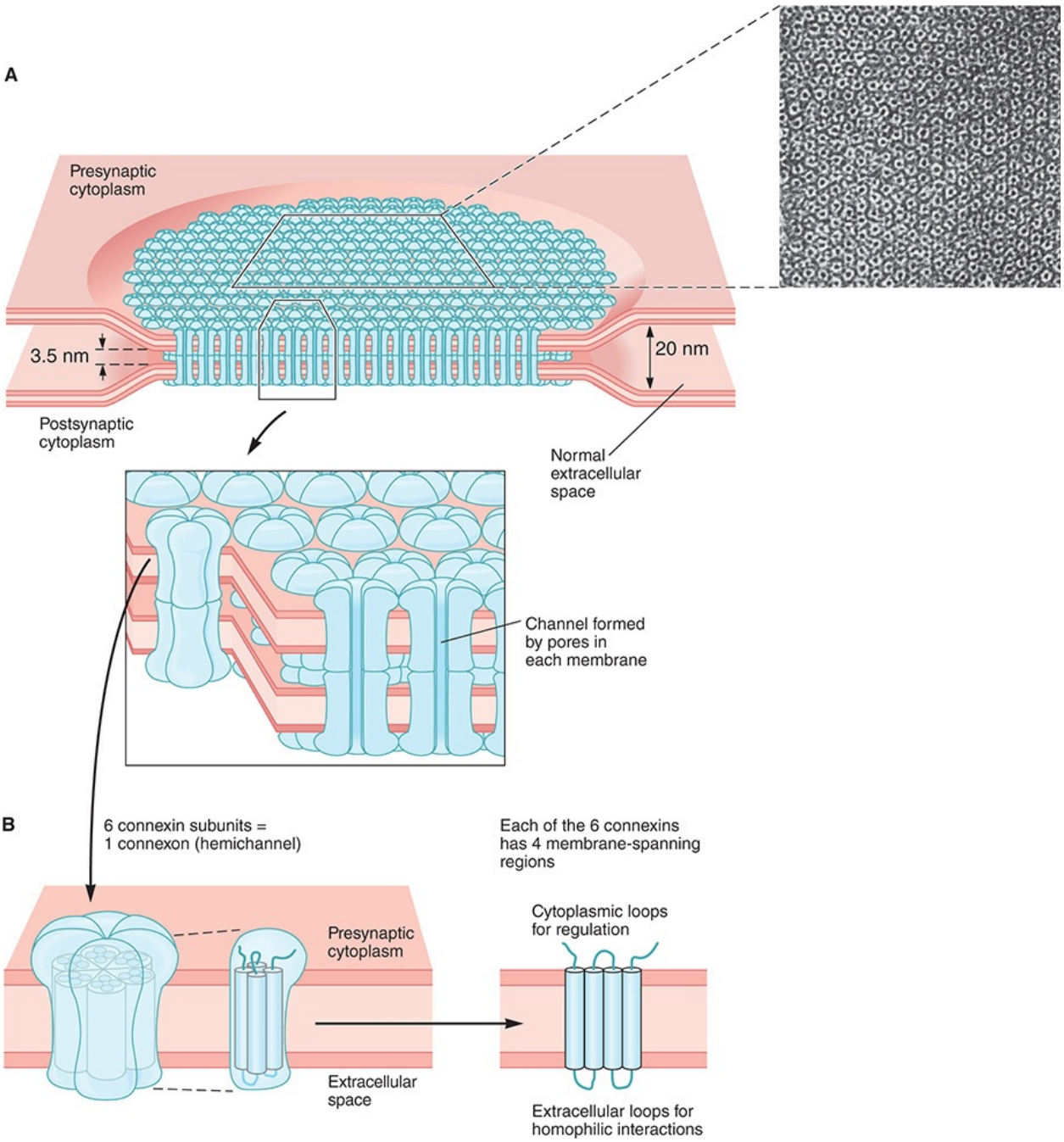


FIGURE 2–9 Gap junction connecting the cytoplasm of two cells. A) A gap

junction plaque, or collection of individual gap junctions, is shown to form multiple pores between cells that allow for the transfer of small molecules. Inset is an electron micrograph from rat liver. **B)** Topographic depiction of individual connexon and corresponding six connexin proteins that traverse the membrane. Note that each connexin traverses the membrane four times. (Reproduced with permission from Kandel ER, Schwartz JH, Jessell TM, Siegelbaum SA, Hudspeth AJ (editors): *Principles of Neural Science*, 5th ed. New York, NY: McGraw-Hill; 2013.)

CLINICAL BOX 2–4

Connexins in Disease

There is extensive information related to the in vivo functions of connexins, growing out of work on connexin knockouts in mice and the analysis of mutations in human connexins. The mouse knockouts demonstrated that connexin deletions lead to electrophysiological defects in the heart and predisposition to sudden cardiac death, female sterility, abnormal bone development, abnormal growth in the liver, cataracts, hearing loss, and a host of other abnormalities. Information from these and other studies has allowed for the identification of several connexin mutations known to be responsible for almost 20 different human diseases. These diseases include several skin disorders such as Clouston syndrome (a connexin 30 (Cx30) defect) and erythrokeratoderma variabilis (Cx30.3 and Cx31); inherited deafness (Cx26, Cx30, and Cx31); predisposition to myoclonic epilepsy (Cx36); predisposition to arteriosclerosis (Cx37); cataract (Cx46 and Cx50); idiopathic atrial fibrillation (Cx40); and X-linked Charcot-Marie-Tooth disease (Cx32). It is interesting to note that each of these target tissues for disease contains other connexins that do not fully compensate for loss of the crucial connexins in disease development. Understanding how loss of individual connexins alters cell physiology to contribute to these and other human diseases is an area of intense research.

NUCLEUS & RELATED STRUCTURES

A nucleus is present in all eukaryotic cells that divide (Figure 2–1). The nucleus is made up in large part of the **chromosomes,** the structures in the nucleus that

carry a complete blueprint for all the heritable species and individual characteristics of the animal. Except in germ cells, the chromosomes occur in pairs, one originally from each parent. Each chromosome is made up of a giant molecule of **DNA.** The DNA strand is about 2 m long, but it can fit in the nucleus because at intervals it is wrapped around a core of histone proteins to form a **nucleosome.** There are about 25 million nucleosomes in each nucleus. Thus, the structure of the chromosomes has been likened to a string of beads. The beads are the nucleosomes, and the linker DNA between them is the string. The whole complex of DNA and proteins is called **chromatin.** During cell division, the coiling around histones is loosened, probably by acetylation of the histones, and pairs of chromosomes become visible, but between cell divisions only clumps of chromatin can be discerned in the nucleus. The ultimate units of heredity are the **genes** on the chromosomes. As discussed in Chapter 1, each gene is a portion of the DNA molecule.

The nucleus of most cells contains a **nucleolus** (Figure 2–1), a patchwork of granules rich in **RNA.** In some cells, the nucleus contains several of these structures. Nucleoli are most prominent and numerous in growing cells. They are the site of synthesis of ribosomes, the structures in the cytoplasm in which proteins are synthesized.

The interior of the nucleus has a skeleton of fine filaments that are attached to the **nuclear membrane,** or **envelope** (Figure 2–1), which surrounds the nucleus. This membrane is a double membrane, and spaces between the twofolds are called **perinuclear cisterns.** The membrane is permeable only to small molecules. However, it contains **nuclear pore complexes.** Each complex has eightfold symmetry and is made up of about 100 proteins organized to form a tunnel through which transport of proteins and mRNA occurs. There are many transport pathways; many proteins that participate in these pathways, including **importins** and **exportins,** have been isolated and characterized. Much current research is focused on transport into and out of the nucleus, and a more detailed understanding of these processes should emerge in the near future.

ENDOPLASMIC RETICULUM

The **endoplasmic reticulum** is a complex series of tubules in the cytoplasm of the cell (Figure 2–1; **Figure 2–10**; and **Figure 2–11**). The inner limb of its membrane is continuous with a segment of the nuclear membrane, so in effect this part of the nuclear membrane is a cistern of the endoplasmic reticulum. The tubule walls are made up of membrane. In **rough (granular) endoplasmic**

reticulum, ribosomes are attached to the cytoplasmic side of the membrane, whereas in **smooth (agranular) endoplasmic reticulum,** ribosomes are absent. Free ribosomes are also found in the cytoplasm. The rough endoplasmic reticulum is concerned with protein synthesis and the initial folding of polypeptide chains with the formation of disulfide bonds. The smooth endoplasmic reticulum is the site of steroid synthesis in steroid-secreting cells and the site of detoxification processes in other cells. A modified endoplasmic reticulum, the sarcoplasmic reticulum, plays an important role in skeletal muscle, cardiac muscle, and smooth muscle cells. In particular, the endoplasmic or sarcoplasmic reticulum can sequester cytosolic Ca^{2+} ions and function as an intracellular Ca^{2+} store, which then allow for Ca^{2+} release into the cytosol as signaling molecules to stimulate cell contraction, proliferation, and migration.

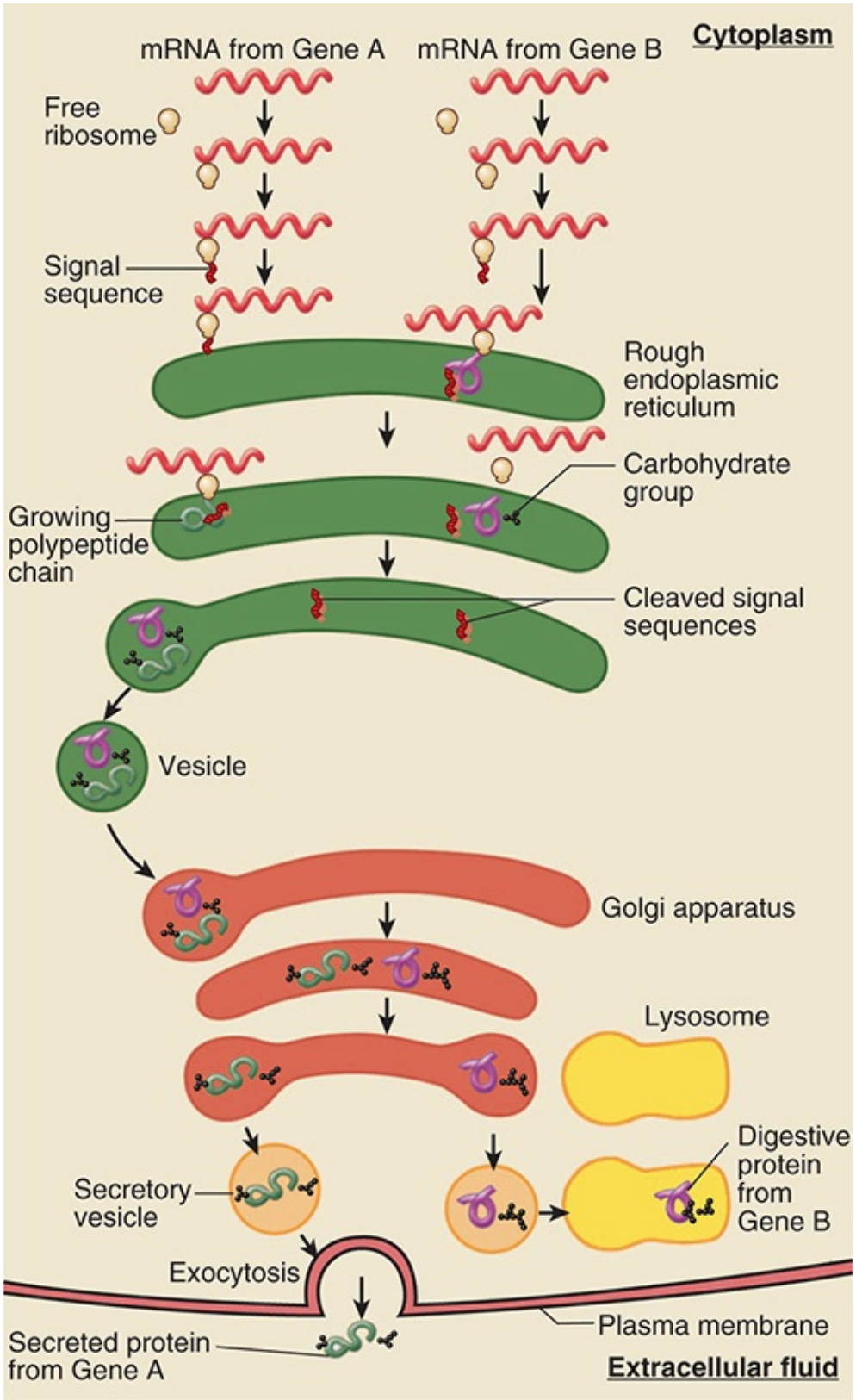


FIGURE 2–10 Rough endoplasmic reticulum and protein translation.

Messenger RNA and ribosomes meet up in the cytosol for translation. Proteins that have appropriate signal peptides begin translation, and then associate with the endoplasmic reticulum (ER) to complete translation. The association of ribosomes is what gives the ER its “rough” appearance. (Reproduced with permission from Widmaier EP, Raff H, Strang KT: *Vander’s Human Physiology: The Mechanisms of Body Function*, 11th ed. New York, NY: McGraw-Hill; 2008.)

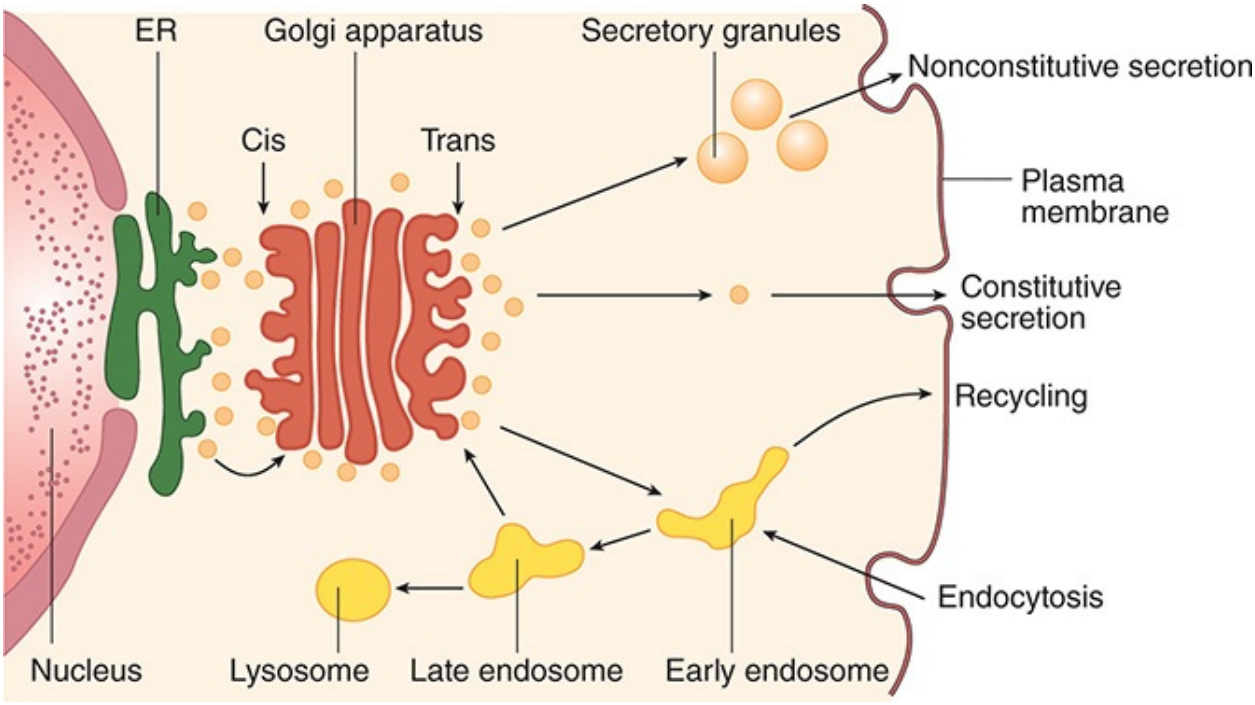


FIGURE 2–11 Cellular structures involved in protein processing. Structures involved in protein processing, from transcription to secretion, are shown. See text for details.

RIBOSOMES

The ribosomes in eukaryotes measure approximately 22 × 32 nm. Each is made up of a large and a small subunit called, on the basis of their rates of sedimentation in the ultracentrifuge, the 60S and 40S subunits. The ribosomes are complex structures, containing many different proteins and at least three ribosomal RNAs. They are the sites of protein synthesis. The ribosomes that become attached to the endoplasmic reticulum synthesize all transmembrane proteins, most secreted proteins, and most proteins that are stored in the Golgi apparatus, lysosomes, and endosomes. These proteins typically have a

hydrophobic **signal peptide** at one end (Figure 2–10). The polypeptide chains that form these proteins are extruded into the endoplasmic reticulum. The free ribosomes synthesize cytoplasmic proteins such as hemoglobin and the proteins found in peroxisomes and mitochondria.

GOLGI APPARATUS & VESICULAR TRAFFIC

The Golgi apparatus is a collection of membrane-enclosed sacs (cisternae) that are stacked like dinner plates (Figure 2–1). One or more Golgi apparati are present in all eukaryotic cells, usually near the nucleus. Much of the organization of the Golgi is directed at proper glycosylation of proteins and lipids. There are more than 200 enzymes that function to add, remove, or modify sugars from proteins and lipids in the Golgi apparatus.

The Golgi apparatus is a polarized structure with two distinct sides or faces, cis and trans sides (Figures 2–1, 2–10, 2–11). The cis face is near the endoplasmic reticulum, while the opposite trans face is near the plasma membrane (Figure 2–11). Membranous vesicles containing newly synthesized proteins bud off from the rouph endoplasmic reticulum and fuse with the cistern on the cis side of the Golgi apparatus. The proteins are then passed via other vesicles to the middle cisterns and finally to the cistern on the trans side, from which vesicles branch off into the cytoplasm. From the trans side of Golgi, vesicles shuttle to the lysosomes and to the cell exterior via constitutive and nonconstitutive pathways, both involving **exocytosis.** Conversely, vesicles are pinched off from the cell membrane by **endocytosis** and pass to endosomes. From there, they are recycled.

Vesicular traffic in the Golgi, and between other membranous compartments in the cell, is regulated by a combination of common mechanisms along with special mechanisms that determine where inside the cell they will go. One prominent feature is the involvement of a series of regulatory proteins controlled by GTP or guanosine diphosphate (GDP) binding **(small G-proteins)** associated with vesicle assembly and delivery. A second prominent feature is the presence of proteins called **SNAREs** (for soluble N-ethylmaleimide-sensitive factor attachment receptor). The v- (for vesicle) SNAREs on vesicle membranes interact in a lock-and-key fashion with t- (for target) SNAREs. Individual vesicles also contain structural protein or lipids in their membrane that help target them for specific membrane compartments (eg, Golgi sacs, cell membranes).

QUALITY CONTROL

The processes involved in protein synthesis, folding, and migration to the various parts of the cell are so complex that it is remarkable that more errors and abnormalities do not occur. The fact that these processes work as well as they do is because of mechanisms at each level that are responsible for "quality control." Damaged DNA is detected and repaired or bypassed. The various RNAs are also checked during the translation process. Finally, when the protein chains are in the endoplasmic reticulum and Golgi apparatus, defective structures are detected and the abnormal proteins are degraded in lysosomes and proteasomes. The net result is a remarkable accuracy in the production of the proteins needed for normal cell function.

APOPTOSIS

In addition to dividing and growing under genetic control, cells can die and be absorbed under genetic control. This process is called **programmed cell death,** or **apoptosis** (Gr. apo "away" + ptosis "fall"). It can be called "cell suicide" in the sense that the cell's own genes play an active role in its demise. It should be distinguished from necrosis ("cell murder"), in which healthy cells are destroyed by external processes such as inflammation.

Apoptosis is a very common process during development and in adulthood. In the central nervous system (CNS), large numbers of neurons are produced and then die during the remodeling that occurs during development and synapse formation. In the immune system, apoptosis gets rid of inappropriate clones of immunocytes and is responsible for the lytic effects of glucocorticoids on lymphocytes. Apoptosis is also an important factor in processes such as removal of the webs between the fingers in fetal life and regression of duct systems in the course of sexual development in the fetus. In adults, it participates in the cyclic breakdown of the endometrium that leads to menstruation. In epithelia, cells that lose their connections to the basal lamina and neighboring cells undergo apoptosis. This is responsible for the death of the enterocytes sloughed off the tips of intestinal villi. Abnormal apoptosis probably occurs in autoimmune diseases, neurodegenerative diseases, and cancer. It is interesting that apoptosis occurs in invertebrates, including nematodes and insects. However, its molecular mechanism is much more complex than that in vertebrates.

One final common pathway bringing about apoptosis is activation of **caspases,** a group of cysteine proteases. Many of these have been characterized

to date in mammals; 14 have been found in humans. They exist in cells as inactive proenzymes (procaspases) until activated by the cellular machinery. The net result is DNA fragmentation, cytoplasmic and chromatin condensation, and eventually membrane bleb formation, with cell breakup and removal of the debris by phagocytes **(Clinical Box 2–5)**.

CLINICAL BOX 2–5

Cellular and Molecular Medicine

Fundamental research on molecular aspects of genetics, regulation of gene expression, and protein synthesis has been paying off in clinical medicine at a rapidly accelerating rate.

One early dividend was an understanding of the mechanisms by which antibiotics exert their effects. Almost all act by inhibiting protein synthesis at one or another of the steps described previously. Antiviral drugs act in a similar way; for example, acyclovir and ganciclovir act by inhibiting DNA polymerase. Some of these drugs have this effect primarily in bacteria, but others inhibit protein synthesis in the cells of other animals, including mammals. This fact makes antibiotics of great value for research as well as for treatment of infections. Single genetic abnormalities that cause over 600 human diseases have been identified. Many of the diseases are rare, but others are more common and some cause conditions that are severe and eventually fatal. Examples include the defectively regulated Cl^- channel in cystic fibrosis and the unstable **trinucleotide repeats** in various parts of the genome that cause Huntington disease, the fragile X syndrome, and several other neurologic diseases. Abnormalities in mitochondrial DNA can also cause human diseases such as Leber hereditary optic neuropathy and some forms of cardiomyopathy. Not surprisingly, genetic aspects of cancer are probably receiving the greatest current attention. Some cancers are caused by **oncogenes,** genes that are carried in the genomes of cancer cells and are responsible for producing their malignant properties. These genes are derived by somatic mutation from closely related **proto-oncogenes,** which are normal genes that control growth. Over 100 oncogenes have been described. Another group of genes produce proteins that suppress tumors, and more than 10 of these **tumor suppressor genes** have been described. The most studied of these is the p53 gene on human chromosome 17. The p53 protein produced by this gene triggers apoptosis. It is also a nuclear transcription factor that

appears to increase production of a 21-kDa protein that blocks two cell cycle enzymes, slowing the cycle and permitting repair of mutations and other defects in DNA. The p53 gene is mutated in up to 50% of human cancers, with the production of p53 proteins that fail to slow the cell cycle and permit other mutations in DNA to persist. The accumulated mutations eventually cause cancer.

TRANSPORT ACROSS CELL MEMBRANES

There are several mechanisms of transport across cellular membranes. Primary pathways include exocytosis, endocytosis, membrane permeability and ion channels, and primary and secondary active transport. Each of these is discussed below.

EXOCYTOSIS

Vesicles containing material for export, such as secretory granules, are targeted to the cell membrane (Figure 2–11), where they bond in a similar manner to that discussed in vesicular traffic between Golgi stacks, via the v-SNARE/t-SNARE arrangement. The area of fusion then breaks down, leaving the contents of the vesicle outside and the cell membrane intact. This is the Ca^{2+}-dependent process of **exocytosis** **(Figure 2–12)**.

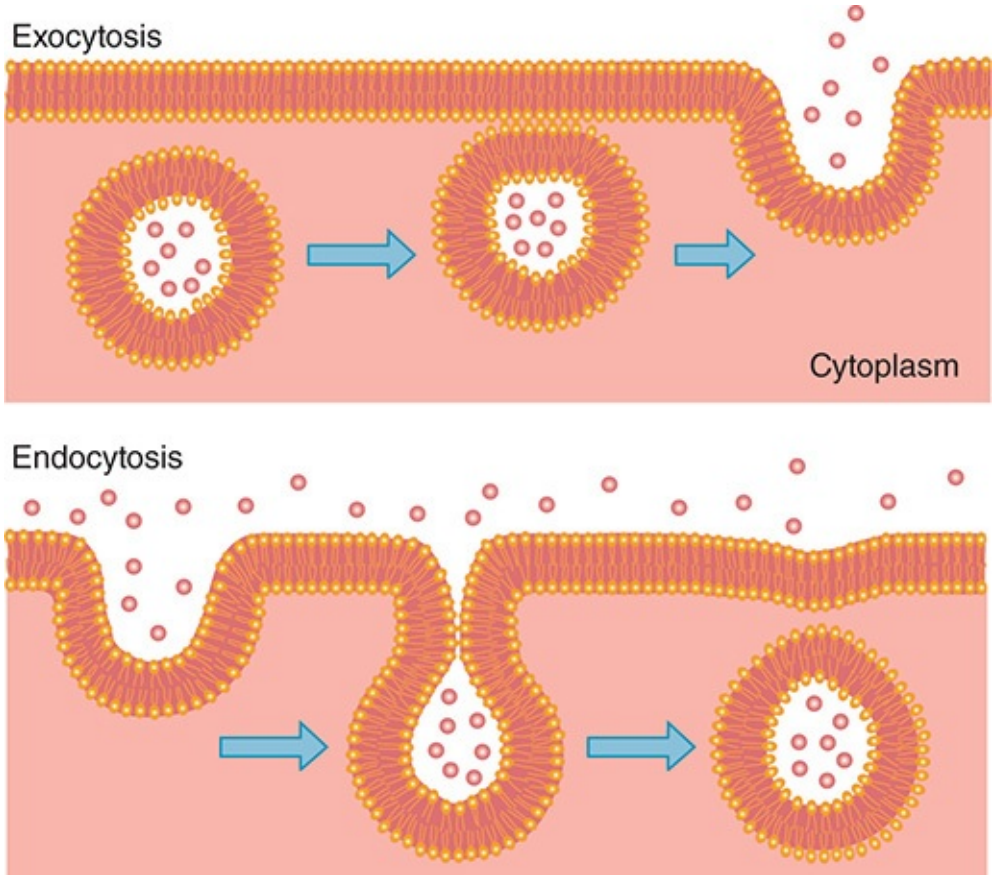


FIGURE 2–12 Exocytosis and endocytosis. Note that in exocytosis the cytoplasmic sides of two membranes fuse, whereas in endocytosis two extracellular sides fuse.

Note that secretion from the cell occurs via two pathways (Figure 2–11). In the **nonconstitutive (or regulated) pathway,** proteins from the Golgi apparatus initially enter secretory granules, where processing of prohormones to the mature hormones occurs before exocytosis. The other pathway, the **constitutive pathway,** involves the prompt transport of proteins to the cell membrane in vesicles, with little or no processing or storage. The nonconstitutive pathway is sometimes called the **regulated pathway,** but this term is misleading because the output of proteins by the constitutive pathway is also regulated.

ENDOCYTOSIS

Endocytosis is the reverse of exocytosis. There are various types of endocytosis named for the size of particles being ingested as well as the regulatory requirements for the particular process. These include **phagocytosis,**

pinocytosis, clathrin-mediated endocytosis, caveolae-dependent uptake, and **nonclathrin/noncaveolae endocytosis.**

Phagocytosis ("cell eating") is the process by which bacteria, dead cells, or other bits of microscopic material are engulfed by cells such as the polymorphonuclear leukocytes of the blood. The material makes contact with the cell membrane, which then invaginates. The invagination is pinched off, leaving the engulfed material in the membrane-enclosed vacuole and the cell membrane intact. **Pinocytosis** ("cell drinking") is a similar process with the vesicles much smaller in size and the substances ingested are in solution. The small size membrane that is ingested with each event should not be misconstrued; cells undergoing active pinocytosis (eg, macrophages) can ingest the equivalent of their entire cell membrane in just 1 h.

Clathrin-mediated endocytosis occurs at membrane indentations where the protein **clathrin** accumulates. Clathrin molecules have the shape of triskelions, with three "legs" radiating from a central hub **(Figure 2–13)**. As endocytosis progresses, the clathrin molecules form a geometric array that surrounds the endocytotic vesicle. At the neck of the vesicle, the GTP-binding protein **dynamin** is involved, either directly or indirectly, in pinching off the vesicle. Once the complete vesicle is formed, the clathrin falls off and the three-legged proteins recycle to form another vesicle. The vesicle fuses with and dumps its contents into an **early endosome** (Figure 2–11). From the early endosome, a new vesicle can bud off and return to the cell membrane. Alternatively, the early endosome can become a **late endosome** and fuse with a lysosome (Figure 2–11) in which the contents are digested by the lysosomal proteases. Clathrin-mediated endocytosis is responsible for the internalization of many receptors and the ligands bound to them—including, for example, nerve growth factor (NGF) and low-density lipoproteins. It also plays a major role in synaptic function.

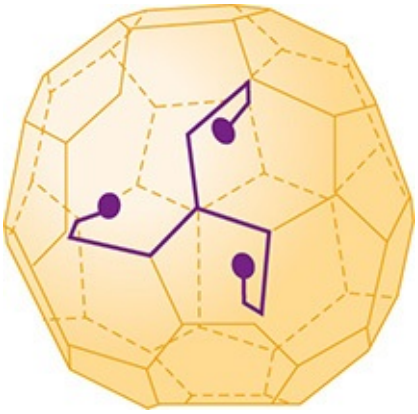


FIGURE 2–13 Clathrin molecule on the surface of an endocytotic vesicle.

Note the characteristic triskelion shape and the fact that with other clathrin molecules it forms a net supporting the vesicle.

It is apparent that exocytosis adds to the total amount of membrane surrounding the cell, and if membrane were not removed elsewhere at an equivalent rate, the cell would enlarge. However, removal of cell membrane occurs by endocytosis, and such exocytosis–endocytosis coupling maintains the surface area of the cell at its normal size.

RAFTS & CAVEOLAE

Some areas of the cell membrane are especially rich in cholesterol and sphingolipids and have been called **rafts.** These rafts are probably the precursors of flask-shaped membrane depressions called **caveolae** (little caves) when their walls become infiltrated with a protein called **caveolin** that resembles clathrin **(Figure 2–14)**. There is considerable debate about the functions of rafts and caveolae, with evidence that they are involved in cholesterol regulation and transcytosis. It is clear, however, that cholesterol can interact directly with caveolin, effectively limiting the protein's ability to move around in the membrane. Internalization via caveolae involves binding of cargo to caveolin and regulation by dynamin. Caveolae are prominent in endothelial cells, where they help in the uptake of nutrients from the blood.

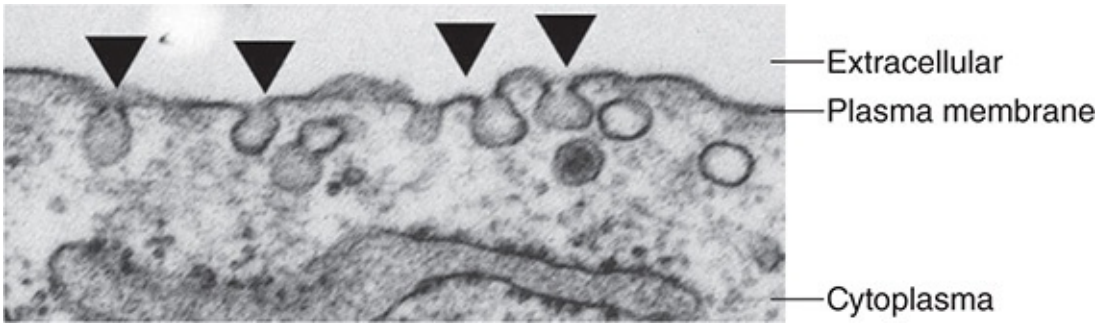


FIGURE 2–14 Caveolae in vascular smooth muscle cell. Ultrastructure of the plasma membrane of a human pulmonary arterial smooth muscle cell is assessed by electron microscopy. Arrow heads indicate individual caveolae.

COATS & VESICLE TRANSPORT

It now appears that all vesicles involved in transport have protein coats. In

humans, over 50 coat complex subunits have been identified. Vesicles that transport proteins from the trans Golgi to lysosomes have **assembly protein 1 (AP-1)** clathrin coats, and endocytotic vesicles that transport to endosomes have AP-2 clathrin coats. Vesicles that transport between the endoplasmic reticulum and the Golgi have coat proteins I and II (COPI and COPII). Certain amino acid sequences or attached groups on the transported proteins target the proteins for particular locations. For example, the amino acid sequence Asn–Pro–any amino acid–Tyr targets transport from the cell surface to the endosomes, and mannose-6-phosphate groups target transfer from the Golgi to mannose-6-phosphate receptors (MPR) on the lysosomes.

Various small G-proteins of the Rab family are especially important in vesicular traffic. They appear to guide and facilitate orderly attachments of these vesicles. To illustrate the complexity of directing vesicular traffic, humans have 60 Rab proteins and 35 SNARE proteins.

MEMBRANE PERMEABILITY & MEMBRANE TRANSPORT PROTEINS

Small, nonpolar molecules (including O_2 and N_2) and small uncharged polar molecules such as CO_2 diffuse across the lipid membranes of cells. However, the membranes have very limited permeability to other substances. Instead, they cross the membranes by endocytosis and exocytosis and by passage through highly specific **transport proteins,** transmembrane proteins that form channels for ions or transport substances such as glucose, urea, and amino acids. The limited permeability applies even to water, with simple diffusion being supplemented throughout the body with various water channels **(aquaporins).** For reference, the sizes of ions and other biologically important substances are summarized in **Table 2–2**.

TABLE 2–2 Size of hydrated ions and other substances of biologic interest.

Substance	Atomic or Molecular Weight	Radius (nm)
Cl^-	35	0.12
K^+	39	0.12
H_2O	18	0.12
Ca^{2+}	40	0.15
Na^+	23	0.18
Urea	60	0.23
Li^+	7	0.24
Glucose	180	0.38
Sucrose	342	0.48
Inulin	5000	0.75
Albumin	69,000	7.50

Data from Moore EW: *Physiology of Intestinal Water and Electrolyte Absorption.* American Gastroenterological Association, 1976.

Some transport proteins are simple aqueous **ion channels,** though many of these have special features that make them selective for a given substance such as Ca^{2+} or, in the case of aquaporins, for water. These membrane-spanning proteins (or collections of proteins) have tightly regulated pores that can be **gated** opened or closed in response to local changes **(Figure 2–15)**. Some are gated by alterations in membrane potential **(voltage-gated),** whereas others are opened or closed in response to a ligand **(ligand-gated).** The ligand is often external (eg, a neurotransmitter, a hormone, or an agonist). However, it can also be internal; intracellular Ca^{2+}, cyclic adenosine 3′,5′-monophosphate (cAMP), lipids, or one of the G-proteins produced in cells can all bind directly to channels and activate them. Some channels are also opened by mechanical stretch, and these mechanosensitive channels play an important role in cell movement.

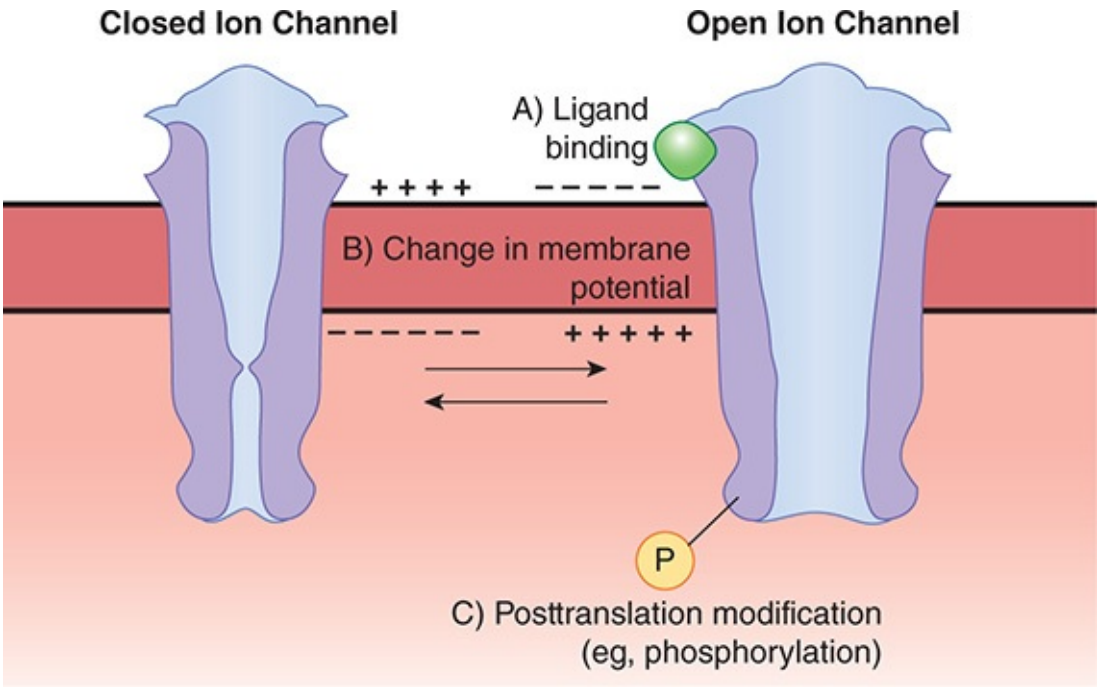


FIGURE 2–15 Regulation of gating in ion channels. Ion channels can gate open or closed in response to several environmental signals. Some typical examples are shown in an idealized channel. **A)** Ligand gating: Channel opens in response to ligand binding. **B)** Voltage gating: Channel opens in response to a change in membrane potential. **C)** Posttranslational modification: Channel gates in response to modification such as phosphorylation.

Other transport proteins are **carriers** that bind ions and other molecules and then change their configuration, moving the bound molecule from one side of the cell membrane to the other. Molecules move from areas of high concentration to areas of low concentration (down their **chemical gradient)**, and cations move to negatively charged areas whereas anions move to positively charged areas ("down" their **electrical gradient)**. When carrier proteins move substances in the direction of their chemical or electrical gradients, no energy input is required and the process is called **facilitated diffusion.** A typical example is glucose transport by the glucose transporter, which moves glucose down its concentration gradient from the ECF to the cytoplasm of the cell. Other carriers transport substances against their electrical and chemical gradients. This form of transport requires energy and is called **active transport.** In animal cells, the energy is provided almost exclusively by hydrolysis of ATP. Not surprisingly, therefore, many carrier molecules are ATPases, enzymes that catalyze the hydrolysis of ATP. One of these ATPases is **sodium–potassium adenosine triphosphatase (Na, K ATPase),** which is also known as the **Na, K pump.**

There are also H, K ATPases in the gastric mucosa and the renal tubules. Ca^{2+} ATPase pumps Ca^{2+} out of cells. Proton ATPases acidify many intracellular organelles, including parts of the Golgi complex and lysosomes.

Some of the transport proteins are called **uniports** because they transport only one substance. Others are called **symports** because transport requires the binding of more than one substance to the transport protein and the substances are transported across the membrane together. An example is the symport in the intestinal mucosa that is responsible for the cotransport of Na^+ and glucose from the intestinal lumen into mucosal cells. Other transporters are called **antiports** because they exchange one substance for another.

ION CHANNELS

There are ion channels specific for K^+, Na^+, Ca^{2+}, and Cl^-, as well as channels that are nonselective for cations or anions. Each type of channel exists in multiple forms with diverse properties. Most are made up of identical or very similar subunits. **Figure 2–16** shows the multiunit structure of various channels in diagrammatic cross-section.

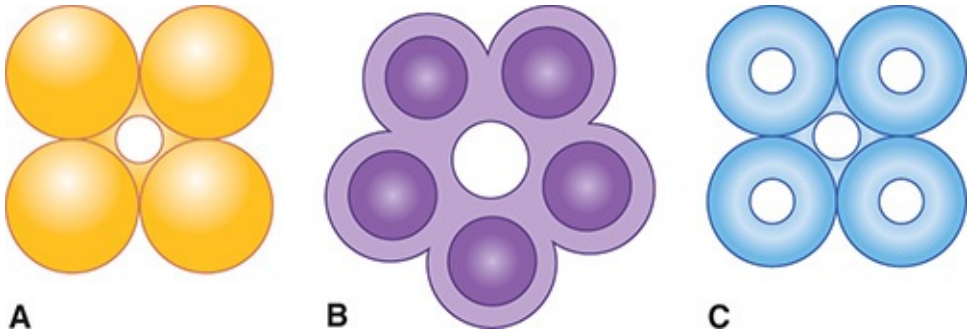


FIGURE 2–16 Different ways in which ion channels form pores. A) Four subunits form a central pore in typical K^+ channels. **B)** Some ligand-gated channels such as the acetylcholine receptor contain five similar subunits that form a central pore in the channel. **C)** Aquaporin, or water channels, contain four subunits that each contain a pore that can allow for water movement. Additionally, these subunits form a central pore that can allow for ion movement. It should be noted that other formations also occur.

Most K^+ channels are tetramers, with each of the four subunits forming part of the pore through which K^+ ions pass. Structural analysis of a bacterial

voltage-gated K^+ channel indicates that each of the four subunits has a paddle-like extension containing four charges. When the channel is closed, these extensions are near the negatively charged interior of the cell. When the membrane potential is reduced, the paddles containing the charges bend through the membrane to its exterior surface, causing the channel to open. The bacterial K^+ channel is very similar to the voltage-gated K^+ channels in a wide variety of species, including mammals and humans. In the acetylcholine ion channel and other ligand-gated cation or anion channels, five subunits make up the pore. Members of the ClC family of Cl^- channels are dimers, but they have two pores, one in each subunit. Finally, aquaporins are tetramers with a water pore in each of the subunits. Recently, a number of ion channels with intrinsic enzyme activity have been cloned. More than 30 different voltage-gated or cyclic nucleotide-gated Na^+ and Ca^{2+} channels of this type have been described. Representative Na^+, Ca^{2+}, and K^+ channels are shown in extended diagrammatic form in **Figure 2–17**.

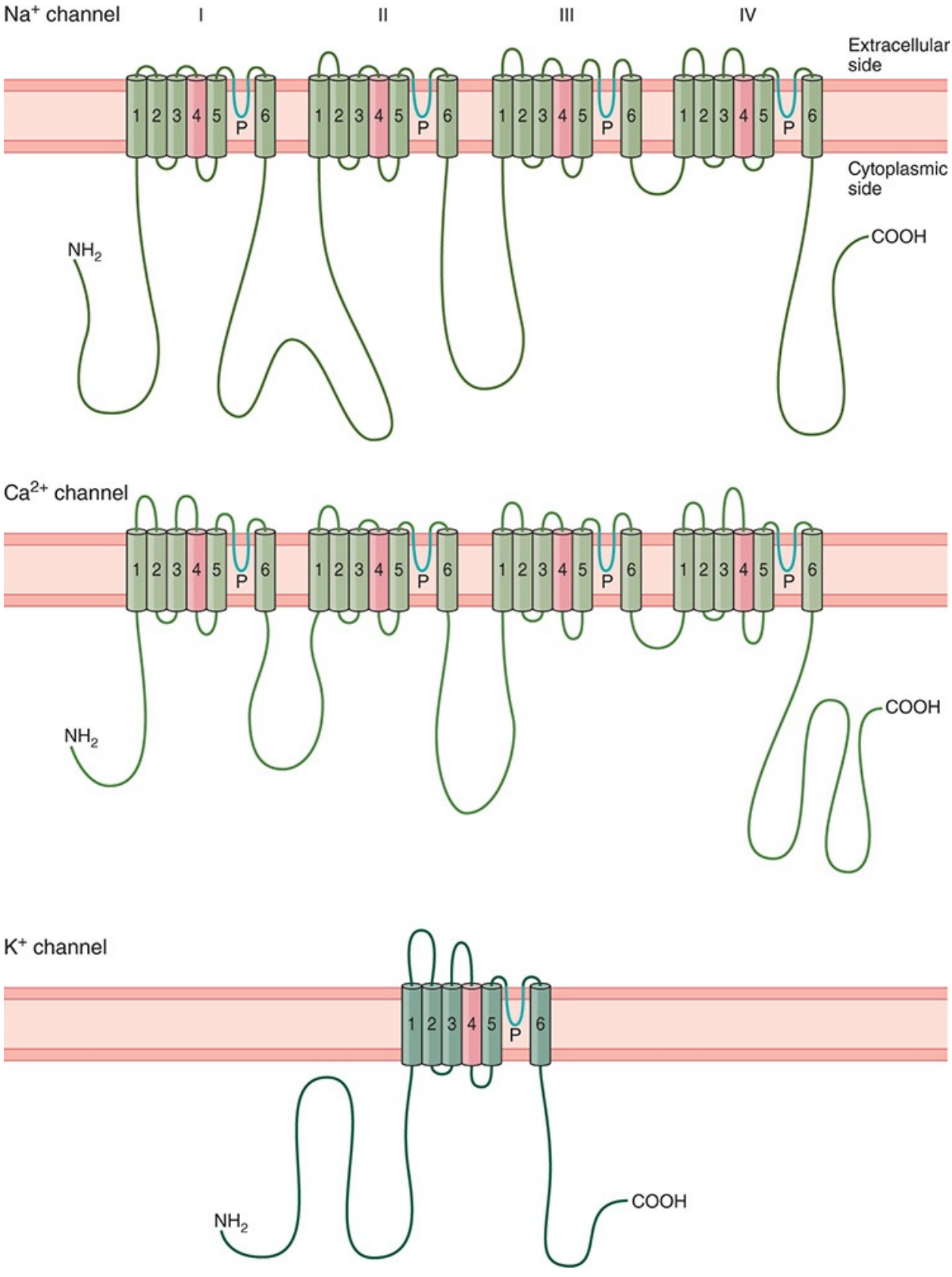

Na+ channel
I
II
III
IV
Extracellular side
Cytoplasmic side
1 2 3 4 5 P 6
NH2
COOH
Ca2+ channel
K+ channel

FIGURE 2–17 Diagrammatic representation of the pore-forming subunits of three ion channels. The α subunit of the Na^+ and Ca^{2+} channels traverses the membrane 24 times in four repeats of six membrane-spanning units. Each repeat has a "P" loop between membrane spans 5 and 6 that does not traverse the membrane. These P loops are thought to form the pore. Note that span 4 of each repeat is colored in red, representing its net "+" charge. The K^+ channel has only a single repeat of the six spanning regions and P loop. Four K^+ subunits are assembled for a functional K^+ channel. (Reproduced with permission from Kandel ER, Schwartz JH, Jessell TM, Siegelbaum SA, Hudspeth AJ (editors): *Principles of Neural Science*, 5th ed. New York, NY: McGraw-Hill; 2013.)

Another family of Na^+ channels with a different structure has been found in the apical membranes of epithelial cells in the kidneys, colon, lungs, and brain. The **epithelial sodium channels (ENaCs)** are made up of three subunits encoded by three different genes. Each of the subunits probably spans the membrane twice, and the amino terminal and carboxyl terminal are located inside the cell. The α subunit transports Na^+, whereas the β and γ subunits do not. However, the addition of the β and γ subunits increases Na^+ transport through the α subunit. ENaCs are inhibited by the diuretic amiloride, which binds to the α subunit, and they used to be called **amiloride inhibitable Na^+ channels.** The ENaCs in the kidney play an important role in the regulation of ECF volume by aldosterone. ENaC knockout mice are born alive but promptly die because they cannot move Na^+, and hence water, out of their lungs.

Humans have several types of Cl^- channels. The ClC dimeric channels are found in plants, bacteria, and animals, and there are nine different ClC genes in humans. Other Cl^- channels have the same pentameric form as the acetylcholine receptor; examples include the γ-aminobutyric acid A ($GABA_A$) and glycine receptors in the CNS. The cystic fibrosis transmembrane conductance regulator (CFTR) that is mutated in patients with cystic fibrosis is also a Cl^- channel. Ion channel mutations cause a variety of **channelopathies**—diseases that mostly affect muscle and brain tissue and produce episodic paralyses or convulsions, but are also observed in nonexcitable tissues **(Clinical Box 2–6)**.

PATCH CLAMP TECHNIQUE

Activity of ion channels and transporters in the plasma membrane plays an important role in the regulation of cell excitability and muscle contraction, as

well as cell proliferation, migration, and apoptosis. An important technique that has permitted major advances in our knowledge about ion transport proteins (channels and transporters) is **patch clamp technique.** The technique is used to measure and study ionic currents through membrane ion channels and via electrogenic membrane transporters in isolated living cells and tissue sections. Depending on the research interest, there are four recording configurations using patch clamp technique. A micropipette or glass pipette is placed on the membrane of a cell and forms a tight seal to the membrane. The patch of membrane under the pipette tip usually contains only a few transport or channel proteins or a single channel protein (eg, a K^+ channel). The cell can be left intact to record single-channel activity using the **cell-attached patch** mode **(Figure 2–18)**. Alternatively, the membrane patch sealed to the pipette can be pulled loose from the cell, forming an **inside-out patch** to record single-channel currents. If the membrane patch is then disrupted by briefly applying a strong suction, the interior of the pipette becomes continuous with the cytoplasm. This **whole-cell recording** configuration is used to record the whole-cell currents from the entire cell (Figure 2–18) and to record electrical potentials (eg, the resting membrane potential and action potentials) in the whole cell. The pipette can be retracted during the whole-cell recording to cause a rupture and reformation of the membrane to form an **outside-out patch** to study the effect of extracellular agonists on the single-channel currents.

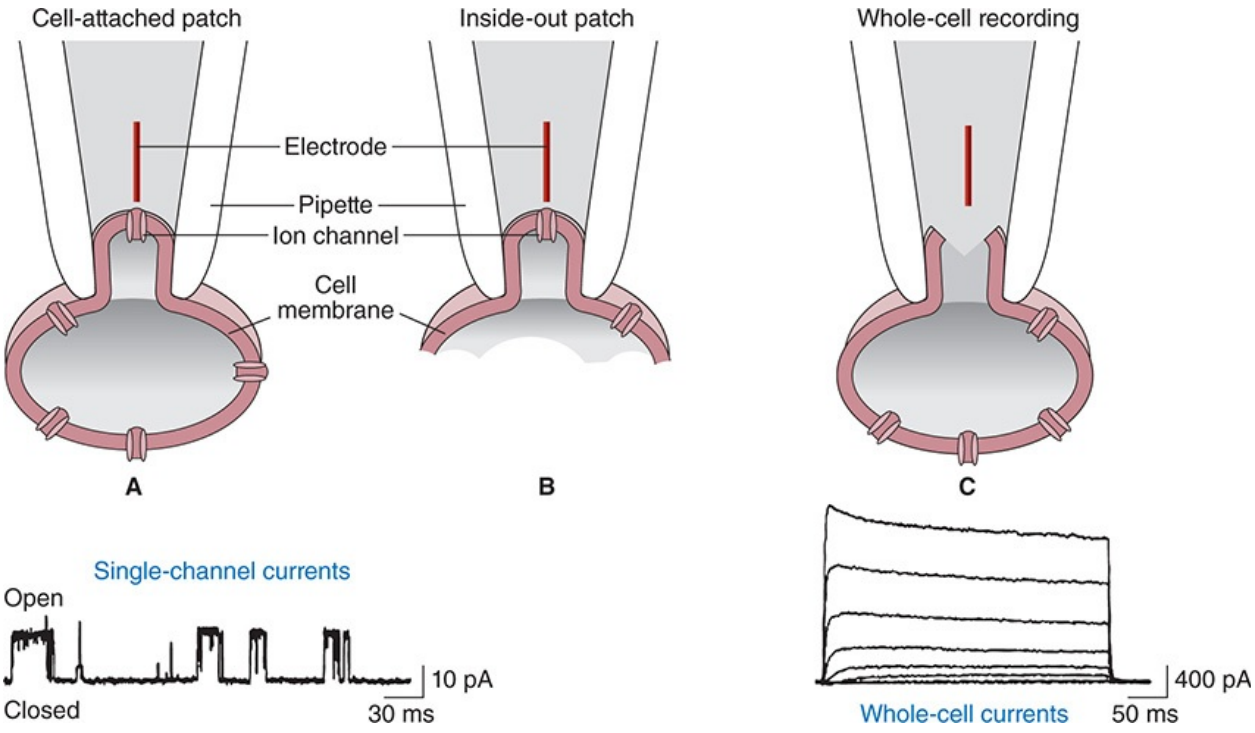


FIGURE 2–18 Patch clamp technique for studying ion transport. In a patch clamp experiment, a small glass pipette filled with extracellular (or intracellular) solution is carefully maneuvered to seal off a portion of a cell membrane (cell-attached patch, **A**). The pipette filled with an appropriate solution has an electrode that allows for recording of electrical changes through any pore or channel in the membrane patch. A typical single-channel recording of the Ca^{2+}-activated K^+ currents in cell-attached patch is shown below in panel **A**. The inside-out patch configuration is achieved by rapidly pulling up the pipette from the attached cell, without breaking the seal (**B**). The patch sealed to the pipette can also be ruptured off by a brief suction to form the whole-cell recording mode (**C**). This mode allows to record ionic currents in the entire cell. A typical whole-cell recording of the superimposed voltage-gated K^+ currents is shown in panel **C**.

CLINICAL BOX 2–6

Channelopathies

Channelopathies include a wide range of diseases that can affect both excitable (eg, neurons and muscle) and nonexcitable cells. Using molecular genetic tools, many of the pathologic defects in channelopathies have been traced to mutations in single ion channels. Examples of channelopathies in excitable cells include periodic paralysis (eg, Kir2.6/KCNJ18, a K^+ channel, $Ca_V1.1$/CACNA1S, a Ca^{2+} channel, or $Na_V1.4$/SCN4A, a Na^+ channel), myasthenia (eg, nicotinic acetyl choline receptor, a ligand-gated nonspecific cation channel), myotonia (eg, Kir1.1/KCNJ1, a K^+ channel), malignant hyperthermia (ryanodine receptor/RYR1, a Ca^{2+} channel), long QT syndrome (both Na^+ and K^+ channels, such as $K_V7.1$/KCNQ1, $K_V11.1$/KCNH2/hHERG, Kir2.1/KCNJ2 and $Na_V1.5$/SCN5A) and several other disorders. Examples of channelopathies in nonexcitable cells include the underlying cause for cystic fibrosis (CFTR/ABCC7, a Cl^- channel) and a form of Bartter syndrome (Kir1.1/KCNJ1, a K^+ channel). Importantly, advances in treatment of these disorders can come from the understanding of the basic defect and tailoring drugs that act to alter the mutated properties of the affected channel.

Na, K ATPase

As noted previously, Na, K ATPase catalyzes the hydrolysis of ATP to adenosine diphosphate (ADP) and uses the energy to extrude three Na^+ from the cell and take two K^+ into the cell for each molecule of ATP hydrolyzed. It is an **electrogenic pump** in that it moves three positive charges ($3Na^+$) out of the cell for each two ($2K^+$) that it moves in, and it is therefore said to have a **coupling ratio** of 3:2. It is found in all parts of the body. Its activity is inhibited by ouabain and related digitalis glycosides used in the treatment of heart failure. It is a heterodimer made up of an α subunit with a molecular weight of approximately 100,000 and a β subunit with a molecular weight of approximately 55,000. Both extend through the cell membrane **(Figure 2–19)**. Separation of the subunits eliminates activity. The β subunit is a glycoprotein, whereas Na^+ and K^+ transport occur through the α subunit. The β subunit has a single membrane-spanning domain and three extracellular glycosylation sites, all of which appear to have attached carbohydrate residues. These residues account for one-third of its molecular weight. The α subunit probably spans the cell membrane 10 times, with the amino and carboxyl terminals both located intracellularly. This subunit has intracellular Na^+- and ATP- binding sites and a phosphorylation site; it also has extracellular binding sites for K^+ and ouabain. The endogenous ligand of the ouabain-binding site, or endogenous ouabain, is identified in blood plasma and various organs and tissues of humans, especially in patients with essential hypertension. When Na^+ binds to the α subunit, ATP also binds and is converted to ADP, with a phosphate being transferred to Asp 376, the phosphorylation site. This causes a change in the configuration of the protein, extruding Na^+ into the ECF. K^+ then binds extracellularly, dephosphorylating the α subunit, which returns to its previous conformation, releasing K^+ into the cytoplasm.

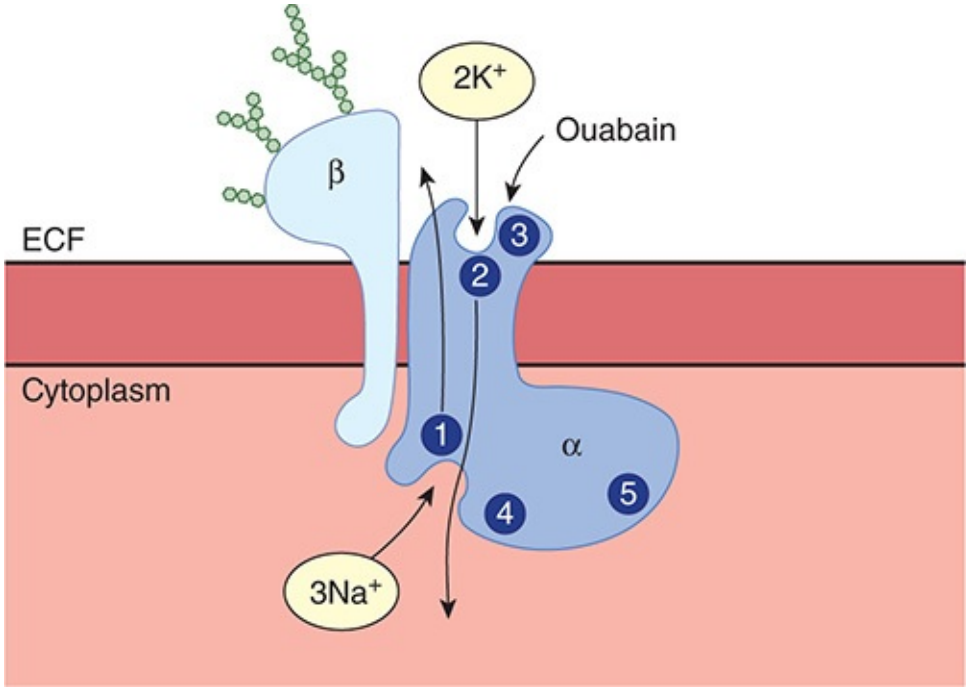


FIGURE 2–19 Na, K ATPase. The intracellular portion of the α subunit has a Na^+-binding site **(1)**, a phosphorylation site **(4)**, and an ATP-binding site **(5)**. The extracellular portion has a K^+-binding site **(2)** and an ouabain-binding site **(3)**. (Reproduced with permission from Horisberger JD, Lemas V, Kraehenbühl JP et al: Structure–function relationship of Na, K-ATPase. Annu Rev Physiol 1991;53:565-584.)

The α and β subunits are heterogeneous, with α_1, α_2, and α_3 subunits and β_1, β_2, and β_3 subunits described so far. The α_1 isoform is found in the membranes of most cells, whereas α_2 is present in muscle, heart, adipose tissue, and brain, and α_3 is present in heart and brain. The β_1 subunit is widely distributed but is absent in certain astrocytes, vestibular cells of the inner ear, and glycolytic fast-twitch muscles. The fast-twitch muscles contain only β_2 subunits. The different α and β subunit structures of Na, K ATPase in various tissues probably represent specialization for specific tissue functions.

REGULATION OF Na, K ATPase

The amount of Na^+ normally found in cells is not enough to saturate the pump, so if the Na^+ increases, more is pumped out. Pump activity is affected by second messenger molecules (eg, cAMP and diacylglycerol [DAG]). The magnitude and

direction of the altered pump effects vary with the experimental conditions. Thyroid hormones increase pump activity by a genomic action to increase the formation of Na, K ATPase molecules. Aldosterone also increases the number of pumps, although this effect is probably secondary. Dopamine in the kidney inhibits the pump by phosphorylating it, causing a natriuresis. Insulin increases pump activity, probably by a variety of different mechanisms.

SECONDARY ACTIVE TRANSPORT

In many situations, the active transport of Na^+ is coupled to the transport of other substances **(secondary active transport).** For example, the luminal membranes of mucosal cells in the small intestine contain a symport that transports glucose into the cell only if Na^+ binds to the protein and is transported into the cell at the same time. From the cells, the glucose enters the blood. The electrochemical gradient for Na^+ is maintained by the active transport of Na^+ out of the mucosal cell into ECF. Other examples are shown in **Figure 2–20**. In the heart, Na, K ATPase indirectly affects Ca^{2+} transport. An antiport in the membranes of cardiac muscle cells normally exchanges intracellular Ca^{2+} for extracellular Na^+.

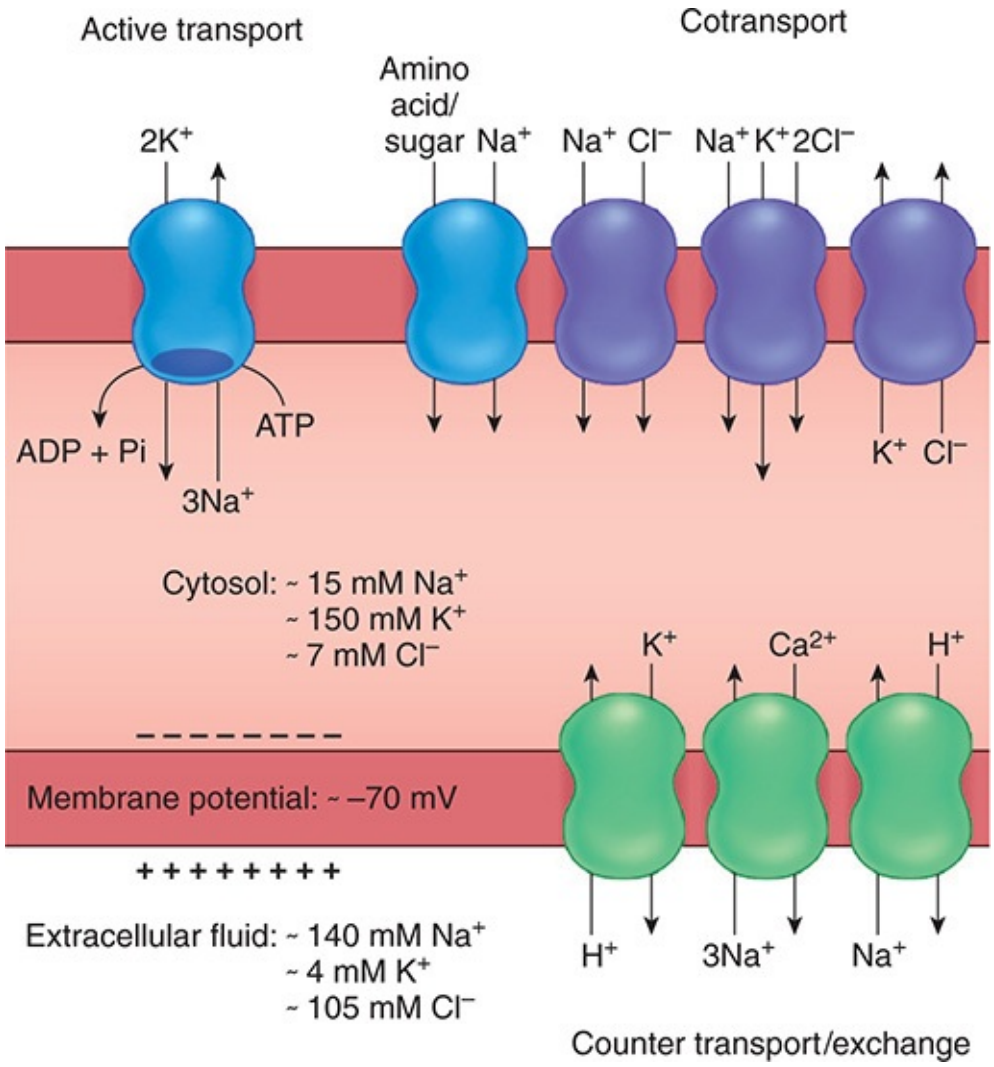


FIGURE 2–20 Composite diagram of main secondary effects of active transport of Na^+ and K^+. Na, K ATPase converts the chemical energy of ATP hydrolysis into maintenance of an inward gradient for Na^+ and an outward gradient for K^+. The energy of the gradients is used for countertransport, cotransport, and maintenance of the membrane potential. Some examples of cotransport and countertransport that use these gradients are shown.

Active transport of Na^+ and K^+ is one of the major energy-using processes in the body. On the average, it accounts for about 24% of the energy utilized by cells, and in neurons it accounts for 70%. Thus, it accounts for a large part of the basal metabolism. A major payoff for this energy use is the establishment of the electrochemical gradient in cells.

TRANSPORT ACROSS EPITHELIA

In the gastrointestinal tract, the pulmonary airways, the renal tubules, and other structures lined with polarized epithelial cells, substances enter one side of a cell and exit another, producing movement of the substance from one side of the epithelium to the other (Figure 2–8). For transepithelial transport to occur, the cells need to be bound by tight junctions and, obviously, have different ion channels and transport proteins in different parts of their membranes. Most of the instances of secondary active transport cited in the preceding paragraph involve transepithelial movement of ions and other molecules.

SPECIALIZED TRANSPORT ACROSS THE CAPILLARY WALL

The capillary wall separating plasma from interstitial fluid is different from the cell membranes separating interstitial fluid from intracellular fluid because the pressure difference across it makes **filtration** a significant factor in producing movement of water and solute. By definition, filtration is the process by which fluid is forced through a membrane or other barrier because of a difference in pressure on the two sides.

The structure of the capillary wall varies from one vascular bed to another. However, near skeletal muscle and many other organs, water and relatively small solutes are the only substances that cross the wall with ease. The apertures in the junctions between the endothelial cells are too small to permit plasma proteins and other colloids to pass through in significant quantities. The colloids have a high molecular weight but are present in large amounts. Small amounts cross the capillary wall by vesicular transport, but their effect is slight. Therefore, the capillary wall behaves like a membrane impermeable to colloids, and these exert an osmotic pressure of about 25 mmHg. The colloid osmotic pressure due to the plasma colloids is called the **oncotic pressure.** Filtration across the capillary membrane as a result of the hydrostatic pressure head in the vascular system is opposed by the oncotic pressure. The way the balance between the hydrostatic and oncotic pressures controls exchanges across the capillary wall is considered in detail in Chapter 31.

TRANSCYTOSIS

Vesicles are present in the cytoplasm of endothelial cells, and tagged protein molecules injected into the bloodstream have been found in the vesicles and in the interstitium. This indicates that small amounts of protein are transported out of capillaries across endothelial cells by endocytosis on the capillary side followed by exocytosis on the interstitial side of the cells. The transport mechanism makes use of coated vesicles that appear to be coated with caveolin and is called **transcytosis, vesicular transport,** or **cytopempsis.**

INTERCELLULAR COMMUNICATION

Cells communicate with one another via chemical messengers. Within a given tissue, some messengers move from cell to cell via gap junctions without entering the ECF. In addition, cells are affected by chemical messengers secreted into the ECF, or by direct cell–cell contacts. Chemical messengers typically bind to protein receptors on the surface of the cell or, in some instances, in the cytoplasm or the nucleus, triggering sequences of intracellular changes that produce their physiologic effects. Three general types of intercellular communication are mediated by messengers in the ECF: (1) **neural communication,** in which neurotransmitters are released at synaptic junctions from nerve cells and act across a narrow synaptic cleft on a postsynaptic cell; (2) **endocrine communication,** in which hormones and growth factors reach cells via the circulating blood or the lymph; and (3) **paracrine communication,** in which the products of cells diffuse in the ECF to affect neighboring cells that may be some distance away **(Figure 2–21)**. In addition, cells secrete chemical messengers that in some situations bind to receptors on the same cell, that is, the cell that secreted the messenger **(autocrine communication).** The chemical messengers include amines, amino acids, steroids, polypeptides, and in some instances, lipids, purine nucleotides, and pyrimidine nucleotides. It is worth noting that in various parts of the body, the same chemical messenger can function as a neurotransmitter, a paracrine mediator, a hormone secreted by neurons into the blood (neural hormone), and a hormone secreted by gland cells into the blood.

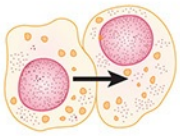
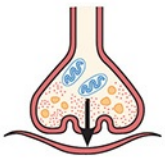
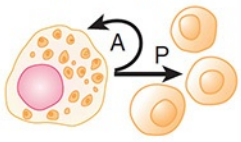
	GAP JUNCTIONS	SYNAPTIC	PARACRINE AND AUTOCRINE	ENDOCRINE
				
Message transmission	Directly from cell to cell	Across synaptic cleft	By diffusion in interstitial fluid	By circulating body fluids
Local or general	Local	Local	Locally diffuse	General
Specificity depends on	Anatomic location	Anatomic location and receptors	Receptors	Receptors

FIGURE 2–21 Intercellular communication by chemical mediators. Several common forms of cellular communication are illustrated. A, autocrine; P, paracrine.

An additional form of intercellular communication is called **juxtacrine communication.** Some cells express multiple repeats of growth factors such as **transforming growth factor alpha (TGF-α)** extracellularly on transmembrane proteins that provide an anchor to the cell. Other cells have TGF-α receptors. Consequently, TGF-α anchored to a cell can bind to a TGF-α receptor on another cell, linking the two. This could be important in producing local foci of growth in tissues. Notch-mediated juxtacrine signaling between adjacent cells is another good example of juxtacrine communication. Notch ligands, such as Jagged (Jag1/2) and Delta-like (DLL1/3/4), in signal sending cells first bind to Notch receptors, such as Notch1-4, in signal receiving cells, through cell-to-cell contact. The activated Notch receptors mediate a series of signal transduction cascades involved in regulating cell fate decisions, cell proliferation and differentiation, and tissue and organ development.

RECEPTORS FOR CHEMICAL MESSENGERS

The recognition of chemical messengers by cells typically begins by interaction with a receptor at that cell. There have been over 20 families of receptors for chemical messengers characterized. These proteins are not static components of the cell, but their numbers increase and decrease in response to various stimuli, and their properties change with changes in physiologic conditions. When a hormone or neurotransmitter is present in excess, the number of active receptors generally decreases **(downregulation),** whereas in the presence of a deficiency

of the chemical messenger, there is an increase in the number of active receptors **(upregulation).** In its actions on the adrenal cortex, angiotensin II is an exception; it increases rather than decreases the number of its receptors in the adrenal. In the case of receptors in the membrane, receptor-mediated endocytosis is responsible for downregulation in some instances; ligands bind to their receptors, and the ligand-receptor complexes move laterally in the membrane to coated pits, where they are taken into the cell by endocytosis **(internalization).** This decreases the number of receptors in the membrane. Some receptors are recycled after internalization, whereas others are replaced by de novo synthesis in the cell. Another type of downregulation is **desensitization,** in which receptors are chemically modified in ways that make them less responsive.

MECHANISMS BY WHICH CHEMICAL MESSENGERS ACT

Receptor–ligand interaction is usually just the beginning of the cell response. This event is transduced into secondary responses within the cell that can be divided into four broad categories: (1) ion channel activation, (2) **G-protein** activation, (3) activation of enzyme activity within the cell, or (4) direct activation of transcription. Within each of these groups, responses can be quite varied. Some of the common mechanisms by which chemical messengers exert their intracellular effects are summarized in **Table 2–3.** Ligands such as acetylcholine bind directly to ion channels in the cell membrane, changing their conductance. Thyroid and steroid hormones, 1,25-dihydroxycholecalciferol, and retinoids enter cells and act on one or another member of a family of structurally related cytoplasmic or nuclear receptors. The activated receptor binds to DNA and increases transcription of selected mRNAs. Many other ligands in the ECF bind to receptors on the surface of cells and trigger the release of intracellular mediators such as cAMP, inositol trisphosphate (IP_3), and DAG that initiate changes in cell function. Consequently, the extracellular ligands are called **“first messengers”** and the intracellular mediators are called **“second messengers.”** Second messengers bring about many short-term changes in cell function by altering enzyme function, triggering exocytosis, and so on, but they also can lead to the alteration of transcription of various genes. A variety of enzymatic changes, protein–protein interactions, or second messenger changes can be activated within a cell in an orderly fashion following receptor recognition of the primary messenger. The resulting **cell signaling pathway** provides amplification

of the primary signal and distribution of the signal to appropriate targets within the cell. Extensive cell signaling pathways also provide opportunities for feedback and regulation that can fine-tune the signal for the correct physiologic response by the cell.

TABLE 2–3 Common mechanisms by which chemical messengers in the ECF bring about changes in cell function.

Mechanism	Examples
Open or close ion channels in cell membrane	Acetylcholine on nicotinic cholinergic receptor; norepinephrine on K^+ channel in the heart; capsaicin on transient receptor potential cation channel vanilloid member 1 (TRPV1)
Act via cytoplasmic or nuclear receptors to increase transcription of selected mRNAs	Thyroid hormones, retinoic acid, steroid hormones
Activate phospholipase C with intracellular production of DAG, IP_3, and other inositol phosphates	Angiotensin II, norepinephrine via α_1-adrenergic receptor, vasopressin via V_1 receptor
Activate or inhibit adenylyl cyclase, causing increased or decreased intracellular production of cAMP	Norepinephrine via β_1-adrenergic receptor (increased cAMP), norepinephrine via α_2-adrenergic receptor (decreased cAMP)
Increase cGMP in cell	Atrial natriuretic peptide, nitric oxide
Increase tyrosine kinase activity of cytoplasmic portions of transmembrane receptors	Insulin, EGF, PDGF, M-CSF
Increase serine or threonine kinase activity	TGF-β, activin, inhibin

cAMP, cyclic adenosine 3′,5′-monophosphate; cGMP, cyclic guanosine monophosphate; DAG, diacylglycerol; ECF, extracellular fluid; EGF, epidermal growth factor; IP_3, inositol triphosphate; M-CSF, monocyte colony-stimulating factor; PDGF, platelet-derived growth factor; TGF-β, transforming growth factor β.

The most predominant posttranslation modification of proteins, phosphorylation, is a common theme in cell signaling pathways. Cellular phosphorylation is under the control of two groups of proteins: **kinases,** enzymes that catalyze the phosphorylation of tyrosine or serine and threonine residues in proteins (or in some cases, in lipids); and **phosphatases,** proteins that remove phosphates from proteins (or lipids). Some of the larger receptor families are themselves kinases. Tyrosine kinase receptors initiate phosphorylation on tyrosine residues on complementary receptors following ligand binding. Serine/threonine kinase receptors initiate phosphorylation on serines or threonines in complementary receptors following ligand binding. Cytokine receptors are directly associated with a group of protein kinases that are activated following cytokine binding. Alternatively, second messenger changes can lead to phosphorylation further downstream in the signaling pathway. More than 500 protein kinases have been described. Some of the principal ones that are important in mammalian cell signaling are summarized in **Table 2–4**. In general, addition of phosphate groups changes the conformation of the proteins, altering their functions and consequently the functions of the cell. The close relationship between phosphorylation and dephosphorylation of cellular proteins allows for a temporal control of activation of cell signaling pathways. This is sometimes referred to as a **"phosphate timer."** The dysregulation of the phosphate timer and subsequent cellular signaling in a cell can lead to disease **(Clinical Box 2–7)**.

TABLE 2–4 Sample protein kinases.

Phosphorylate serine or threonine residues, or both
Calmodulin-dependent
Myosin light-chain kinase
Phosphorylase kinase
Ca^{2+}/calmodulin kinase I
Ca^{2+}/calmodulin kinase II
Ca^{2+}/calmodulin kinase III
Calcium-phospholipid-dependent
Protein kinase C (more than 10 subtypes)
Cyclic nucleotide-dependent
cAMP-dependent kinase (protein kinase A; two subtypes)
cGMP-dependent kinase
Phosphorylate tyrosine residues
Insulin receptor, EGF receptor, PDGF receptor, and M-CSF receptor

cAMP, cyclic adenosine 3′,5′-monophosphate; cGMP, cyclic guanosine monophosphate; EGF, epidermal growth factor; M-CSF, monocyte colony-stimulating factor; PDGF, platelet-derived growth factor.

STIMULATION OF TRANSCRIPTION

The activation of transcription, and subsequent translation, is a common outcome of cellular signaling. There are three distinct pathways for primary messengers to alter transcription of cells. First, as is the case with steroid or thyroid hormones, the primary messenger is able to cross the cell membrane and bind to a nuclear receptor, which then can directly interact with DNA to alter gene expression. A second pathway to gene transcription is the activation of cytoplasmic protein kinases that can move to the nucleus to phosphorylate a latent transcription factor for activation. This pathway is a common end point of signals that go through the **mitogen-activated protein (MAP) kinase** cascade. MAP kinases can be activated following a variety of receptor–ligand interactions through second messenger signaling. They comprise a series of three kinases that coordinate a stepwise phosphorylation to activate each protein in series in the cytosol. Phosphorylation of the last MAP kinase in series allows it to migrate to the nucleus where it phosphorylates a latent transcription factor. A third common pathway is the activation of a latent transcription factor in the cytosol, which

then migrates to the nucleus and alters transcription. This pathway is shared by a diverse set of transcription factors that include **nuclear factor kappa B (NF-κB;** activated following tumor necrosis family receptor binding and others) and **signal transducers of activated transcription (STATs;** activated following cytokine receptor binding). In all cases, the binding of the activated transcription factor to DNA increases (or in some cases, decreases) the transcription of mRNAs encoded by the gene to which it binds. The mRNAs are translated in the ribosomes, with the production of increased quantities of proteins that alter cell function.

CLINICAL BOX 2–7

Kinases in Cancer: Chronic Myeloid Leukemia

Kinases frequently play important roles in regulating cellular physiology outcomes, including cell growth and cell death. Dysregulation of cell proliferation or cell death is a hallmark of cancer. Although cancer can have many causes, a role for kinase dysregulation is exemplified in chronic myeloid leukemia (CML). CML is a pluripotent hematopoietic stem cell disorder characterized by the Philadelphia (Ph) chromosome translocation. The Ph chromosome is formed following a translocation of chromosomes 9 and 22, resulting in a shortened chromosome 22 (Ph chromosome). At the point of fusion, a novel gene (*bcr-abl*) encoding the active tyrosine kinase domain from a gene on chromosome 9 (Abelson tyrosine kinase; c-Abl) is fused to novel regulatory region of a separate gene on chromosome 22 (breakpoint cluster region; bcr). The *bcr-abl* fusion gene encodes a cytoplasmic protein with constitutively active tyrosine kinase. The dysregulated kinase activity in bcr-abl protein effectively limits white blood cell death signaling pathways while promoting cell proliferation and genetic instability. Experimental models have shown that translocation to produce the fusion bcr-abl protein is sufficient to produce CML in animal models.

THERAPEUTIC HIGHLIGHTS

The identification of *bcr-abl* as the initial transforming event in CML provided an ideal target for drug discovery. The drug imatinib was developed to specifically block the tyrosine kinase activity of the bcr-abl protein. Imatinib has proven to be an effective agent for treating chronic phase CML.

INTRACELLULAR CA^{2+} AS A SECOND MESSENGER

Ca^{2+} regulates a very large number of physiological processes that are as diverse as proliferation, neural signaling, learning, contraction, secretion, and fertilization, so regulation of intracellular Ca^{2+} is of great importance. The free Ca^{2+} concentration in the cytoplasm at rest is maintained at about 100 nmol/L. The Ca^{2+} concentration in the interstitial fluid is about 12,000 to 18,000 times the cytoplasmic concentration (ie, 1,200,000 to 1,8,00,000 nmol/L), so there is a marked inwardly directed concentration gradient as well as an inwardly directed electrical gradient. Much of the intracellular Ca^{2+} is stored at relatively high concentrations in the endoplasmic reticulum (approximately 100,000 to 1,200,000 nmol/L) and other organelles **(Figure 2–22)**, and these organelles provide a store from which Ca^{2+} can be mobilized via ligand-gated channels in the endoplasmic reticulum (eg, ryanodine receptors and inositol trisphosphate receptors) to increase the concentration of free Ca^{2+} in the cytoplasm. Increased cytoplasmic Ca^{2+} binds to and activates calcium-binding proteins. These proteins can have direct effects in cellular physiology, or can activate other proteins, commonly protein kinases, to further cell signaling pathways.

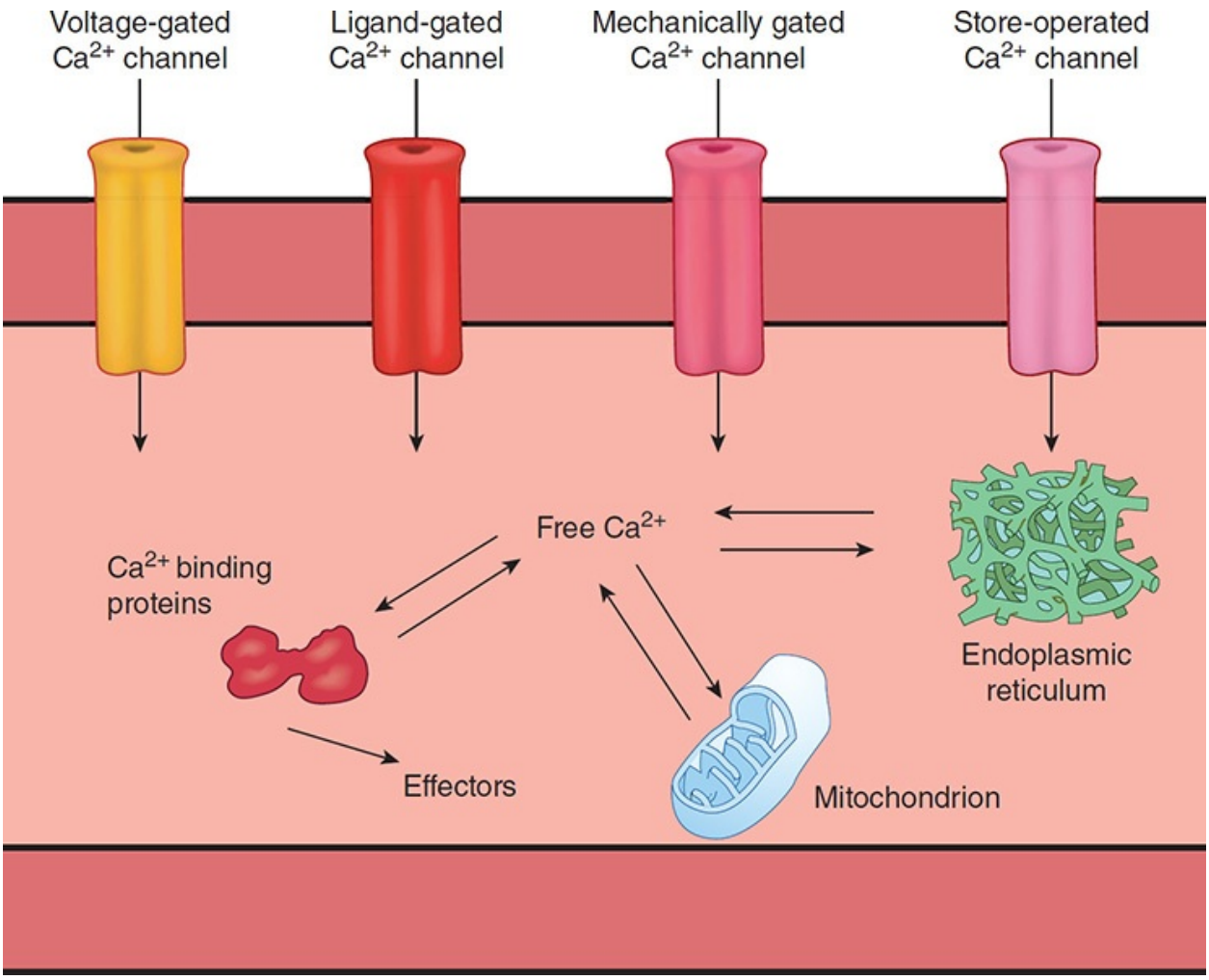


FIGURE 2–22 Ca^{2+} handling in mammalian cells. Ca^{2+} can enter the cell via a variety of channel types. In addition, Ca^{2+} is stored in the endoplasmic reticulum (and, to a lesser extent in the mitochondrion) where it can be released to alter free Ca^{2+} concentration in the cytoplasm. Free Ca^{2+} can be bound by proteins that then have a variety of downstream physiologic effects. Ca^{2+} can be removed from the cytoplasm by ATPases in the endoplasmic reticulum or at the plasma membrane, or via Na, Ca exchangers (not shown).

Ca^{2+} can enter the cell from the ECF, down its electrochemical gradient, through many different Ca^{2+} channels. Some of these are ligand-gated and others are voltage-gated. Stretch-activated channels exist in some cells as well.

Many second messengers act by increasing the cytoplasmic Ca^{2+} concentration. The increase is produced by releasing Ca^{2+} from intracellular stores—primarily the endoplasmic reticulum—or by increasing the entry of Ca^{2+}

into cells, or by both mechanisms. IP_3 is the major second messenger that causes Ca^{2+} release from the endoplasmic reticulum through the direct activation of a ligand-gated channel, the IP_3 receptor. In effect, the generation of one second messenger (IP_3) can lead to the release of another second messenger (Ca^{2+}). In many tissues, transient release of Ca^{2+} from internal stores into the cytoplasm triggers opening of a population of Ca^{2+} channels in the cell membrane **(store-operated Ca^{2+} channels; SOCCs).** The resulting Ca^{2+} influx replenishes the total intracellular Ca^{2+} supply and refills the endoplasmic reticulum. Recent research has identified the physical relationships between SOCCs (eg, Orai channels) and regulatory interactions of proteins (eg, stromal interaction molecules 1 and 2) from the endoplasmic reticulum that facilitate the recruitment of Orai to form SOCCs.

As with other second messenger molecules, the increase in Ca^{2+} within the cytosol is rapid, and is followed by a rapid decrease. Because the movement of Ca^{2+} outside of the cytosol (ie, across the plasma membrane or the membrane of the internal store) requires that it move up its electrochemical gradient, it requires energy. Ca^{2+} movement out of the cell is facilitated by the plasma membrane Ca^{2+} ATPase. Alternatively, it can be transported by an antiport (ie, Na^+/Ca^{2+} exchanger) that exchanges three Na^+ for each Ca^{2+} driven by the energy stored in the Na^+ electrochemical gradient. Ca^{2+} movement into the internal stores is through the action of the **sarcoplasmic** or **endoplasmic reticulum Ca^{2+} ATPase,** also known as the **SERCA pump.**

CALCIUM-BINDING PROTEINS

Many different Ca^{2+}-binding proteins have been described, including **troponin, calmodulin,** and **calbindin.** Troponin is the Ca^{2+}-binding protein involved in contraction of skeletal muscle (Chapter 5). Calmodulin contains 148 amino acid residues **(Figure 2–23)** and has four Ca^{2+}-binding domains. It is unique in that amino acid residue 115 is trimethylated, and it is extensively conserved, being found in plants as well as animals. When calmodulin binds Ca^{2+}, it is capable of activating five different calmodulin-dependent kinases (CaMKs; Table 2–4), among other proteins. One of the kinases is **myosin light-chain kinase,** which phosphorylates myosin. This brings about contraction in smooth muscle. CaMKI and CaMKII are concerned with synaptic function, and CaMKIII is concerned with protein synthesis. Another calmodulin-activated protein is **calcineurin,** a

phosphatase that inactivates Ca^{2+} channels by dephosphorylating them. It also plays a prominent role in activating T cells and is inhibited by some immunosuppressants.

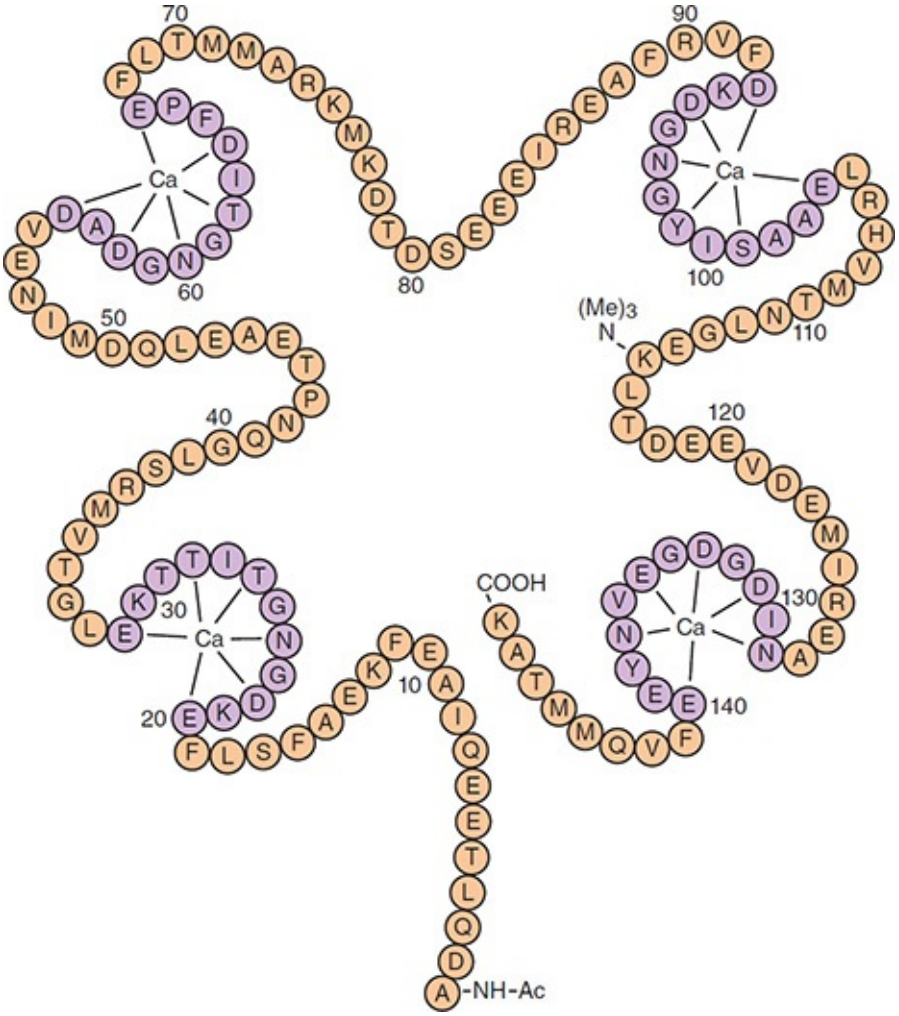


FIGURE 2–23 Secondary structure of calmodulin from bovine brain. Single-letter abbreviations are used for the amino acid residues. Note the four calcium domains (purple residues) flanked on either side by stretches of amino acids that form α-helices in tertiary structure. (Reproduced with permission from Cheung WY: Calmodulin: An overview. Fed Proc 1982;May;41(7):2253-2257.)

MECHANISMS OF DIVERSITY OF Ca^{2+} ACTIONS

It may seem difficult to understand how intracellular Ca^{2+} can have so many

varied effects as a second messenger. Part of the explanation is that Ca^{2+} may have different effects at low and at high concentrations. The ion may be at high concentration at the site of its release from an organelle or a channel **(Ca2+ sparks)** and at a subsequent lower concentration after it diffuses throughout the cell. Some of the changes it produces can outlast the rise in intracellular Ca^{2+} concentration because of the way it binds to some of the Ca^{2+}-binding proteins. In addition, once released, intracellular Ca^{2+} concentrations frequently oscillate at regular intervals, and there is evidence that the frequency and, to a lesser extent, the amplitude of those oscillations codes information for effector mechanisms. Finally, increases in intracellular Ca^{2+} concentration can spread from cell to cell in waves, producing coordinated events such as the rhythmic beating of cilia in airway epithelial cells.

G-PROTEINS

A common way to translate a signal to a biological effect inside cells is by way of nucleotide regulatory proteins that are activated after binding GTP **(G-proteins).** When an activating signal reaches a G-protein, the protein exchanges GDP for GTP. The GTP–protein complex brings about the activating effect of the G-protein. The inherent GTPase activity of the protein then converts GTP to GDP, restoring the G-protein to an inactive resting state. G-proteins can be divided into two principal groups involved in cell signaling: **small G-proteins** and **heterotrimeric G-proteins.** Other groups that have similar regulation and are also important to cell physiology include elongation factors, dynamin, and translocation GTPases.

There are several different families of small G-proteins (or **small GTPases**) that are all highly regulated. **GTPase activating proteins (GAPs)** tend to inactivate small G-proteins by encouraging hydrolysis of GTP to GDP in the central binding site. **Guanine exchange factors (GEFs)** tend to activate small G-proteins by encouraging exchange of GDP for GTP in the active site. Some of the small G-proteins contain lipid modifications that help anchor them to membranes, while others are free to diffuse throughout the cytosol. Small G-proteins are involved in many cellular functions. Members of the Rab family regulate the rate of vesicle traffic between the endoplasmic reticulum, the Golgi apparatus, lysosomes, endosomes, and the cell membrane. Another family of small GTP-binding proteins, the Rho/Rac family, mediates interactions between the cytoskeleton and cell membrane. The Ras family regulates growth by transmitting signals from the cell membrane to the nucleus.

Another family of G-proteins, the larger **heterotrimeric G-proteins,** couple cell surface receptors to catalytic units that catalyze the intracellular formation of second messengers or couple the receptors directly to ion channels. Despite the knowledge of the small G-proteins described above, the heteromeric G-proteins are frequently referred to in the shortened "G-protein" form because they were the first to be identified. Heterotrimeric G-proteins are made up of three subunits designated α, β, and γ **(Figure 2–24)**. Both the α and the γ subunits have lipid modifications that anchor these proteins to the plasma membrane. The α subunit is bound to GDP. When a ligand binds to a G-protein-coupled receptor (GPCR, discussed below), this GDP is exchanged for GTP and the α subunit separates from the combined β and γ subunits. The separated α subunit brings about many biologic effects. The β and γ subunits are tightly bound in the cell and together form a signaling molecule that can also activate a variety of effectors. The intrinsic GTPase activity of the α subunit then converts GTP to GDP, and this leads to reassociation of the α with the βγ subunit and termination of effector activation. The GTPase activity of the α subunit can be accelerated by a family of **regulators of G-protein signaling (RGS).**

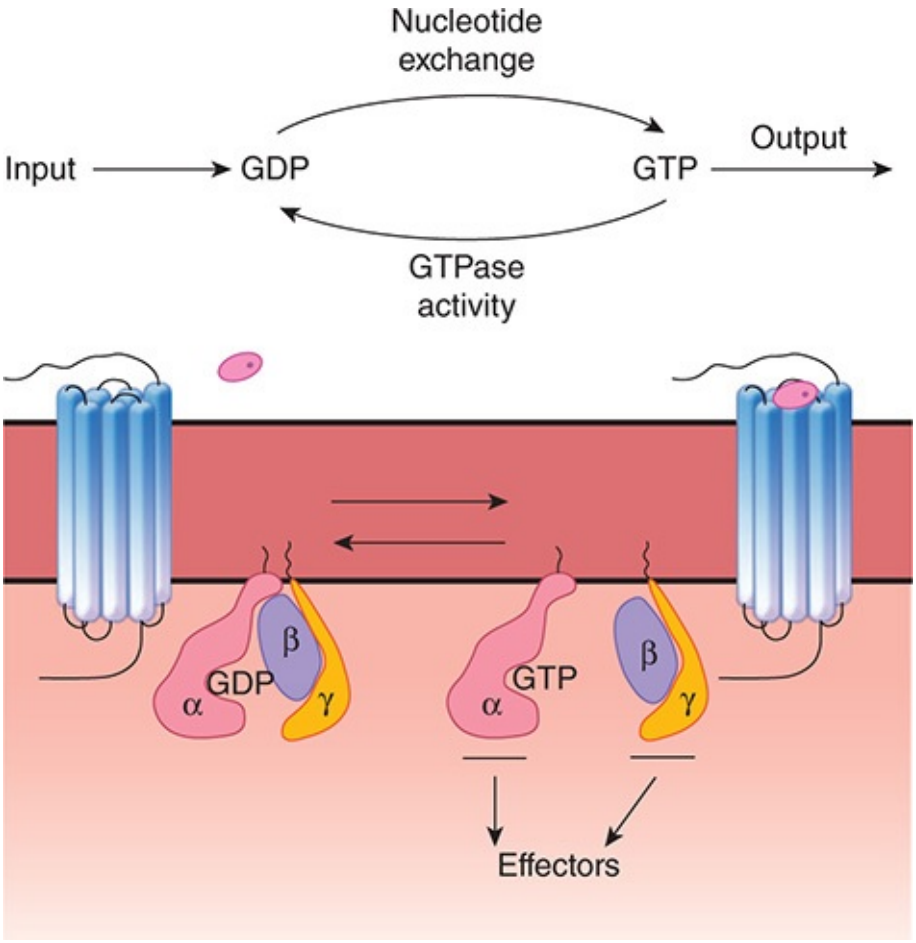


FIGURE 2–24 Heterotrimeric G-proteins. Top: Summary of overall reaction that occurs in the Gα subunit. **Bottom:** When the ligand (red oval) binds to the G-protein-coupled receptor in the cell membrane, GTP replaces GDP on the α subunit. GTP-α separates from the βγ subunit and GTP-α and βγ both activate various effectors, producing physiologic effects. The intrinsic GTPase activity of GTP-α then converts GTP to GDP, and the α, β, and γ subunits reassociate.

Heterotrimeric G-proteins relay signals from over 1000 GPCRs, and their effectors in the cells include ion channels and enzymes. There are 20 α, 6 β, and 12 γ genes, which allow for over 1400 α, β, and γ combinations. Not all combinations occur in the cell, but over 20 different heterotrimeric G-proteins have been well documented in cell signaling. They can be divided into five families, each with a relatively characteristic set of effectors.

G-PROTEIN-COUPLED RECEPTORS

All the **GPCRs** that have been characterized to date are proteins that span the cell membrane seven times. Because of this structure they are alternatively referred to as **seven-helix receptors** or **serpentine receptors.** A very large number have been cloned, and their functions are multiple and diverse. This is emphasized by the extensive variety of ligands that target GPCRs **(Table 2–5)**. The structures of four GPCRs are shown in **Figure 2–25**. These receptors assemble into a barrel-like structure. Upon ligand binding, a conformational change activates a resting heterotrimeric G-protein associated with the cytoplasmic leaf of the plasma membrane. Activation of a single receptor can result in 1, 10, or more active heterotrimeric G-proteins, providing amplification as well as transduction of the first messenger. Bound receptors can be inactivated to limit the amount of cellular signaling. This frequently occurs through phosphorylation of the cytoplasmic side of the receptor. Because of their diversity and importance in cellular signaling pathways, GPCRs are prime targets for drug discovery **(Clinical Box 2–8)**.

TABLE 2–5 Examples of ligands for G-protein-coupled receptors.

Class	Ligand
Neurotransmitters	Epinephrine Norepinephrine Dopamine 5-Hydroxytryptamine Histamine Acetylcholine Adenosine Opioids
Tachykinins	Substance P Neurokinin A Neuropeptide K
Other peptides	Angiotensin II Arginine vasopressin Oxytocin VIP, GRP, TRH, PTH
Glycoprotein hormones	TSH, FSH, LH, hCG
Arachidonic acid derivatives	Thromboxane A_2
Other	Odorants Tastants Endothelins Platelet-activating factor Cannabinoids Light Cations (eg, Ca^{2+}, Mg^{2+})

FSH, follicle-stimulating hormone; GRP, gastrin-releasing hormone; hCG, human chorionic gonadotropin; LH, luteinizing hormone; PTH, parathyroid hormone; TRH, thyrotropin-releasing hormone; TSH, thyroid-stimulating hormone; VIP, vasoactive intestinal peptide.

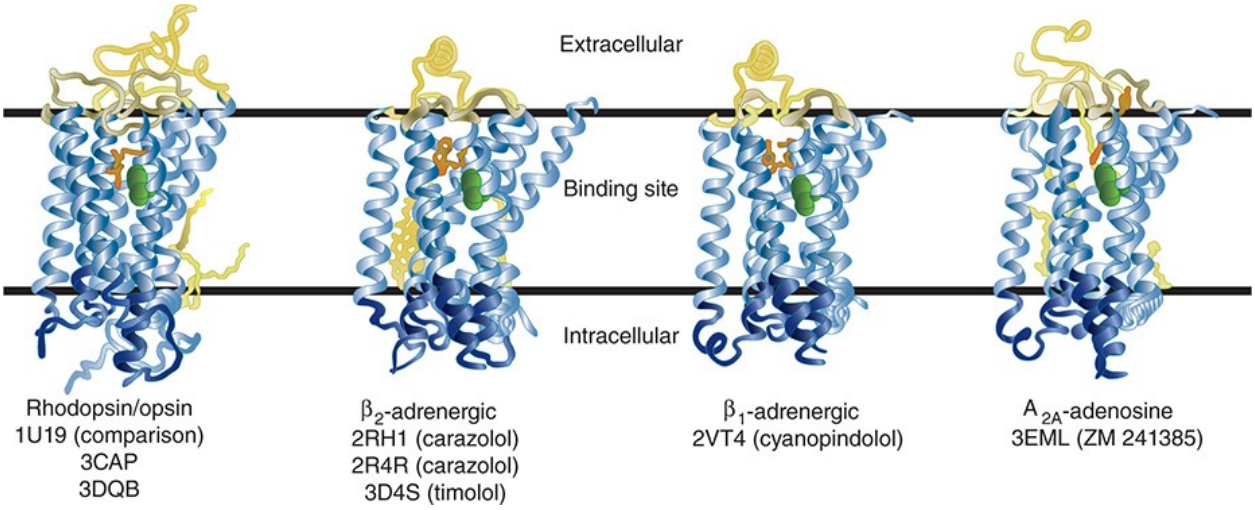


FIGURE 2–25 Representative structures of four G-protein-coupled receptors from solved crystal structures. Each group of receptors is represented by one structure, all rendered with the same orientation and color scheme: transmembrane helices are colored light blue, intracellular regions are colored darker blue, and extracellular regions are brown. Each ligand is colored orange and rendered as sticks, bound lipids are colored yellow, and the conserved tryptophan residue is rendered as spheres and colored green. This figure highlights the observed differences seen in the extracellular and intracellular domains as well as the small differences seen in the ligand binding orientations among the four GPCRs various ligands. (Reproduced with permission from Hanson MA, Stevens RC: Discovery of new GPCR biology: one receptor structure at a time. Structure 1988;Jan 14;17(1):8-14.)

INOSITOL TRISPHOSPHATE & DIACYLGLYCEROL AS SECOND MESSENGERS

The link between membrane binding of a ligand that acts via Ca^{2+} and the prompt increase in the cytoplasmic Ca^{2+} concentration is often **IP_3**. When one of these ligands binds to its receptor, activation of the receptor produces activation of phospholipase C (PLC) on the inner surface of the membrane. Ligands bound to GPCR can do this through the G_q heterotrimeric G-proteins, while ligands bound to tyrosine kinase receptors can do this through other cell signaling pathways. PLC has at least eight isoforms; PLC_β is activated by heterotrimeric G-proteins, while PLCγ forms are activated through tyrosine kinase receptors. PLC isoforms can catalyze the hydrolysis of the membrane lipid

phosphatidylinositol 4,5-diphosphate (PIP_2) to form IP_3 and **DAG** **(Figure 2–26)**. The IP_3 diffuses to the endoplasmic reticulum where it triggers the release of Ca^{2+} into the cytoplasm by binding the IP_3 receptor, a ligand-gated Ca^{2+} channel **(Figure 2–27)**. DAG is also a second messenger; it stays in the cell membrane where it activates one of several isoforms of **protein kinase C** and activates receptor-operated Ca^{2+} channels (ROCCs) in the plasma membrane to further increase intracellular Ca^{2+} concentration.

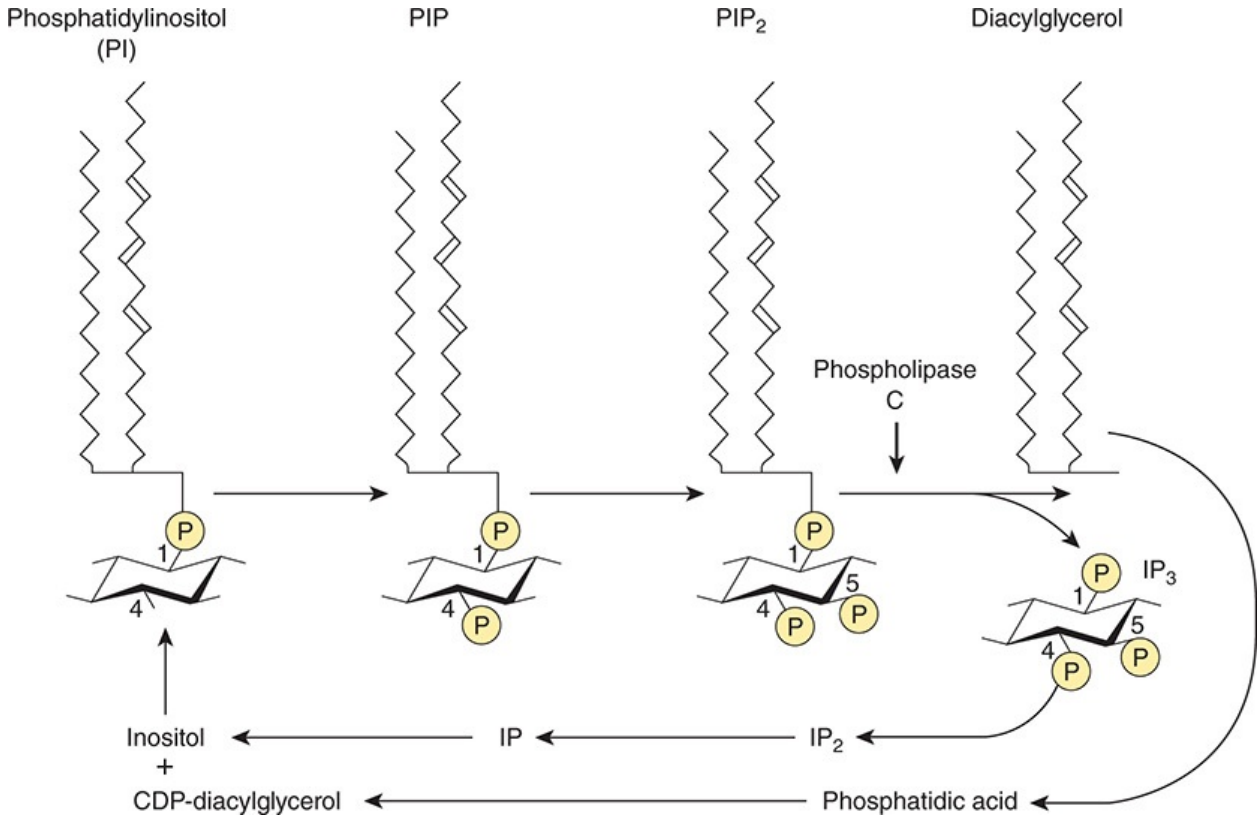


FIGURE 2–26 Metabolism of phosphatidylinositol in cell membranes. Phosphatidylinositol is successively phosphorylated to form phosphatidylinositol 4-phosphate (PIP), then phosphatidylinositol 4,5-bisphosphate (PIP_2). Phospholipase C_β and phospholipase $C\gamma$ catalyze the breakdown of PIP_2 to inositol 1,4,5-trisphosphate (IP_3) and diacylglycerol. Other inositol phosphates and phosphatidylinositol derivatives can also be formed. IP_3 is dephosphorylated to inositol, and diacylglycerol is metabolized to cytosine diphosphate (CDP)-diacylglycerol. CDP-diacylglycerol and inositol then combine to form phosphatidylinositol, completing the cycle. (Modified with permission from Berridge MJ: Inositol triphosphate and diacylglycerol as second messengers.

Biochem J 1984;June 1;220(2):345-360.)

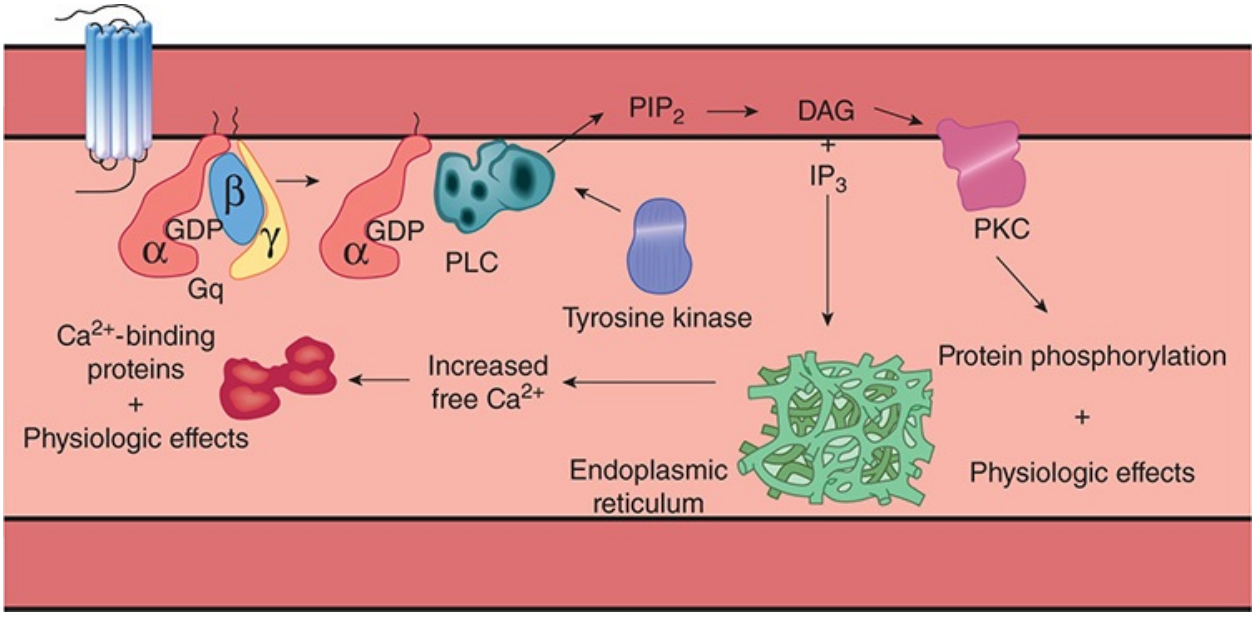


FIGURE 2–27 Diagrammatic representation of release of inositol trisphosphate (IP_3) and diacylglycerol (DAG) as second messengers. Binding of ligand to G-protein-coupled receptor activates phospholipase C (PLC). Alternatively, activation of receptors with intracellular tyrosine kinase domains can activate PLCγ. The resulting hydrolysis of phosphatidylinositol 4,5-diphosphate (PIP_2) produces IP_3, which releases Ca^{2+} from the endoplasmic reticulum (ER), and DAG, which activates protein kinase C (PKC).

CLINICAL BOX 2–8

Drug Development: Targeting the G-Protein-Coupled Receptors (GPCRs)

GPCRs are among the most heavily investigated drug targets in the pharmaceutical industry, representing approximately 40% of all the drugs in the marketplace today. These proteins are active in just about every organ system and present a wide range of opportunities as therapeutic targets in areas including cancer, cardiac dysfunction, diabetes, central nervous system disorders, obesity, inflammation, and pain. Features of GPCRs that allow them to be drug targets are their specificity in recognizing extracellular ligands to initiate cellular response, the cell surface location of GPCRs that

make them accessible to novel ligands or drugs, and their prevalence in leading to human pathology and disease.

Specific examples of successful GPCR drug targets are noted with two types of **histamine receptors.**

Histamine-1 receptor (H_1-receptor) antagonists: allergy therapy. Allergens can trigger local mast cells or basophils to release histamine in the airway. A primary target for histamine is the H_1-receptor in several airway cell types and this can lead to transient itching, sneezing, rhinorrhea, and nasal congestion. There are a variety of medications with improved peripheral H_1-receptor selectivity that are currently used to block histamine activation of the H_1-receptor and thus limit allergen effects in the upper airway. H_1-receptor antagonists on the market include loratadine, fexofenadine, cetirizine, and desloratadine. These "second" and "third" generation medications have improved specificity and reduced adverse side effects (eg, drowsiness and central nervous system dysfunction) associated with some of the "first" generation drugs first introduced in the late 1930s and widely developed over the next 40 years.

Histamine-2 receptor (H_2-receptor) antagonists: treating excess stomach acid. Excess stomach acid can result in gastroesophageal reflux disease or even peptic ulcer symptoms. The parietal cell in the stomach can be stimulated to produce acid via histamine action at the H_2-receptor. Excess stomach acid results in heartburn. Antagonists or H_2-receptor blockers, reduce acid production by preventing H_2-receptor signaling that leads to production of stomach acid. There are several drugs (eg, ranitidine, famotidine, cimetidine, and nizatidine) that specifically block the H_2-receptor and thus reduce excess acid production.

CYCLIC AMP

Another important second messenger is **cAMP** (Figure 2–28). cAMP is formed from ATP by the action of the enzyme **adenylyl cyclase** and converted to physiologically inactive 5' AMP by the action of the enzyme **phosphodiesterase.** Some of the phosphodiesterase isoforms that break down cAMP are inhibited by methylxanthines such as caffeine and theophylline. Consequently, these compounds can augment hormonal and transmitter effects

mediated via cAMP. cAMP activates one of the cyclic nucleotide-dependent protein kinases **(protein kinase A, PKA)** that, like protein kinase C, catalyzes the phosphorylation of proteins, changing their conformation and altering their activity. In addition, the active catalytic subunit of PKA moves to the nucleus and phosphorylates the **cAMP-responsive element-binding protein (CREB).** This transcription factor then binds to DNA and alters transcription of a number of genes.

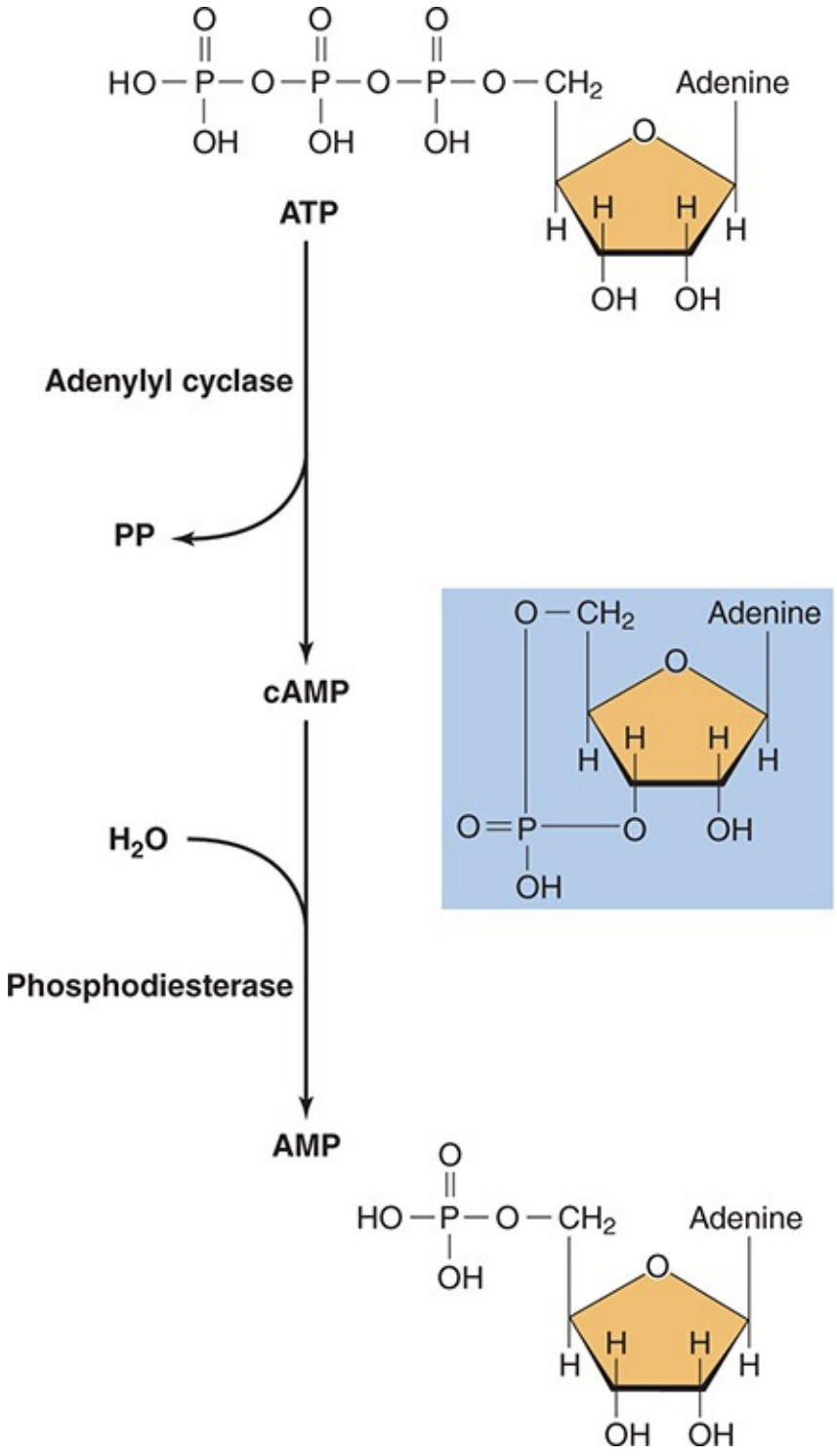


FIGURE 2–28 Formation and metabolism of cAMP. The second messenger cAMP is made from ATP by adenylyl cyclase and broken down into AMP by phosphodiesterase.

PRODUCTION OF cAMP BY ADENLYL CYCLASE

Adenylyl cyclase is a membrane bound protein with 12 transmembrane regions. Ten isoforms of this enzyme have been described and each can have distinct regulatory properties, permitting the cAMP pathway to be customized to specific tissue needs. Notably, stimulatory heterotrimeric G-proteins (G_s) activate, while inhibitory heterotrimeric G-proteins (G_i) inactivate adenylyl cyclase **(Figure 2–29)**. When the appropriate ligand binds to a stimulatory receptor, a G_s α subunit activates one of the adenylyl cyclases. Conversely, when the appropriate ligand binds to an inhibitory receptor, a G_i α subunit inhibits adenylyl cyclase. The receptors are specific, responding at low threshold to only one or a select group of related ligands. However, heterotrimeric G-proteins mediate the stimulatory and inhibitory effects produced by many different ligands. In addition, cross-talk occurs between the phospholipase C system and the adenylyl cyclase system, as several of the isoforms of adenylyl cyclase are stimulated by calmodulin. Finally, the effects of protein kinase A and protein kinase C are very widespread and can also affect directly, or indirectly, the activity at adenylyl cyclase. The close relationship between activation of G-proteins and adenylyl cyclases also allows for spatial regulation of cAMP production. All of these events, and others, allow for fine-tuning the cAMP response for a particular physiologic outcome in the cell.

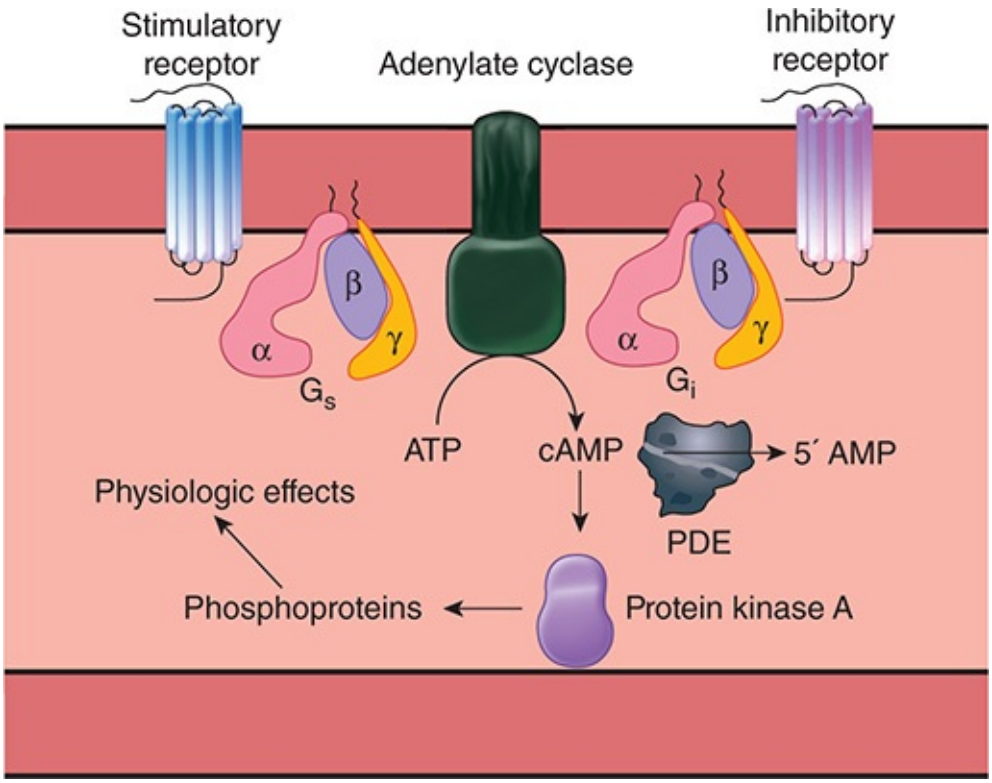


FIGURE 2–29 The cAMP system. Activation of adenylyl cyclase catalyzes the conversion of ATP to cAMP. Cyclic AMP activates protein kinase A, which phosphorylates proteins, producing physiologic effects. Stimulatory ligands bind to stimulatory receptors and activate adenylyl cyclase via G_s. Inhibitory ligands inhibit adenylyl cyclase via inhibitory receptors and G_i.

Two bacterial toxins have important effects on adenylyl cyclase that are mediated by G-proteins. The A subunit of **cholera toxin** catalyzes the transfer of ADP ribose to an arginine residue in the middle of the α subunit of G_s. This inhibits its GTPase activity, producing prolonged stimulation of adenylyl cyclase. **Pertussis toxin** catalyzes ADP-ribosylation of a cysteine residue near the carboxyl terminal of the α subunit of G_i. This inhibits the function of G_i. In addition to the implications of these alterations in disease, both toxins are used for fundamental research on G-protein function. The compound forskolin also stimulates adenylyl cyclase activity by a direct action on the enzyme, and is commonly used in research studies to evaluate adenylyl cyclase/cAMP contributions to cellular physiology.

GUANYLYL CYCLASE

Another cyclic nucleotide of physiologic importance is **cyclic guanosine monophosphate (cyclic GMP** or **cGMP).** cGMP is important in vision in both rod and cone cells. In addition, there are cGMP-regulated ion channels, and cGMP activates cGMP-dependent kinase, producing a number of physiologic effects.

Guanylyl cyclases are a family of enzymes that catalyze the formation of cGMP. They exist in two forms **(Figure 2–30)**. One form has an extracellular amino terminal domain that is a receptor, a single transmembrane domain, and a cytoplasmic portion with guanylyl cyclase catalytic activity. Several such guanylyl cyclases have been characterized. Two are receptors for atrial natriuretic peptide (ANP; also known as atrial natriuretic factor), and a third binds an *Escherichia coli* enterotoxin and the gastrointestinal polypeptide guanylin. The other form of guanylyl cyclase is soluble (sGC), contains heme, and is not bound to the membrane. There appear to be several isoforms of the intracellular enzyme. They are activated by nitric oxide (NO) and NO-containing compounds.

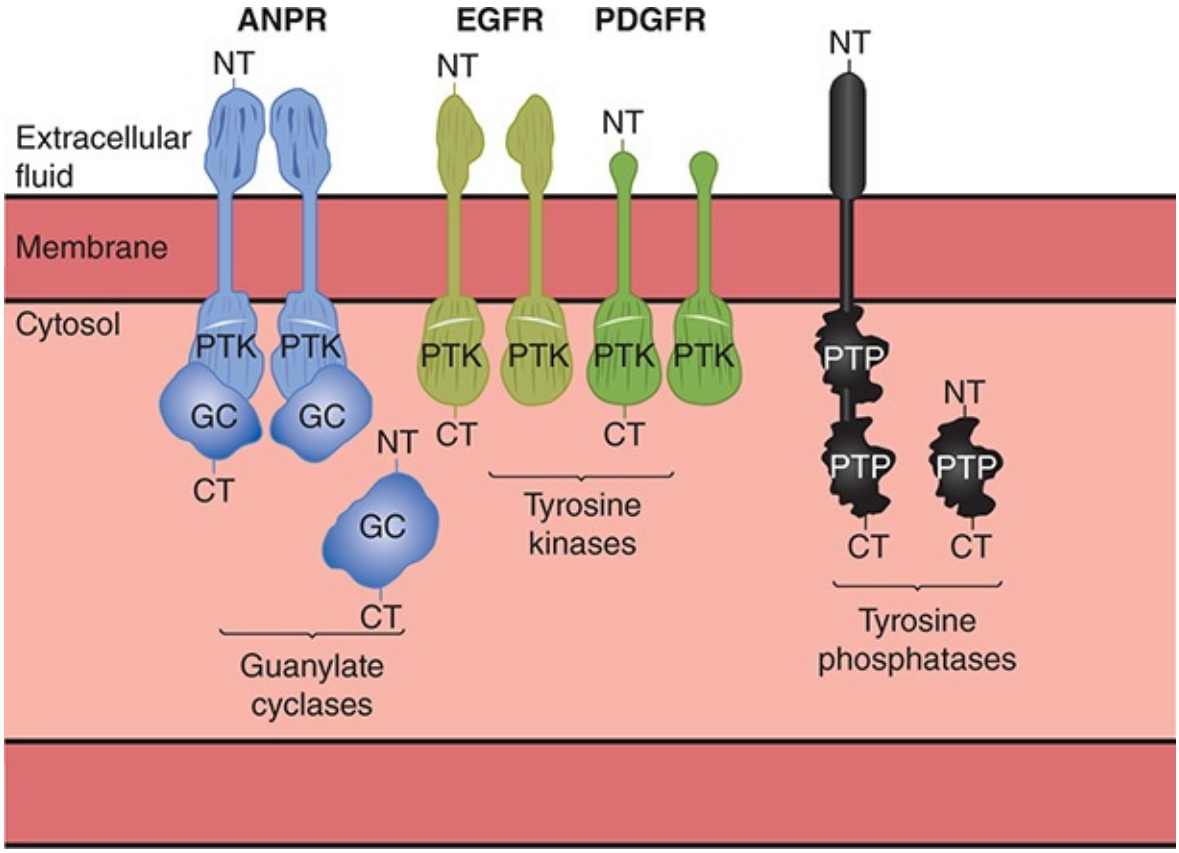


FIGURE 2–30 Diagrammatic representation of guanylyl cyclases, tyrosine kinases, and tyrosine phosphatases. NT refers to the amino (NH_2) terminus

and CT to the carboxyl terminus of each protein. Individual molecules are as follows: ANP, atrial natriuretic peptide; GC, guanylyl cyclase domain; EGFR, epidermal growth factor receptor; PDGFR, platelet-derived growth factor receptor; PTK, protein tyrosine kinase domain (PTK is inactive in guanylyl cyclase); PTP, tyrosine phosphatase domain.

GROWTH FACTORS

Growth factors have become increasingly important in many different aspects of physiology. They are polypeptides and proteins that are conveniently divided into three groups. One group is made up of agents that foster the multiplication or development of various types of cells; NGF, insulin-like growth factor I (IGF-I), activins and inhibins, and epidermal growth factor (EGF) are examples. More than 20 have been described. The cytokines are a second group. These factors are produced by macrophages and lymphocytes, as well as other cells, and are important in regulation of the immune system (see Chapter 3). Again, more than 20 have been described. The third group is made up of the colony-stimulating factors that regulate proliferation and maturation of red and white blood cells.

Receptors for EGF, platelet-derived growth factor (PDGF), and many of the other factors that foster cell multiplication and growth have a single membrane-spanning domain with an intracellular tyrosine kinase domain (Figure 2–29). When ligand binds to a tyrosine kinase receptor, it first causes a dimerization of two similar receptors. The dimerization results in partial activation of the intracellular tyrosine kinase domains and a cross-phosphorylation to fully activate each other. One of the pathways activated by phosphorylation leads, through the small G-protein Ras, to MAP kinases, and eventually to the production of transcription factors in the nucleus that alter gene expression **(Figure 2–31)**.

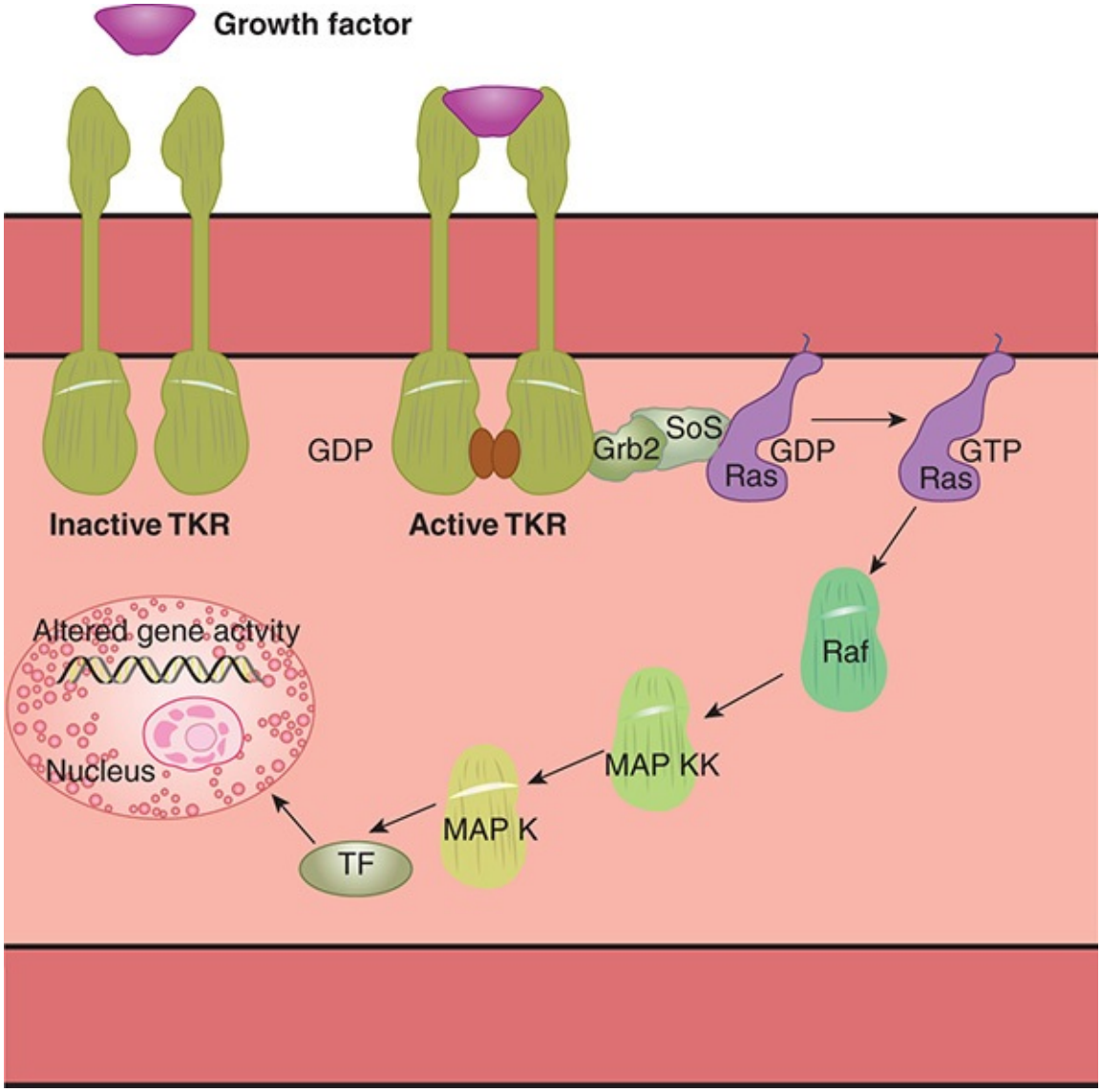


FIGURE 2–31 One of the direct pathways by which growth factors alter gene activity. Grb2, Ras activator/controller; MAP K, mitogen-activated protein kinase; MAP KK, MAP kinase kinase; Ras, product of the *ras* gene; Sos, Ras activator; TF, transcription factors; TKR, tyrosine kinase domain. There is a cross-talk between this pathway and the cAMP pathway, as well as a cross-talk with the IP_3–DAG pathway.

Receptors for cytokines and colony-stimulating factors differ from the other growth factors in that most of them do not have tyrosine kinase domains in their cytoplasmic portions and some have little or no cytoplasmic tail. However, they initiate tyrosine kinase activity in the cytoplasm. In particular, they activate the so-called Janus tyrosine kinases **(JAKs)** in the cytoplasm **(Figure 2–32)**. These

in turn phosphorylate **STAT** proteins. The phosphorylated STATs form homo- and heterodimers and move to the nucleus, where they act as transcription factors. There are four known mammalian JAKs and seven known STATs. Interestingly, the JAK–STAT pathway can also be activated by growth hormone and is another important direct path from the cell surface to the nucleus. However, it should be emphasized that both the Ras and the JAK–STAT pathways are complex and there is cross-talk between them and other signaling pathways discussed previously.

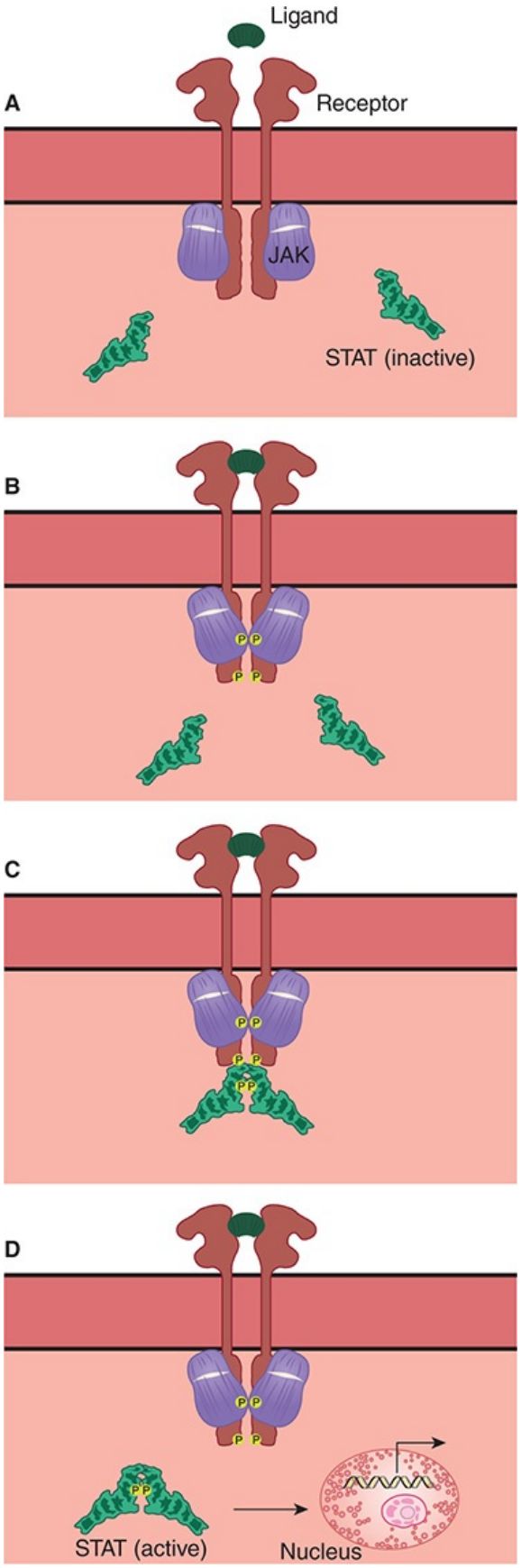

Ligand
A
Receptor
JAK
STAT (inactive)
B
P
C
D
STAT (active)
Nucleus

FIGURE 2–32 Signal transduction via the JAK–STAT pathway. A) Inactive JAKs are associated with individual receptors. **B)** Ligand binding leads to dimerization of receptor and activation JAKs that phosphorylate tyrosine residues on opposing receptors and their associated JAK. **C)** STATs then associate with the phosphorylated receptors and JAKs in turn phosphorylate these STATs. **D)** Phosphorylated STATs dimerize and move to nucleus, where they bind to response elements on DNA.

Finally, note that the whole subject of second messengers and intracellular signaling has become immensely complex, with multiple pathways and interactions. It is only possible in a book such as this to list highlights and present general themes that will aid the reader in understanding the rest of physiology **(Clinical Box 2–9).**

CLINICAL BOX 2–9

Receptor & G-Protein Diseases

Many diseases are being traced to mutations in the genes for receptors. For example, loss-of-function receptor mutations that cause disease have been reported for the 1,25-dihydroxycholecalciferol receptor and the insulin receptor. Certain other diseases are caused by production of antibodies against receptors. Thus, antibodies against thyroid-stimulating hormone (TSH) receptors cause Graves disease, and antibodies against nicotinic acetylcholine receptors cause myasthenia gravis.

An example of loss of function of a receptor is the type of nephrogenic diabetes insipidus that is due to loss of the ability of mutated V_2 vasopressin receptors to mediate concentration of the urine. Mutant receptors can gain as well as lose function. A gain-of-function mutation of the calcium-sensing receptor (CaSR) causes excess inhibition of parathyroid hormone secretion and familial hypercalciuric hypocalcemia. G-proteins can also undergo loss-of-function or gain-of-function mutations that cause disease **(Table 2–6)**. In one form of pseudohypoparathyroidism, a mutated $G_s\alpha$ fails to respond to parathyroid hormone, producing the symptoms of hypoparathyroidism without any decline in circulating parathyroid hormone. Testotoxicosis is an interesting disease that combines gain and loss of function. In this condition, an activating mutation of $G_s\alpha$ causes excess testosterone secretion and

prepubertal sexual maturation. However, this mutation is temperature-sensitive and is active only at the relatively low temperature of the testes (33°C). At 37°C, the normal temperature of the rest of the body, it is replaced by loss of function, with the production of hypoparathyroidism and decreased responsiveness to TSH. A different activating mutation in $G_s\alpha$ is associated with the rough-bordered areas of skin pigmentation and hypercortisolism of the McCune–Albright syndrome. This mutation occurs during fetal development, creating a mosaic of normal and abnormal cells. A third mutation in $G_s\alpha$ reduces its intrinsic GTPase activity. As a result, it is much more active than normal, and excess cAMP is produced. This causes hyperplasia and eventually neoplasia in somatotrope cells of the anterior pituitary. Forty percent of somatotrope tumors causing acromegaly have cells containing a somatic mutation of this type.

TABLE 2–6 Examples of abnormalities caused by loss- or gain-of-function mutations of heterotrimeric G-protein-coupled receptors and G-proteins.

Site	Type of Mutation	Disease
Receptor		
Cone opsins	Loss	Color blindness
Rhodopsin	Loss	Congenital night blindness; two forms of retinitis pigmentosa
Vasopressin receptor 2 (V2R) or arginine vasopressin receptor 2	Loss	X-linked nephrogenic diabetes insipidus
Adrenocorticotropic hormone (ACTH) receptor or melanocortin receptor 2	Loss	Familial glucocorticoid deficiency
Luteinizing hormone receptor (LHR)	Gain	Familial male precocious puberty
Thyrotropin receptor (TSH receptor)	Gain	Familial nonautoimmune hyperthyroidism
TSH receptor	Loss	Familial hypothyroidism
Calcium-sensing receptor (CaSR)	Gain	Familial hypercalciuric hypocalcemia
Thromboxane A_2 receptor (TP) or prostanoid TP receptor	Loss	Congenital bleeding
Endothelin receptor B (ET_B)	Loss	Hirschsprung disease
G-protein		
G_s α	Loss	Pseudohypothyroidism type 1a
G_s α	Gain/loss	Testotoxicosis
G_s α	Gain (mosaic)	McCune-Albright syndrome
G_s α	Gain	Somatotroph adenomas with acromegaly
G_i α	Gain	Ovarian and adrenocortical tumors

ACTH, adrenocorticotropic hormone; LH, luteinizing hormone; TSH, thyroid-stimulating hormone.

CHAPTER SUMMARY

- The cell and the intracellular organelles are surrounded by semipermeable membranes. Biological membranes have a lipid bilayer core that is populated by structural and functional proteins. These proteins contribute greatly to the semipermeable properties of biological membrane.
- Cells contain a variety of organelles that perform specialized cell functions. The nucleus is an organelle that contains DNA and is the site of gene transcription. The endoplasmic reticulum and the Golgi apparatus are important in protein processing and the targeting of proteins to correct compartments within the cell. Lysosomes and peroxisomes are membrane-bound organelles that contribute to protein and lipid processing. Mitochondria are organelles that allow for oxidative phosphorylation in eukaryotic cells and also are important in specialized cellular signaling.
- The cytoskeleton is a network of three types of filaments that provide structural integrity to the cell as well as a means for trafficking of organelles and other structures around the cell. Actin filaments are important in cellular contraction, migration, and signaling. Actin filaments also provide the backbone for muscle contraction. Intermediate filaments are primarily structural. Microtubules provide a dynamic structure in cells that allows for the movement of cellular components around the cell.
- There are three superfamilies of molecular motor proteins in the cell that use the energy of ATP to generate force, movement, or both. Myosin is the force generator for muscle cell contraction. Cellular myosins can also interact with the cytoskeleton (primarily thin filaments) to participate in contraction as well as movement of cell contents. Kinesins and cellular dyneins are motor proteins that primarily interact with microtubules to move cargo around the cells.
- Cellular adhesion molecules aid in tethering cells to each other or to the extracellular matrix as well as providing for initiation of cellular signaling. There are four main families of these proteins: integrins, immunoglobulins, cadherins, and selectins.
- Cells contain distinct protein complexes that serve as cellular connections to other cells or the extracellular matrix. Tight junctions provide intercellular connections that link cells into a regulated tissue barrier and also provide a barrier to movement of proteins in the cell membrane. Gap junctions

provide contacts between cells that allow for direct passage of small molecules between two cells. Desmosomes and adherens junctions are specialized structures that hold cells together. Hemidesmosomes and focal adhesions attach cells to their basal lamina.

- Exocytosis and endocytosis are vesicular fusion events that allow for movement of proteins and lipids between the cell interior, the plasma membrane, and the cell exterior. Exocytosis can be constitutive or nonconstitutive; both are regulated processes that require specialized proteins for vesicular fusion. Endocytosis is the formation of vesicles at the plasma membrane to take material from the extracellular space into the cell interior.
- Cells can communicate with one another via chemical messengers. Individual messengers (or ligands) typically bind to a plasma membrane receptor to initiate intracellular changes that lead to physiological changes. Plasma membrane receptor families include ion channels, G-protein-coupled receptors, or a variety of enzyme-linked receptors (eg, tyrosine kinase receptors). There are additional cytosolic receptors (eg, steroid receptors) that can bind membrane-permeant compounds. Activation of receptors leads to cellular changes that include changes in membrane potential, activation of heterotrimeric G-proteins, increase in second messenger molecules, or initiation of transcription.
- Second messengers are molecules that undergo a rapid concentration changes in the cell following primary messenger recognition. Common second messenger molecules include Ca^{2+}, cyclic adenosine monophosphate (cAMP), and cyclic guanine monophosphate (cGMP), inositol trisphosphate (IP_3).

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. The electrogenic Na, K ATPase plays a critical role in cellular physiology by

A. using the energy in ATP to extrude 3 Na^+ out of the cell in exchange for taking two K^+ into the cell.

B. using the energy in ATP to extrude 3 K^+ out of the cell in exchange for taking two Na^+ into the cell.

C. using the energy in moving Na^+ into the cell or K^+ outside the cell to make

ATP.

D. using the energy in moving Na^+ outside of the cell or K^+ inside the cell to make ATP.

2. Cell membranes

A. contain relatively few protein molecules.

B. contain many carbohydrate molecules.

C. are freely permeable to electrolytes but not to proteins.

D. have variable protein and lipid contents depending on their location in the cell.

E. have a stable composition throughout the life of the cell.

3. Second messengers

A. are substances that interact with first messengers outside cells.

B. are substances that bind to first messengers in the cell membrane.

C. are hormones secreted by cells in response to stimulation by another hormone.

D. mediate the intracellular responses to many different hormones and neurotransmitters.

E. are not formed in the brain.

4. The Golgi complex

A. is an organelle that participates in the breakdown of proteins and lipids.

B. is an organelle that participates in posttranslational processing of proteins.

C. is an organelle that participates in energy production.

D. is an organelle that participates in transcription and translation.

E. is a subcellular compartment that stores proteins for trafficking to the nucleus.

5. Endocytosis

A. includes phagocytosis and pinocytosis, but not clathrin-mediated or caveolae-dependent uptake of extracellular contents.

B. refers to the merging of an intracellular vesicle with the plasma membrane to deliver intracellular contents to the extracellular milieu.

C. refers to the invagination of the plasma membrane to uptake extracellular contents into the cell.

D. refers to vesicular trafficking between Golgi stacks.

6. G-protein-coupled receptors

A. are intracellular membrane proteins that help regulate movement within the cell.

B. are plasma membrane proteins that couple the extracellular binding of primary signaling molecules to exocytosis.

C. are plasma membrane proteins that couple the extracellular binding of primary signaling molecules to the activation of heterotrimeric G-proteins.

D. are intracellular proteins that couple the binding of primary messenger molecules with transcription.

7. Gap junctions are intercellular connections that

A. primarily serve to keep cells separated and allow for transport across a tissue barrier.

B. serve as a regulated cytoplasmic bridge for sharing of small molecules between cells.

C. serve as a barrier to prevent protein movement within the cellular membrane.

D. are cellular components for constitutive exocytosis that occurs between adjacent cells.

8. F-actin is a component of the cellular cytoskeleton that

A. provides a structural component for cell movement.

B. is defined as the “functional” form of actin in the cell.

C. refers to the actin subunits that provide the molecular building blocks of the extended actin molecules found in the cell.

D. provides the molecular architecture for cell to cell communication.

CHAPTER 3

Immunity, Infection, & Inflammation

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OBJECTIVES

After studying this chapter, you should be able to:

- Understand the significance of immunity, particularly with respect to defending the body against microbial invaders.
- Delineate the role and mechanisms of innate immunity.
- Identify the roles of the humoral and cellular arms of acquired immunity, the cellular mediators of these responses, and the molecular basis for the recognition of a diverse set of antigens.
- Understand the general principles governing regulation of immunity by soluble cytokines, chemokines and hematopoietic growth factors, as well as the complement system.
- Define the roles of additional circulating and tissue cell types that contribute to immune and inflammatory responses, including granulocytes, mast cells, monocytes, and platelets. Describe how phagocytes are able to kill internalized bacteria.
- Understand the basis of inflammatory responses and wound healing.

INTRODUCTION

As an open system, the body is continuously called upon to defend itself from potentially harmful invaders such as bacteria, viruses, and other microbes. This is accomplished by the immune system, which is subdivided into innate and adaptive (or acquired) branches. The immune system is composed of specialized effector cells that sense and respond to foreign antigens and other molecular patterns not found in human tissues, as well as various regulatory molecules, including the large collection of soluble mediators known as cytokines. Likewise, the immune system clears the body's own cells that have become senescent or abnormal, such as cancer cells. Finally, normal host tissues occasionally become the subject of inappropriate immune attack, such as in autoimmune diseases or in settings where normal cells are harmed as innocent bystanders when the immune system mounts an inflammatory response to an invader. It is beyond the scope of this volume to provide a full treatment of all aspects of modern immunology. Nevertheless, the student of physiology should have a working knowledge of immune functions and their regulation, due to a growing appreciation for the ways in which the immune system can contribute to normal physiologic regulation in a variety of tissues, as well as contributions of immune effectors to pathophysiology.

IMMUNITY

OVERVIEW

All multicellular organisms, including invertebrates and plants, express the ancient protective mechanism of **innate immunity.** This system is triggered by receptors that bind sequences of sugars, lipids, amino acids, or nucleic acids that are common in bacteria and other microorganisms, but are not found in eukaryotic cells. These receptors, in turn, activate various defense mechanisms. The receptors are coded in the germ line, and their fundamental structure is not modified by exposure to antigen. The activated defenses include, in various species, release of cytokines known as interferons, phagocytosis, production of antibacterial peptides, activation of the complement system, and several proteolytic cascades. This primitive immune system is important in vertebrates, particularly in the early response to infection. However, innate immunity is also complemented in vertebrates by **adaptive** or **acquired immunity,** a system in which T and B lymphocytes are activated by specific antigens. Activated B lymphocytes form clones that produce secreted antibodies, which attack foreign proteins. T cells bear receptors that are related to antibody molecules, but which

remain cell-bound. When these receptors encounter their cognate antigen, the T cell is stimulated to proliferate and produce cytokines that orchestrate the immune response, including that of B cells. After the invasion is repelled, small numbers of lymphocytes persist as memory cells so that a second exposure to the same antigen provokes a prompt and magnified immune attack. The genetic event that led to acquired immunity occurred 450 million years ago and was probably insertion of a transposon into the genome in a way that made possible the generation of the immense repertoire of T cell receptors and antibodies that can be produced by the body.

In vertebrates, including humans, innate immunity provides the first line of defense against infections, but it also triggers the slower but more specific acquired immune response **(Figure 3–1)**. In vertebrates, natural and acquired immune mechanisms also attack tumors and tissue transplanted from other animals.

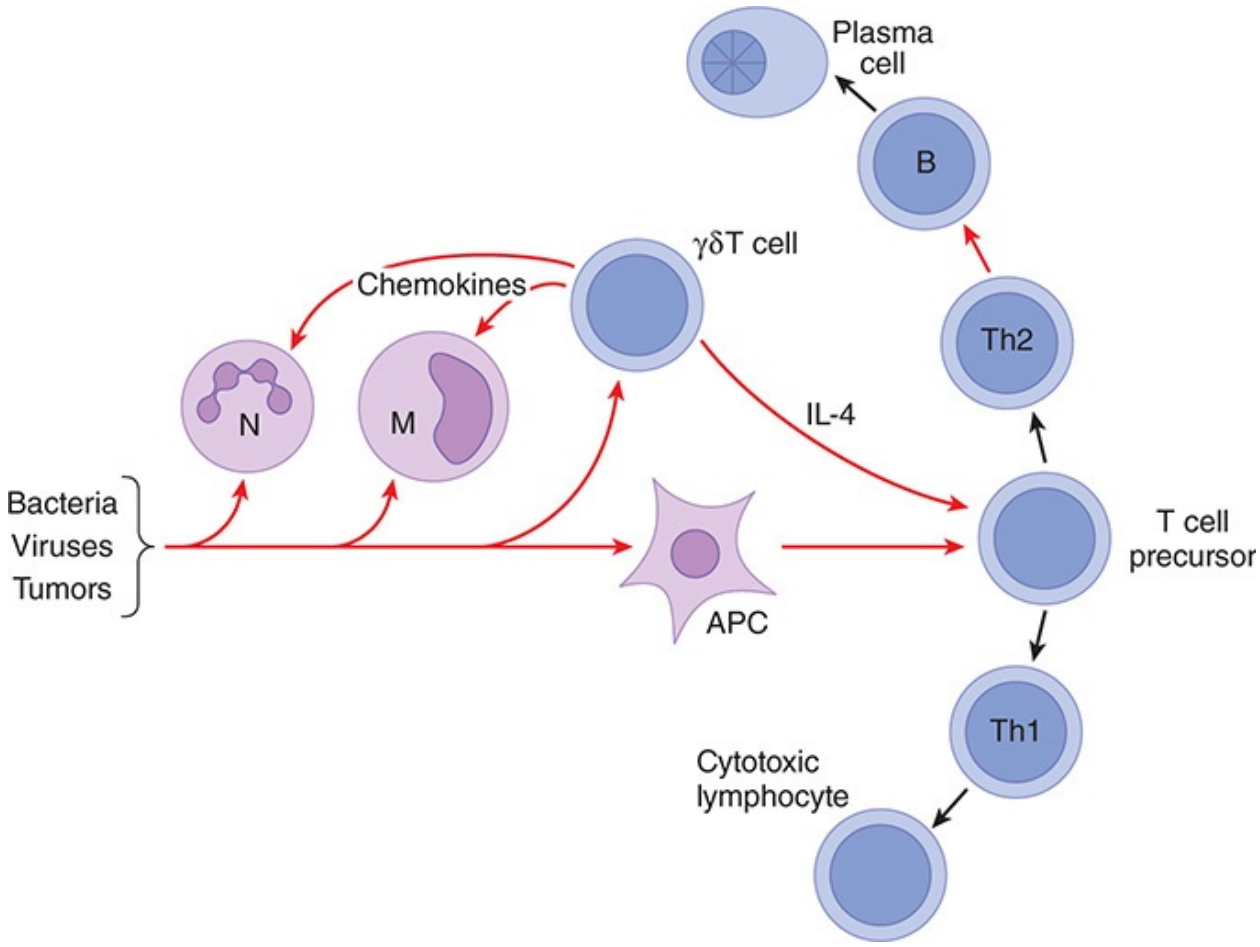


FIGURE 3–1 How bacteria, viruses, and tumors trigger innate immunity and initiate the acquired immune response. Red arrows indicate

mediators/cytokines that act on the target cell shown; black arrows show pathways of differentiation. APC, antigen-presenting cell; M, monocyte; N, neutrophil; Th1 and Th2, type 1 and type 2 helper T cells, respectively.

Once activated, immune cells communicate by means of cytokines and chemokines. They kill viruses, bacteria, and other foreign cells by secreting other cytokines and activating the complement system.

INNATE IMMUNITY

A wide variety of innate immune cells respond to molecular patterns produced by bacteria and to other substances characteristic of viruses, tumors, and transplanted cells. Many cells that are not professional immunocytes contribute to innate immune responses, such as endothelial and epithelial cells. The activated cells produce their effects via the release of cytokines, as well as, in some cases, complement and other systems.

Innate immunity in *Drosophila* centers around a receptor protein named **toll,** which binds fungal antigens and triggers activation of genes coding for antifungal proteins. An expanding list of toll-like receptors (TLRs) has now been identified in humans and other vertebrates. One of these, TLR4, binds bacterial lipopolysaccharide and a protein called CD14, and this initiates intracellular events that activate transcription of genes for a variety of proteins involved in innate immune responses. This is important because bacterial lipopolysaccharide produced by gram-negative organisms is the cause of septic shock. TLR2 mediates the response to microbial lipoproteins, TLR6 cooperates with TLR2 in recognizing certain peptidoglycans, TLR5 recognizes a molecule known as flagellin in bacterial flagellae, and TLR9 recognizes bacterial DNA. TLRs are referred to as **pattern recognition receptors (PRRs)** because they recognize and respond to the molecular patterns expressed by pathogens. Other PRRs may be intracellular, such as the so-called NOD proteins. One NOD protein, NOD2, has received attention as the product of a candidate gene leading to the intestinal inflammatory condition, Crohn disease **(Clinical Box 3–1)**.

CLINICAL BOX 3–1

Crohn Disease

Crohn disease is a chronic, relapsing, and remitting disease that involves

transmural inflammation of the intestine that can occur at any point along the length of the gastrointestinal tract but most commonly is confined to the distal small intestine and colon. Patients with this condition suffer from changes in bowel habits, bloody diarrhea, severe abdominal pain, weight loss, and malnutrition. Evidence is accumulating that the disease reflects a failure to downregulate inflammatory responses to the normal gut commensal microbiota. In genetically susceptible individuals, mutations in genes controlling innate immune responses (eg, *NOD2*) or regulators of acquired immunity appear to predispose to disease when individuals are exposed to appropriate environmental factors, which can include a change in the microbiota, or stress.

THERAPEUTIC HIGHLIGHTS

During flares of Crohn disease, the mainstay of treatment remains high-dose corticosteroids to suppress inflammation nonspecifically. Surgery is often required to treat complications such as strictures, fistulas, and abscesses. Some patients with severe disease also benefit from ongoing treatment with immunosuppressive drugs, or from treatment with antibodies targeted against tumor necrosis factor-α (TNF-α) or other inflammatory cytokines. Probiotics, therapeutic microorganisms designed to restore a “healthy” microbiota, may have some role in prophylaxis. The pathogenesis of Crohn disease, as well as the related inflammatory bowel disease, ulcerative colitis, remains the subject of intense investigation, and therapies that target specific facets of the inflammatory cascade that may be selectively implicated in individual patients with differing genetic backgrounds are under development.

ACQUIRED IMMUNITY

As noted previously, the key to acquired immunity is the ability of lymphocytes to produce antibodies (in the case of B cells) or cell-surface receptors (in the case of T cells) that are specific for one of the many millions of foreign agents that may invade the body. The antigens stimulating production of T cell receptors or antibodies are usually proteins and polypeptides, but antibodies can also be formed against nucleic acids and lipids if these are presented as nucleoproteins and lipoproteins. Antibodies to small molecules can also be produced experimentally if the molecules are bound to protein. Acquired

immunity has two components: humoral immunity and cellular immunity. **Humoral immunity** is mediated by circulating immunoglobulin antibodies in the γ-globulin fraction of the plasma proteins. Immunoglobulins are produced by differentiated forms of B lymphocytes known as **plasma cells,** and they activate the complement system and attack and neutralize antigens (Figure 3–1). Humoral immunity is a major defense against bacterial infections. **Cellular immunity** is mediated by T lymphocytes. It is responsible for delayed allergic reactions and rejection of transplants of foreign tissue. Cytotoxic T cells attack and destroy cells bearing the antigen that activated them. They kill by inserting proteins known as perforins that form pores in the membrane of target cells, and by initiating apoptosis. Cellular immunity constitutes a major defense against infections due to viruses, fungi, and a few bacteria such as the tubercle bacillus. It also helps defend against tumors.

LYMPHOCYTES

Lymphocytes are the key elements in the production of acquired immunity. After birth, some lymphocytes are formed in the bone marrow. However, most are formed in the lymph nodes **(Figure 3–2)**, thymus, and spleen from precursor cells that originally came from the bone marrow and were processed in the thymus (T cells) or bone marrow (B cells, see below). Lymphocytes enter the bloodstream for the most part via the lymphatics. At any given time, only about 2% of the body lymphocytes are in the peripheral blood. Most of the rest are in the lymphoid organs.

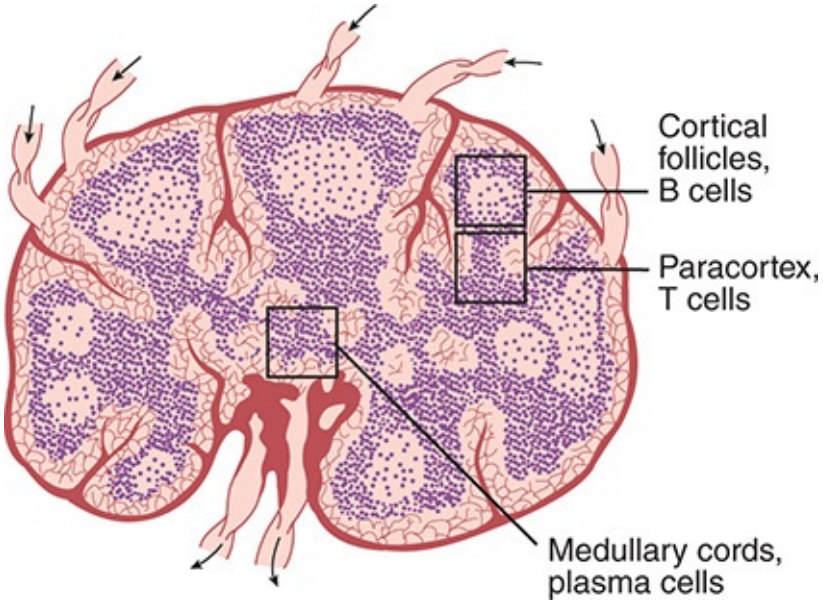


FIGURE 3–2 Anatomy of a normal lymph node. (Reproduced with permission from Chandrasoma. Reproduced with permission from McPhee SJ, Lingappa VR, Ganong WF [editors]: *Pathophysiology of Disease*, 4th ed. New York, NY: McGraw-Hill; 2003.)

During fetal development, and to a much lesser extent during adult life, lymphocyte precursors come from the bone marrow. Those that populate the thymus **(Figure 3–3)** become transformed by the environment in this organ into T lymphocytes. Another lymphoid subset that forms in the thymus is the NKT cell, so-called because it shares features of both T lymphocytes and **natural killer** (NK) cells. Transformation to B lymphocytes, on the other hand, occurs in the fetal liver and, after birth, the bone marrow. NK cells also form in these sites. After residence in the thymus, liver, or bone marrow, many of the T and B lymphocytes migrate to the lymph nodes.

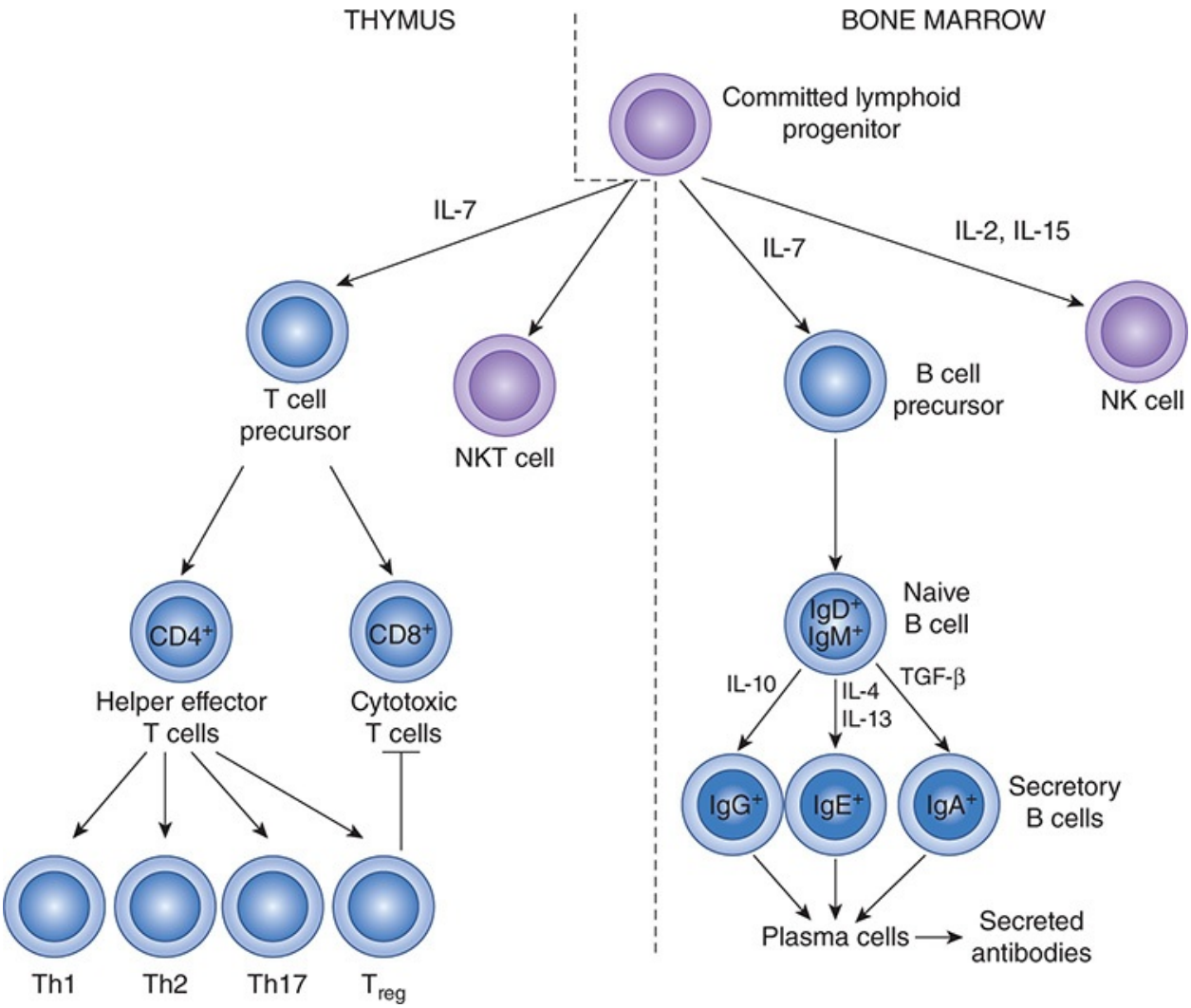


FIGURE 3–3 Development of the system mediating acquired immunity. Committed lymphoid progenitors arise in the bone marrow. Maturation into the B and NK cell lineages occurs in this site, whereas development into T and NKT cells takes place after lymphoid progenitors migrate to the thymus. NK, natural killer.

T and B lymphocytes are morphologically indistinguishable but can be identified by markers on their cell membranes. B cells differentiate sequentially into cells capable of producing the various classes of immunoglobulins and thereafter into **plasma cells.** There are two major types of T cells: **cytotoxic T cells** and **helper/effector T cells.** There are at least four subtypes of helper T cells: T helper 1 (Th1) cells secrete interleukin (IL)-2 and γ-interferon and are concerned primarily with stimulating cellular immunity; Th2 cells secrete IL-4 and IL-5 and interact primarily with B cells in relation to humoral immunity. Th17 cells are induced in response to bacterial infections, produce IL-6 and IL-17, and help recruit neutrophils. They are also implicated in the generation of harmful inflammatory responses that occur in autoimmune diseases. Finally, T_{reg} cells produce IL-10 to dampen T cell–driven responses. Cytotoxic T cells destroy transplanted cells and those expressing foreign antigens (eg, virally infected targets), with their development aided and directed by helper T cells.

Markers on the surface of lymphocytes are assigned CD (clusters of differentiation) numbers on the basis of their reactions to a panel of monoclonal antibodies. Most cytotoxic T cells display the glycoprotein CD8, and helper T cells display the glycoprotein CD4. These proteins are closely associated with the T cell receptors and may function as coreceptors. On the basis of differences in their receptors and functions, cytotoxic T cells are divided into αβ and γδ types (see below). NK and NKT cells (see above) are also cytotoxic lymphocytes. Thus, there are four main types of cytotoxic lymphocytes in the body: αβ T cells, γδ T cells, NK cells, and NKT cells.

MEMORY B CELLS & T CELLS

After exposure to a given antigen, a small number of activated B and T cells persist as memory B and T cells. These cells are readily converted to effector cells by a later encounter with the same antigen. This ability to produce an accelerated response to a second exposure to an antigen is a key characteristic of acquired immunity. The ability persists for long periods of time, and in some instances (eg, immunity to measles) it can be life-long.

After activation in lymph nodes, lymphocytes disperse widely throughout the body and are especially plentiful in areas where invading organisms enter the body (eg, the mucosa of the respiratory and gastrointestinal tracts). This puts memory cells close to sites of reinfection and may account in part for the rapidity and strength of their response. Chemotactic cytokines known as **chemokines** (see below) are involved in guiding activated lymphocytes to these locations.

ANTIGEN RECOGNITION

The number of different antigens recognized by lymphocytes in the body is extremely large. The repertoire develops initially without exposure to the antigen. Stem cells differentiate into many millions of different T and B lymphocytes, each with the ability to respond to a particular antigen. When the antigen first enters the body, it can bind directly to the appropriate receptors on B cells. However, a full antibody response requires that the B cells contact helper T cells. In the case of T cells, the antigen is taken up by an antigen-presenting cell (APC) and partially digested. A peptide fragment of it is presented to the appropriate receptors on T cells. In either case, the cells are stimulated to divide, forming **clones** of cells that respond to this antigen **(clonal selection).** Effector cells are also subject to **negative selection,** during which lymphocyte precursors that are reactive with self-antigens are normally deleted. This results in immune **tolerance.** It is this latter process that presumably goes awry in autoimmune diseases, where the body reacts to and destroys cells expressing normal proteins, with accompanying inflammation that may lead to tissue destruction.

ANTIGEN PRESENTATION

APCs include specialized cells called **dendritic cells** in the lymph nodes and spleen and the Langerhans dendritic cells in the skin. Macrophages and B cells themselves, and likely many other cell types, can also function as APCs. For example, in the intestine, the epithelial cells that line the tract are likely important in presenting antigens derived from commensal bacteria. In APCs, polypeptide products of antigen digestion are coupled to the HLA protein products of the **major histocompatibility complex (MHC)** genes and presented on the surface of the cell.

The genes of the MHC encode glycoproteins that are divided into two classes

on the basis of structure and function. Class I antigens are composed of a 45-kDa heavy chain associated noncovalently with β_2-microglobulin encoded by a gene outside the MHC **(Figure 3–4)**. They are found on all nucleated cells. Class II antigens are heterodimers made up of a 29- to 34-kDa α chain associated noncovalently with a 25- to 28-kDa β chain. They are present in "professional" APCs, including B cells, and in activated T cells.

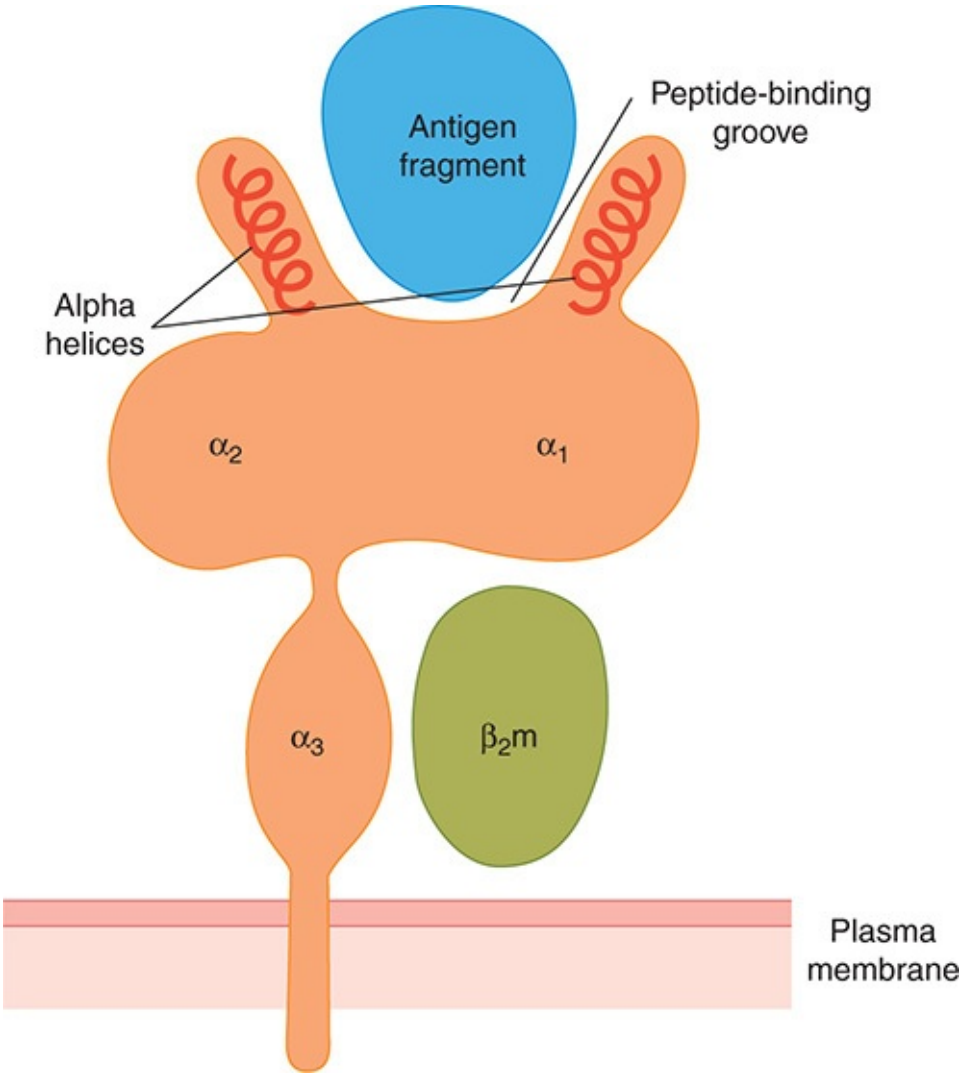


FIGURE 3–4 Schematic structure of a class I MHC protein bound to an antigen fragment, based on the structure of the human histocompatibility antigen HLA-A2. The antigen-binding pocket is at the top and is formed by the α_1 and α_2 parts of the molecule. The α_3 portion and the associated β_2-microglobulin (β_2m) are close to the membrane. MHC, major histocompatibility complex. (Reproduced with permission from Bjorkman PJ, et al: Structure of the

human histocompatibility antigen HLA-A2, Nature 1987;October 8-14;329(6139):506-512.)

The class I MHC proteins (MHC-I proteins) are coupled primarily to peptide fragments generated from proteins synthesized within cells. Peptides to which the host is not tolerant (eg, those from mutant or viral proteins) are recognized by T cells. The digestion of these proteins occurs in complexes of proteolytic enzymes known as **proteasomes,** and the peptide fragments bind to MHC proteins in the endoplasmic reticulum. The class II MHC proteins (MHC-II proteins) interact primarily with peptide products of extracellular antigens, such as bacteria, that enter the cell by endocytosis and are digested in the late endosomes.

T CELL RECEPTORS

The MHC protein–peptide complexes on the surface of the APCs bind to appropriate T cells. Therefore, receptors on the T cells must recognize a very wide variety of complexes. Most of the receptors on circulating T cells are made up of two polypeptide units designated α and β. They form heterodimers that recognize the MHC proteins and the antigen fragments with which they are combined **(Figure 3–5)**. These cells are called αβ T cells. On the other hand, about 10% of circulating T cells have two different polypeptides designated γ and δ in their receptors, and they are called γδ T cells. These T cells are prominent in the mucosa of the gastrointestinal tract, and there is evidence that they form a link between the innate and acquired immune systems by way of the cytokines they secrete (Figure 3–3).

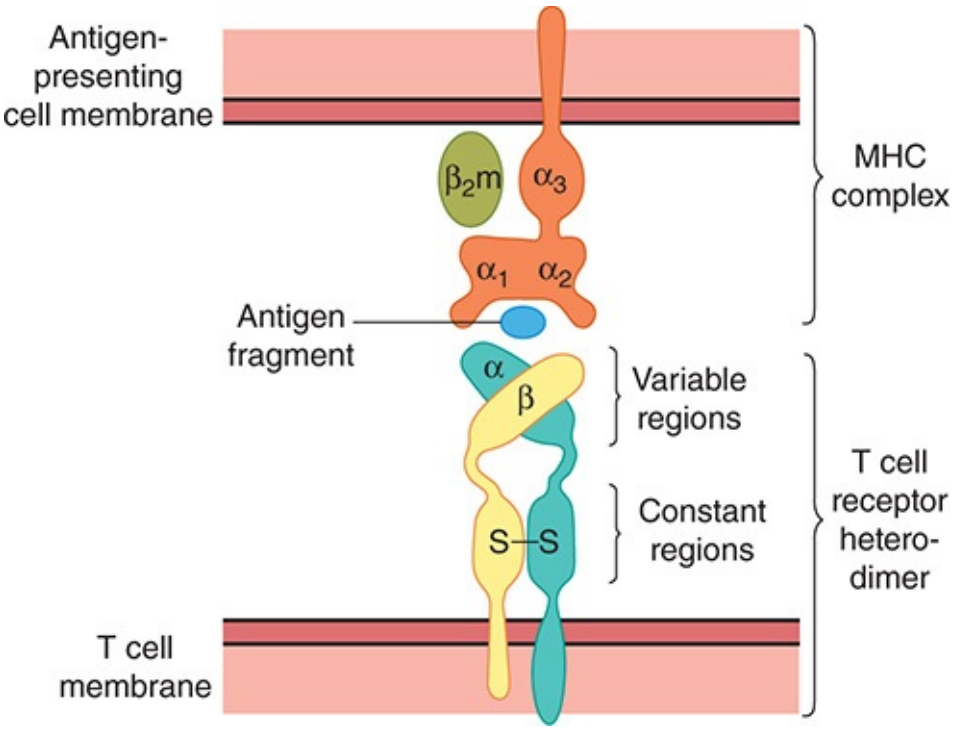


FIGURE 3–5 Interaction between antigen-presenting cell (top) and αβ T lymphocyte (bottom). The MHC proteins (in this case, MHC-I) and their peptide antigen fragment bind to the α and β units that combine to form the T cell receptor. Some experts refer to this set of protein–protein interactions as the "immune synapse." MHC, major histocompatibility complex.

CD8 occurs on the surface of cytotoxic T cells that bind MHC-I proteins, and CD4 occurs on the surface of helper T cells that bind MHC-II proteins **(Figure 3–6)**. The CD8 and CD4 proteins facilitate the binding of the MHC proteins to the T cell receptors, and they also foster lymphocyte development. The activated CD8 cytotoxic T cells kill their targets directly, whereas the activated CD4 helper T cells secrete cytokines that activate other lymphocytes.

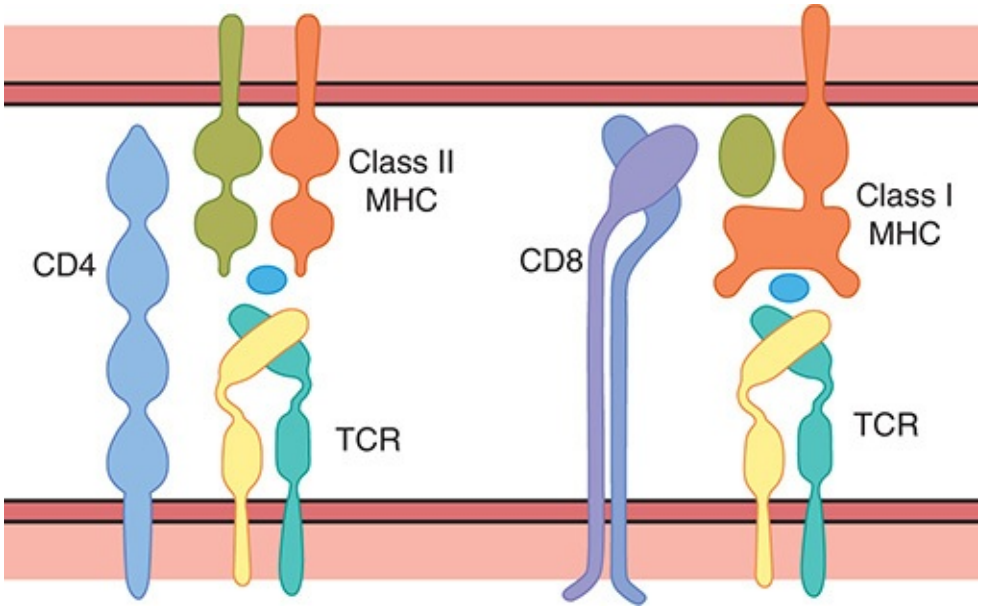


FIGURE 3–6 Diagrammatic summary of the structure of CD4 and CD8, and their relation to MHC-I and MHC-II proteins. Note that CD4 is a single protein, whereas CD8 is a heterodimer. MHC, major histocompatibility complex.

The T cell receptors are surrounded by adhesion molecules and proteins that bind to complementary proteins in the APC when the two cells transiently join to form the "immunologic synapse" that permits T cell activation to occur **(Figure 3–7)**. It is now generally accepted that two signals are necessary to produce activation. One is produced by the binding of the digested antigen to the T cell receptor. The other is produced by the joining of the surrounding proteins in the "synapse." If the first signal occurs but the second does not, the T cell is inactivated and becomes unresponsive.

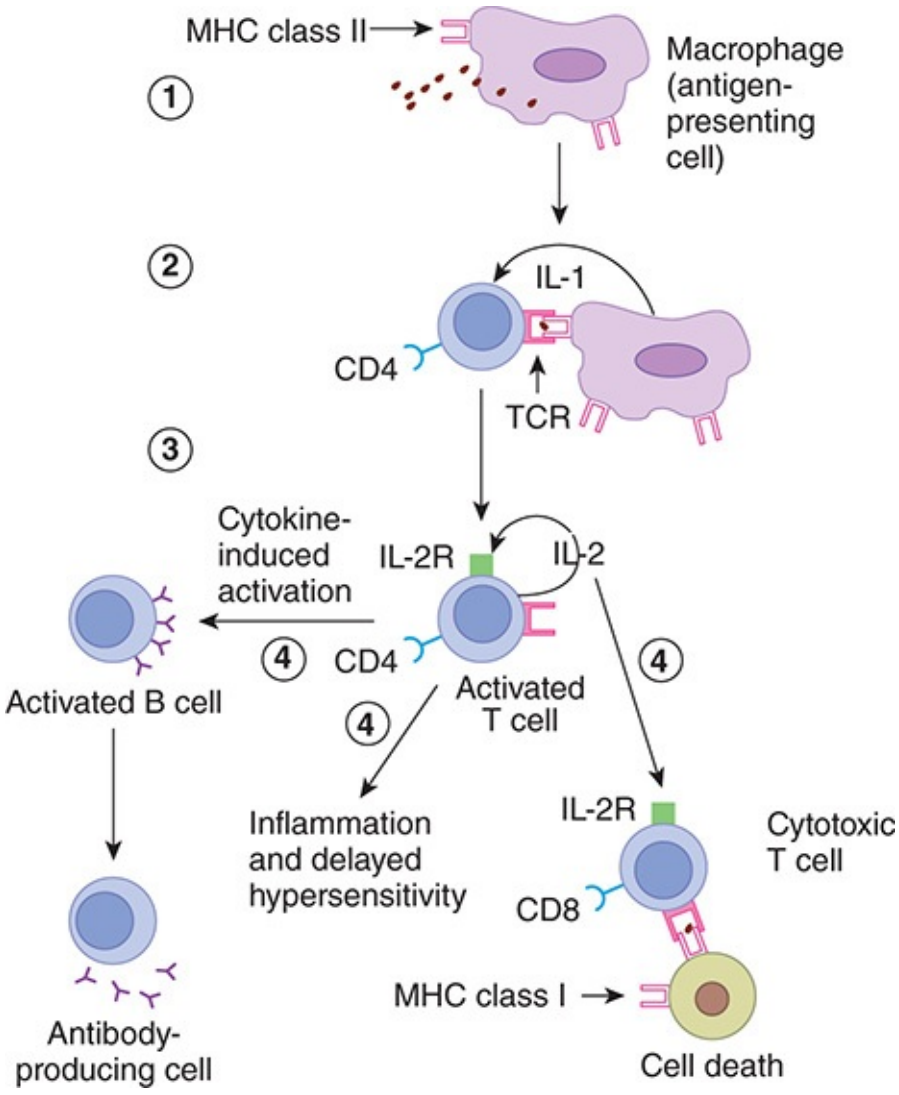


FIGURE 3–7 Summary of acquired immunity. (1) An antigen-presenting cell (depicted here as a macrophage, but other cell types, including dendritic cells, are important) ingests and partially digests an antigen, then presents part of the antigen along with MHC peptides (in this case, MHC II peptides on the cell surface). **(2)** An "immune synapse" forms with a naive CD4 T cell, which is activated to produce IL-2. **(3)** IL-2 acts in an autocrine fashion to cause the cell to multiply, forming a clone. **(4)** The activated CD4 cell may promote B cell activation and the proliferation of plasma cells that produce antibodies or it may activate a cytotoxic CD8 cell. The CD8 cell can also be activated by forming a synapse with an MCH I antigen-presenting cell. MHC, major histocompatibility complex. (Reproduced with permission from McPhee SJ, Lingappa VR, Ganong WF [editors]: *Pathophysiology of Disease*, 6th ed. New York, NY: McGraw-Hill; 2010.)

B CELLS

As noted above, B cells can bind antigens directly, but they must contact helper T cells to produce full activation and antibody formation. It is the Th2 subtype that is mainly involved. Helper T cells develop along the T_h2 lineage in response to IL-4 (see below). On the other hand, IL-12 promotes the Th1 phenotype. IL-2 acts in an autocrine fashion to cause activated T cells to proliferate. The role of various cytokines in B cell and T cell activation is summarized in Figure 3–7.

The activated B cells proliferate and transform into **memory B cells** (see above) and **plasma cells.** The plasma cells secrete large quantities of antibodies into the general circulation. The antibodies circulate in the globulin fraction of the plasma and, like antibodies elsewhere, are called **immunoglobulins.** The immunoglobulins are actually the secreted form of antigen-binding receptors on the B cell membrane.

IMMUNOGLOBULINS

Circulating antibodies protect their host by binding to and neutralizing some protein toxins, by blocking the attachment of some viruses and bacteria to cells, by opsonizing bacteria (see below), and by activating complement (see below). Five general types of immunoglobulin antibodies are produced by plasma cells. The basic component of each is a symmetric unit containing four polypeptide chains **(Figure 3–8)**. The two long chains are called **heavy chains,** whereas the two short chains are called **light chains.** There are two types of light chains, k and λ, and nine types of heavy chains. The chains are joined by disulfide bridges that permit mobility, and there are intrachain disulfide bridges as well. In addition, the heavy chains are flexible in a region called the hinge. Each heavy chain has a variable (V) segment in which the amino acid sequence is highly variable, a diversity (D) segment in which the amino acid segment is also highly variable, a joining (J) segment in which the sequence is moderately variable, and a constant (C) segment in which the sequence is constant. Each light chain has a V, J, and C segment. The V segments form part of the antigen-binding sites (Fab portion of the molecule [Figure 3–8]). The Fc portion of the molecule is the effector portion, which mediates the reactions initiated by antibodies.

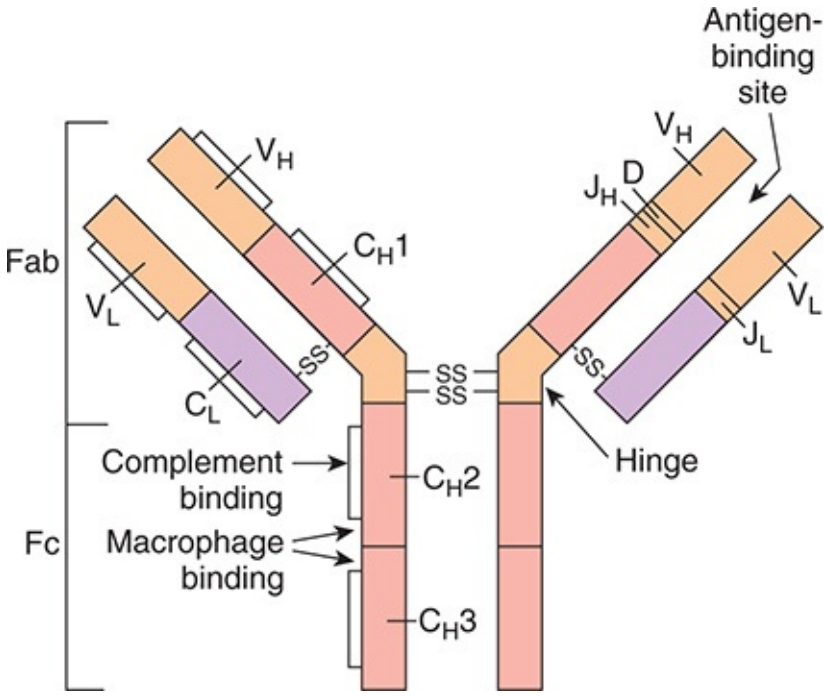


FIGURE 3–8 Typical immunoglobulin G molecule. Fab, portion of the molecule that is concerned with antigen binding; Fc, effector portion of the molecule. The constant regions are pink and purple, and the variable regions are orange. The constant segment of the heavy chain is subdivided into CH1, CH2, and CH3. SS lines indicate intersegmental disulfide bonds. On the right side, the C labels are omitted to show regions J_H, D, and J_L.

Two of the classes of immunoglobulins contain additional polypeptide components **(Table 3–1)**. In IgM, five of the basic immunoglobulin units join around a polypeptide called the J chain to form a pentamer. In IgA, the **secretory immunoglobulin,** the immunoglobulin units form dimers and trimers around a J chain and a polypeptide that comes from epithelial cells, the secretory component (SC).

TABLE 3–1 Human immunoglobulins.[a]

Immunoglobulin	Function	Heavy Chain	Additional Chain	Structure	Plasma Concentration (mg/dL)
IgG	Complement activation	$\gamma_1, \gamma_2, \gamma_3, \gamma_4$		Monomer	1000
IgA	Localized protection in external secretions (tears, intestinal secretions, etc.)	α_1, α_2	J, SC	Monomer; dimer with J or SC chain; trimer with J chain	200
IgM	Complement activation	μ	J	Pentamer with J chain	120
IgD	Antigen recognition by B cells	δ		Monomer	3
IgE	Reagin activity; releases histamine from basophils and mast cells	ε		Monomer	0.05

[a]In all instances, the light chains are k or γ.

In the intestine, bacterial and viral antigens are taken up by M cells (see Chapter 26) and passed on to underlying aggregates of lymphoid tissue **(Peyer patches),** where they activate naive T cells. These lymphocytes then form B cells that infiltrate mucosa of the gastrointestinal, respiratory, genitourinary, and female reproductive tracts and the breast. There they secrete large amounts of IgA when exposed again to the antigen that initially stimulated them. The epithelial cells produce the SC, which acts as a receptor for, and binds to, IgA. The resulting secretory immunoglobulin passes through the epithelial cell and is secreted by exocytosis. This system of **secretory immunity** is an important and effective defense mechanism at all mucosal surfaces. Because IgA is secreted into the breast milk, it also accounts for the immune protection that is conferred by the breastfeeding of infants whose immune systems are otherwise immature.

CLINICAL BOX 3–2

Autoimmunity

Sometimes, the processes that eliminate antibodies against self-antigens fail and a variety of different **autoimmune diseases** are produced. These can be B cell– or T cell– mediated and can be organ-specific or systemic. They include type 1 diabetes mellitus (antibodies against pancreatic islet B cells), myasthenia gravis (antibodies against nicotinic cholinergic receptors), and multiple sclerosis (antibodies against myelin basic protein and several other components of myelin). In some instances, the antibodies are against receptors and are capable of activating those receptors; for example, antibodies against TSH receptors increase thyroid activity and cause Graves disease (see Chapter 20). Other conditions are due to the production of

antibodies against invading organisms that cross-react with normal body constituents **(molecular mimicry)**. An example is rheumatic fever following a streptococcal infection; a portion of cardiac myosin resembles a portion of the streptococcal M protein, and antibodies induced by the latter attack the former and damage the heart. Some conditions may be due to **bystander effects,** in which inflammation sensitizes T cells in the neighborhood, causing them to become activated when otherwise they would not respond.

THERAPEUTIC HIGHLIGHTS

The therapy of autoimmune disorders rests on efforts to replace or restore the damaged function (eg, provision of exogenous insulin in type 1 diabetes) as well as nonspecific efforts to reduce inflammation (using corticosteroids) or to suppress immunity. Recently, agents that deplete or dampen the function of B cells have been shown to have some efficacy in a range of autoimmune disorders, including rheumatoid arthritis, most likely by interrupting the production of autoantibodies that contribute to disease pathogenesis.

GENETIC BASIS OF DIVERSITY IN THE IMMUNE SYSTEM

The genetic mechanism for the production of the immensely large number of different configurations of immunoglobulins produced by human B cells, as well as T cell receptors, is a fascinating biologic problem. Diversity is brought about in part by the fact that in immunoglobulin molecules there are two kinds of light chains and nine kinds of heavy chains. As noted previously, there are areas of great variability **(hypervariable regions)** in each chain. The variable portion of the heavy chains consists of the V, D, and J segments. In the gene family responsible for this region, there are several hundred different coding regions for the V segment, about 20 for the D segment, and four for the J segment. During B cell development, one V coding region, one D coding region, and one J coding region are selected at random and recombined to form the gene that produces that particular variable portion. A similar variable recombination takes place in the coding regions responsible for the two variable segments (V and J) in the light chain. In addition, the J segments are variable because the gene segments join in an imprecise and variable fashion (junctional site diversity) and nucleotides are sometimes added (junctional insertion diversity). It has been

calculated that these mechanisms permit the production of about 10^{15} different immunoglobulin molecules. Additional variability is added by somatic mutation.

Similar gene rearrangement and joining mechanisms operate to produce the diversity in T cell receptors. In humans, the α subunit has a V region encoded by 1 of about 50 different genes and a J region encoded by 1 of another 50 different genes. The β subunits have a V region encoded by 1 of about 50 genes, a D region encoded by 1 of 2 genes, and a J region encoded by 1 of 13 genes. These variable regions permit the generation of up to an estimated 10^{15} different T cell receptors (**Clinical Box 3–2** and **Clinical Box 3–3**).

More than 300 primary immunodeficiency states are now known to arise from defects in these various stages of B and T lymphocyte maturation **(Clinical Box 3–4)**. A few important ones are shown in **Figure 3–9**.

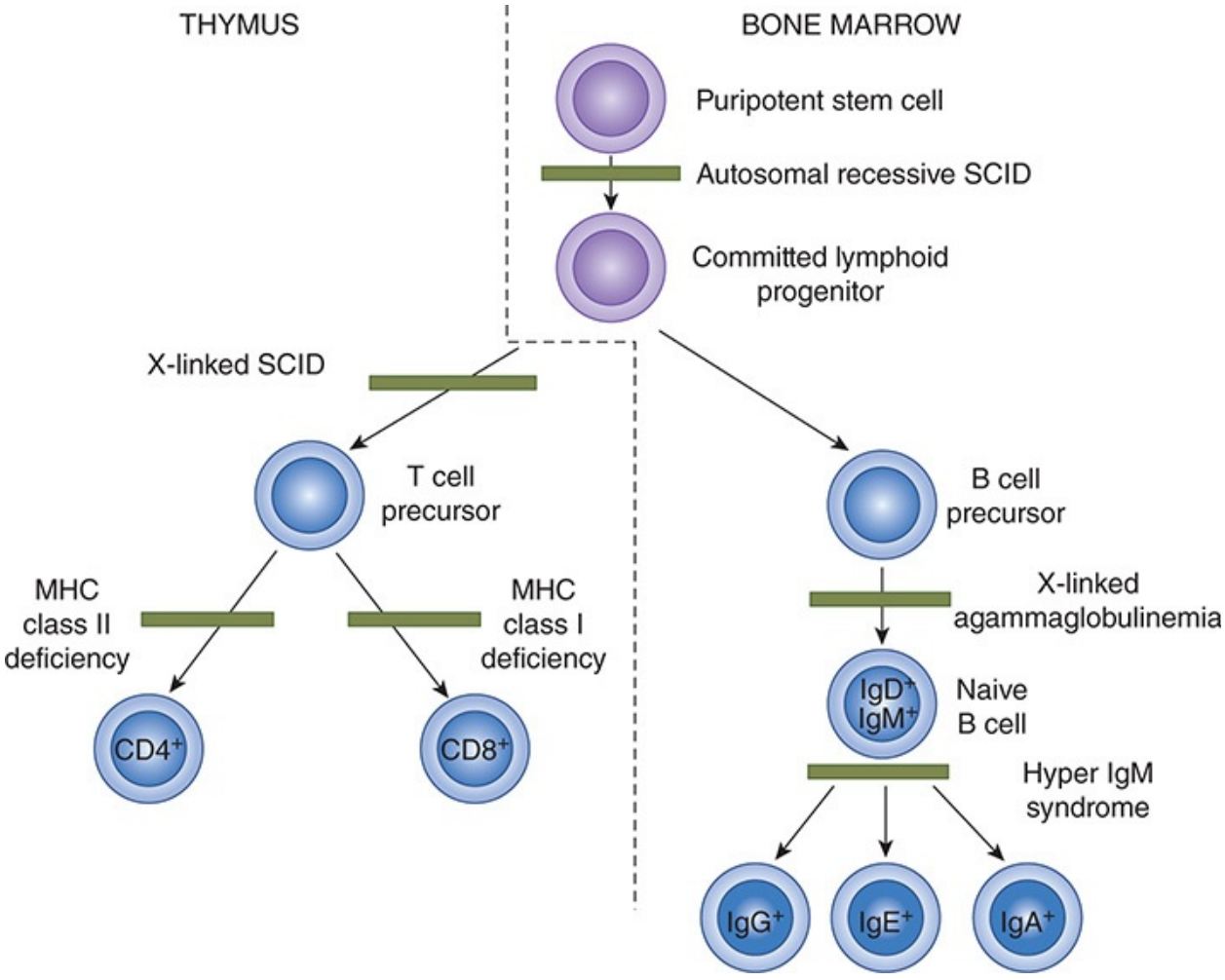


FIGURE 3–9 Sites of congenital blockade of B and T lymphocyte maturation in various primary immunodeficiency states. SCID, severe

combined immune deficiency.

SOLUBLE REGULATORS OF IMMUNITY

CYTOKINES

Cytokines are hormone-like molecules that act—generally in a paracrine fashion—to regulate immune responses. They are secreted not only by lymphocytes and macrophages but also by endothelial cells, neurons, glial cells, and other types of cells. Most of the cytokines were initially named for their actions, for example, B cell–differentiating factor, or B cell–stimulating factor 2. However, the nomenclature has since been rationalized by international agreement to that of the **interleukins.** For example, the name of B cell–differentiating factor was changed to interleukin-4. A number of cytokines selected for their biologic and clinical relevance are listed in Table 3–2, but it would be beyond the scope of this text to list all cytokines, which now number more than 100.

TABLE 3–2 Examples of cytokines and their clinical relevance.

Cytokine	Cellular Sources	Major Activities	Clinical Relevance
IL-1	Macrophages	Activation of T cells and macrophages; promotion of inflammation	Implicated in the pathogenesis of septic shock, rheumatoid arthritis, and atherosclerosis
IL-2	Type 1 (Th1) helper T cells	Activation of lymphocytes, natural killer cells, and macrophages	Used to induce lymphokine-activated killer cells; used in the treatment of metastatic renal cell carcinoma, melanoma, and various other tumors
IL-4	Type 2 (Th2) helper T cells, mast cells, basophils, and eosinophils	Activation of lymphocytes, monocytes, and IgE class switching	As a result of its ability to stimulate IgE production, plays a part in mast-cell sensitization and thus in allergy and in defense against nematode infections
IL-5	Type 2 (Th2) helper T cells, mast cells, and eosinophils	Differentiation of eosinophils	Monoclonal antibody against interleukin-5 used to inhibit the antigen-induced late-phase eosinophilia in animal models of allergy
IL-6	Type 2 (Th2) helper T cells and macrophages	Activation of lymphocytes; differentiation of B cells; stimulation of the production of acute-phase proteins	Overproduced in Castleman disease; acts as an autocrine growth factor in myeloma and in mesangial proliferative glomerulonephritis
IL-8	T cells and macrophages	Chemotaxis of neutrophils, basophils, and T cells	Levels are increased in diseases accompanied by neutrophilia, making it a potentially useful marker of disease activity
IL-11	Bone marrow stromal cells	Stimulation of the production of acute-phase proteins	Used to reduce chemotherapy-induced thrombocytopenia
IL-12	Macrophages and B cells	Stimulation of the production of interferon γ by type 1 (Th1) helper T cells and by natural killer cells; induction of type 1 (Th1) helper T cells	May be useful as an adjuvant for vaccines
IL-17	T cells	Promotion of inflammatory cell chemotaxis and inflammation	Implicated in many immune/autoimmune diseases such as rheumatoid arthritis, asthma, and psoriasis
Tumor necrosis factor-α	Macrophages, natural killer cells, T cells, B cells, and mast cells	Promotion of inflammation	Treatment with antibodies against tumor necrosis factor-α beneficial in rheumatoid arthritis and Crohn disease
Lymphotoxin (tumor necrosis factor-β)	Type 1 (Th1) helper T cells and B cells	Promotion of inflammation	Implicated in the pathogenesis of multiple sclerosis and insulin-dependent diabetes mellitus
Transforming growth factor-β	T cells, macrophages, B cells, and mast cells	Immunosuppression	May be useful therapeutic agent in multiple sclerosis and myasthenia gravis
GM-CSF	T cells, macrophages, natural killer cells, and B cells	Promotion of the growth of granulocytes and monocytes	Used to reduce neutropenia after chemotherapy for tumors and in ganciclovir-treated patients with AIDS; used to stimulate cell production after hematopoietic stem cell transplantation
Interferon-α	Virally infected cells	Induction of resistance of cells to viral infection	Used to treat AIDS-related Kaposi sarcoma, melanoma, chronic hepatitis B infection, and chronic hepatitis C infection
Interferon-β	Virally infected cells	Induction of resistance of cells to viral infection	Used to reduce the frequency and severity of relapses in multiple sclerosis
Interferon-γ	Type 1 (Th1) helper T cells and natural killer cells	Activation of macrophages; inhibition of type 2 (Th2) helper T cells	Used to enhance the killing of phagocytosed bacteria in chronic granulomatous disease

Modified with permission from Delves PJ, Roitt IM: The immune system. First of two parts, N Engl J Med 2000; July 6:343(1):37-49.

CLINICAL BOX 3–3

Tissue Transplantation

The T lymphocyte system is responsible for the rejection of transplanted tissue. When tissues such as skin and kidneys are transplanted from a donor to a recipient of the same species, the transplants "take" and function for a while but then become necrotic and are "rejected" because an immune response to the transplanted tissue develops in the recipient. This is generally true even if the donor and recipient are close relatives, and the only transplants that are never rejected are those from an identical twin. Nevertheless, organ transplantation remains the only viable option in a number of end-stage diseases.

THERAPEUTIC HIGHLIGHTS

A number of treatments have been developed to overcome the rejection of transplanted organs in humans. The goal of treatment is to stop rejection without leaving the patient vulnerable to massive infections. One approach is to kill T lymphocytes by killing all rapidly dividing cells with drugs such as azathioprine, a purine antimetabolite, but this makes patients susceptible to infections and cancer. Another is to administer corticosteroids, which inhibit cytotoxic T cell proliferation by inhibiting production of IL-2, but these cause osteoporosis, mental changes, and the other facets of Cushing syndrome (see Chapter 19). More recently, immunosuppressive drugs such as **cyclosporine** or **tacrolimus (FK-506)** have found favor. Activation of the T cell receptor normally increases intracellular Ca^{2+}, which acts via calmodulin to activate calcineurin. Calcineurin dephosphorylates the transcription factor NF-AT, which moves to the nucleus and increases the activity of genes coding for IL-2 and related stimulatory cytokines. Cyclosporine and tacrolimus prevent the dephosphorylation of NF-AT. However, these drugs inhibit all T cell–mediated immune responses, and cyclosporine causes kidney damage and cancer. A new and promising approach to transplant rejection is the production of T cell unresponsiveness by using drugs that block the costimulation that is required for normal activation (see text). Clinically effective drugs that act in this fashion could be of great value to transplant surgeons.

CLINICAL BOX 3–4

Primary Immunodeficiencies

Mutations in enzymes, transcription factors, receptors, and other regulators of acquired or innate immunity lead to primary immunodeficiency disorders. Typically, patients experience frequent infections or have an increased susceptibility to unusual infections, but they may also be subject to autoimmune problems, spontaneous inflammation, and malignancy. The implications depend on the level at which the immune system is disrupted, with mutations that prevent the development of many lymphoid lineages having the most serious consequences (Figure 3–9).

THERAPEUTIC HIGHLIGHTS

Mild immunodeficiencies may require only minimal or supportive therapies, but the most serious cause substantial morbidity and early mortality. Patients suffering from the latter conditions, therefore, are candidates for early and definitive therapy, and many have benefitted greatly from the transplant of allogeneic hematopoietic stem cells (HSCs), provided they can tolerate the procedure and a suitable donor can be found. On the other hand, the monogenic nature of many serious primary immunodeficiencies has made them an attractive target for trials of gene therapy approaches. There have been a number of notable successes with transplant of HSCs transduced with retroviral or lentiviral vectors carrying a wild-type copy of the affected gene, such as in X-linked severe combined immune deficiency (SCID), which results in mutations in the gamma chain of the receptor for IL-2.

Many of the receptors for cytokines and hematopoietic growth factors (see above), as well as the receptors for prolactin (see Chapter 22), and growth hormone (see Chapter 18) are members of a cytokine-receptor superfamily that has three subfamilies **(Figure 3–10)**. The members of subfamily 1, which includes the receptors for IL-4 and IL-7, are homodimers. The members of subfamily 2, which includes the receptors for IL-3, IL-5, and IL-6, are heterodimers. The receptor for IL-2 (and several other cytokines) consists of a heterodimer plus an unrelated protein, the so-called Tac antigen. The other members of subfamily 3 have the same γ chain as the IL-2 receptor. The extracellular domain of the homodimer and heterodimer subunits all contain four conserved cysteine residues plus a conserved Trp-Ser-X-Trp-Ser domain, and although the intracellular portions do not contain tyrosine kinase catalytic

domains, they activate cytoplasmic tyrosine kinases when ligand binds to the receptors.

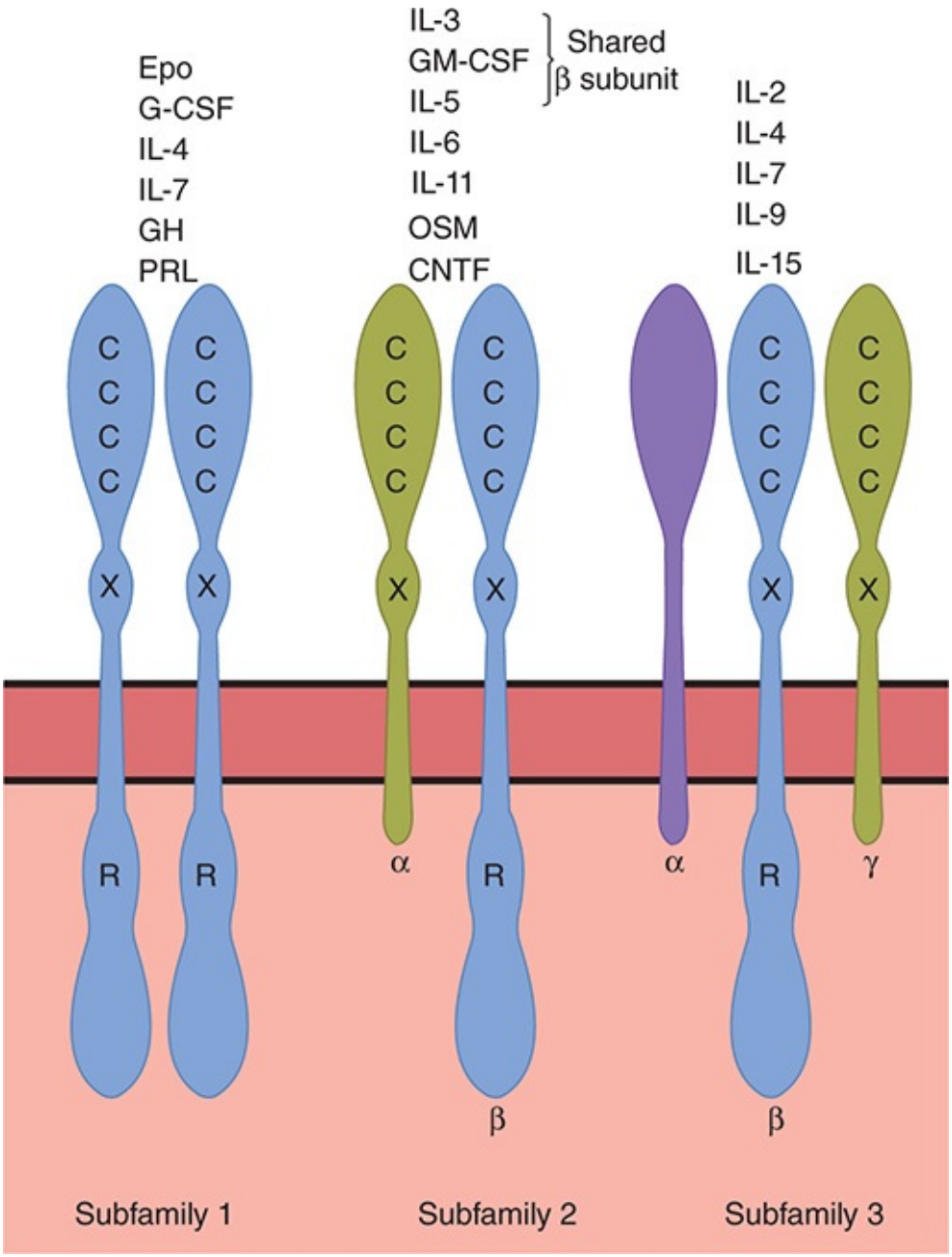


FIGURE 3–10 Members of an important cytokine receptor superfamily, showing shared structural elements. Note that all subunits except the α subunit in subfamily 3 have four conserved cysteine residues (C) and a Trp-Ser-X-Trp-Ser motif (X). Many subunits also contain a critical regulatory domain in their cytoplasmic portions (R). CNTF, ciliary neurotrophic factor; Epo, erythropoietin; GH, growth hormone; OSM, oncostatin M; PRL, prolactin.

The effects of the principal cytokines are listed in Table 3–2. Some of them have systemic as well as local paracrine effects. For example, IL-1, IL-6, and TNF-α cause fever, and IL-1 increases slow-wave sleep and reduces appetite.

Another superfamily of cytokines is the **chemokine** family. Chemokines are substances that attract neutrophils and other immune effector cells to areas of inflammation or an immune response. Over 40 have now been identified, and it is clear that they also play a role in the regulation of cell growth and angiogenesis. The chemokine receptors are G-protein-coupled receptors that cause, among other things, extension of pseudopodia with migration of the cell toward the source of the chemokine.

HEMATOPOIETIC GROWTH FACTORS

The production of immune effector cells is regulated with great precision in healthy individuals, and the production of granulocytes, in particular, is rapidly and dramatically increased in infections. Many cytokines participate in this regulation, together with other soluble mediators, many of which are referred to as colony-stimulating factors (CSF) because they stimulate progenitors to grow as colonies in soft agar. The overall proliferation and self-renewal of HSCs depends on **stem cell factor (SCF).** The proliferation and maturation of the cells that enter the blood from the marrow are then regulated by growth factors that cause cells in one or more of the committed cell lines to proliferate and mature **(Table 3–3)**. The factors stimulating the production of committed stem cells include granulocyte/macrophage **(GM)-CSF, granulocyte CSF (G-CSF),** and **macrophage CSF (M-CSF).** Interleukins **IL-1** and **IL-6** followed by **IL-3** (Table 3–3) act in sequence to convert pluripotential uncommitted stem cells to committed progenitor cells. IL-3 is also known as **multi-CSF.** Each of the CSFs has a predominant action, but all the CSFs and interleukins also have other overlapping actions. In addition, they activate and sustain mature blood cells. In fact, it is interesting that basal hematopoiesis is normal in mice in which the GM-CSF gene is knocked out, indicating that loss of one factor can be compensated for by others. On the other hand, the absence of GM-CSF causes accumulation of surfactant in the lungs (see Chapter 34).

TABLE 3–3 Hematopoietic growth factors.

Cytokine	Cell Types Stimulated	Source
IL-1	Erythrocyte Granulocyte Megakaryocyte Monocyte	Multiple cell types
IL-3	Erythrocyte Granulocyte Megakaryocyte Monocyte	T lymphocytes
IL-4	Basophil	T lymphocytes
IL-5	Eosinophil	T lymphocytes
IL-6	Erythrocyte Granulocyte Megakaryocyte Monocyte	Endothelial cells, fibroblasts, macrophages
IL-11	Erythrocyte Granulocyte Megakaryocyte	Fibroblasts, osteoblasts
Erythropoietin	Erythrocyte	Kidney Kupffer cells of liver
SCF	Erythrocyte Granulocyte Megakaryocyte Monocyte	Multiple cell types
G-CSF	Granulocyte	Endothelial cells, fibroblasts, monocytes
GM-CSF	Erythrocyte Granulocyte Megakaryocyte	Endothelial cells, fibroblasts, monocytes, T lymphocytes
M-CSF	Monocyte	Endothelial cells, fibroblasts, monocytes,
Thrombopoietin	Megakaryocyte	Liver, kidney

CSF, colony-stimulating factor; G, granulocyte; IL, interleukin; M, macrophage; SCF, stem cell factor.

Reproduced with permission from McPhee SJ, Lingappa VR, Ganong WF (editors): *Pathophysiology of Disease*, 6th ed. New York, NY: McGraw-Hill; 2010.

The other factors listed are produced by macrophages, activated T cells, fibroblasts, and endothelial cells. For the most part, the factors act locally in the bone marrow.

THE COMPLEMENT SYSTEM

The cell-killing effects of innate and acquired immunity are mediated in part by a system of more than 30 plasma proteins originally named the **complement system** because they "complemented" the effects of antibodies. Three different pathways or enzyme cascades activate the system: the **classic pathway,** triggered by immune complexes; the **mannose-binding lectin pathway,** triggered when this lectin binds mannose groups in bacteria; and the **alternative** or **properdin pathway,** triggered by contact with various viruses, bacteria, fungi, and tumor cells. The proteins that are produced have three functions: they help kill invading organisms by opsonization (coating of the organism), chemotaxis, and eventual lysis of the cell; they serve in part as a bridge from innate to acquired immunity by activating B cells and aiding immune memory; and they help dispose of waste products after apoptosis. Cell lysis, one of the principal ways the complement system kills cells, is brought about by inserting proteins called **perforins** into their cell membranes. These create holes, which permit free flow of ions and thus disruption of membrane polarity.

OTHER IMMUNE/INFLAMMATORY EFFECTOR CELLS

Many additional immune effector cells circulate in the blood as the white blood cells. In addition, the blood is the conduit for the precursor cells that eventually develop into the immune cells of the tissues. The circulating immunologic cells include **granulocytes (polymorphonuclear leukocytes, PMNs),** comprising **neutrophils, eosinophils,** and **basophils;** lymphocytes (discussed above); and **monocytes.** Immune responses in the tissues are further amplified by these cells following their extravascular migration, as well as tissue **macrophages** (derived from monocytes) and **mast cells** (related to basophils). Acting together, these cells provide the body with powerful defenses against tumors and viral, bacterial, and parasitic infections.

GRANULOCYTES

All granulocytes have cytoplasmic granules that contain biologically active substances involved in inflammatory and allergic reactions.

The average half-life of a neutrophil in the circulation is 6 h. To maintain the normal circulating blood level, it is necessary to produce over 100 billion neutrophils per day. Many neutrophils enter the tissues, particularly if triggered to do so by an infection or by inflammatory cytokines. They are attracted to the endothelial surface by cell adhesion molecules known as selectins, and they roll along it. They then bind firmly to neutrophil adhesion molecules of the integrin family. They next insinuate themselves through the walls of the capillaries between endothelial cells by a process called **diapedesis.** Many of those that leave the circulation enter the gastrointestinal tract and are eventually lost from the body.

Invasion of the body by bacteria triggers the **inflammatory response.** The bone marrow is stimulated to produce and release large numbers of neutrophils. Bacterial products interact with plasma factors and cells to produce chemokines and other agents that attract neutrophils to the infected area **(chemotaxis).** The chemotactic agents include a component of the complement system (C5a); leukotrienes; and polypeptides from lymphocytes, mast cells, and basophils. Other plasma factors act on the bacteria to make them “tasty” to the phagocytes **(opsonization).** The principal opsonins that coat the bacteria are IgG immunoglobulins and complement proteins. The coated bacteria then bind to G-protein-coupled receptors on the neutrophil cell membrane. This triggers increased motor activity of the cell, exocytosis, and the so-called respiratory burst. The increased motor activity leads to prompt ingestion of the bacteria by endocytosis **(phagocytosis).** By **exocytosis,** neutrophil granules discharge their contents into the phagocytic vacuoles containing the bacteria and also into the interstitial space **(degranulation).** The granules contain various proteases plus antimicrobial proteins called **defensins.** In addition, the cell membrane-bound enzyme **nicotinamide adenine dinucleotide phosphate (NADPH) oxidase** is activated, with the production of toxic oxygen metabolites. The combination of the toxic oxygen metabolites and the proteolytic enzymes from the granules makes the neutrophil a very effective killing machine. Disorders that affect the neutrophil’s killing ability increase the susceptibility to infections **(Clinical Box 3–5)**.

Activation of NADPH oxidase is associated with a sharp increase in O_2 uptake and metabolism in the neutrophil (the **respiratory burst**) and generation

of superoxide (O_2^-) by the following reaction:

$$NADPH + H^+ + 2O_2^+ \rightarrow NADP^+ + 2H^+ + 2O_2^-$$

O_2^- is a **free radical** formed by the addition of one electron to O_2. Two O_2^- react with two H^+ to form H_2O_2 in a reaction catalyzed by the cytoplasmic form of superoxide dismutase (SOD-1):

$$2O_2^- + 2H^+ \xrightarrow{\text{SOD-1}} H_2O_2 + O_2$$

O_2^- and H_2O_2 are both oxidants that are effective bactericidal agents, but H_2O_2 is converted to H_2O and O_2 by the enzyme **catalase.** The cytoplasmic form of SOD-1 contains both Zn and Cu. It is found in many parts of the body. It is defective as a result of genetic mutation in a familial form of **amyotrophic lateral sclerosis** (ALS; see Chapter 15). Therefore, it may be that O_2^- accumulates in motor neurons and kills them in at least one form of this progressive, fatal disease. Two other forms of SOD encoded by at least one different gene are also found in humans.

Neutrophils also discharge the enzyme **myeloperoxidase,** which catalyzes the conversion of Cl^-, Br^-, I^-, and SCN^- to the corresponding acids (HOCl, HOBr, etc). These acids are also potent oxidants. Because Cl^- is present in greatest abundance in body fluids, the principal product is HOCl.

In addition to myeloperoxidase and defensins, neutrophil granules contain elastase, metalloproteinases that attack collagen, and a variety of other proteases that help destroy invading organisms. These enzymes act in a cooperative fashion with O_2^-, H_2O_2, and HOCl to produce a killing zone around the activated neutrophil. This zone is effective in killing invading organisms, but in certain diseases (eg, rheumatoid arthritis) the neutrophils may also cause local destruction of host tissue.

Like neutrophils, **eosinophils** have a short half-life in the circulation, are attracted to the surface of endothelial cells by selectins, bind to integrins that attach them to the vessel wall, and enter the tissues by diapedesis. Like neutrophils, they release proteins, cytokines, and chemokines that produce inflammation but are capable of killing invading organisms. However, eosinophils have some selectivity in the way in which they respond and in the killing molecules they secrete. Their maturation and activation in tissues is particularly stimulated by IL-3, IL-5, and GM-CSF. They are especially

abundant in the mucosa of the gastrointestinal tract, where they defend against parasites, and in the mucosa of the respiratory and urinary tracts. Circulating eosinophils are increased in allergic diseases such as asthma and in various other respiratory and gastrointestinal diseases.

CLINICAL BOX 3–5

Disorders of Phagocytic Function

More than 15 primary defects in neutrophil function have been described, along with at least 30 other conditions in which there is a secondary depression of the function of neutrophils. Patients with these diseases are prone to infections that are relatively mild when only the neutrophil system is involved, but which can be severe when the monocyte-tissue macrophage system is also involved. In one syndrome (neutrophil hypomotility), actin in the neutrophils does not polymerize normally, and the neutrophils move slowly. In another, there is a congenital deficiency of leukocyte integrins. In a more serious disease (chronic granulomatous disease), there is a failure to generate O_2^- in both neutrophils and monocytes and consequent inability to kill many phagocytosed bacteria. In severe congenital glucose-6-phosphate dehydrogenase deficiency, there are multiple infections because of failure to generate the NADPH necessary for O_2^- production. In congenital myeloperoxidase deficiency, microbial killing power is reduced because hypochlorous acid is not formed.

THERAPEUTIC HIGHLIGHTS

The cornerstones of treatment in disorders of phagocytic function include scrupulous efforts to avoid exposure to infectious agents, and antibiotic and antifungal prophylaxis. Antimicrobial therapies must also be implemented aggressively if infections occur. Sometimes, surgery is needed to excise and/or drain abscesses and relieve obstructions. HSC transplantation may offer the hope of a definitive cure for severe conditions, such as chronic granulomatous disease. Sufferers of this condition have a significantly reduced life expectancy due to recurrent infections and their complications, and so the risks of bone marrow transplantation may be acceptable. Gene therapy, on the other hand, remains a distant goal.

Basophils also enter tissues and release proteins and cytokines. They resemble but are not identical to mast cells, and like mast cells they contain histamine (see below). They release histamine and other inflammatory mediators when activated by binding of specific antigens to cell-fixed IgE molecules, and participate in immediate-type hypersensitivity (allergic) reactions. These range from mild urticaria and rhinitis to severe anaphylactic shock. The antigens that trigger IgE formation and basophil (and mast cell) activation are innocuous to most individuals and are referred to as allergens.

MAST CELLS

Mast cells are heavily granulated cells of the connective tissue that are abundant in tissues that come into contact with the external environment, such as beneath epithelial surfaces. Their granules contain proteoglycans, histamine, and many proteases. Like basophils, they degranulate when allergens bind to cell-bound IgE molecules directed against them. They are involved in inflammatory responses initiated by immunoglobulins IgE and IgG. The inflammation combats invading parasites. In addition to this involvement in acquired immunity, they release TNF-α in response to bacterial products by an antibody-independent mechanism, thus participating in the nonspecific **innate immunity** that combats infections prior to the development of an adaptive immune response (see following text). Marked mast cell degranulation produces clinical manifestations of allergy up to and including anaphylaxis.

MONOCYTES

Monocytes enter the blood from the bone marrow and circulate for about 72 h. They then enter the tissues and become **tissue macrophages** **(Figure 3–11)**. Their life span in the tissues is unknown, but bone marrow transplantation data in humans suggest that they persist for about 3 months. It appears that they do not reenter the circulation. Some may end up as the multinucleated giant cells seen in chronic inflammatory diseases such as tuberculosis. The tissue macrophages include the Kupffer cells of the liver, pulmonary alveolar macrophages (see Chapter 34), and microglia in the brain, all of which come originally from the circulation.

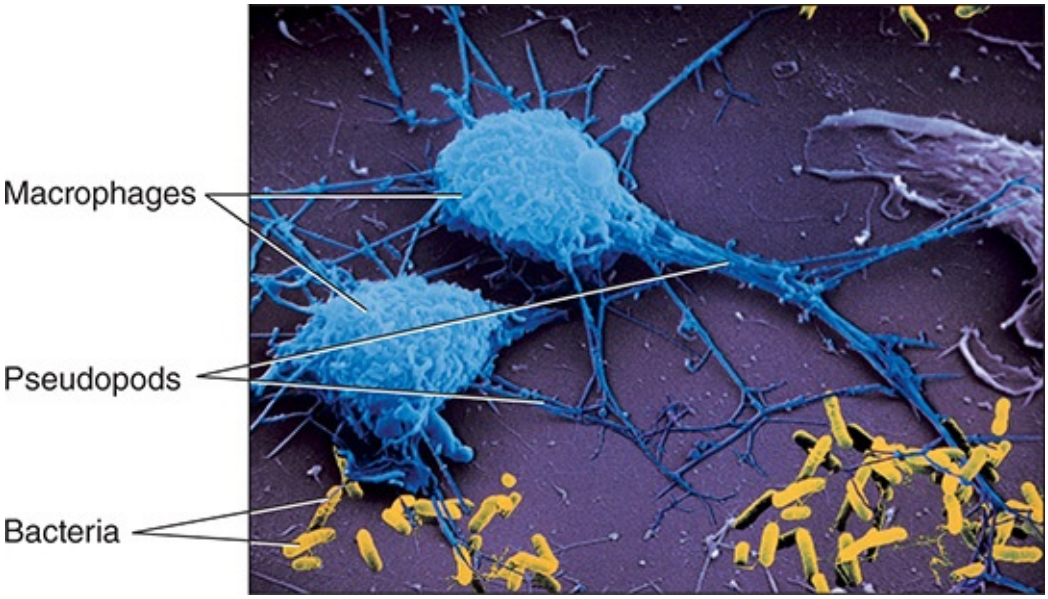


FIGURE 3–11 Macrophages contacting bacteria and preparing to engulf them. Figure is a colorized version of a scanning electron micrograph.

Macrophages are activated by cytokines released from T lymphocytes, among others. Activated macrophages migrate in response to chemotactic stimuli and engulf and kill bacteria by processes generally similar to those occurring in neutrophils. They play a key role in innate immunity (see below). They also secrete up to 100 different substances, including factors that affect lymphocytes and other cells, prostaglandins of the E series, and clot-promoting factors.

PLATELETS

Platelets are non-nucleated circulating blood elements that are important mediators of hemostasis. While not immune cells, per se, they often participate in the response to tissue injury in cooperation with inflammatory cell types. They form in the bone marrow as pinched-off pieces of precursor cells known as megakaryocytes. **Thrombopoietin** facilitates megakaryocyte maturation (see Figure 31–3) and is produced constitutively by the liver and kidneys, and there are also thrombopoietin receptors on platelets. Consequently, when the number of platelets is low, less is bound and more is available to stimulate production of platelets. Conversely, when the number of platelets is high, more is bound and less is available, producing a form of feedback control of platelet production.

Platelets have a ring of microtubules around their periphery and an extensively invaginated membrane with an intricate canalicular system in

contact with the extracellular fluid. Their membranes contain receptors for collagen, ADP, vessel wall von Willebrand factor (see below), and fibrinogen. Their cytoplasm contains actin, myosin, glycogen, lysosomes, and two types of granules: (1) dense granules, which contain nonprotein substances that are secreted in response to platelet activation, including serotonin, ADP, and other adenine nucleotides; and (2) α-granules, which contain secreted proteins. These proteins include clotting factors and **platelet-derived growth factor (PDGF).** PDGF stimulates wound healing and is a potent mitogen for vascular smooth muscle. Blood vessel walls as well as platelets contain von Willebrand factor, which, in addition to its role in adhesion, regulates circulating levels of the clotting factor, factor VIII (see below).

When a blood vessel wall is injured, platelets adhere to the exposed collagen and **von Willebrand factor** in the wall via receptors on the platelet membrane (see also Chapter 31). von Willebrand factor is a large circulating molecule that is produced by endothelial cells. Binding produces platelet activation, which releases the contents of their granules. The released ADP acts on the ADP receptors in the platelet membranes to produce further accumulation of more platelets **(platelet aggregation).** Humans have at least three different types of platelet ADP receptors: $P2Y_1$, $P2Y_2$, and $P2X_1$. These are obviously attractive targets for drug development, and several new inhibitors have shown promise in the prevention of myocardial infarctions and strokes. Aggregation is also fostered by **platelet-activating factor (PAF)**, a cytokine secreted by neutrophils and monocytes as well as platelets. This compound also has inflammatory activity. It is an ether phospholipid, 1-alkyl-2-acetylglyceryl-3-phosphorylcholine, which is produced from membrane lipids. It acts via a G-protein-coupled receptor to increase the production of arachidonic acid derivatives, including thromboxane A_2. The role of this latter compound in the balance between clotting and anticlotting activity at the site of vascular injury is discussed in Chapter 31.

When the platelet count is low, clot retraction is deficient and there is poor constriction of ruptured vessels. The resulting clinical syndrome **(thrombocytopenic purpura)** is characterized by easy bruisability and multiple subcutaneous hemorrhages. Purpura may also occur when the platelet count is normal, and in some of these cases, the circulating platelets are abnormal **(thrombasthenic purpura).** Individuals with thrombocytosis are predisposed to thrombotic events.

INFLAMMATION & WOUND HEALING

LOCAL INJURY

Inflammation is a complex localized response to foreign substances such as bacteria or, in some instances, to internally produced substances. It includes a sequence of reactions initially involving cytokines, neutrophils, adhesion molecules, complement, and IgG. PAF also plays a role. Later, monocytes and lymphocytes are involved. Arterioles in the inflamed area dilate, and capillary permeability is increased (see Chapters 32 and 33). When the inflammation occurs in or just under the skin **(Figure 3–12)**, it is characterized by redness, swelling, tenderness, and pain. Elsewhere, it is a key component of asthma, ulcerative colitis, Crohn disease (Clinical Box 3–1), rheumatoid arthritis, and many other diseases.

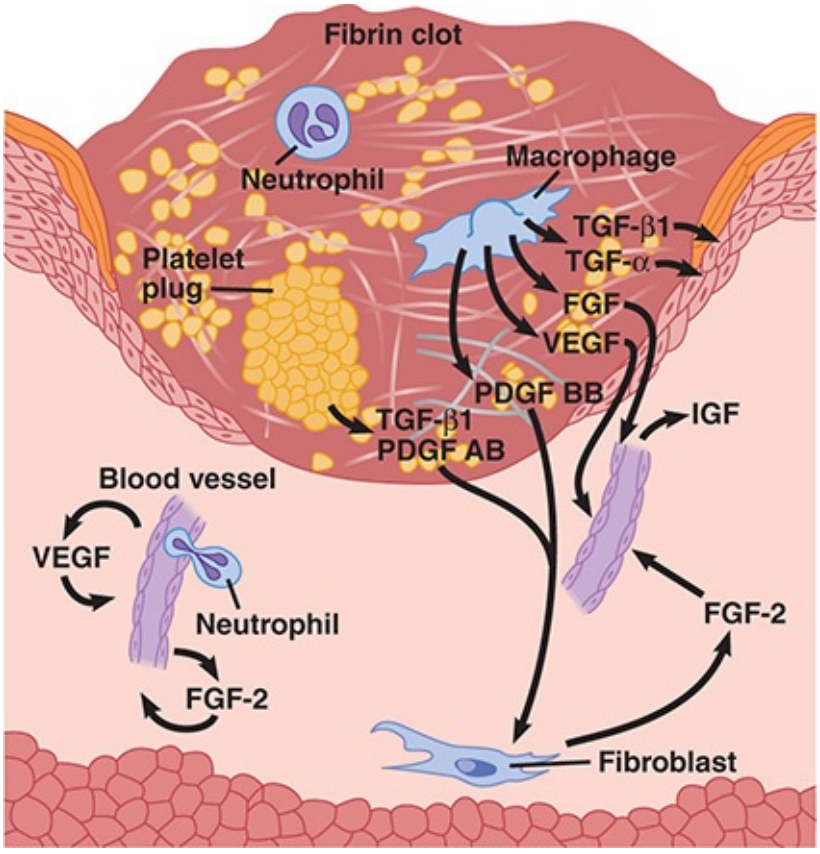


FIGURE 3–12 Cutaneous wound 3 days after injury, showing the multiple cytokines and growth factors affecting the repair process. FGF, fibroblast growth factor; IGF, insulin-like growth factor; PDGF, platelet-derived growth factor; TGF, transforming growth factor; VEGF, vascular endothelial growth

factor. Note the epidermis growing down under the fibrin clot, restoring skin continuity. (Modified from Singer AJ, Clark RAF: Cutaneous wound healing. N Engl J Med 1999;September 2;341(10):738-746.)

A transcription factor known as **nuclear factor-κB** (NF-κB) plays the pivotal role in orchestrating the inflammatory response. NF-κB is a heterodimer that normally exists in the cytoplasm of cells bound to IκBα, which renders it inactive. Stimuli such as cytokines, viruses, and oxidants induce signals that allow NF-κB to dissociate from IκBα, which is then degraded. NF-κB moves to the nucleus, where it binds to the DNA of the genes for numerous inflammatory mediators, resulting in their increased production and secretion. Corticosteroids inhibit the activation of NF-κB by increasing the production of IκBα, and this is probably the main basis of their anti-inflammatory action (see Chapter 19).

SYSTEMIC RESPONSE TO INJURY

Cytokines produced in response to inflammation and other injuries, as well as disseminated infection, also produce systemic responses. These include alterations in plasma **acute phase proteins,** defined as proteins whose concentration is increased or decreased by at least 25% following injury. Many of the proteins are of hepatic origin. A number of them are shown in **Figure 3–13**. The causes of the changes in concentration are incompletely understood, but it can be said that many of the changes make homeostatic sense. Thus, for example, an increase in C-reactive protein activates monocytes and causes further production of cytokines. Other changes that occur in response to injury include somnolence, negative nitrogen balance, and fever.

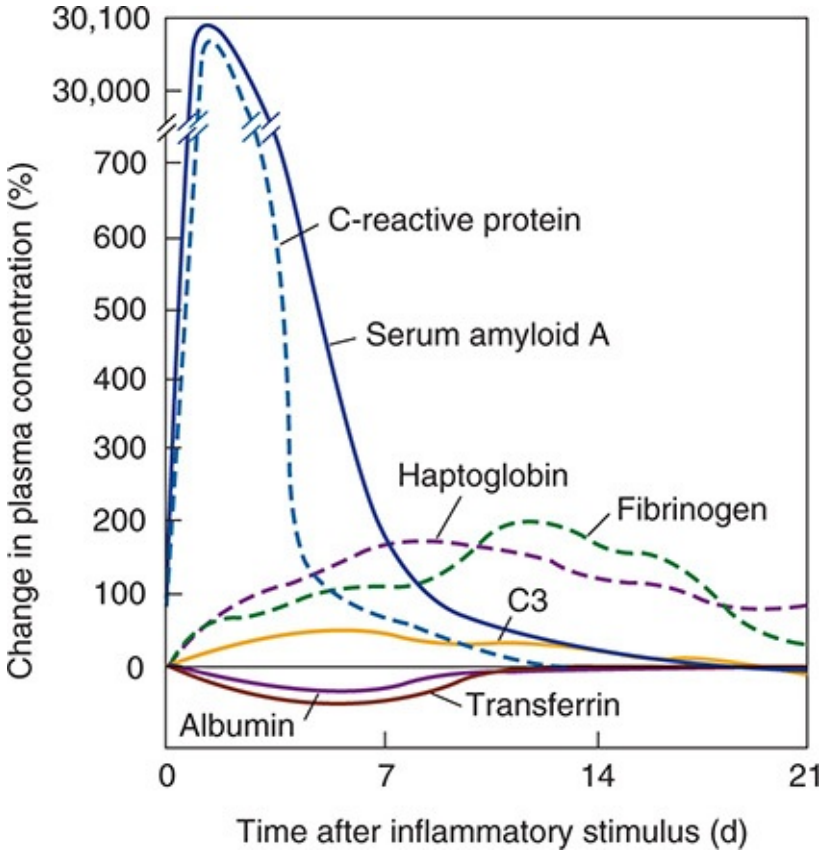


FIGURE 3–13 Time course of changes in some major acute phase proteins. C3, C3 component of complement. (Modified with permission from McAdam KP, Elin RJ, Sipe JD, Wolff SM: Changes in human serum amyloid A and C-reactive protein after etiocholanolone-induced inflammation. J Clin Invest 1978;February;61(2):390-394.)

WOUND HEALING

When tissue is damaged, platelets adhere to exposed matrix via integrins that bind to collagen and laminin (Figure 3–12). Blood coagulation produces thrombin, which promotes platelet aggregation and granule release. The platelet granules generate an inflammatory response. White blood cells are attracted by selectins and bind to integrins on endothelial cells, leading to their extravasation through the blood vessel walls. Cytokines released by the white blood cells and platelets upregulate integrins on macrophages, which migrate to the area of injury, and on fibroblasts and epithelial cells, which mediate wound healing and scar formation. Plasmin aids healing by removing excess fibrin. This aids the migration of keratinocytes into the wound to restore the epithelium under the

scab. Collagen synthesis is upregulated, producing the scar. Wounds gain 20% of their ultimate strength in 3 weeks and later gain more strength, but they never reach more than about 70% of the strength of normal skin.

CHAPTER SUMMARY

- Immunity is the means by which the body is protected from foreign invaders or from abberent host cells, such as tumor cells.
- Innate immunity represents an evolutionarily conserved, primitive response to stereotypical microbial components.
- Acquired immunity is slower to develop than innate immunity, but long-lasting and often more effective due to its greater specificity.
- Genetic rearrangements endow B and T lymphocytes with a vast array of receptors capable of recognizing billions of foreign antigens.
- Self-reactive lymphocytes are normally deleted; a failure of this process leads to autoimmune disease.
- A variety of soluble mediators orchestrate the development of immunologic effector cells, their migration, and their subsequent immune and inflammatory reactions.
- Immune and inflammatory responses are mediated by several additional cell types—granulocytes, monocytes, mast cells, tissue macrophages, and antigen-presenting cells—that arise predominantly from the bone marrow and may circulate or reside in connective tissues.
- Granulocytes mount phagocytic responses that engulf and destroy bacteria. These are accompanied by the release of reactive oxygen species and other mediators into adjacent tissues that may cause tissue injury. Disease can result from abnormal function or development of granulocytes. Deficient immune responses to microbial threats usually result.
- Mast cells and basophils underpin allergic reactions to substances that would be treated as innocuous by nonallergic individuals.
- Inflammatory responses occur in response to infection or injury, and serve to resolve the threat, although they may cause damage to otherwise healthy tissue. A number of chronic diseases reflect excessive inflammatory responses that persist even once the threat is controlled, or are triggered by stimuli that healthy individuals would not respond to.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. Cells responsible for innate immunity are activated most commonly by

   A. glucocorticoids

   B. pollen

   C. carbohydrate sequences in bacterial cell walls

   D. eosinophils

   E. thrombopoietin

2. A biotechnology company is working to design a new therapeutic strategy for cancer that involves triggering an enhanced immune response to cellular proteins that are mutated in the disease. Which of the following immune cells or processes will most likely **not** be required for a successful therapy?

   A. Cytotoxic T cells

   B. Antigen presentation in the context of MHC-II

   C. Proteosomal degradation

   D. Gene rearrangements producing T cell receptors

   E. The immune synapse

3. A six-year-old boy is brought to the pediatrician with complaints of progressive excessive thirst, increased urination, weight loss, and severe fatigue. A glucose tolerance test results in a diagnosis of type 1 diabetes mellitus and the patient is treated effectively with injected insulin. The emergent symptoms in this patient reflect a failure of which immune process?

   A. Antibody synthesis

   B. Neutrophil chemotaxis

   C. Antigen presentation

   D. Tolerance

   E. Complement activation

4. A 20-year-old college student comes to the student health center in April complaining of runny nose and congestion, itchy eyes, and wheezing. She reports that similar symptoms have occurred at the same time each year, and that she obtains some relief from over-the-counter antihistamine drugs, although they make her too drowsy to study. Her symptoms can most likely be attributed to inappropriate synthesis of which of the following antibodies specific for tree pollen?

A. IgA
B. IgD
C. IgE
D. IgG
E. IgM

5. The ability of the blood to phagocytose pathogens and mount a respiratory burst is increased by

A. interleukin-2 (IL-2)
B. granulocyte colony-stimulating factor (G-CSF)
C. erythropoietin
D. interleukin-4 (IL-4)
E. interleukin-5 (IL-5)

6. In an experiment, a scientist treats a group of mice with an antiserum that substantially depletes the number of circulating neutrophils. Compared with untreated control animals, the mice with reduced numbers of neutrophils were found to be significantly more susceptible to death induced by bacterial inoculation. The increased mortality can be ascribed to a relative deficit in which of the following?

A. Acquired immunity
B. Oxidants
C. Platelets
D. Granulocyte/macrophage colony stimulating factor (GM-CSF)
E. Integrins

7. If a nasal biopsy were performed on the patient described in Question 4 while symptomatic, histologic examination of the tissue would most likely reveal degranulation of which of the following cell types?

A. Dendritic cells
B. Lymphocytes
C. Neutrophils
D. Monocytes
E. Mast cells

8. A patient suffering from an acute flare in his rheumatoid arthritis undergoes a procedure where fluid is removed from his swollen and inflamed knee joint. Biochemical analysis of the inflammatory cells recovered from the removed

fluid would most likely reveal a decrease in which of the following proteins?

A. Interleukin-1

B. Tumor necrosis factor-α

C. Nuclear factor-κB

D. IκBα

E. von Willebrand factor

CHAPTER 4

Excitable Tissue: Nerve

OBJECTIVES

After studying this chapter, you should be able to:

- Draw a typical neuron and identify the role played by soma, dendrites, axon, and initial segment in impulse generation and conduction.
- Explain the basis for the resting membrane potential of a neuron and the effect of hyperkalemia and hypokalemia on the resting potential.
- Explain the ionic fluxes that occur during an action potential.
- Compare and contrast how unmyelinated and myelinated neurons propagate impulses.
- Compare the conduction velocity and other properties of different types of sensory and motor nerve fibers.
- Explain the importance of orthograde and retrograde axonal transport.
- Compare the functions of the various types of glia found in the nervous system.
- Identify neuropathologies related to dysfunction of myelin proteins or the loss of myelin.
- Describe the function of neurotrophins.

INTRODUCTION

The basic working unit of the central and peripheral nervous system is the nerve cell or **neuron.** Neurons are identified as excitable cells because they have the ability to be electrically excited resulting in the generation of action potentials. Other examples of excitable cells are skeletal, smooth, and cardiac muscle cells (Chapter 5) and secretory cells of the pancreas. Neurons come in many different shapes and sizes, but they share features that impart in them an ability to receive, process, integrate, and transmit information from external and internal sources to initiate most physiological behaviors. This chapter describes the ionic mechanisms that enable neurons to generate and conduct impulses; Chapter 6 explains how neurons communicate with other neurons and effector organs (synaptic transmission). This chapter also describes the roles played by neuroglia cells and neurotrophins in physiological and pathophysiological processes.

THE NEURON: BASIC WORKING UNIT OF THE NERVOUS SYSTEM

Figure 4–1 shows the basic components of a neuron for the prototypical spinal motor neuron. The cell body **(soma)** contains the nucleus that is the metabolic center of the neuron and stores the hereditary material or DNA. Neurons have several processes called **dendrites** that extend outward from the cell body and arborize extensively to aid their role in receiving incoming signals, processing the information, and then transmitting the information to the soma of the neuron. A typical neuron also has a long fibrous **axon** that originates from a thickened area of the cell body **(axon hillock).** The first portion of the axon is called the **initial segment**. The axon divides into **presynaptic terminals,** each ending in a number of **synaptic knobs** that are also called **terminal buttons** or **boutons.** They contain granules or vesicles in which the synaptic transmitters released by the nerves are stored. Based on the number of processes that emanate from their cell body, neurons can be classified as **unipolar, bipolar, pseudounipolar,** and **multipolar (Figure 4–2)**.

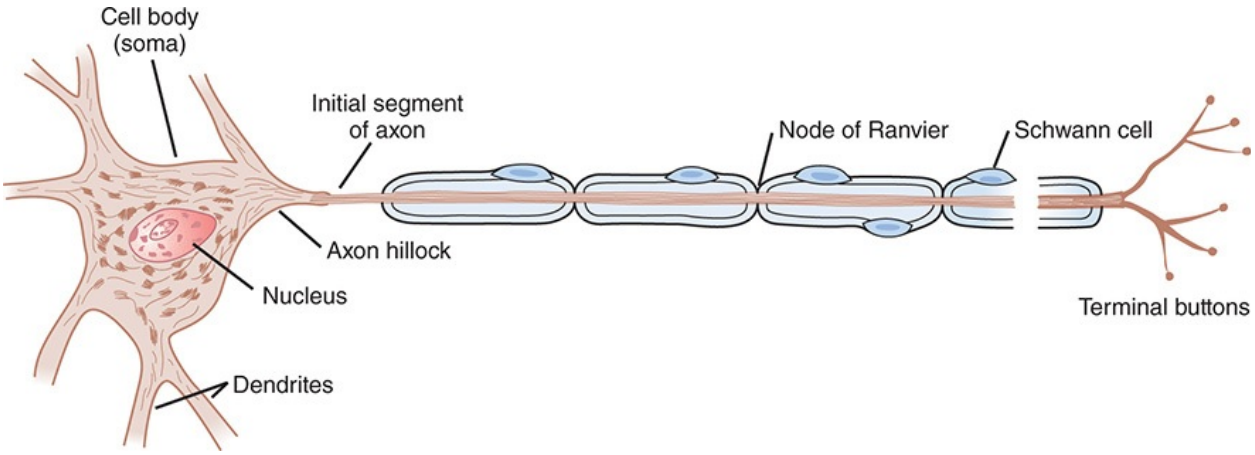


FIGURE 4–1 Motor neuron with a myelinated axon. A motor neuron is composed of a cell body (soma) with a nucleus, several processes called dendrites, and a long fibrous axon that originates from the axon hillock. The first portion of the axon is called the initial segment. A myelin sheath forms from Schwann cells and surrounds the axon except at its ending and at the nodes of Ranvier. Terminal buttons (boutons) are located at the terminal endings.

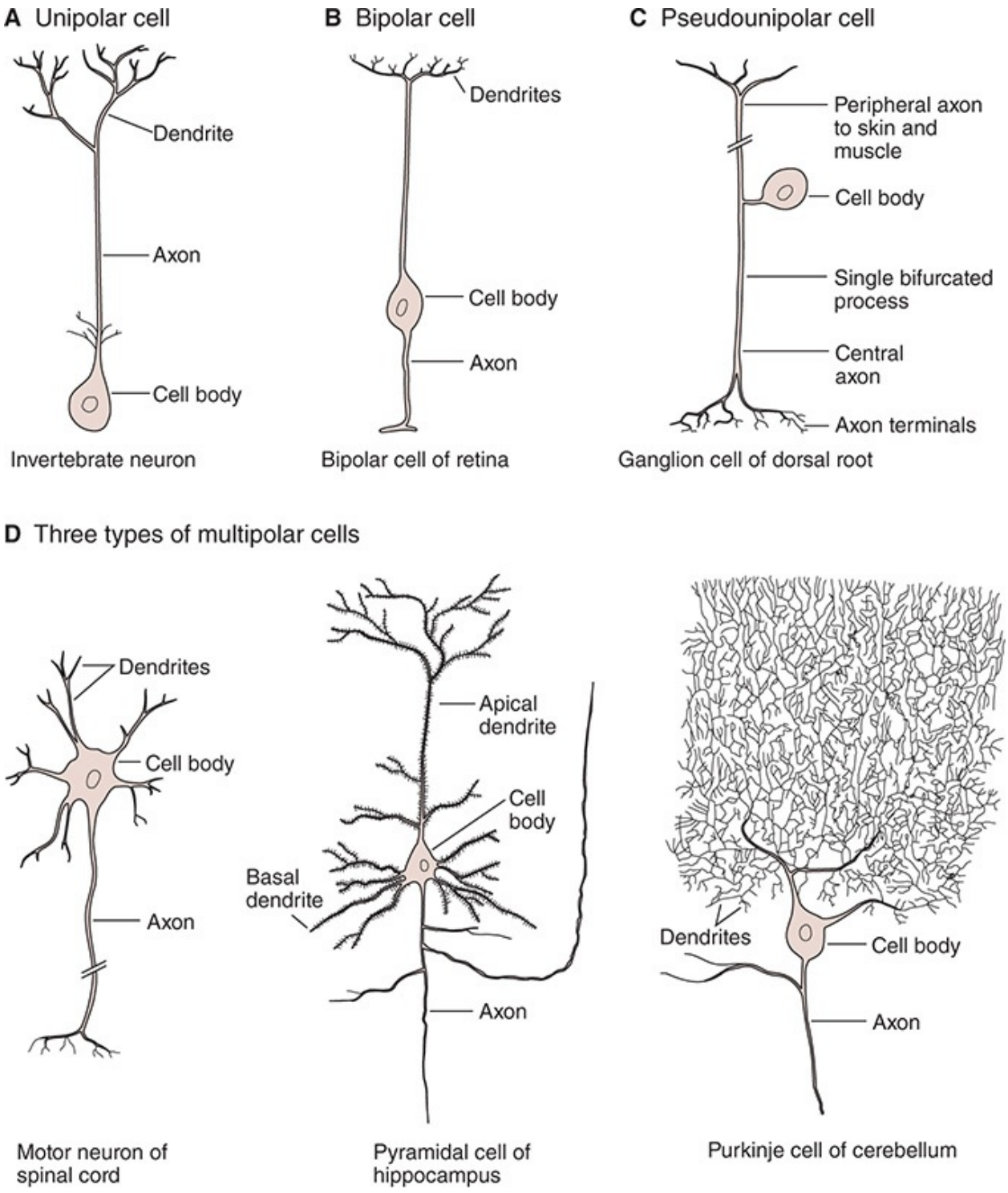


FIGURE 4–2 Some of the types of neurons in the mammalian nervous system. A) Unipolar neurons have one process, with different segments serving as receptive surfaces and releasing terminals. **B)** Bipolar neurons have two specialized processes: a dendrite that carries information to the cell and an axon that transmits information from the cell. **C)** Some sensory neurons are in a

subclass of bipolar cells called pseudounipolar cells. As the cell develops, a single process splits into two, both of which function as axons—one going to skin or muscle and another to the spinal cord. **D)** Multipolar cells have one axon and many dendrites. Examples include motor neurons, hippocampal pyramidal cells with dendrites in the apex and base, and cerebellar Purkinje cells with an extensive dendritic tree in a single plane. (Reproduced with permission from Kandel ER, Schwartz JH, Jessell TM, Siegelbaum SA, Hudspeth AJ (editors): *Principles of Neural Science*, 5th ed. New York, NY: McGraw-Hill; 2013.)

From a functional point of view, neurons generally have four important zones: (1) a dendritic zone where multiple local potential changes generated by synaptic connections are integrated; (2) a site where propagated action potentials are generated (the initial segment in spinal motor neurons, the initial node of Ranvier in cutaneous sensory neurons); (3) an axonal process that transmits propagated impulses to the nerve endings; and (4) the nerve endings, where action potentials cause the release of synaptic transmitters.

The axons of many neurons acquire a **myelin sheath,** a protein–lipid complex that is wrapped around the axon. In the peripheral nervous system, myelin forms when a **Schwann cell** (a type of glia) wraps its membrane around an axon up to 100 times (Figure 4–1). The myelin sheath envelops the axon except at the **nodes of Ranvier**, periodic 1-µm gaps where the axon is unmyelinated (Figure 4–1). The insulating function of myelin and its role in axonal conduction are discussed later in this chapter.

EXCITATION & CONDUCTION

A hallmark of nerve cells is their excitable membrane. Neurons respond to electrical, chemical, or mechanical stimuli by producing local (nonpropagated) or propagated potentials reflecting changes in the conduction of ions across the cell membrane. Depending on their location, the nonpropagated potentials are called **synaptic, generator,** or **electrotonic potentials.** The propagated **action potentials** are the primary electrical responses of neurons; they are the main form of communication within the nervous system. The electrical events in neurons are rapid, measured in milliseconds (ms); and the potential changes are small, measured in millivolts (mV).

The impulse is normally transmitted (conducted) along the axon to its termination. Conduction is an active, self-propagating process, and the impulse moves along the nerve at a constant amplitude and velocity. The process is often

compared to what happens when a match is applied to one end of a trail of gunpowder; by igniting the powder particles immediately in front of it, the flame moves steadily down the trail to its end as it is extinguished in its wake.

RESTING MEMBRANE POTENTIAL

The voltage difference across the cell membrane of a neuron is called a **membrane potential;** it is the difference between the electrical potential in the cytoplasm of the cell and the electrical potential in the extracellular space. The membrane potential results from the separation of positive and negative charges across the cell membrane **(Figure 4–3)**. In order for a potential difference to be present across a membrane lipid bilayer, two conditions must be met. First, there must be an unequal distribution of ions of one or more type across the membrane (ie, a **concentration gradient**). Second, the membrane must be permeable to these ions. The permeability is provided by the existence of channels or pores in the bilayer; these channels are usually permeable to a single type of ion.

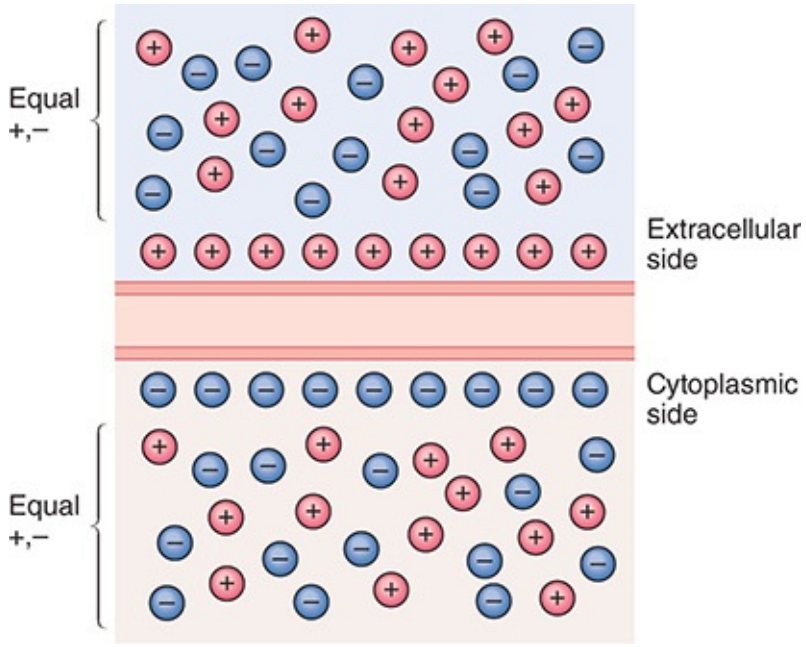


FIGURE 4–3 A membrane potential results from separation of positive and negative charges across the cell membrane. The excess of positive charges (red circles) outside the cell and negative charges (blue circles) inside the cell at rest represents a small fraction of the total number of ions present. (Reproduced with permission from Kandel ER, Schwartz JH, Jessell TM, Siegelbaum SA, Hudspeth AJ (editors): *Principles of Neural Science*, 5th ed. New York, NY:

McGraw-Hill; 2013.)

The **resting membrane potential** represents an equilibrium situation at which the driving force for the membrane-permeant ions down their concentration gradients across the membrane is equal and opposite to the driving force for these ions down their **electrical gradients.** In neurons, the concentration of K^+ is much higher inside than outside the cell, while the reverse is the case for Na^+. This concentration difference is established by **Na, K ATPase**. The outward K^+ concentration gradient results in passive movement of K^+ out of the cell when K^+-selective channels are open. Similarly, the inward Na^+ concentration gradient results in passive movement of Na^+ into the cell when Na^+-selective channels are open.

The resting membrane potential of neurons is usually about –70 mV (step 1 in **Figure 4–4**). Because there are more open K^+ channels than Na^+ channels at rest, the membrane permeability to K^+ is greater. Consequently, the intracellular and extracellular K^+ concentrations are the prime determinants of the resting membrane potential, which is therefore close to the equilibrium potential for K^+. Steady ion leaks cannot continue forever without eventually dissipating the ion gradients. Na, K ATPase prevents this from occurring by actively moving Na^+ and K^+ against their electrochemical gradients.

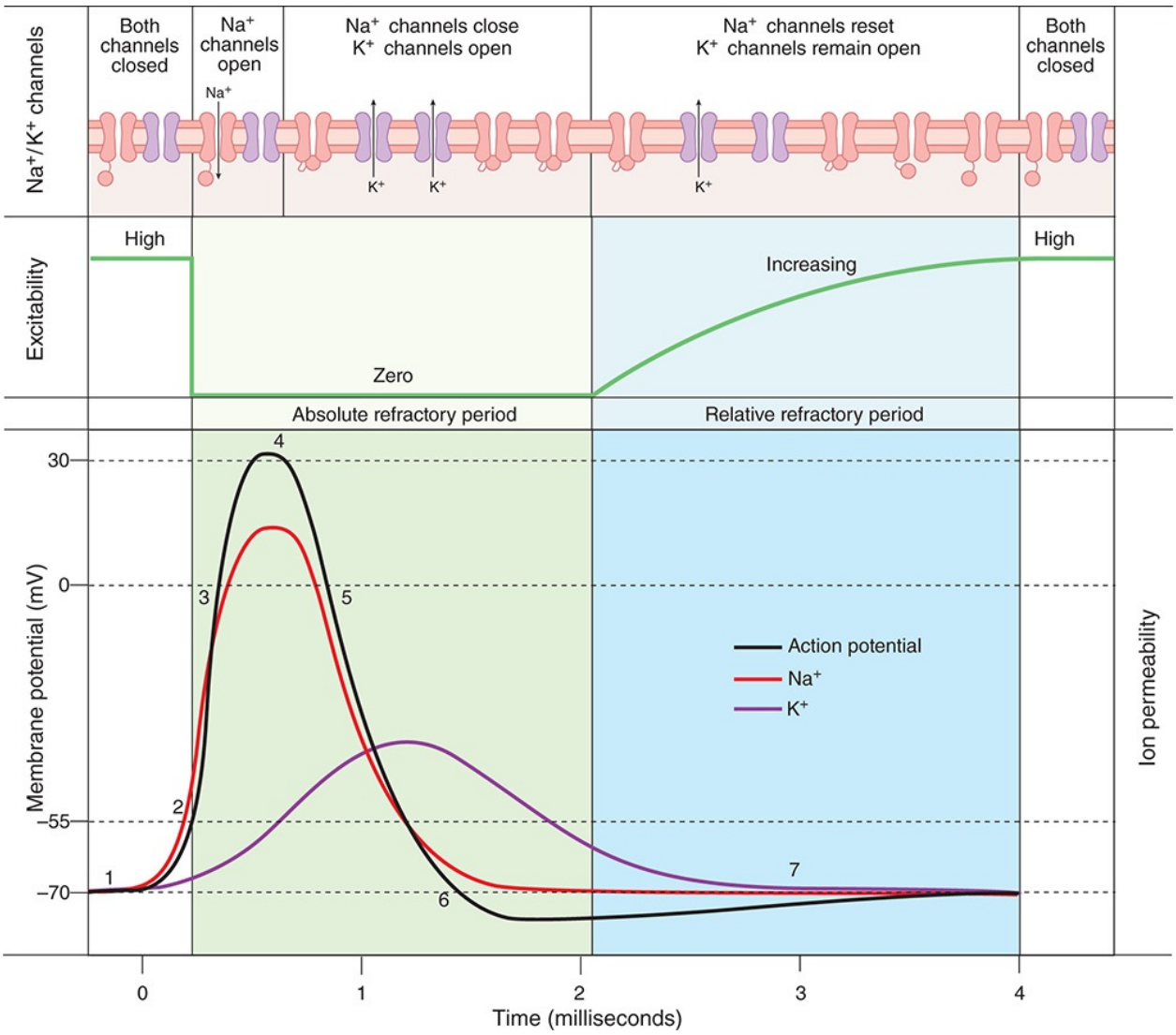


FIGURE 4–4 Changes in membrane potential and relative membrane permeability to Na^+ and K^+ during an action potential. Steps 1 through 7 are detailed in the text. These changes in threshold for activation (excitability) are correlated with the phases of the action potential. (Modified with permission from Silverthorn DU: *Human Physiology: An Integrated Approach*, 5th ed. Pearson, 2010.)

IONIC FLUXES DURING THE ACTION POTENTIAL

Neuronal cell membranes contain many types of ion channels, including both ligand-gated and voltage-gated ion channels. **Ligand-gated ion channels** open

when a ligand (eg, neurotransmitter) binds to them, and **voltage-gated ion channels** open when there is a change in the voltage gradient across the membrane. The behavior of these channels, particularly Na^+ and K^+ channels, explains the electrical events in neurons.

The changes in **membrane conductance** of Na^+ and K^+ that occur during an action potential are shown by steps 1 through 7 in Figure 4–4. The conductance of an ion is the reciprocal of its electrical resistance in the membrane and is a measure of the membrane permeability to that ion. In response to a depolarizing stimulus, some of the voltage-gated Na^+ channels open and Na^+ enters the cell and the membrane is brought to its **threshold potential** (step 2) and the voltage-gated Na^+ channels overwhelm the K^+ and other channels. The entry of Na^+ causes the opening of more voltage-gated Na^+ channels and further depolarization, setting up a **positive feedback loop.** The rapid upstroke in the membrane potential ensues (step 3). The membrane potential moves toward the equilibrium potential for Na^+ (+60 mV) but does not reach it during the action potential (step 4), primarily because the increase in Na^+ conductance is short-lived. The Na^+ channels rapidly enter a closed state called the **inactivated state** and remain in this state for a few milliseconds before returning to the resting state, when they again can be activated. The direction of the electrical gradient for Na^+ is reversed during the **overshoot** because the membrane potential is reversed which limits Na^+ influx; also the voltage-gated K^+ channels open. These factors contribute to **repolarization.** The opening of voltage-gated K^+ channels is slower and more prolonged than the opening of the Na^+ channels; consequently, much of the increase in K^+ conductance comes after the increase in Na^+ conductance (step 5). The net movement of positive charge out of the cell due to K^+ efflux at this time helps complete the process of repolarization. The slow return of the K^+ channels to the closed state also explains the **after-hyperpolarization** (step 6), followed by a return to the resting membrane potential (step 7). Thus, voltage-gated K^+ channels bring the action potential to an end and cause closure of their gates through a **negative feedback process.** **Figure 4–5** shows the sequential feedback control in voltage-gated K^+ and Na^+ channels during the action potential.

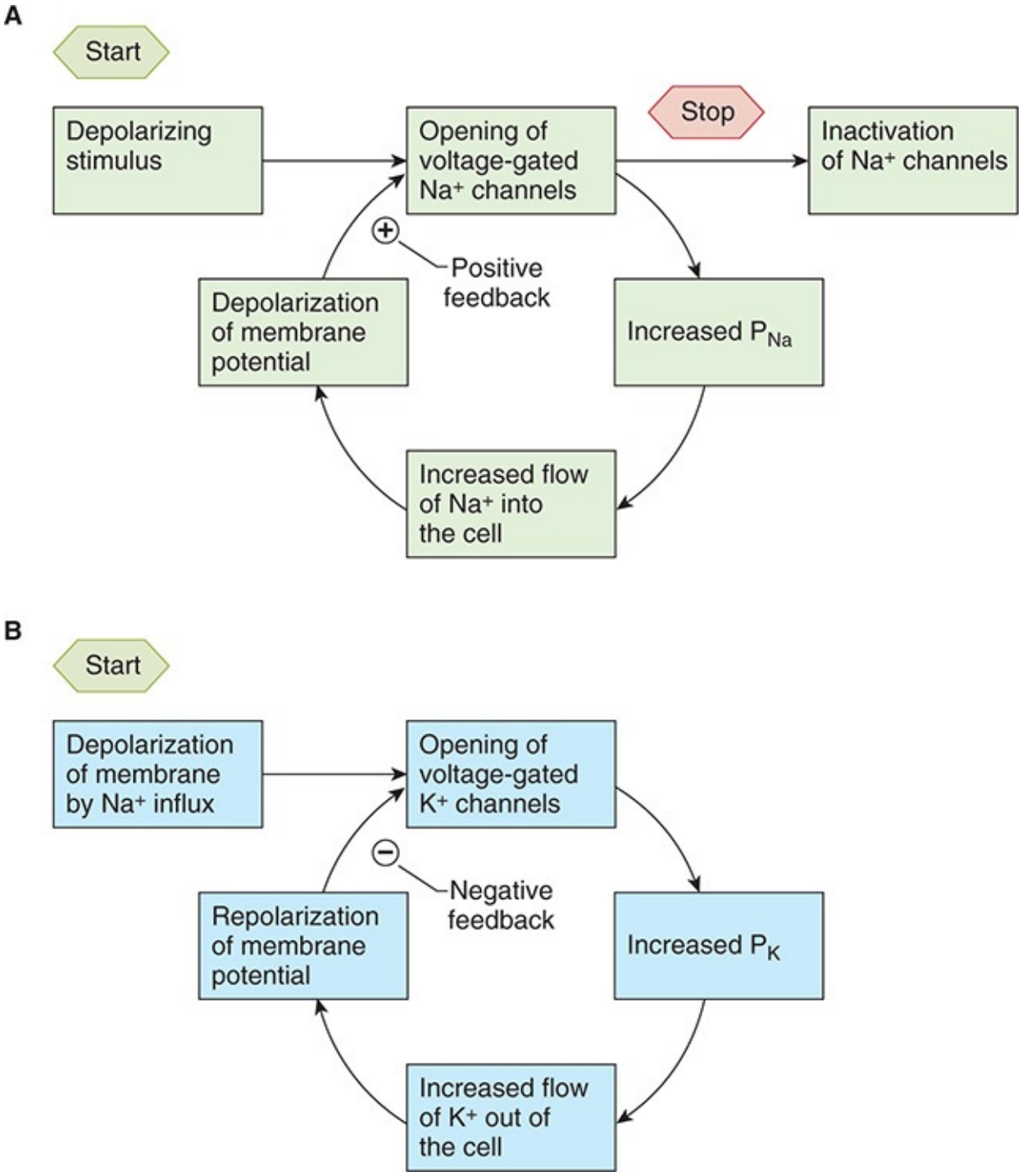


FIGURE 4–5 Feedback control in voltage-gated ion channels in the membrane. A) Na^+ channels exert positive feedback. **B)** K^+ channels exert negative feedback. P_{Na}, P_K is permeability to Na^+ and K^+, respectively. (Reproduced with permission from Widmaier EP, Raff H, Strang KT: *Vander's Human Physiology*. New York, NY: McGraw-Hill; 2008.)

Decreasing the external Na^+ concentration reduces the size of the action potential but has little effect on the resting membrane potential. The lack of much effect on the resting membrane potential is predicted because the permeability of the membrane to Na^+ at rest is relatively low. In contrast, since the resting membrane potential is close to the equilibrium potential for K^+, changes in the external concentration of this ion can have major effects on the resting membrane potential. If the extracellular level of K^+ is increased **(hyperkalemia),** the resting potential moves closer to the threshold for eliciting an action potential and the neuron becomes more excitable. If the extracellular level of K^+ is decreased **(hypokalemia),** the membrane potential is reduced and the neuron is hyperpolarized. **Clinical Box 4–1** describes some of the common causes, signs, and treatments for the deviation of plasma levels of K^+ away from the normal value of 3.5–5.0 mEq/L on excitable cells (eg, neurons, heart, skeletal muscle, smooth muscle).

Other ions, notably Ca^{2+}, can affect the membrane potential through both channel movement and membrane interactions. A decrease in extracellular Ca^{2+} concentration increases the excitability of nerve and muscle cells by decreasing the amount of depolarization needed to initiate the changes in the Na^+ and K^+ conductance that produce the action potential. Conversely, an increase in extracellular Ca^{2+} concentration can stabilize the membrane by reducing excitability.

ALL-OR-NONE ACTION POTENTIALS

The minimal intensity of stimulating current needed to produce an action potential is called the **threshold intensity**; it varies with the stimulus duration. A weak stimulus needs a long duration, and a strong stimulus is sufficient at a short duration. The relation between the strength and the duration of a threshold stimulus is called the **strength–duration curve.** Slowly rising currents fail to induce an action potential in the nerve because the nerve undergoes **adaptation.** The action potential is **all-or-none** in character. That is, no action potential occurs if the stimulus is subthreshold in magnitude, and the action potential has a constant amplitude and form at any stimulus strength above the threshold intensity.

CLINICAL BOX 4–1

Hyperkalemia and Hypokalemia

Potassium homeostasis is critical for the normal functioning of nerves, skeletal muscle, smooth muscle, and the heart. A primary cause of **hyperkalemia** is impairment in the ability of the kidney to excrete K^+ due to advanced renal failure, adrenal insufficiency, distal renal tubular acidosis, type 1 diabetes, and dehydration. Drug-induced hyperkalemia can result from the use of angiotensin II receptor blockers, angiotensin-converting enzyme (ACE) inhibitors, nonsteroidal anti-inflammatory drugs (NSAIDs), and potassium sparing diuretics. Symptoms of hyperkalemia include muscle pain and weakness, numbness, cardiac arrhythmias, and nausea. Extremely high levels of plasma K^+ can lead to cardiac arrest and death. **Hyperkalemic periodic paralysis** is a rare inherited disorder (prevalence of 1 per 100,000 individuals) in which patients experience transient episodes of muscle paralysis due to hyperkalemia. The resting membrane potential of skeletal muscle in affected individuals shifts from a normal value of −90 mV to a value of −60 mV which inactivates Na^+ channels and prevents action potential generation.

The most common cause of **hypokalemia** is increased excretion of K^+, but it can also occur if there is a shift of K^+ from the extracellular to the intracellular space. Hypokalemia can be a side effect of rare genetic disorders of the kidney (**Bartter syndrome** [see Chapter 37] and **Gitelman syndrome), Cushing syndrome,** the use of K^+-wasting diuretics, diabetic **ketoacidosis,** renal tubular acidosis, and familial hypokalemia. Symptoms are somewhat nonspecific but can include weakness and fatigue, constipation, muscle cramping, palpitations, and psychological symptoms such as depression or psychosis. Severe hypokalemia (below 2.5 mEq/L) mainly leads to problems with cardiac rhythmicity and can manifest as bradycardia, tachycardia, premature beats, and atrial or ventricular fibrillation.

THERAPEUTIC HIGHLIGHTS

Treatment of hyperkalemia typically includes treatment of the underlying cause. It may also include a low-potassium diet, intravenous administration of calcium to protect the heart and muscles, or administration of sodium bicarbonate to promote movement of K^+ from the extracellular space back into the cells. Treatment of hypokalemia focuses on ways to reduce K^+ loss (eg, discontinue the use of a diuretic or use a K^+-sparing diuretic), replenish K^+ stores (oral or

intravenous administration of K^+), and determining and treating the cause of the hypokalemia.

ELECTROTONIC POTENTIALS, LOCAL RESPONSE, & FIRING LEVEL

Although subthreshold stimuli do not produce a propagating action potential, they do have an effect on the membrane potential. Applying subthreshold stimuli of fixed duration leads to a localized depolarizing potential change that rises sharply and decays exponentially with time **(Figure 4–6)**. The magnitude of this response drops off rapidly as the distance between the stimulating and recording electrode is increased. Conversely, an anodal current produces a hyperpolarizing potential change of similar duration. These potential changes are called **electrotonic potentials.** As the strength of the current is increased, the response is greater due to the increasing addition of a local response of the membrane. Finally, at 7–15 mV of depolarization (potential of –55 mV), the threshold potential is reached and an action potential occurs.

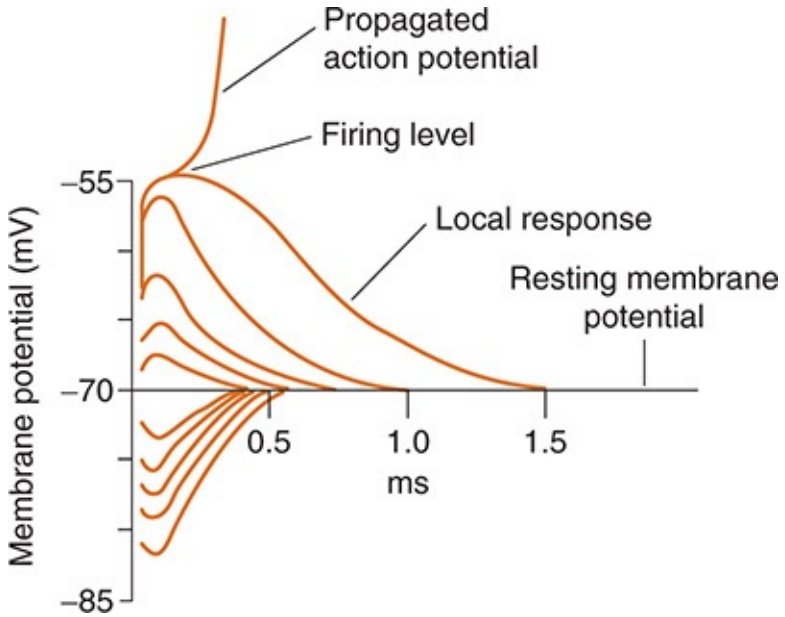


FIGURE 4–6 Electrotonic potentials and local response. The changes in the membrane potential of a neuron following application of stimuli of 0.2, 0.4, 0.6, 0.8, and 1.0 times threshold intensity are shown superimposed on the same time scale. The responses below the horizontal line are those recorded near the anode,

and the responses above the line are those recorded near the cathode. The stimulus of threshold intensity was repeated twice. Once it caused a propagated action potential (top line), and once it did not.

REFRACTORY PERIODS

During the action potential, as well as during electrotonic potentials and the local response, the threshold of the neuron to stimulation changes (Figure 4–4). Hyperpolarizing responses elevate the threshold, and depolarizing potentials lower it as they move the membrane potential closer to the threshold potential. During the local response, the threshold is lowered, but during the rising and much of the falling phases of the spike potential, the neuron is refractory to stimulation. The **refractory period** is divided into an **absolute refractory period,** corresponding to the period from the time the firing level is reached until repolarization is about one-third complete, and a **relative refractory period,** lasting from this point to the start of after-depolarization. During the absolute refractory period, no stimulus, no matter how strong, will excite the nerve. However, during the relative refractory period, stronger than normal stimuli can cause excitation. These changes in threshold are correlated with the phases of the action potential in Figure 4–4.

CONDUCTION OF THE ACTION POTENTIAL

The nerve cell membrane is polarized at rest, with positive charges lined up along the outside of the membrane and negative charges along the inside. During the action potential, this polarity is abolished and for a brief period is actually reversed **(Figure 4–7)**. Positive charges from the membrane ahead of and behind the action potential flow into the area of negativity represented by the action potential (“current sink”). By drawing off positive charges, this flow decreases the polarity of the membrane ahead of the action potential. Such electrotonic depolarization initiates a local response, and when the firing level is reached, a propagated response occurs that in turn electrotonically depolarizes the membrane in front of it.

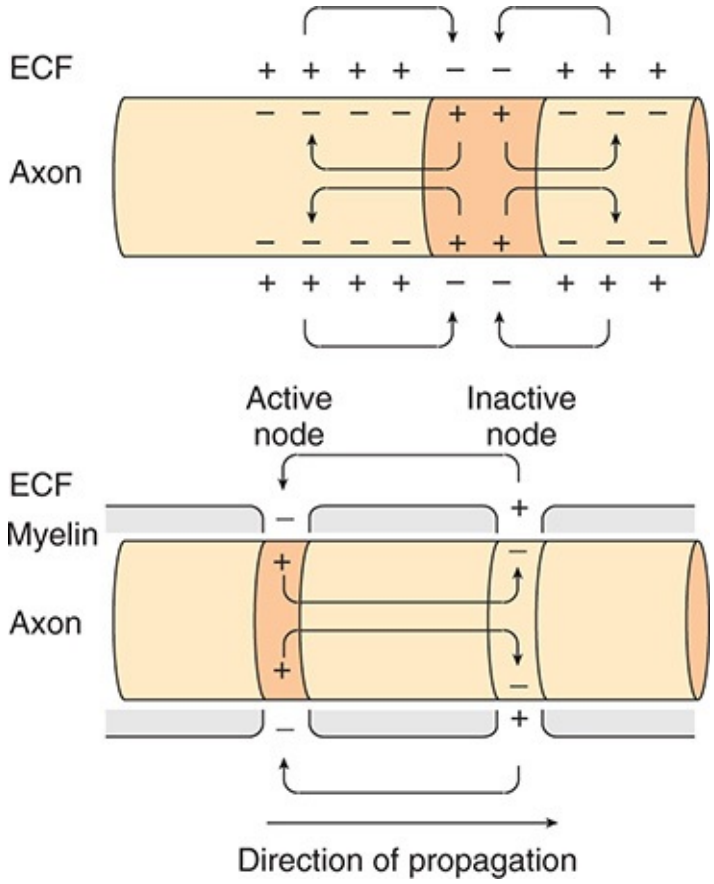


FIGURE 4–7 Local current flow (movement of positive charges) around an impulse in an axon. Top: Unmyelinated axon. **Bottom:** Myelinated axon. Positive charges from the membrane ahead of and behind the action potential flow into the area of negativity represented by the action potential ("current sink"). In myelinated axons, depolarization appears to "jump" from one node of Ranvier to the next (saltatory conduction).

The spatial distribution of ion channels along the axon plays a key role in the initiation and regulation of the action potential. Voltage-gated Na^+ channels are highly concentrated in the nodes of Ranvier and the initial segment in myelinated neurons. The number of Na^+ channels per square micrometer of membrane in myelinated mammalian neurons is 50–75 in the cell body, 350–500 in the initial segment, less than 25 on the surface of the myelin, 2000–12,000 at the nodes of Ranvier, and 20–75 at the axon terminals. Along the axons of unmyelinated neurons, the number is about 110. In many myelinated neurons, the Na^+ channels are flanked by K^+ channels that are involved in repolarization.

Conduction in myelinated axons depends on a similar pattern of circular current flow as just described. However, myelin is an effective insulator, and

current flow through it is negligible. Instead, depolarization in myelinated axons travels from one node of Ranvier to the next, with the current sink at the active node serving to electrotonically depolarize the node ahead of the action potential to the firing level (Figure 4–7). This "jumping" of depolarization from node to node is called **saltatory conduction.** It is a rapid process that allows myelinated axons to conduct up to 50 times faster than the fastest unmyelinated fibers.

NERVE FIBER TYPES & FUNCTION

Mammalian nerve fibers are divided into A, B, and C groups, and the A group can be subdivided into α, β, γ, and δ fibers. In **Table 4–1**, the various fiber types are listed with their diameters, electrical characteristics, and functions. After a stimulus is applied to a nerve, there is a latent period before the start of the action potential. This interval corresponds to the time it takes the impulse to travel along the axon from the site of stimulation to the recording electrodes. Its duration is proportionate to the distance between the stimulating and recording electrodes and inversely proportionate to the speed of conduction. If the duration of the latent period and the distance between the stimulating and recording electrodes are known, **axonal conduction velocity** can be calculated. In general, the greater the diameter of a given nerve fiber, the greater its speed of conduction. The large axons are concerned primarily with proprioceptive sensation, somatic motor function, conscious touch, and pressure, while the smaller axons subserve pain and temperature sensations and autonomic function.

TABLE 4–1 Types of mammalian nerve fibers.

Fiber Type	Function	Fiber Diameter (μm)	Conduction Velocity (m/s)	Spike Duration (ms)	Absolute Refractory Period (ms)
Aα	Proprioception; somatic motor	12–20	70–120		
Aβ	Touch, pressure	5–12	30–70	0.4–0.5	0.4–1
Aγ	Motor to muscle spindles	3–6	15–30		
Aδ	Pain, temperature	2–5	12–30		
B	Preganglionic autonomic	< 3	3–15	1.2	1.2
C, Dorsal root	Pain, temperature	0.4–1.2	0.5–2	2	2
C, Sympathetic	Postganglionic sympathetic	0.3–1.3	0.7–2.3	2	2

Although the letter classification is commonly used to describe motor fibers, a numerical system (Ia, Ib, II, III, and IV) is often used to classify sensory fibers based on their axonal diameter and conduction velocity. **Table 4–2** shows the corresponding classification of the number system and the letter system.

TABLE 4–2 Numerical classification of sensory nerve fibers.

Number	Origin	Fiber Type
Ia	Muscle spindle, annulo-spiral ending	Aα
Ib	Golgi tendon organ	Aα
II	Muscle spindle, flower-spray ending; touch, pressure	Aβ
III	Pain and cold receptors; some touch receptors	Aδ
IV	Pain, temperature, and other receptors	C

In addition to variations in speed of conduction and fiber diameter, the various classes of fibers in peripheral nerves differ in their sensitivity to hypoxia and anesthetics **(Table 4–3)**. This fact has clinical as well as physiologic significance. Local anesthetics depress transmission in the unmyelinated group C fibers before they affect the myelinated group A fibers **(Clinical Box 4–2)**. Conversely, pressure on a nerve can cause loss of conduction in large-diameter motor, touch, and pressure fibers while pain sensation remains relatively intact. Patterns of this type are sometimes seen in individuals who sleep with their arms under their heads for long periods, causing compression of the nerves in the arms. Because of the association of deep sleep with alcoholic intoxication, the syndrome is most common on weekends and has acquired the interesting name Saturday night or Sunday morning paralysis.

TABLE 4–3 Relative susceptibility of mammalian A, B, and C nerve fibers to conduction block produced by various agents.

Agent	Most Susceptible	Intermediate	Least Susceptible
Hypoxia	B	A	C
Pressure	A	B	C
Local anesthetics	C	B	A

AXONAL TRANSPORT

The apparatus for protein synthesis in neurons is located primarily in the soma, with transport of proteins and polypeptides to the axonal ending by **axoplasmic flow.** Thus, the cell body maintains the functional and anatomic integrity of the axon. **Orthograde transport** occurs along microtubules located along the length of the axon; it requires two molecular motors, dynein and kinesin **(Figure 4–8)**. Orthograde transport moves from the cell body toward the axon terminals. It has both fast and slow components, **fast axonal transport** occurs at about 400 mm/day, and **slow axonal transport** occurs at 0.5–10 mm/day. **Retrograde transport** from the nerve ending to the cell body occurs at about 200 mm/day. Synaptic vesicles recycle in the membrane, but some used vesicles are carried back to the cell body and deposited in lysosomes. Some materials taken up at the ending by endocytosis, including **nerve growth factor (NGF)** and some viruses, are also transported back to the cell body.

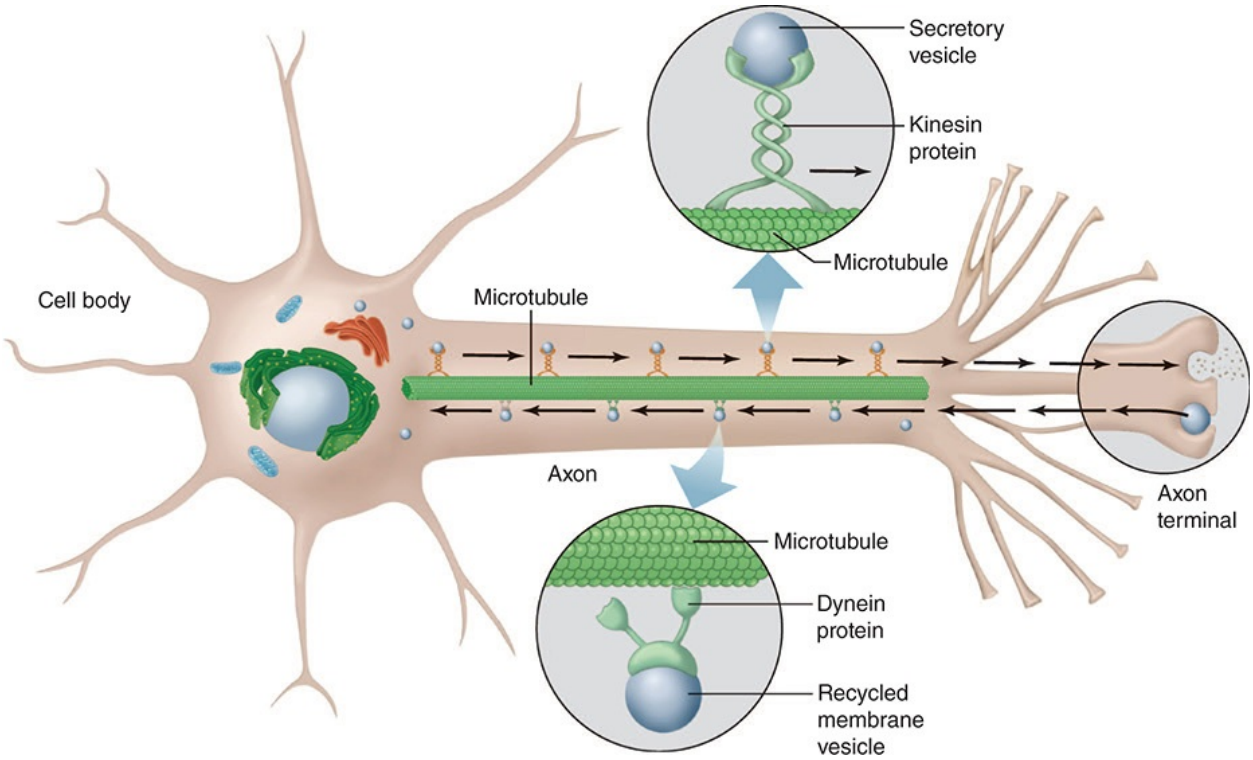


FIGURE 4–8 Axonal transport along microtubules by dynein and kinesin. Fast (400 mm/day) and slow (0.5–10 mm/day) axonal orthograde transport occurs along microtubules that run along the length of the axon from the cell body to the terminal. Retrograde transport (200 mm/day) occurs from the terminal to the cell body. (Reproduced with permission from Widmaier EP, Raff

H, Strang KT: *Vander's Human Physiology*. New York, NY: McGraw-Hill; 2008.)

GLIA

The word *glia* is Greek for *glue*. For many years following their discovery, neuroglia cells were viewed as connective tissue. Today these cells are recognized for their role in communication within the central nervous system (CNS) in partnership with neurons. Unlike most neurons, glia continue to undergo cell division in adulthood and their ability to proliferate is particularly noticeable after brain injury (eg, stroke).

Glia are classified as **microglia** and **macroglia** depending on their size. Microglia are part of the immune system; they are scavenger cells that resemble tissue macrophages and remove debris resulting from injury, infection, and disease (eg, multiple sclerosis [MS], AIDS-related dementia, Parkinson disease, and Alzheimer disease). Microglia arise from macrophages outside of the nervous system and are physiologically and embryologically unrelated to other neural cell types.

There are three types of macroglia: oligodendrocytes, Schwann cells, and astrocytes **(Figure 4–9)**. **Oligodendrocytes** and **Schwann cells** are involved in myelin formation around axons in the CNS and peripheral nervous system, respectively. Unlike the Schwann cell, which forms the myelin on a single axon (Figure 4–1), oligodendrocytes emit multiple processes that form myelin on many neighboring axons. In MS, patchy destruction of myelin occurs in the CNS **(Clinical Box 4–3)**. The loss of myelin is associated with delayed or blocked conduction in the demyelinated axons. Myelin protein zero (P0) and a hydrophobic protein PMP22 are components of the myelin sheath in the peripheral nervous system. Autoimmune reactions to these proteins cause **Guillain–Barré syndrome**, a peripheral demyelinating neuropathy. Mutations in myelin protein genes cause peripheral neuropathies that disrupt myelin and cause axonal degeneration (eg, **Charcot-Marie-Tooth disease**).

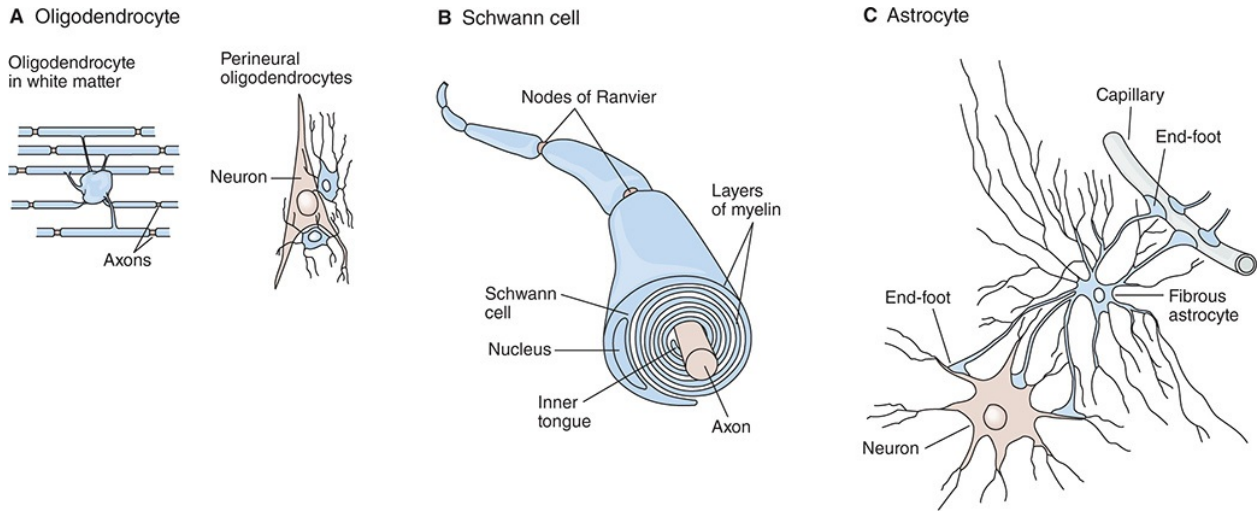


FIGURE 4–9 The principal types of macroglia in the nervous system. A) Oligodendrocytes are small with relatively few processes. Those in the white matter provide myelin, and those in the gray matter support neurons. **B)** Schwann cells provide myelin to the peripheral nervous system. Each cell forms a segment of myelin sheath about 1 mm long; the sheath assumes its form as the inner tongue of the Schwann cell turns around the axon several times, wrapping in concentric layers. Intervals between segments of myelin are the nodes of Ranvier. **C)** Astrocytes are the most common glia in the CNS and are characterized by their starlike shape. They contact both capillaries and neurons and are thought to have a nutritive function. They are also involved in forming the blood-brain barrier. (Reproduced with permission from Kandel ER, Schwartz JH, Jessell TM, Siegelbaum SA, Hudspeth AJ (editors): *Principles of Neural Science*, 5th ed. New York, NY: McGraw-Hill; 2013.)

Astrocytes are found throughout the brain and are subdivided into two groups. **Fibrous astrocytes** are found primarily in white matter and contain many intermediate filaments; **protoplasmic astrocytes** are found in gray matter and have a granular cytoplasm. Both types of astrocytes send processes to blood vessels, where they induce capillaries to form the tight junctions making up the **blood-brain barrier.** They also send processes that envelop synapses and the surface of nerve cells. Protoplasmic astrocytes have a membrane potential that varies with the external K^+ concentration but do not generate propagated potentials. They produce substances that are tropic to neurons, and they help maintain the appropriate concentration of ions and neurotransmitters by taking up K^+ and the neurotransmitters glutamate and γ-aminobutyrate (GABA).

CLINICAL BOX 4–2

Local Anesthesia

Local or regional anesthesia is used to block the conduction of action potentials in sensory and motor nerve fibers. This usually occurs as a result of blockade of voltage-gated Na^+ channels on the nerve cell membrane. This causes a gradual increase in the threshold for electrical excitability of the nerve, a reduction in the rate of rise of the action potential, and a slowing of axonal conduction velocity. There are two major categories of local anesthetics: **ester-linked** (eg, **cocaine, procaine, tetracaine**) or **amide-linked** (eg, **lidocaine, bupivacaine**). In addition to either the ester or amide, all local anesthetics contain an aromatic and an amine group. The structure of the aromatic group determines the drug's hydrophobic characteristics, and the amine group determines its latency to onset of action and its potency. Application of these drugs into the vicinity of a central (eg, **epidural, spinal anesthesia**) or peripheral nerve can lead to rapid, temporary, and near complete interruption of neural traffic to allow a surgical or other potentially noxious procedure to be done without eliciting pain. Cocaine (from the coca shrub, *Erythroxylan coca*) was the first chemical to be identified as having local anesthetic properties and remains the only naturally occurring local anesthetic. Its addictive and toxic properties prompted the development of other local anesthetics. Nociceptive fibers (unmyelinated C fibers) are the most sensitive to the blocking effect of local anesthetics. This is followed by sequential loss of sensitivity to temperature, touch, and deep pressure. Motor nerve fibers are the most resistant to the actions of local anesthetics.

NEUROTROPHINS: THEIR FUNCTION & RECEPTORS

Neurotrophins are necessary for survival and growth of neurons. Some of them are products of muscles or other structures that the neurons innervate, but many in the CNS are produced by astrocytes. These proteins bind to receptors at the endings of a neuron. They are internalized and then transported by retrograde transport to the neuronal cell body where they foster the production of proteins associated with neuronal development, growth, and survival. Other

neurotrophins are produced in soma and transported to the nerve ending where they maintain the integrity of the postsynaptic neuron.

NGF, the first neurotrophin identified, is a protein growth factor that is needed for the growth and maintenance of sympathetic neurons and some sensory neurons. NGF is made up of two *α*, two β, and two γ subunits. The β subunits, each of which has a molecular mass of 13,200 Da, have all the nerve growth-promoting activity, the α subunits have trypsin-like activity, and the γ subunits are serine proteases. The function of the proteases is unknown. The structure of the β subunit of NGF resembles that of insulin.

NGF is picked up by neurons and transported in a retrograde fashion from the nerve endings to the cell body. NGF may be responsible for the growth and maintenance of cholinergic neurons in the basal forebrain and the striatum. NGF-mediated survival of neurons is due to suppression of apoptosis rather than promotion of cell metabolism.

In addition to NGF, there are other neurotrophins, including **brain-derived neurotrophic factor (BDNF), neurotrophin 3 (NT-3),** and **NT-4/5.** They each maintain a different pattern of neurons, although there is some overlap. NT-3 is important for proprioceptor neurons that innervate the muscle spindle and mechanoreceptors in the skin; NT-4/5 is important for neurons that innervate the hair follicle; NGF is important for skin nociceptive neurons. Sympathetic neurons depend on both NGF and NT-3. BDNF acts rapidly and can actually depolarize neurons. BDNF-deficient mice lose peripheral sensory neurons and have severe degenerative changes in their vestibular ganglia and blunted long-term potentiation.

These four established neurotrophins and their three high-affinity **tyrosine kinase associated (Trk) receptors** are listed in **Table 4–4**. Each of these Trk receptors dimerizes and initiates phosphorylation in the cytoplasmic tyrosine kinase domains of the receptors. An additional low-affinity NGF receptor that is a 75-kDa protein is called the **p75 receptor.** This receptor binds all four of the listed neurotrophins with equal affinity. Interestingly, if a p75 receptor becomes activated in the absence of exposure to a neurotrophin, it causes apoptosis or cell death, an effect opposite to the usual growth-promoting and nurturing effects of neurotrophins. Research is ongoing to characterize the distinct roles of p75 and Trk receptors and factors that influence their expression in neurons.

TABLE 4–4 Neurotrophins.

Neurotrophin	Receptor
Nerve growth factor (NGF)	Trk A
Brain-derived neurotrophic factor (BDNF)	Trk B
Neurotrophin 3 (NT-3)	Trk C, Trk B
Neurotrophin 4/5 (NT-4/5)	Trk B

CLINICAL BOX 4–3

Demyelinating Diseases

Normal conduction of action potentials relies on the insulating properties of myelin. Thus, defects in myelin can have major adverse neurologic consequences. One example is **MS,** an autoimmune disease that affects over 3 million people worldwide, usually striking between the ages of 20 and 50 and affecting women about twice as often as men. The causes of MS include both genetic and environmental factors. It is most common among whites living in countries with temperate climates including Europe, southern Canada, northern United States, and southeastern Australia. Environmental triggers include early exposure to viruses such as Epstein-Barr virus and those that cause measles, herpes, chickenpox, or influenza. In MS, antibodies and white blood cells in the immune system attack myelin, causing inflammation and injury to the sheath and eventually the nerves that it surrounds. Loss of myelin leads to leakage of K^+ through voltage-gated channels, hyperpolarization, and failure to conduct action potentials. Initial presentation commonly includes reports of **paraparesis** (weakness in lower extremities) that may be accompanied by mild spasticity and hyperreflexia; **paresthesia;** numbness; urinary incontinence; and heat intolerance. Clinical assessment often reports **optic neuritis,** characterized by blurred vision, a change in color perception, visual field defect (**central scotoma**), and pain with eye movements; **dysarthria;** and **dysphagia.** Symptoms are often exacerbated by increased body temperature or increased ambient temperature. Progression of the disease is variable. In the most common form called **relapsing-remitting MS,** transient episodes appear suddenly, last a few weeks or months, and then gradually disappear. Subsequent episodes can appear years later, and eventually full recovery does not occur. A steadily worsening course with

only minor periods of remission (**secondary-progressive MS**) develops later in many individuals. Others have a progressive form of the disease in which there are no periods of remission (**primary-progressive MS**). Diagnosing MS is very difficult and generally is delayed until multiple episodes occur with deficits separated in time and space. **Nerve conduction tests** can detect slowed conduction in motor and sensory pathways. Cerebral spinal fluid analysis can detect the presence of **oligoclonal** bands indicative of an abnormal immune reaction against myelin. The most definitive assessment is **magnetic resonance imaging (MRI)** to visualize multiple scarred (sclerotic) areas or plaques in the brain. These plaques often appear in the periventricular regions of the cerebral hemispheres.

THERAPEUTIC HIGHLIGHTS

Although there is no cure for MS, **corticosteroids** (eg, **prednisone**) are the most common treatment used to reduce the inflammation that is accentuated during a relapse. Some drug treatments are designed to modify the course of the disease. For example, daily injections of **β-interferons** suppress the immune response to reduce the severity and slow the progression of the disease. **Glatiramer acetate** may block the immune system's attack on the myelin. **Natalizumab** interferes with the ability of potentially damaging immune cells to move from the bloodstream to the CNS. A clinical trial using B cell-depleting therapy with **rituximab,** an anti-CD20 monoclonal antibody, showed that the progression of the disease was slowed in patients younger than 51 years in whom the primary-progressive form of MS was diagnosed. Another clinical trial has shown that oral administration of **fingolimod** slowed the progression of the relapsing- remitting form of MS. This immunosuppressive drug acts by sequestering lymphocytes in the lymph nodes, thereby limiting their access to the CNS. In 2017, the Food and Drug Administration approved the use of **ocrelizumab** for the treatment of primary-progressive MS; it is an immunosuppressive agent that targets the CD20 marker on B lymphocytes.

CLINICAL BOX 4–4

Axonal Regeneration

Peripheral nerve damage is often reversible. Although the axon will

degenerate distal to the damage, connective elements of the so-called **distal stump** often survive. **Axonal sprouting** occurs from the proximal stump, growing toward the nerve ending. This results from **growth-promoting factors** secreted by **Schwann cells** that attract axons toward the distal stump. Adhesion molecules of the immunoglobulin superfamily (eg, the neuron-glia cell adhesion molecule or NgCAM/L1) promote axon growth along cell membranes and extracellular matrices. Inhibitory molecules in the perineurium ensure that the regenerating axons grow in a correct trajectory. Denervated distal stumps are able to upregulate production of **neurotrophins** that promote growth. Once the regenerated axon reaches its target, a new functional connection (eg, neuromuscular junction) is formed, allowing for considerable, although not full, recovery. For example, fine motor control may be permanently impaired because some motor neurons are guided to an inappropriate motor fiber. Nonetheless, recovery of peripheral nerves from damage far surpasses that of central nerve pathways. The proximal stump of a damaged axon in the CNS will form short sprouts, but distant stump recovery is rare, and the damaged axons are unlikely to form new synapses. This is in part because CNS neurons do not have the growth-promoting chemicals needed for regeneration. In fact, CNS myelin is a potent inhibitor of axonal growth. In addition, after a CNS injury, **astrocytic proliferation, activation of microglia, scar formation, inflammation,** and **invasion of immune cells** create an inappropriate environment for regeneration. Thus, treatment of brain and spinal cord injuries focuses on rehabilitation rather than reversing the nerve damage. New research is aiming to identify ways to initiate and maintain axonal growth, to direct regenerating axons to reconnect with their target neurons, and to reconstitute original neuronal circuitry.

THERAPEUTIC HIGHLIGHTS

There is evidence showing that the use of NSAIDs such as ibuprofen can overcome the factors that inhibit axonal growth following injury. This effect is thought to be mediated by the ability of NSAIDs to inhibit RhoA, a small GTPase protein that normally prevents repair of neural pathways and axons. Growth cone collapse in response to myelin-associated inhibitors after nerve injury is prevented by drugs (such as **pertussis toxin**) that interfere with signal transduction via trimeric G-protein. Experimental drugs that inhibit the **phosphoinositide 3-kinase (PI3) pathway** or the **inositol triphosphate (IP3) receptor** have also been shown to promote regeneration after nerve injury.

OTHER FACTORS AFFECTING NEURONAL GROWTH

The regulation of neuronal growth is a complex process. Schwann cells and astrocytes produce **ciliary neurotrophic factor (CNTF).** This factor promotes the survival of damaged and embryonic spinal cord neurons and may prove to be of value in treating human diseases in which motor neurons degenerate. **Glial cell line-derived neurotrophic factor (GDNF)** maintains the survival of midbrain dopaminergic neurons and prevents the apoptosis of spinal motor neurons. Another factor that enhances the growth of neurons is **leukemia inhibitory factor (LIF).** In addition, neurons as well as other cells respond to **insulin-like growth factor I (IGF-I)** and the various forms of **transforming growth factor (TGF), fibroblast growth factor (FGF),** and **platelet-derived growth factor (PDGF).**

Clinical Box 4–4 compares the ability to regenerate neurons after central and peripheral nerve injury.

CHAPTER SUMMARY

- Neurons are composed of a cell body (soma) that is the metabolic center of the neuron, dendrites that extend arborize extensively and are a common zone for receiving input to the neuron, the initial segment of the axon where the action potential is initiated, and a long axon that conducts the action potentials.
- The resting membrane potential of neurons is about –70 mV, which is close to the equilibrium potential for K^+. During hyperkalemia, the resting potential moves closer to the threshold for eliciting an action potential, thus the neuron becomes more excitable. During hypokalemia, the resting membrane potential is reduced and the neuron is hyperpolarized.
- In response to a depolarizing stimulus, voltage-gated Na^+ channels are activated; when the threshold potential is reached, an action potential results. The membrane potential moves toward the equilibrium potential for Na^+. The Na^+ channels are rapidly inactivated before returning to the resting state.
- The direction of the electrical gradient for Na^+ is reversed during the overshoot because the membrane potential is reversed; this limits Na^+

influx. Voltage-gated K^+ channels open to complete the process of repolarization. The slow return of the K^+ channels to the closed state explains after-hyperpolarization, followed by a return to the resting membrane potential.

- The axons of many neurons are wrapped in myelin, a protein-lipid complex that is an effective insulator; depolarization of myelinated axons travels from one node of Ranvier to the next, with the current sink at the active node serving to electrotonically depolarize to the firing level the node ahead of the action potential.
- Nerve fibers are divided into different categories (A, B, and C) based on axonal diameter, conduction velocity, and function. A numerical classification (Ia, Ib, II, III, and IV) is also used for sensory afferent fibers. These various classes differ in their sensitivity to hypoxia and anesthetics.
- Orthograde transport occurs along microtubules that run the length of the axon and requires two molecular motors: dynein and kinesin. It moves from the cell body toward the axon terminals and has both fast (400 mm/day) and slow (0.5–10 mm/day) components. Retrograde transport goes in the opposite direction (from nerve ending to cell body) at a rate of about 200 mm/day.
- There are two main types of glia: microglia and macroglia. Microglia are scavenger cells. Macroglia include oligodendrocytes, Schwann cells, and astrocytes. Oligodendrocytes and Schwann cells are involved in myelin formation; astrocytes produce substances that are tropic to neurons and help maintain the appropriate concentration of ions and neurotransmitters.
- Patchy destruction of myelin within the CNS associated with multiple sclerosis delays the conduction in axons. Autoimmune reactions to the myelin proteins P0 and PMP22 cause Guillain–Barré syndrome, a peripheral demyelinating neuropathy. Mutations in myelin protein genes cause peripheral neuropathies (eg, Charcot-Marie-Tooth disease).
- Neurotrophins (eg, NGF) are carried by retrograde transport to the neuronal cell body where they produce proteins associated with neuronal development, growth, and survival and suppress neuronal apoptosis.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. Glia are critical for the development of the nervous system and have important roles in some neurodegenerative disorder. Which of the following statements correctly describe a property of a type of glia?

   A. Microglia arise from macrophages outside of the nervous system and are physiologically and embryologically similar to other neural cell types.

   B. Fibrous astrocytes are found primarily in the gray matter and induce capillaries to form tight junctions to form the blood-brain barrier.

   C. Protoplasmic astrocytes produce substances that are tropic to neurons to maintain the appropriate concentration of ions and neurotransmitters by taking up K^+ and the neurotransmitters glutamate and GABA.

   D. Oligodendrocytes and Schwann cells are involved in myelin formation around axons in the peripheral nervous system and central nervous system, respectively.

   E. Macroglia are scavenger cells that resemble tissue macrophages and remove debris resulting from injury, infection, and disease.

2. Primary erythromelalgia due to a peripheral nerve sodium channelopathy was diagnosed in a 13-year-old girl who was experiencing frequent episodes of red, painful, warm extremities. Which part of a neuron has the highest concentration of Na^+ channels per square micrometer of cell membrane?

   A. dendrites

   B. cell body near dendrites

   C. initial segment

   D. axonal membrane under myelin

   E. node of Ranvier

3. A 45-year-old woman who works in an office had been experiencing tingling in her index and middle fingers and thumb of her right hand. Recently, her wrist and hand had become weak. Her physician ordered a nerve conduction test to evaluate her for carpal tunnel syndrome. Which one of the following nerves has the slowest conduction velocity?

   A. Aα fibers

   B. Aβ fibers

   C. Aγ fibers

   D. B fibers

   E. C fibers

4. Axoplasmic flow is a cellular process responsible for movement of proteins

and polypeptides within a neuron. Which of the following statements correctly describes a property of orthograde or retrograde axonal transport?

A. Synaptic transmission: Antidromic conduction

B. Molecular motors for orthograde and retrograde transport are dynein and kinesin, respectively.

C. The rate of retrograde fast axonal transport is ∼400 mm/day

D. The rate of orthograde slow axonal transport is ∼200 mm/day

E. Some viruses use retrograde transport to move from the nerve terminal to the soma.

5. For a summer research project, a second-year medical student is introduced to the patch clamp technique in a neurophysiology laboratory. As part of her training, she learns to monitor both membrane potential and individual channel function. Which of the following ionic changes is correctly matched with a component of the action potential?

A. Opening of voltage-gated K^+ channels: After-hyperpolarization

B. A decrease in extracellular Ca^{2+}: Repolarization

C. Opening of voltage-gated Na^+ channels: Depolarization

D. Rapid closure of voltage-gated Na^+ channels: Resting membrane potential

E. Rapid closure of voltage-gated K^+ channels: Relative refractory period

6. A man falls into a deep sleep with one arm under his head. This arm is paralyzed when he awakens, but it tingles, and pain sensation in it is still intact. The reason for the loss of motor function without loss of pain sensation is:

A. A fibers are more susceptible to hypoxia than B fibers.

B. A fibers are more sensitive to pressure than C fibers.

C. C fibers are more sensitive to pressure than A fibers.

D. Motor nerves are more affected by sleep than sensory nerves.

E. Sensory nerves are nearer the bone than motor nerves and hence are less affected by pressure.

7. Neurotrophins foster the production of proteins associated with neuronal development, growth, and survival. Which of the following statements about nerve growth factor (NGF) is true?

A. NGF is made up of one α, two β, and one γ polypeptide subunits.

B. NGF is responsible for the growth and maintenance of adrenergic neurons

in the basal forebrain and the striatum.

C. NGF uses orthograde transport to move from the soma to the nerve terminal of a neuron.

D. NGF is important for the growth of sensory neurons that innervate the muscle spindle.

E. NGF binds to both p75 receptor and Trk A receptors.

8. A 20-year-old female student awakens one morning with severe pain and blurry vision in her left eye; the symptoms abate over several days. About 6 months later, on a morning after playing volleyball with friends, she notices weakness but not pain in her right leg; the symptoms intensify while taking a hot shower. Which of the following is most likely to be the case?

A. The two episodes described are not likely to be related.

B. She may have primary-progressive multiple sclerosis.

C. She may have relapsing-remitting multiple sclerosis.

D. She may have a lumbar disk rupture.

E. She may have Guillain–Barré syndrome.

9. A medical student was working in a neurophysiology lab and was learning factors that determine the resting membrane potential of a neuron. Which of the following statements correctly explains how a change in concentration of an ion inside or outside of the neuron would change its resting membrane potential?

A. A decrease in extracellular Ca^{2+} concentration would stabilize the membrane and reduce its excitability.

B. A decrease in the extracellular Na^{+} concentration would reduce the size of the resting membrane potential.

C. An increase in the extracellular K^{+} concentration would move the resting membrane potential from a normal value of −90 mV to −70 mV.

D. A decrease in the extracellular K^{+} concentration increases the gradient for K^{+} to leak out of the neuron, making the cell more hyperpolarized.

E. A decrease in intracellular Na^{+} concentration would make the resting membrane potential more negative.

10. A precocious 6-year-old girl asked her aunt who was a neurologist at a local medical school to explain how a neuron works. Her aunt explained how different parts of the neuron have a specific role in generating and transmitting an action potential. She identified the part of the neuron where

the action potential is initiated, a part that receives input from other neurons, and a part that conducts the action potential. These would have been as follows:

A. Soma, dendrite, axon

B. Axon hillock, soma, myelin sheath

C. Cell membrane, soma, dendrite

D. Initial segment, dendrite, axon

E. Axon hillock, soma, myelin sheath

CHAPTER 5

Excitable Tissue: Muscle

OBJECTIVES

After studying this chapter, you should be able to:

- Differentiate the major classes of muscle in the body.
- Describe the molecular and electrical makeup of muscle cell excitation–contraction coupling.
- Define elements of the sarcomere that underlie striated muscle contraction.
- Differentiate the role(s) for Ca^{2+} in skeletal, cardiac, and smooth muscle contraction.
- Appreciate muscle cell diversity and function.

INTRODUCTION

Muscle cells, like neurons, can be excited chemically, electrically, and mechanically to produce an action potential that is transmitted along their cell membranes. Unlike neurons, they respond to stimuli by activating a contractile mechanism. The contractile protein myosin and the cytoskeletal protein actin are abundant in muscle, where they are the primary structural components that bring about contraction.

Muscle is generally divided into three types: **skeletal muscle, cardiac muscle,** and **smooth muscle,** although smooth muscle is not a homogeneous single category. Skeletal muscle makes up the great mass of the somatic musculature. It has well-developed cross-striations, does not normally contract in the absence of nervous stimulation, lacks anatomic and functional connections between individual muscle fibers, and is generally under voluntary control. Cardiac muscle also has cross-striations, but it is functionally syncytial and, although it can be modulated via the autonomic nervous system, it can contract rhythmically in the absence of external innervation owing to the presence in the myocardium of pacemaker cells that discharge spontaneously (see Chapter 29). Smooth muscle lacks cross-striations and can be further subdivided into two broad types: unitary (or visceral) smooth muscle and multiunit smooth muscle. The type found in most hollow viscera is functionally syncytial and contains pacemakers that discharge irregularly. The multiunit type found in the eye and in some other locations is not spontaneously active and resembles skeletal muscle in graded contractile ability. Smooth muscle is also classified on the basis of its function or activity pattern to rhythmic/phasic smooth muscle and continuous/tonic smooth muscle.

SKELETAL MUSCLE MORPHOLOGY

ORGANIZATION

Skeletal muscle is made up of individual muscle fibers that are the “building blocks” of the muscular system in the same sense that the neurons are the building blocks of the nervous system. Most skeletal muscles begin and end in tendons, and the muscle fibers are arranged in parallel between the tendinous ends, so that the force of contraction of the units is additive. Each muscle fiber is a single cell that is multinucleated, long, cylindrical, and surrounded by a cell membrane, the **sarcolemma** **(Figure 5–1)**. There are no syncytial bridges between cells. The muscle fibers are made up of myofibrils, which are divisible into individual filaments. These myofilaments contain several proteins that together make up the contractile machinery of the skeletal muscle.

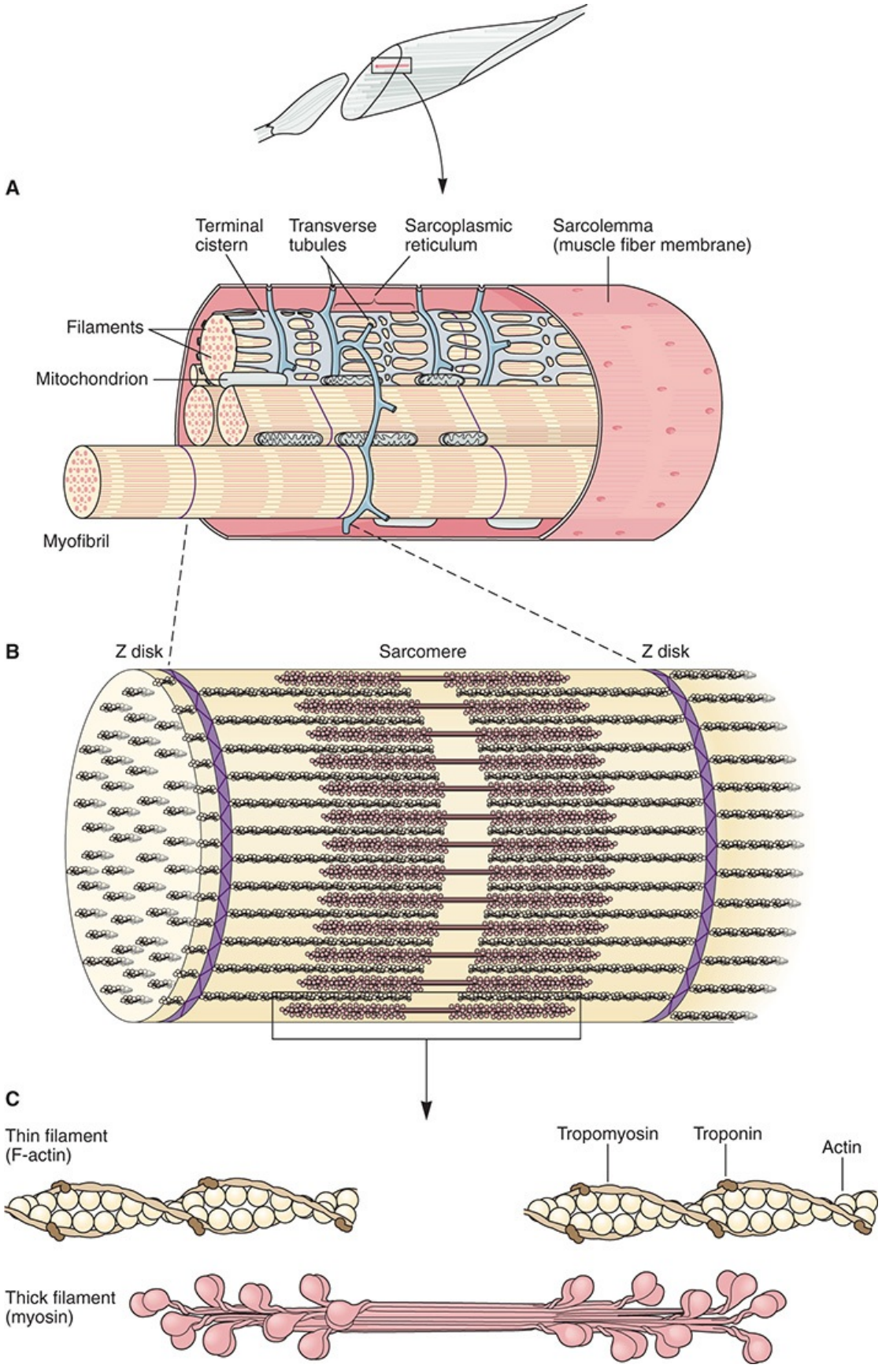

A
Terminal cistern
Transverse tubules
Sarcoplasmic reticulum
Sarcolemma (muscle fiber membrane)
Filaments
Mitochondrion
Myofibril
B
Z disk
Sarcomere
Z disk
C
Thin filament (F-actin)
Tropomyosin
Troponin
Actin
Thick filament (myosin)

FIGURE 5–1 Mammalian skeletal muscle. A) A single muscle fiber surrounded by its sarcolemma has been cut away to show individual myofibrils. The cut surface of the myofibrils shows the arrays of thick and thin filaments. The sarcoplasmic reticulum with its transverse (T) tubules and terminal cisterns surrounds each myofibril. The T tubules invaginate from the sarcolemma and contact the myofibrils twice in every sarcomere. Mitochondria are found between the myofibrils and a basal lamina surrounds the sarcolemma. **B** and **C)** Structural elements of myofibril shown in detail (see also Figure 5–2).

The contractile mechanism in skeletal muscle largely depends on the proteins **myosin-II, actin, tropomyosin,** and **troponin.** Troponin is made up of three subunits: **troponin I, troponin T,** and **troponin C.** Other important proteins in muscle are involved in maintaining the proteins that participate in contraction in appropriate structural relation to one another and to the extracellular matrix.

STRIATIONS

Differences in the refractive indexes of the various parts of the muscle fiber are responsible for the characteristic cross-striations seen in skeletal muscle when viewed under the microscope. The parts of the cross-striations are frequently identified by letters A, H, I, M, and Z **(Figure 5–2)**. The light I band is divided by the dark Z-line, and the dark A band has the lighter H band in its center. A transverse M-line is seen in the middle of the H band, and this line plus the narrow light areas on either side of it are sometimes called the pseudo-H zone. The area between two adjacent Z-lines is called a **sarcomere.** The orderly arrangement of actin, myosin, and related proteins that produces this pattern can also be seen in Figures 5–1 and 5–2. The thick filaments, which are about twice the diameter of the thin filaments, are made up of myosin; the thin filaments are made up of actin, tropomyosin, and troponin. The thick filaments are lined up to form the A bands, whereas the array of thin filaments extends out of the A band and into the less dense staining I bands. The lighter H bands in the center of the A bands are the regions where, when the muscle is relaxed, the thin filaments do not overlap the thick filaments. The Z-lines allow for anchoring of the thin filaments. If a transverse section through the A band is examined under the electron microscope, each thick filament is seen to be surrounded by six thin filaments in a regular hexagonal pattern.

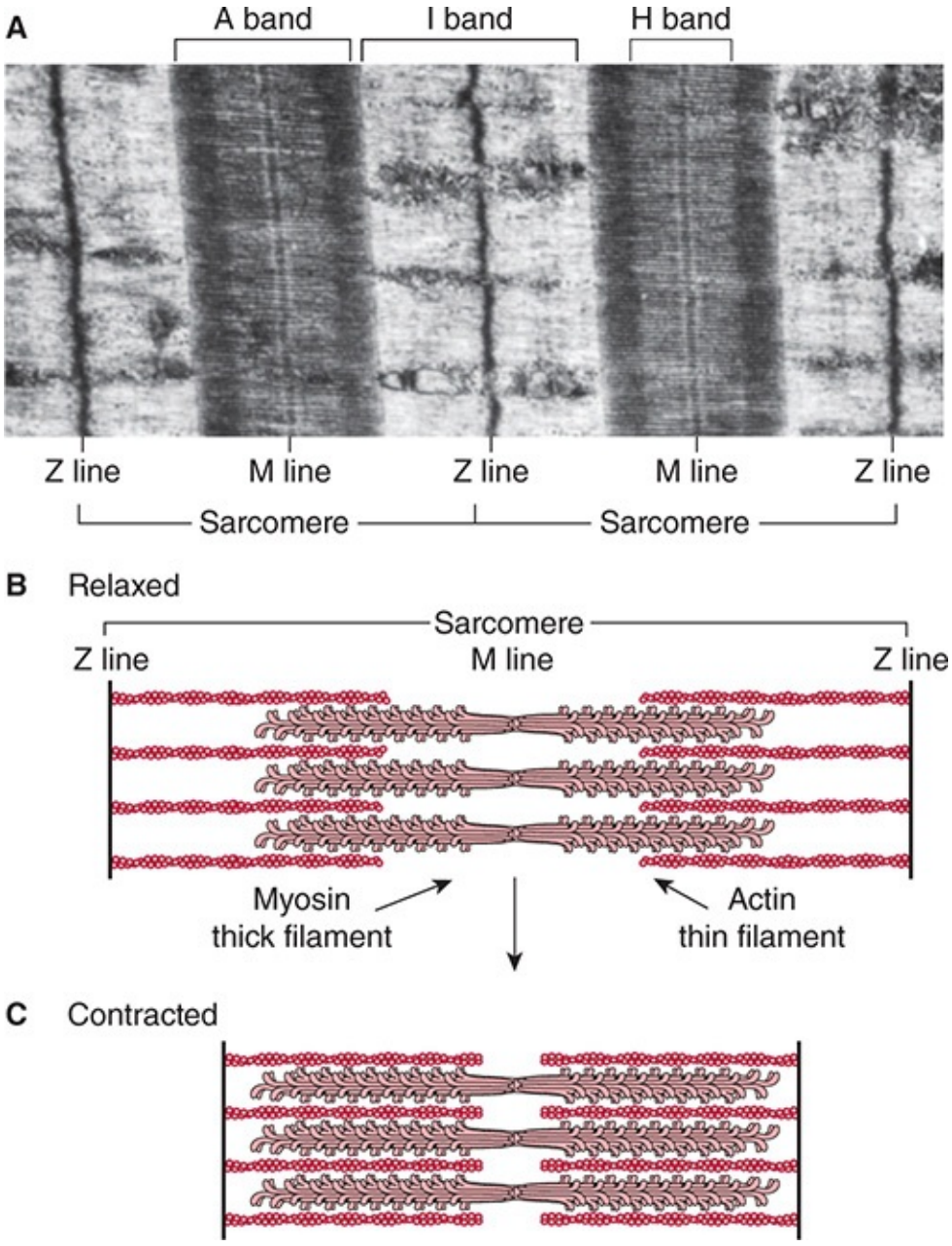


FIGURE 5–2 Skeletal muscle sarcomere. A) Electron micrograph of human gastrocnemius muscle (× 13,500). The sarcomere, named bands, and lines are shown. (Used with permission from GM Walker and GR Schrodt.) **B)** Arrangement of thin (actin) and thick (myosin) filaments and the Z-line in a relaxed skeletal muscle. **C)** Arrangement of thin and thick filaments and the Z-line in a contracted skeletal muscle. Note that the Z-lines come together as the thick and thin filaments slide next to each other during contraction. The thick and thin filaments do not change in size.

The form of myosin found in muscle is myosin-II, with two globular heads and a long tail. The heads of the myosin molecules form cross-bridges with actin. Myosin contains heavy chains and light chains, and its heads are made up of the light chains and the amino terminal portions of the heavy chains. These heads contain an actin-binding site and a catalytic site that hydrolyzes adenosine triphosphate (ATP). The myosin molecules are arranged symmetrically on either side of the center of the sarcomere, and it is this arrangement that creates the light areas in the pseudo-H zone. The M-line is the site of the reversal of polarity of the myosin molecules in each of the thick filaments. At these points, there are slender cross-connections that hold the thick filaments in proper array. Each thick filament contains several hundred myosin molecules.

The thin filaments are polymers made up of two chains of actin that form a long double helix. Tropomyosin molecules are long filaments located in the groove between the two chains in the actin. Each thin filament contains 300–400 actin molecules and 40–60 tropomyosin molecules. Troponin molecules are small globular units located at intervals along the tropomyosin molecules. Each of the three troponin subunits has a unique function: Troponin T binds the troponin components to tropomyosin, troponin I inhibits the interaction of myosin with actin, and troponin C contains the binding sites for the Ca^{2+} that initiates contraction.

Some additional structural proteins that are important in skeletal muscle function include **actinin, titin,** and **desmin.** Actinin binds actin to the Z-lines. Titin, the largest known protein (with a molecular mass near 3,000,000 Da), connects the Z-lines to the M-lines and provides scaffolding for the sarcomere. It contains two kinds of folded domains that provide muscle with its elasticity. At first when the muscle is stretched, there is relatively little resistance as the domains unfold, but with further stretch, there is a rapid increase in resistance that protects the structure of the sarcomere. Desmin adds structure to the Z-lines in part by binding the Z-lines to the plasma membrane. Some muscle disorders associated with these structural components are described in **Clinical Box 5–1**. It should be noted that although these proteins are important in muscle structure/function, by no means do they represent an exhaustive list.

CLINICAL BOX 5–1

Structural & Metabolic Disorders in Muscle Disease

The term **muscular dystrophy** is applied to diseases that cause progressive

weakness of skeletal muscle. About 50 such diseases have been described, some of which include cardiac as well as skeletal muscle. They range from mild to severe and some are eventually fatal. They have multiple causes, but mutations in the genes for the various components of the dystrophin–glycoprotein complex are a prominent cause. The dystrophin gene is one of the largest in the body, and mutations can occur at many different sites in it. **Duchenne muscular dystrophy** is a serious form of dystrophy in which the dystrophin protein is absent from muscle. It is X-linked and usually fatal by the age of 30. In a milder form of the disease, **Becker muscular dystrophy,** dystrophin is present but altered or reduced in amount. Limb-girdle muscular dystrophies of various types are associated with mutations of the genes coding for the sarcoglycans or other components of the dystrophin–glycoprotein complex.

Due to its enormous size and structural role in the sarcomere, **titin** is a prominent target for mutations that give rise to muscle disease. Mutations that encode for shorter titin structure have been associated with dilated cardiomyopathy, while other mutations have been associated with hypertrophic cardiomyopathy. The skeletal muscle-associated tibialis muscular dystrophy is a genetic muscle disease of titin that is predicted to destabilize the folded state of the protein. Interestingly, many of the titin mutations identified thus far are in regions of titin that are expressed in all striated muscles, yet not all muscles are affected in the same way. Such muscle type-specific phenotypes underscore the need to study titin's multiple functions in different muscles, under both normal and pathologic conditions.

Desmin-related myopathies are a very rare heterogeneous group of muscle disorders that typically result in cellular aggregates of desmin. Common symptoms of these diseases are failing and wasting in the distal muscles of the lower limbs that can later be identified in other body areas. Studies in desmin knockout mice have revealed defects in skeletal, smooth, and cardiac muscle, notably in the diaphragm and heart.

Metabolic Myopathies

Mutations in genes that code for enzymes involved in the metabolism of carbohydrates, fats, and proteins to CO_2 and H_2O in muscle and the production of ATP can cause **metabolic myopathies** (eg, McArdle syndrome). Metabolic myopathies all have in common exercise intolerance and the possibility of muscle breakdown due to accumulation of toxic metabolites.

THERAPEUTIC HIGHLIGHTS

Although acute muscle pain and soreness can be treated with anti-inflammatory drugs and rest, the genetic dysfunctions described above are not as easily addressed. The overall goals are to slow muscle function/structure loss and, when possible relieve symptoms associated with the disease. Extensive monitoring, physical therapy, and appropriate drugs including corticosteroids can aid to slow disease progression. Assistive devices and surgery are not uncommon as the diseases progress.

SARCOTUBULAR SYSTEM

The muscle fibrils are surrounded by structures made up of membranes that appear in electron micrographs as vesicles and tubules. These structures form the **sarcotubular system,** which is made up of a **T system** and a **sarcoplasmic reticulum.** The T system of transverse tubules, which is continuous with the sarcolemma of the muscle fiber, forms a grid perforated by the individual muscle fibrils (Figure 5–1). The space between the two layers of the T system is an extension of the extracellular space. The sarcoplasmic reticulum, which forms an irregular curtain around each of the fibrils, has enlarged **terminal cisterns** in close contact with the T system at the junctions between the A and I bands. At these points of contact, the arrangement of the central T system with a cistern of the sarcoplasmic reticulum on either side has led to the use of the term **triads** to describe the system. The T system, which is continuous with the sarcolemma, provides a path for the rapid transmission of the action potential from the cell plasma membrane to all the fibrils in the muscle. The sarcoplasmic reticulum is an important store of Ca^{2+} and also participates in muscle metabolism.

DYSTROPHIN–GLYCOPROTEIN COMPLEX

The large **dystrophin** protein (molecular mass 427,000 Da) forms a rod that connects the thin actin filaments to the transmembrane protein **β-dystroglycan** in the sarcolemma by smaller proteins in the cytoplasm, **syntrophins.** β-Dystroglycan is connected to **merosin** (merosin refers to laminins that contain the α_2 subunit in their trimeric makeup) in the extracellular matrix by **α-dystroglycan** **(Figure 5–3)**. The dystroglycans are in turn associated with a

complex of four transmembrane glycoproteins: α-, β-, γ-, and δ-**sarcoglycan.** This **dystrophin–glycoprotein complex** adds strength to the muscle by providing a scaffolding for the fibrils and connecting them to the extracellular environment. Disruption of these important structural features can result in several different muscular dystrophies (see Clinical Box 5–1).

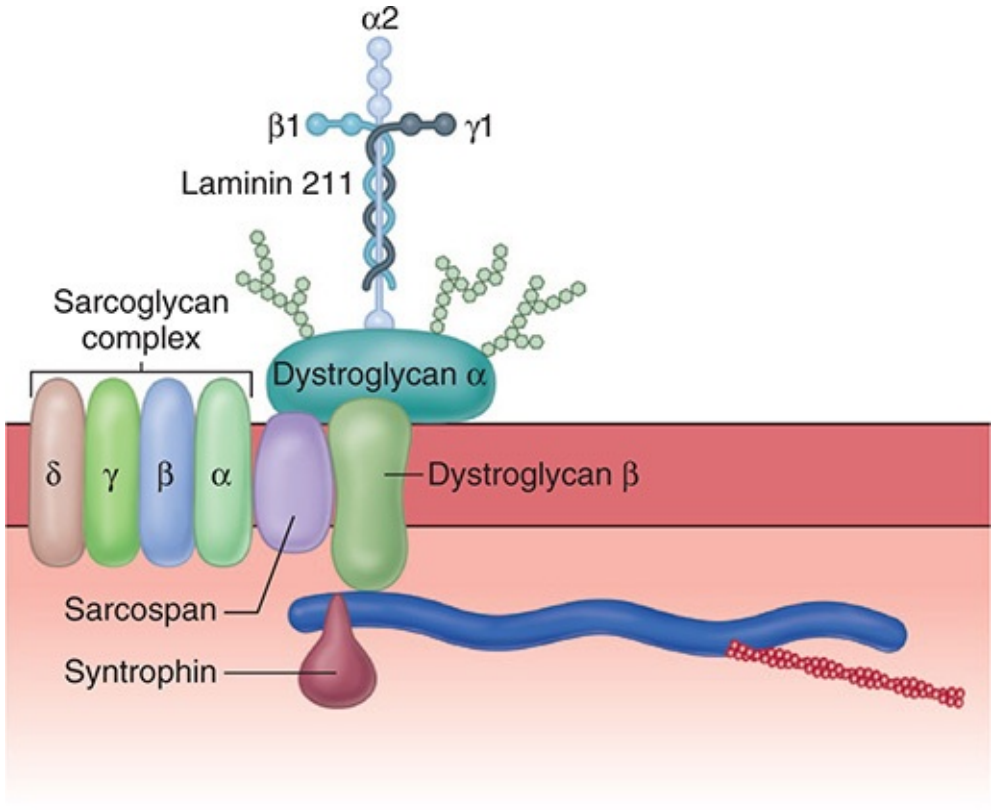


FIGURE 5–3 The dystrophin–glycoprotein complex. Dystrophin connects F-actin to the two members of the dystroglycan (DG) complex, α and β-dystroglycan, and these in turn connect to the merosin subunit of laminin 211 in the extracellular matrix. The sarcoglycan complex of four glycoproteins, α-, β-, γ-, and δ-sarcoglycan, sarcospan, and syntropins are all associated with the dystroglycan complex. There are muscle disorders associated with loss, abnormalities, or both of the sarcoglycans and merosin. (Used with permission of Justin Fallon and Kevin Campbell.)

ELECTRICAL PHENOMENA & IONIC FLUXES

ELECTRICAL CHARACTERISTICS OF

SKELETAL MUSCLE

The electrical events in skeletal muscle and the ionic fluxes that underlie them share distinct similarities to those in nerve, with quantitative differences in timing and magnitude. The resting membrane potential of skeletal muscle is about −90 mV. The action potential lasts 2–4 ms and is conducted along the muscle fiber at about 5 m/s. The absolute refractory period is 1–3 ms long, and the after-polarizations, with their related changes in threshold to electrical stimulation, are relatively prolonged. The initiation of impulses at the myoneural junction is discussed in Chapter 6.

ION DISTRIBUTION & FLUXES

The distribution of ions across the muscle fiber membrane is similar to that across the nerve cell membrane. Approximate values for the various ions and their equilibrium potentials are shown in **Table 5–1**. As in nerves, membrane depolarization is largely a manifestation of Na^+ influx, and repolarization is largely a manifestation of K^+ efflux.

TABLE 5–1 Steady-state distribution of ions in the intracellular and extracellular compartments of mammalian skeletal muscle, and the equilibrium potentials for these ions.

	Concentration (mmol/L)		
Ion[a]	Intracellular Fluid	Extracellular Fluid	Equilibrium Potential (mV)
Na^+	12	145	+65
K^+	155	4	–95
H^+	13×10^{-5}	3.8×10^{-5}	–32
Cl^-	3.8	120	–90
HCO_3^-	8	27	–32
A^-	155	0	…
Membrane potential = –90 mV			

[a] A^- represents organic anions. The value for intracellular Cl^- is calculated from the membrane potential, using the Nernst equation.

CONTRACTILE RESPONSES

It is important to distinguish between the electrical and mechanical events in skeletal muscle. Although one response does not normally occur without the other, their physiological bases and characteristics are different. Muscle fiber membrane depolarization normally starts at the motor endplate, the specialized structure under the motor nerve ending. The action potential is transmitted along the muscle fiber and initiates the contractile response by inducing Ca^{2+} influx and release.

THE MUSCLE TWITCH

A single action potential causes a brief contraction followed by relaxation. This response is called a **muscle twitch.** In **Figure 5–4**, the action potential and the twitch are plotted on the same time scale. The twitch starts about 2 ms after the start of membrane depolarization, before repolarization is complete. The duration of the twitch varies with the type of muscle being tested. “Fast” muscle fibers, primarily those concerned with fine, rapid, precise movement, have twitch durations as short as 7.5 ms. “Slow” muscle fibers, principally those involved in strong, gross, sustained movements, have twitch durations up to 100 ms.

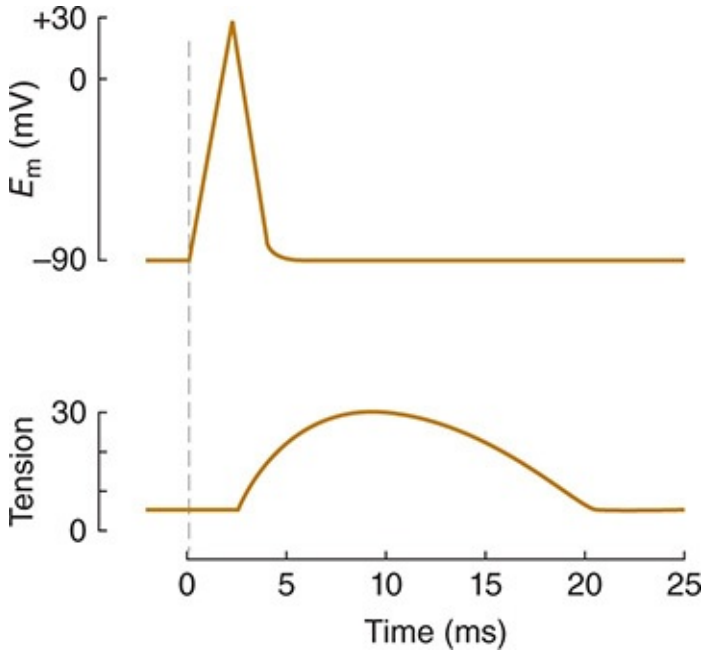


FIGURE 5–4 The electrical and mechanical responses of a mammalian

skeletal muscle fiber to a single maximal stimulus. The electrical response or change in membrane potential (E_m) in mV and the mechanical response or tension (T) in in arbitrary units are plotted on the same abscissa (time in ms). The mechanical response is relatively long-lived compared to the electrical response that initiates contraction.

MOLECULAR BASIS OF CONTRACTION

The process by which the contraction of muscle is brought about is a sliding of the thin filaments over the thick filaments. Note that this shortening is not due to changes in the actual lengths of the thick and thin filaments, rather, due to their increased overlap within the muscle cell. The width of the A bands is constant, whereas the Z-lines move closer together when the muscle contracts and farther apart when it relaxes (Figure 5–2).

The sliding during muscle contraction occurs when the myosin heads bind firmly to actin, bend at the junction of the head with the neck, and then detach. This "power stroke" depends on the simultaneous hydrolysis of ATP. Myosin-II molecules are dimers that have two heads, but only one attaches to actin at any given time. The probable sequence of events of the power stroke is outlined in **Figure 5–5**. In resting muscle, troponin I is bound to actin and tropomyosin and covers the sites where myosin heads interact with actin. Also at rest, the myosin head contains tightly bound adenosine phosphate (ADP) and inorganic phosphate (Pi). Following an action potential, cytosolic Ca^{2+} is increased (due to Ca^{2+} release from the sarcoplasmic reticulum and Ca^{2+} influx through voltage-gated Ca^{2+} channels) and free Ca^{2+} binds to troponin C. This binding results in a weakening of the troponin I interaction with actin and exposes the actin binding site for myosin to allow for formation of myosin/actin cross-bridges. Upon formation of the cross-bridge, ADP is released, causing a conformational change in the myosin head that moves the thin filament relative to the thick filament, comprising the cross-bridge "power stroke." ATP quickly binds to the free site on the myosin, which leads to a detachment of the myosin head from the thin filament. ATP is hydrolyzed and P_i released, causing a "re-cocking" of the myosin head and completing the cycle. As long as cytosolic Ca^{2+} remains elevated and sufficient ATP is available, this cycle repeats. Many myosin heads cycle at or near the same time, and they cycle repeatedly, producing gross muscle contraction. Each power stroke shortens the sarcomere about 10 nm. Each thick filament has about 500 myosin heads, and each head cycles about

five times per second during a rapid contraction.

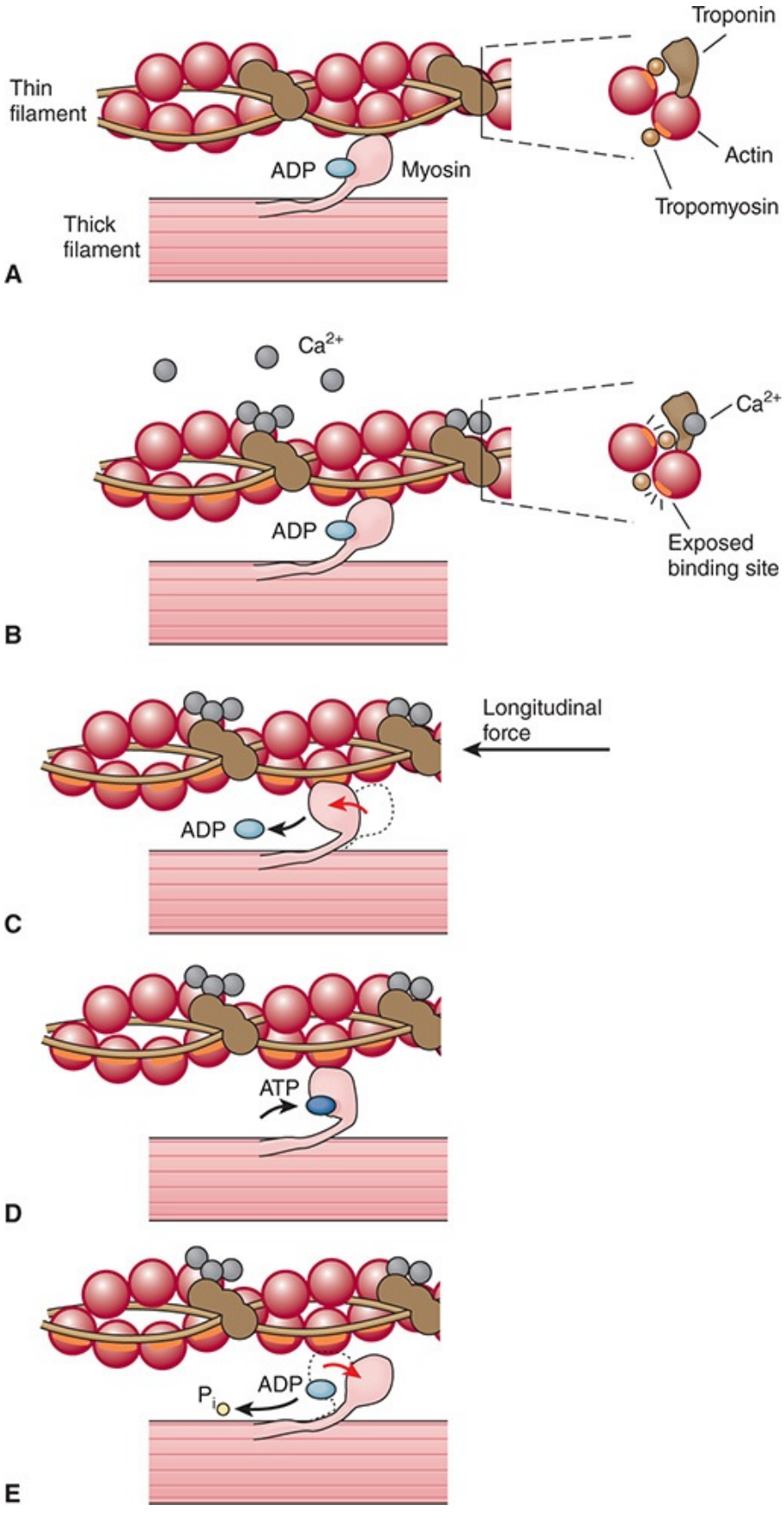
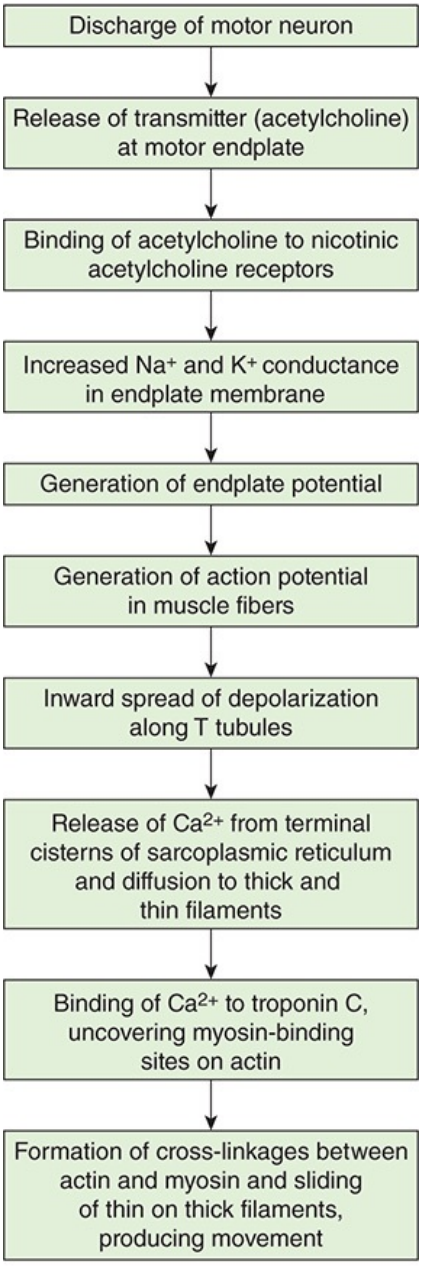

Troponin
Thin filament
Actin
ADP
Myosin
Tropomyosin
Thick filament
A
Ca^{2+}
Ca^{2+}
ADP
Exposed binding site
B
Longitudinal force
ADP
C
ATP
D
P_i
ADP
E

FIGURE 5–5 Power stroke of myosin in skeletal muscle. A) At rest, myosin heads are bound to adenosine diphosphate and are said to be in a "cocked" position in relation to the thin filament, which does not have Ca^{2+} bound to the troponin—tropomyosin complex. **B)** Ca^{2+} bound to the troponin—tropomyosin complex induces a conformational change in the thin filament that allows for myosin heads to cross-bridge with thin filament actin. **C)** Myosin heads rotate, move the attached actin, and shorten the muscle fiber, forming the power stroke. **D)** At the end of the power stroke, ATP binds to a now exposed site and causes a detachment from the actin filament. **E)** ATP is hydrolyzed into ADP and inorganic phosphate (P_i) and this chemical energy is used to "re-cock" the myosin head. (Data from Huxley AF, Simmons RM: Proposed mechanism of force generation in striated muscle. Nature Oct 22;233(5321):533–538, 1971 and Squire JM: Molecular mechanisms in muscular contraction. Trends Neurosci 6:409–413, 1983.)

The process by which membrane depolarization of the muscle fiber initiates contraction is called **excitation–contraction coupling.** The action potential is transmitted to all the fibrils in the fiber via the T system **(Figure 5–6)**. It triggers the release of Ca^{2+} from the terminal cisterns, the lateral sacs of the sarcoplasmic reticulum next to the T system. Depolarization of the T tubule membrane activates the sarcoplasmic reticulum via **dihydropyridine receptors (DHPRs),** named for the drug dihydropyridine, which blocks them **(Figure 5–7)**. DHPRs are voltage-gated Ca^{2+} channels in the T tubule membrane. In cardiac muscle, influx of Ca^{2+} via these channels triggers the release of Ca^{2+} stored in the sarcoplasmic reticulum (ie, Ca^{2+}-induced Ca^{2+} release) by activating the **ryanodine receptor (RyR).** The RyR is named after the plant alkaloid ryanodine that was used in its discovery. The RyR is a ligand-gated Ca^{2+} channel with Ca^{2+} as its natural ligand. In skeletal muscle, Ca^{2+} entry from the extracellular fluid (ECF) by this route is not required for Ca^{2+} release. Instead, the DHPR that serves as the voltage sensor triggers release of Ca^{2+} from the nearby sarcoplasmic reticulum via physical interaction with the RyR. The released Ca^{2+} is quickly amplified through Ca^{2+}-induced Ca^{2+} release. Cytosolic Ca^{2+} concentration is reduced in the muscle cell by the sarcoplasmic or endoplasmic reticulum Ca^{2+} ATPase (SERCA). The SERCA pump uses energy from ATP hydrolysis to remove Ca^{2+} from the cytosol back into the terminal cisterns against its concentration gradient, where it is stored until released by the next

action potential. Once cytosolic Ca^{2+} concentration, or the Ca^{2+} concentration outside the sarcoplasmic reticulum, has been lowered sufficiently, chemical interaction between myosin and actin ceases and the muscle relaxes. Note that ATP provides the energy for both contraction (at the myosin head) and relaxation (via SERCA). If transport or uptake of Ca^{2+} into the sarcoplasmic reticulum is inhibited, relaxation does not occur even though there are no more action potentials; the resulting sustained contraction is called a **contracture.** Alterations in the excitable response in muscle underscore many different pathologic conditions **(Clinical Box 5–2)**.

Steps in contraction



Steps in relaxation

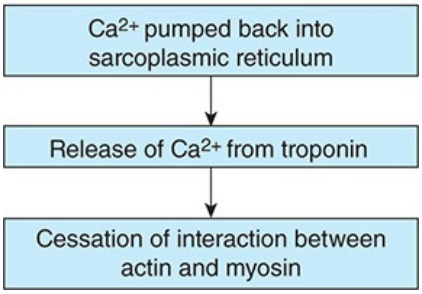


FIGURE 5–6 Flow of information that leads to muscle contraction.

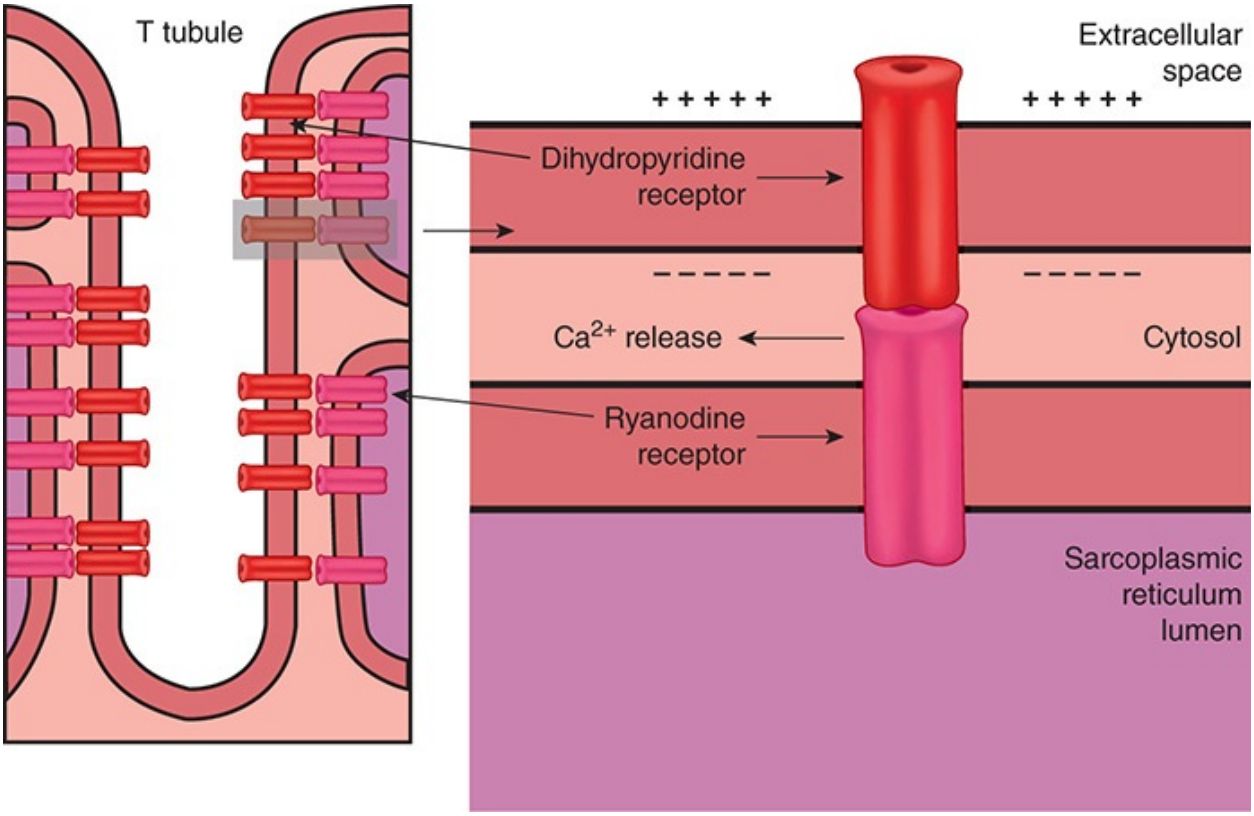


FIGURE 5–7 Relation of the T tubule to the sarcoplasmic reticulum in Ca^{2+} transport. In skeletal muscle, the voltage-gated dihydropyridine receptor in the T tubule triggers Ca^{2+} release from the sarcoplasmic reticulum (SR) via the ryanodine receptor (RyR). Upon sensing a voltage change, there is a physical interaction between the sarcolemmal-bound DHPR and the SR-bound RyR. This interaction gates the RyR and allows for Ca^{2+} release from the SR.

TYPES OF CONTRACTION

Muscular contraction involves shortening of the contractile elements, but because muscles have elastic and viscous elements in series with the contractile mechanism, it is possible for contraction to occur without an appreciable decrease in the length of the whole muscle **(Figure 5–8)**. Such a contraction is called **isometric** (“same measure” or length). Contraction against a constant load with a decrease in muscle length is **isotonic** (“same tension”). Note that because work is the product of force times distance, isotonic contractions do work,

whereas isometric contractions do not. In other situations, muscle can do negative work while lengthening against a constant weight.

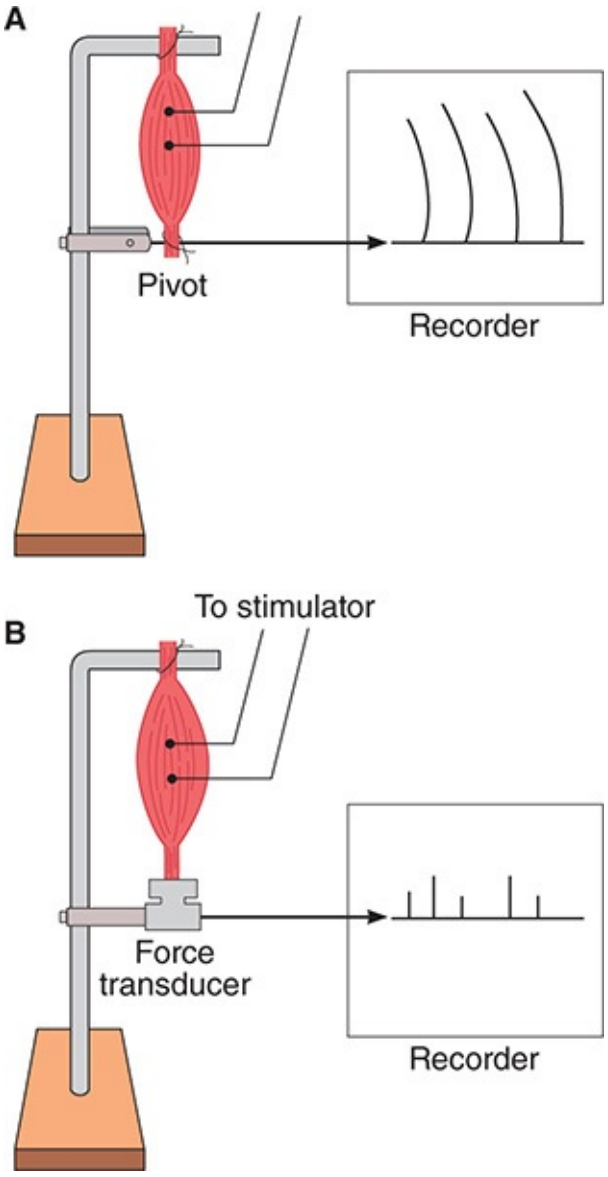


FIGURE 5–8 Isotonic and isometric contractions. A) Muscle preparation arranged for recording isotonic contractions. **B)** Preparation arranged for recording isometric contractions. In **A,** the muscle is fastened to a writing lever that swings on a pivot. In **B,** it is attached to an electronic transducer that measures the force generated without permitting the muscle to shorten.

SUMMATION OF CONTRACTIONS

The electrical response of a muscle fiber to repeated stimulation is like that of nerve. The fiber is electrically refractory only during the rising phase and part of the falling phase of the spike potential. At this time, the contraction initiated by the first stimulus is just beginning. However, because the contractile mechanism does not have a refractory period, repeated stimulation before relaxation has occurred produces additional activation of the contractile elements and a response that is added to the contraction already present. This phenomenon is known as **summation of contractions.** The tension developed during summation is considerably greater than that during the single muscle twitch. With rapidly repeated stimulation, activation of the contractile mechanism occurs repeatedly before any relaxation has occurred, and the individual responses fuse into one continuous contraction. Such a response is called **tetanus (tetanic contraction).** It is a **complete tetanus** when no relaxation occurs between stimuli and an **incomplete tetanus** when periods of incomplete relaxation take place between the summated stimuli. During a complete tetanus, the tension developed is about four times that developed by the individual twitch contractions. The development of an incomplete and a complete tetanus in response to stimuli of increasing frequency is shown in **Figure 5–9**.

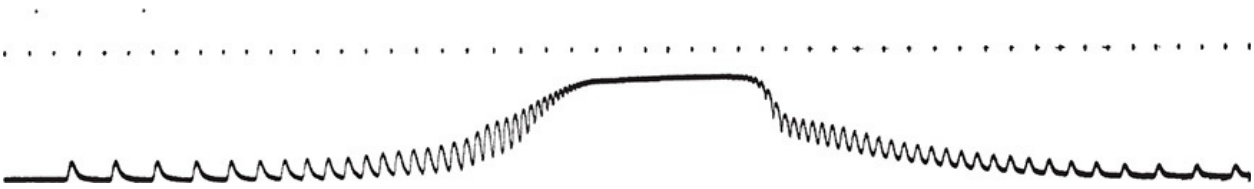


FIGURE 5–9 Tetanus. Isometric tension of a single muscle fiber during continuously increasing and decreasing stimulation frequency. Dots at the top are at intervals of 0.2 s. Note the development of incomplete and then complete tetanus as stimulation is increased, and the return of incomplete tetanus, then full response, as stimulation frequency is decreased.

CLINICAL BOX 5–2

Muscle Channelopathies

Channelopathies are diseases that have as their underlying feature mutations or dysregulation of ion channels. Such diseases are frequently associated with excitable cells, including muscle. In the various forms of clinical **myotonia,** muscle relaxation is prolonged after voluntary contraction. The molecular bases of myotonias are due to dysfunction of channels that shape the action

potential. Myotonia dystrophy is caused by an autosomal dominant mutation that leads to overexpression of a K^+ channel (although the mutation is *not* at the K^+ channel). A variety of myotonias are associated with mutations in Na^+ channels (eg, hyperkalemic periodic paralysis, paramyotonia congenita, or Na^+ channel congenita) or Cl^- channels (eg, dominant or recessive myotonia congenita). **Myasthenia,** defined as abnormal muscle weakness or disease, can also be related to loss of ion channel function in the muscle. In **congenital myasthenia,** the patient has an inheritable disorder of one of ion channels necessary for the transmission of neuronal signaling to muscle response. Mutations in Ca^{2+} channels that allow for neuronal transmitter release or in the acetylcholine receptor nonspecific cation channels, important in recognition of neuronal transmitters, have both been shown to cause congenital myasthenia. Alterations of channel functions can also occur via autoimmune disease, such as that observed in **myasthenia gravis.** In this disease, antibodies to the nicotinic acetylcholine receptor can reduce its functional presence at the muscle membrane by up to 80%, and thus limit muscle response to neuronal transmitter release.

Channelopathies can also occur in the Ca^{2+} release channels in muscle (eg, ryanodine receptors) that amplify the Ca^{2+} response within the cell. Such mutations can cause **malignant hyperthermia.** Patients with this condition display normal muscle function under normal conditions. However, certain anesthetic agents, or in rare cases exposure to high environmental heat or strenuous exercise, can trigger abnormal release of Ca^{2+} from the sarcoplasmic reticulum in the muscle cell, resulting in sustained muscle contraction and heat production. In severe cases, fatality can occur.

THERAPEUTIC HIGHLIGHTS

Although the symptoms associated with each individual channelopathy may be similar, treatments for the individual diseases include a wide variety of drugs that are targeted to the defect in the individual ion channel (or proteins associated with ion channel). Appropriate drug therapy helps improve symptoms and maintain acceptable muscle function. Further interventions related to individual diseases are to avoid muscle movements that exacerbate the disease.

The stimulation frequency at which summation of contractions occurs is

determined by the twitch duration of the particular muscle being studied. For example, if the twitch duration is 10 ms, frequencies less than 1/10 ms (100/s) cause discrete responses interrupted by complete relaxation, and frequencies greater than 100/s cause summation.

RELATION BETWEEN MUSCLE LENGTH, TENSION, & VELOCITY OF CONTRACTION

Both the tension that a muscle develops when stimulated to contract isometrically (the **total tension)** and the **passive tension** exerted by the unstimulated muscle vary with the length of the muscle fiber. This relationship can be studied in a whole skeletal muscle preparation such as that shown in Figure 5–8. The length of the muscle can be varied by changing the distance between its two attachments. At each length, the passive tension is measured, the muscle is then stimulated electrically, and the total tension is measured. The difference between the two values at any length is the amount of tension actually generated by the contractile process, the **active tension.** The records obtained by plotting passive tension and total tension against muscle length are shown in **Figure 5–10**. Similar curves are obtained when single muscle fibers are studied. The length of the muscle at which the active tension is maximal is usually called its optimal **resting length.** The term comes originally from experiments demonstrating that the length of many of the muscles in the body at rest is the length at which they develop maximal tension.

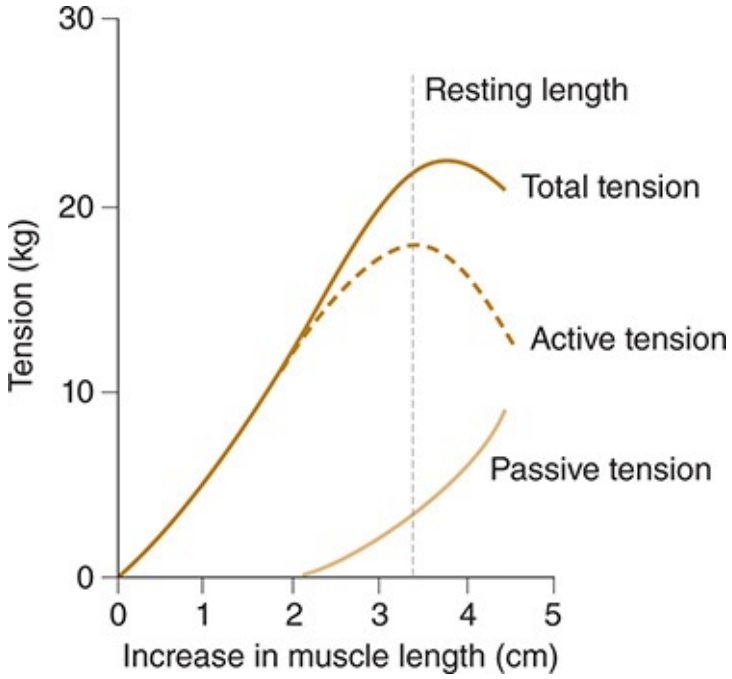


FIGURE 5–10 Length–tension relationship for the human triceps muscle. The passive tension curve measures the tension exerted by this skeletal muscle at each length when it is not stimulated. The total tension curve represents the tension developed when the muscle contracts isometrically in response to a maximal stimulus. The active tension is the difference between the two.

The observed length—tension relation in skeletal muscle can be explained by the sliding filament mechanism of muscle contraction. When the muscle fiber contracts isometrically, the tension developed is proportional to the number of cross-bridges between the actin and the myosin molecules. When muscle is stretched, the overlap between actin and myosin is reduced and the number of cross-linkages is therefore reduced. Conversely, when the muscle is appreciably shorter than resting length, the distance the thin filaments can move is reduced.

The velocity of muscle contraction varies inversely with the load on the muscle. At a given load, the velocity is maximal at the resting length and declines if the muscle is shorter or longer than this length.

FIBER TYPES

Although skeletal muscle fibers resemble one another in a general way, skeletal muscle is a heterogeneous tissue made up of fibers that vary in myosin ATPase activity, contractile speed, and other properties. Muscles are frequently classified

into two types: “slow” and “fast.” These muscles can contain a mixture of three fiber types: type I (for slow-oxidative or SO), type IIA (for fast-oxidative-glycolytic or FOG), or type IIB (for fast glycolytic or FG). Some of the properties associated with types I, IIA, and IIB fibers are summarized in **Table 5–2**. Although these classification schemes are valid for muscles across many mammalian species, there are significant variations of fibers within and between muscles. For example, type I fibers in a given muscle can be larger than type IIA fibers from a different muscle in the same animal. Many of the differences in the fibers that make up muscles stem from differences in the proteins within them. Most of these are encoded by multigene families. Ten different **isoforms** of the myosin heavy chains (MHCs) have been characterized. Each of the two types of light chains also has isoforms. It appears that there is only one form of actin, but multiple isoforms of tropomyosin and all three components of troponin.

TABLE 5–2 Classification of fiber types in skeletal muscles.

	Type I	Type IIA	Type IIB
Other names	Slow, Oxidative (SO)	Fast, Oxidative, Glycolytic (FOG)	Fast, Glycolytic (FG)
Color	Red	Red	White
Myosin ATPase activity	Slow	Fast	Fast
Ca^{2+}-pumping capacity of sarcoplasmic reticulum	Moderate	High	High
Diameter	Small	Large	Large
Glycolytic capacity	Moderate	High	High
Oxidative capacity	High	Moderate	Low
Associated Motor Unit Type	Slow (S)	Fast Resistant to Fatigue (FR)	Fast Fatigable (FF)
Membrane potential = –90 mV			

ENERGY SOURCES & METABOLISM

Muscle contraction requires energy, and muscle has been called “a machine for converting chemical energy into mechanical work.” The immediate source of this energy is ATP, and this is formed by the metabolism of carbohydrates and lipids (see Chapters 1 and 2).

PHOSPHORYLCREATINE

ATP is resynthesized from ADP by the addition of a phosphate group. Some of the energy for this endothermic reaction is supplied by the breakdown of glucose to CO_2 and H_2O, but there also exists in muscle another energy-rich phosphate compound that can supply this energy for short periods. This compound is **phosphorylcreatine,** which is hydrolyzed to creatine and phosphate groups with the release of considerable energy **(Figure 5–11)**. At rest, some ATP in the mitochondria transfers its phosphate to creatine, so that a phosphorylcreatine store is built up. During exercise, the phosphorylcreatine is hydrolyzed at the junction between the myosin heads and actin, forming ATP from ADP and thus permitting contraction to continue.

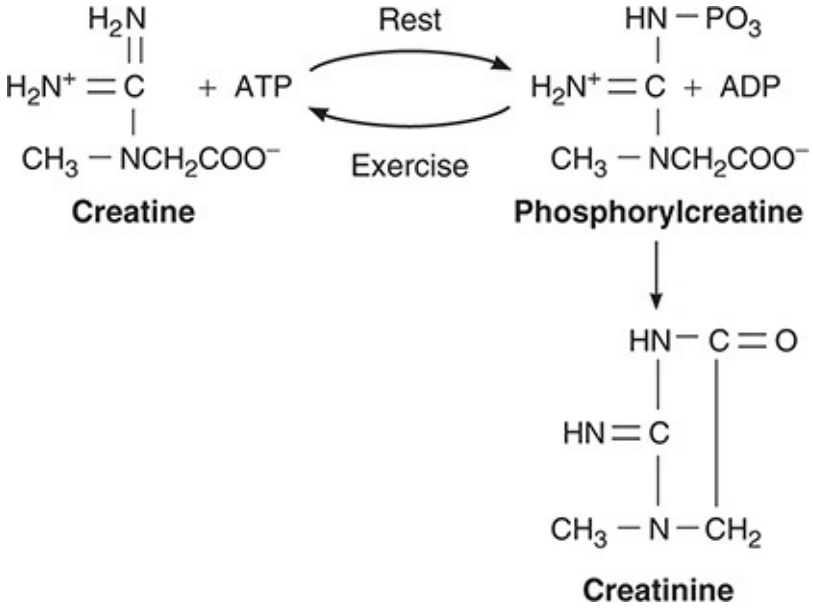


FIGURE 5–11 Creatine, phosphorylcreatine, and creatinine cycling in muscle. During periods of high activity, cycling of phosphorylcreatine allows for quick release of ATP to sustain muscle activity.

CARBOHYDRATE & LIPID BREAKDOWN

At rest and during light exercise, muscles utilize lipids in the form of free fatty acids as their energy source. As the intensity of exercise increases, lipids alone cannot supply energy fast enough and so use of carbohydrate becomes the predominant component in the muscle fuel mixture. Thus, during exercise, much of the energy for phosphorylcreatine and ATP resynthesis comes from the breakdown of glucose to CO_2 and H_2O. Glucose in the bloodstream enters cells, where it is degraded through a series of chemical reactions to pyruvate. Another source of intracellular glucose, and consequently of pyruvate, is glycogen, the carbohydrate polymer that is especially abundant in liver and skeletal muscle. When adequate O_2 is present, pyruvate enters the citric acid cycle and is metabolized—through this cycle and the so-called respiratory enzyme pathway —to CO_2 and H_2O. This process is called **aerobic glycolysis.** The metabolism of glucose or glycogen to CO_2 and H_2O forms large quantities of ATP from ADP. If O_2 supplies are insufficient, the pyruvate formed from glucose does not enter the tricarboxylic acid cycle but is reduced to lactate. This process of **anaerobic glycolysis** is associated with the net production of much smaller quantities of energy-rich phosphate bonds, but it does not require the presence of O_2. A brief overview of the various reactions involved in supplying energy to skeletal muscle is shown in **Figure 5–12**.

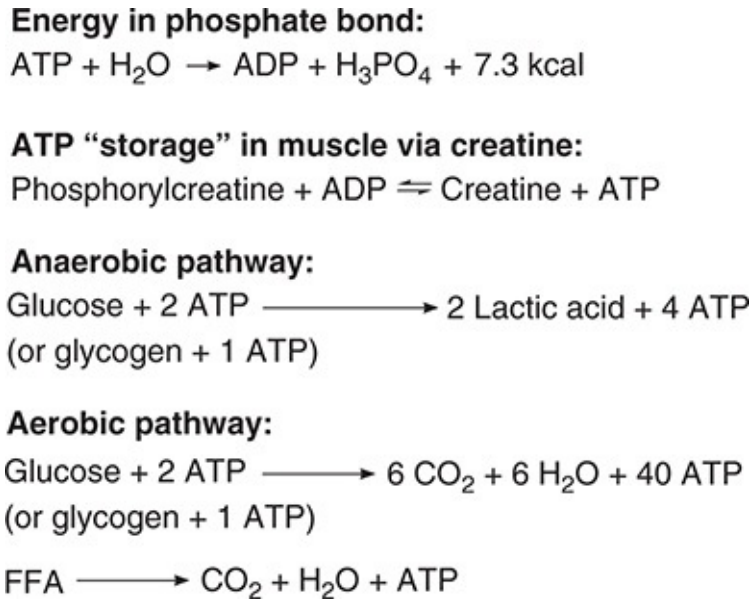


FIGURE 5–12 ATP turnover in muscle cells. Energy released by hydrolysis of

1 mol of ATP and reactions responsible for resynthesis of ATP. The amount of ATP formed per mole of free fatty acid (FFA) oxidized is large but varies with the size of the FFA. For example, complete oxidation of 1 mol of palmitic acid generates 140 mol of ATP.

THE OXYGEN DEBT MECHANISM

During exercise, the muscle blood vessels dilate and blood flow is increased so that the available O_2 supply is increased. Up to a point, the increase in O_2 consumption is proportional to the energy expended, and all the energy needs are met by aerobic processes. However, when muscular exertion is very great, aerobic resynthesis of energy stores cannot keep pace with their utilization. Under these conditions, phosphorylcreatine is still used to resynthesize ATP. In addition, some ATP synthesis is accomplished by using the energy released by the anaerobic breakdown of glucose to lactate. Use of the anaerobic pathway is self-limiting because in spite of rapid diffusion of lactate into the bloodstream, enough accumulates in the muscles to eventually exceed the capacity of the tissue buffers and produce an enzyme-inhibiting decline in pH. However, for short periods, the presence of an anaerobic pathway for glucose breakdown permits muscular exertion of a far greater magnitude than would be possible without it. For example, in a 100-m dash that takes 10 seconds, 85% of the energy consumed is derived anaerobically; in a 2-mile race that takes 10 minutes, 20% of the energy is derived anaerobically; and in a long-distance race that takes 60 minutes, only 5% of the energy comes from anaerobic metabolism.

After a period of exertion is over, extra O_2 is consumed to remove the excess lactate, replenish the ATP and phosphorylcreatine stores, and replace the small amounts of O_2 that were released by myoglobin. Without replenishment of ATP, muscles enter a state of rigor **(Clinical Box 5–3)**. The amount of extra O_2 consumed is proportional to the extent to which the energy demands during exertion exceeded the capacity for the aerobic synthesis of energy stores, that is, the extent to which an **oxygen debt** was incurred. The O_2 debt is measured experimentally by determining O_2 consumption after exercise until a constant, basal consumption is reached and subtracting the basal consumption from the total. The amount of this debt may be six times the basal O_2 consumption, which indicates that the subject is capable of six times the exertion that would have been possible without it.

HEAT PRODUCTION IN MUSCLE

Thermodynamically, the energy supplied to a muscle must equal its energy output. The energy output appears in work done by the muscle, in energy-rich phosphate bonds formed for later use, and in heat. The overall mechanical efficiency of skeletal muscle (work done/total energy expenditure) ranges up to 50% while lifting a weight during isotonic contraction and is essentially 0% during isometric contraction. Energy storage in phosphate bonds is a small factor. Consequently, heat production is considerable. The heat produced in muscle can be measured accurately with suitable thermocouples.

CLINICAL BOX 5–3

Muscle Rigor

When muscle fibers are completely depleted of ATP and phosphorylcreatine, they develop a state of rigidity called rigor. When this occurs after death, the condition is called **rigor mortis.** In rigor, almost all of the myosin heads attach to actin but in an abnormal, fixed, and resistant way. The muscles effectively are locked into place and become quite stiff to the touch.

Resting heat, the heat given off at rest, is the external manifestation of basal metabolic processes. The heat produced in excess of resting heat during contraction is called the **initial heat.** This is made up of **activation heat,** the heat that muscle produces whenever it is contracting, and **shortening heat,** which is proportional in amount to the distance the muscle shortens. Shortening heat is apparently due to some change in the structure of the muscle during shortening.

Following contraction, heat production in excess of resting heat continues for as long as 30 min. This **recovery heat** is the heat liberated by the metabolic processes that restore the muscle to its precontraction state. The recovery heat of muscle is approximately equal to the initial heat; that is, the heat produced during recovery is equal to the heat produced during contraction.

If a muscle that has contracted isotonically is restored to its previous length, extra heat in addition to recovery heat is produced **(relaxation heat).** External work must be done on the muscle to return it to its previous length, and relaxation heat is mainly a manifestation of this work.

PROPERTIES OF SKELETAL MUSCLES IN THE INTACT ORGANISM

THE MOTOR UNIT

Innervation of muscle fibers is critical to muscle function **(Clinical Box 5–4)**. Because the axons of the spinal motor neurons supplying skeletal muscle each branch to innervate several muscle fibers, the smallest possible amount of muscle that can contract in response to the excitation of a single motor neuron is not one muscle fiber but all the fibers supplied by the neuron. Each single motor neuron and the muscle fibers it innervates constitute a **motor unit.** The number of muscle fibers in a motor unit varies. In muscles such as those of the hand and those concerned with motion of the eye (ie, muscles concerned with fine, graded, precise movement), each motor unit innervates very few (on the order of three to six) muscle fibers. On the other hand, values of 600 muscle fibers per motor unit can occur in human leg muscles. The group of muscle fibers that contribute to a motor unit can be intermixed within a muscle. That is, although they contract as a unit, they are not necessarily "neighboring" fibers within the muscle.

Each spinal motor neuron innervates only one kind of muscle fiber, so that all the muscle fibers in a motor unit are of the same type. On the basis of the type of muscle fiber they innervate, and thus on the basis of the duration of their twitch contraction, motor units are divided into S (slow), FR (fast, resistant to fatigue), and FF (fast, fatigable) units. Interestingly, there is also a gradation of innervation of these fibers, with S fibers tending to have a low innervation ratio (ie, small units) and FF fibers tending to have a high innervation ratio (ie, large units). The recruitment of motor units during muscle contraction is not random; rather it follows a general scheme, the **size principle.** In general, a specific muscle action is developed first by the recruitment of S muscle units that contract relatively slowly to produce controlled contraction. Next, FR muscle units are recruited, resulting in more powerful response over a shorter period of time. Lastly, FF muscle units are recruited for the most demanding tasks. For example, in muscles of the leg, the small, slow units are first recruited for standing. As walking motion is initiated, their recruitment of FR units increases. As this motion turns to running or jumping, the FF units are recruited. Of course, there is overlap in recruitment, but, in general, this principle holds true.

CLINICAL BOX 5–4

Denervation of Muscle

In the intact animal healthy skeletal muscle does not contract except in response to stimulation of its motor nerve supply. Destruction of this nerve supply causes muscle atrophy. It also leads to abnormal excitability of the muscle and increases its sensitivity to circulating acetylcholine (**denervation hypersensitivity;** see Chapter 6). Fine, irregular contraction of individual fibers (**fibrillations**) appears. This is the classic picture of a **lower motor neuron lesion.** If the motor nerve regenerates, the fibrillations disappear. Usually, the contractions are not visible grossly, and they should not be confused with **fasciculations,** which are jerky, visible contractions of groups of muscle fibers that occur as a result of pathologic discharge of spinal motor neurons.

The differences between types of muscle units are not inherent but are determined by, among other things, their activity. When the nerve to a slow muscle is cut and the nerve to a fast muscle is spliced to the cut end, the fast nerve grows and innervates the previously slow muscle. However, the muscle becomes fast and corresponding changes take place in its muscle protein isoforms and myosin ATPase activity. This change is due to changes in the pattern of activity of the muscle; in stimulation experiments, changes in the expression of *MHC* genes and consequently of MHC isoforms can be produced by changes in the pattern of electrical activity used to stimulate the muscle. More commonly, muscle fibers can be altered by a change in activity initiated through exercise (or lack thereof). Increased activity can lead to muscle cell hypertrophy, which allows for increase in contractile strength. Types IIA and IIB fibers are most susceptible to these changes. Alternatively, inactivity can lead to muscle cell atrophy and a loss of contractile strength. Type I fibers—that is, the ones used most often—are most susceptible to these changes.

ELECTROMYOGRAPHY

Activation of motor units can be studied by electromyography, the process of recording the electrical activity of muscle. This may be done in unanaesthetized humans by using small metal disks on the skin overlying the muscle as the pick-up electrodes or by using needle or fine wire electrodes inserted into the muscle. The record obtained with such electrodes is the **electromyogram (EMG).** With

needle or fine wire electrodes, it is usually possible to pick up the activity of single muscle fibers. The measured EMG depicts the potential difference between the two electrodes, which is altered by the activation of muscles in between the electrodes. A typical EMG is shown in **Figure 5–13**.

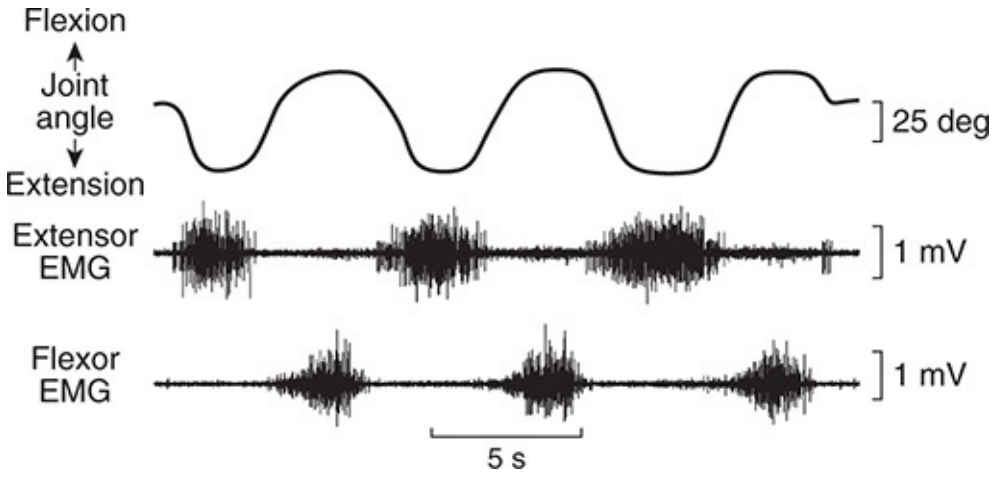


FIGURE 5–13 Relative joint angle and electromyographic tracings from human extensor pollicis longus and flexor pollicis longus during alternate flexion and extension of the distal joint of the thumb. The extensor pollicis longus and flexor pollicis longus extend and flex the distal joint of the thumb, respectively. The distal thumb joint angle (top) is superimposed over the extensor pollicis longus (middle) and flexor pollicus longus (bottom) EMGs. Note the alternate activation and rest patterns as one muscle is used for extension and the other for flexion. (Used with permission of Andrew J. Fuglevand.)

It has been shown by electromyography that little if any spontaneous activity occurs in the skeletal muscles of normal individuals at rest. With minimal voluntary activity a few motor units discharge, and with increasing voluntary effort, more and more are brought into play to monitor the **recruitment of motor units.** Gradation of muscle response is therefore in part a function of the number of motor units activated. In addition, the frequency of discharge in the individual nerve fibers plays a role, the tension developed during a tetanic contraction being greater than that during individual twitches. The length of the muscle is also a factor. Finally, the motor units fire asynchronously, that is, out of phase with one another. This asynchronous firing causes the individual muscle fiber responses to merge into a smooth contraction of the whole muscle. In summary, EMGs can be used to quickly (and roughly) monitor abnormal electrical activity associated with muscle responses.

THE STRENGTH OF SKELETAL MUSCLES

Human skeletal muscle can exert 3–4 kg of tension per square centimeter of cross-sectional area. This figure is about the same as that obtained in a variety of experimental animals and seems to be constant for mammalian species. Because many of the muscles in humans have a relatively large cross-sectional area, the tension they can develop is quite large. The gastrocnemius, for example, not only supports the weight of the whole body during climbing but resists a force several times this great when the foot hits the ground during running or jumping. An even more striking example is the gluteus maximus, which can exert a tension of 1200 kg. The total tension that could be developed if all muscles in the body of an adult man pulled together is approximately 22,000 kg (nearly 25 tons).

BODY MECHANICS

Body movements are generally organized in such a way that they take maximal advantage of the physiological principles outlined above. For example, the attachments of the muscles in the body are such that many of them are normally at or near their resting length when they start to contract. In muscles that extend over more than one joint, movement at one joint may compensate for movement at another in such a way that relatively little shortening of the muscle occurs during contraction. Nearly isometric contractions of this type permit development of maximal tension per contraction. The hamstring muscles extend from the pelvis over the hip joint and the knee joint to the tibia and fibula. Hamstring contraction produces flexion of the leg on the thigh. If the thigh is flexed on the pelvis at the same time, the lengthening of the hamstrings across the hip joint tends to compensate for the shortening across the knee joint. In the course of various activities, the body moves in a way that takes advantage of this. Such factors as momentum and balance are integrated into body movement in ways that make possible maximal motion with minimal muscular exertion. One net effect is that the stress put on tendons and bones rarely exceeds 50% of their failure strength, protecting them from damage.

In walking, each limb passes rhythmically through a support or stance phase when the foot is on the ground and a swing phase when the foot is off the ground. The support phases of the two legs overlap, so that two periods of double support occur during each cycle. There is a brief burst of activity in the leg flexors at the start of each step, and then the leg is swung forward with little more active muscular contraction. Therefore, the muscles are active for only a

fraction of each step, and walking for long periods causes relatively little fatigue.

A young adult walking at a comfortable pace moves at a velocity of about 80 m/min and generates a power output of 150–175 W per step. A group of young adults asked to walk at their most comfortable rate selected a velocity close to 80 m/min, and it was found that they had selected the velocity at which their energy output was minimal. Walking more rapidly or more slowly took more energy.

CARDIAC MUSCLE MORPHOLOGY

The striations in cardiac muscle are similar to those in skeletal muscle, and Z-lines are present. Large numbers of elongated mitochondria are in close contact with the muscle fibrils. The muscle fibers branch and interdigitate, but each is a complete unit surrounded by a cell membrane. Where the end of one muscle fiber abuts on another, the membranes of both fibers parallel each other through an extensive series of folds. These areas, which always occur at Z-lines, are called **intercalated disks** **(Figure 5–14)**. They provide a strong union between fibers, maintaining cell-to-cell cohesion, so that the pull of one contractile cell can be transmitted along its axis to the next. Along the sides of the muscle fibers next to the disks, the cell membranes of adjacent fibers fuse for considerable distances, forming gap junctions. These junctions provide low-resistance bridges for the spread of excitation from one fiber to another. They permit cardiac muscle to function as if it were a syncytium, even though no protoplasmic bridges are present between cells. The T system in cardiac muscle is located at the Z-lines rather than at the A–I junction, where it is located in mammalian skeletal muscle.

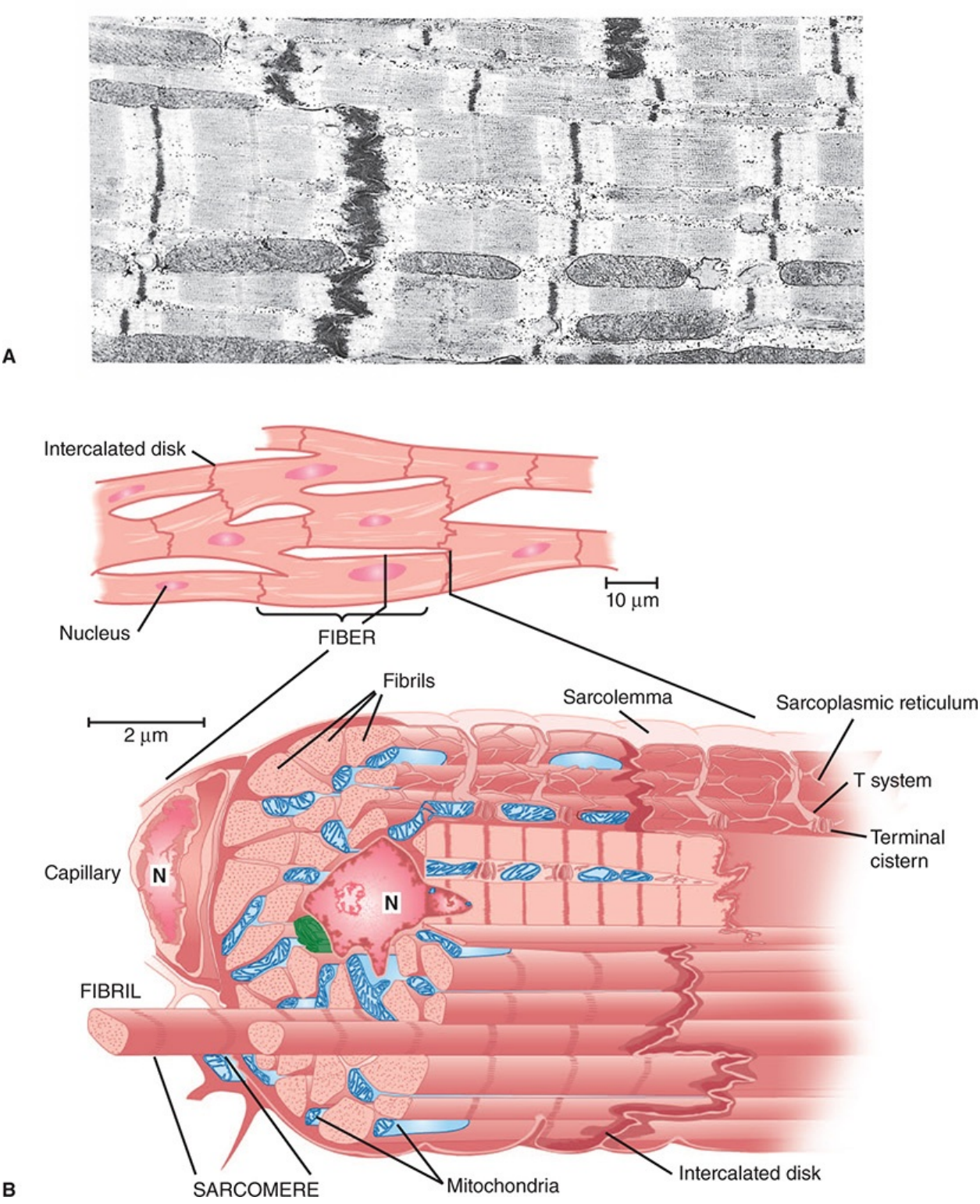


FIGURE 5–14 Cardiac muscle. A) Electron micrograph of cardiac muscle. Note the similarity of the A-I regions seen in the skeletal muscle EM of Figure 5–2. The fuzzy thick lines are intercalated disks and function similarly to the Z-

lines but occur at cell membranes (× 12,000). (Reproduced with permission from Bloom W, Fawcett DW: *A Textbook of Histology*, 10th ed. Saunders, 1975.) **B)** Artist interpretation of cardiac muscle as seen under the light microscope (top) and the electron microscope (bottom). Again, note the similarity to skeletal muscle structure. N, nucleus. (Reproduced with permission from Braunwald E, Ross J, Sonnenblick EH: Mechanisms of contraction of the normal and failing heart. N Engl J Med 1967 October 12;277(15):794–800.)

ELECTRICAL PROPERTIES

RESTING MEMBRANE & ACTION POTENTIALS

The resting membrane potential of individual mammalian cardiac muscle cells is about −90 mV. Stimulation produces a propagated action potential that is responsible for initiating contraction. Although action potentials vary among the cardiomyocytes in different regions of the heart (discussed in Chapter 29), the action potential of a typical ventricular cardiomyocyte can be used as an example **(Figure 5–15)**. Depolarization proceeds rapidly and an overshoot of the zero potential is present, as in skeletal muscle and nerve, but this is followed by a plateau before the membrane potential returns to the baseline. In mammalian hearts, depolarization lasts about 2 ms, but the plateau phase and repolarization last 200 ms or more. Repolarization is therefore not complete until the contraction is half over.

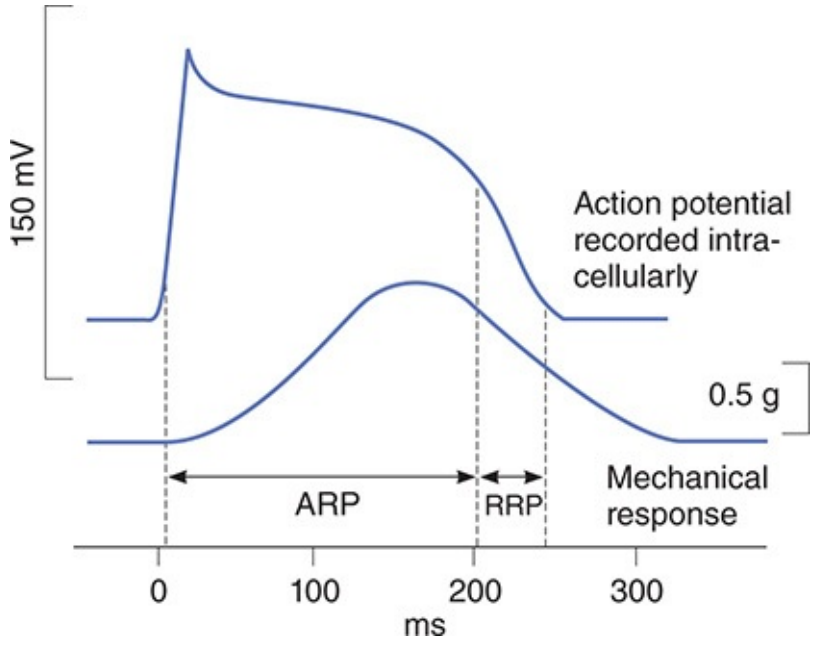


FIGURE 5–15 Comparison of action potentials and contractile response of a mammalian cardiac muscle fiber in a typical ventricular cell. In the top trace, the intracellular recording of the action potential shows the quick depolarization and extended recovery. In the bottom trace, the mechanical response is matched to the extracellular and intracellular electrical activities. Note that in the absolute refractory period (ARP), the cardiac myocyte cannot be excited, whereas in the relative refractory period (RRP) minimal excitation can occur.

As in other excitable tissues, changes in the external K^+ concentration affect the resting membrane potential of cardiac muscle, whereas changes in the external Na^+ concentration affect the magnitude of the action potential. The initial rapid depolarization and the overshoot (phase 0) are due to opening of voltage-gated Na^+ channels similar to that occurring in nerve and skeletal muscle **(Figure 5–16)**. The initial rapid repolarization (phase 1) is due to closure of Na^+ channels and opening of one type of K^+ channel (ie, Kv4.2/KCND2 and/or Kv4.3/CCND3, and KV1.4/KCNA4). The subsequent prolonged plateau (phase 2) is due to a slower but prolonged opening of voltage-gated Ca^{2+} channels. Final repolarization (phase 3) to the resting membrane potential (phase 4) is due to closure of the Ca^{2+} channels and a slow, delayed increase of K^+ efflux through various types of K^+ channels. Cardiac myocytes contain at least two types of voltage-gated Ca^{2+} channels (T- and L-types), but the Ca^{2+} current is mostly due to opening of the slower L-type Ca^{2+} channels. Mutations or dysfunctions in any of these channels lead to serious pathologies of the heart **(eg, Clinical Box 5–5**).

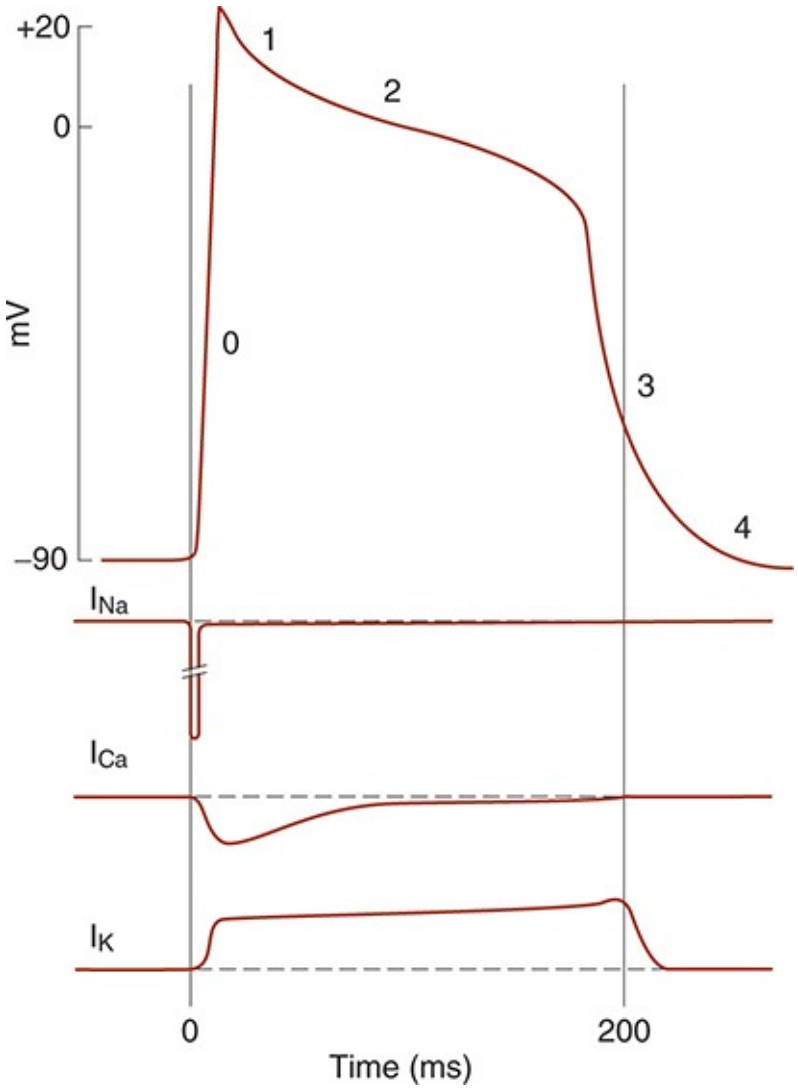


FIGURE 5–16 Dissection of the cardiac action potential. Top: The action potential of a cardiac muscle fiber can be broken down into several phases: 0, depolarization; 1, initial rapid repolarization; 2, plateau phase; 3, late rapid repolarization; 4, baseline. **Bottom:** Diagrammatic summary of Na^+, Ca^{2+}, and cumulative K+ currents during the action potential. As is convention, inward currents are downward, and outward currents are upward.

MECHANICAL PROPERTIES

CONTRACTILE RESPONSE

The contractile response of cardiac muscle begins just after the start of depolarization and lasts about 1.5 times as long as the action potential (Figure 5–

15). The role of Ca^{2+} in excitation–contraction coupling is similar to its role in skeletal muscle (see above). However, it is the influx of extracellular Ca^{2+} through the voltage-sensitive DHPR or voltage-gated Ca^{2+} channels in the T system that triggers Ca^{2+}-induced Ca^{2+} release through the RyR at the sarcoplasmic reticulum. Because there is a net influx of Ca^{2+} during activation, there is also a more prominent role for plasma membrane Ca^{2+} ATPases and the Na^{+}/Ca^{2+} exchanger in recovery of cytosolic Ca^{2+} concentrations. Specific effects of drugs that indirectly alter intracellular Ca^{2+} concentrations are discussed in **Clinical Box 5–6**.

CLINICAL BOX 5–5

Long QT Syndrome

Long QT syndrome (LQTS) is defined as a prolongation of the QT interval observed on an electrocardiogram. LQTS can lead to irregular heartbeats and subsequent fainting, seizure, cardiac arrest, or even death. Although certain medications can lead to LQTS, it is more frequently associated with genetic mutations in a variety of cardiac-expressed ion channels. Mutations in cardiac-expressed voltage-gated K^{+} channel genes (*KCNQ1* or *KCNH2*) account for most of the mutation-based cases of LQTS (~90%). Mutations in cardiac-expressed voltage-gated Na^{+} channel genes (eg, *SCN5A*) or cardiac-expressed Ca^{2+} channel genes (eg, *CACNA1C*) have also been associated with the disease. The fact that mutations in diverse channels all can result in the prolongation of the QT interval and subsequent pathophysiological changes underlies the intricate interplay of these channels in shaping the heart's electrical response.

THERAPEUTIC HIGHLIGHTS

Patients with LQTS should avoid drugs that prolong the QT interval or reduce their serum K^{+} or Mg^{2+} levels; any K^{+} or Mg^{2+} deficiencies should be corrected. Drug interventions in asymptomatic patients remain somewhat controversial, although patients with congenital defects that lead to LQTS are considered candidates for intervention independent of symptoms. In general, β-blockers have been used for LQTS to reduce the risk of cardiac arrhythmias. More specific and effective treatments can be introduced once the underlying cause of LQTS is identified.

During phases 0 to 2 and about half of phase 3 (until the membrane potential reaches approximately −50 mV during repolarization), cardiac muscle cannot be excited again; that is, it is in its **absolute refractory period.** It remains relatively refractory until phase 4. Therefore, tetanus of the type seen in skeletal muscle cannot occur. Of course, tetanization of cardiac muscle for any length of time would have lethal consequences, and in this sense, the fact that cardiac muscle cannot be tetanized is a safety feature.

CLINICAL BOX 5–6

Glycosidic Drugs & Cardiac Contraction

Ouabain and other digitalis glycosides are commonly used to treat failing hearts. These drugs have the effect of increasing the strength of cardiac contraction. Although there is discussion as to full mechanisms, a working hypothesis is based on the ability of these drugs to inhibit the Na, K ATPase in cell membranes of the cardiomyocytes. The block of the Na, K ATPase in cardiomyocytes would result in an increased intracellular Na^+ concentration. Such an increase would result in a decreased inward Na^+ transportation and hence outward Ca^{2+} transportation via the Na^+-Ca^{2+} exchange antiport during the Ca^{2+} recovery period. The resulting increase in intracellular Ca^{2+} concentration in turn increases the strength of contraction of the cardiac muscle. With this mechanism in mind, these drugs can also be quite toxic. Over inhibition of the Na, K ATPase would result in a depolarized cell that could slow conduction, or even spontaneously activate contraction. Alternatively, an overly increased Ca^{2+} concentration could also have ill effects on cardiomyocyte function.

ISOFORMS

Cardiac muscle is generally slow and has relatively low ATPase activity. Its fibers are dependent on oxidative metabolism and hence on a continuous supply of O_2. The human heart contains both the α and the β isoforms of the myosin heavy chain (α MHC and β MHC). β MHC has lower myosin ATPase activity

than α MHC. Both are present in the atria, with the α isoform predominating, whereas the β isoform predominates in the ventricle. The spatial differences in expression contribute to the well-coordinated contraction of the heart.

CORRELATION BETWEEN MUSCLE FIBER LENGTH & TENSION

The relation between initial fiber length and total tension in cardiac muscle is similar to that in skeletal muscle; there is an optimal resting length at which the tension developed on stimulation is maximal. In the body, the initial length of the fibers is determined by the degree of diastolic filling of the heart, and the pressure developed in the ventricle is proportional to the volume of the ventricle at the end of the filling phase **(Starling law of the heart).** The developed tension **(Figure 5–17)** increases as the diastolic volume increases until it reaches a maximum, then tends to decrease. However, unlike skeletal muscle, the decrease in developed tension at high degrees of stretch is not due to a decrease in the number of cross-bridges between actin and myosin, because even severely dilated hearts are not stretched to this degree. The decrease is instead due to beginning disruption of the myocardial fibers.

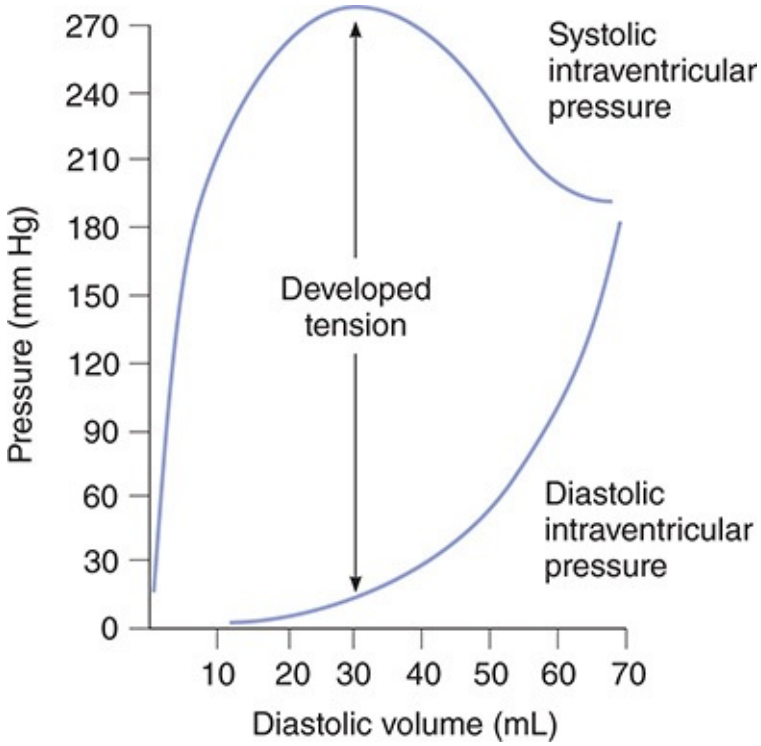


FIGURE 5–17 Length–tension relationship for cardiac muscle. Comparison of the systolic intraventricular pressure (top trace) and diastolic intraventricular pressure (bottom trace) display the developed tension in the cardiomyocyte. Values shown are for canine heart.

The force of contraction of cardiac muscle can be also increased by catecholamines, and this increase occurs without a change in muscle length. This positive ionotropic effect of catecholamines is mediated via innervated β_1-adrenergic receptors, cyclic adenosine monophosphate (AMP), and their effects on Ca^{2+} homeostasis. The heart also contains noninnervated β_2-adrenergic receptors, which also act via cyclic AMP, but their ionotropic effect is smaller and is maximal in the atria. Cyclic AMP activates protein kinase A, and this leads to phosphorylation of the voltage-gated Ca^{2+} channels, causing them to spend more time in the open state. Cyclic AMP also increases the active transport of Ca^{2+} to the sarcoplasmic reticulum, thus accelerating relaxation and consequently shortening systole. This is important when the cardiac rate is increased because it permits adequate diastolic filling (see Chapter 30).

METABOLISM

Mammalian hearts have an abundant blood supply, numerous mitochondria, and a high content of myoglobin, a muscle pigment that can function as an O_2 storage mechanism. Normally, less than 1% of the total energy liberated is provided by anaerobic metabolism. During hypoxia, this figure may increase to nearly 10%, but under totally anaerobic conditions, the energy liberated is inadequate to sustain ventricular contractions. Under basal conditions, 35% of the caloric needs of the human heart are provided by carbohydrate, 5% by ketones and amino acids, and 60% by fat. However, the proportions of substrates utilized vary greatly with the nutritional state. After ingestion of large amounts of glucose, more lactate and pyruvate are used; during prolonged starvation, more fat is used. Circulating free fatty acids normally account for almost 50% of the lipid utilized. In untreated diabetics, the carbohydrate utilization of cardiac muscle is reduced and that of fat is increased.

SMOOTH MUSCLE MORPHOLOGY

Smooth muscle is distinguished anatomically from skeletal and cardiac muscle

because it lacks visible cross-striations. Actin and myosin-II are present, and they slide on each other to produce contraction. However, they are not arranged in regular arrays, as in skeletal and cardiac muscle, and so the striations are absent. Instead of Z-lines, there are **dense bodies** in the cytoplasm and attached to the cell membrane, and these are bound by α-actinin to actin filaments. Smooth muscle also contains tropomyosin, but troponin appears to be absent. The isoforms of actin and myosin differ from those in skeletal muscle. A sarcoplasmic reticulum is present, but it is less extensive than those observed in skeletal or cardiac muscle. In general, smooth muscles contain few mitochondria and depend, to a large extent, on glycolysis for their metabolic needs.

TYPES

There is considerable variation in the structure and function of smooth muscle in different parts of the body. In general, smooth muscle can be divided into **unitary** (or **visceral**) **smooth muscle** and **multiunit smooth muscle.** Unitary smooth muscle occurs in large sheets, has many low-resistance gap junctional connections between individual muscle cells, and functions in a syncytial fashion. Unitary smooth muscle is found primarily in the walls of hollow viscera. The musculature of the intestine, the uterus, and the ureters are examples. Multiunit smooth muscle is made up of individual units with few (or no) gap junctional bridges. It is found in structures such as the iris of the eye, in which fine, graded contractions occur. It is not under voluntary control, but it has many functional similarities to skeletal muscle. Each multiunit smooth muscle cell has en passant endings of nerve fibers, but in unitary smooth muscle there are en passant junctions on fewer cells, with excitation spreading to other cells by gap junctions. In addition, these cells respond to hormones and other circulating substances. Blood vessels have both unitary and multiunit smooth muscle in their walls.

ELECTRICAL & MECHANICAL ACTIVITY

Unitary smooth muscle is characterized by the instability of its membrane potential and by the fact that it shows continuous, irregular contractions that are independent of its nerve supply. This maintained state of partial contraction is called **tonus,** or **tone.** The membrane potential has no true “resting” value, being relatively low (or less negative) when the tissue is active and higher (or more negative) when it is inhibited, but in periods of relative quiescence values for

resting potential are on the order of −20 to −65 mV. Smooth muscle cells can display divergent electrical activity **(eg, Figure 5–18).** There are slow sine wave-like fluctuations a few millivolts in magnitude and spikes that sometimes overshoot the zero potential line and sometimes do not. In many tissues, the spikes have a duration of about 50 ms, whereas in some tissues the action potentials have a prolonged plateau during repolarization, like the action potentials in cardiac muscle. As in the other muscle types, there are significant contributions of K^+, Na^+, and Ca^{2+} channels and Na, K ATPase to this electrical activity. However, discussion of contributions to individual smooth muscle types is beyond the scope of this text.

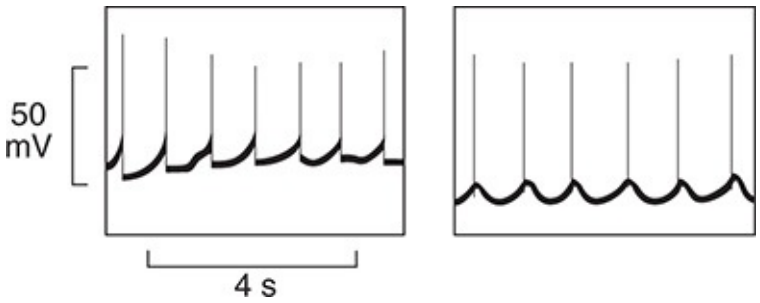


FIGURE 5–18 Electrical activity of individual smooth muscle cells in the guinea pig taenia coli. Left: Pacemaker-like activity with spikes firing at each peak. **Right:** Sinusoidal fluctuation of membrane potential with firing on the rising phase of each wave. In other fibers, spikes can occur on the falling phase of sinusoidal fluctuations and there can be mixtures of sinusoidal and pacemaker potentials in the same fiber.

Because of the continuous activity, it is difficult to study the relation between the electrical and mechanical events in unitary smooth muscle, but in some relatively inactive preparations, a single spike can be generated. In such preparations, the excitation–contraction coupling in unitary smooth muscle can occur with as much as a 500-ms delay. Thus, it is a very slow process compared with that in skeletal and cardiac muscle, in which the time from initial depolarization to initiation of contraction is less than 10 ms. Unlike unitary smooth muscle, multiunit smooth muscle is nonsyncytial and contractions do not spread widely through it. Because of this, the contractions of multiunit smooth muscle are more discrete, fine, and localized than those of unitary smooth muscle.

MOLECULAR BASIS OF CONTRACTION

As in skeletal and cardiac muscle, an increase in cytosolic Ca^{2+} plays a prominent role in the initiation of contraction of smooth muscle. However, the source of Ca^{2+} increase can be quite different in unitary smooth muscle. Depending on the activating stimulus, cytosolic Ca^{2+} increase can be due to Ca^{2+} influx through voltage- or ligand-gated channels in the plasma membrane, Ca^{2+} release or mobilization from intracellular stores (eg, sarcoplasmic reticulum) through the RyR and the **inositol trisphosphate receptor (IP3R)**, Ca^{2+} release channels in the sarcoplasmic reticulum. In addition, the lack of troponin in smooth muscle prevents Ca^{2+} activation via troponin binding. Rather, myosin in smooth muscle must be phosphorylated for activation of the myosin ATPase. Phosphorylation and dephosphorylation of myosin also occur in skeletal muscle, but phosphorylation is not necessary for activation of the ATPase. In smooth muscle, Ca^{2+} binds to calmodulin, and the resulting complex activates **calmodulin-dependent myosin light chain kinase.** This enzyme catalyzes the phosphorylation of the myosin light chain on serine at position 19, increasing its ATPase activity.

Myosin is dephosphorylated by **myosin light chain phosphatase** in the cell. However, dephosphorylation of myosin light chain kinase does not necessarily lead to relaxation of the smooth muscle. Various mechanisms are involved. One appears to be a latch bridge mechanism by which myosin cross-bridges remain attached to actin for some time after the cytoplasmic Ca^{2+} concentration falls. This produces sustained contraction with little expenditure of energy, which is especially important in vascular smooth muscle. Relaxation of the muscle presumably occurs when the Ca^{2+}–calmodulin complex finally dissociates or when some other mechanism comes into play. The events leading to contraction and relaxation of unitary smooth muscle are summarized in **Figure 5–19**. The events in multiunit smooth muscle are generally similar.

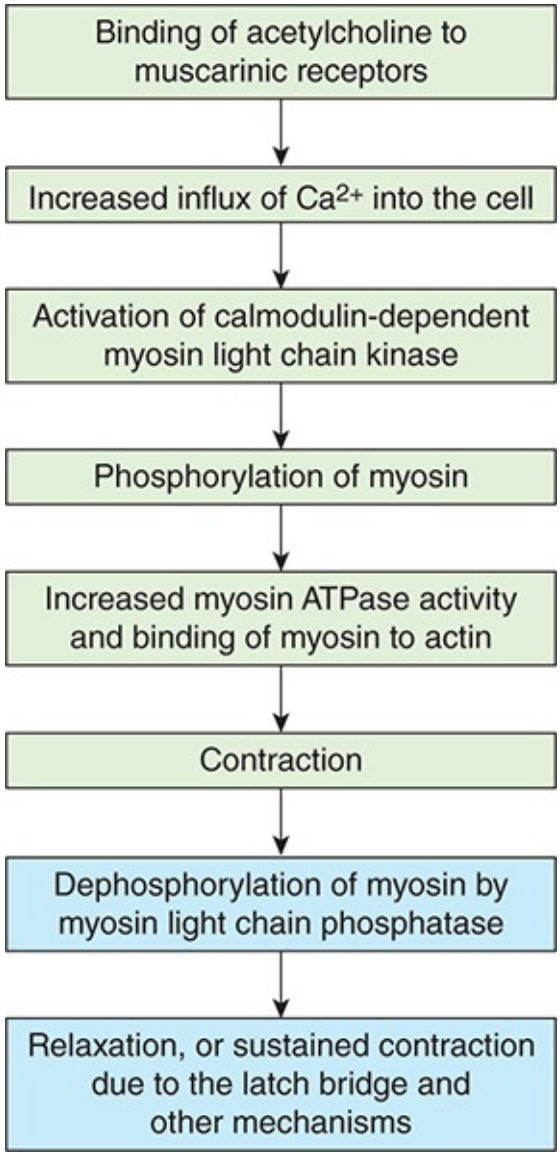


FIGURE 5–19 Sequence of events in contraction and relaxation of smooth muscle. Flow chart illustrates many of the molecular changes that occur from the initiation of contraction to its relaxation. Note the distinct differences from skeletal and cardiac muscle excitation.

Unitary smooth muscle is unique in that, unlike other types of muscle, it contracts when stretched in the absence of any extrinsic innervation. Stretch is followed by a decline (depolarization) in membrane potential, an increase in the frequency of spikes, and a general increase in tone.

If epinephrine or norepinephrine is added to a preparation of intestinal smooth muscle arranged for recording of intracellular potentials in vitro, the membrane potential usually becomes larger (more negative or hyperpolarization), the spikes

decrease in frequency, and the muscle relaxes **(Figure 5–20)**. Norepinephrine is the chemical mediator released at noradrenergic nerve endings, and stimulation of the noradrenergic nerves to the preparation produces inhibitory potentials. Acetylcholine has an effect opposite to that of norepinephrine on the membrane potential and contractile activity of intestinal smooth muscle. If acetylcholine is added to the fluid bathing a smooth muscle preparation in vitro, the membrane potential decreases (less negative or depolarization) and the spikes become more frequent. The muscle becomes more active, with an increase in tonic tension and the number of rhythmic contractions. The effect is mediated by phospholipase C, which produces IP_3 and allows for Ca^{2+} release through IP_3 receptors. In the intact animal, stimulation of cholinergic nerves causes release of acetylcholine, excitatory potentials, and increased intestinal contractions.

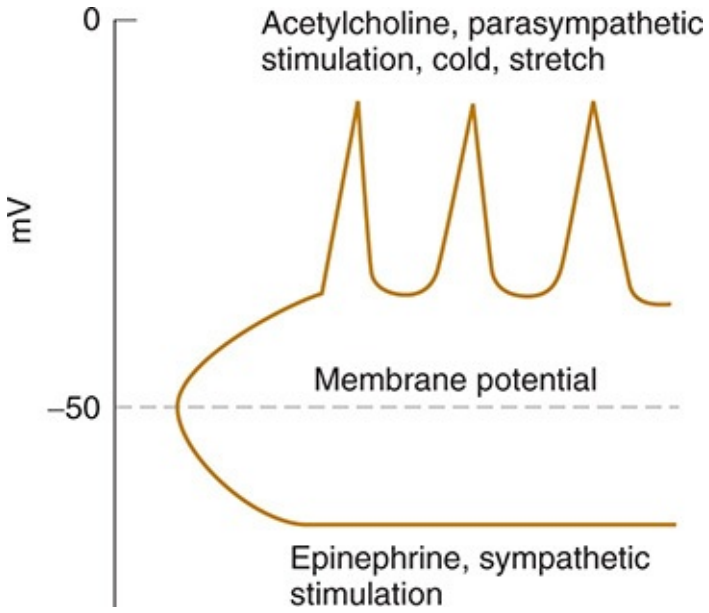


FIGURE 5–20 Effects of various agents on the membrane potential of intestinal smooth muscle. Drugs and hormones can alter firing of smooth muscle action potentials by raising (top trace) or lowering (bottom trace) resting membrane potential.

Like unitary smooth muscle, multiunit smooth muscle is very sensitive to circulating chemical substances and is normally activated by chemical mediators (acetylcholine and norepinephrine) released at the endings of its motor nerves. Norepinephrine in particular tends to persist in the muscle and to cause repeated firing of the muscle after a single stimulus rather than a single action potential. Therefore, the contractile response produced is usually an irregular tetanus rather

than a single twitch. When a single twitch response is obtained, it resembles the twitch contraction of skeletal muscle except that its duration is 10 times long.

RELAXATION

In addition to cellular mechanisms that increase contraction of smooth muscle, there are cellular mechanisms that lead to its relaxation **(Clinical Box 5–7)**. This is especially important in smooth muscle that forms the blood vessels or arteries to increase blood flow. The arterial wall consists of three layers: the intima containing mainly the endothelium or endothelial cells, the media containing mainly smooth muscle cells, and the externa or adventitia containing mainly connective tissues, extracellular matrix and fibroblasts. It was long known that endothelial cells that line the inside (the endothelium) of blood vessels could release a substance that relaxed smooth muscle **(endothelium-derived relaxing factor, EDRF).** EDRF was later identified as the gaseous second messenger molecule, **nitric oxide (NO).** NO produced in endothelial cells is free to diffuse into the smooth muscle for its effects. Once in muscle, NO directly activates a soluble guanylyl cyclase to produce another second messenger molecule, **cyclic guanosine monophosphate (cGMP).** This molecule can activate cGMP-specific protein kinases that can affect ion channels, Ca^{2+} homeostasis, or phosphatases, or all of those mentioned, leading to smooth muscle relaxation (see Chapter 32).

CLINICAL BOX 5–7

Common Drugs That Act on Smooth Muscle

Overexcitation of smooth muscle in the airways, such as that observed during an asthma attack, can lead to bronchoconstriction or bronchospasm. Inhalers that deliver drugs to the conducting airway are commonly used to offset this bronchial smooth muscle contraction, as well as other symptoms in the asthmatic airways. The rapid effects of drugs in inhalers are related to smooth muscle relaxation. Rapid response inhaler drugs (eg, ventolin, albuterol, sambuterol) target β-adrenergic receptors in the airway smooth muscle to elicit a relaxation. Although these β-adrenergic receptor agonists targeting the smooth muscle do not treat all symptoms associated with asthma (eg, inflammation and increased mucus), they act rapidly and frequently allow for sufficient opening of the conducting airway to restore airflow, and thus allow for other treatments to reduce airway obstruction.

Smooth muscle is also a target for drugs developed to increase blood flow and decrease blood pressure. As discussed in the text, NO is a natural signaling molecule that relaxes smooth muscle by raising cyclic guanosine monophosphate (cGMP). This signaling pathway is naturally downregulated by the action of **phosphodiesterase (PDE),** which transforms cGMP into a nonsignaling form, GMP. The drugs sildenafil, tadalafil, and vardenafil are all specific inhibitors of PDE V, an isoform found mainly in the smooth muscle in the corpus cavernosum of the penis (see Chapters 25 and 32) and the pulmonary vasculature. Thus, oral administration of these drugs can block the action of PDE V, increasing blood flow in a very limited region in the body and offsetting erectile dysfunction. These drugs are also used to treat pulmonary arterial hypertension.

FUNCTION OF THE NERVE SUPPLY TO SMOOTH MUSCLE

The effects of acetylcholine and norepinephrine on unitary smooth muscle serve to emphasize two of its important properties: (1) its spontaneous activity in the absence of nervous stimulation and (2) its sensitivity to chemical agents released from nerves locally or brought to it in the circulation. In mammals, unitary muscle usually has a dual nerve supply from the two divisions of the autonomic nervous system. The function of the nerve supply is not to initiate activity in the muscle but rather to modify it. Stimulation of one division of the autonomic nervous system usually increases smooth muscle activity, whereas stimulation of the other decreases it. In some organs, noradrenergic stimulation increases and cholinergic stimulation decreases smooth muscle activity; in others, the reverse is true.

FORCE GENERATION & PLASTICITY OF SMOOTH MUSCLE

Smooth muscle displays a unique economy when compared to skeletal muscle. Despite approximately 20% of the myosin content and a 100-fold difference in ATP use when compared with skeletal muscle, they can generate similar force per cross-sectional area. One of the tradeoffs of obtaining force under these conditions is the noticeably slower contractions when compared to skeletal

muscle. There are several known reasons for these noticeable changes, including unique isoforms of myosin and contractile-related proteins expressed in smooth muscle and their distinct regulation. The unique architecture of the smooth cell and its coordinated units also likely contribute to these changes.

Another special characteristic of smooth muscle is the variability of the tension it exerts at any given length. If a unitary smooth muscle is stretched, it first exerts increased tension. However, if the muscle is held at the greater length after stretching, the tension gradually decreases. Sometimes the tension falls to or below the level exerted before the muscle was stretched. It is consequently impossible to correlate length and developed tension accurately, and no resting length can be assigned. In some ways, therefore, smooth muscle behaves more like a viscous mass than a rigidly structured tissue, and it is this property that is referred to as the **plasticity** of smooth muscle.

The consequences of plasticity can be demonstrated in humans. For example, the tension exerted by the smooth muscle of the bladder wall can be measured at different degrees of distension as fluid is infused into the bladder via a catheter. Initially, tension increases relatively little as volume is increased because of the plasticity of the bladder wall. However, a point is eventually reached at which the bladder contracts forcefully (see Chapter 37).

CHAPTER SUMMARY

- There are three main types of muscle cells: skeletal, cardiac, and smooth.
- Skeletal muscle is a true syncytium under voluntary control. Skeletal muscles receive electrical stimuli from nerve cells to elicit contraction: "excitation–contraction coupling." Action potentials in muscle cells are developed largely through coordination of Na^+ and K^+ channels. Contraction in skeletal muscle cells is coordinated through Ca^{2+} regulation of the actomyosin system that gives the muscle its classic striated pattern under the microscope.
- There are several different types of skeletal muscle fibers (I, IIA, IIB) that have distinct properties in terms of protein makeup and force generation. Skeletal muscle fibers are arranged into motor units of like fibers within a muscle. Skeletal motor units are recruited in a specific pattern as the need for more force is increased.
- Cardiac muscle is a collection of individual cells (cardiomyocytes) that are linked as a syncytium by gap junctional communication. Cardiac muscle

cells also undergo excitation–contraction coupling. Pacemaker cells in the heart can initiate propagated action potentials. Cardiac muscle cells also have a striated, actomyosin system that underlies contraction.

- Smooth muscle exists as individual cells and are frequently under control of the autonomic nervous system.
- There are two broad categories of smooth muscle cells: unitary and multiunit. Unitary smooth muscle contraction is synchronized by gap junctional communication to coordinate contraction among many cells. Multiunit smooth muscle contraction is coordinated by motor units, functionally similar to skeletal muscle.
- Smooth muscle cells contract through an actomyosin system, but do not have well-organized striations. Unlike skeletal and cardiac muscle, Ca^{2+} regulation of contraction is primarily through phosphorylation–dephosphorylation reactions.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. The action potential of skeletal muscle
   A. has a prolonged plateau phase.
   B. spreads inward to all parts of the muscle via the T tubules.
   C. causes the immediate uptake of Ca^{2+} into the lateral sacs of the sarcoplasmic reticulum.
   D. is longer than the action potential of cardiac muscle.
   E. is not essential for contraction.
2. The functions of tropomyosin in skeletal muscle include
   A. sliding on actin to produce shortening.
   B. releasing Ca^{2+} after initiation of contraction.
   C. binding to myosin during contraction.
   D. acting as a “relaxing protein” at rest by covering up the sites where myosin binds to actin.
   E. generating ATP, which it passes to the contractile mechanism.
3. The cross-bridges of the sarcomere in skeletal muscle are made up of
   A. actin.

B. myosin.
C. troponin.
D. tropomyosin.
E. myelin.

4. The contractile response in skeletal muscle
A. starts after the action potential is over.
B. does not last as long as the action potential.
C. produces more tension when the muscle contracts isometrically than when the muscle contracts isotonically.
D. produces more work when the muscle contracts isometrically than when the muscle contracts isotonically.
E. decreases in magnitude with repeated stimulation.

5. Gap junctions
A. are absent in cardiac muscle.
B. are present but of little functional importance in cardiac muscle.
C. are present and provide the pathway for rapid spread of excitation from one cardiac muscle fiber to another.
D. are absent in smooth muscle.
E. connect the sarcotubular system to individual skeletal muscle cells.

CHAPTER 6

Synaptic & Junctional Transmission

OBJECTIVES

After studying this chapter, you should be able to:

- Describe the major location and functional components of a neuron-to-neuron synapse.
- Contrast the ionic fluxes responsible for the fast and slow excitatory and inhibitory postsynaptic potentials.
- Compare and contrast the terms temporal summation and spatial summation and their role in action potential generation in a postsynaptic neuron.
- Explain postsynaptic inhibition, presynaptic inhibition, and presynaptic facilitation.
- Identify the components of the neuromuscular junction and the sequence of events that leads to a propagated action potential in the skeletal muscle fiber.
- Explain how autonomic neurons communicate with their effector organs at a neuroeffector junction.
- Define denervation hypersensitivity.
- Describe some pathologies associated with dysfunction at the neuromuscular junction.

INTRODUCTION

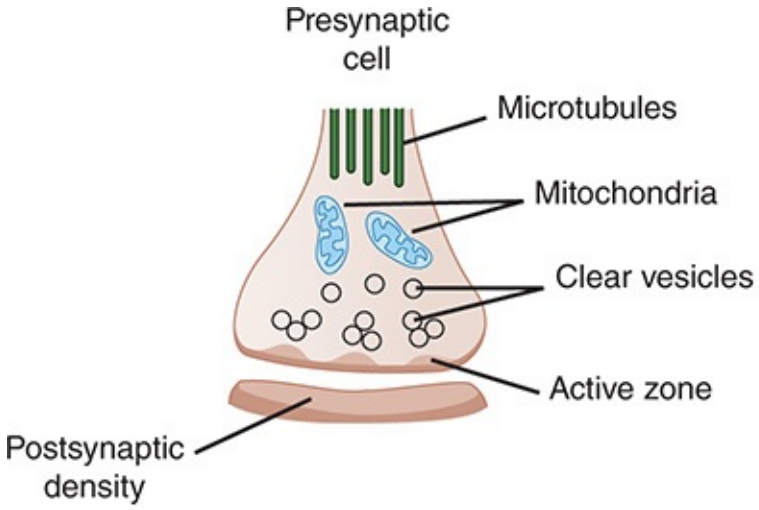
The “all-or-none” type of conduction seen in axons and skeletal muscle has been discussed in Chapters 4 and 5. Impulses are transmitted from one neuron to another at a **synapse.** This is the region where the axon or some other portion of one neuron (**presynaptic neuron**) terminates on the dendrites, soma, or axon of another neuron (**postsynaptic neuron**). Neuron-to-neuron communication occurs across either a **chemical** or an **electrical synapse.** Because most synaptic transmission is chemical, consideration in this chapter is primarily related to chemical neurotransmission.

When a neuron terminates on a muscle, the connection is properly called a **neuromuscular junction** rather than a synapse. Transmission from a nerve to a muscle resembles chemical synaptic transmission from one neuron to another. The contacts between autonomic neurons and smooth and cardiac muscle or glands are less specialized than those between a neuron and skeletal muscle, and transmission in these locations is a more diffuse process. The region where the neuron communicates with the effector organ is called the **neuroeffector junction**. These forms of transmission are also considered in this chapter.

SYNAPTIC TRANSMISSION: FUNCTIONAL ANATOMY

The anatomic structure of synapses varies considerably in the different parts of the mammalian nervous system. The ends of the presynaptic fibers are generally enlarged to form **terminal boutons** or **synaptic knobs** **(Figure 6–1)**. In the cerebral and cerebellar cortex, endings are commonly located on dendrites **(axodendritic synapse)** and frequently on **dendritic spines,** which are small knobs projecting from dendrites **(Figure 6–2)**. Some presynaptic nerves terminate on the soma **(axosomatic synapse)** or axons **(axoaxonal synapses)** of postsynaptic neurons. On average, each neuron divides to form over 2000 synaptic endings; thus, communication between neurons is very complex. Synapses are dynamic structures, increasing and decreasing in complexity and number with use and experience.

FIGURE 6–1 Electron micrograph of synaptic knob (S) ending on the shaft of a dendrite (D) in the central nervous system. P, postsynaptic density; M, mitochondrion (×56,000).


Presynaptic cell
Microtubules
Mitochondria
Clear vesicles
Active zone
Postsynaptic density

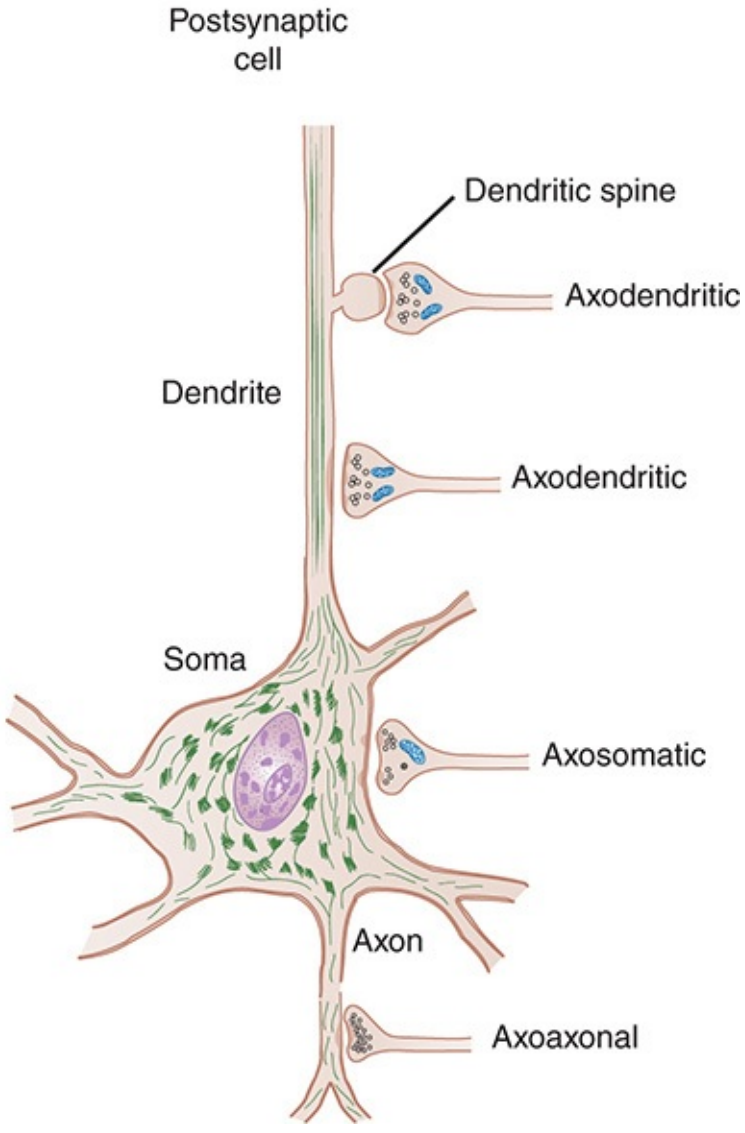

Postsynaptic cell
Dendritic spine
Axodendritic
Dendrite
Axodendritic
Soma
Axosomatic
Axon
Axoaxonal

FIGURE 6–2 Axodendritic, axoaxonal, and axosomatic synapses. Many presynaptic neurons terminate on dendritic spines, as shown at the top, but some also end directly on the shafts of dendrites. Note the presence of clear and granulated synaptic vesicles in endings and clustering of clear vesicles at active zones.

FUNCTIONS OF SYNAPTIC ELEMENTS

The dendrites function as a major site that receives electrical signals from other neurons, processes these signals, and transfers the information to the soma of the neuron. When the dendritic tree of a neuron is extensive and has multiple presynaptic knobs ending on it, there is room for a great interplay of inhibitory and excitatory activity. Dendritic spines can appear, change, and even disappear over a time scale of minutes and hours, not days and months. Also, although protein synthesis occurs mainly in the soma with its nucleus, strands of mRNA migrate into the dendrites. There, each can become associated with a single ribosome in a dendritic spine and produce proteins to alter the effects of input from individual synapses on the spine. These changes in dendritic spines have been implicated in motivation, learning, and long-term memory (see Chapter 15).

Each presynaptic terminal of a chemical synapse is separated from the postsynaptic structure by a **synaptic cleft** that is 20–40 nm wide. Across the synaptic cleft are many neurotransmitter receptors in the postsynaptic membrane, and usually a postsynaptic thickening called the **postsynaptic density** (Figure 6–2). The postsynaptic density is an ordered complex of specific receptors, binding proteins, and enzymes induced by postsynaptic effects.

Inside the presynaptic terminal are many mitochondria, as well as many membrane-enclosed vesicles, which contain neurotransmitters (see Chapter 7). There are three kinds of **synaptic vesicles:** small, clear synaptic vesicles contain acetylcholine, glycine, GABA, or glutamate; small dense-core vesicles contain catecholamines; and large dense-core vesicles contain neuropeptides. The vesicles and the proteins contained within them are synthesized in the neuronal cell body and transported along the axon to the endings by **fast axoplasmic transport**. The neuropeptides in the large dense-core vesicles must also be produced by the protein-synthesizing machinery in the cell body. However, the small clear vesicles and the small dense-core vesicles recycle in the nerve ending **(Figure 6–3)**. These vesicles fuse with the cell membrane and release transmitters through **exocytosis** and are then recovered by endocytosis to be

refilled locally. In some instances, they enter endosomes and are budded off the endosome and refilled, starting the cycle over again. More commonly, however, the synaptic vesicle discharges its contents through a small hole in the cell membrane, then the opening reseals rapidly and the main vesicle stays inside the cell **(kiss-and-run discharge).** In this way, the full endocytotic process is short-circuited.

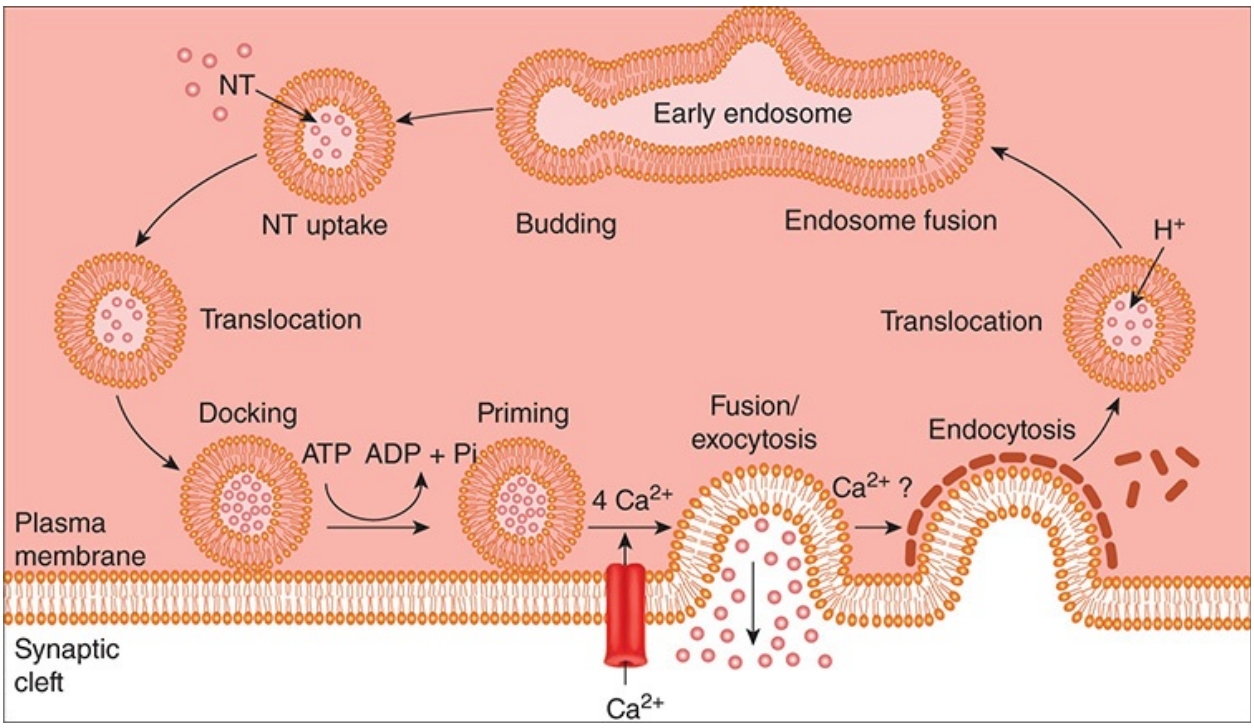


FIGURE 6–3 Small synaptic vesicle cycle in presynaptic nerve terminals. Vesicles bud off the early endosome and then are filled with a neurotransmitter (NT). The vesicles then translocate toward release sites on the plasma membrane where they dock, and then become primed. Following arrival of an action potential at the nerve ending, Ca^{2+} influx triggers fusion and exocytosis of the vesicular contents into the synaptic cleft. The vesicle wall then is coated with clathrin, which likely facilitates endocytosis. Once back inside the cytoplasm, the vesicle becomes uncoated and fuses with the early endosome to start the cycle again.

The large dense-core vesicles are found throughout some presynaptic nerve terminals and release their neuropeptide contents by exocytosis from all parts of the terminal. On the other hand, the small vesicles are located near the synaptic cleft and fuse to the membrane, discharging their contents very rapidly into the cleft at areas of membrane thickening called **active zones** (Figure 6–2). Voltage-

gated Ca^{2+} channels are very close to the release sites at the active zones.

The Ca^{2+} that triggers exocytosis of transmitters enters the presynaptic neurons, and transmitter release starts within 200 μs. In addition, the transmitter must be released close to the postsynaptic receptors to be effective on the postsynaptic neuron. This orderly organization of the synapse depends in part on **neurexins**, presynaptic cell-adhesion molecules that bind to **neuroligins** on the membrane of postsynaptic neurons. Neurexin–neuroligin interactions not only hold synapses together, but they also provide a mechanism for the production of synaptic specificity.

As noted in Chapter 2, vesicle budding, fusion, and discharge of contents with subsequent retrieval of vesicle membrane are fundamental processes occurring in most, if not all, cells. Thus, neurotransmitter secretion at synapses and the accompanying membrane retrieval are specialized forms of the general processes of exocytosis and endocytosis. The fusion of synaptic vesicles with the cell membrane involves the action of several proteins, including the N-ethylmaleimide–sensitive fusion protein (NSF), SNAPs (soluble NSF attachment proteins), and SNAREs (SNAP receptors; **Figure 6–4**). **Synaptobrevin** in the vesicle membrane locks with **syntaxin** and **SNAPs** in the cell membrane; GTPases regulate a multiprotein complex that includes Rab and Sec1/Munc18-like proteins as part of the fusion process. The one-way gate at the synapses is necessary for orderly neural function.

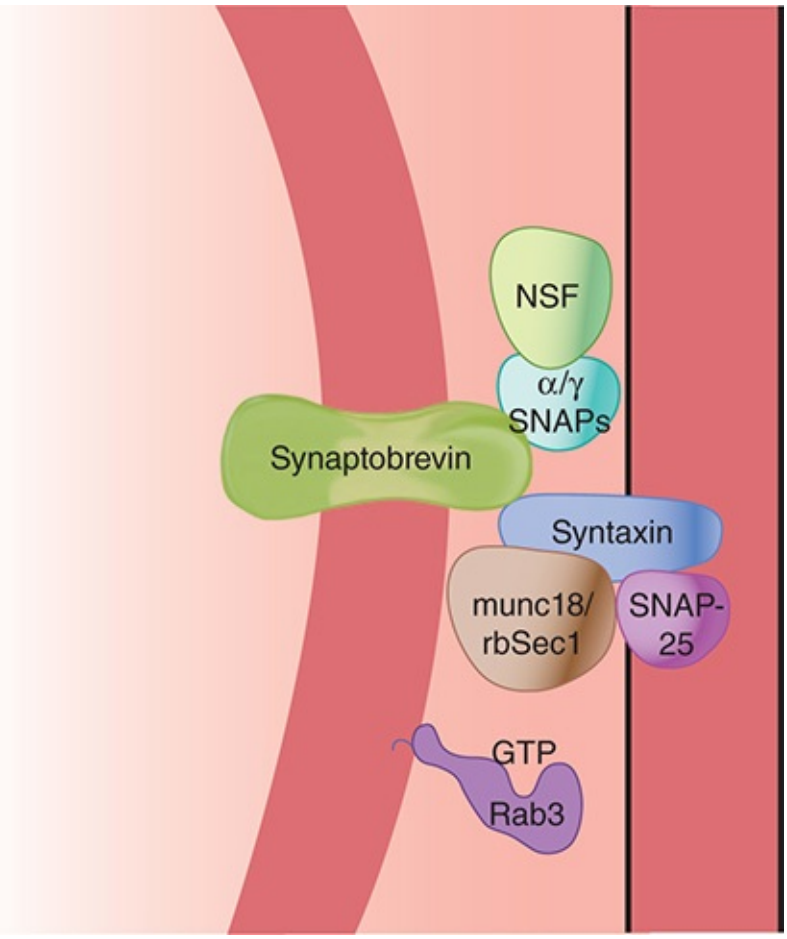


FIGURE 6–4 Main proteins that interact to produce synaptic vesicle docking and fusion in nerve endings. Several proteins are involved in synaptic vesicle docking and fusion, including the N-ethylmaleimide–sensitive fusion protein (NSF), SNAPs (soluble NSF attachment proteins), and SNAREs (SNAP receptors). Synaptobrevin in the vesicle membrane links with syntaxin and SNAPs in the cell membrane; GTPases regulate a multiprotein complex that includes Rab and Sec1/Munc18-like proteins.

Several deadly toxins that block neurotransmitter release are zinc endopeptidases that cleave and hence inactivate proteins in the fusion-exocytosis complex. **Clinical Box 6–1** describes how neurotoxins from bacteria called *Clostridium tetani* and *Clostridium botulinum* can disrupt neurotransmitter release in either the central nervous system (CNS) or at the neuromuscular junction.

ELECTRICAL EVENTS IN POSTSYNAPTIC

NEURONS

EXCITATORY & INHIBITORY POSTSYNAPTIC POTENTIALS

Stimulation of a dorsal root afferent (sensory neuron) can be used to study both excitatory and inhibitory events in α-motor neurons **(Figure 6–5)**. Once an impulse reaches the presynaptic terminals of the sensory neuron, a response can be obtained in the postsynaptic neuron after a **synaptic delay.** The delay is due to the time it takes for the synaptic mediator to be released and to act on the receptors on the membrane of the postsynaptic neuron. Because of it, conduction along a chain of neurons is slower if there are many synapses compared to if there are only a few synapses. Because the minimum time for transmission across one synapse is 0.5 ms, it is possible to determine whether a given reflex pathway is **monosynaptic** or **polysynaptic** (contains more than one synapse) by measuring the synaptic delay.

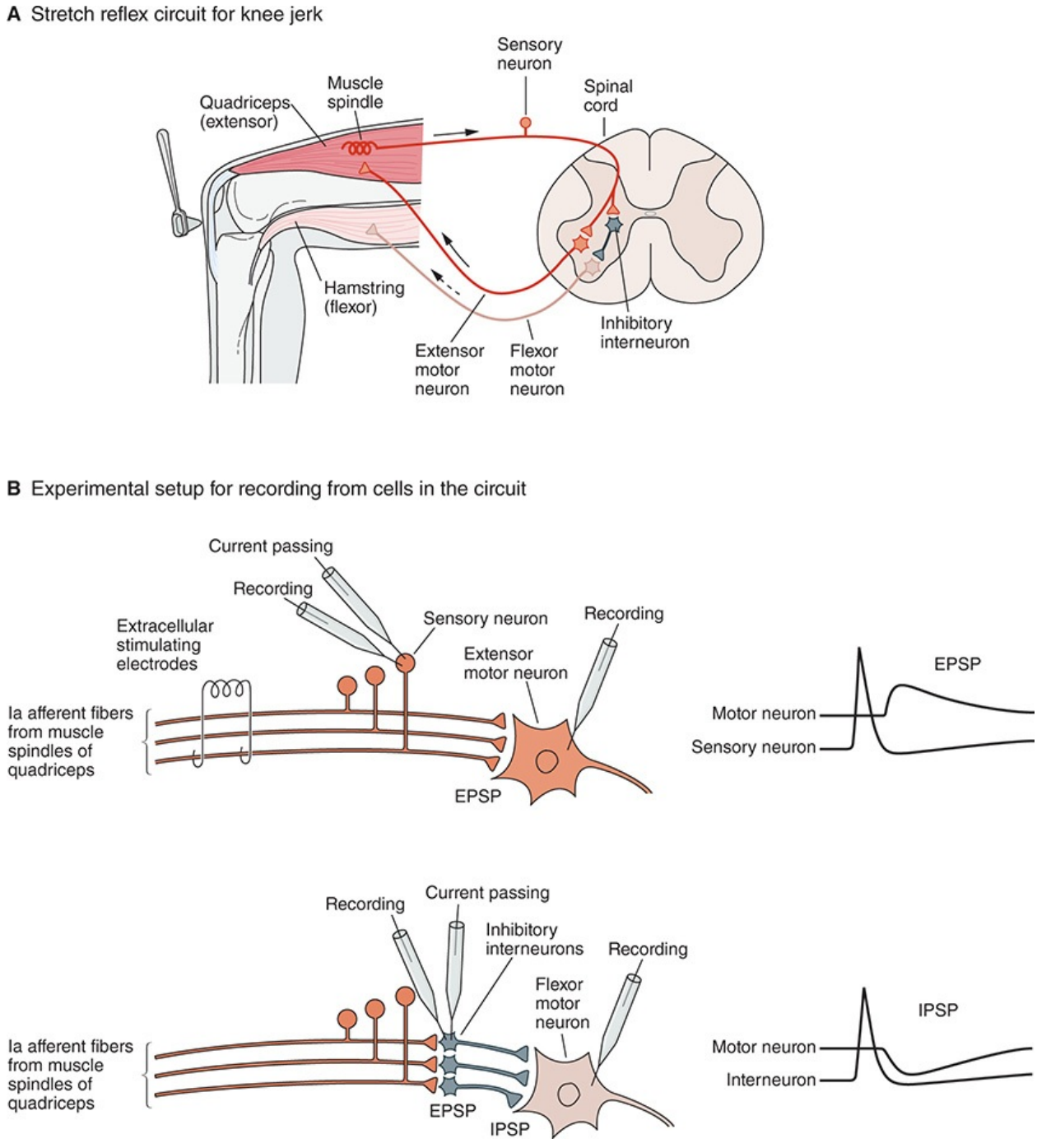


FIGURE 6–5 Excitatory and inhibitory synaptic connections mediating the stretch reflex provide an example of typical circuits within the CNS. A) The stretch receptor sensory neuron of the quadriceps muscle makes an excitatory connection with the extensor motor neuron of the same muscle and an inhibitory interneuron projecting to flexor motor neurons supplying the antagonistic hamstring muscle. **B)** Experimental setup to study excitation and inhibition of the extensor motor neuron. Top panel shows two approaches to elicit an

excitatory (depolarizing) postsynaptic potential or EPSP in the extensor motor neuron–electrical stimulation of the whole Ia afferent nerve using extracellular electrodes and intracellular current passing through an electrode inserted into the cell body of a sensory neuron. Bottom panel shows that current passing through an inhibitory interneuron elicits an inhibitory (hyperpolarizing) postsynaptic potential or IPSP in the flexor motor neuron. EPSP, excitatory postsynaptic potential; IPSP, inhibitory postsynaptic potential. (Reproduced with permission from Kandel ER, Schwartz JH, Jessell TM [editors]: *Principles of Neural Science*, 4th ed. New York, NY: McGraw-Hill; 2000.)

CLINICAL BOX 6–1

Botulinum and Tetanus Toxins

Clostridia are gram-positive bacteria. Two varieties, *Clostridium tetani* and *Clostridium botulinum*, produce some of the most potent biologic toxins **(tetanus toxin** and **botulinum toxin)** known to affect humans. These neurotoxins act by preventing the release of neurotransmitters in the CNS and at the neuromuscular junction. Tetanus toxin binds irreversibly to the presynaptic membrane of the neuromuscular junction and uses retrograde axonal transport to travel to the cell body of the motor neuron in the spinal cord. From there it is picked up by the terminals of presynaptic inhibitory interneurons. The toxin attaches to **gangliosides** in these terminals and blocks the release of glycine and GABA. As a result, the activity of motor neurons is markedly increased. Clinically, tetanus toxin causes spastic paralysis; the characteristic symptom of “lockjaw” is spasms of the masseter muscle. Botulism can result from ingestion of contaminated food, colonization of the gastrointestinal tract in an infant, or wound infection. Botulinum toxins are a family of seven neurotoxins, but it is mainly botulinum toxins A, B, and E that are toxic to humans. Botulinum toxins A and E cleave synaptosome-associated protein-25 **(SNAP-25).** This is a presynaptic membrane protein needed for fusion of synaptic vesicles containing acetylcholine to the terminal membrane, an important step in transmitter release. Botulinum toxin B cleaves **synaptobrevin,** a vesicle-associated membrane protein **(VAMP).** By blocking acetylcholine release at the neuromuscular junction, these toxins cause flaccid paralysis. Symptoms can include ptosis (drooping eyelid), diplopia (double vision), dysarthria (slurred speech), dysphonia (difficulty speaking), and dysphagia (difficulty swallowing).

THERAPEUTIC HIGHLIGHTS

Tetanus can be prevented by treatment with **tetanus toxoid vaccine.** The widespread use of this vaccine in the United States beginning in the mid-1940s led to a marked decline in the incidence of tetanus toxicity. The incidence of botulinum toxicity is also low (about 100 cases per year in the United States), but in those individuals who are affected, the fatality rate is 5–10%. An antitoxin is available for treatment, and those who are at risk for respiratory failure are placed on a ventilator. On the positive side, local injection of small doses of botulinum toxin (**botox**) has proven to be effective in the treatment of a wide variety of conditions characterized by muscle hyperactivity. Examples include injection into the lower esophageal sphincter to relieve achalasia, injection into facial muscles to remove wrinkles, and injection into leg muscles to reduce spasticity in individuals with cerebral palsy.

A single stimulus applied to the sensory nerves characteristically does not lead to the formation of a propagated action potential in the postsynaptic neuron. Instead, the stimulation produces either a transient partial depolarization or a transient hyperpolarization. The initial depolarizing response produced by a single stimulus to the proper input begins about 0.5 ms after the afferent impulse enters the spinal cord. It reaches its peak 1–1.5 ms later and then declines exponentially (Figure 6–5). During this potential, the excitability of the neuron to other stimuli is increased; thus, it is called a fast **excitatory postsynaptic potential (EPSP)**.

The fast EPSP is produced by depolarization of the postsynaptic cell membrane immediately under the presynaptic ending. The excitatory transmitter opens Na^+ or Ca^{2+} channels in the postsynaptic membrane, producing an inward current. The area of current flow thus created is so small that it does not drain off enough positive charge to depolarize the whole membrane. Instead, an EPSP is induced. The EPSP due to activity in one synaptic knob is small, but the depolarizations produced by each of the active knobs summate.

EPSPs are produced by stimulation of some inputs, but stimulation of other inputs produces hyperpolarizing responses. Like the EPSPs, they peak 1–1.5 ms after the stimulus and decrease exponentially (Figure 6–5). During this potential, the excitability of the neuron to other stimuli is decreased; thus, it is called a fast **inhibitory postsynaptic potential (IPSP)**.

A fast IPSP can be produced by a localized increase in Cl^- transport. When an

inhibitory synaptic knob becomes active, the released neurotransmitter triggers the opening of Cl^- channels in the area of the postsynaptic cell membrane under the knob. Cl^- moves down its concentration gradient. The net effect is the transfer of negative charge into the cell, increasing the membrane potential.

The decreased excitability of the nerve cell during the IPSP is due to movement of the membrane potential away from the firing level. Consequently, more excitatory (depolarizing) activity is necessary to reach the firing level. The fact that an IPSP is mediated by Cl^- can be demonstrated by repeating the stimulus while varying the resting membrane potential of the postsynaptic cell. When the membrane potential is at the equilibrium potential for chloride (E_{Cl}), the postsynaptic potential disappears **(Figure 6–6)**, and at more negative membrane potentials, it becomes positive **(reversal potential).**

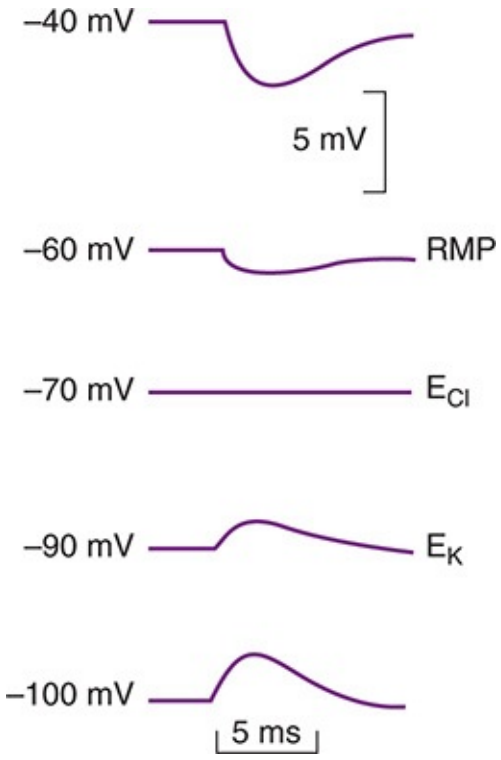


FIGURE 6–6 IPSP is due to increased Cl^- influx during stimulation. This can be demonstrated by repeating the stimulus while varying the resting membrane potential (RMP) of the postsynaptic cell. When the membrane potential is at E_{Cl}, the potential disappears, and at more negative membrane potentials (eg, E_K and below), it becomes positive (reversal potential).

Because IPSPs are net hyperpolarizations, they can be produced by alterations

in other ion channels in the neuron. For example, they can be produced by opening K^+ channels, with movement of K^+ out of the postsynaptic cell, or by closure of Na^+ or Ca^{2+} channels.

SLOW POSTSYNAPTIC POTENTIALS

In addition to the fast EPSPs and IPSPs, slow EPSPs and IPSPs occur in autonomic ganglia, cardiac and smooth muscle, and cortical neurons. These postsynaptic potentials have a latency of 100–500 ms and last several seconds. The slow EPSPs are generally due to decreases in K^+ conductance, and the slow IPSPs are due to increases in K^+ conductance.

ELECTRICAL TRANSMISSION

In electrical synapses, the membranes of the presynaptic and postsynaptic neurons come close together, and gap junctions are formed between the cells (see Chapter 2). Like the intercellular junctions in other tissues, these junctions form low-resistance bridges through which ions can pass with relative ease. At these junctions the impulse reaching the presynaptic terminal generates an EPSP in the postsynaptic cell that, because of the low-resistance bridge between the two, has a much shorter latency than the EPSP at a synapse where transmission is chemical.

GENERATION OF AN ACTION POTENTIAL IN THE POSTSYNAPTIC NEURON

The constant interplay of excitatory and inhibitory activity on the postsynaptic neuron produces a fluctuating membrane potential that is the algebraic summation of the hyperpolarizing and depolarizing potentials. The soma of the neuron thus acts as an integrator. When the level of depolarization reaches the threshold voltage, a propagated action potential will occur. However, the discharge of the neuron is slightly more complicated than this. In motor neurons, the portion of the cell with the lowest threshold for the production of an action potential is the initial segment, the portion of the axon at and just beyond the axon hillock (see Chapter 4). This unmyelinated segment is depolarized or hyperpolarized electrotonically by the current sinks and sources under the

excitatory and inhibitory synaptic knobs. It is the first part of the neuron to fire, and its discharge is propagated in two directions: down the axon and back into the soma. Retrograde firing of the soma in this fashion probably has value in wiping the slate clean for subsequent renewal of the interplay of excitatory and inhibitory activity on the cell.

TEMPORAL & SPATIAL SUMMATION OF POSTSYNAPTIC POTENTIALS

Two passive membrane properties of a neuron affect the ability of postsynaptic potentials to summate to elicit an action potential **(Figure 6–7)**. The **time constant** of a neuron determines the time course of the synaptic potential, and the **length constant** of a neuron determines the degree to which a depolarizing current is reduced as it spreads passively. Figure 6–7 also shows how the time constant of the postsynaptic neuron can affect the amplitude of the depolarization caused by consecutive EPSPs produced by a single presynaptic neuron. The longer the time constant, the greater is the chance for two potentials to summate to induce an action potential. If a second EPSP is elicited before the first EPSP decays, the two potentials summate and, as in this example, their additive effects are sufficient to induce an action potential in the postsynaptic neuron **(temporal summation).** Figure 6–7 also shows how the length constant of a postsynaptic neuron can affect the amplitude of two EPSPs produced by different presynaptic neurons in a process called **spatial summation.** If a neuron has a long length constant, the membrane depolarization induced by input arriving at two points on the neuron can spread to the trigger zone of the neuron with minimal decrement. The two potentials can summate and induce an action potential.

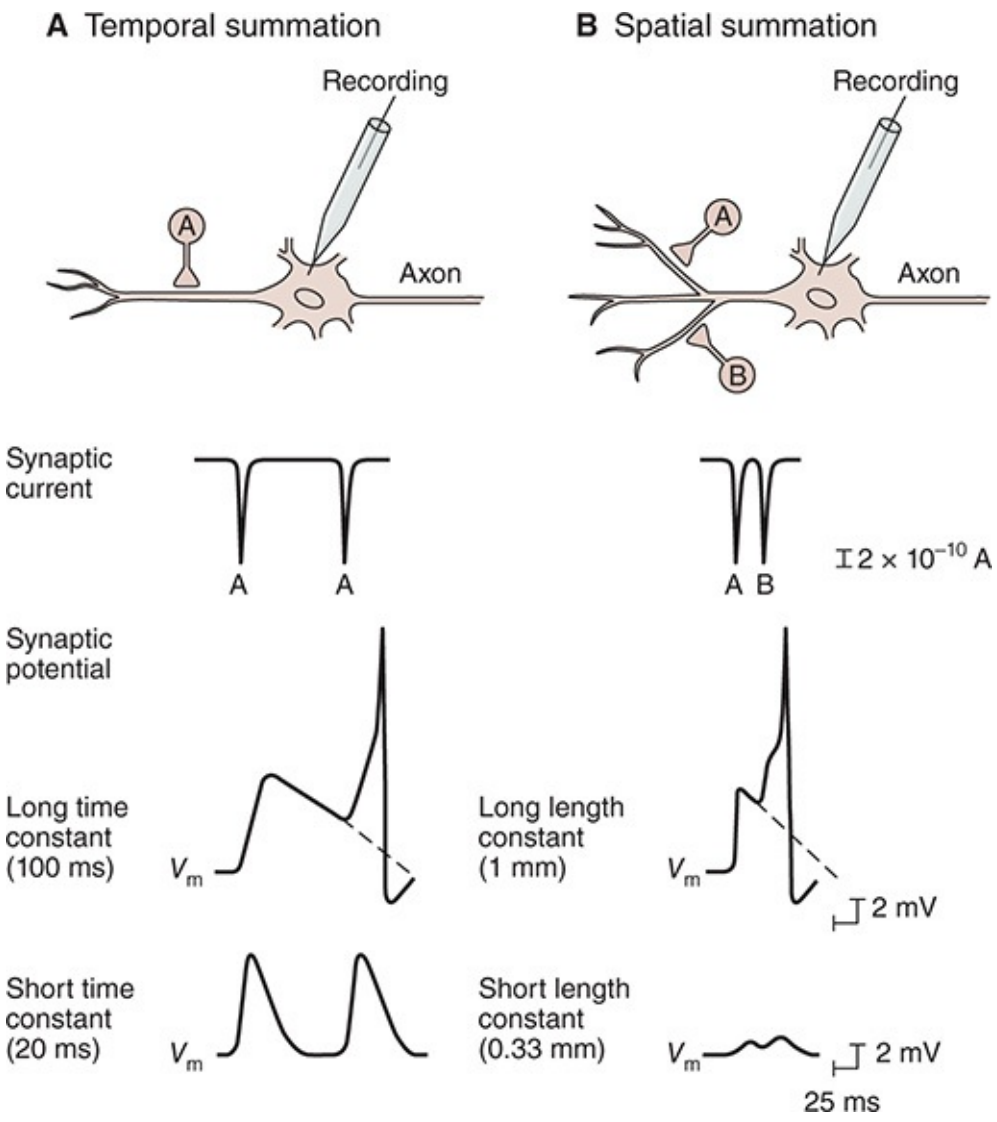


FIGURE 6–7 Central neurons integrate a variety of synaptic inputs through temporal and spatial summation. A) The time constant of the postsynaptic neuron affects the amplitude of the depolarization caused by consecutive EPSPs produced by a single presynaptic neuron. In cases of a long time constant, if a second EPSP is elicited before the first EPSP decays, the two potentials summate to induce an action potential. **B)** The length constant of a postsynaptic cell affects the amplitude of two EPSPs produced by two presynaptic neurons, A andB. If the length constant is long, the depolarization induced at two points on the neuron can spread to the trigger zone with minimal decrement so that the two potentials summate and an action potential is elicited. EPSP, excitatory postsynaptic potential. (Reproduced with permission from Kandel ER, Schwartz JH, Jessell TM [editors]: *Principles of Neural Science*, 4th ed. New York, NY: McGraw-Hill; 2000.)

INHIBITION & FACILITATION AT SYNAPSES

Inhibition within the CNS can be either postsynaptic or presynaptic. Examples of the neuronal connections that can mediate **presynaptic inhibition** and **postsynaptic inhibition** are compared in **Figure 6–8**. Postsynaptic inhibition during the course of an IPSP is called **direct inhibition** because it is not a consequence of previous discharges of the postsynaptic neuron. There are various forms of **indirect inhibition** that is due to the effects of previous postsynaptic neuron discharge. For example, the postsynaptic cell can be refractory to excitation because it has just fired and is in its refractory period. During after-hyperpolarization it is also less excitable. In spinal neurons, especially after repeated firing, this after- hyperpolarization may be large and prolonged.

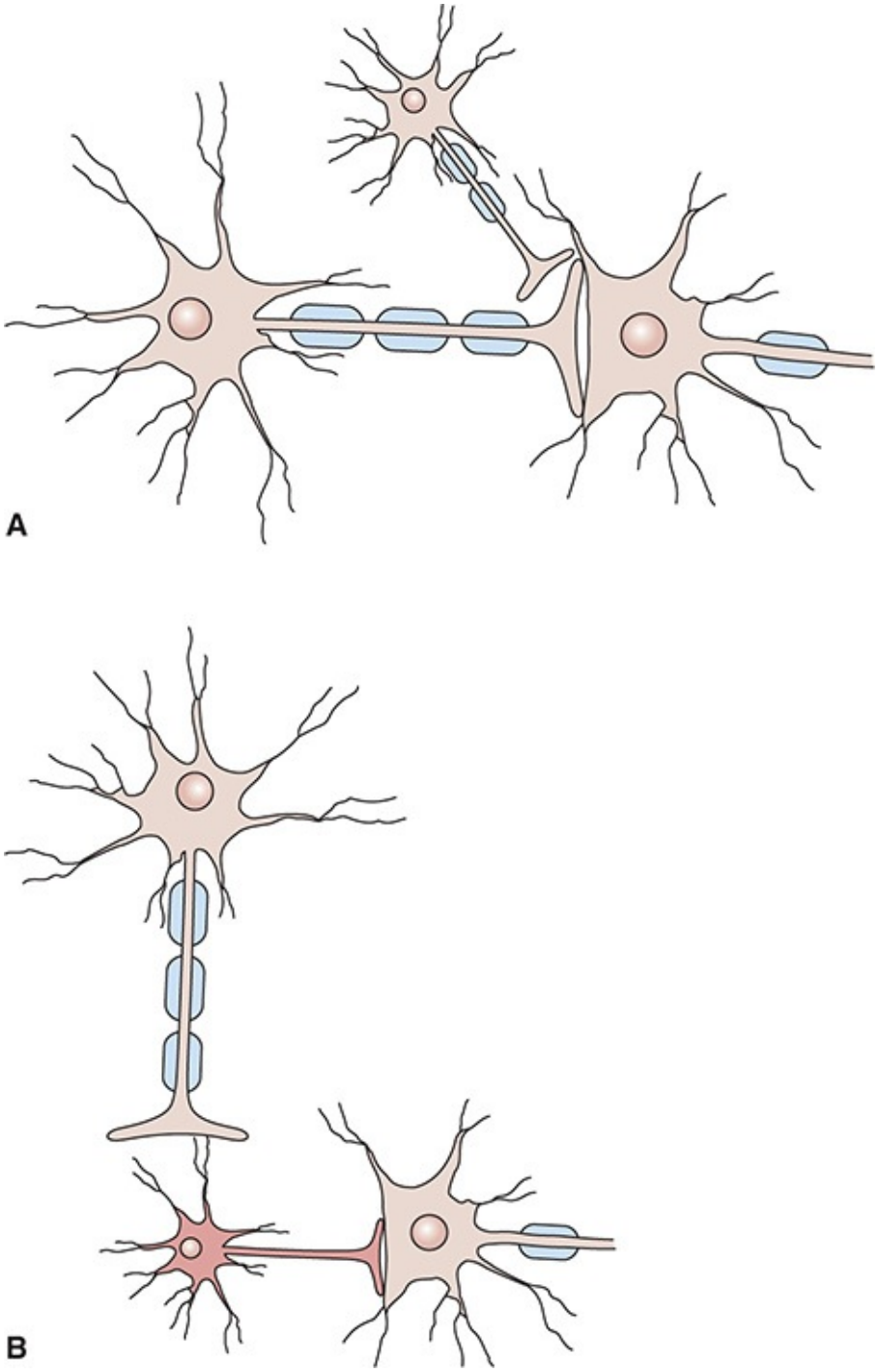


FIGURE 6–8 Comparison of neuronal connections that can produce presynaptic and postsynaptic inhibition. A) Presynaptic inhibition is a process mediated by neurons whose terminals are on excitatory nerve endings, forming axoaxonal synapses and reducing transmitter release from the excitatory neuron. **B)** Postsynaptic inhibition occurs when an inhibitory transmitter such as GABA is released from the nerve terminals of an inhibitory interneuron (dark) that

synapses on a postsynaptic neuron.

POSTSYNAPTIC INHIBITION

Postsynaptic inhibition occurs when an inhibitory transmitter such as glycine or GABA is released from a presynaptic nerve terminal onto the postsynaptic neuron to induce an IPSP in the postsynaptic neuron (Figure 6–8B). Various pathways in the nervous system mediate postsynaptic inhibition, and one illustrative example is presented in Figure 6–5. Afferent fibers from the muscle spindles (stretch receptors) in skeletal muscle project directly to the spinal motor neurons supplying the same muscle. Impulses in this afferent fiber cause an EPSP and, with summation, propagated action potentials in the postsynaptic motor neuron. At the same time, an IPSP is produced in motor neurons supplying the antagonistic muscle, which has an inhibitory interneuron interposed between the afferent fiber and the motor neuron. Therefore, activity in the afferent fibers from the muscle spindles excites the motor neurons supplying the muscle from which the impulses come and inhibits the motor neurons supplying its antagonists **(reciprocal innervation).** These reflexes are considered in more detail in Chapter 12.

PRESYNAPTIC INHIBITION & FACILITATION

Actions at the presynaptic nerve terminal can either reduce (presynaptic inhibition) or enhance (presynaptic facilitation) neurotransmitter release as a mechanism to fine-tune the strength of synaptic transmission. Presynaptic inhibition is a process mediated by neurons whose terminals are on excitatory endings, forming axoaxonal synapses (Figure 6–2 and 6–8A). Activation of the presynaptic receptors increases Cl^- conductance, and this decreases the size of the action potential reaching the excitatory ending **(Figure 6–9)**. This in turn reduces Ca^{2+} entry and consequently the amount of excitatory transmitter released. Voltage-gated K^+ channels are also opened, and the resulting K^+ efflux causes a decrease in Ca^{2+} influx.

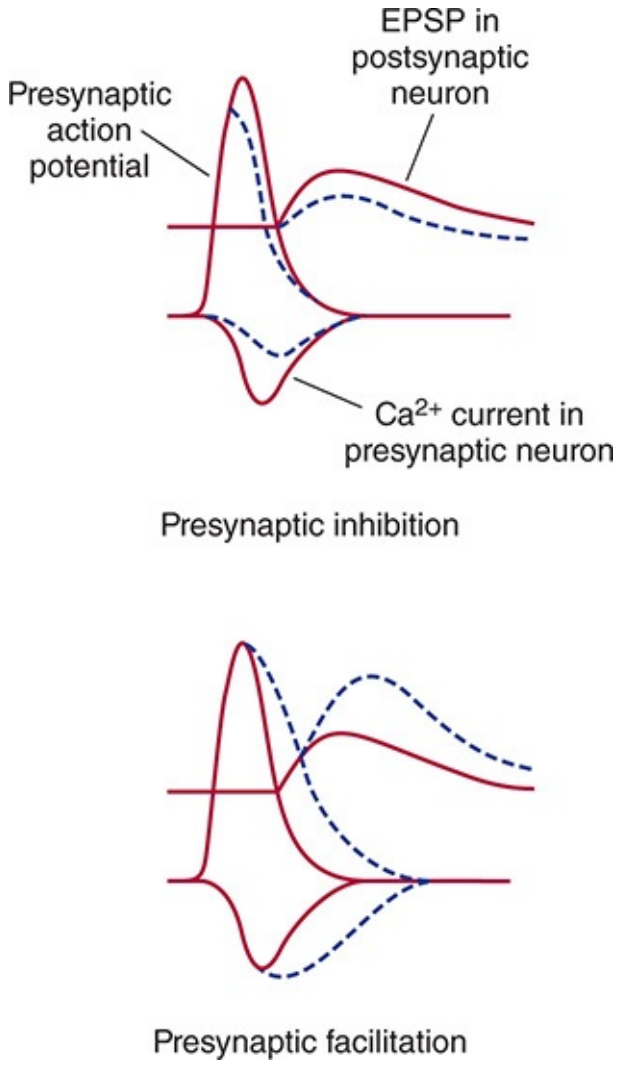


FIGURE 6–9 Effects of presynaptic inhibition and facilitation on the action potential and the Ca^{2+} current in the presynaptic neuron and the EPSP in the postsynaptic neuron. In each case, the solid lines are the controls and the dashed lines the records obtained during inhibition or facilitation. Presynaptic inhibition occurs when activation of presynaptic receptors increases Cl^- conductance, which decreases the size of the action potential. This reduces Ca^{2+} entry and thus the amount of excitatory transmitter released. Presynaptic facilitation is produced when the action potential is prolonged and the Ca^{2+} channels are open for a longer duration. EPSP, excitatory postsynaptic potential. (Reproduced with permission from Kandel ER, Schwartz JH, Jessell TM [editors]: *Principles of Neural Science*, 4th ed. New York, NY: McGraw-Hill; 2000.)

The first transmitter shown to produce presynaptic inhibition was GABA.

Acting via $GABA_A$ receptors, GABA increases Cl^- conductance. $GABA_B$ receptors are also present in the spinal cord and mediate presynaptic inhibition via a G-protein that produces an increase in K^+ conductance. Baclofen, a $GABA_B$ agonist, is effective in the treatment of the spasticity of spinal cord injury and multiple sclerosis, particularly when administered intrathecally via an implanted pump. Other transmitters also mediate presynaptic inhibition by G-protein-mediated effects on Ca^{2+} channels and K^+ channels.

Conversely, **presynaptic facilitation** is produced when the action potential is prolonged (Figure 6–9) and the Ca^{2+} channels are open for a longer period. The molecular events responsible for the production of presynaptic facilitation mediated by serotonin in the sea snail *Aplysia* have been worked out in detail. Serotonin released at an axoaxonal ending increases intraneuronal cyclic adenosine monophosphate (cAMP) levels, and the resulting phosphorylation of one group of K^+ channels closes the channels, slowing repolarization and prolonging the action potential.

ORGANIZATION OF INHIBITORY SYSTEMS

Presynaptic inhibition and postsynaptic inhibition are usually produced by stimulation of certain systems converging on a given postsynaptic neuron. Neurons may also inhibit themselves in a negative feedback manner (negative feedback inhibition). For instance, a spinal motor neuron emits a recurrent collateral that synapses on an inhibitory interneuron **(Renshaw cell)** that terminates on the cell body of spinal motor neuron **(Figure 6–10)**. The Renshaw cell releases **glycine** to reduce the activity of the motor neuron. Similar inhibition via recurrent collaterals is seen in the cerebral cortex and limbic system. Presynaptic inhibition due to descending pathways that terminate on afferent pathways in the dorsal horn may be involved in the gating of pain transmission.

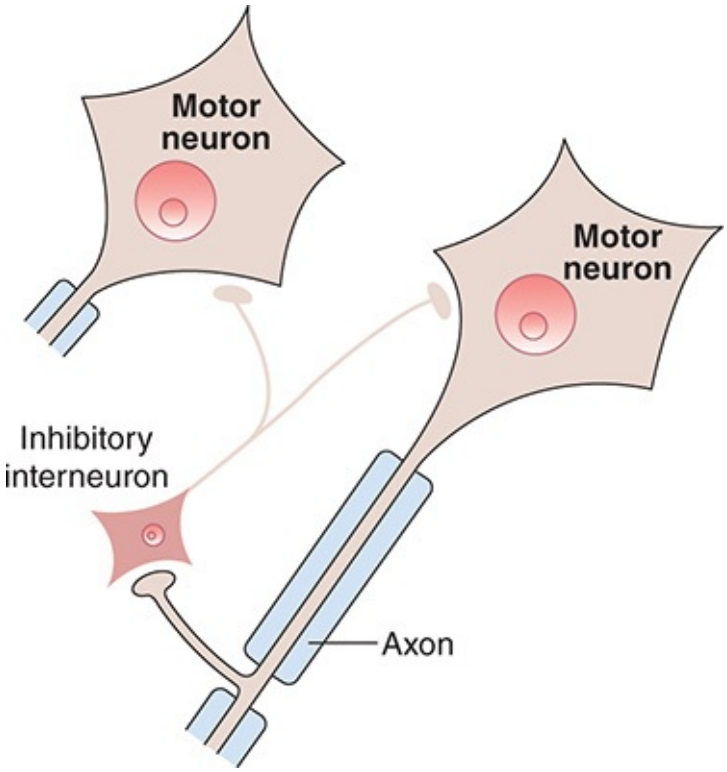


FIGURE 6–10 Negative feedback inhibition of a spinal motor neuron via an inhibitory interneuron. The axon of a spinal motor neuron has a recurrent collateral that synapses on an inhibitory interneuron that terminates on the cell body of the same and other motor neurons. The inhibitory interneuron is called a Renshaw cell and its neurotransmitter is glycine.

Another example of inhibition is that resulting from stimulation of cerebellar basket cells that produces IPSPs in the Purkinje cells. The basket cells and Purkinje cells are both excited by the same parallel-fiber input (see Chapter 12). This arrangement is called **feed-forward inhibition** and may limit the duration of the excitation produced by any given afferent volley.

NEUROMUSCULAR TRANSMISSION

NEUROMUSCULAR JUNCTION

Figure 6–11 shows the components of the neuromuscular junction, the specialized area where a motor nerve terminates on a skeletal muscle fiber. As the axon supplying a skeletal muscle fiber approaches its termination, it loses its myelin sheath and divides into a number of terminal boutons. The terminal

contains many small, clear vesicles that contain acetylcholine, the transmitter at these junctions. The endings fit into **junctional folds** or depressions in the **motor endplate,** the thickened portion of the muscle membrane at this junction. Each endplate receives input from a single nerve fiber. The space between the nerve and the thickened muscle membrane is comparable to the synaptic cleft at a neuron-to-neuron synapse. The whole structure is known as the **neuromuscular junction.**

A

B

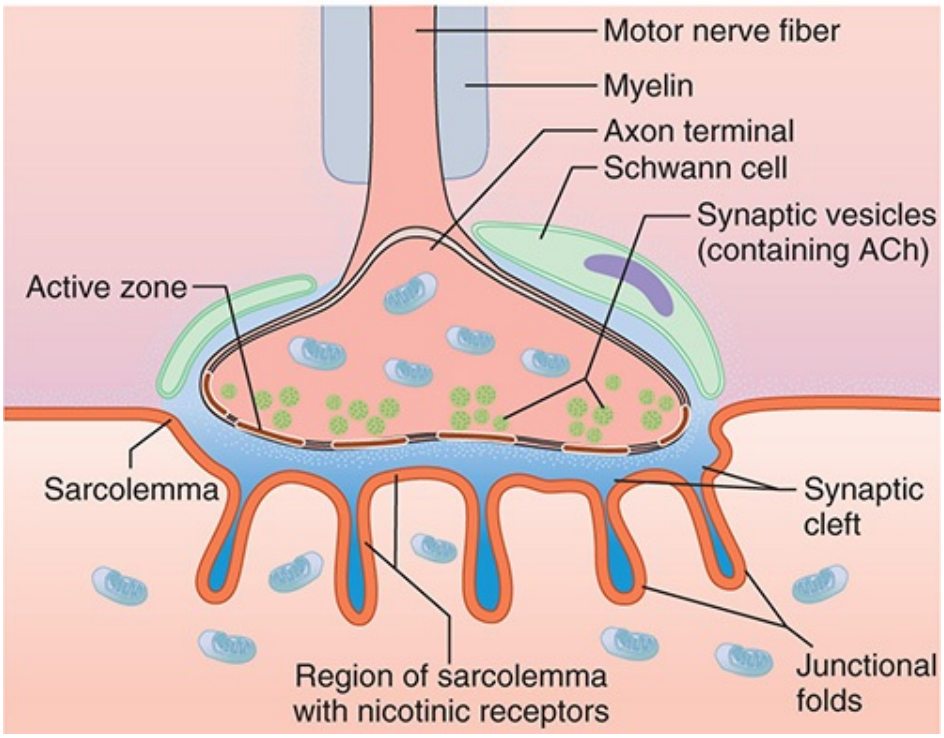

Motor nerve fiber
Myelin
Axon terminal
Schwann cell
Synaptic vesicles (containing ACh)
Active zone
Sarcolemma
Synaptic cleft
Region of sarcolemma with nicotinic receptors
Junctional folds

FIGURE 6–11 The neuromuscular junction. A) Scanning electron micrograph showing branching of motor axons with terminals embedded in grooves in the muscle fiber's surface. **B)** Structure of a neuromuscular junction. ACh, acetylcholine. (Modified with permission from Widmaier EP, Raff H, Strang KT: *Vanders Human Physiology*. New York, NY: McGraw-Hill; 2008.)

SEQUENCE OF EVENTS DURING TRANSMISSION

The events occurring during transmission of impulses from the motor nerve to the muscle are somewhat similar to those occurring at neuron-to-neuron synapses **(Figure 6–12)**. The impulse arriving in the terminal of the motor neuron increases its permeability to Ca^{2+}. Ca^{2+} enters the nerve endings and triggers a marked increase in exocytosis of the acetylcholine-containing synaptic vesicles. The acetylcholine diffuses to nicotinic cholinergic (N_M) receptors that are concentrated at the tops of the junctional folds of the membrane of the motor endplate. Binding of acetylcholine to these receptors increases the Na^+ and K^+ conductance, and the resultant influx of Na^+ produces a depolarizing potential, the **endplate potential.** The current sink created by this local potential depolarizes the adjacent muscle membrane to its firing level. Action potentials are generated on either side of the endplate and are conducted away from the endplate in both directions along the muscle fiber. The muscle action potential, in turn, initiates muscle contraction (see Chapter 5). Acetylcholine is then removed from the synaptic cleft by acetylcholinesterase, which is present in high concentration at the neuromuscular junction.

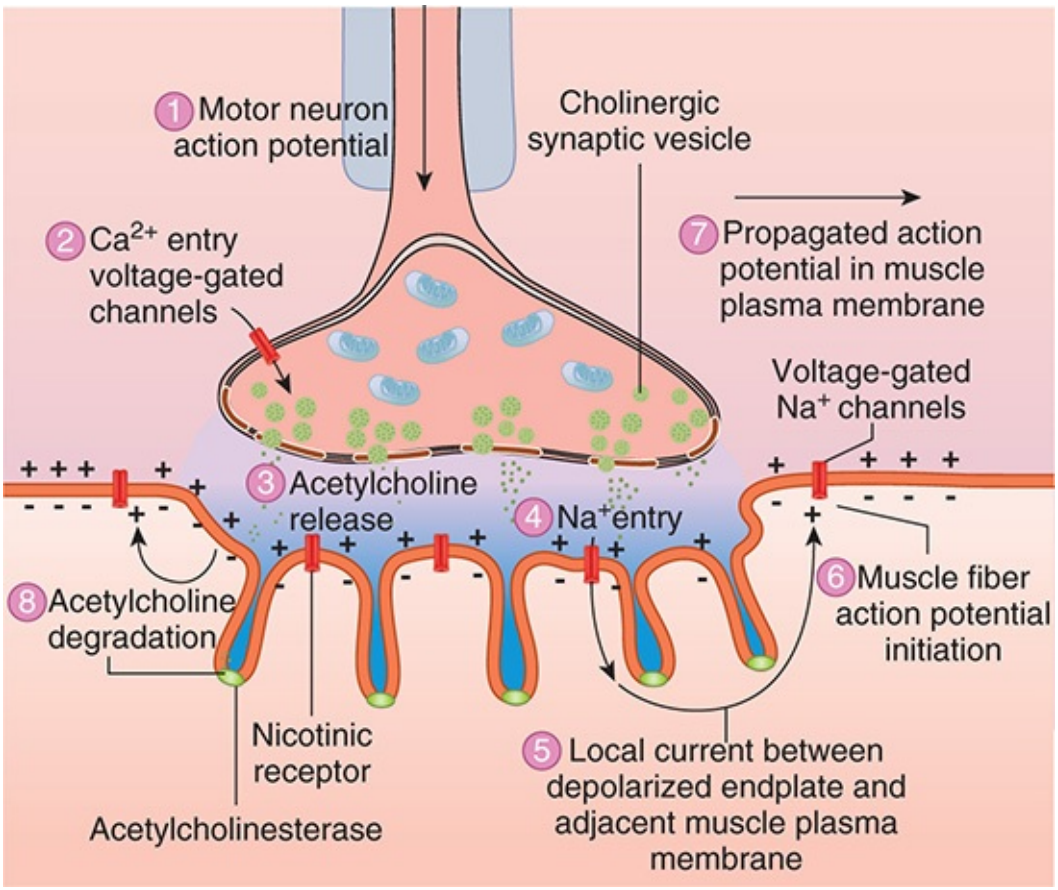


FIGURE 6–12 Events at the neuromuscular junction that lead to an action potential in the muscle fiber plasma membrane. The impulse arriving in the end of the motor neuron increases the permeability of its endings to Ca^{2+}, which enters the endings and triggers exocytosis of the acetylcholine (ACh)-containing synaptic vesicles. ACh diffuses and binds to nicotinic cholinergic (N_M) receptors in the motor endplate, which increases Na^+ and K^+ conductance. The resultant influx of Na^+ produces the endplate potential. The current sink created by this local potential depolarizes the adjacent muscle membrane to its firing level. Action potentials are generated on either side of the endplate and are conducted away from the endplate in both directions along the muscle fiber and the muscle contracts. ACh is then removed from the synaptic cleft by acetylcholinesterase. (Modified with permission from Widmaier EP, Raff H, Strang KT: *Vanders Human Physiology*. New York, NY: McGraw-Hill; 2008.)

QUANTAL RELEASE OF TRANSMITTER

Each nerve impulse releases acetylcholine from about 60 synaptic vesicles, and

each vesicle contains about 10,000 molecules of the neurotransmitter. Small quanta (packets) of acetylcholine are released randomly from the nerve cell membrane at rest. Each produces a minute depolarizing spike called a **miniature endplate potential** that is about 0.5 mV in amplitude. The size of the quanta of acetylcholine released in this way varies directly with the Ca^{2+} concentration and inversely with the Mg^{2+} concentration at the endplate. When a nerve impulse reaches the ending, the number of quanta released increases by several orders of magnitude, and the result is the large endplate potential that exceeds the firing threshold of the muscle fiber. Quantal release of a neurotransmitter also occurs at noradrenergic, glutamatergic, and other synaptic junctions. Two diseases of the neuromuscular junction, myasthenia gravis and Lambert-Eaton myasthenic syndrome, are described in **Clinical Box 6–2** and **Clinical Box 6–3**, respectively.

NERVE ENDINGS IN SMOOTH & CARDIAC MUSCLE

The postganglionic autonomic nerve fibers supplying smooth muscle in various organs branch extensively and come in close contact with the muscle cells **(Figure 6–13)**. Some of these nerve fibers have clear vesicles containing acetylcholine; others have dense-core vesicles containing norepinephrine. There are no recognizable endplates or other postsynaptic specializations at these **neuroeffector junctions**. The nerve fibers run along the membranes of the muscle cells and sometimes groove their surfaces. The nerve branches are beaded with enlargements **(varicosities)** that contain synaptic vesicles. In noradrenergic neurons, the varicosities are about 5 µm apart, with up to 20,000 varicosities per neuron. Neurotransmitter can be released at each of the varicosities along the axon; this arrangement permits one neuron to innervate many effector cells. The type of contact in which a neuron forms a synapse on the surface of a smooth muscle cell and then passes on to make similar contacts with other cells is called a **synapse en passant.**

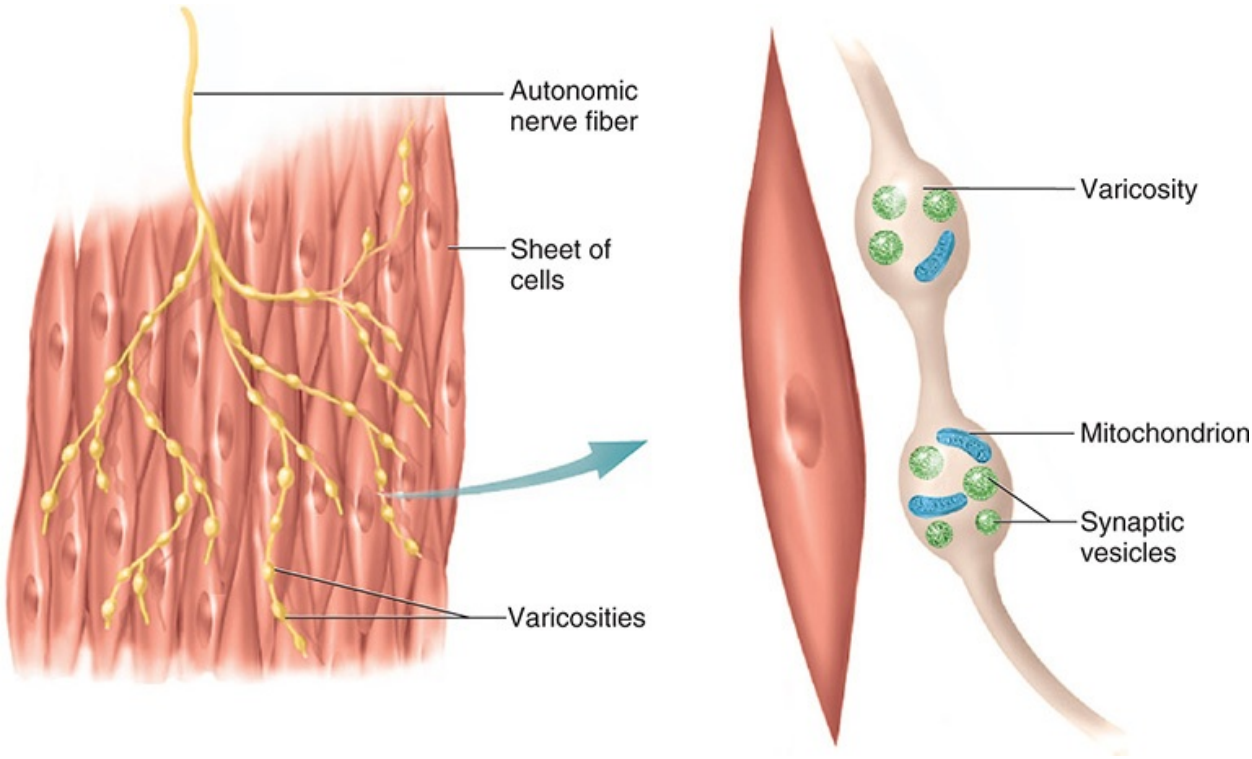


FIGURE 6–13 Endings of postganglionic autonomic neurons on smooth muscle. The nerve fibers run along the membranes of the smooth muscle cells and sometimes groove their surfaces. The multiple branches of postganglionic neurons are beaded with enlargements (varicosities) and contain synaptic vesicles. Neurotransmitter is released from the varicosities and diffuses to receptors on smooth muscle cell plasma membranes. (Reproduced with permission from Widmaier EP, Raff H, Strang KT: *Vanders Human Physiology*. New York, NY: McGraw-Hill; 2008.)

In the heart, cholinergic and noradrenergic nerve fibers end on the sinoatrial node, the atrioventricular node, and the bundle of His (see Chapter 29). Noradrenergic fibers also innervate the ventricular muscle. The exact nature of the endings on nodal tissue is not known. In the ventricle, the contacts between the noradrenergic fibers and the cardiac muscle fibers resemble those found in smooth muscle.

CLINICAL BOX 6–2

Myasthenia Gravis

Myasthenia gravis is a serious and sometimes fatal disease in which skeletal

muscles are weak and tire easily. It occurs in 25 to 125 of every 1 million people worldwide and can occur at any age but seems to have a bimodal distribution, with peak occurrences in individuals in their 20s (mainly women) and 60s (mainly men). It is caused by the formation of circulating antibodies to the skeletal muscle type of **nicotinic cholinergic receptors.** These antibodies destroy some of the receptors and bind others to neighboring receptors, triggering their removal by endocytosis. Normally, the number of quanta released from the motor nerve terminal declines with successive repetitive stimuli. In myasthenia gravis, neuromuscular transmission fails at these low levels of quantal release. This leads to the major clinical feature of the disease, muscle fatigue with sustained or repeated activity. There are two major forms of the disease. In one form, the extraocular muscles are primarily affected. In the second form, there is a generalized skeletal muscle weakness. In severe cases, all muscles, including the diaphragm, can weaken and respiratory failure and death can ensue. The major structural abnormality in myasthenia gravis is the appearance of sparse, shallow, and abnormally wide or absent synaptic clefts in the motor endplate. The postsynaptic membrane has a reduced response to acetylcholine, and there is a 70–90% decrease in the number of receptors per endplate in affected muscles. Patients with myasthenia gravis have a greater than normal tendency to also have rheumatoid arthritis, systemic lupus erythematosus, and polymyositis. About 30% of patients with myasthenia gravis have a maternal relative with an autoimmune disorder. These associations suggest that individuals with myasthenia gravis share a genetic predisposition to autoimmune disease. The thymus may play a role in the pathogenesis of the disease by supplying helper T cells sensitized against thymic proteins that cross-react with acetylcholine receptors. In most patients, the thymus is hyperplastic; and 10–15% have a thymoma.

THERAPEUTIC HIGHLIGHTS

Muscle weakness due to myasthenia gravis improves after a period of rest or after administration of an **acetylcholinesterase inhibitor** such as **neostigmine** or **pyridostigmine.** Cholinesterase inhibitors prevent metabolism of acetylcholine and can thus compensate for the normal decline in released neurotransmitters during repeated stimulation. **Immunosuppressive drugs** (eg, **prednisone, azathioprine,** or **cyclosporine**) can suppress antibody production and have been shown to improve muscle strength in some patients with myasthenia gravis. **Thymectomy** is indicated especially if a thymoma is

suspected in the development of myasthenia gravis. Even in those without thymoma, thymectomy induces remission in 35% and improves symptoms in another 45% of patients.

JUNCTIONAL POTENTIALS

In smooth muscles in which noradrenergic activity is excitatory, stimulation of the noradrenergic nerves produces discrete partial depolarizations that look like small endplate potentials and are called **excitatory junction potentials (EJPs).** These potentials summate with repeated stimuli. Similar EJPs are seen in tissues excited by cholinergic discharges. In tissues inhibited by noradrenergic stimuli, hyperpolarizing **inhibitory junction potentials (IJPs)** are produced by stimulation of the noradrenergic nerves. Junctional potentials spread in an electrotonic fashion.

AXONAL INJURY & DENERVATION SUPERSENSITIVITY

Figure 6–14 shows some of the reactions triggered by injury or section of an axon. Orthograde degeneration **(Wallerian degeneration)** occurs from the point of injury to the nerve terminal, interrupting neural transmission. Distal to the injury, the membrane breaks down and the myelin sheath degenerates. The proximal axon also is affected and may undergo retrograde degeneration and die. The cell body of the injured neuron swells, the nucleus moves to an eccentric position, and the rough endoplasmic reticulum gets fragmented **(chromatolytic reaction).**

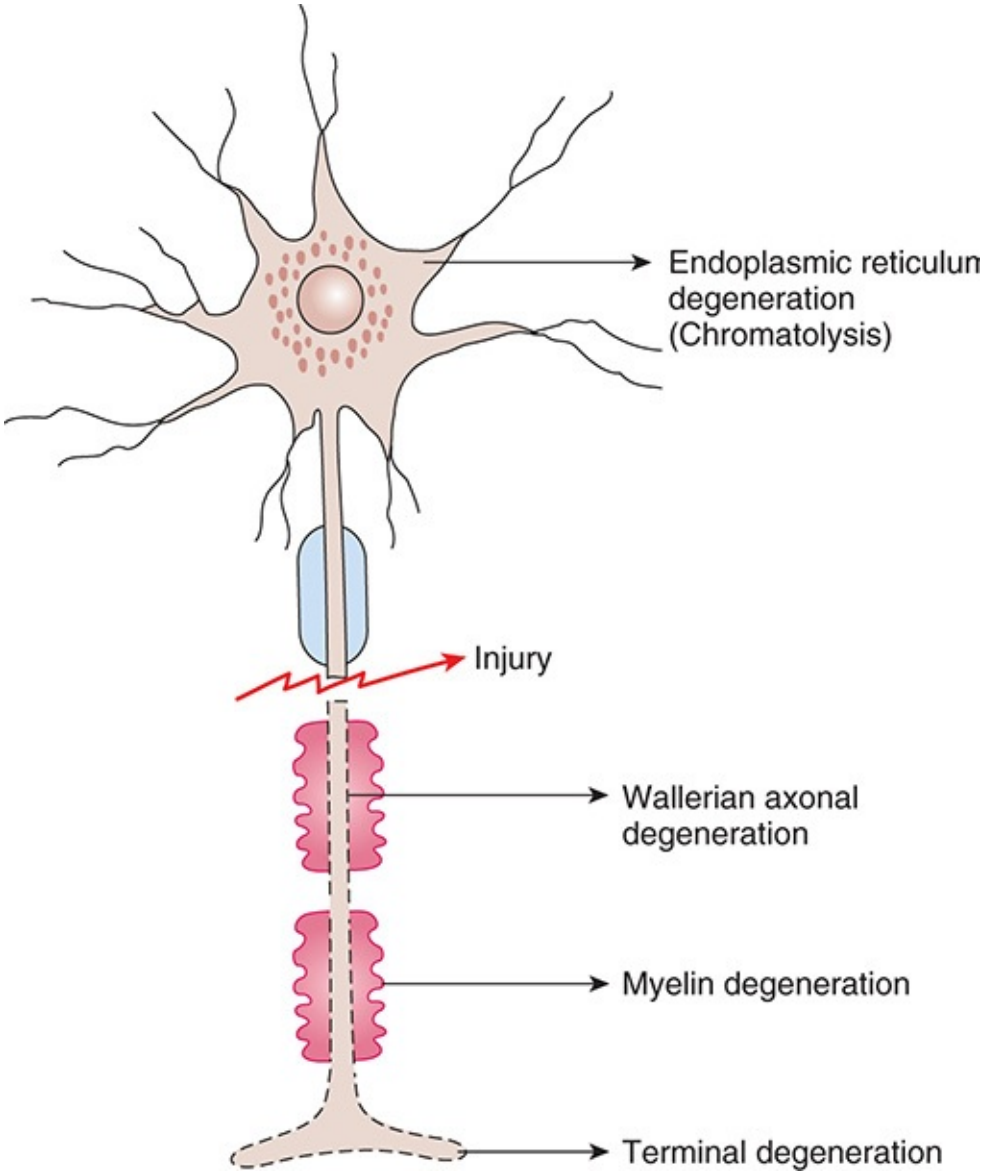


FIGURE 6–14 Changes occurring in a neuron when its axon is crushed or injured. The distal axon stump separates from the cell body, orthograde (Wallerian) degeneration occurs from the point of damage to the terminal, and the myelin sheath degenerates. The cell body of the injured neuron swells and the endoplasmic reticulum is fragmented as part of the chromatolytic reaction.

The nerve then starts to regrow, with multiple small branches projecting along the path the axon previously followed **(regenerative sprouting).** Axons sometimes grow back to their original targets, especially in locations like the neuromuscular junction. However, nerve regeneration is generally limited because axons often become entangled in the area of tissue damage at the site where they were disrupted. This difficulty has been reduced by administration of

neurotrophins (see Chapter 4).

When the motor nerve to skeletal muscle is cut and allowed to degenerate, the muscle gradually becomes extremely sensitive to acetylcholine. This is called **denervation hypersensitivity** or **supersensitivity.** Normally nicotinic receptors are located only in the vicinity of the motor endplate where the axon of the motor nerve terminates. When the motor nerve is severed, there is a marked proliferation of nicotinic receptors over a wide region of the neuromuscular junction. Denervation supersensitivity also occurs at autonomic junctions. Smooth muscle, unlike skeletal muscle, does not atrophy when denervated, but it becomes hyperresponsive to the chemical mediator that normally activates it.

CLINICAL BOX 6–3

Lambert-Eaton Syndrome

In a relatively rare condition called **Lambert-Eaton myasthenic syndrome (LEMS),** muscle weakness is caused by an autoimmune attack against one of the voltage-gated Ca^{2+} channels in the nerve endings at the neuromuscular junction. This decreases the normal Ca^{2+} influx that causes acetylcholine release. The incidence of LEMS in the United States is about 1 case per 100,000 people; it is usually an adult-onset disease that has a similar occurrence in men and women. Proximal muscles of the lower extremities are primarily affected, producing a waddling gait and difficulty raising the arms. Repetitive stimulation of the motor nerve facilitates accumulation of Ca^{2+} in the nerve terminal and increases acetylcholine release, leading to an increase in muscle strength. This is in contrast to myasthenia gravis in which symptoms are exacerbated by repetitive stimulation. About 40% of patients with LEMS also have cancer, especially small cell cancer of the lung. One theory is that antibodies that have been produced to attack the cancer cells may also attack Ca^{2+} channels, leading to LEMS. LEMS has also been associated with lymphosarcoma; malignant thymoma; and cancer of the breast, stomach, colon, prostate, bladder, kidney, or gallbladder. Clinical signs usually precede the diagnosis of cancer. A syndrome similar to LEMS can occur after the use of **aminoglycoside antibiotics,** which also impair Ca^{2+} channel function.

THERAPEUTIC HIGHLIGHTS

Since there is a high comorbidity with small cell lung cancer, the first treatment strategy is to determine if the individual also has cancer and, if so, to treat that appropriately. In patients without cancer, **immunotherapy** is initiated. **Prednisone** administration, **plasmapheresis,** and **intravenous immunoglobulin** are some examples of effective therapies for LEMS. The use of **aminopyridines** facilitates the release of acetylcholine in the neuromuscular junction and can improve muscle strength in LEMS patients. This class of drugs causes blockade of presynaptic K^+ channels and promotes activation of voltage-gated Ca^{2+} channels. Acetylcholinesterase inhibitors can be used but often do not ameliorate the symptoms of LEMS.

Denervation hypersensitivity has multiple causes. As noted in Chapter 2, a deficiency of a given chemical messenger generally produces an upregulation of its receptors. Another factor is a lack of reuptake of secreted neurotransmitters.

CHAPTER SUMMARY

- Most synapses occur on dendrites (axodendritic), but some presynaptic nerves terminate on the soma (axosomatic) or axon (axoaxonic) of a postsynaptic neuron. At chemical synapses, an impulse in the presynaptic axon causes release of a neurotransmitter from synaptic vesicles that diffuses across a synaptic cleft and binds to receptors on the postsynaptic density, triggering events that open or close ion channels in the membrane of the postsynaptic neuron. At electrical synapses, gap junctions between the presynaptic and postsynaptic neurons form low-resistance bridges through which ions pass with relative ease from one neuron to the next.
- An excitatory stimulus opens Na^+ or Ca^{2+} ion channels to produce an EPSP (inward current) in a postsynaptic neuron after a latency of 0.5 ms. An IPSP is produced by a localized increase in Cl^- transport. Slow EPSPs and IPSPs are due to decreases and increases in K^+ conductance, respectively. They occur after a latency of 100–500 ms in autonomic ganglia, cardiac and smooth muscle, and cortical neurons.
- The time constant of a neuron determines the time course of the synaptic potential; the longer the time constant, the greater is the chance for two potentials to summate to induce an action potential (temporal summation).

The length constant of a neuron determines the degree to which a depolarizing current is reduced as it spreads passively; if a neuron has a long length constant, the depolarization induced at two points on the neuron can summate and induce an action potential (spatial summation).

- Postsynaptic inhibition occurs when an inhibitory transmitter (glycine, GABA) is released from a presynaptic nerve terminal and induces an IPSP in the postsynaptic neuron. Presynaptic inhibition is mediated at axoaxonal synapses; activation of presynaptic receptors increases Cl^- conductance, decreasing the size of the action potentials reaching the excitatory ending, and reducing Ca^{2+} entry and the amount of excitatory transmitter released. Presynaptic facilitation occurs when the action potential is prolonged and the Ca^{2+} channels are open for a longer period.
- The axon terminal of motor neurons synapses on the motor endplate on the skeletal muscle membrane to form the neuromuscular junction. An impulse arriving in the motor nerve terminal leads to the entry of Ca^{2+} that triggers the exocytosis of the acetylcholine-containing synaptic vesicles. The acetylcholine diffuses and binds to nicotinic cholinergic receptors on the motor endplate, causing an increase in Na^+ and K^+ conductance; the influx of Na^+ induces the endplate potential and depolarization of the adjacent muscle membrane.
- Autonomic nerve terminals in smooth muscle form a neuroeffector junction; nerve branches are beaded with enlargements (varicosities) that contain synaptic vesicles. Release of neurotransmitter mediates EJPs or IJPs that spread electrotonically.
- When a nerve is damaged and then degenerates, the postsynaptic structure gradually becomes extremely sensitive to the transmitter released by the nerve (denervation hypersensitivity).
- Neurotoxins from bacteria can disrupt neurotransmitter release at the neuromuscular junction (botulinum toxin, tetanus toxin). Autoimmune diseases of the neuromuscular junction include myasthenia gravis (antibodies against nicotinic cholinergic receptors) and Lambert-Eaton myasthenic syndrome (antibodies against voltage-gated Ca^{2+} channels).

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. A neurology resident was conducting research in an electrophysiology laboratory and was studying the roles of cation channels in mediating nonpropagated potentials. Which of the following electrophysiologic events is correctly paired with the change in ionic currents causing the event?

   A. Fast inhibitory postsynaptic potentials (IPSPs) and closing of Cl^- channels

   B. Fast excitatory postsynaptic potentials (EPSPs) and an increase in Ca^{2+} conductance

   C. Endplate potential and an increase in Na^+ conductance

   D. Excitatory junctional potentials and closure of voltage-gated K^+ channels

   E. Slow EPSPs and an increase in K^+ conductance

2. A neurology resident was conducting research in an electrophysiology laboratory and was studying the functional anatomy of synaptic transmission. Which of the following physiologic processes is correctly paired with a structure?

   A. Electrical transmission: synaptic cleft

   B. Positive feedback inhibition: Renshaw cell

   C. Synaptic vesicle docking and fusion: presynaptic nerve terminal

   D. Endplate potential: muscarinic cholinergic receptor

   E. Action potential initiation: axon hillock

3. A medical student was studying the passive membrane properties of neurons and their ability to affect the amplitude of an EPSP recorded from the neuron. She found that in one neuron, applying two stimuli separated by 25 ms to one presynaptic input induced two EPSPs of identical amplitude. In a second neuron, the same type of stimulation induced an EPSP followed an action potential. What can she conclude from this experiment?

   A. The second neuron had a longer time constant than the first neuron.

   B. The second neuron had a shorter time constant than the first neuron.

   C. The second neuron had a longer length constant than the first neuron.

   D. The second neuron had a shorter length constant than the first neuron.

4. A medical student was doing research in a laboratory that studies the neuromuscular junction. Which of the following events at the junction are listed in correct sequential order?

   A. Motor neuron action potential, acetylcholine release, Na^+ entry at the endplate

B. Ca^{2+} entry into the motor nerve terminal, Na^{+} entry at the endplate, muscle fiber action potential

C. Na^{+} entry at the endplate, acetylcholine release, muscle fiber action potential

D. Motor neuron action potential, Na^{+} entry at the endplate, formation of an endplate potential

E. Ca^{2+} entry into the motor nerve terminal, acetylcholine release, Na^{+} entry at the endplate

5. Activation of a sensory nerve from the muscle spindle caused contraction of the extensor muscle and relaxation of the flexor muscle. The relaxation of the flexor muscle is an example of

A. negative feedback inhibition.

B. postsynaptic inhibition.

C. Renshaw cell-mediated inhibition.

D. presynaptic inhibition.

E. indirect inhibition.

6. A medical student was working in a laboratory that studied autonomic neurotransmission. He was recording a potential in a vascular smooth muscle from stimulation of the postganglionic sympathetic nerve. What is the name of the structure on the nerve where the neurotransmitter is stored and what is the name of the response in the smooth muscle?

A. Synaptic vesicle and endplate potential

B. Varicosity and inhibitory postsynaptic potential

C. Large dense-core vesicle and inhibitory junctional potential

D. Varicosity and excitatory junctional potential

E. Small dense-core vesicle and excitatory postsynaptic potential

7. A 35-year-old woman sees her physician to report weakness in the extraocular eye muscles and muscles of the extremities. She states that she feels fine when she gets up in the morning, but the weakness begins soon after she becomes active. The weakness is improved by rest. The physician treats her with an anticholinesterase inhibitor, and she notes immediate return of muscle strength. Her physician diagnoses her with

A. Lambert-Eaton syndrome.

B. myasthenia gravis.

C. multiple sclerosis.

D. Parkinson disease.

E. muscular dystrophy.

8. A 55-year-old woman had an autonomic neuropathy that disrupted the sympathetic nerve supply to the pupillary dilator muscle of her right eye. While having her eyes examined, the ophthalmologist placed phenylephrine in her eyes. The right eye became much more dilated than the left eye. This suggests that

A. the sympathetic nerve to the right eye had regenerated.

B. the parasympathetic nerve supply to the right eye remained intact and compensated for the loss of the sympathetic nerve.

C. phenylephrine blocked the pupillary constrictor muscle of the right eye.

D. denervation hypersensitivity had developed.

E. the left eye also had nerve damage and so was not responding as expected.

9. A 47-year-old woman was admitted to the hospital after experiencing nausea and vomiting for about 2 days followed by severe muscle weakness and neurologic symptoms, including ptosis and dysphagia. She indicated she had eaten at a restaurant the evening before the symptoms began. Laboratory tests were positive for *Clostridium botulinum*. The basis for the muscle weakness in this case was most likely because the toxin

A. blocked the reuptake of neurotransmitter into presynaptic terminals.

B. bound irreversibly to the receptor on the postsynaptic membrane at the neuromuscular junction.

C. reached the cell body of the motor neuron by diffusion into the spinal cord.

D. exerted its adverse effects by a direct action on the skeletal muscle.

E. prevented the release of acetylcholine from motor neurons.

CHAPTER 7

Neurotransmitters & Neuromodulators

OBJECTIVES

After studying this chapter, you should be able to:

- List the major types of neurotransmitters and neuromodulators that are broadly characterized as small-molecule transmitters, large-molecule transmitters, and gas transmitters.
- Summarize the five common steps involved in the biosynthesis, release, action, and removal from the synaptic cleft of the major small-molecule and large-molecule neurotransmitters.
- Compare the actions initiated by binding of a neurotransmitter to an ionotropic (ligand-gated) versus metabotropic (G-protein-coupled, GPCR) receptor and identify the second messengers involved in mediating the actions of neurotransmitters that act on GPCRs.
- Recognize the major distribution of the various types of receptors that mediate the functional responses of the common neurotransmitters: amino acids (glutamate and GABA), acetylcholine, monoamines (norepinephrine, epinephrine, dopamine, and serotonin), and opioid peptides.
- List receptor antagonists for each of the common neurotransmitters.
- Describe the role of nitric oxide and carbon monoxide (CO) in modulating synaptic transmission.

- Provide examples of how neurotransmitter dysfunction contributes to some neuropathological disorders.

INTRODUCTION

The dominant form of neuron-to-neuron or neuron-to-effector organ communication within the mammalian nervous system is mediated by the release of a chemical **neurotransmitter** that induces excitation or inhibition of the postsynaptic target. **Neuromodulators** are chemicals released by neurons that have little or no direct effects on their own but can modify the effects of neurotransmitters. This chapter provides a summary of the major properties of some of the most common chemical neurotransmitters, including excitatory and inhibitory amino acids, acetylcholine, monoamines, and neuropeptides. For many of these chemicals, there are some common steps involved in the process of neurotransmission. These steps include uptake of a neurotransmitter precursor into a nerve terminal, biosynthesis of the neurotransmitter, its storage within **synaptic vesicles**, its release into the **synaptic cleft** in response to the arrival of a wave of depolarization into the presynaptic nerve terminal, binding of the neurotransmitter to **receptors** on the membrane of the postsynaptic target, and finally termination of its actions via diffusion away from the synapse, reuptake into the nerve terminal, or enzymatic degradation.

CHEMISTRY OF TRANSMITTERS

Many neurotransmitters and the enzymes involved in their synthesis and catabolism are localized in nerve endings. There are three main classes of chemical substances that serve as neurotransmitters and neuromodulators: small-molecule transmitters, large-molecule transmitters, and gas transmitters. Small-molecule transmitters include amino acids (eg, **glutamate, γ-aminobutyric acid [GABA]**, and **glycine**), **acetylcholine,** and monoamines (eg, **norepinephrine, epinephrine, dopamine,** and **serotonin**). Large-molecule transmitters include neuropeptides such as **substance P, enkephalin,** and **vasopressin**. Neuropeptides are often co-localized with one of the small-molecule neurotransmitters **(Table 7–1)**. Gas transmitters include nitric oxide (NO) and carbon monoxide (CO).

TABLE 7–1 Examples of co-localization of small-molecule transmitters with neuropeptides.

Small-Molecule Transmitter	Neuropeptide
Glutamate	Substance P
GABA	Cholecystokinin, enkephalin, somatostatin, substance P, thyrotropin-releasing hormone
Glycine	Neurotensin
Acetylcholine	Calcitonin gene-related protein, enkephalin, galanin, gonadotropin-releasing hormone, neurotensin, somatostatin, substance P, vasoactive intestinal polypeptide
Dopamine	Cholecystokinin, enkephalin, neurotensin
Norepinephrine	Enkephalin, neuropeptide Y, neurotensin, somatostatin, vasopressin
Epinephrine	Enkephalin, neuropeptide Y, neurotensin, substance P
Serotonin	Cholecystokinin, enkephalin, neuropeptide Y, substance P, vasoactive intestinal polypeptide

Figure 7–1 shows the biosynthesis of some common small-molecule transmitters released by neurons in the central nervous system (CNS) or peripheral nervous system. **Figure 7–2** shows the location of major groups of neurons that contain norepinephrine, serotonin, dopamine, and acetylcholine. These are some of the major central neurotransmitter and neuromodulatory systems.

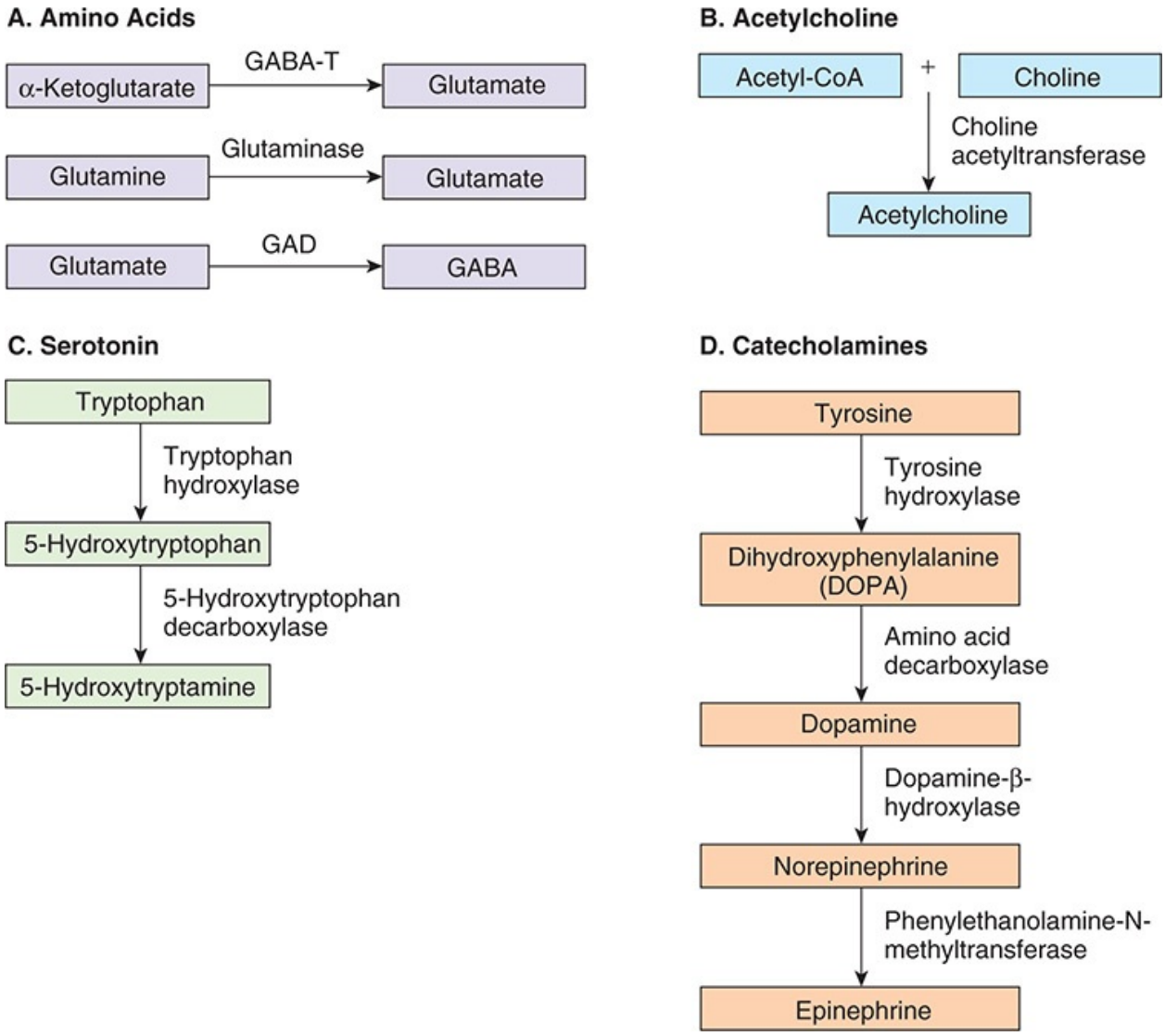


FIGURE 7–1 Biosynthesis of some common small-molecule neurotransmitters. A) Glutamate is synthesized in the Krebs cycle by the conversion of α-ketoglutarate to the amino acid via the enzyme γ-aminobutyric acid transferase (GABA-T) or in nerve terminals by the hydrolysis of glutamine by the enzyme glutaminase. GABA is synthesized by the conversion of glutamate by the enzyme glutamic acid decarboxylase (GAD). **B)** Acetylcholine is synthesized in the cytoplasm of a nerve terminal from acetyl-Co-A and choline by the enzyme choline acetyltransferase. **C)** Serotonin is synthesized from the amino acid tryptophan in a two-step process: the enzymatic hydroxylation of tryptophan to 5-hydroxytryptophan and the enzymatic decarboxylation of this intermediate to form 5-hydroxytryptamine (also called serotonin). **D)** Catecholamines are synthesized from the amino acid tyrosine by a multi-step process. Tyrosine is oxidized to dihydroxyphenylalanine (DOPA) by the enzyme

tyrosine hydroxylase in the cytoplasm of the neuron; DOPA is then decarboxylated to dopamine. In dopaminergic neurons, the process stops there. In noradrenergic neurons, the dopamine is transported into synaptic vesicles where it is converted to norepinephrine by dopamine-β-hydroxylase. In neurons that also contain the enzyme phenylethanolamine-N-methyltransferase, norepinephrine is converted to epinephrine.

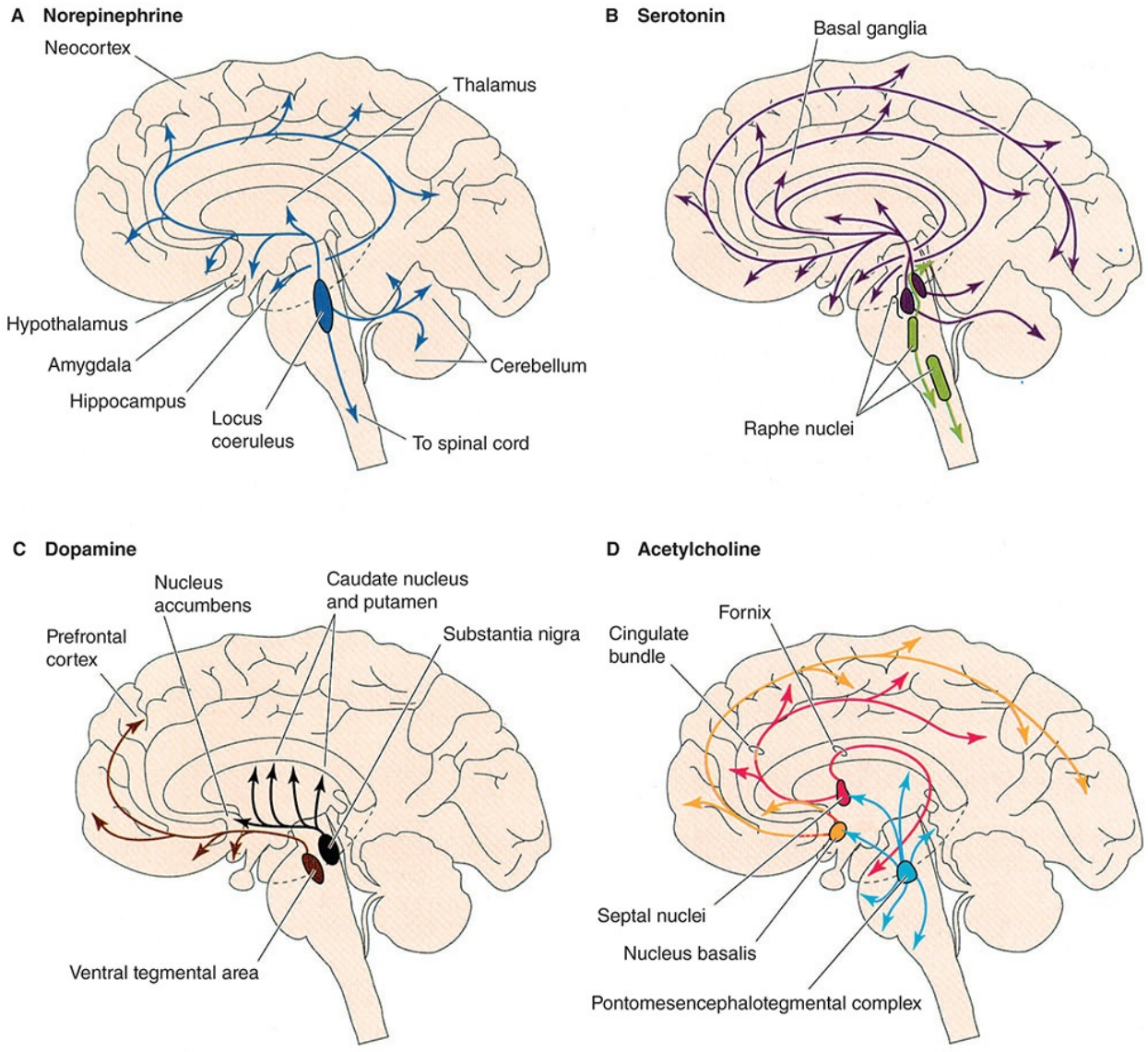


FIGURE 7–2 Four diffusely connected systems of central neuromodulators. **A)** Noradrenergic neurons in the locus coeruleus innervate the spinal cord, cerebellum, several nuclei of the hypothalamus, thalamus, basal telencephalon, and neocortex. **B)** Serotonergic neurons in the raphe nuclei project to the hypothalamus, limbic system, neocortex, cerebellum, and spinal cord. **C)**

Dopaminergic neurons in the substantia nigra project to the striatum and those in the ventral tegmental area of the midbrain project to the prefrontal cortex of the limbic system. **D)** Cholinergic neurons in the basal forebrain complex project to the hippocampus and the neocortex and those in the pontomesencephalotegmental cholinergic complex project to the dorsal thalamus and the forebrain. (Reproduced with permission from Boron WF, Boulpaep EL: *Medical Physiology*. St. Louis, MO: Elsevier; 2005.)

RECEPTORS

The action of a chemical mediator on its target structure is more dependent on the type of receptor on which it acts than on the properties of the mediator per se. The individual receptors, along with their ligands (the molecules that bind to them), are discussed in the following parts of this chapter. However, five common themes about the actions of ligands on receptors have emerged.

First, each chemical mediator has the potential to act on many subtypes of receptors. For example, norepinephrine acts on α_1-, α_2-, β_1-, β_2-, and β_3-adrenergic receptors. This multiplies the possible effects of a given ligand and makes its effects in each cell more selective.

Second, receptors for many neurotransmitters are located on both presynaptic and postsynaptic elements. A presynaptic receptor called an **autoreceptor** often inhibits further release of the transmitter, providing feedback control. For example, norepinephrine acts on α_2-presynaptic receptors to inhibit additional norepinephrine release. A presynaptic **heteroreceptor** is one whose ligand is a chemical other than the transmitter released by the nerve ending on which the receptor is located. For example, norepinephrine acts on a heteroreceptor on a cholinergic nerve terminal to inhibit the release of acetylcholine. In some cases, presynaptic receptors facilitate the release of neurotransmitters.

Third, receptors are grouped into two large families based on structure and function: **ligand-gated channels** (also known as **ionotropic receptors**) and **metabotropic receptors** (also known as **G-protein-coupled receptors [GPCRs]).** In the case of ionotropic receptors, a membrane channel is opened when a ligand binds to the receptor; and activation of the channel usually elicits a brief (few to tens of milliseconds) increase in ionic conductance. Thus, these receptors are important for fast synaptic transmission. Metabotropic receptors are 7-transmembrane GPCRs, and binding of a neurotransmitter to these receptors initiates the production of a second messenger that modulates the

voltage-gated channels on neuronal membranes. The receptors for some neurotransmitters and neuromodulators are listed in **Table 7–2**, along with their principal second messengers and, where established, their net effect on ion channels. This table is an over-simplification. For example, activation of α_2-adrenergic receptors decreases intracellular cyclic adenosine monophosphate (cAMP) concentrations, but there is evidence that the G-protein activated by α_2-adrenergic presynaptic receptors also acts directly on Ca^{2+} channels to inhibit norepinephrine release.

TABLE 7–2 Pharmacology of a selection of receptors for some small-molecule neurotransmitters.

Neurotransmitter	Receptor	Second Messenger	Net Channel Effects	Agonists	Antagonists
Glutamate	AMPA		↑Na^+, K^+	AMPA	CNQX, DNQX
	Kainate		↑Na^+, K^+	Kainate	CNQX, DNQX
	NMDA		↑Na^+, K^+, Ca^{2+}	NMDA	AP5, AP7
	$mGluR_1$	↑cAMP, IP_3, DAG	↓K^+, ↑Ca^{2+}	DHPG	
	$mGluR_5$	↑IP_3, DAG	↓K^+, ↑Ca^{2+}	Quisqualate	
	$mGluR_2$, $mGluR_3$	↓cAMP	↑K^+, ↓Ca^{2+}	DCG-IV	
	$mGluR_4$, $mGluR_{6-7}$	↓cAMP	↓Ca^{2+}	L-AP4	
GABA	$GABA_A$		↑Cl^-	Muscimol	Bicuculline, Gabazine, Picrotoxin
	$GABA_B$	↑IP_3, DAG	↑K^+, ↓Ca^{2+}	Baclofen	Saclofen
Glycine	Glycine		↑Cl^-	Taurine, β-alanine	Strychnine
Acetylcholine	N_M		↑Na^+, K^+	Nicotine	Tubocurarine, Gallamine triethiodide
	N_N		↑Na^+, K^+	Nicotine, lobeline	Trimethaphan
	M_1, M_3, M_5	↑IP_3, DAG	↑Ca^{2+}	Muscarine, Bethanechol, Oxotremorine (M_1)	Atropine, Pirenzipine (M_1)
	M_2, M_4	↓cAMP	↑K^+	Muscarine, Bethanechol (M_2)	Atropine, Tropicamide (M_4)
Norepinephrine	α_1	↑IP_3, DAG	↓K^+	Phenylephrine	Prazosin, Tamsulosin
	α_2	↓cAMP	↑K^+, ↓Ca^{2+}	Clonidine	Yohimbine
	β_1	↑cAMP	↓K^+	Isoproterenol, Dobutamine	Atenolol, Esmolol
	β_2	↑cAMP		Albuterol	Butoxamine
Serotonin	$5HT_{1A}$	↓cAMP	↑K^+	8-OH-DPAT	Metergoline, Spiperone
	$5HT_{1B}$	↓cAMP		Sumatriptan	
	$5HT_{1D}$	↓cAMP	↓K^+	Sumatriptan	
	$5HT_{2A}$	↑IP_3, DAG	↓K^+	Dobutamine	Ketanserin
	$5HT_{2C}$	↑IP_3, DAG		α-Methyl-5-HT	
	$5HT_3$		↑Na^+	α-Methyl-5-HT	Ondansetron
	$5HT_4$	↑cAMP	↓K^+	5-Methoxytryptamine	

8-OH-DPAT, 8-hydroxy-N, N-dipropyl-2-aminotetralin; AMPA, α-amino-3-hydroxy-5-methyl-4-isoxazole-propionate; DAG, diacylglycerol; DCG-IV, 2-(2, 3-dicarboxycyclopropyl) glycine; DHPG, 3-5 dihydroxyphenylglycine; IP_3, inositol triphosphate; L-AP4, 2-amino-4-phosphonobutyrate; NMDA, N-methyl-D-aspartate.

Fourth, receptors are concentrated in clusters on the postsynaptic membrane close to the endings of neurons that secrete the neurotransmitters specific for them. This is generally due to the presence of specific binding proteins for them.

Fifth, in response to prolonged exposure to their ligands, most receptors become unresponsive; that is, they undergo **desensitization.** This can be of two types: **homologous desensitization,** with loss of responsiveness only to the ligand and maintained responsiveness of the cell to other ligands; and **heterologous desensitization,** in which the cell becomes unresponsive to other ligands as well.

REUPTAKE

Neurotransmitters are rapidly transported from the synaptic cleft back into the cytoplasm of the neurons that released them via a process called **reuptake,** which involves a high-affinity, Na^+-dependent membrane transporter. **Figure 7–3** illustrates the reuptake of norepinephrine in a sympathetic postganglionic nerve ending. After release of norepinephrine into the synaptic cleft, it is rapidly routed back into the sympathetic nerve terminal by a **norepinephrine transporter (NET).** A portion of the norepinephrine that re-enters the neuron is sequestered into the synaptic vesicles through the **vesicular monoamine transporter (VMAT).** There are analogous membrane and vesicular transporters for other small-molecule neurotransmitters released at other synapses in the CNS and peripheral nervous system.

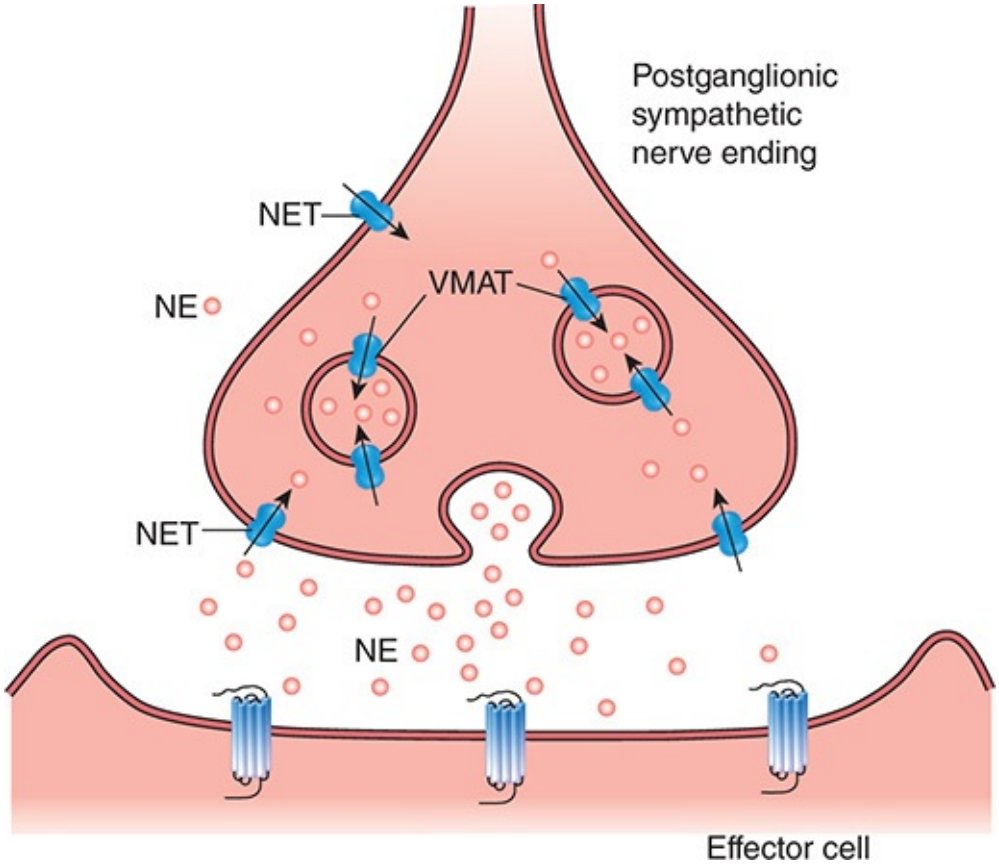


FIGURE 7–3 Fate of monoamines released at synaptic junctions. In each monoamine-secreting neuron, the monoamine is synthesized in the cytoplasm and the secretory granules and its concentration in secretory granules is maintained by the two vesicular monoamine transporters (VMAT). The monoamine is released by exocytosis of the granules, and it acts on G-protein-coupled receptors. In this example, the monoamine is norepinephrine acting on adrenoceptors. Many of these receptors are postsynaptic, but some are presynaptic and some are located on glia. In addition, there is extensive reuptake of the monoamine into the cytoplasm of the presynaptic terminal via a monoamine transporter, in this case the norepinephrine transporter (NET). (Modified with permission from Katzung BG, Masters SB, Trevor AJ: *Basic and Clinical Pharmacology*, 11th ed. New York, NY: McGraw-Hill; 2009.)

Reuptake is a major factor in terminating the action of neurotransmitters, and when it is inhibited, the effects of neurotransmitter release are increased and prolonged. This has clinical consequences. For example, several effective antidepressant drugs are inhibitors of the reuptake of amine transmitters. Glutamate reuptake into neurons and glia is important because glutamate is an excitotoxin that can kill cells by overstimulating them **(see Clinical Box 7–1)**.

During ischemia and anoxia, loss of neurons is increased because glutamate reuptake is inhibited.

SMALL-MOLECULE TRANSMITTERS: EXCITATORY & INHIBITORY AMINO ACIDS, ACETYLCHOLINE, AND MONOAMINES

Glutamate

The amino acid **glutamate** is the main excitatory neurotransmitter in the brain and spinal cord and may be responsible for 75% of the excitatory transmission in the CNS. There are two distinct pathways involved in the synthesis of glutamate (Figure 7–1). In one pathway, **α-ketoglutarate** produced by the Krebs cycle is converted to glutamate by the enzyme **GABA transaminase (GABA-T).** In the second pathway, glutamate is released from the nerve terminal into the synaptic cleft by Ca^{2+}-dependent exocytosis and transported via a **glutamate reuptake transporter** into glia, where it is converted to **glutamine** by the enzyme **glutamine synthetase (Figure 7–4)**. Glutamine then diffuses back into the nerve terminal where it is hydrolyzed back to glutamate by the enzyme **glutaminase.** A membrane transporter also returns glutamate directly into the nerve terminal. Within glutamatergic neurons, glutamate is highly concentrated in synaptic vesicles by a **vesicular glutamate transporter.**

Glutamate Receptors

Glutamate acts on both ionotropic and metabotropic receptors in the CNS (Figure 7–4). There are three subtypes of ionotropic glutamate receptors, each named for its relatively specific agonist. These are the **AMPA** (α-amino-3-hydroxy-5-methylisoxazole-4-propionate), **kainate**, and **NMDA receptors.** Table 7–2 summarizes some of the major properties of these receptors. Ionotropic glutamate receptors are tetramers composed of different subunits whose helical domains span the membranes three times and a short sequence that forms the channel pore. Four AMPA (GluR1–GluR4), five kainate (GluR5–GluR7, KA1, KA2), and six NMDA (NR1, NR2A–NR2D) subunits have been identified, and each is coded by a different gene.

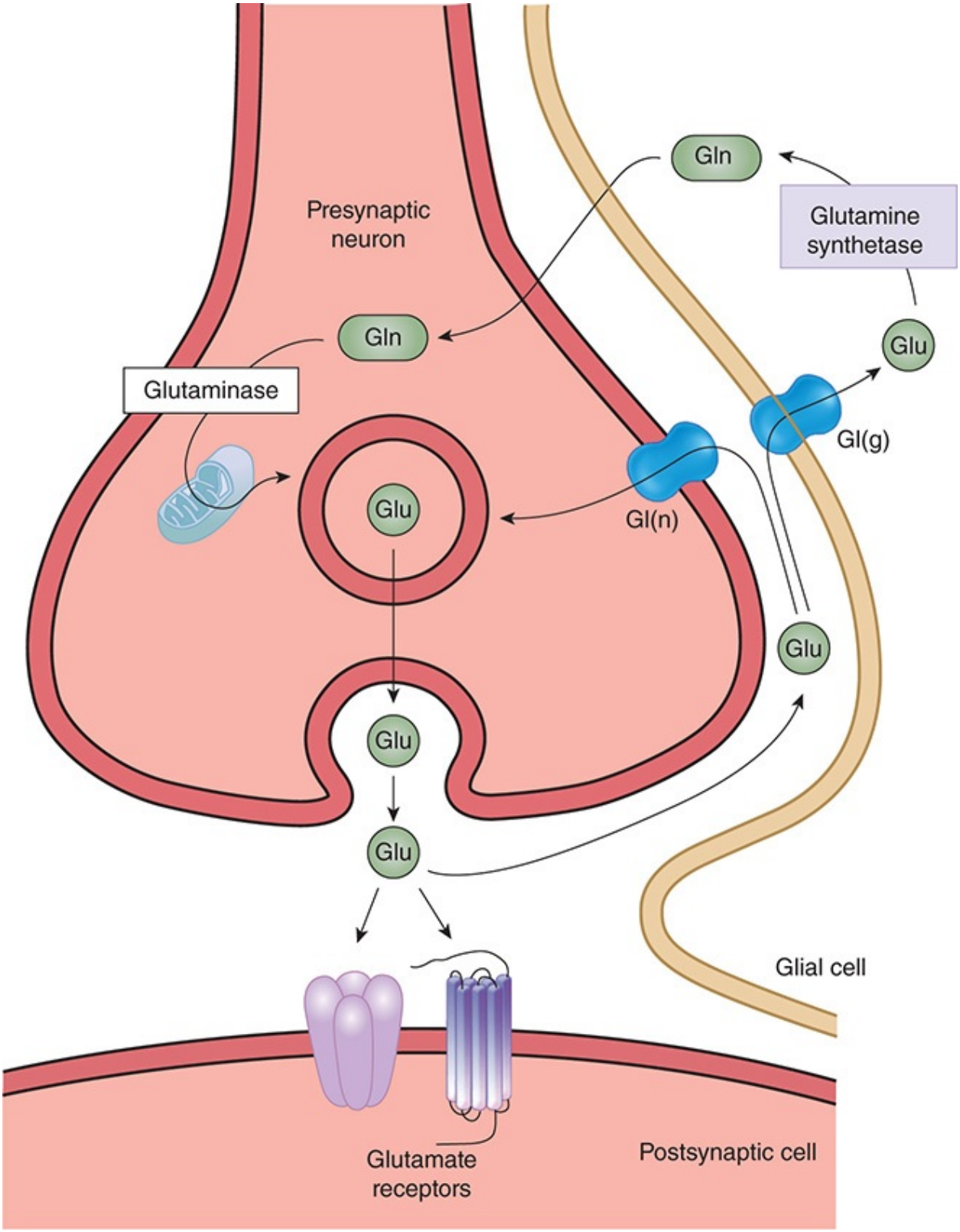


FIGURE 7–4 Biochemical events at a glutamatergic synapse. Glutamate

(Glu) released into the synaptic cleft by Ca^{2+}-dependent exocytosis. Released Glu can act on ionotropic and G-protein-coupled receptors on the postsynaptic neuron. Synaptic transmission is terminated by the active transport of Glu via by a Na^{+}-dependent glutamate transporters located on membranes of the presynaptic terminal [Gl(n)] and glia [Gl(g)]. In glia, Glu is converted to glutamine (Gln) by the enzyme glutamine synthetase; Gln then diffuses into the nerve terminal where it is hydrolyzed back to Glu by the enzyme glutaminase. In the nerve terminal, Glu is highly concentrated in synaptic vesicles by a vesicular glutamate transporter.

The release of glutamate and its binding to AMPA or kainate receptors primarily permits the influx of Na^{+} and the efflux of K^{+}, accounting for a fast-excitatory postsynaptic potential (EPSP). Most AMPA receptors have low Ca^{2+} permeability, but the absence of certain subunits in the receptor complex at some sites allows for the influx of Ca^{2+}, which may contribute to the excitotoxic effect of glutamate (see Clinical Box 7–1).

Activation of the NMDA receptor permits the influx of relatively large amounts of Ca^{2+} along with Na^{+}. When glutamate is in excess in the synaptic cleft, the NMDA receptor-induced influx of Ca^{2+} into neurons is the major basis for the excitotoxic actions of glutamate. The NMDA receptor is unique in several ways **(Figure 7–5)**. First, glycine binding to the NMDA receptor is essential for the receptor to respond to glutamate. Second, when glutamate binds to the NMDA receptor, it opens, but at normal membrane potentials, the channel is blocked by extracellular Mg^{2+}. This block is removed only when the neuron containing the receptor is partially depolarized by the activation of adjacent AMPA and kainate receptors. Third, the EPSP induced by activation of NMDA receptors is slower than that elicited by activation of the AMPA and kainate receptors.

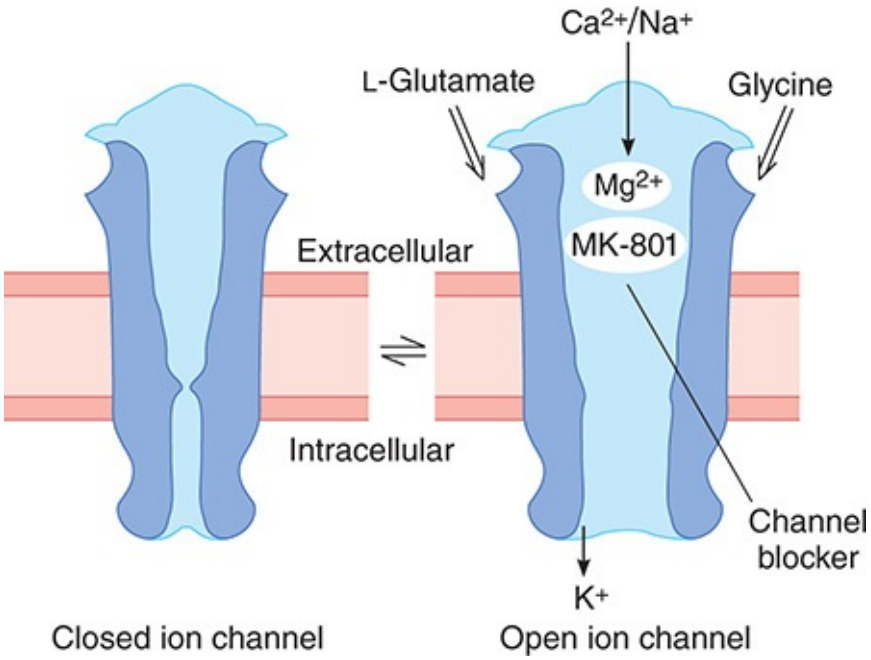


FIGURE 7–5 Diagrammatic representation of the NMDA receptor. When glycine and glutamate bind to the receptor, the closed ion channel (left) opens, but at the resting membrane potential, the channel is blocked by Mg^{2+} (right). This block is removed if partial depolarization is produced by other inputs to the neuron containing the receptor, and Ca^{2+} and Na^{+} enter the neuron. Blockade can also be produced by the drug dizocilpine maleate (MK-801).

Essentially all neurons in the CNS have both AMPA and NMDA receptors. Kainate receptors are located presynaptically on GABA-secreting nerve endings and postsynaptically at various sites, most notably in the hippocampus, cerebellum, and spinal cord. Kainate and AMPA receptors are also found in glia. The concentration of NMDA receptors in the hippocampus is high, and blockade of these receptors prevents **long-term potentiation,** a long-lasting facilitation of transmission in neural pathways following a brief period of high-frequency stimulation. Thus, these receptors may be involved in memory and learning (see Chapter 15).

Activation of the metabotropic glutamate receptors (mGluR) leads to either an increase in intracellular inositol 1,4,5-triphosphate (IP_3) and diacylglycerol (DAG) levels or a decrease in intracellular cAMP levels (Table 7–2). There are eight known subtypes of mGluR located at either presynaptic ($mGluR_{2\text{-}4,6\text{-}8}$) or postsynaptic sites ($mGluR_{1,5}$) and are widely distributed in the brain. They may be involved in the production of **synaptic plasticity,** particularly in the

hippocampus and the cerebellum. Activation of presynaptic mGluR autoreceptors on hippocampal neurons limits the release of glutamate from these neurons. Knockout of the gene *mGluR1* causes severe motor incoordination and deficits in spatial learning. Dysregulation of brain levels of the $mGluR_5$ has been linked to neurological disorders including schizophrenia, major depressive disorders, and autism.

CLINICAL BOX 7–1

Excitotoxins

Glutamate is usually cleared from the brain's extracellular fluid by Na^+-dependent uptake systems in neurons and glia, keeping only micromolar levels of the chemical in the extracellular fluid despite millimolar levels inside neurons. However, excessive levels of glutamate occur in response to ischemia, anoxia, hypoglycemia, or trauma. Glutamate and some of its synthetic agonists are unique in that when they act on neuronal cell bodies, they can produce so much Ca^{2+} influx that the neurons die. Excitotoxins play a significant role in the damage done to the brain by a **stroke.** When a cerebral artery is occluded, the cells in the severely ischemic area die. The surrounding partially ischemic cells may survive but lose their ability to maintain the transmembrane Na^+ gradient. The elevated levels of intracellular Na^+ prevent the ability of **astrocytes** to remove glutamate from the brain's extracellular fluid. Therefore, glutamate accumulates to the point that excitotoxic damage and cell death occurs in the **penumbra,** the region around the completely infarcted area. In addition, excessive glutamate receptor activation may contribute to the pathophysiology of some neurodegenerative diseases such as **amyotrophic lateral sclerosis (ALS), Parkinson disease,** and **Alzheimer disease**.

THERAPEUTIC HIGHLIGHTS

Riluzole is a voltage-gated channel blocker that may antagonize N-methyl-D-aspartate (NMDA) receptors. It slows the progression of impairment and modestly improves the life expectancy of patients with ALS. Another NMDA receptor antagonist **memantine** has been used to slow the progressive decline in patients with Alzheimer disease. A third NMDA receptor antagonist, **amantadine,** in conjunction with **levodopa,** improves function in patients with

Parkinson disease.

Pharmacology of Glutamate Synapses

Table 7–2 shows some of the pharmacologic properties of glutamate receptors, examples of the agonists that bind to these receptors, and some of the antagonists that prevent activation of the receptors. The clinical applications of drugs that modulate glutamatergic transmission are still in their infancy. This is because the role of glutamate as a neurotransmitter was discovered much later than that of most other small- molecule transmitters. Intraspinal or extradural administration of NMDA receptor antagonists is used for the treatment of chronic pain. The $mGluR_5$ is a target for development of drugs in the treatment of neuropsychiatric disorders.

GABA

GABA is the major inhibitory mediator in the brain and mediates both presynaptic and postsynaptic inhibition. GABA is formed by decarboxylation of glutamate (Figure 7–1) by the enzyme **glutamate decarboxylase (GAD),** in nerve endings in many parts of the brain. A **vesicular GABA transporter (VGAT)** transports GABA into secretory vesicles. GABA is metabolized primarily by transamination to succinic semialdehyde and then to succinate in the citric acid cycle. **GABA-T** is the enzyme that catalyzes the transamination. After GABA is released from a neuron, a high-affinity GABA transporter allows for its reuptake.

GABA Receptors

Three subtypes of GABA receptors have been identified: $GABA_A$, $GABA_B$, and $GABA_C$ (Table 7–2). The $GABA_A$ and $GABA_B$ receptors **(Figure 7–6)** are widely distributed in the CNS, $GABA_C$ receptors are found almost exclusively in the retina. The $GABA_A$ and $GABA_C$ receptors are ionotropic receptors, activation of which allows the entry of Cl^- into neurons to mediate fast inhibitory postsynaptic potentials (IPSP). The $GABA_B$ receptors are GPCRs that are linked via G-proteins to alter the influx of K^+ and Ca^{2+}; G_i inhibits adenylyl

cyclase to open a K^+ channel, and G_o inhibits or delays Ca^{2+} influx. Activation of $GABA_B$ receptors mediates both presynaptic and slow postsynaptic inhibition.

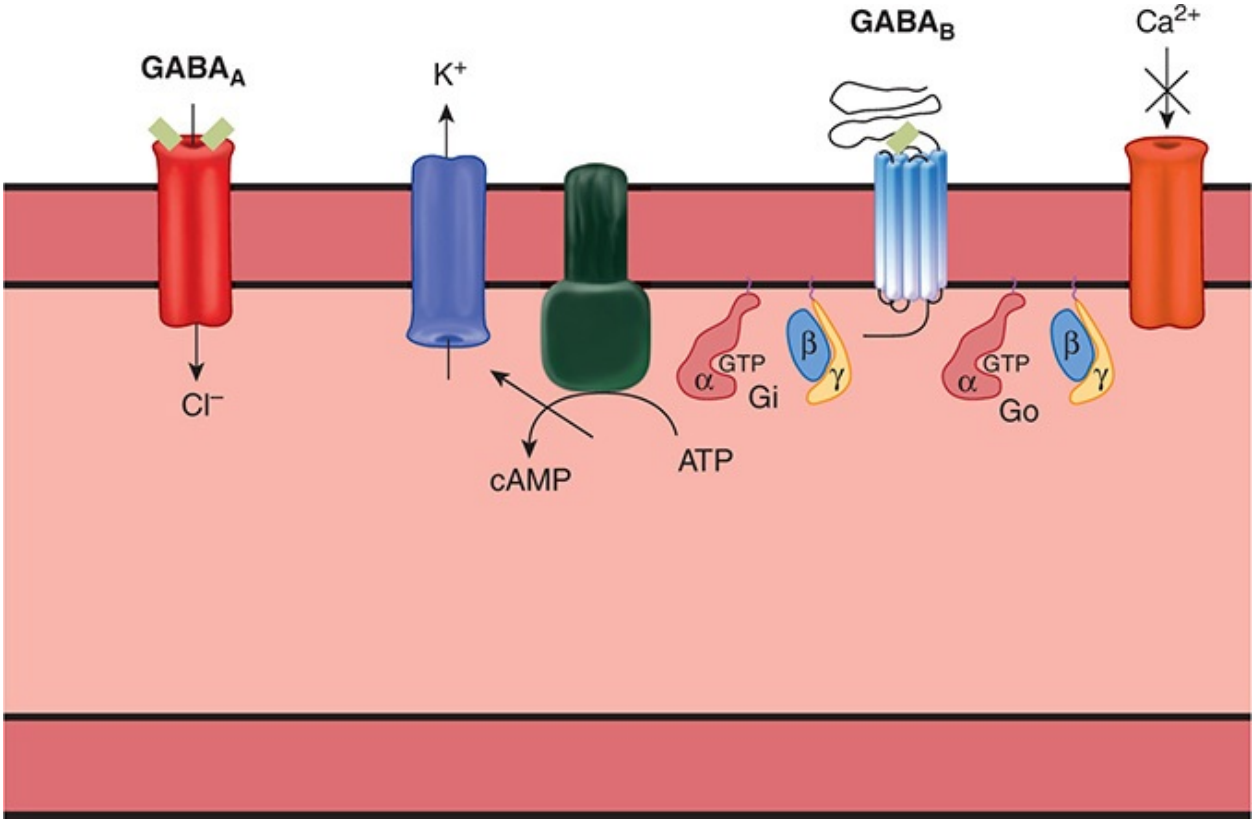


FIGURE 7–6 Diagram of $GABA_A$ and $GABA_B$ receptors, showing their principal actions. Two molecules of GABA (*squares*) bind to the $GABA_A$ receptor to allow an influx of Cl^- to mediate fast inhibitory postsynaptic potentials. One molecule of GABA binds to the $GABA_B$ receptor, which couples to the α-subunit of a G-protein; G_i inhibits adenylyl cyclase to open a K^+ channel and G_o inhibits Ca^{2+} influx.

The $GABA_A$ receptors are pentamers made up of various combinations of six α subunits, four β, four γ, one δ, and one ε. This endows them with considerably different properties from one location to another. However, most synaptic $GABA_A$ receptors have two α, two β, and one γ subunit (Figure 7–6). $GABA_A$ receptors on dendrites, axons, or somas often contain δ and ε subunits in place of the γ subunit. The $GABA_C$ receptors are relatively simple in that they are pentamers of three ρ subunits in various combinations.

Pharmacology of GABA Synapses

Table 7–2 shows some of the pharmacologic properties of GABA receptors, including examples of agonists that bind to receptors and some of the antagonists that prevent activation of these receptors. The increase in Cl^- conductance produced by $GABA_A$ receptors is potentiated by **benzodiazepines** (eg, **diazepam**). Thus, these are examples of neuromodulators. These drugs have marked antianxiety activity and are also effective muscle relaxants, anticonvulsants, and sedatives. Benzodiazepines bind to α subunits of $GABA_A$ receptors. **Barbiturates** such as **phenobarbital** are effective anticonvulsants because they enhance $GABA_A$ receptor-mediated inhibition and suppress AMPA receptor-mediated excitation. The anesthetic actions of barbiturates (**thiopental, pentobarbital,** and **methoxital**) result from their actions as agonists at $GABA_A$ receptors as well as by acting as neuromodulators of GABA transmission. Regional variation in anesthetic actions in the brain parallels the variation in subtypes of $GABA_A$ receptors.

Glycine

Glycine has both excitatory and inhibitory effects in the CNS. When it binds to NMDA receptors, it makes them more sensitive to the actions of glutamate. Glycine may spill over from synaptic junctions into the interstitial fluid and in the spinal cord; for example, it may facilitate pain transmission by NMDA receptors in the dorsal horn. However, glycine mediates direct inhibition in the brainstem and spinal cord. The glycine receptor that mediates inhibition is a Cl^- channel, a pentamer made up of two subunits: the ligand-binding α subunit and the structural β subunit. Like GABA, glycine acts by increasing Cl^- conductance. Its action is antagonized by strychnine. The clinical picture of convulsions and muscular hyperactivity produced by strychnine emphasizes the importance of postsynaptic inhibition in normal neuronal function.

There are three kinds of spinal cord neurons that mediate direct inhibition: neurons that release glycine, neurons that release GABA, and neurons that release both. Neurons that release only glycine have the glycine transporter GLYT2, those that release only GABA have VGAT, and those that release glycine and GABA have both. Glycine and GABA can be found in the same vesicles in these latter neurons.

Acetylcholine

Acetylcholine is the transmitter at the neuromuscular junction, in autonomic ganglia, and in postganglionic parasympathetic nerve-target organ junctions and some postganglionic sympathetic nerve-target junctions (see Chapter 13). In fact, acetylcholine is the transmitter released by all neurons that exit the CNS (cranial nerves, motor neurons, and preganglionic neurons). Acetylcholine is also found in the basal forebrain complex (septal nuclei and nucleus basalis), which projects to the hippocampus and neocortex, and the pontomesencephalic cholinergic complex, which projects to the dorsal thalamus and forebrain (Figure 7–2). These systems may be involved in regulation of sleep-wake states, learning, and memory (see Chapters 14 and 15).

Acetylcholine is found in small, clear synaptic vesicles in high concentration in the terminals of **cholinergic neurons.** It is synthesized in the nerve terminal from choline and acetyl-CoA by the enzyme **choline acetyltransferase (ChAT)** (Figure 7–1 and **Figure 7–7**). Choline used in the synthesis of acetylcholine is transported from the extracellular space into the nerve terminal via a Na^+-dependent **choline transporter (CHT).** Following its synthesis, acetylcholine is transported from the cytoplasm into vesicles by a **vesicle-associated transporter (VAT).** Acetylcholine is released when a nerve impulse triggers the influx of Ca^{2+} into the nerve terminal.

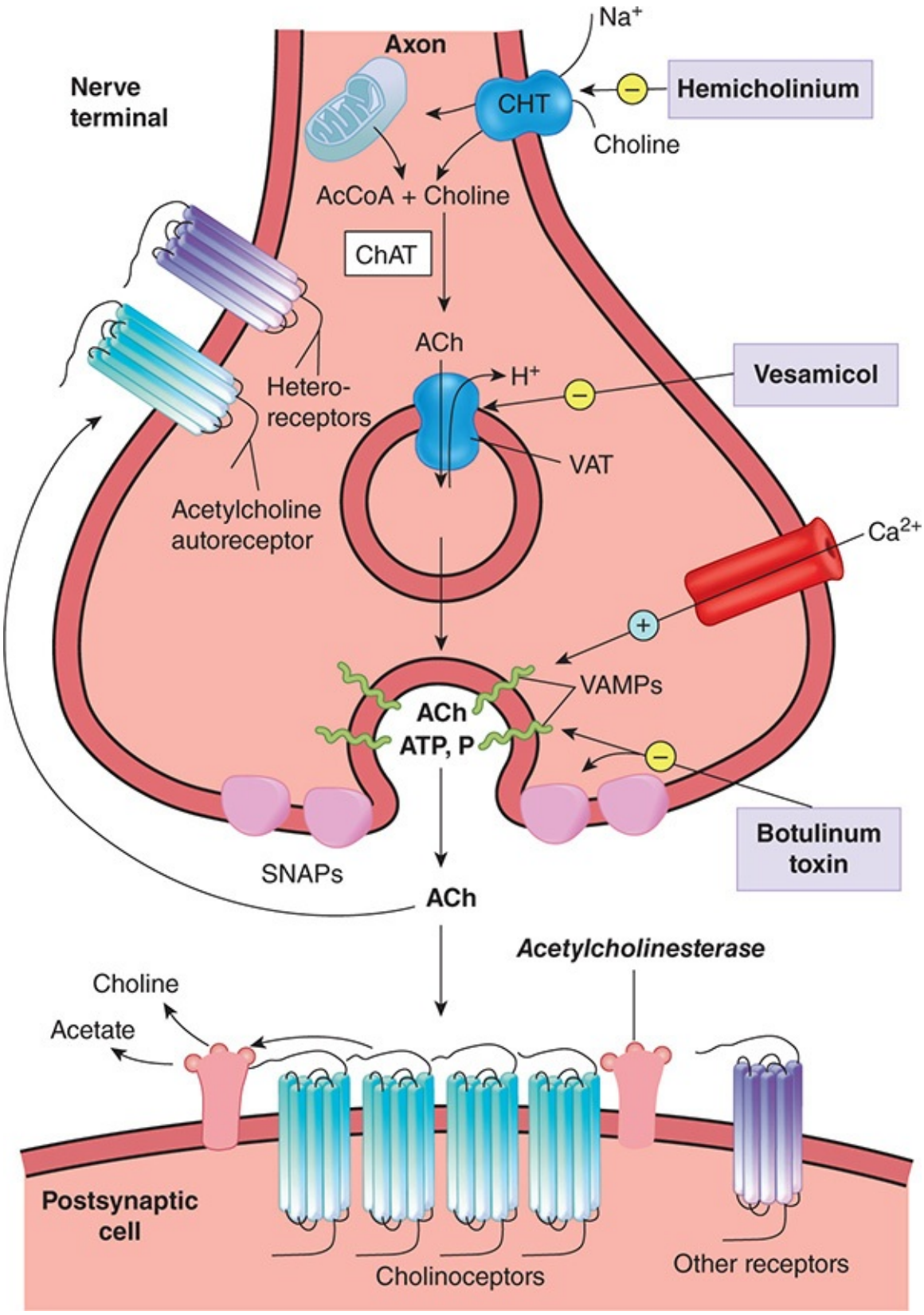


FIGURE 7–7 Biochemical events at a cholinergic synapse. Choline is

transported into the presynaptic nerve terminal by a Na^+-dependent choline transporter (CHT), which can be blocked by the drug hemicholinium. Acetylcholine (ACh) is synthesized from choline and acetyl Co-A (AcCoA) by the enzyme choline acetyltransferase (ChAT) in the cytoplasm. ACh is then transported from the cytoplasm into vesicles by the vesicle-associated transporter (VAT) along with peptides (P) and adenosine triphosphate (ATP). This step can be blocked by the drug vesamicol. ACh is released from the nerve terminal when voltage-sensitive Ca^{2+} channels open, allowing an influx of Ca^{2+}, which leads to fusion of vesicles with the surface membrane and expulsion of ACh and co-transmitters into the synaptic cleft. This process involves synaptosome-associated proteins (SNAPs) and vesicle-associated membrane proteins (VAMPs) and can be prevented by the drug botulinum toxin. The released ACh can act on muscarinic G-protein-coupled receptors on the postsynaptic target (eg, smooth muscle) or on nicotinic ionotropic receptors in autonomic ganglia or the endplate of skeletal muscle (not shown). In the synaptic junction, ACh is readily metabolized by the enzyme acetylcholinesterase. Autoreceptors and heteroreceptors on the presynaptic nerve ending modulate neurotransmitter release.

Acetylcholine must be rapidly removed from the synapse if repolarization is to occur. The removal occurs by way of hydrolysis of acetylcholine to choline and acetate, a reaction catalyzed by the enzyme **acetylcholinesterase** in the synaptic cleft. Acetylcholinesterase molecules are clustered in the postsynaptic membrane of cholinergic synapses. Hydrolysis of acetylcholine by this enzyme is rapid enough to explain the observed changes in Na^+ conductance and electrical activity during synaptic transmission.

Acetylcholine Receptors

Acetylcholine receptors are divided into two main types based on their pharmacologic properties. In sympathetic ganglia and skeletal muscle, nicotine mimics the stimulatory actions of acetylcholine. These actions of acetylcholine are called **nicotinic actions,** and the receptors involved are **nicotinic cholinergic receptors.** Nicotinic receptors are subdivided into those found at the neuromuscular junction (N_M) and those found in the CNS and autonomic ganglia (N_N). Muscarine, the alkaloid responsible for the toxicity of toadstools, mimics the stimulatory action of acetylcholine on smooth muscle and glands. These actions of acetylcholine are called **muscarinic actions,** and the receptors

involved are **muscarinic cholinergic receptors.** Both muscarinic and nicotinic acetylcholine receptors are found in the brain.

The nicotinic acetylcholine receptors are members of a superfamily of ligand-gated ion channels (ionotropic receptors). Each nicotinic cholinergic receptor is made up of five subunits that form a central channel which, when the receptor is activated, permits the passage of Na^+ and other cations. The five subunits are designated as α, β, γ, δ, and ε that are each coded by different genes. The N_M receptor is comprised of two α, one β, one δ, and either one γ or one ε subunit **(Figure 7–8)**. The N_N receptors are comprised of only α and β subunits. Each α subunit has a binding site for acetylcholine, and binding of an acetylcholine molecule to each of them induces a conformational change in the protein so that the channel opens. This increases the conductance of Na^+, and the resulting influx of Na^+ produces a depolarizing potential. A prominent feature of N_N receptors is their high permeability to Ca^{2+}. Many of the N_N receptors in the brain are located presynaptically on glutamate-secreting axon terminals, and they facilitate the release of this neurotransmitter.

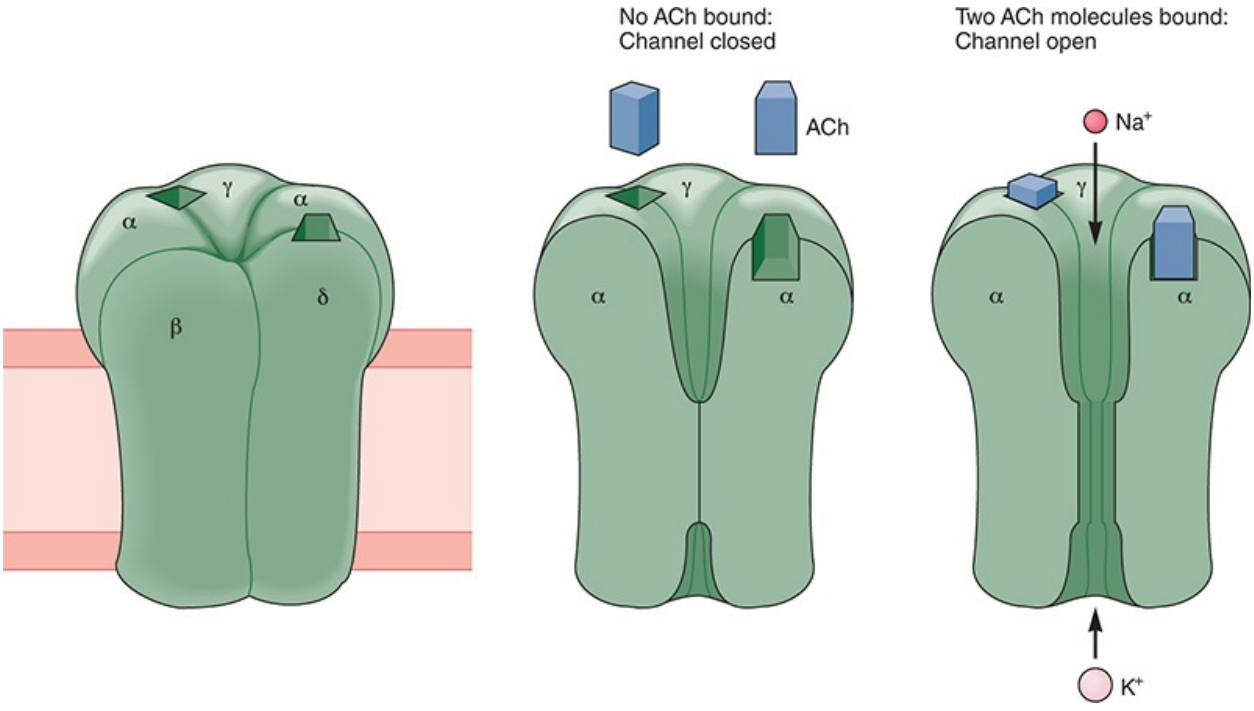


FIGURE 7–8 Three-dimensional model of the nicotinic acetylcholine-gated ion channel. The receptor–channel complex consists of five subunits, all of which contribute to forming the pore. When two molecules of acetylcholine bind to portions of the α-subunits exposed to the membrane surface, the receptor-

channel changes conformation. This opens the pore in the portion of the channel embedded in the lipid bilayer, and both K^+ and Na^+ flow through the open channel down their electrochemical gradient. (Reproduced with permission from Kandel ER, Schwartz JH, Jessell TM [editors]: *Principles of Neural Science*, 4th ed. New York, NY: McGraw-Hill; 2000.)

There are five types of muscarinic cholinergic receptors (M_1–M_5), which are encoded by five separate genes. These are metabotropic receptors that are coupled via G-proteins to adenylyl cyclase, K^+ channels, and/or phospholipase C (Table 7–2). M_1, M_4, and M_5 receptors are located in the CNS; M_2 receptors are in the heart; M_3 receptors are on glands and smooth muscle. M_1 receptors are also located on autonomic ganglia where they can modulate neurotransmission.

Pharmacology of Cholinergic Synapses

Table 7–2 shows some of the major agonists that bind to cholinergic receptors as well as some of the cholinergic receptor antagonists. Figure 7–7 also shows the site of action of various drugs that alter cholinergic transmission. For example, **hemicholinium** blocks the CHT that moves choline into the nerve terminal, and **vesamicol** blocks the VAT that moves acetylcholine into the synaptic vesicle. Also, **botulinum toxin** prevents the release of acetylcholine from the nerve terminal. Chapter 13 includes more information regarding the pharmacology of acetylcholine as related to its role in autonomic neurotransmission.

Norepinephrine & Epinephrine

Norepinephrine is the transmitter released from most sympathetic postganglionic endings where it is found in characteristic small vesicles that have a dense core (granulated vesicles). Norepinephrine and its methyl derivative, epinephrine, are also released by the adrenal medulla (see Chapter 20), but epinephrine is not a mediator at postganglionic sympathetic endings. As discussed in Chapter 6, each sympathetic postganglionic neuron has multiple varicosities along its course, and each of these varicosities is a site at which norepinephrine is released.

Norepinephrine and epinephrine are also released from neurons in the brain. Norepinephrine-containing neurons are properly called **noradrenergic neurons,** although the term **adrenergic neurons** is often used. However, the latter term should be reserved for epinephrine-containing neurons. The cell bodies of the

norepinephrine-containing neurons are found in the locus coeruleus and other medullary and pontine nuclei (Figure 7–2). From the locus coeruleus, the axons of the noradrenergic neurons descend into the spinal cord, enter the cerebellum, and ascend to innervate the paraventricular, supraoptic, and periventricular nuclei of the hypothalamus, the thalamus, the basal telencephalon, and the entire neocortex. The action of norepinephrine in these regions is primarily to modulate neurotransmission.

Biosynthesis & Release of Catecholamines

The principal **catecholamines** found in the body (norepinephrine, epinephrine, and dopamine) are formed by hydroxylation and decarboxylation of the amino acid tyrosine (Figure 7–1 and **Figure 7–9**). Some of the tyrosine is formed from phenylalanine, but most is of dietary origin. **Phenylalanine hydroxylase** is found primarily in the liver **(see Clinical Box 7–2)**. Tyrosine is transported into catecholamine-secreting neurons via a Na^+-dependent carrier. It is converted to dihydroxy-phenylalanine (dopa) and then to dopamine in the cytoplasm of the cells by **tyrosine hydroxylase** and **dopa decarboxylase,** respectively. The decarboxylase is also called **amino acid decarboxylase.** The rate-limiting step in synthesis of catecholamines is the conversion of tyrosine to dopa. Tyrosine hydroxylase is subject to feedback inhibition by dopamine and norepinephrine, thus providing internal control of the synthetic process.

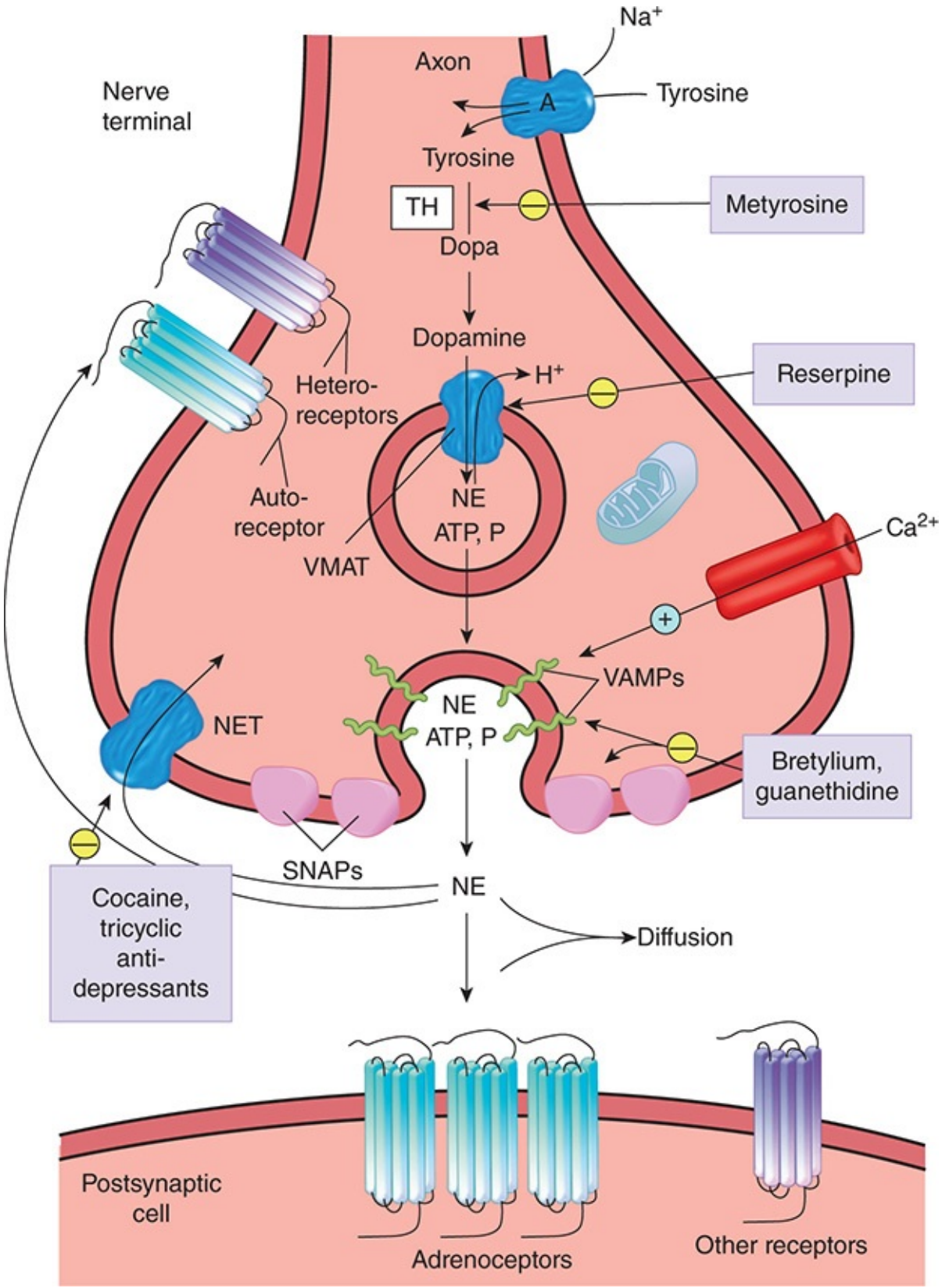


FIGURE 7–9 Biochemical events at a noradrenergic synapse. Tyrosine (Tyr)

is transported into the noradrenergic nerve terminal by a Na^+-dependent carrier (A). The steps involved in the conversion of Tyr to dopamine and dopamine to norepinephrine (NE) are described in Figure 7–1. Metyrosine blocks the action of tyrosine hydroxylase (TH), the rate-limiting step in the production of catecholamines. Dopamine is transported from the cytoplasm into the vesicle by the vesicular monoamine transporter (VMAT), which can be blocked by the drug reserpine. NE and other amines can also be carried by VMAT. Dopamine is converted to NE in the vesicle. An action potential opens voltage-sensitive Ca^{2+} channels to allow an influx of Ca^{2+}, and the vesicles then fuse with the surface membrane to trigger expulsion of NE along with peptides (P) and adenosine triphosphate (ATP). This process involves synaptosome-associated proteins (SNAPs) and vesicle-associated membrane proteins (VAMPs); it can be blocked by drugs such as guanethidine and bretylium. NE released into the nerve terminal can act on G-protein-coupled receptors on the postsynaptic neuron or neuroeffector organ (eg, blood vessels). NE can also diffuse out of the cleft or be transported back into the nerve terminal by the norepinephrine transporter (NET). NET can be blocked by cocaine and tricyclic antidepressants. Autoreceptors and heteroreceptors on the presynaptic nerve ending modulate neurotransmitter release.

Once dopamine is synthesized, it is transported into the vesicle by the **VMAT.** Here the dopamine is converted to norepinephrine by **dopamine β-hydroxylase.** Norepinephrine is the only small-molecule transmitter that is synthesized in synaptic vesicles instead of being transported into the vesicle after its synthesis in the cytoplasm.

Some neurons in the CNS and adrenal medullary cells also contain the cytoplasmic enzyme **phenylethanolamine-N-methyltransferase,** which catalyzes the conversion of norepinephrine to epinephrine. In these cells, norepinephrine leaves the vesicles, is converted to epinephrine in the cytoplasm, and then enters other vesicles for storage until it is released by exocytosis.

Catabolism of Catecholamines

Norepinephrine, like other amine and amino acid transmitters, is removed from the synaptic cleft by binding to postsynaptic receptors, binding to presynaptic receptors, reuptake into the presynaptic neurons, or catabolism. Reuptake via a **NET** is a major mechanism to terminate the actions of norepinephrine (Figure 7–3), and the hypersensitivity of sympathetically denervated structures is explained

in part on this basis. After the noradrenergic neurons are cut, their endings degenerate with loss of NET to remove norepinephrine from the synaptic cleft. Consequently, more norepinephrine from other sources is available to stimulate the receptors on the autonomic effectors.

Epinephrine and norepinephrine are metabolized to biologically inactive products by oxidation and methylation. The former reaction is catalyzed by **monoamine oxidase (MAO)** and the latter by **catechol-O-methyltransferase (COMT).** MAO is located on the outer surface of the mitochondria. MAO is widely distributed and is plentiful in the nerve endings from which catecholamines are released. COMT is also widely distributed, particularly in the liver, kidneys, and smooth muscles. In the brain, it is present in glial cells, and small amounts are found in postsynaptic neurons, but none is found in presynaptic noradrenergic neurons.

Extracellular epinephrine and norepinephrine are for the most part O-methylated, and measurement of the concentrations of the O-methylated derivatives normetanephrine and metanephrine in the urine is a good index of their rate of release. The O-methylated derivatives that are not excreted are largely oxidized, and vanillylmandelic acid is the most plentiful catecholamine metabolite in the urine.

CLINICAL BOX 7–2

Phenylketonuria

Phenylketonuria (PKU) is an example of an inborn error of metabolism. PKU is characterized by severe mental deficiency and the accumulation in the blood, tissues, and urine of large amounts of **phenylalanine** and its keto acid derivatives. It is usually due to decreased function resulting from mutation of the gene for **phenylalanine hydroxylase.** This gene is located on the long arm of chromosome 12. Catecholamines are still formed from tyrosine, and the cognitive impairment is largely due to accumulation of phenylalanine and its derivatives in the blood. The condition can also be caused by **tetrahydrobiopterin (BH4) deficiency.** Because BH4 is a cofactor for tyrosine hydroxylase and tryptophan hydroxylase, as well as phenylalanine hydroxylase, PKU cases due to BH4 deficiency have catecholamine and serotonin deficiencies in addition to hyperphenylalaninemia. These cause hypotonia, inactivity, and developmental problems. BH4 is also essential for the synthesis of NO by NO synthase. Severe BH4 deficiency can lead to

impairment of NO formation, and the CNS may be subjected to increased oxidative stress. Blood phenylalanine levels are usually determined in newborns in North America, Australia, and Europe; if PKU is diagnosed, dietary interventions should be started before the age of 3 weeks to prevent the development of mental retardation.

THERAPEUTIC HIGHLIGHTS

PKU can usually be treated successfully by markedly reducing the amount of phenylalanine in the diet. This means restricting the intake of high-protein foods such as milk, eggs, cheese, meats, and nuts. In individuals with a BH4 deficiency, treatment can include BH4, **levodopa,** and **5-hydroxytryptophan** in addition to a low- phenylalanine diet. The US Food and Drug Administration approved the drug **sapropterin,** a synthetic BH4, for the treatment of some people with PKU.

In the noradrenergic nerve terminals, some of the norepinephrine is constantly being converted by intracellular MAO to the physiologically inactive deaminated derivatives, 3,4- dihydroxymandelic acid and its corresponding glycol. These are then converted to their corresponding O-methyl derivatives, vanillylmandelic acid and 3-methoxy-4-hydroxyphenylglycol.

α- & β-Adrenoceptors

Epinephrine and norepinephrine both act on α- and β-adrenergic receptors **(adrenoceptors),** with norepinephrine having a greater affinity for α-adrenoceptors and epinephrine for β-adrenoceptors. These receptors are GPCRs, and each has multiple subtypes (α_{1A}, α_{1B}, α_{1D}, α_{2A}, α_{2B}, $\alpha_{2C,}$ and $\beta_{1\text{-}3}$). Most α_1-adrenoceptors are coupled via G_q proteins to phospholipase C, leading to the formation of IP_3 and DAG, which mobilizes intracellular Ca^{2+} stores and activates protein kinase C, respectively. Thus, at many synapses, activation of α_1-adrenoceptors is excitatory to the postsynaptic target. In contrast, α_2-adrenoceptors activate G_i inhibitory proteins to inhibit adenylyl cyclase and decrease cAMP. Other actions of α_2-adrenoceptors are to activate G-protein-coupled inward rectifier K^+ channels to cause membrane hyperpolarization and to inhibit neuronal Ca^{2+} channels. Thus, at many synapses, activation of α_2-

adrenoceptors inhibits the postsynaptic target. Presynaptic α_2-adrenoceptors are autoreceptors which, when activated, inhibit further release of norepinephrine from postganglionic sympathetic nerve terminals. β-Adrenoceptors activate a stimulatory G_S protein to activate adenylyl cyclase to increase cAMP.

α_1-Adrenoceptors are located on smooth muscle and the heart, and α_2-adrenoceptors are located in the CNS, pancreatic islets cells, and nerve terminals. β_1-Adrenoceptors are found in the heart and renal juxtaglomerular cells, β_2-adrenoceptors are located in bronchial smooth muscle and some vascular smooth muscle, and β_3-adrenoceptors are located in adipose tissue.

Pharmacology of Noradrenergic Synapses

Table 7–2 shows some of the common agonists that bind to adrenoceptors as well as some of the common adrenoceptor antagonists. Figure 7–9 also shows the site of action of various drugs that alter noradrenergic transmission. For example, **metyrosine** blocks the action of tyrosine hydroxylase, the rate-limiting step in the synthetic pathway for catecholamine production in the nerve terminal. **Reserpine** blocks the VMAT that moves dopamine into the synaptic vesicle. Also, **bretylium** and **guanethidine** prevent the release of norepinephrine from the nerve terminal. **Cocaine** and **tricyclic antidepressants** block the NET. In addition to the agonists listed in Table 7–2, some drugs mimic the actions of norepinephrine by releasing stored transmitter from the noradrenergic endings. These are called **sympathomimetics** and include **amphetamines** and **ephedrine**. Chapter 13 includes more information regarding the pharmacology of norepinephrine as related to its role in autonomic neurotransmission.

Dopamine

In some parts of the brain, catecholamine synthesis stops at dopamine (Figure 7–1), which can then be released into the synaptic cleft. Active reuptake of dopamine occurs via a Na^+- and Cl^--dependent **dopamine transporter.** Dopamine is metabolized to inactive compounds by MAO and COMT in a manner analogous to the inactivation of norepinephrine. 3,4-Dihydroxyphenylacetic acid and homovanillic acid are conjugated, primarily to sulfate.

Dopaminergic neurons are found in several brain regions (Figure 7–2). One region is the **nigrostriatal system,** which projects from the midbrain substantia

nigra to the striatum in the basal ganglia and is involved in motor control. Another dopaminergic system is the **mesocortical system;** it arises primarily in the ventral tegmental area, which projects to the nucleus accumbens and limbic subcortical areas. The mesocortical system is involved in reward behavior and addiction and in psychiatric disorders such as schizophrenia **(see Clinical Box 7–3)**. Studies using **positive emission tomography (PET)** scanning in healthy humans show that a steady loss of dopamine receptors occurs in the basal ganglia with age.

Dopamine Receptors

Five dopamine receptors have been cloned, but they fall into two major categories: D_1-like (D_1 and D_5) and D_2-like (D_2, D_3, and D_4). All dopamine receptors are GPCRs. Activation of D_1-type receptors leads to an increase in cAMP, whereas activation of D_2-like receptors reduces cAMP levels. Overstimulation of D_2 receptors may contribute to the pathophysiology of schizophrenia (Clinical Box 7–3). D_3 receptors are highly localized, especially to the nucleus accumbens (Figure 7–2). D_4 receptors have a greater affinity than the other dopamine receptors for the "atypical" antipsychotic drug **clozapine,** which is used primarily to treat schizophrenia in individuals who do not respond to other therapies.

Serotonin

Serotonin (5-hydroxytryptamine; 5-HT) is present in highest concentration in blood platelets and in the gastrointestinal tract, where it is found in the enterochromaffin cells and the myenteric plexus. It is also found within the brainstem in the midline raphe nuclei, which project to a wide portion of the CNS including the hypothalamus, limbic system, neocortex, cerebellum, and spinal cord (Figure 7–2).

Serotonin is synthesized from the essential amino acid **tryptophan** (Figure 7–1 and **Figure 7–10**). The rate-limiting step is the conversion of the amino acid to **5-hydroxytryptophan** by **tryptophan hydroxylase.** This is then converted to serotonin by the **aromatic L-amino acid decarboxylase.** Serotonin is transported into the vesicles by the VMAT. After release from serotonergic neurons, serotonin is recaptured by the relatively selective **serotonin transporter (SERT).** Once serotonin is returned to the nerve terminal, it is

either taken back into the vesicles or is inactivated by MAO to form 5-hydroxyindoleacetic acid (5-HIAA). This substance is the principal urinary metabolite of serotonin, and urinary output of 5-HIAA is used as an index of the rate of serotonin metabolism.

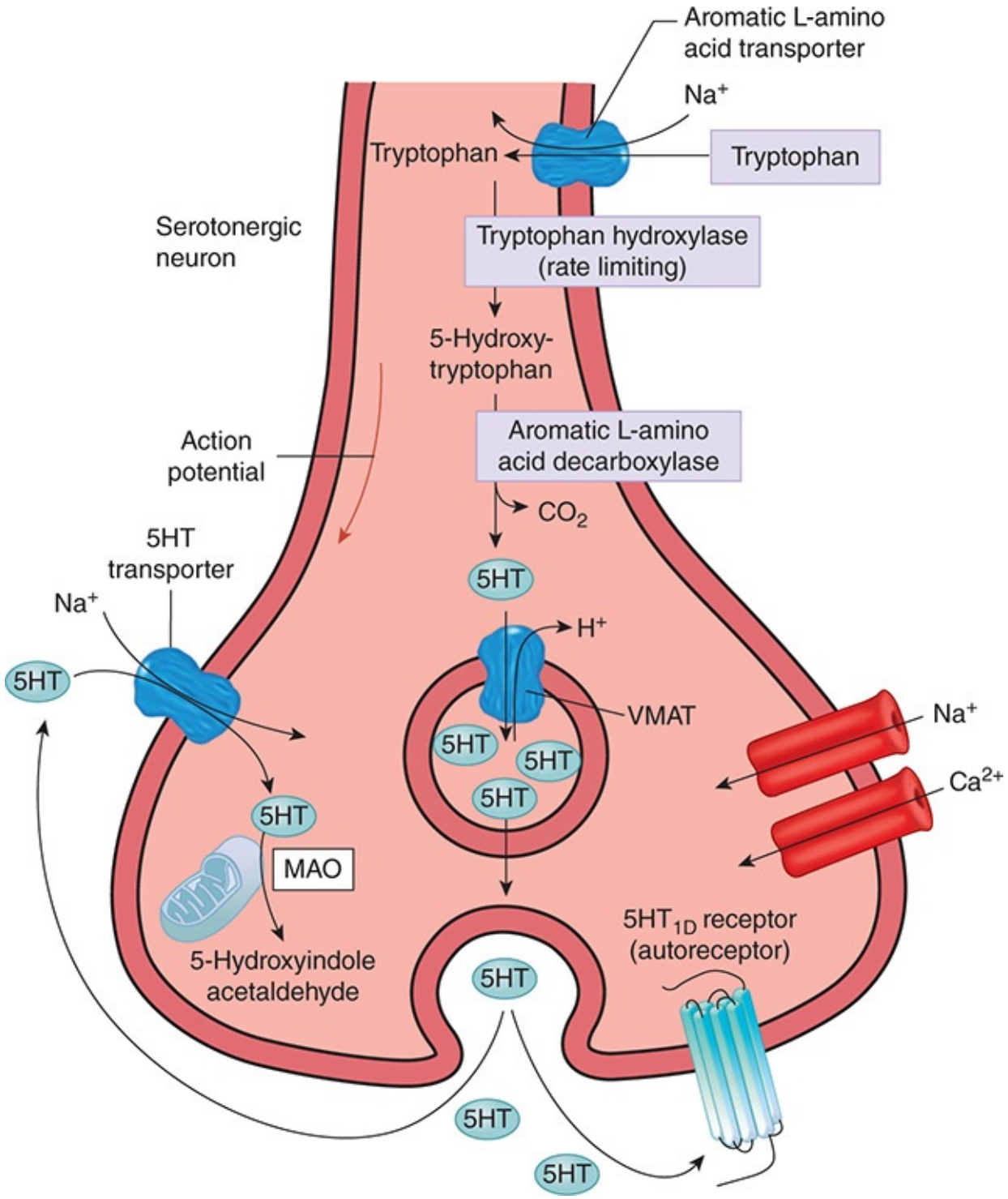


FIGURE 7–10 Biochemical events at a serotonergic synapse. Tryptophan is transported into the serotonergic nerve terminal by a Na^+-dependent aromatic L-amino acid transporter. The steps involved in the conversion of tryptophan to serotonin (5-hydroxytryptamine, 5-HT) are described in Figure 7–1. 5-HT is transported from the cytoplasm into vesicles by the vesicular monoamine transporter (VMAT). 5-HT release occurs when an action potential opens voltage-sensitive Ca^{2+} channels to allow an influx of Ca^{2+} and fusion of vesicles with the surface membrane. 5-HT released into the nerve terminal can act on G-protein-coupled receptors on the postsynaptic neuron (not shown). 5-HT can also diffuse out of the cleft or be transported back into the nerve terminal by the 5-HT transporter. 5-HT can act on presynaptic autoreceptors to inhibit further neurotransmitter release. Cytoplasmic 5-HT is either sequestered in vesicles as described or metabolized to 5-hydroxyindole acetaldehyde by mitochondrial monoamine oxidase (MAO).

CLINICAL BOX 7–3

Schizophrenia

Schizophrenia is an illness involving deficits of multiple brain systems that alter an individual's inner thoughts as well as their interactions with others. Individuals with schizophrenia suffer from hallucinations, delusions, and racing thoughts (positive symptoms); and they experience apathy, difficulty dealing with novel situations, and little spontaneity or motivation (negative symptoms). Worldwide, about 1–2% of the population lives with schizophrenia. A combination of genetic, biologic, cultural, and psychologic factors contributes to the illness. A defect in the **mesocortical system** is responsible for the development of at least some of the symptoms of schizophrenia. Attention was initially focused on overstimulation of limbic **D_2 dopamine receptors. Amphetamine,** which causes release of dopamine and norepinephrine in the brain, causes a schizophrenia-like psychosis; brain levels of D_2 receptors are elevated in individuals; and there is a positive correlation between the anti-schizophrenic activity of many drugs and their ability to block D_2 receptors. However, several recently developed drugs are effective antipsychotic agents but bind to D_2 receptors to a limited degree. Instead, they bind to D_4 receptors, and there is ongoing research into the possibility that these receptors are abnormal in individuals with

schizophrenia.

THERAPEUTIC HIGHLIGHTS

Since the mid-1950s numerous antipsychotic drugs (eg, **chlorpromazine, haloperidol, perphenazine,** and **fluphenazine**) have been used to treat schizophrenia. In the 1990s, new "atypical" antipsychotics were developed. These include **clozapine,** which reduces psychotic symptoms, hallucinations, suicidal behavior, and breaks with reality. However, a potential adverse side effect is agranulocytosis (a loss of the white blood cells), which impairs the ability to fight infections. Other atypical antipsychotics do not cause agranulocytosis, including **risperidone, olanzapine, quetiapine, ziprasidone, aripiprazole,** and **paliperidone.**

Serotonergic Receptors

There are seven classes of 5-HT receptors, and all except one (5-HT_3) are GPCRs and affect adenylyl cyclase or phospholipase C (Table 7–2). Within the 5-HT_1 group are the 5-HT_{1A}, 5-HT_{1B}, 5-HT_{1D}, 5-HT_{1E}, and 5-HT_{1F} subtypes. Within the 5-HT_2 group there are 5-HT_{2A}, 5-HT_{2B}, and 5-HT_{2C} subtypes. There are two 5-HT_5 subtypes: 5-HT_{5A} and 5-HT_{5B}. Some of the serotonin receptors are presynaptic and others are postsynaptic.

5-HT_{2A} receptors mediate platelet aggregation and smooth muscle contraction. 5-HT_3 receptors are present in the gastrointestinal tract and the area postrema and are related to vomiting. 5-HT_4 receptors are also present in the gastrointestinal tract, where they facilitate secretion and peristalsis, and in the brain. 5-HT_6 and 5-HT_7 receptors are distributed throughout the limbic system, and the 5-HT_6 receptors have a high affinity for antidepressant drugs.

Pharmacology of Serotonergic Synapses

Table 7–2 shows some of the common agonists that bind to 5-HT receptors as well as some of the common 5-HT receptor antagonists. In addition, tricyclic antidepressants inhibit the uptake of serotonin by blockade of SERT. **Selective serotonin uptake inhibitors (SSRIs)** such as **fluoxetine** are widely used in the

treatment of depression **(see Clinical Box 7–4)**.

CLINICAL BOX 7–4

Major Depression

According to the National Institutes of Mental Health, nearly 21 million Americans over the age of 18 have a mood disorder that includes **major depressive disorder**, **dysthymia**, and **bipolar disease.** The largest group is those with major depression. Major depression has a median age of onset of 32 years and is more prevalent in women than in men. Symptoms of major depression include depressed mood, anhedonia, loss of appetite, insomnia or hypersomnia, restlessness, fatigue, feelings of worthlessness, diminished ability to think or concentrate, and recurrent thoughts of suicide. **Typical depression** is characterized by feelings of sadness, early-morning awakenings, decreased appetite, restlessness, and anhedonia. Symptoms of **atypical depression** include pleasure-seeking behavior and hypersomnia.

The precise cause of depression is unknown, but genetic factors likely contribute. There is strong evidence for a role of central monoamines, including norepinephrine, serotonin, and dopamine. The hallucinogenic agent **lysergic acid diethylamide (LSD)** is a central 5-HT_2 receptor agonist. The transient hallucinations produced by this drug were discovered when the chemist who synthesized it inhaled some by accident. Its discovery drew attention to the correlation between behavior and variations in brain serotonin content. **2,5-Dimethoxy-4-methyl-amphetamine** and **mescaline** and other true hallucinogens are phenylethylamines. However, each of these may exert its effect by binding to 5-HT_2 receptors. **3,4-Methylenedioxymethamphetamine** (MDMA or **Ecstasy**) is a popular drug of abuse that produces euphoria followed by difficulty in concentrating and depression. The drug causes release of serotonin followed by serotonin depletion; the euphoria may be due to the release and the later symptoms to the depletion.

THERAPEUTIC HIGHLIGHTS

In cases of typical depression, drugs such as **fluoxetine (Prozac),** which are selective **serotonin reuptake inhibitors (SSRIs),** are effective as antidepressants. SSRIs are also used to treat **anxiety disorders.** In atypical

depression, SSRIs are often ineffective. Instead, **monoamine oxidase inhibitors (MAOIs)** such as **phenelzine** and **selegiline** have been shown to be effective as antidepressants. However, they have adverse consequences including hypertensive crisis if the patient ingests large quantities of products high in **tyramine,** which include aged cheese, processed meats, avocados, dried fruits, and red wines (especially Chianti). Based on evidence that atypical depression may result from a decrease in both serotonin and dopamine, drugs acting more generally on monoamines have been developed. These drugs, called **atypical antidepressants,** include **bupropion,** which resembles amphetamine and increases both serotonin and dopamine levels in the brain. Bupropion is also used as **smoking cessation therapy.**

LARGE-MOLECULE TRANSMITTERS: NEUROPEPTIDES

Substance P

Substance P is a polypeptide containing 11 amino acid residues that is found in the intestine, various peripheral nerves, and many parts of the CNS. It is one of a family of polypeptides called tachykinins that differ at the amino terminal end but have in common the carboxyl terminal sequence of Phe-X-Gly-LeuMet-NH_2, where X is Val, His, Lys, or Phe. Other members of the family include **neurokinin A** and **neurokinin B.**

There are three **neurokinin receptors (**$\mathbf{NK_1}$**–**$\mathbf{NK_3}$**)**, which are GPCRs. Substance P is the preferred ligand for NK_1 receptors in the CNS, and activation of this receptor leads to increased formation of IP_3 and DAG.

Substance P is found in high concentrations in the endings of primary afferent neurons in the spinal cord, and it is a mediator at the first synapse in pain transmission pathway in the dorsal horn. It is also found in high concentrations in the nigrostriatal system, where its concentration is proportional to that of dopamine, and in the hypothalamus, where it may play a role in neuroendocrine regulation. In the intestine, it is involved in peristalsis. Several recently developed centrally active NK-1 receptor antagonists have been shown to have antidepressant activity. They have also been used as antiemetics in patients undergoing chemotherapy.

Opioid Peptides

The brain and the gastrointestinal tract contain receptors that bind morphine. The search for endogenous ligands for these receptors led to the discovery of two closely related pentapeptides (**enkephalins**) that bind to these opioid receptors: **met-enkephalin** and **leu-enkephalin.** These and other peptides that bind to opioid receptors are called **opioid peptides.** The enkephalins are found in nerve endings in the gastrointestinal tract and many different parts of the brain, and they function as synaptic transmitters. They are found in the substantia gelatinosa and have analgesic activity when injected into the brainstem. They also decrease intestinal motility. Enkephalins are metabolized primarily by two peptidases: enkephalinase A, which splits the Gly-Phe bond, and enkephalinase B, which splits the Gly-Gly bond. Aminopeptidase, which splits the Tyr-Gly bond, also contributes to their metabolism.

Like other small peptides, the endogenous opioid peptides are synthesized as part of larger precursor molecules. More than 20 active opioid peptides have been identified. Each opioid peptide has a prepro form and a pro form from which the signal peptide has been cleaved. **Proenkephalin** is the precursor for met-enkephalin and leu-enkephalin in the brain and adrenal medulla. Each proenkephalin molecule contains four met-enkephalins, one leuenkephalin, one octapeptide, and one heptapeptide. **Proopiomelanocortin,** a large precursor molecule found in the anterior and intermediate lobes of the pituitary gland and the brain, contains **β-endorphin,** a polypeptide of 31 amino acid residues that has metenkephalin at its amino terminal. There are separate enkephalin-secreting and β-endorphin–secreting systems of neurons in the brain. β-Endorphin is also secreted into the bloodstream by the pituitary gland. A third precursor molecule is **prodynorphin,** a protein that contains three leuenkephalin residues associated with dynorphin and neoendorphin. Different types of dynorphins are found in the duodenum and the posterior pituitary and hypothalamus; β-neoendorphins are also found in the hypothalamus.

There are three classes of **opioid receptors:** μ, κ, and δ with various subtypes of each of these. As shown in **Table 7–3**, they differ in physiologic effects and affinity for various opioid peptides. All three are GPCRs, and all inhibit adenylyl cyclase. Activation of μ receptors increases K^+ conductance, hyperpolarizing central neurons and primary afferents. Activation of κ receptors and δ receptors closes Ca^{2+} channels.

TABLE 7–3 Physiologic effects produced by stimulation of opioid receptors.

Receptor	Endogenous Opioid Peptide Affinity	Effect
μ	Endorphins > Enkephalins > Dynorphins	Supraspinal and spinal analgesia Respiratory depression Constipation Euphoria Sedation Increased secretion of growth hormone and prolactin Miosis
κ	Enkephalins > Endorphins and Dynorphins	Supraspinal and spinal analgesia Diuresis Sedation Miosis Dysphoria
δ	Dynorphins > > Endorphins and Enkephalins	Supraspinal and spinal analgesia

Other Polypeptides

Somatostatin is found in various parts of the brain, where it may function as a neurotransmitter with effects on sensory input, locomotor activity, and cognitive function. In the hypothalamus, this growth hormone-inhibiting hormone is secreted into the portal hypophysial vessels; in the endocrine pancreas, it inhibits insulin secretion and the secretion of other pancreatic hormones; and in the gastrointestinal tract, it is an important inhibitory gastrointestinal regulator. A family of five somatostatin receptors have been identified (SSTR1 through SSTR5). All are GPCRs that inhibit adenylyl cyclase and exert various other effects on intracellular messenger systems. SSTR2 mediates cognitive effects and inhibition of growth hormone secretion, whereas SSTR5 mediates the inhibition of insulin secretion.

Vasopressin and **oxytocin** are not only secreted as hormones but also are

released by neurons that project to the brainstem and spinal cord. **Bradykinin, angiotensin II,** and **endothelin** are also released by neurons in the brain. The gastrointestinal hormones, including **vasoactive intestinal polypeptide (VIP), cholecystokinin (CCK-4** and **CCK-8),** are also found in the brain. There are two kinds of CCK receptors in the brain, CCK-A and CCK-B. CCK-8 acts at both binding sites, whereas CCK-4 acts at the CCK-B sites. **Gastrin, neurotensin, galanin,** and **gastrin-releasing peptide** are also found in the gastrointestinal tract and brain. Neurotensin, VIP, and CCK receptors are GPCRs. The hypothalamus contains both gastrin 17 and gastrin 34. VIP produces vasodilation and is found in vasomotor nerve fibers. The functions of these peptides in the nervous system are unknown, although some of the peptides also expressed in the gastrointestinal system have been implicated in satiety (see Chapter 26).

Calcitonin gene-related peptide (CGRP) is present in the CNS and peripheral nervous system, gastrointestinal tract, cardiovascular system, and urogenital system. CGRP is colocalized with either substance P or acetylcholine. CGRP-like immunoreactivity is present in the circulation, and injection of CGRP causes vasodilation. CGRP and the calcium-lowering hormone calcitonin are both products of the calcitonin gene. CGRP has little effect on Ca^{2+} metabolism, and calcitonin is only a weak vasodilator. Release of CGRP from trigeminal afferent fibers may contribute to the pathophysiology of migraine. Actions of this peptide are mediated by two types of metabotropic CGRP receptors.

Neuropeptide Y is a polypeptide that is very abundant throughout the brain and the autonomic nervous system, and it acts on eight receptors: Y_1–Y_8; except for Y_3, these are GPCRs. Activation of these receptors mobilizes Ca^{2+} and inhibits adenylyl cyclase. It acts within the CNS to increase food intake, and Y_1 and Y_5 receptor antagonists may be used to treat obesity. It also acts in the periphery to cause vasoconstriction. It acts on heteroreceptors on postganglionic sympathetic nerve terminals to reduce release of norepinephrine.

GAS TRANSMITTERS

NO, a compound released by the endothelium of blood vessels as endothelium-derived relaxing factor, is also produced in the brain and the enteric nervous system. It is synthesized from arginine, a reaction catalyzed in the brain by one of the three forms of NO synthase. It activates guanylyl cyclase and, unlike most other neurotransmitters, it is a gas, which crosses cell membranes with ease and

binds directly to guanylyl cyclase. NO is not stored in vesicles like other classic neurotransmitters; it is synthesized on demand at postsynaptic sites and diffuses to adjacent sites on the neuron. Synthesis may be triggered by activation of NMDA receptors, which leads to an influx of Ca^{2+} and activation of **neuronal nitric oxide synthase.** It may be the signal by which postsynaptic neurons communicate with presynaptic endings to enhance release of glutamate. It may also play a role in synaptic plasticity and thus in memory and learning.

CO is another diffusible gas that is endogenously formed in CNS neurons and enteric nervous system neurons via the enzymatic degradation of heme by heme oxygenase-2. Like NO it can act as an intracellular messenger to stimulate soluble guanylyl cyclase. It can modulate neurotransmission and has been implicated in having a role in olfaction, pain, and long-term potentiation. Endogenously formed CO may also act centrally to attenuate endotoxin-induced arginine vasopressin release.

OTHER CHEMICAL TRANSMITTERS

Two types of **endocannabinoids** have been identified as neurotransmitters: **2-arachidonyl glycerol** and **anandamide.** They are rapidly synthesized in response to Ca^{2+} influx after a neuron is depolarized. Both act on a cannabinoid receptor (CB_1) with a high affinity for Δ^9-tetrahydrocannabinol, the psychoactive ingredient in marijuana. These receptors are primarily located on presynaptic nerve terminals. The CB_1 receptor triggers a G-protein-mediated decrease in intracellular cAMP levels and is common in central pain pathways as well as in parts of the cerebellum, hippocampus, and cerebral cortex. In addition to inducing euphoria, CB_1 receptor agonists have an anti-nociceptive effect, and CB_1 receptor antagonists enhance nociception. Endocannabinoids also act as retrograde synaptic messengers; they travel back across a synapse after release and bind to presynaptic CB_1 receptors to inhibit further transmitter release. A CB_2 receptor, which also couples to G-proteins, is located primarily in the periphery. Agonists of this class of receptor do not induce the euphoric effects of activation of CB_1 receptors, and they may have potential for use in the treatment of chronic pain.

CHAPTER SUMMARY

- Neurotransmitters and neuromodulators can be divided into three categories: small-molecule transmitters, large-molecule transmitters (neuropeptides), and gas transmitters. Usually neuropeptides are colocalized with one of the small-molecule neurotransmitters.
- For many major neurotransmitters, the common steps involved in the process of neurotransmission include uptake of a neurotransmitter precursor into a nerve terminal; biosynthesis of the neurotransmitter; its storage in synaptic vesicles; its release into the synaptic cleft in response to depolarization of the nerve terminal; binding of the neurotransmitter to receptors on the membrane of the postsynaptic target; and rapid termination of its actions via diffusion away from the synapse, reuptake into the nerve terminal, or enzymatic degradation.
- When a neurotransmitter binds to an ionotropic receptor, the membrane channel is opened and activation of the channel usually elicits a brief increase in ionic conductance. When a neurotransmitter binds to a GPCR, it initiates the production of a second messenger (eg, cAMP, IP_3, and DAG) that modulates the voltage-gated channels on neuronal membranes.
- Major neurotransmitters and neuromodulators include amino acids (glutamate, GABA, and glycine), acetylcholine, monoamines (norepinephrine, epinephrine, dopamine, and serotonin), neuropeptides (substance P and opioids), and gasses (nitric oxide and carbon monoxide).
- Glutamate is the main excitatory neurotransmitter in the CNS. There are two major types of glutamate receptors: metabotropic (GPCR) and ionotropic (ligand-gated ion channels receptors, including kainate, AMPA, and NMDA). Riluzole and memantine are examples of clinically relevant NMDA receptor antagonists.
- GABA is the major inhibitory neurotransmitter in the brain. There are three subtypes of GABA receptors: $GABA_A$ and $GABA_C$ (ligand-gated ion channel) and $GABA_B$ (GPCR). The $GABA_A$ and $GABA_B$ receptors are widely distributed in the CNS. Bicuculline and gabazine are examples of clinically relevant $GABA_A$ receptor antagonists.
- Acetylcholine is found at the neuromuscular junction, in autonomic ganglia, and in postganglionic parasympathetic nerve-target organ junctions and a few postganglionic sympathetic nerve-target junctions. It is also found in the basal forebrain complex and pontomesencephalic cholinergic complex. There are two major types of cholinergic receptors: muscarinic (GPCR) and

nicotinic (ligand-gated ion channel receptors). Atropine is an example of a clinically relevant muscarinic receptor antagonist.

- Norepinephrine-containing neurons are found in the locus coeruleus and other medullary and pontine nuclei. Some neurons also contain phenylethanolamine-N-methyltransferase, which catalyzes the conversion of norepinephrine to epinephrine. Epinephrine and norepinephrine act on α- and β-adrenoceptors, with norepinephrine having a greater affinity for α-adrenoceptors and epinephrine for β-adrenoceptors. They are GPCR, and each has multiple subtypes. Prazosin and atenolol are examples of clinically relevant α- and β-adrenoceptor antagonists, respectively.
- Serotonin (5-HT) is found within the brainstem in the midline raphe nuclei, which project to portions of the hypothalamus, the limbic system, the neocortex, the cerebellum, and the spinal cord. There are at least seven types of 5-HT receptors, and many of these have subtypes. Most are GPCRs.
- The three types of opioid receptors (μ, κ, and δ with various subtypes of each) are all GPCRs that differ in physiologic effects, distribution in the brain and elsewhere, and affinity for various opioid peptides.
- Nitric oxide is synthesized on demand in the brain from arginine, a reaction catalyzed by nitric oxide synthase; it readily diffuses across cell membranes and binds to guanylyl cyclase. It may be the signal by which postsynaptic neurons communicate with presynaptic endings to enhance release of glutamate. Carbon monoxide is endogenously formed in neurons via the enzymatic degradation of heme; it also stimulates soluble guanylyl cyclase to modulate neurotransmission.
- Examples of neuropathologies linked to neurotransmitter dysfunction include ALS (excessive glutamate release), schizophrenia (abnormal activation of dopamine receptors), and major depression (abnormal central actions of monoamines).

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. Which of the following statements about neurotransmitters is true?

A. All neurotransmitters are derived from amino acid precursors.

B. Small-molecule neurotransmitters include dopamine, glycine, acetylcholine, enkephalin, and norepinephrine.

C. Large-molecule transmitters include GABA, endocannabinoids, substance P, and vasopressin.

D. Norepinephrine can act as a neurotransmitter in the periphery and a neuromodulator in the CNS.

E. Nitrous oxide is a neurotransmitter in the CNS.

2. Which of the following statements correctly describes the processes involved in the synthesis, storage, release, binding to a receptor, and termination of action of a common neurotransmitter?

A. Glutamate is synthesized in glia by the enzymatic conversion from glutamine and then diffuses into the neuronal terminal where it is sequestered into vesicles until released by an influx of Ca^{2+} into the cytoplasm after an action potential reaches the nerve terminal, it binds exclusively to ligand-gated ion channel receptors, and is inactivated by reuptake into the nerve terminal.

B. Serotonin is synthesized from trytophan, stored in synaptic vesicles until its release into the synaptic cleft; it then acts primarily on GPCRs and its actions are terminated primarily by reuptake into the presynaptic nerve terminal.

C. Norepinephrine is the only small-molecule transmitter that is synthesized in synaptic vesicles instead of being transported into the vesicle after its synthesis from the amino acid phenylalanine. After its release in response to depolarization, it binds to ligand-gated ion channels or GPCRs and its action is terminated by reuptake into the nerve terminal.

D. Acetylcholine is synthesized from acetlyene, transported from the cytoplasm into vesicles by a vesicle-associated membrane protein, released into the synaptic cleft in response to neuronal depolarization, acts on GPCRs, and its actions are terminated primarily by enzymatic degradation.

3. Which of the following receptors is correctly identified as an ionotropic or a GPCR and correctly paired with the ionic changes and/or second messenger induced by the binding of an agonist?

A. $5\text{-}HT_{1A}$ receptors are GPCRs whose activation increases IP_3 and DAG and increases K^+ conduction.

B. Nicotinic receptors are ionotropic receptors whose activation decreases Na^+ and K^+ conduction.

C. $GABA_A$ receptors are GPCRs whose activation increases cAMP and decreases K^+ conduction.

D. NMDA receptors are ionotropic receptors whose activation increases Na^+, K^+, and Ca^{2+} conductance.

E. Glycine receptors are GPCRs whose activation increases IP_3 and DAG and increases K^+ conductance.

4. A medical student is studying transmission through autonomic ganglia. She studied the effects of two drugs on the activity of a postganglionic neuron. Drug A induced an EPSP in the postganglionic neuron, and drug B blocked the EPSP produced by electrical stimulation of a preganglionic nerve. Drugs A and B might be the following drugs, respectively.

A. Glutamate and glycine

B. Nicotine and atropine

C. Strychnine and atenolol

D. Nicotine and trimethaphan

E. Acetylcholine and phenylephrine

5. A 38-year-old woman was referred to a psychiatrist after telling her primary care physician that she had difficulty sleeping (awakening at 4 AM frequently for the past few months) and a lack of appetite causing a weight loss of over 20 lb. She also said she no longer enjoyed going out with her friends or doing volunteer service for underprivileged children. What type of drug is her doctor most likely to suggest as an initial step in her therapy?

A. A serotonergic receptor antagonist

B. A selective serotonin reuptake inhibitor

C. An inhibitor of monoamine oxidase

D. An amphetamine-like drug

E. A drug that causes an increase in both serotonin and dopamine levels

6. A 55-year-old woman had been receiving long-term treatment with phenelzine for her depression. After she consumed Chianti wine, aged cheddar cheese, processed meats, and dried fruits one night at a party, the following symptoms developed: a severe headache, chest pain, rapid heartbeat, enlarged pupils, increased sensitivity to light, and nausea. What is the most likely cause of these symptoms?

A. The foods were contaminated with botulinum toxin.

B. She had a myocardial infarction.

C. She experienced a migraine headache.

D. She had an unexpected adverse reaction to the mixture of alcohol with her

antidepressant.

E. She had a hypertensive crisis from eating foods high in tyramine while taking a monoamine oxidase inhibitor for her depression.

7. A 27-year-old man was brought to the emergency department by a friend who suspected he had overdosed on a drug. Upon arrival at the emergency department, he had depressed respiration, miosis, and reduced consciousness. Based on these symptoms, what type of drug did the individual likely take and what is its mechanism of action?

A. A drug acting as a D_2 receptor agonist

B. A 5-HT_2 receptor antagonist

C. A δ- and κ-opioid receptor agonist

D. A serotonin reuptake inhibitor

E. A μ-opioid receptor agonist

8. Which of the following statements correctly describes properties of a gas neurotransmitter?

A. Nitric oxide is synthesized by enzymatic degradation of arginine, actively transported across cell membranes, stored in vesicles, released by neuronal depolarization, and binds to nitric oxide receptors.

B. Nitric oxide is synthesized from arginine by nitric oxide synthase, diffuses across cell membranes, is released by neuronal depolarization, and acts on presynaptic nitric oxide receptors to enhance the release of glutamate.

C. Nitric oxide is synthesized from arginine by nitric oxide synthase, diffuses across cell membranes, activates guanylyl cyclase, and may enhance release of glutamate.

D. Carbon monoxide is endogenously formed in neurons by the action of carbon monoxide synthase, stimulates soluble guanylyl cyclase to enhance endotoxin-induced arginine vasopressin release.

E. Carbon monoxide is synthesized in the periphery by enzymatic degradation of heme by heme oxygenase-2, transported to the brain where it activates guanylyl cyclase, and has a role in long-term potentiation.

9. A full-term infant boy is delivered without complications and has normal APGAR scores. Routine newborn screening tests reveal elevated levels of phenylalanine in his blood, leading to a diagnosis of phenylketonuria (PKU). What is the likely outcome if a dietary intervention with restricted intake of high-protein foods is not initiated by the age of 3 weeks?

A. Development of cholestasis
B. Development of neonatal seizures
C. Malformation of the enteric nervous system
D. Autism
E. Profound intellectual disability

SECTION II

Central & Peripheral Neurophysiology

INTRODUCTION TO NEUROPHYSIOLOGY

The central nervous system (CNS) can be likened to a computer processor that is the command center for most if not all of the functions of the body. The peripheral nervous system is like a set of cables that transfers critical data from the CNS to the body and then feeds back information from the body to the CNS. This “computer system” is very sophisticated and is designed to continually make appropriate adjustments to its inputs and outputs in order to allow one to react and adapt to changes in the external and internal environment (sensory systems), to maintain posture, permit locomotion, and use the fine motor control in our hands to create pieces of art (somatomotor system), to maintain homeostasis (autonomic nervous system), to regulate the transitions between sleep and wakefulness (consciousness), and to allow us to recall past events and to communicate with the outside world (higher cortical functions). This section on neurophysiology will describe the fundamental properties and integrative capabilities of neural systems that allow for the exquisite control of this vast array of physiologic functions. Medical fields such as neurology, neurosurgery, and clinical psychology build on the foundation of neurophysiology.

This next series of chapters on neurophysiology contains information that is relevant to many clinical problems. One of the most common reasons that an individual seeks medical advice is because they are in pain. Severe chronic pain involves the rewiring of neural circuits that can result in an unpleasant sensation from even simple touch of the skin. Chronic pain is a devastating health problem that affects nearly 1 in 10 Americans (more than 25 million people). Within the past decade or so there have been considerable advancements in understanding

how pain alters neural activity and in identifying receptors types that are unique to nociceptive pathways. These findings have led to an expanding research effort to develop novel therapies that specifically target synaptic transmission in central nociceptive pathways and in peripheral sensory transduction. This is welcomed by the many individuals who do not get pain relief from nonsteroidal anti-inflammatory drugs or even opioids. These kinds of research breakthroughs depend on a thorough understanding of how the brain and body communicate with each other.

In addition to chronic pain, there are over 600 known neurologic disorders. Nearly 50 million people in the United States alone and an estimated 1 billion people worldwide suffer from the effects of damage to the central or peripheral nervous system. Each year, nearly 7 million people die of a neurologic disorder or its complications. Neurologic disorders include genetic disorders (eg, Huntington disease), demyelinating diseases (eg, multiple sclerosis), developmental disorders (eg, cerebral palsy), degenerative diseases that target specific types of neurons (eg, Parkinson disease and Alzheimer disease), an imbalance of neurotransmitters (eg, depression, anxiety, and eating disorders), trauma (eg, spinal cord and head injury), and convulsive disorders (eg, epilepsy). In addition, there are neurologic complications associated with cerebrovascular problems (eg, stroke) and exposure to neurotoxic chemicals (eg, nerve gases, mushroom poisoning, and pesticides).

Identifying the pathophysiologic basis for neurologic disorders has benefitted from advances in stem cell biology and brain imaging techniques, a greater understanding of the basis for synaptic plasticity of the brain, knowledge about the regulation of receptors and the release of neurotransmitters, and the detection of genetic and molecular defects that lead to neurologic problems. They have also set the stage for identifying better therapies to prevent, reverse, or stabilize the physiologic deficits that more than 600 neurologic disorders cause.

CHAPTER 8

Somatosensory Neurotransmission: Touch, Pain, & Temperature

OBJECTIVES

After studying this chapter, you should be able to:

- Describe the location, type, and function of receptors that mediate the sensations of touch, temperature, and pain.
- Describe the steps involved in sensory transduction and action potential generation in cutaneous mechanoreceptors and nociceptors.
- Explain the basic elements of sensory coding including modality, location, intensity, and duration and how these properties relate to receptor specificity, receptive field, receptor sensitivity, and receptor adaptation.
- Explain the differences between pain and nociception, first and second pain, acute and chronic pain, and hyperalgesia and allodynia.
- Describe and explain the basis for visceral and referred pain.
- Compare the pathway that mediates sensory input from touch, proprioceptive, and vibratory senses to that mediating information from nociceptors and thermoreceptors.
- Describe the deficits caused by lesions of ascending sensory pathways that mediate touch, pain, and temperature.

- Describe processes involved in modulation of transmission in pain pathways.
- Identify drugs used for relief of pain and give the rationale for their use and their clinical effectiveness.

INTRODUCTION

Table 8–1 provides a list of the principle sensory modalities. Sensory receptors convert specific forms of energy into action potentials in sensory neurons. Cutaneous **mechanoreceptors** mediate responses to touch and pressure. **Proprioceptors** in muscles, tendons, and joints relay information about muscle length and tension. **Thermoreceptors** detect the sensations of warmth and cold. **Nociceptors** respond to potentially harmful stimuli such as pain, extreme heat, and extreme cold. **Chemoreceptors** are stimulated by a change in the chemical composition of the local environment. These include receptors for taste and smell as well as visceral receptors that are sensitive to changes in the plasma level of O_2, pH, and osmolality. **Photoreceptors** in the rods and cones in the retina respond to light.

TABLE 8–1 Principle sensory modalities.

Sensory System	Modality	Stimulus	Receptor Class	Receptor Cell Type
Somatosensory	Touch	Tap, flutter 5–40 Hz	Cutaneous mechanoreceptor	Meissner corpuscle
Somatosensory	Touch	Motion	Cutaneous mechanoreceptor	Hair follicle receptor
Somatosensory	Touch	Vibration 60–500 Hz; deep pressure	Cutaneous mechanoreceptor	Pacinian corpuscle
Somatosensory	Touch	Touch, sustained pressure	Cutaneous mechanoreceptor	Merkel cell
Somatosensory	Touch	Skin stretch, vibration	Cutaneous mechanoreceptor	Ruffini corpuscle
Somatosensory	Proprioception	Stretch	Mechanoreceptor	Muscle spindle
Somatosensory	Proprioception	Tension	Mechanoreceptor	Golgi tendon organ
Somatosensory	Temperature	Thermal	Thermoreceptor	Cold and warm receptors
Somatosensory	Pain	Chemical, thermal, and mechanical	Chemoreceptor, thermoreceptor, and mechanoreceptor	Polymodal receptors or chemical, thermal, and mechanical nociceptors
Somatosensory	Itch	Chemical	Chemoreceptor	Chemical nociceptor
Visual	Vision	Light	Photoreceptor	Rods, cones
Auditory	Hearing	Sound	Mechanoreceptor	Hair cells (cochlea)
Vestibular	Balance	Angular acceleration	Mechanoreceptor	Hair cells (semicircular canals)
Vestibular	Balance	Linear acceleration, gravity	Mechanoreceptor	Hair cells (otolith organs)
Olfactory	Smell	Chemical	Chemoreceptor	Olfactory sensory neuron
Gustatory	Taste	Chemical	Chemoreceptor	Taste buds

This chapter describes primarily the characteristics of cutaneous receptors that mediate the sensory modalities of touch, pain, and temperature; the way they generate impulses in afferent neurons; and the central pathways that mediate or modulate information from these receptors. Since pain is one of the main reasons an individual seeks the advice of a clinician, this topic gets considerable attention in this chapter. Receptors involved in the somatosensory modality of proprioception are described in Chapter 12 as they play key roles in the control of balance, posture, and limb movement.

SENSORY RECEPTORS

CUTANEOUS MECHANORECEPTORS

Touch and pressure are sensed by four types of mechanoreceptors **(Figure 8–1).** **Meissner corpuscles** are dendrites encapsulated in connective tissue beneath the epidermis of glabrous (non-hairy) skin and respond to slow vibration. **Merkel cells** are expanded dendritic endings in epidermis of glabrous skin that respond to sustained pressure and touch. **Ruffini corpuscles** are enlarged dendritic endings with elongated capsules in the dermis of glabrous and hairy skin; they respond to stretch and fluttering vibration. **Pacinian corpuscles** are the largest cutaneous mechanoreceptor, 2 mm long and about 1 mm in diameter, in the dermis of glabrous and hairy skin. They are comprised of a nerve ending encapsulated by concentric layers of connective tissue that give it an onion-like appearance. These receptors respond to fast vibration and deep pressure. The sensory nerves from cutaneous mechanoreceptors are myelinated Aα and Aβ fibers whose conduction velocities range from ~70–120 to ~40–75 m/s, respectively.

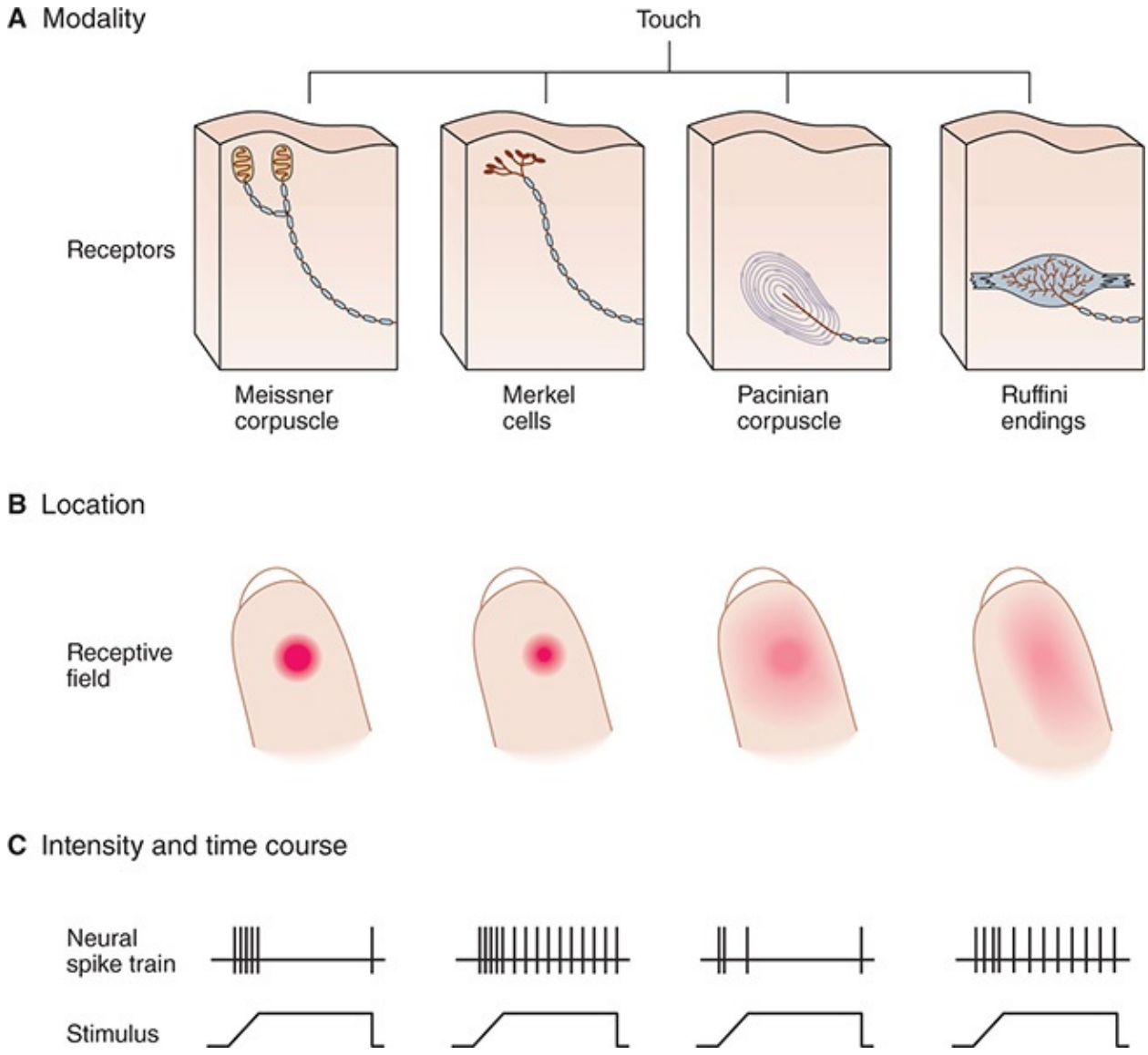


FIGURE 8–1 Sensory systems encode four elementary attributes of stimuli: modality, location, intensity, and duration. A) The human hand has four types of mechanoreceptors; their combined activation produces the sensation of contact with an object. Selective activation of Merkel cells and Ruffini endings causes sensation of steady pressure; selective activation of Meissner and Pacinian corpuscles causes tingling and vibratory sensation. **B)** Location of a stimulus is encoded by spatial distribution of the population of receptors activated. A receptor fires only when the skin close to its sensory terminals is touched. The receptive fields of mechanoreceptors (shown as red areas on fingertips) differ in size and response to touch. Merkel cells and Meissner corpuscles provide the most precise localization as they have the smallest receptive fields and are most sensitive to pressure applied by a small probe. **C)**

Stimulus intensity is signaled by firing rates of individual receptors; duration of stimulus is signaled by time course of firing. The spike trains indicate action potentials elicited by pressure from a small probe at the center of each receptive field. Meissner and Pacinian corpuscles adapt rapidly, the others adapt slowly. (Reproduced with permission from Kandel ER, Schwartz JH, Jessell TM [editors]: *Principles of Neural Science*, 4th ed. New York, NY: McGraw-Hill; 2000.)

NOCICEPTORS

Pain and temperature sensations arise from receptors located on unmyelinated dendrites of sensory neurons located throughout the glabrous and hairy skin as well as deep tissue. **Mechanical nociceptors** respond to strong pressure (eg, from a sharp object). **Thermal nociceptors** are activated by skin temperatures above 45°C or by severe cold (<20°C). **Chemically sensitive nociceptors** respond to chemicals such as bradykinin, histamine, high acidity, and environmental irritants. **Polymodal nociceptors** respond to combinations of these stimuli.

Impulses from nociceptors are transmitted via thinly myelinated Aδ fibers (2–5 µm in diameter) that conduct at rates of ∼12–35 m/s and unmyelinated C fibers (0.4–1.2 µm in diameter) that conduct at low rates of ∼0.5–2 m/s. Activation of Aδ fibers, which release **glutamate,** is responsible for **first pain** (also called **fast pain**), which is a rapid response and mediates the discriminative aspect of pain or the ability to localize the site and intensity of the noxious stimulus. Activation of C fibers, which release a combination of glutamate and **substance P,** is responsible for the delayed **second pain** (also called **slow pain**), which is the dull, intense, diffuse, and unpleasant feeling associated with a noxious stimulus. **Itch** is also related to pain sensation **(see Clinical Box 8–1)**.

A variety of receptors on the endings of nociceptive sensory nerves respond to noxious thermal, mechanical, or chemical stimuli **(Figure 8–2)**. Many of these are part of a family of nonselective cation channels called **transient receptor potential** (TRP) channels. This includes **TRPV1** receptors (the V refers to a group of chemicals called **vanilloids**) that are activated by intense heat, acids, and chemicals such as **capsaicin** (the active ingredient in hot peppers and an example of a vanilloid). TRPV1 receptors can also be activated indirectly by initial activation of TRPV3 receptors in keratinocytes in the skin. Noxious mechanical, cold, and chemical stimuli may activate **TRPA1** receptors (A, for **ankyrin**) on sensory nerve terminals. Sensory nerve endings also have **acid-**

sensing ion channel (ASIC) receptors that are activated by pH changes within a physiologic range and may be the dominant receptors mediating acid-induced pain. In addition to direct activation of receptors on nerve endings, some nociceptive stimuli release intermediate molecules that then activate receptors on the nerve ending. For example, nociceptive mechanical stimuli cause the release of adenosine triphosphate (ATP) that acts on **purinergic receptors** (eg, **P2X**, an ionotropic receptor, and **P2Y**, a G-protein-coupled receptor). **Tyrosine receptor kinase A (TrkA)** is activated by **nerve growth factor (NGF)** that is released due to tissue damage.

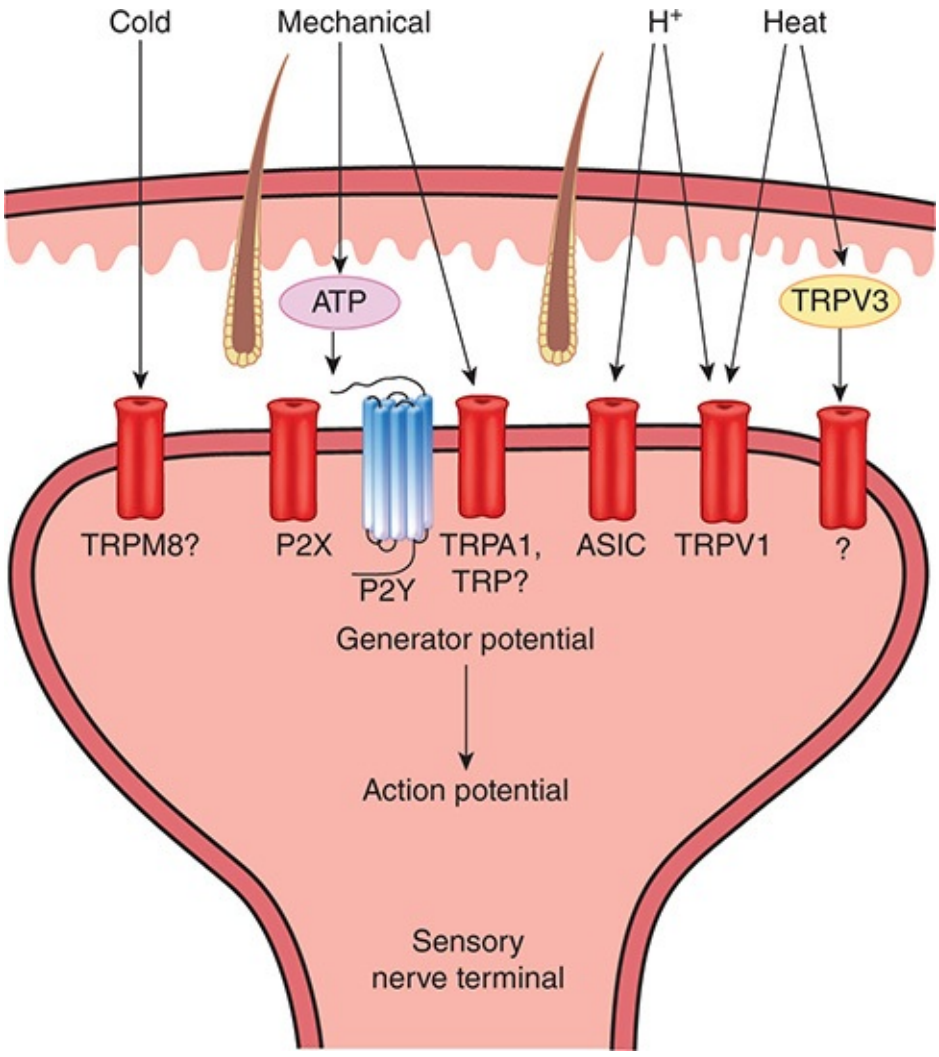


FIGURE 8–2 Receptors on nociceptive unmyelinated nerve terminals in the skin. Nociceptive stimuli (eg, heat) can activate some receptors directly due to transduction of the stimulus energy by receptors (eg, transient receptor potential (TRP) channel TRPV1) or indirectly by activation of TRP channels on

keratinocytes (eg, TRPV3). Nociceptors (eg, mechanoreceptors) can also be activated by the release of intermediate molecules (eg, ATP). ASIC, acid-sensitive ion channel; P2X, ionotropic purinoceptor; P2Y, G-protein-coupled purinergic receptor.

CLINICAL BOX 8–1

Itch

Itching (**pruritus**) is not a major problem for healthy individuals, but severe itching that is difficult to treat occurs in diseases such as chronic kidney disease, some forms of liver disease, atopic dermatitis, and HIV infection. There may be an itch- specific sensory pathway since itch spots have been mapped on the skin and itch-specific fibers have been identified in the ventrolateral spinothalamic tract. Itching can be produced not only by repeated local mechanical stimulation of the skin but also by a variety of chemical agents, including **histamine** and kinins such as **bradykinin**, that are released in the skin in response to tissue damage. Kinins exert their effects by activation of two types of G-protein-coupled receptors, B_1 and B_2. Activation of bradykinin B_2 receptors is a downstream event in **protease-activated receptor-2 (PAR-2)** activation, which induces both a nociceptive and a pruritogenic response.

THERAPEUTIC HIGHLIGHTS

Simple scratching relieves itching because it activates large, fast-conducting afferents that gate transmission in the dorsal horn in a manner analogous to the inhibition of pain by stimulation of similar afferents. **Antihistamines** are primarily effective in reducing pruritus associated with an allergic reaction. In a mouse model exhibiting scratching behavior in response to activation of PAR-2, treatment with a B_2 receptor antagonist reduced the scratching behavior. B_2 receptor antagonists may be a useful therapy for treating pruritic conditions.

Nerve endings also have a variety of receptors that respond to immune mediators that are released in response to tissue injury and mediate inflammatory pain (see Chapter 3). These include B_1 and B_2 receptors (bradykinin), prostanoid receptors (prostaglandins), and cytokine receptors (interleukins).

THERMORECEPTORS

Innocuous **cold receptors** are on dendritic endings of Aδ and C fibers; innocuous **warm receptors** are on C fibers. Mapping experiments show that the skin has discrete cold and heat-sensitive spots. There are 4–10 times as many cold-sensitive as heat-sensitive spots.

TRPM8 receptors are activated by moderate cold; the M refers to **menthol,** the ingredient in mint that gives it its "cool" taste. Cold receptors are inactive at temperatures of 40°C, but then steadily increase their firing rate as skin temperature falls to about 24°C. As skin temperature further decreases, the firing rate of cold receptors decreases until the temperature reaches 10°C. Below that temperature, they are inactive and cold, which becomes an effective local anesthetic.

TRPV3 and **TRPV4** receptors on sensory nerve endings are activated when skin temperatures reach 33–39°C and 25–34°C, respectively. The firing rate of warm receptors can further increase as the skin temperature reaches about 45°C; they become silent if skin temperature is further increased to induce a sensation of pain.

GENERATION OF IMPULSES IN CUTANEOUS RECEPTORS

The way that activation of a sensory receptors leads to an action potential in the nerve that innervates them varies based on the complexity of the sensory receptor. **Figure 8–3** shows how this occurs in the Pacinian corpuscle. A myelinated portion of the sensory nerve along with the first node of Ranvier of the sensory nerve is inside the receptor; the second node of Ranvier is usually near the point at which the nerve fiber leaves the corpuscle.

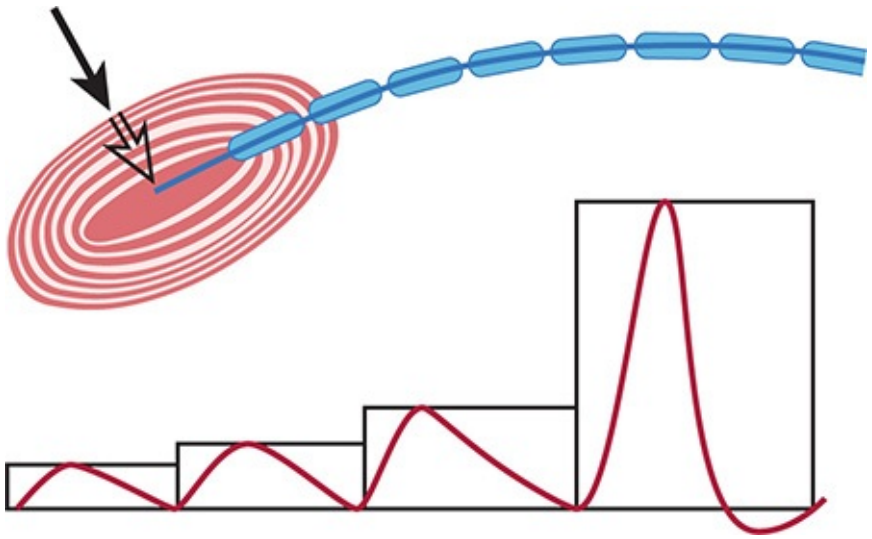


FIGURE 8–3 Demonstration that the generator potential in a Pacinian corpuscle originates in the unmyelinated nerve terminal. The electrical responses to pressures (black arrow) of 1, 2×, 3×, and 4× are shown. The strongest stimulus produced an action potential in the sensory nerve, originating in the center of the corpuscle. (Reproduced with permission from Waxman SG: *Clinical Neuroanatomy*, 26th ed. New York, NY: McGraw-Hill; 2010.)

When a small amount of pressure is applied to a Pacinian corpuscle, a nonpropagated depolarizing potential resembling an excitatory postsynaptic potential is induced (Figure 8–3). The receptor converts mechanical energy into an electrical response, the magnitude of which is proportional to the intensity of the stimulus **(graded receptor potential).** When the magnitude of the generator potential reaches about 10 mV, an action potential is produced at the first node of Ranvier. The nerve then repolarizes. If the generator potential is great enough, the neuron fires again as soon as it repolarizes, and it continues to fire if the generator potential is large enough to bring the membrane potential of the node to the firing level. Thus, the node converts the graded response of the receptor into action potentials, the frequency of which is proportional to the magnitude of the applied stimulus.

SENSORY CODING

Converting a receptor stimulus to a recognizable sensation is termed **sensory coding.** All sensory systems code for four elementary attributes of a stimulus: modality, location, intensity, and duration. **Modality** is the type of energy transmitted by the stimulus. **Location** is the site on the body or space where the

stimulus originated. **Intensity** is signaled by the response amplitude or frequency of action potential generation. **Duration** refers to the time from start to end of a response in the receptor. These attributes of sensory coding are shown for the modality of touch in Figure 8–1.

MODALITY

A sensory receptor is specialized to respond to either mechanical, chemical, thermal, or electromagnetic stimuli **(receptor specificity).** The form of energy to which a receptor is most sensitive is called its **adequate stimulus.** The adequate stimulus for the rods and cones in the eye, for example, is light. Sensory receptors can respond to forms of energy other than their adequate stimuli, but the threshold for these nonspecific responses is much higher. Pressure on the eyeball will stimulate the rods and cones, for example, but the threshold of these receptors to pressure is much higher than the threshold of the pressure receptors in the skin.

LOCATION

A **sensory unit** is a single sensory axon and all its peripheral branches. The **receptive field** of a sensory unit is the spatial distribution from which a stimulus produces a response in the unit (Figure 8–1). Representation of the senses in the skin is punctate. If the skin is carefully mapped, millimeter by millimeter, with a fine hair, a sensation of touch is evoked from spots overlying touch receptors. None is evoked from the intervening areas. Similarly, temperature sensations and pain are produced by stimulation of the skin only over the spots where the receptors for these modalities are located.

One of the most important mechanisms that enable localization of a stimulus site is **lateral inhibition.** Information from sensory neurons whose receptors are at the peripheral edge of the stimulus is inhibited compared to information from the sensory neurons at the center of the stimulus. Thus, lateral inhibition enhances the contrast between the center and periphery of a stimulated area and increases the ability of the brain to localize a sensory input. Lateral inhibition underlies **two-point discrimination** **(see Clinical Box 8–2)**.

CLINICAL BOX 8–2

Neurologic Exam

The **two-point threshold test** is used to measure the size of the receptive fields for light touch. In this procedure, the two points on a pair of calipers are simultaneously positioned on the skin and one determines the minimum distance between the two caliper points that can be perceived as separate points of stimulation **(two-point discrimination threshold).** If the distance is very small, each caliper point is touching the receptive field of only one sensory neuron. If the distance between stimulation points is less than this threshold, only one point of stimulation can be felt. Thus, the two-point discrimination threshold is a measure of **tactile acuity.** The magnitude of two-point discrimination thresholds varies from place to place on the body and is smallest where touch receptors are most abundant. Stimulus points on the back, for instance, must be separated by at least 65 mm before they can be distinguished as separate, whereas on the fingertips two stimuli are recognized if they are separated by as little as 2 mm. Blind individuals benefit from the tactile acuity of fingertips to facilitate the ability to read Braille; the dots forming Braille symbols are separated by 2.5 mm. Two-point discrimination is used to test the integrity of the **dorsal column (medial lemniscus) system,** the central pathway for touch and proprioception.

Vibratory sensibility is tested by applying a vibrating (128-Hz) tuning fork to the skin on the fingertip, tip of the toe, or bony prominences of the toes. The normal response is a “buzzing” sensation that is most marked over bones. **Pallesthesia** is the ability to feel mechanical vibrations. The receptors involved are those for touch, especially **Pacinian corpuscles,** but a time factor is also necessary. A pattern of rhythmic pressure stimuli is interpreted as vibration. The impulses responsible for the vibrating sensation are carried in the dorsal columns. Degeneration of this part of the spinal cord occurs in poorly controlled diabetes, pernicious anemia, vitamin B_{12} deficiencies, or early tabes dorsalis. An increased threshold for vibratory stimuli is an early symptom of this degeneration. Vibratory sensation and proprioception are closely related; when one is diminished, so is the other.

Stereognosis is the perception of the form and nature of an object without looking at it. Healthy persons can readily identify objects such as keys and coins of various denominations. This ability depends on intact touch and pressure sensation and is compromised if the dorsal columns are damaged. **Tactile agnosia** is the inability to identify an object by touch. Impaired stereognosis can be an early sign of damage to the cerebral cortex and sometimes occurs in the absence of any detectable defect in touch and

pressure sensation when there is a lesion in the primary sensory cortex. Stereognosia can also be expressed by the failure to identify an object by sight **(visual agnosia),** the inability to identify sounds or words **(auditory agnosia)** or color **(color agnosia),** or the inability to identify the position of an extremity **(position agnosia).**

INTENSITY

The intensity of sensation is determined by the amplitude of the stimulus applied to the receptor **(Figure 8–4)**. As a greater pressure is applied to the skin, the receptor potential in the mechanoreceptor increases (not shown), and the frequency of the action potentials in a single axon transmitting information to the CNS is increased. Weak stimuli activate the receptors with the lowest thresholds, and stronger stimuli activate those with higher thresholds **(receptor sensitivity).** As the stimulus strength is increased, it also spreads over a large area to "recruit" fibers in the surrounding area. Some of the receptors activated are part of the same sensory unit, and impulse frequency in the unit therefore increases. Because of overlap and interdigitation of one unit with another, receptors of other units are also stimulated and more sensory fibers fire. In this way, more afferent pathways are activated, which is interpreted in the brain as an increase in intensity of the sensation.

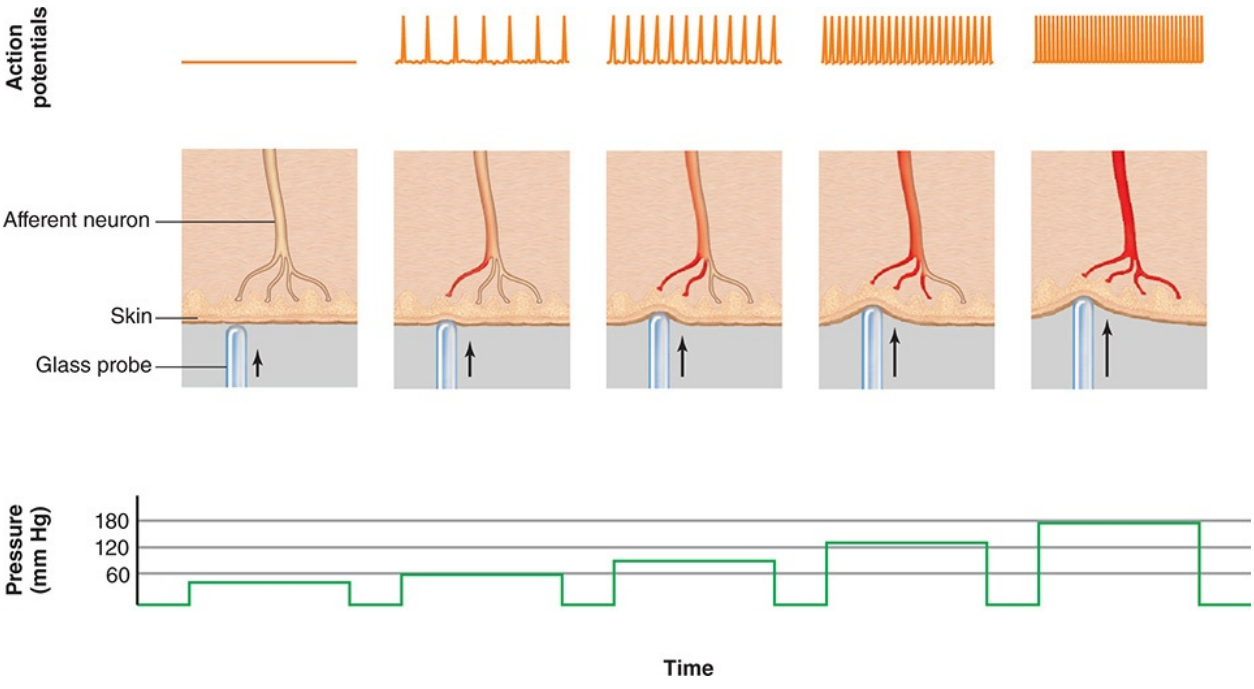


FIGURE 8–4 Relationship between stimulus and impulse frequency in an afferent fiber. Action potentials in an afferent fiber from a mechanoreceptor of a single sensory unit increase in frequency as branches of the afferent neuron are stimulated by pressure of increasing magnitude. (Reproduced with permission from Widmaier EP, Raff H, Strang KT: *Vanders Human Physiology*. 11th ed. New York, NY: McGraw-Hill; 2008.)

DURATION

If a stimulus of constant strength is maintained on a sensory receptor, the frequency of the action potentials in its sensory nerve declines over time. This phenomenon is known as **receptor adaptation.** The degree to which adaptation occurs varies from one sensation to another. Receptors can be classified as **rapidly adapting (phasic) receptors** or **slowly adapting (tonic) receptors.** This is illustrated for the four types of touch receptors in Figure 8–1; Meissner and Pacinian corpuscles are rapidly adapting receptors, and Merkel cells and Ruffini endings are slowly adapting receptors. Muscle spindles and nociceptors are also slowly adapting receptors. Different types of sensory adaptation have some value to the individual. Light touch would be distracting if it were persistent; and, conversely, slow adaptation of spindle input is needed to maintain posture. Similarly, input from nociceptors provides a warning that would be lost if the receptor adapted rapidly.

NEUROLOGIC EXAM

The sensory component of a neurologic exam includes an assessment of sensory modalities including touch, proprioception, vibratory sense, and pain. Cortical sensory function can be tested by placing familiar objects in a patient's hands and asking him or her to identify it with the eyes closed. Clinical Box 8–2 describes some of the common assessments made in a neurologic exam.

PAIN

One of the most common reasons an individual seeks the advice of a clinician is because he or she is in pain. Painful stimuli generally initiate potent withdrawal and avoidance responses. Pain differs from other sensations in that it warns that something is wrong, preempts other signals, and is associated with an unpleasant

affect. Pain is complex because when tissue is damaged, central nociceptive pathways are sensitized and reorganized, which leads to persistent or **chronic pain** **(see Clinical Box 8–3)**.

CLINICAL BOX 8–3

Chronic Pain

A 2009 report in *Scientific American* indicated that 10–20% of the US and European populations experience **chronic pain;** 59% of these individuals are women. Based on a survey of primary care clinicians, only 15% indicated that they felt comfortable treating patients with chronic pain; and 41% said they waited until patients specifically requested opioid pain killers before prescribing them. Nearly 20% of adults with chronic pain indicated that they have visited an alternative medicine therapist. Risk factors for chronic neck and back pain include aging, being female, anxiety, repetitive work, obesity, depression, heavy lifting, and nicotine use. One example of chronic pain is **neuropathic pain** that occurs when nerve fibers are injured. Nerve damage can cause an inflammatory response due to activation of microglia in the spinal cord. Neuropathic pain is excruciating and a difficult condition to treat. The resulting pain lasts much longer than the injury itself. For example, in **causalgia,** a spontaneous burning pain occurs long after seemingly trivial injuries. The pain is often accompanied by **hyperalgesia** and **allodynia. Reflex sympathetic dystrophy** is often present as well. In this condition, the skin in the affected area is thin and shiny, and there is increased hair growth. This results from sprouting and eventual overgrowth of sympathetic nerve fibers into the dorsal root ganglia of the sensory nerves from the injured area. Sympathetic discharge then brings on pain. Thus, it appears that the periphery has been short-circuited and that the relevant altered fibers are being stimulated by norepinephrine at the dorsal root ganglion level.

THERAPEUTIC HIGHLIGHTS

Chronic pain is often refractory to most conventional therapies such as **nonsteroidal anti-inflammatory drugs** and **opioids.** In new efforts to treat chronic pain, some therapies focus on synaptic transmission in central nociceptive pathways and peripheral sensory transduction mechanisms. **TRPV1,** a capsaicin receptor, is activated by noxious stimuli such as heat, protons, and products of inflammation. **Capsaicin transdermal patches** or

creams reduce pain by exhausting the supply of substance P in nerves. Nav1.8 (a tetrodotoxin-resistant voltage-gated sodium channel) is uniquely associated with nociceptive neurons in dorsal root ganglia. **Lidocaine** and **mexiletine** are useful in some cases of chronic pain and may act by blocking this channel. **Ziconotide,** a voltage-gated N-type Ca^{2+} channel blocker, has been approved for intrathecal analgesia in patients with refractory chronic pain. **Gabapentin** is an anticonvulsant drug that is an analog of GABA; it has been shown to be effective in treatment of neuropathic and **inflammatory pain** by acting on voltage-gated Ca^{2+} channels. Two other anticonvulsant drugs, **topiramate** and **valproate (valproic acid),** block voltage-gated Na^{+} channels and are used to treat **migraine** headaches. **NMDA (N-methyl-D-aspartate) receptor antagonists** can be co-administered with an opioid to reduce tolerance to an opioid. Endogenous **cannabinoids** have analgesic actions in addition to their euphoric effects. G-protein-coupled receptor (CB_2) receptor agonists that are devoid of euphoric effects are under development for the treatment of inflammatory and neuropathic pain.

CLASSIFICATION OF PAIN

For scientific and clinical purposes, **pain** is defined by the International Association for the Study of Pain (IASP) as, “an unpleasant sensory and emotional experience associated with actual or potential tissue damage or described in terms of such damage.” This is to be distinguished from the term **nociception** that the IASP defines as the unconscious activity induced by a harmful stimulus applied to sense receptors.

Pain is frequently classified as **physiologic** or **acute pain** and **pathologic** or **chronic pain.** Acute pain typically has a sudden onset and recedes during the healing process; it can be regarded as “good pain” as it serves an important protective mechanism. The withdrawal reflex (Chapter 12) is an example of the expression of this protective role of pain. Chronic pain can be considered “bad pain” because it persists long after recovery from an injury and is often refractory to common analgesic agents, including **nonsteroidal anti-inflammatory drugs (NSAIDs**) and **opioids.** Chronic pain can result from inflammation **(inflammatory pain)** or nerve injury (**neuropathic pain**) including diabetic neuropathy, toxin-induced nerve damage, and ischemia (see Clinical Box 8–3).

HYPERALGESIA & ALLODYNIA

Pain is often accompanied by **hyperalgesia** and **allodynia.** Hyperalgesia is an exaggerated response to a noxious stimulus, and allodynia is a sensation of pain in response to a normally innocuous stimulus. An example of the latter is the painful sensation from a warm shower when the skin is damaged by sunburn.

Hyperalgesia and allodynia signify increased sensitivity of nociceptive afferent fibers. **Figure 8–5** shows how chemicals released at the site of injury can further directly activate receptors on sensory nerve endings leading to inflammatory pain. Injured tissues also release chemicals such as K^+ that directly depolarize nerve terminals, making nociceptors more responsive **(sensitization).** Injured tissues also release bradykinin and substance P to further sensitize nociceptive terminals. Histamine is released from mast cells, serotonin (5-HT) from platelets, and prostaglandins from cell membranes, all contributing to the inflammatory process and they activate or sensitize the nociceptors. Some released substances act by releasing another one (eg, bradykinin activates both Aδ and C nerve endings and increases synthesis and release of prostaglandins). Prostaglandin E_2 (a cyclooxygenase metabolite of arachidonic acid) is released from damaged cells and produces hyperalgesia. This explains why aspirin and other NSAIDs (nonselective inhibitors of cyclooxygenase) alleviate pain.

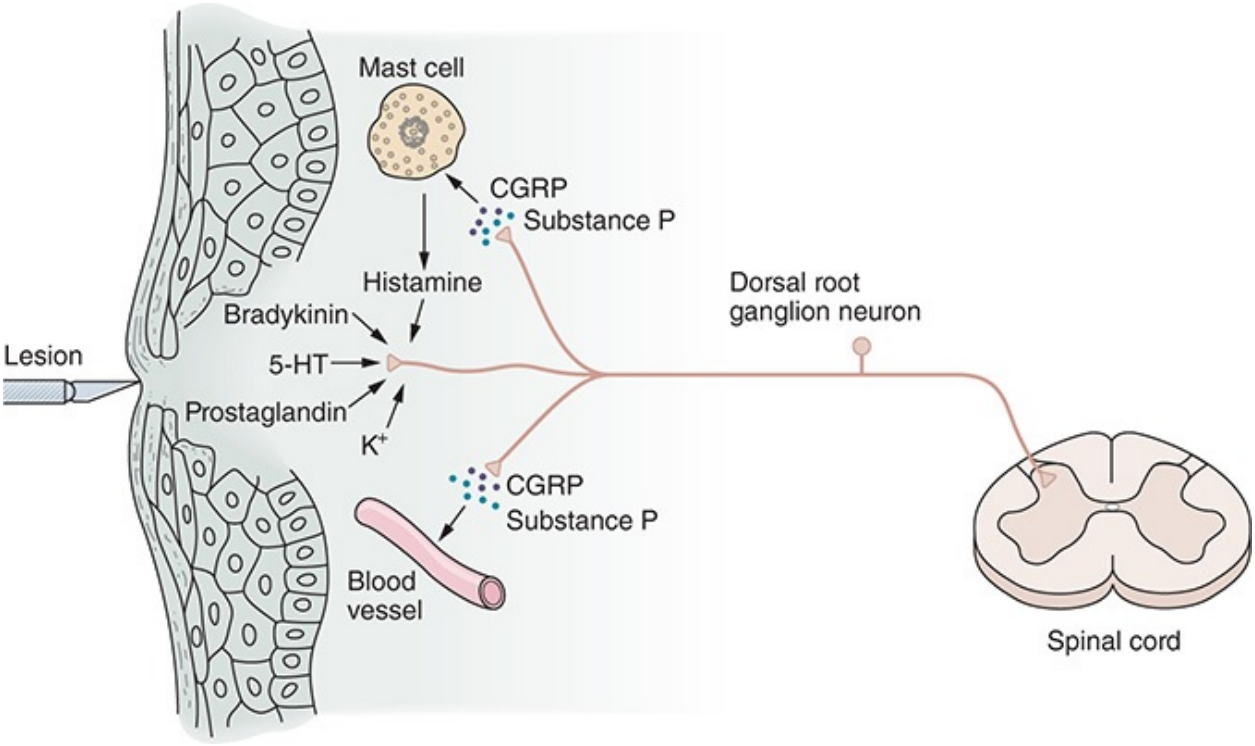


FIGURE 8–5 Chemical mediators are released in response to tissue damage and can sensitize or directly activate nociceptors. These factors contribute to hyperalgesia and allodynia. Tissue injury releases bradykinin and prostaglandins that sensitize or activate nociceptors, which in turn release substance P and calcitonin gene-related peptide (CGRP). Substance P acts on mast cells to cause degranulation and release histamine, which activates nociceptors. Substance P causes plasma extravasation and CGRP dilates blood vessels; the resulting edema causes additional release of bradykinin. Serotonin (5-HT) is released from platelets and activates nociceptors. (Reproduced with permission from Lembeck F: CIBA Foundation Symposium. Summit, NJ: Pitman Medical; 1981.)

In addition to sensitization of nerve endings by chemical mediators, several other changes occur within the periphery and the CNS that can contribute to chronic pain. The NGF released by tissue damage is picked up by nerve terminals and transported retrogradely to cell bodies in dorsal root ganglia where it can alter gene expression. Transport may be facilitated by the activation of TrkA receptors on the nerve endings. In the dorsal root ganglia, the NGF increases production of substance P and converts non-nociceptive neurons to nociceptive neurons (a phenotypic change). The NGF also influences expression of a tetrodotoxin-resistant voltage-gated sodium channel (**Nav1.8**) on dorsal root ganglia cells, further increasing activity.

Damaged nerve fibers undergo sprouting; fibers from touch receptors synapse on spinal dorsal horn neurons that normally receive only nociceptive input (see below). This can explain why innocuous stimuli can induce pain after injury. The combined release of substance P and glutamate from nociceptive afferents in the spinal cord causes excessive activation of **NMDA receptors** on spinal neurons, a phenomenon called “wind-up” that leads to increased activity in pain transmitting pathways. Another change in the spinal cord is due to the activation of **microglia** near afferent nerve terminals in the spinal cord by the release of transmitters from sensory afferents. This, in turn, leads to the release of pro-inflammatory cytokines and chemokines that modulate pain processing by affecting presynaptic release of neurotransmitters and postsynaptic excitability. There are P2X receptors on microglia; antagonists of these receptors may be a useful therapy for treatment of chronic pain.

DEEP & VISCERAL PAIN

The main difference between superficial (cutaneous) and deep or visceral pain is

the nature of the pain evoked by noxious stimuli. This may be due to a relative deficiency of Aδ nerve fibers in deep structures, so there is little rapid, sharp pain. Also, deep pain and visceral pain are poorly localized, nauseating, and frequently accompanied by sweating and changes in blood pressure. Muscle spasms can result from injuries to bones, tendons, and joints. The steadily contracting muscles become ischemic, and ischemia stimulates the pain receptors in the muscles. The pain in turn initiates more spasms, setting up a vicious cycle.

Nociceptors are present in visceral organs, but they are more sparsely distributed than in somatic structures. Afferent fibers from visceral structures reach the CNS via sympathetic and parasympathetic nerves. Their cell bodies are in the dorsal root ganglia and cranial nerve ganglia. Specifically, there are visceral afferents in the facial, glossopharyngeal, and vagus nerves and in the thoracic, upper lumbar, and sacral dorsal roots.

Visceral pain can be severe. The receptors in the walls of the viscera are especially sensitive to distension of these organs. **Intestinal colic** is a pain that waxes and wanes due to muscle spasms that occur after an intestinal obstruction. When a visceral organ is inflamed, or hyperemic, relatively minor stimuli cause severe pain, a form of hyperalgesia.

REFERRED PAIN

Visceral pain often radiates or is referred to other areas. Irritation of a visceral organ frequently produces pain that is felt not at that site but in a somatic structure that may be some distance away **(referred pain).** Knowledge of the common sites of pain referral from each of the visceral organs is of importance to a clinician. One of the best-known examples is referral of cardiac pain to the inner aspect of the left arm. Other examples include pain in the tip of the shoulder caused by irritation of the central portion of the diaphragm and pain in the testicle due to distension of the ureter. Sites of reference are not stereotyped, and unusual reference sites occur with considerable frequency. Ischemic cardiac pain, for instance, may be referred to the right arm, the abdominal region, or even the back, neck, or jaw.

When pain is referred, it is usually to a structure that developed from the same embryonic segment or dermatome as the structure in which the pain originates. For example, the heart and the arm have the same segmental origin, and the testicle migrated with its nerve supply from the primitive urogenital ridge from which the kidney and ureter also developed.

The basis for referred pain may be convergence of somatic and visceral pain fibers on the same second-order neurons in the dorsal horn that project to the thalamus and then to the somatosensory cortex **(Figure 8–6)**. This is called the **convergence-projection theory.** Somatic and visceral neurons converge in the ipsilateral dorsal horn. When the visceral stimulus is prolonged, facilitation of activity from the somatic fiber endings occurs. They now stimulate the second-order neurons, and of course the brain cannot determine whether the stimulus came from the viscera or from the area of referral.

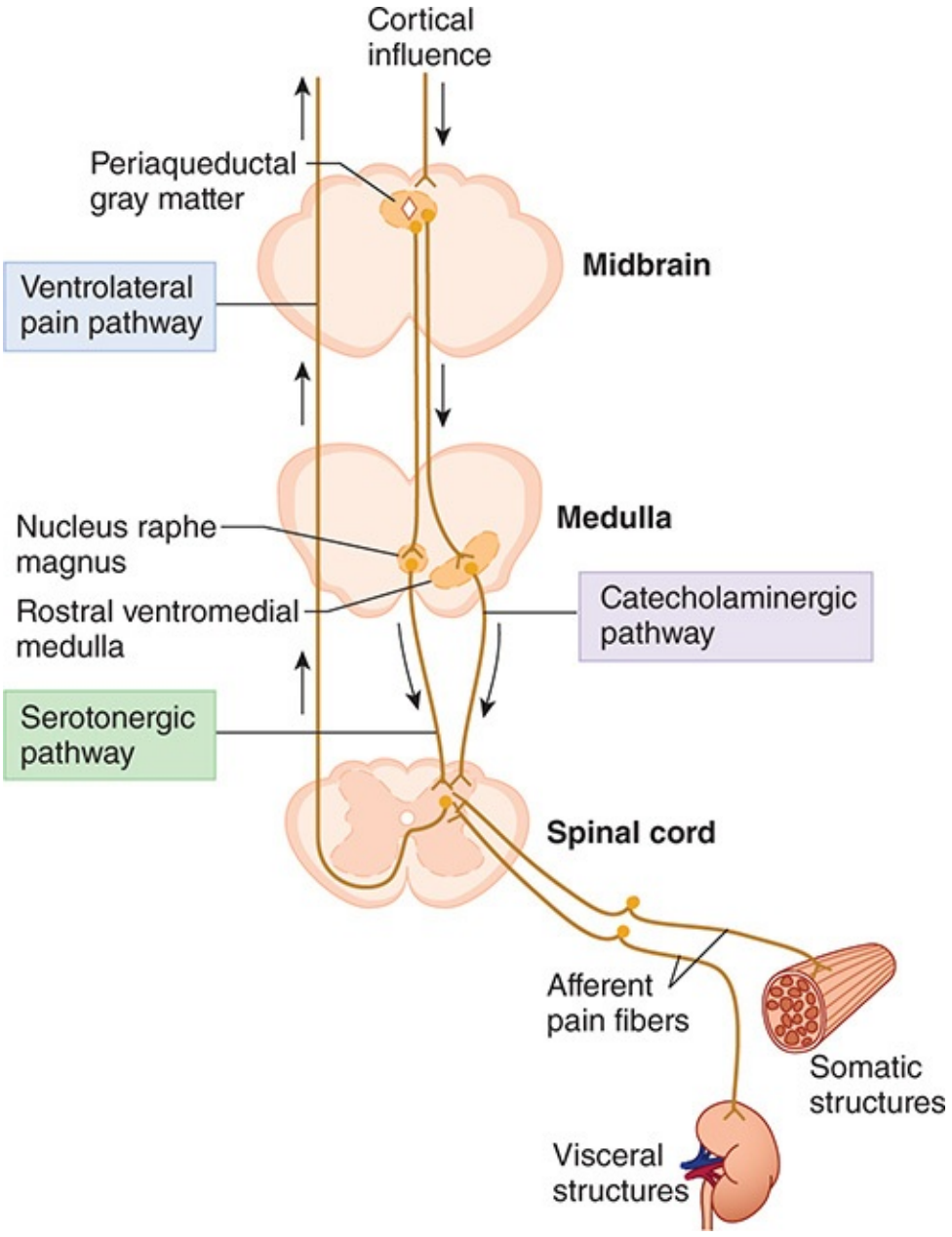


FIGURE 8–6 Schematic illustration of the convergence-projection theory

for referred pain and descending pathways involved in pain control. The basis for referred pain may be convergence of somatic and visceral pain fibers on the same second-order neurons in the dorsal horn of the spinal cord that project higher brain regions. The periaqueductal gray is a part of a descending pathway that includes serotonergic neurons in the nucleus raphe magnus and catecholaminergic neurons in the rostral ventromedial medulla to modulate pain transmission by inhibition of primary afferent transmission in the dorsal horn.

SOMATOSENSORY PATHWAYS

The sensation evoked by impulses generated in a sensory receptor depends in part on the specific part of the brain they ultimately activate. Below is a comparison of the ascending sensory pathway that mediates touch, vibratory sense, and proprioception **(dorsal column** or **medial lemniscal pathway)** and that which mediates pain and temperature **(ventrolateral spinothalamic pathway).**

DORSAL COLUMN PATHWAY

The principal pathways to the cerebral cortex for touch, vibratory sense, and proprioception are shown in **Figure 8–7**. Fibers mediating these sensations ascend ipsilaterally in the dorsal columns of the spinal cord to the medulla, where they synapse in the **gracilus** and **cuneate nuclei.** The second-order neurons from these nuclei cross the midline and ascend in the **medial lemniscus** to end in the contralateral **ventral posterior lateral (VPL) nucleus**. This ascending system is called the dorsal column or medial lemniscal system. The fibers within the dorsal column pathway are joined in the brainstem by fibers mediating sensation from the head. Touch and proprioception from the head are relayed mostly via the main sensory and mesencephalic nuclei of the trigeminal nerve.

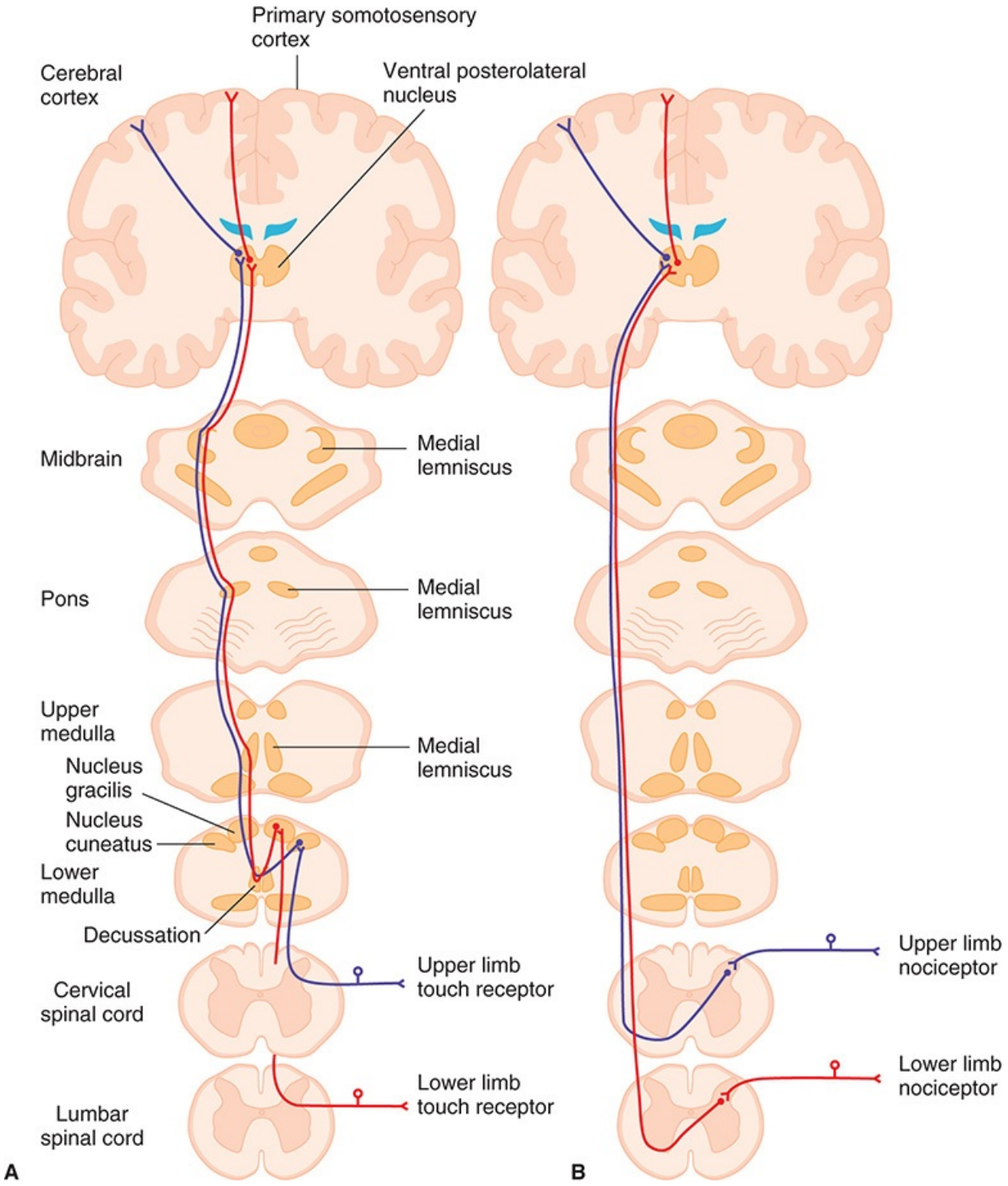


FIGURE 8–7 Ascending tracts carrying sensory information from peripheral receptors to the cerebral cortex. A) Dorsal column pathway mediates touch, vibratory sense, and proprioception. Sensory fibers ascend ipsilaterally via the spinal dorsal columns to medullary gracilus and cuneate nuclei; from there the fibers cross the midline and ascend in the medial

lemniscus to the contralateral thalamic ventral posterior lateral (VPL) and then to the primary somatosensory cortex. **B)** Ventrolateral spinothalamic tract mediates pain and temperature. These sensory fibers terminate in the dorsal horn and projections from there cross the midline and ascend in the ventrolateral quadrant of the spinal cord to the VPL and then to the primary somatosensory cortex.

Somatotopic Organization

Within the dorsal columns, fibers arising from different levels of the cord are somatotopically organized (Figure 8–7). Specifically, fibers from the sacral cord are positioned most medially and those from the cervical cord are positioned most laterally. This arrangement continues in the medulla with lower body (eg, foot) representation in the gracilus nucleus and upper body (eg, finger) representation in cuneate nucleus. The medial lemniscus is organized dorsal to ventral representing from neck to foot.

Somatotopic organization continues through the thalamus and cortex. The VPL thalamic neurons carrying sensory information project in a highly specific way to the **primary somatosensory cortex** in the **postcentral gyrus** of the **parietal lobe** **(Figure 8–8)**. The arrangement of projections to this region is such that the parts of the body are represented in order along the postcentral gyrus, with the legs on top and the head at the foot of the gyrus. Not only is there detailed localization of the fibers from the various parts of the body in the postcentral gyrus but also the size of the cortical receiving area for impulses from a part of the body is proportional to the use of the part. The relative sizes of the cortical receiving areas are shown dramatically in **Figure 8–9**, in which the proportions of the **homunculus** have been distorted to correspond to the size of the cortical receiving areas for each. Note that the cortical areas for sensation from the trunk and back are small, whereas very large areas are concerned with impulses from the hand and the parts of the mouth concerned with speech.